



**Simulation Interoperability
Standards Organization**

"Simulation Interoperability & Reuse through Standards"

SISO-STD-001.1-2025

**Standard for
Real-time Platform Reference
Federation Object Model**

Annex B

Version 3.0

4 November 2025

**Real-time Platform Reference
Federation Object Model
Product Development Group**

This is an **informative** document.

The RPR FOM 3.0 in HLA 1516-2010 modular format is the normative version of SISO-STD-001.1-2025.

SISO-STD-001.1-2025

Standard for Real-time Platform Reference Federation Object Model

Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc.

7901 4th St N, Suite 300-4043
St Petersburg, FL 33702, USA

All rights reserved.

Permission is hereby granted for this document to be used for production of both commercial and non-commercial products. Removal of this copyright statement and claiming rights to this document is prohibited. In addition, permission is hereby granted for this document to be distributed in its original or modified format (e.g., as part of a database) provided that no charge is invoked for the provision. Modification only applies to format and does not apply to the content of this document.

SISO Inc. Board of Directors
7901 4th St N, Suite 300-4043
St. Petersburg, FL 33702, USA

Table of contents

<u>Foundation</u>	1
This module contains the foundation for all other RPR FOM modules, including generic datatypes that can be considered independent from the RPR FOM purpose.	
<u>Enumerations</u>	7
This module collects all enumerations (Enumerated data types) used in RPR FOM modules that are defined in "Enumerations for Simulation Interoperability" reference document (SISO-REF-010).	
<u>Base</u>	502
This module provides a common base for RPR based FOM modules. It contains common datatypes and the BaseEntity and EmbeddedSystem object class definitions.	
<u>Physical</u>	548
This module provides object class definitions for representing physical entities including aircraft, ground vehicles, ships, life forms, ammunition, etc. In addition, it provides interaction classes to signal collisions between physical entities.	
<u>Aggregate</u>	705
This module provides the object class definition for representing aggregates of entities.	
<u>Synthetic Environment</u>	716
The Synthetic Environment FOM module relates to the simulation of environmental information both under the form of (point, linear, areal) objects and processes.	
<u>Minefield</u>	846
The Minefield FOM module relates to the simulation of minefields at both an aggregate and individual level simultaneously. In addition, it implements the two modes defined for exchanging the data of individual mines within minefields, depending on the number of mines.	

<u>Communication</u>	882
The Communication module is used to simulate radio transmitters and receivers as well as the radio signals that are transmitted between them.	
<u>Distributed Emission Regeneration</u>	908
Represent electromagnetic emissions using the regeneration principle.	
<u>Underwater Acoustics</u>	1052
Definitions of the objects and interactions required to express acoustic signatures of surface and subsurface vessels for passive sonar sensors. Includes sounds produced as byproducts of propulsion and other ship board systems and activities as well as sounds produced by acoustic transducers for active sonar sensing and depth measurement.	
<u>Warfare</u>	1070
Represents the firing of munitions and expendables and detonation, explosion, burst, or ignition of munitions, non-munitions, and expendables.	
<u>Logistics</u>	1091
The Logistics module defines interactions that represent repair and resupply logistic services.	
<u>Simulation Management</u>	1102
Simulation Management interactions.	
<u>Information Operations</u>	1156
Information operations interactions.	
<u>Switches</u>	1167
This module contains the switches, required by the HLA standard to be part of a complete FOM.	

1. Module Foundation



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	This module contains the foundation for all other RPR FOM modules, including generic datatypes that can be considered independent from the RPR FOM purpose.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

References

Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

1.1. Datatypes

1.1.1. Basic Datatypes

RPRunsignedInteger16BE

Size in bits: 16

Interpretation: *Integer in the range $[0, 2^{16}-1]$*

Endian: Big

Encoding: *16-bit unsigned integer.*

RPRunsignedInteger32BE

Size in bits: 32

Interpretation: *Integer in the range $[0, 2^{32}-1]$*

Endian: Big

Encoding: *32-bit unsigned integer.*

RPRunsignedInteger64BE

Size in bits: 64

Interpretation: *Integer in the range $[0, 2^{64}-1]$*

Endian: Big

Encoding: *64-bit unsigned integer.*

RPRunsignedInteger8BE

Size in bits: 8

Interpretation: *Integer in the range $[0, 2^8-1]$*

Endian: Big

Encoding: *8-bit unsigned integer.*

1.1.2. Enumerated Datatypes

RPRboolean

Representation: HLAoctet

Semantics: Standard Boolean type used for 8-bit compatibility.

Enumerator	Value
False	0
True	1

1.1.3. Array Datatypes

RPRnoData

Element HLAbyte
Type:
Cardinality: 0
Encoding: HLAfixedArray
Semantics: *Indicates that no data is provided.*

RTIobjectId [RPRnoteFoundation1](#)

Element HLAunicodeChar
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *An RTI object instance identification string.*

RTIobjectIdArray

Element [RTIobjectId](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Set of ID's of registered object instances.*

1.2. Notes

RPRnoteFoundation1

Semantics: *This is the unique ObjectName associated with each object instance. The user can define the name to be used in the registerObjectInstance RTI call. If user does not define the name then the RTI will generate a unique name for the object. RTI generated names may be fairly long, so federations wishing to conserve bandwidth may wish to implement their own object naming scheme.*

The ObjectName is provided by the RTI in the discoverObjectInstance call. The user can also obtain the ObjectName for a particular object instance using the getObjectInstanceName call.

2. Module Enumerations



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	This module collects all enumerations (Enumerated data types) used in RPR FOM modules that are defined in "Enumerations for Simulation Interoperability" reference document (SISO-REF-010).
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012
Text Document	Reference for Enumerations for Simulation Interoperability Version 35 27 April 2025

Dependencies

Foundation

2.1. Datatypes

2.1.1. Enumerated Datatypes

APAPParameterIndexAPAStatusEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *APA Parameter Index-APA Status [UID 281]*

Enumerator	Value
Deselected_Off	0
APA_Value_Change_Only	1
State_Change	2
Record_Activation	3

AcknowledgeAcknowledgeFlagEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Acknowledge-Acknowledge Flag [UID 69]*

Enumerator	Value
Create_Entity	1
Remove_Entity	2
Start_Resume	3
Stop_Freeze	4
Transfer_Ownership	5

AcknowledgeResponseFlagEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Acknowledge-Response Flag [UID 70]*

Enumerator	Value
Other	0
Able_to_comply	1
Unable_to_comply	2

Enumerator	Value
Pending_Operator_Action	3

ActionRequestActionIDEnum32

Representation: [RPRunsignedInteger32BE](#)

Semantics: *Action Request-Action ID [UID 71]*

Enumerator	Value
Other	0
Local_storage_of_the_requested_information	1
Inform_SM_of_event_ran_out_of_ammunition_	2
Inform_SM_of_event_killed_in_action_	3
Inform_SM_of_event_damage_	4
Inform_SM_of_event_mobility_disabled_	5
Inform_SM_of_event_fire_disabled_	6
Inform_SM_of_event_ran_out_of_fuel_	7
Recall_checkpoint_data	8
Recall_initial_parameters	9
Initiate_tether_lead	10
Initiate_tether_follow	11
Untether	12
Initiate_service_station_resupply	13
Initiate_tailgate_resupply	14
Initiate_hitch_lead	15
Initiate_hitch_follow	16
Unhitch	17
Mount	18
Dismount	19
Start_DRC_Daily_Readiness_Check_	20
Stop_DRC	21

Enumerator	Value
Data_Query	22
Status_Request	23
Send_Object_State_Data	24
Reconstitute	25
Lock_Site_Configuration	26
Unlock_Site_Configuration	27
Update_Site_Configuration	28
Query_Site_Configuration	29
Tethering_Information	30
Mount_Intent	31
Accept_Subscription	33
Unsubscribe	34
Teleport_entity	35
Change_aggregate_state	36
Request_Start_PDU	37
Wakeup_get_ready_for_initialization	38
Initialize_internal_parameters	39
Send_plan_data	40
Synchronize_internal_clocks	41
Run	42
Save_internal_parameters	43
Simulate_malfunction	44
Join_exercise	45
Resign_exercise	46
Time_advance	47
TACCSF_LOS_Request_Type_1	100
TACCSF_LOS_Request_Type_2	101
Airmount_Mount_Request	4303

Enumerator	Value
Airmount_Dismount_Request	4304
Airmount_Information_Request	4305

ActionResponseRequestStatusEnum32

Representation: [RPRunsignedInteger32BE](#)

Semantics: *Action Response-Request Status [UID 72]*

Enumerator	Value
Other	0
Pending	1
Executing	2
Partially_Complete	3
Complete	4
Request_rejected	5
Retransmit_request_now	6
Retransmit_request_later	7
Invalid_time_parameters	8
Simulation_time_exceeded	9
Request_done	10
TACCSF_LOS_Reply_Type_1	100
TACCSF_LOS_Reply_Type_2	101
Join_Exercise_Request_Rejected	201

AddSpecificDimensionEnumerationsForSubsurfaceAreaSizeEnum8 [RPRnoteEnumerations1](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Add Specific Dimension Enumerations for Subsurface Area Size [UID 133]*

Enumerator	Value
Small_School_5km_long_x_250m_wide_x_83m_deep_1km_x_5km_x_166m_	222
Small_School_Dense_5km_long_x_250m_wide_x_83m_deep_1km_x_5km_x_166m_	223
Medium_School_5km_x_1km_x_125m_1km_x_2km_x_500m_	224

Enumerator	Value
Medium_School_Dense_5km_x_1km_x_125m_1km_x_2km_x_500m_	225
Large_School_5km_x_2km_x_500m_10km_x_4km_x_1km_	226
Large_School_Dense_5km_x_2km_x_500m_10km_x_4km_x_1km_	227

AddVariantsForAirCategory201InsectEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Add Variants for Air Category 201 - Insect [UID 137]*

Enumerator	Value
Vertical_Shaped_Insect_Swarm	1
Circular_Shaped_Insect_Swarm	2
Irregular_Shaped_Insect_Swarm	3

AddVariantsForLandCategory200MammalEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Add Variants for Land Category 200 - Mammal [UID 134]*

Enumerator	Value
Animal_with_a_Male_Child_Rider	1
Animal_with_a_Female_Child_Rider	2
Animal_with_an_Adult_Male_Rider	3
Animal_with_an_Adult_Female_Rider	4
Animal_Harnessed_to_a_Plow	5
Animal_Harnessed_to_a_Cart	6

AddVariantsForSubsurfaceCategoriesFishMolluskCrustaceanAndInsectEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Add Variants for Subsurface Categories Fish, Mollusk, Crustacean, and Insect [UID 138]*

Enumerator	Value
Black	1
Green	2

Enumerator	Value
Spotted	3
Red	4
Brown	5
Blue	6
Silver	7
Grey	8

AggregateStateAggregateKindEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Aggregate State-Aggregate Kind [UID 206]*

Enumerator	Value
Other	0
Military_Hierarchy	1
Common_Type	2
Common_Mission	3
Similar_Capabilities	4
Common_Location	5

AggregateStateAggregateStateEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Aggregate State-Aggregate State [UID 204]*

Enumerator	Value
Other	0
Aggregated	1
Disaggregated	2
Fully_disaggregated	3
Pseudo_disaggregated	4
Partially_disaggregated	5

AggregateStateFormationEnum32

Representation: [RPRunsignedInteger32BE](#)

Semantics: *Aggregate State-Formation [UID 205]*

Enumerator	Value
Other	0
Assembly	1
Vee	2
Wedge	3
Line	4
Column	5

AggregateStateSpecificEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Aggregate State-Specific [UID 209]*

Enumerator	Value
No_headquarters	0
Yes_aggregate_unit_contains_a_headquarters	1

AggregateStateSubcategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Aggregate State-Subcategory [UID 208]*

Enumerator	Value
Other	0
Cavalry_Troop	1
Armor	2
Infantry	3
Mechanized_Infantry	4
Cavalry	5
Armored_Cavalry	6

Enumerator	Value
Artillery	7
Self_Propelled_Artillery	8
Close_Air_Support	9
Engineer	10
Air_Defense_Artillery	11
Anti_Tank	12
Army_Aviation_Fixed_wing	13
Army_Aviation_Rotary_wing	14
Army_Attack_Helicopter	15
Air_Cavalry	16
Armor_Heavy_Task_Force	17
Motorized_Rifle	18
Mechanized_Heavy_Task_Force	19
Command_Post	20
CEWI	21
Tank_only	22

AircraftAddressSourceEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Aircraft Address Source [UID 364]*

Enumerator	Value
Mode_S_Aircraft_Address_Field_Value	0
GICB_IFF_Data_Record_Available	1

AircraftIDSourceEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Aircraft ID Source [UID 801]*

Enumerator	Value
Mode_S_Aircraft_Identification_Field_Value	0

Enumerator	Value
GICB_IFF_Data_Record_Available	1

AircraftIdentificationTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Aircraft Identification Type [UID 357]*

Enumerator	Value
No_Statement	0
Flight_Number	1
Tail_Number	2

AircraftPresentDomainEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Aircraft Present Domain [UID 356]*

Enumerator	Value
No_Statement	0
Airborne	1
On_Ground_Surface RPRnoteEnumerations2	2

AircraftTypeWakeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize03](#)

Semantics: *Aircraft Type/Wake [UID 368]*

Enumerator	Value
No_Statement	0

AnimalLifeFormGroupSizeRangeEnumerationForAllDomainsEnum8 [RPRnoteEnumerations1](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Animal Life Form Group Size Range Enumeration for all Domains [UID 130]*

Enumerator	Value
Number_of_animals_range_from_201_to_249	201

Enumerator	Value
Number_of_animals_range_from_250_to_299	202
Number_of_animals_range_from_300_to_399	203
Number_of_animals_range_from_400_to_499	204
Number_of_animals_range_from_500_to_999	205
Number_of_animals_range_from_1_000_to_1_499	206
Number_of_animals_range_from_1_500_to_1_999	207
Number_of_animals_range_from_2_000_to_2_999	208
Number_of_animals_range_from_3_000_to_4_999	210
Number_of_animals_range_from_5_000_to_6_999	212
Number_of_animals_range_from_7_000_to_9_999	214
Number_of_animals_range_from_10_000_to_19_999	216
Number_of_animals_range_from_20_000_to_50_000	218
Number_of_animals_range_greater_than_50_000	220

AntennaSelectionEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize02

Semantics: *Antenna Selection [UID 351]*

Enumerator	Value
No_Statement	0
Top	1
Bottom	2
Diversity	3

AntennaStatusEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Antenna Status [UID 371]*

Enumerator	Value
No_Statement	0
Not_Able_to_Emit	1

Enumerator	Value
Able_to_Emit	2

AppearanceAntiCollisionDayNightEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Appearance-Anti-Collision Day/Night [UID 397]*

Enumerator	Value
Day	0
Night	1

AppearanceCamouflageTypeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Appearance-Camouflage Type [UID 384]*

Enumerator	Value
Desert_Camouflage	0
Winter_Camouflage	1
Forest_Camouflage	2
Other	3

AppearanceCanopyEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize03](#)

Semantics: *Appearance-Canopy [UID 387]*

Enumerator	Value
Not_Applicable	0
Single_Canopy_Single_Troop_Door_Closed	1
Front_and_Rear_Canopy_Left_and_Right_Troop_Door_Closed	2
Front_Canopy_Left_Troop_Door_Open	3
Single_Canopy_Single_Troop_Door_Open	4
Rear_Canopy_Right_Troop_Door_Open	5
Front_and_Rear_Canopy_Left_and_Right_Troop_Door_Open	6

AppearanceConcealedMovementEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Appearance-Concealed Movement [UID 394]*

Enumerator	Value
Open_Movement	0
Rushes_Between_Covered_Positions	1

AppearanceConcealedPositionEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Appearance-Concealed Position [UID 385]*

Enumerator	Value
Not_Concealed	0
Prepared_Concealed_Position	1

AppearanceDamageEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Appearance-Damage [UID 379]*

Enumerator	Value
No_Damage	0
Slight_Damage	1
Moderate_Damage	2
Destroyed	3

AppearanceEntityOrObjectStateEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Appearance-Entity or Object State [UID 386]*

Enumerator	Value
Active	0
Deactivated	1

AppearanceEnvironmentalDensityEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize04

Semantics: *Appearance-Environmental Density [UID 395]*

Enumerator	Value
Clear	0
Hazy	1
Dense	2
Very_Dense	3
Opaque	4

AppearanceHatchEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize03

Semantics: *Appearance-Hatch [UID 382]*

Enumerator	Value
Not_Applicable	0
Closed	1
Popped	2
Popped_and_Person_Is_Visible	3
Open	4
Open_and_Person_Is_Visible	5

AppearanceLauncherOperationalEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize01

Semantics: *Appearance-Launcher/Operational [UID 383]*

Enumerator	Value
Not_Raised_Not_Operational	0
Raised_Operational	1

AppearanceLifeFormComplianceStatusEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize04](#)Semantics: *Appearance-Life Form Compliance Status [UID 391]*

Enumerator	Value
Not_Specified	0
Detained	1
Surrender	2
Using_Fists	3
Verbal_Abuse_Level_1	4
Verbal_Abuse_Level_2	5
Verbal_Abuse_Level_3	6
Passive_Resistance_Level_1	7
Passive_Resistance_Level_2	8
Passive_Resistance_Level_3	9
Using_Non_Lethal_Weapon_1	10
Using_Non_Lethal_Weapon_2	11
Using_Non_Lethal_Weapon_3	12
Using_Non_Lethal_Weapon_4	13
Using_Non_Lethal_Weapon_5	14
Using_Non_Lethal_Weapon_6	15

AppearanceLifeFormHealthEnum8Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)Semantics: *Appearance-Life Form Health [UID 390]*

Enumerator	Value
No_Injury	0
Slight_Injury	1
Moderate_Injury	2
Fatal_Injury	3

AppearanceLifeFormPostureEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize04

Semantics: *Appearance-Life Form Posture [UID 392]*

Enumerator	Value
Not_Specified	0
Upright_Standing_Still	1
Upright_Walking	2
Upright_Running	3
Kneeling	4
Prone	5
Crawling	6
Swimming	7
Parachuting	8
Jumping	9
Sitting	10
Squatting	11
Crouching	12
Wading	13
Surrender	14
Detained	15

AppearanceLifeFormWeaponImplementEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize02

Semantics: *Appearance-Life Form Weapon/Implement [UID 393]*

Enumerator	Value
Not_Present	0
Stowed	1
Deployed_Active	2
Firing_Position_In_Use	3

AppearanceLinearObjectLaneMarkerVisibleEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Appearance-Linear Object Lane Marker Visible [UID 410]*

Enumerator	Value
Left_Side_Is_Visible	0
Right_Side_Is_Visible	1
Both_Sides_Are_Visible	2

AppearanceLinearObjectTankDitchBreachEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Appearance-Linear Object Tank Ditch Breach [UID 409]*

Enumerator	Value
No_Breaching	0
Slight_Breaching	1
Moderate_Breached	2
Cleared	3

AppearanceNVGModeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Appearance-NVG Mode [UID 400]*

Enumerator	Value
Overt_Lighting	0
Covert_Lighting_Night_Vision_Goggles_	1

AppearanceNavigationPositionBrightnessEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Appearance-Navigation/Position Brightness [UID 398]*

Enumerator	Value
Dim	0

Enumerator	Value
Bright	1

AppearanceObjectGeneralDamageEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Appearance-Object General-Damage [UID 405]*

Enumerator	Value
No_Damage	0
Damaged	1
Destroyed	2

AppearanceObjectGeneralIEDPresentEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Appearance-Object General-IED Present [UID 411]*

Enumerator	Value
None	0
Visible	1
Partially_Hidden	2
Completely_Hidden	3

AppearanceObjectGeneralPredistributedEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Appearance-Object General-Predistributed [UID 406]*

Enumerator	Value
Object_Created_During_the_Exercise	0
Object_Predistributed_Prior_to_Exercise_Start	1

AppearanceObjectSpecificBreachStateEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Appearance-Object Specific-Breach State [UID 407]*

Enumerator	Value
No_Breaching	0
Breached	1
Cleared	2

AppearanceObjectSpecificChemicalTypeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Appearance-Object Specific-Chemical Type [UID 408]*

Enumerator	Value
Other	0
Hydrochloric	1
White_Phosphorous	2
Red_Phosphorous	3

AppearancePaintSchemeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Appearance-Paint Scheme [UID 378]*

Enumerator	Value
Uniform_Color	0
Camouflage	1

AppearanceSubsurfaceHatchEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize03](#)

Semantics: *Appearance-Subsurface Hatch [UID 388]*

Enumerator	Value
Not_Applicable	0
Hatch_Is_Closed	1
Hatch_Is_Open	4

AppearanceSupplyDeployedEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Appearance-Supply Deployed [UID 399]*

Enumerator	Value
Not_Applicable	0
Stowed	1
Deployed	2
Deployed_and_Active	3

AppearanceTrailingEffectsEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Appearance-Trailing Effects [UID 381]*

Enumerator	Value
None	0
Small	1
Medium	2
Large	3

ArticulatedPartsTypeClassEnum32

Representation: [RPRunsignedInteger32BE](#)

Semantics: *Articulated Parts-Type Class [UID 59]*

Enumerator	Value
Not_Specified	0
Rudder	1024
Left_Flap	1056
Right_Flap	1088
Left_Aileron	1120
Right_Aileron	1152
Helicopter_Main_Rotor	1184
Helicopter_Tail_Rotor	1216

Enumerator	Value
Other_Aircraft_Control_Surfaces_Defined_as_Needed	1248
Propeller_Number_1	1280
Propeller_Number_2	1312
Propeller_Number_3	1344
Propeller_Number_4	1376
Left_Stabilator_Stabilator_Number_1_	1408
Right_Stabilator_Stabilator_Number_2_	1440
Left_Ruddervator_Ruddervator_Number_1_	1472
Right_Ruddervator_Ruddervator_Number_2_	1504
Left_Leading_Edge_Flap_Slat	1536
Right_Leading_Edge_Flap_Slat	1568
Left_Elevator	1600
Right_Elevator	1632
Canard_left_	1664
Canard_right_	1696
Elevon_Inner_left_	1728
Elevon_Inner_right_	1760
Elevon_Middle_left_	1792
Elevon_Middle_right_	1824
Elevon_Outer_left_	1856
Elevon_Outer_right_	1888
Canopy_Aircraft_	1920
Spoiler_left_	1952
Spoiler_right_	1984
Periscope	2048
Generic_Antenna	2080
Snorkel	2112
Other_Extendible_Parts_Defined_as_Needed	2144

Enumerator	Value
Dive_Plane_Sail_Left	2176
Dive_Plane_Sail_Right	2208
Dive_Plane_Bow_Left	2240
Dive_Plane_Bow_Right	2272
Dive_Plane_Stern_Left	2304
Dive_Plane_Stern_Right	2336
Left_Rudder	2368
Right_Rudder	2400
Landing_Gear_Door_Nose	2976
Landing_Gear_Door_Left_Main	3008
Landing_Gear_Door_Right_Main	3040
Landing_Gear RPRnoteEnumerations3	3072
Tail_Hook	3104
Speed_Brake	3136
Left_Door_of_Primary_Weapon_Bay	3168
Right_Door_of_Primary_Weapon_Bay	3200
Tank_or_APC_Hatch	3232
Wingsweep	3264
Bridge_Launcher	3296
Bridge_Section_1	3328
Bridge_Section_2	3360
Bridge_Section_3	3392
Primary_Blade_1	3424
Primary_Blade_2	3456
Primary_Boom	3488
Primary_Launcher_Arm	3520
Other_Fixed_Position_Parts_Defined_as_Needed	3552
Landing_Gear_Nose	3584

Enumerator	Value
Landing_Gear_Left_Main	3616
Landing_Gear_Right_Main	3648
Doors_of_Left_Side_Weapon_Bay	3680
Doors_of_Right_Side_Weapon_Bay	3712
Spot_Search_Light_1	3744
Spot_Search_Light_2	3776
Spot_Search_Light_3	3808
Spot_Search_Light_4	3840
Landing_Light	3872
Primary_Turret_Number_1	4096
Primary_Turret_Number_2	4128
Primary_Turret_Number_3	4160
Primary_Turret_Number_4	4192
Primary_Turret_Number_5	4224
Primary_Turret_Number_6	4256
Primary_Turret_Number_7	4288
Primary_Turret_Number_8	4320
Primary_Turret_Number_9	4352
Primary_Turret_Number_10	4384
Primary_Gun_Number_1	4416
Primary_Gun_Number_2	4448
Primary_Gun_Number_3	4480
Primary_Gun_Number_4	4512
Primary_Gun_Number_5	4544
Primary_Gun_Number_6	4576
Primary_Gun_Number_7	4608
Primary_Gun_Number_8	4640
Primary_Gun_Number_9	4672

Enumerator	Value
Primary_Gun_Number_10	4704
Primary_Launcher_1	4736
Primary_Launcher_2	4768
Primary_Launcher_3	4800
Primary_Launcher_4	4832
Primary_Launcher_5	4864
Primary_Launcher_6	4896
Primary_Launcher_7	4928
Primary_Launcher_8	4960
Primary_Launcher_9	4992
Primary_Launcher_10	5024
Primary_Defense_Systems_1 RPRnoteEnumerations4	5056
Primary_Defense_Systems_2	5088
Primary_Defense_Systems_3	5120
Primary_Defense_Systems_4	5152
Primary_Defense_Systems_5	5184
Primary_Defense_Systems_6	5216
Primary_Defense_Systems_7	5248
Primary_Defense_Systems_8	5280
Primary_Defense_Systems_9	5312
Primary_Defense_Systems_10	5344
Primary_Radar_1 RPRnoteEnumerations5	5376
Primary_Radar_2	5408
Primary_Radar_3	5440
Primary_Radar_4	5472
Primary_Radar_5	5504
Primary_Radar_6	5536
Primary_Radar_7	5568

Enumerator	Value
Primary_Radar_8	5600
Primary_Radar_9	5632
Primary_Radar_10	5664
Secondary_Turret_Number_1	5696
Secondary_Turret_Number_2	5728
Secondary_Turret_Number_3	5760
Secondary_Turret_Number_4	5792
Secondary_Turret_Number_5	5824
Secondary_Turret_Number_6	5856
Secondary_Turret_Number_7	5888
Secondary_Turret_Number_8	5920
Secondary_Turret_Number_9	5952
Secondary_Turret_Number_10	5984
Secondary_Gun_Number_1	6016
Secondary_Gun_Number_2	6048
Secondary_Gun_Number_3	6080
Secondary_Gun_Number_4	6112
Secondary_Gun_Number_5	6144
Secondary_Gun_Number_6	6176
Secondary_Gun_Number_7	6208
Secondary_Gun_Number_8	6240
Secondary_Gun_Number_9	6272
Secondary_Gun_Number_10	6304
Secondary_Launcher_1	6336
Secondary_Launcher_2	6368
Secondary_Launcher_3	6400
Secondary_Launcher_4	6432
Secondary_Launcher_5	6464

Enumerator	Value
Secondary_Launcher_6	6496
Secondary_Launcher_7	6528
Secondary_Launcher_8	6560
Secondary_Launcher_9	6592
Secondary_Launcher_10	6624
Secondary_Defense_Systems_1	6656
Secondary_Defense_Systems_2	6688
Secondary_Defense_Systems_3	6720
Secondary_Defense_Systems_4	6752
Secondary_Defense_Systems_5	6784
Secondary_Defense_Systems_6	6816
Secondary_Defense_Systems_7	6848
Secondary_Defense_Systems_8	6880
Secondary_Defense_Systems_9	6912
Secondary_Defense_Systems_10	6944
Secondary_Radar_1	6976
Secondary_Radar_2	7008
Secondary_Radar_3	7040
Secondary_Radar_4	7072
Secondary_Radar_5	7104
Secondary_Radar_6	7136
Secondary_Radar_7	7168
Secondary_Radar_8	7200
Secondary_Radar_9	7232
Secondary_Radar_10	7264
Deck_Elevator_1	7296
Deck_Elevator_2	7328
Catapult_1	7360

Enumerator	Value
Catapult_2	7392
Jet_Blast_Deflector_1	7424
Jet_Blast_Deflector_2	7456
Arrestor_Wires_1	7488
Arrestor_Wires_2	7520
Arrestor_Wires_3	7552
Wing_or_Rotor_Fold	7584
Fuselage_Fold	7616
Main_Cargo_Door	7648
Cargo_Ramp	7680
Air_to_Air_Refueling_Boom	7712
Primary_Aerial_Refueling_Receptacle_Door	7744
Secondary_Aerial_Refueling_Receptacle_Door	7776
Aerial_Refueling_Receptacle_Latch	7808
Cargo_Door_1	7840
Cargo_Door_2	7872
Cargo_Door_3	7904
Cargo_Door_4	7936
Cargo_Door_5	7968
Cargo_Door_6	8000
Cargo_Door_7	8032
Cargo_Door_8	8064
Cargo_Door_9	8096
Cargo_Door_10	8128
Centre_Refuelling_Drogue	8160
Port_Refuelling_Drogue	8192
Starboard_Refuelling_Drogue	8224
Submarine_Engine_Exhaust_Mast	8256

Enumerator	Value
Submarine_Mast_1 RPRnoteEnumerations6	8288
Submarine_Mast_2	8320
Submarine_Mast_3	8352
Submarine_Mast_4	8384
Submarine_Mast_5	8416
Submarine_Mast_6	8448
Submarine_Mast_7	8480
Submarine_Mast_8	8512
Submarine_Mast_9	8544
Submarine_Mast_10	8576
Vectored_Thrust_Nozzle	8608
Left_Door_of_the_Left_Weapon_Bay	8640
Right_Door_of_the_Left_Weapon_Bay	8672
Left_Door_of_the_Right_Weapon_Bay	8704
Right_Door_of_the_Right_Weapon_Bay	8736
Gun_Door	8768
Countermeasure_Door_Left	8800
Countermeasure_Door_Right	8832
Hook_Door_Forward	8864
Hook_Door_Aft	8896
Lift_Fan_Upper_Door	8928
Lift_Fan_Lower_Door_Left	8960
Lift_Fan_Lower_Door_Right	8992
Refuel_Probe_Door	9024
Left_Engine_Nacelle	9056
Right_Engine_Nacelle	9088
_1st_Left_Wheel RPRnoteEnumerations7	9120
_1st_Right_Wheel	9152

Enumerator	Value
_2nd_Left_Wheel	9184
_2nd_Right_Wheel	9216
_3rd_Left_Wheel	9248
_3rd_Right_Wheel	9280
_4th_Left_Wheel	9312
_4th_Right_Wheel	9344
_5th_Left_Wheel	9376
_5th_Right_Wheel	9408
_6th_Left_Wheel	9440
_6th_Right_Wheel	9472
_7th_Left_Wheel	9504
_7th_Right_Wheel	9536
_8th_Left_Wheel	9568
_8th_Right_Wheel	9600
_9th_Left_Wheel	9632
_9th_Right_Wheel	9664
_10th_Left_Wheel	9696
_10th_Right_Wheel	9728
Refueling_Probe	9760
Steering_Wheel	9792
Crane_Body	9824
Crane_Arm_1	9856
Crane_Arm_2	9888
Crane_Arm_3	9920
Crane_Boom	9952
Crane_Hook	9984
Trailer	10016
Roller_Left	10048

Enumerator	Value
Roller_Right	10080
Primary_Gun_Recoil	10112
Secondary_Gun_Recoil	10144

ArticulatedPartsTypeMetricEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize05

Semantics: *Articulated Parts-Type Metric [UID 58]*

Enumerator	Value
Not_Specified	0
Position	1
Position_Rate	2
Extension	3
Extension_Rate	4
X	5
X_Rate	6
Y	7
Y_Rate	8
Z	9
Z_Rate	10
Azimuth	11
Azimuth_Rate	12
Elevation	13
Elevation_Rate	14
Rotation	15
Rotation_Rate	16

AttachedPartDetachedIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Attached Part-Detached Indicator [UID 415]*

Enumerator	Value
Attached	0
Detached	1

AttachedPartsEnum32

Representation: [RPRunsignedInteger32BE](#)

Semantics: *Attached Parts [UID 57]*

Enumerator	Value
Nothing_Empty	0
Model_specific_station_1	1
Model_specific_station_2	2
Model_specific_station_3	3
Model_specific_station_4	4
Model_specific_station_5	5
Model_specific_station_6	6
Model_specific_station_7	7
Model_specific_station_8	8
Model_specific_station_9	9
Model_specific_station_10	10
Model_specific_station_11	11
Model_specific_station_12	12
Model_specific_station_13	13
Model_specific_station_14	14
Model_specific_station_15	15
Model_specific_station_16	16
Model_specific_station_17	17
Model_specific_station_18	18
Model_specific_station_19	19
Model_specific_station_20	20

Enumerator	Value
Model_specific_station_21	21
Model_specific_station_22	22
Model_specific_station_23	23
Model_specific_station_24	24
Model_specific_station_25	25
Model_specific_station_26	26
Model_specific_station_27	27
Model_specific_station_28	28
Model_specific_station_29	29
Model_specific_station_30	30
Model_specific_station_31	31
Model_specific_station_32	32
Model_specific_station_33	33
Model_specific_station_34	34
Model_specific_station_35	35
Model_specific_station_36	36
Model_specific_station_37	37
Model_specific_station_38	38
Model_specific_station_39	39
Model_specific_station_40	40
Model_specific_station_41	41
Model_specific_station_42	42
Model_specific_station_43	43
Model_specific_station_44	44
Model_specific_station_45	45
Model_specific_station_46	46
Model_specific_station_47	47
Model_specific_station_48	48

Enumerator	Value
Model_specific_station_49	49
Model_specific_station_50	50
Model_specific_station_51	51
Model_specific_station_52	52
Model_specific_station_53	53
Model_specific_station_54	54
Model_specific_station_55	55
Model_specific_station_56	56
Model_specific_station_57	57
Model_specific_station_58	58
Model_specific_station_59	59
Model_specific_station_60	60
Model_specific_station_61	61
Model_specific_station_62	62
Model_specific_station_63	63
Model_specific_station_64	64
Model_specific_station_65	65
Model_specific_station_66	66
Model_specific_station_67	67
Model_specific_station_68	68
Model_specific_station_69	69
Model_specific_station_70	70
Model_specific_station_71	71
Model_specific_station_72	72
Model_specific_station_73	73
Model_specific_station_74	74
Model_specific_station_75	75
Model_specific_station_76	76

Enumerator	Value
Model_specific_station_77	77
Model_specific_station_78	78
Model_specific_station_79	79
Model_specific_station_80	80
Model_specific_station_81	81
Model_specific_station_82	82
Model_specific_station_83	83
Model_specific_station_84	84
Model_specific_station_85	85
Model_specific_station_86	86
Model_specific_station_87	87
Model_specific_station_88	88
Model_specific_station_89	89
Model_specific_station_90	90
Model_specific_station_91	91
Model_specific_station_92	92
Model_specific_station_93	93
Model_specific_station_94	94
Model_specific_station_95	95
Model_specific_station_96	96
Model_specific_station_97	97
Model_specific_station_98	98
Model_specific_station_99	99
Model_specific_station_100	100
Model_specific_station_101	101
Model_specific_station_102	102
Model_specific_station_103	103
Model_specific_station_104	104

Enumerator	Value
Model_specific_station_105	105
Model_specific_station_106	106
Model_specific_station_107	107
Model_specific_station_108	108
Model_specific_station_109	109
Model_specific_station_110	110
Model_specific_station_111	111
Model_specific_station_112	112
Model_specific_station_113	113
Model_specific_station_114	114
Model_specific_station_115	115
Model_specific_station_116	116
Model_specific_station_117	117
Model_specific_station_118	118
Model_specific_station_119	119
Model_specific_station_120	120
Model_specific_station_121	121
Model_specific_station_122	122
Model_specific_station_123	123
Model_specific_station_124	124
Model_specific_station_125	125
Model_specific_station_126	126
Model_specific_station_127	127
Model_specific_station_128	128
Model_specific_station_129	129
Model_specific_station_130	130
Model_specific_station_131	131
Model_specific_station_132	132

Enumerator	Value
Model_specific_station_133	133
Model_specific_station_134	134
Model_specific_station_135	135
Model_specific_station_136	136
Model_specific_station_137	137
Model_specific_station_138	138
Model_specific_station_139	139
Model_specific_station_140	140
Model_specific_station_141	141
Model_specific_station_142	142
Model_specific_station_143	143
Model_specific_station_144	144
Model_specific_station_145	145
Model_specific_station_146	146
Model_specific_station_147	147
Model_specific_station_148	148
Model_specific_station_149	149
Model_specific_station_150	150
Model_specific_station_151	151
Model_specific_station_152	152
Model_specific_station_153	153
Model_specific_station_154	154
Model_specific_station_155	155
Model_specific_station_156	156
Model_specific_station_157	157
Model_specific_station_158	158
Model_specific_station_159	159
Model_specific_station_160	160

Enumerator	Value
Model_specific_station_161	161
Model_specific_station_162	162
Model_specific_station_163	163
Model_specific_station_164	164
Model_specific_station_165	165
Model_specific_station_166	166
Model_specific_station_167	167
Model_specific_station_168	168
Model_specific_station_169	169
Model_specific_station_170	170
Model_specific_station_171	171
Model_specific_station_172	172
Model_specific_station_173	173
Model_specific_station_174	174
Model_specific_station_175	175
Model_specific_station_176	176
Model_specific_station_177	177
Model_specific_station_178	178
Model_specific_station_179	179
Model_specific_station_180	180
Model_specific_station_181	181
Model_specific_station_182	182
Model_specific_station_183	183
Model_specific_station_184	184
Model_specific_station_185	185
Model_specific_station_186	186
Model_specific_station_187	187
Model_specific_station_188	188

Enumerator	Value
Model_specific_station_189	189
Model_specific_station_190	190
Model_specific_station_191	191
Model_specific_station_192	192
Model_specific_station_193	193
Model_specific_station_194	194
Model_specific_station_195	195
Model_specific_station_196	196
Model_specific_station_197	197
Model_specific_station_198	198
Model_specific_station_199	199
Model_specific_station_200	200
Model_specific_station_201	201
Model_specific_station_202	202
Model_specific_station_203	203
Model_specific_station_204	204
Model_specific_station_205	205
Model_specific_station_206	206
Model_specific_station_207	207
Model_specific_station_208	208
Model_specific_station_209	209
Model_specific_station_210	210
Model_specific_station_211	211
Model_specific_station_212	212
Model_specific_station_213	213
Model_specific_station_214	214
Model_specific_station_215	215
Model_specific_station_216	216

Enumerator	Value
Model_specific_station_217	217
Model_specific_station_218	218
Model_specific_station_219	219
Model_specific_station_220	220
Model_specific_station_221	221
Model_specific_station_222	222
Model_specific_station_223	223
Model_specific_station_224	224
Model_specific_station_225	225
Model_specific_station_226	226
Model_specific_station_227	227
Model_specific_station_228	228
Model_specific_station_229	229
Model_specific_station_230	230
Model_specific_station_231	231
Model_specific_station_232	232
Model_specific_station_233	233
Model_specific_station_234	234
Model_specific_station_235	235
Model_specific_station_236	236
Model_specific_station_237	237
Model_specific_station_238	238
Model_specific_station_239	239
Model_specific_station_240	240
Model_specific_station_241	241
Model_specific_station_242	242
Model_specific_station_243	243
Model_specific_station_244	244

Enumerator	Value
Model_specific_station_245	245
Model_specific_station_246	246
Model_specific_station_247	247
Model_specific_station_248	248
Model_specific_station_249	249
Model_specific_station_250	250
Model_specific_station_251	251
Model_specific_station_252	252
Model_specific_station_253	253
Model_specific_station_254	254
Model_specific_station_255	255
Model_specific_station_256	256
Model_specific_station_257	257
Model_specific_station_258	258
Model_specific_station_259	259
Model_specific_station_260	260
Model_specific_station_261	261
Model_specific_station_262	262
Model_specific_station_263	263
Model_specific_station_264	264
Model_specific_station_265	265
Model_specific_station_266	266
Model_specific_station_267	267
Model_specific_station_268	268
Model_specific_station_269	269
Model_specific_station_270	270
Model_specific_station_271	271
Model_specific_station_272	272

Enumerator	Value
Model_specific_station_273	273
Model_specific_station_274	274
Model_specific_station_275	275
Model_specific_station_276	276
Model_specific_station_277	277
Model_specific_station_278	278
Model_specific_station_279	279
Model_specific_station_280	280
Model_specific_station_281	281
Model_specific_station_282	282
Model_specific_station_283	283
Model_specific_station_284	284
Model_specific_station_285	285
Model_specific_station_286	286
Model_specific_station_287	287
Model_specific_station_288	288
Model_specific_station_289	289
Model_specific_station_290	290
Model_specific_station_291	291
Model_specific_station_292	292
Model_specific_station_293	293
Model_specific_station_294	294
Model_specific_station_295	295
Model_specific_station_296	296
Model_specific_station_297	297
Model_specific_station_298	298
Model_specific_station_299	299
Model_specific_station_300	300

Enumerator	Value
Model_specific_station_301	301
Model_specific_station_302	302
Model_specific_station_303	303
Model_specific_station_304	304
Model_specific_station_305	305
Model_specific_station_306	306
Model_specific_station_307	307
Model_specific_station_308	308
Model_specific_station_309	309
Model_specific_station_310	310
Model_specific_station_311	311
Model_specific_station_312	312
Model_specific_station_313	313
Model_specific_station_314	314
Model_specific_station_315	315
Model_specific_station_316	316
Model_specific_station_317	317
Model_specific_station_318	318
Model_specific_station_319	319
Model_specific_station_320	320
Model_specific_station_321	321
Model_specific_station_322	322
Model_specific_station_323	323
Model_specific_station_324	324
Model_specific_station_325	325
Model_specific_station_326	326
Model_specific_station_327	327
Model_specific_station_328	328

Enumerator	Value
Model_specific_station_329	329
Model_specific_station_330	330
Model_specific_station_331	331
Model_specific_station_332	332
Model_specific_station_333	333
Model_specific_station_334	334
Model_specific_station_335	335
Model_specific_station_336	336
Model_specific_station_337	337
Model_specific_station_338	338
Model_specific_station_339	339
Model_specific_station_340	340
Model_specific_station_341	341
Model_specific_station_342	342
Model_specific_station_343	343
Model_specific_station_344	344
Model_specific_station_345	345
Model_specific_station_346	346
Model_specific_station_347	347
Model_specific_station_348	348
Model_specific_station_349	349
Model_specific_station_350	350
Model_specific_station_351	351
Model_specific_station_352	352
Model_specific_station_353	353
Model_specific_station_354	354
Model_specific_station_355	355
Model_specific_station_356	356

Enumerator	Value
Model_specific_station_357	357
Model_specific_station_358	358
Model_specific_station_359	359
Model_specific_station_360	360
Model_specific_station_361	361
Model_specific_station_362	362
Model_specific_station_363	363
Model_specific_station_364	364
Model_specific_station_365	365
Model_specific_station_366	366
Model_specific_station_367	367
Model_specific_station_368	368
Model_specific_station_369	369
Model_specific_station_370	370
Model_specific_station_371	371
Model_specific_station_372	372
Model_specific_station_373	373
Model_specific_station_374	374
Model_specific_station_375	375
Model_specific_station_376	376
Model_specific_station_377	377
Model_specific_station_378	378
Model_specific_station_379	379
Model_specific_station_380	380
Model_specific_station_381	381
Model_specific_station_382	382
Model_specific_station_383	383
Model_specific_station_384	384

Enumerator	Value
Model_specific_station_385	385
Model_specific_station_386	386
Model_specific_station_387	387
Model_specific_station_388	388
Model_specific_station_389	389
Model_specific_station_390	390
Model_specific_station_391	391
Model_specific_station_392	392
Model_specific_station_393	393
Model_specific_station_394	394
Model_specific_station_395	395
Model_specific_station_396	396
Model_specific_station_397	397
Model_specific_station_398	398
Model_specific_station_399	399
Model_specific_station_400	400
Model_specific_station_401	401
Model_specific_station_402	402
Model_specific_station_403	403
Model_specific_station_404	404
Model_specific_station_405	405
Model_specific_station_406	406
Model_specific_station_407	407
Model_specific_station_408	408
Model_specific_station_409	409
Model_specific_station_410	410
Model_specific_station_411	411
Model_specific_station_412	412

Enumerator	Value
Model_specific_station_413	413
Model_specific_station_414	414
Model_specific_station_415	415
Model_specific_station_416	416
Model_specific_station_417	417
Model_specific_station_418	418
Model_specific_station_419	419
Model_specific_station_420	420
Model_specific_station_421	421
Model_specific_station_422	422
Model_specific_station_423	423
Model_specific_station_424	424
Model_specific_station_425	425
Model_specific_station_426	426
Model_specific_station_427	427
Model_specific_station_428	428
Model_specific_station_429	429
Model_specific_station_430	430
Model_specific_station_431	431
Model_specific_station_432	432
Model_specific_station_433	433
Model_specific_station_434	434
Model_specific_station_435	435
Model_specific_station_436	436
Model_specific_station_437	437
Model_specific_station_438	438
Model_specific_station_439	439
Model_specific_station_440	440

Enumerator	Value
Model_specific_station_441	441
Model_specific_station_442	442
Model_specific_station_443	443
Model_specific_station_444	444
Model_specific_station_445	445
Model_specific_station_446	446
Model_specific_station_447	447
Model_specific_station_448	448
Model_specific_station_449	449
Model_specific_station_450	450
Model_specific_station_451	451
Model_specific_station_452	452
Model_specific_station_453	453
Model_specific_station_454	454
Model_specific_station_455	455
Model_specific_station_456	456
Model_specific_station_457	457
Model_specific_station_458	458
Model_specific_station_459	459
Model_specific_station_460	460
Model_specific_station_461	461
Model_specific_station_462	462
Model_specific_station_463	463
Model_specific_station_464	464
Model_specific_station_465	465
Model_specific_station_466	466
Model_specific_station_467	467
Model_specific_station_468	468

Enumerator	Value
Model_specific_station_469	469
Model_specific_station_470	470
Model_specific_station_471	471
Model_specific_station_472	472
Model_specific_station_473	473
Model_specific_station_474	474
Model_specific_station_475	475
Model_specific_station_476	476
Model_specific_station_477	477
Model_specific_station_478	478
Model_specific_station_479	479
Model_specific_station_480	480
Model_specific_station_481	481
Model_specific_station_482	482
Model_specific_station_483	483
Model_specific_station_484	484
Model_specific_station_485	485
Model_specific_station_486	486
Model_specific_station_487	487
Model_specific_station_488	488
Model_specific_station_489	489
Model_specific_station_490	490
Model_specific_station_491	491
Model_specific_station_492	492
Model_specific_station_493	493
Model_specific_station_494	494
Model_specific_station_495	495
Model_specific_station_496	496

Enumerator	Value
Model_specific_station_497	497
Model_specific_station_498	498
Model_specific_station_499	499
Model_specific_station_500	500
Model_specific_station_501	501
Model_specific_station_502	502
Model_specific_station_503	503
Model_specific_station_504	504
Model_specific_station_505	505
Model_specific_station_506	506
Model_specific_station_507	507
Model_specific_station_508	508
Model_specific_station_509	509
Model_specific_station_510	510
Model_specific_station_511	511
Fuselage_station_1	512
Fuselage_station_2	513
Fuselage_station_3	514
Fuselage_station_4	515
Fuselage_station_5	516
Fuselage_station_6	517
Fuselage_station_7	518
Fuselage_station_8	519
Fuselage_station_9	520
Fuselage_station_10	521
Fuselage_station_11	522
Fuselage_station_12	523
Fuselage_station_13	524

Enumerator	Value
Fuselage_station_14	525
Fuselage_station_15	526
Fuselage_station_16	527
Fuselage_station_17	528
Fuselage_station_18	529
Fuselage_station_19	530
Fuselage_station_20	531
Fuselage_station_21	532
Fuselage_station_22	533
Fuselage_station_23	534
Fuselage_station_24	535
Fuselage_station_25	536
Fuselage_station_26	537
Fuselage_station_27	538
Fuselage_station_28	539
Fuselage_station_29	540
Fuselage_station_30	541
Fuselage_station_31	542
Fuselage_station_32	543
Fuselage_station_33	544
Fuselage_station_34	545
Fuselage_station_35	546
Fuselage_station_36	547
Fuselage_station_37	548
Fuselage_station_38	549
Fuselage_station_39	550
Fuselage_station_40	551
Fuselage_station_41	552

Enumerator	Value
Fuselage_station_42	553
Fuselage_station_43	554
Fuselage_station_44	555
Fuselage_station_45	556
Fuselage_station_46	557
Fuselage_station_47	558
Fuselage_station_48	559
Fuselage_station_49	560
Fuselage_station_50	561
Fuselage_station_51	562
Fuselage_station_52	563
Fuselage_station_53	564
Fuselage_station_54	565
Fuselage_station_55	566
Fuselage_station_56	567
Fuselage_station_57	568
Fuselage_station_58	569
Fuselage_station_59	570
Fuselage_station_60	571
Fuselage_station_61	572
Fuselage_station_62	573
Fuselage_station_63	574
Fuselage_station_64	575
Fuselage_station_65	576
Fuselage_station_66	577
Fuselage_station_67	578
Fuselage_station_68	579
Fuselage_station_69	580

Enumerator	Value
Fuselage_station_70	581
Fuselage_station_71	582
Fuselage_station_72	583
Fuselage_station_73	584
Fuselage_station_74	585
Fuselage_station_75	586
Fuselage_station_76	587
Fuselage_station_77	588
Fuselage_station_78	589
Fuselage_station_79	590
Fuselage_station_80	591
Fuselage_station_81	592
Fuselage_station_82	593
Fuselage_station_83	594
Fuselage_station_84	595
Fuselage_station_85	596
Fuselage_station_86	597
Fuselage_station_87	598
Fuselage_station_88	599
Fuselage_station_89	600
Fuselage_station_90	601
Fuselage_station_91	602
Fuselage_station_92	603
Fuselage_station_93	604
Fuselage_station_94	605
Fuselage_station_95	606
Fuselage_station_96	607
Fuselage_station_97	608

Enumerator	Value
Fuselage_station_98	609
Fuselage_station_99	610
Fuselage_station_100	611
Fuselage_station_101	612
Fuselage_station_102	613
Fuselage_station_103	614
Fuselage_station_104	615
Fuselage_station_105	616
Fuselage_station_106	617
Fuselage_station_107	618
Fuselage_station_108	619
Fuselage_station_109	620
Fuselage_station_110	621
Fuselage_station_111	622
Fuselage_station_112	623
Fuselage_station_113	624
Fuselage_station_114	625
Fuselage_station_115	626
Fuselage_station_116	627
Fuselage_station_117	628
Fuselage_station_118	629
Fuselage_station_119	630
Fuselage_station_120	631
Fuselage_station_121	632
Fuselage_station_122	633
Fuselage_station_123	634
Fuselage_station_124	635
Fuselage_station_125	636

Enumerator	Value
Fuselage_station_126	637
Fuselage_station_127	638
Fuselage_station_128	639
Left_wing_station_1	640
Left_wing_station_2	641
Left_wing_station_3	642
Left_wing_station_4	643
Left_wing_station_5	644
Left_wing_station_6	645
Left_wing_station_7	646
Left_wing_station_8	647
Left_wing_station_9	648
Left_wing_station_10	649
Left_wing_station_11	650
Left_wing_station_12	651
Left_wing_station_13	652
Left_wing_station_14	653
Left_wing_station_15	654
Left_wing_station_16	655
Left_wing_station_17	656
Left_wing_station_18	657
Left_wing_station_19	658
Left_wing_station_20	659
Left_wing_station_21	660
Left_wing_station_22	661
Left_wing_station_23	662
Left_wing_station_24	663
Left_wing_station_25	664

Enumerator	Value
Left_wing_station_26	665
Left_wing_station_27	666
Left_wing_station_28	667
Left_wing_station_29	668
Left_wing_station_30	669
Left_wing_station_31	670
Left_wing_station_32	671
Left_wing_station_33	672
Left_wing_station_34	673
Left_wing_station_35	674
Left_wing_station_36	675
Left_wing_station_37	676
Left_wing_station_38	677
Left_wing_station_39	678
Left_wing_station_40	679
Left_wing_station_41	680
Left_wing_station_42	681
Left_wing_station_43	682
Left_wing_station_44	683
Left_wing_station_45	684
Left_wing_station_46	685
Left_wing_station_47	686
Left_wing_station_48	687
Left_wing_station_49	688
Left_wing_station_50	689
Left_wing_station_51	690
Left_wing_station_52	691
Left_wing_station_53	692

Enumerator	Value
Left_wing_station_54	693
Left_wing_station_55	694
Left_wing_station_56	695
Left_wing_station_57	696
Left_wing_station_58	697
Left_wing_station_59	698
Left_wing_station_60	699
Left_wing_station_61	700
Left_wing_station_62	701
Left_wing_station_63	702
Left_wing_station_64	703
Left_wing_station_65	704
Left_wing_station_66	705
Left_wing_station_67	706
Left_wing_station_68	707
Left_wing_station_69	708
Left_wing_station_70	709
Left_wing_station_71	710
Left_wing_station_72	711
Left_wing_station_73	712
Left_wing_station_74	713
Left_wing_station_75	714
Left_wing_station_76	715
Left_wing_station_77	716
Left_wing_station_78	717
Left_wing_station_79	718
Left_wing_station_80	719
Left_wing_station_81	720

Enumerator	Value
Left_wing_station_82	721
Left_wing_station_83	722
Left_wing_station_84	723
Left_wing_station_85	724
Left_wing_station_86	725
Left_wing_station_87	726
Left_wing_station_88	727
Left_wing_station_89	728
Left_wing_station_90	729
Left_wing_station_91	730
Left_wing_station_92	731
Left_wing_station_93	732
Left_wing_station_94	733
Left_wing_station_95	734
Left_wing_station_96	735
Left_wing_station_97	736
Left_wing_station_98	737
Left_wing_station_99	738
Left_wing_station_100	739
Left_wing_station_101	740
Left_wing_station_102	741
Left_wing_station_103	742
Left_wing_station_104	743
Left_wing_station_105	744
Left_wing_station_106	745
Left_wing_station_107	746
Left_wing_station_108	747
Left_wing_station_109	748

Enumerator	Value
Left_wing_station_110	749
Left_wing_station_111	750
Left_wing_station_112	751
Left_wing_station_113	752
Left_wing_station_114	753
Left_wing_station_115	754
Left_wing_station_116	755
Left_wing_station_117	756
Left_wing_station_118	757
Left_wing_station_119	758
Left_wing_station_120	759
Left_wing_station_121	760
Left_wing_station_122	761
Left_wing_station_123	762
Left_wing_station_124	763
Left_wing_station_125	764
Left_wing_station_126	765
Left_wing_station_127	766
Left_wing_station_128	767
Right_wing_station_1	768
Right_wing_station_2	769
Right_wing_station_3	770
Right_wing_station_4	771
Right_wing_station_5	772
Right_wing_station_6	773
Right_wing_station_7	774
Right_wing_station_8	775
Right_wing_station_9	776

Enumerator	Value
Right_wing_station_10	777
Right_wing_station_11	778
Right_wing_station_12	779
Right_wing_station_13	780
Right_wing_station_14	781
Right_wing_station_15	782
Right_wing_station_16	783
Right_wing_station_17	784
Right_wing_station_18	785
Right_wing_station_19	786
Right_wing_station_20	787
Right_wing_station_21	788
Right_wing_station_22	789
Right_wing_station_23	790
Right_wing_station_24	791
Right_wing_station_25	792
Right_wing_station_26	793
Right_wing_station_27	794
Right_wing_station_28	795
Right_wing_station_29	796
Right_wing_station_30	797
Right_wing_station_31	798
Right_wing_station_32	799
Right_wing_station_33	800
Right_wing_station_34	801
Right_wing_station_35	802
Right_wing_station_36	803
Right_wing_station_37	804

Enumerator	Value
Right_wing_station_38	805
Right_wing_station_39	806
Right_wing_station_40	807
Right_wing_station_41	808
Right_wing_station_42	809
Right_wing_station_43	810
Right_wing_station_44	811
Right_wing_station_45	812
Right_wing_station_46	813
Right_wing_station_47	814
Right_wing_station_48	815
Right_wing_station_49	816
Right_wing_station_50	817
Right_wing_station_51	818
Right_wing_station_52	819
Right_wing_station_53	820
Right_wing_station_54	821
Right_wing_station_55	822
Right_wing_station_56	823
Right_wing_station_57	824
Right_wing_station_58	825
Right_wing_station_59	826
Right_wing_station_60	827
Right_wing_station_61	828
Right_wing_station_62	829
Right_wing_station_63	830
Right_wing_station_64	831
Right_wing_station_65	832

Enumerator	Value
Right_wing_station_66	833
Right_wing_station_67	834
Right_wing_station_68	835
Right_wing_station_69	836
Right_wing_station_70	837
Right_wing_station_71	838
Right_wing_station_72	839
Right_wing_station_73	840
Right_wing_station_74	841
Right_wing_station_75	842
Right_wing_station_76	843
Right_wing_station_77	844
Right_wing_station_78	845
Right_wing_station_79	846
Right_wing_station_80	847
Right_wing_station_81	848
Right_wing_station_82	849
Right_wing_station_83	850
Right_wing_station_84	851
Right_wing_station_85	852
Right_wing_station_86	853
Right_wing_station_87	854
Right_wing_station_88	855
Right_wing_station_89	856
Right_wing_station_90	857
Right_wing_station_91	858
Right_wing_station_92	859
Right_wing_station_93	860

Enumerator	Value
Right_wing_station_94	861
Right_wing_station_95	862
Right_wing_station_96	863
Right_wing_station_97	864
Right_wing_station_98	865
Right_wing_station_99	866
Right_wing_station_100	867
Right_wing_station_101	868
Right_wing_station_102	869
Right_wing_station_103	870
Right_wing_station_104	871
Right_wing_station_105	872
Right_wing_station_106	873
Right_wing_station_107	874
Right_wing_station_108	875
Right_wing_station_109	876
Right_wing_station_110	877
Right_wing_station_111	878
Right_wing_station_112	879
Right_wing_station_113	880
Right_wing_station_114	881
Right_wing_station_115	882
Right_wing_station_116	883
Right_wing_station_117	884
Right_wing_station_118	885
Right_wing_station_119	886
Right_wing_station_120	887
Right_wing_station_121	888

Enumerator	Value
Right_wing_station_122	889
Right_wing_station_123	890
Right_wing_station_124	891
Right_wing_station_125	892
Right_wing_station_126	893
Right_wing_station_127	894
Right_wing_station_128	895
M16A42_rifle	896
M249_SAW	897
M60_Machine_gun	898
M203_Grenade_Launcher	899
M136_AT4	900
M47_Dragon	901
AAWS_M_Javelin	902
M18A1_Claymore_Mine	903
MK19_Grenade_Launcher	904
M2_Machine_Gun	905
Other_attached_part_1	906
Other_attached_part_2	907
Other_attached_part_3	908
Other_attached_part_4	909
Other_attached_part_5	910
Other_attached_part_6	911
Other_attached_part_7	912
Other_attached_part_8	913
Other_attached_part_9	914
Other_attached_part_10	915
Other_attached_part_11	916

Enumerator	Value
Other_attached_part_12	917
Other_attached_part_13	918
Other_attached_part_14	919
Other_attached_part_15	920
Other_attached_part_16	921
Other_attached_part_17	922
Other_attached_part_18	923
Other_attached_part_19	924
Other_attached_part_20	925
Other_attached_part_21	926
Other_attached_part_22	927
Other_attached_part_23	928
Other_attached_part_24	929
Other_attached_part_25	930
Other_attached_part_26	931
Other_attached_part_27	932
Other_attached_part_28	933
Other_attached_part_29	934
Other_attached_part_30	935
Other_attached_part_31	936
Other_attached_part_32	937
Other_attached_part_33	938
Other_attached_part_34	939
Other_attached_part_35	940
Other_attached_part_36	941
Other_attached_part_37	942
Other_attached_part_38	943
Other_attached_part_39	944

Enumerator	Value
Other_attached_part_40	945
Other_attached_part_41	946
Other_attached_part_42	947
Other_attached_part_43	948
Other_attached_part_44	949
Other_attached_part_45	950
Other_attached_part_46	951
Other_attached_part_47	952
Other_attached_part_48	953
Other_attached_part_49	954
Other_attached_part_50	955
Other_attached_part_51	956
Other_attached_part_52	957
Other_attached_part_53	958
Other_attached_part_54	959
Other_attached_part_55	960
Other_attached_part_56	961
Other_attached_part_57	962
Other_attached_part_58	963
Other_attached_part_59	964
Other_attached_part_60	965
Other_attached_part_61	966
Other_attached_part_62	967
Other_attached_part_63	968
Other_attached_part_64	969
Other_attached_part_65	970
Other_attached_part_66	971
Other_attached_part_67	972

Enumerator	Value
Other_attached_part_68	973
Other_attached_part_69	974
Other_attached_part_70	975
Other_attached_part_71	976
Other_attached_part_72	977
Other_attached_part_73	978
Other_attached_part_74	979
Other_attached_part_75	980
Other_attached_part_76	981
Other_attached_part_77	982
Other_attached_part_78	983
Other_attached_part_79	984
Other_attached_part_80	985
Other_attached_part_81	986
Other_attached_part_82	987
Other_attached_part_83	988
Other_attached_part_84	989
Other_attached_part_85	990
Other_attached_part_86	991
Other_attached_part_87	992
Other_attached_part_88	993
Other_attached_part_89	994
Other_attached_part_90	995
Other_attached_part_91	996
Other_attached_part_92	997
Other_attached_part_93	998
Other_attached_part_94	999
Other_attached_part_95	1000

Enumerator	Value
Other_attached_part_96	1001
Other_attached_part_97	1002
Other_attached_part_98	1003
Other_attached_part_99	1004
Other_attached_part_100	1005
Other_attached_part_101	1006
Other_attached_part_102	1007
Other_attached_part_103	1008
Other_attached_part_104	1009
Other_attached_part_105	1010
Other_attached_part_106	1011
Other_attached_part_107	1012
Other_attached_part_108	1013
Other_attached_part_109	1014
Other_attached_part_110	1015
Other_attached_part_111	1016
Other_attached_part_112	1017
Other_attached_part_113	1018
Other_attached_part_114	1019
Other_attached_part_115	1020
Other_attached_part_116	1021
Other_attached_part_117	1022
Other_attached_part_118	1023

AustralianCategoryOverlayEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Australian Category Overlay [UID 500]*

Enumerator	Value
Australian_Army	11
Royal_Australian_Air_Force_RAAF_	13
Royal_Australian_Navy_RAN_	14
Australian_Special_Operations_Command_SOCOMD_	30
Australian_Department_of_Home_Affairs	51
Australian_Federal_Police_AFP_	52

BeamStatusBeamStateEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize01

Semantics: *Beam Status-Beam State [UID 318]*

Enumerator	Value
Active	0
Deactivated	1

BuildingPaintSchemeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Building Paint Scheme [UID 464]*

Enumerator	Value
Default	0

CCTTSINCGARSClearChannelEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *CCTT SINCGARS-Clear Channel [UID 171]*

Enumerator	Value
Not_clear_channel	0
Clear_channel	1

CCTTSINCGARSStartOfMessageEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *CCTT SINCGARS-Start of Message [UID 170]*

Enumerator	Value
Not_start_of_message	0
Start_of_Message	1

CapabilityReportEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Capability Report [UID 358]*

Enumerator	Value
No_Communications_Capability_CA_	0
Reserved_1	1
Reserved_2	2
Reserved_3	3
Signifies_at_Least_Comm_A_and_Comm_B_Capability_and_Ability_to_Set_CA_Code_7_and_on_the_Ground	4
Signifies_at_Least_Comm_A_and_Comm_B_capability_and_Ability_to_Set_CA_Code_7_and_Airborne	5
Signifies_at_Least_Comm_A_and_Comm_B_capability_and_Ability_to_Set_CA_Code_7_and_Either_Airborne_or_on_the_Ground	6
Signifies_the_Downlink_Request_DR_Field_Is_Not_Equal_To_0_and_The_Flight_Status_FS_Field_Equals_2_3_4_or_5_and_Either_Airborne_or_on_the_Ground	7
No_Statement	255

Class10SupplyCategoryMaterialToSupportNonMilitaryProgramsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 10 Supply Category - Material to Support Non-Military Programs [UID 610]*

Enumerator	Value
Other	1

Class11SupplyCategorySuppliesNonDoctrinalEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 11 Supply Category - Supplies (Non Doctrinal) [UID 611]*

Enumerator	Value
Other	1
Pallets	2
Fuel_Tanks_Drums_and_Bladders	3
Chests	4
Boxes	5

Class12SupplyCategorySlingLoadsNonDoctrinalEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 12 Supply Category - Sling Loads (Non Doctrinal) [UID 612]*

Enumerator	Value
Other	1
Sling_Load_Blivet	2
Sling_Load_Crate	3
Sling_Load_Water_Bucket	4
Sling_Load_Vehicles	5
Sling_Load_Howitzer	6
Sling_Load_Collapsible	7
Sling_Load_Bladder	8
Sling_Load_Pallet_of_Crates	9
Sling_Load_Helicopters	10
Sling_Load_Hoist	11
Sling_Load_Concrete_Block	12
Sling_Load_Trailers	13

Class1SupplyCategorySubsistenceEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 1 Supply Category - Subsistence [UID 601]*

Enumerator	Value
Other	1
A_Non_Perishable	2
C_Combat_Rations	3
R_Refrigerated	4
S_Other_Non_Refrigerated	5
W_Water	6

Class2SupplyCategoryClothingIndividualEquipmentToolsAdminSuppliesEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 2 Supply Category - Clothing, Individual Equipment, Tools, Admin. Supplies [UID 602]*

Enumerator	Value
Other	1
A_Air	2
B_Ground_Support_Materiel	3
E_General_Supplies	4
F_Clothing	5
G_Electronics	6
M_Weapons_Humans_	7
T_Industrial_Supplies	8

Class3SupplyCategoryPetroleumOilsLubricantsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 3 Supply Category - Petroleum, Oils, Lubricants [UID 603]*

Enumerator	Value
Other	1
A_POL_for_Air_Vehicles	2

Enumerator	Value
W_POL_for_Land_Vehicles	3
P_Packaged_POL	4

Class4SupplyCategoryConstructionMaterialsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 4 Supply Category - Construction Materials [UID 604]*

Enumerator	Value
Other	1
A_Construction	2
B_Barrier	3

Class6SupplyCategoryPersonnelDemandItemsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 6 Supply Category - Personnel Demand Items [UID 606]*

Enumerator	Value
Other	1

Class7SupplyCategoryMajorItemsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 7 Supply Category - Major Items [UID 607]*

Enumerator	Value
Other	1
A_Air_Not_used_as_described_in_Air_Domain_	2
B_Ground_Support_Materiel	3
D_Admin_Vehicles_Not_used_as_described_in_Land_Domain_	4
G_Electronics	5
J_Racks_Adaptors_Pylons	6
K_Tactical_Vehicles_Not_used_as_described_in_Land_Domain_	7
L_Missiles_Not_used_as_described_in_Munition_Domain_	8

Enumerator	Value
M_Weapons	9
N_Special_Weapons	10
X_Aircraft_Engines	11
Drop_Tank	20
Conformal_Fuel_Tank	21
Luggage_Pod	22
ECM_Pod	23
Para_Drogue	24
Targeting_Pod	25
Fairing	26
Air_Refuelling_Pod	27
Heavy_Airdrop	28
Container_Delivery_System_CDS_Airdrop	29
Rocket_Pod_Launcher	30
Tactical_Pod	31
RECCE_pod	32
FLIR_pod	33

Class8SupplyCategoryMedicalMaterialEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 8 Supply Category - Medical Material [UID 608]*

Enumerator	Value
Other	1
A_Medical_Materiel	2
B_Blood_Fluids	3

Class9SupplyCategoryRepairPartsAndComponentsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Class 9 Supply Category - Repair Parts and Components [UID 609]*

Enumerator	Value
Other	1
A_Air	2
B_Ground_Support_Materiel	3
D_Admin_Vehicles	4
G_Electronics	5
K_Tactical_Vehicles RPRnoteEnumerations8	6
L_Missiles	7
M_Weapons	8
N_Special_Weapons	9
X_Aircraft_Engines	10

ClothingIRSignatureEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Clothing IR Signature [UID 802]*

Enumerator	Value
Standard_Clothing	0
Camouflage_not_just_Paint_Scheme_	1
Thermal_Blanket	2
Other	3

CollisionTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Collision Type [UID 189]*

Enumerator	Value
Inelastic	0
Elastic	1
The_boom_nozzle_is_in_physical_contact_with_the_receptacle_and_the_booms_signal_system_is_operative_	2
The_boom_trainers_signal_system_has_sent_a_disconnect_signal_that_should_cause_the_receivers_refueling_receptacle_to_unlatch_unless_there_is_a_malfunction_	3

Enumerator	Value
Disconnect_without_a_signal_being_sent_disconnected_by_physical_means_brute_force_disconnect_controlled_tension_disconnect	4
The_boom_nozzle_is_in_physical_contact_with_the_receptacle_and_the_booms_signal_system_is_inoperative	5
Boom_simulator_has_calculated_that_the_receivers_latches_have_been_damaged	6
AR_receptacle_door_1_damaged	7
AR_receptacle_door_2_damaged	8
Pilots_Cockpit_Windshield_damaged	9
Copilots_Cockpit_Windshield_damaged	10
Pilots_Left_Side_Window_damaged RPRnoteEnumerations9	11
Copilots_Right_Side_Window_damaged RPRnoteEnumerations10	12
Pilots_Eyebrow_Window_damaged	13
Copilots_Eyebrow_Window_damaged	14
MLS_Glide_Slope_1_Antenna_damaged	15
MLS_Glide_Slope_2_Antenna_damaged	16
ILS_Glide_1_Slope_Antenna_damaged	17
ILS_Glide_2_Slope_Antenna_damaged	18
SKE_OMNI_Antenna_damaged	19
SKE_Directional_Antenna_damaged	20
Weather_Radar_Antenna_damaged	21
Com_1_ARC_210_Antenna_damaged	22
TACAN_Antenna_damaged	23
MLS_Antenna_damaged	24
AF_SATCOM_Antenna_damaged	25
AERO_I_SATCOM_Antenna_damaged	26
AERO_H_SATCOM_Antenna_damaged	28
UHF_SATCOM_Antenna_damaged	29
HMSA_Antenna_damaged	30
IFF_Antenna_damaged	31
Left_Side_Landing_Taxi_Light_damaged	32

Enumerator	Value
Right_Side_Landing_Taxi_Light_damaged	33
Left_Side_Runway_Turnoff_Light_damaged	34
Right_Side_Runway_Turnoff_Light_damaged	35
Left_Side_Formation_Light_damaged	36
Right_Side_Formation_Light_damaged	37
Left_Side_Nacelle_Scanning_Light_damaged	38
Right_Side_Nacelle_Scanning_Light_damaged	39
Copilot_Pitot_Static_Probe_1_damaged	40
Copilot_Pitot_Static_Probe_2_damaged	41
Pilot_Pitot_Static_Probe_1_damaged	42
Pilot_Pitot_Static_Probe_2_damaged	43
Total_Air_Temperature_Probe_1_damaged	44
Total_Air_Temperature_Probe_2_damaged	45
Angle_of_Attack_Sensor_1_damaged	46
Angle_of_Attack_Sensor_2_damaged	47
Angle_of_Attack_Sensor_3_damaged	48
Angle_of_Attack_Sensor_4_damaged	49
Angle_of_Attack_Sensor_5_damaged	50
Angle_of_Attack_Sensor_6_damaged	51
Left_Side_Spoiler_damaged	52
Right_Side_Spoiler_damaged	53
Upper_TCAS_antenna_KC_135_R_T_damaged	54
Boom_nozzle_has_cleared_the_receiver_s_refueling_receptacle RPRnoteEnumerations11	55

ColorEnum8 [RPRnoteEnumerations12](#)Representation: [RPRunsignedInteger8BE](#)Semantics: *Color [UID 463]*

Enumerator	Value
Not_Specified	0
White_VGA	1
Red_VGA	2
Yellow_VGA	3
Lime_VGA	4
Cyan_VGA	5
Blue_VGA	6
Magenta_VGA	7
Grey_VGA	8
Silver_VGA	9
Maroon_VGA	10
Olive_VGA	11
Green_VGA	12
Teal_VGA	13
Navy_VGA	14
Purple_VGA	15
Reserved_16	16
Reserved_17	17
Reserved_18	18
Reserved_19	19
Black	20
Navy	21
Dark_Blue	22
Medium_Blue	23
Blue	24
Dark_Green	25
Green	26
Teal	27

Enumerator	Value
Dark_Cyan	28
Deep_Sky_Blue	29
Dark_Turquoise	30
Medium_Spring_Green	31
Lime	32
Spring_Green	33
Cyan	34
Midnight_Blue	35
Dodger_Blue	36
Light_Sea_Green	37
Forest_Green	38
Sea_Green	39
Dark_Slate_Gray	40
Lime_Green	41
Medium_Sea_Green	42
Turquoise	43
Royal_Blue	44
Steel_Blue	45
Dark_Slate_Blue	46
Medium_Turquoise	47
Indigo	48
Dark_Olive_Green	49
Cadet_Blue	50
Cornflower_Blue	51
Medium_Aquamarine	52
Dim_Gray	53
Slate_Blue	54
Olive_Drab	55

Enumerator	Value
Slate_Gray	56
Light_Slate_Gray	57
Medium_Slate_Blue	58
Lawn_Green	59
Chartreuse	60
Aquamarine	61
Maroon	62
Purple	63
Olive	64
Gray	65
Grey	66
Sky_Blue	67
Light_Sky_Blue	68
Blue_Violet	69
Dark_Red	70
Dark_Magenta	71
Saddle_Brown	72
Dark_Sea_Green	73
Light_Green	74
Medium_Purple	75
Dark_Violet	76
Pale_Green	77
Dark_Orchid	78
Yellow_Green	79
Sienna	80
Brown	81
Dark_Gray	82
Light_Blue	83

Enumerator	Value
Green_Yellow	84
Pale_Turquoise	85
Light_Steel_Blue	86
Powder_Blue	87
Fire_Brick	88
Dark_Golden_Rod	89
Medium_Orchid	90
Rosy_Brown	91
Dark_Khaki	92
Silver	93
Medium_Violet_Red	94
Indian_Red	95
Peru	96
Chocolate	97
Tan	98
Light_Gray	99
Pale_Violet_Red	100
Thistle	101
Orchid	102
Golden_Rod	103
Crimson	104
Gainsboro	105
Plum	106
Burly_Wood	107
Light_Cyan	108
Lavender	109
Dark_Salmon	110
Violet	111

Enumerator	Value
Pale_Golden_Rod	112
Light_Coral	113
Khaki	114
Alice_Blue	115
Honey_Dew	116
Azure	117
Sandy_Brown	118
Wheat	119
Beige	120
White_Smoke	121
Mint_Cream	122
Ghost_White	123
Salmon	124
Antique_White	125
Linen	126
Light_Golden_Rod_Yellow	127
Old_Lace	128
Red	129
Fuchsia	130
Magenta	131
Deep_Pink	132
Orange_Red	133
Tomato	134
Hot_Pink	135
Coral	136
Dark_Orange	137
Light_Salmon	138
Orange	139

Enumerator	Value
Light_Pink	140
Pink	141
Gold	142
Peach_Puff	143
Navajo_White	144
Moccasin	145
Bisque	146
Misty_Rose	147
Blanched_Almond	148
Papaya_Whip	149
Lavender_Blush	150
Sea_Shell	151
Cornsilk	152
Lemon_Chiffon	153
Floral_White	154
Snow	155
Yellow	156
Light_Yellow	157
Ivory	158
White	159

Component VisualDamageStatusSmokeEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize02

Semantics: *Component Visual Damage Status-Smoke [UID 375]*

Enumerator	Value
No_Smoke	0
Light_Smoke	1
Moderate_Smoke	2

Enumerator	Value
Heavy_Smoke	3

ComponentVisualDamageStatusSurfaceDamageEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Component Visual Damage Status-Surface Damage [UID 376]*

Enumerator	Value
Normal_Appearance	0
Light_Charring	1
Heavy_Charring	2
One_or_More_Holes_Burned_Completely_through_Surface	3

CountryEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Country [UID 29]*

Enumerator	Value
Other	0
Afghanistan_AFG_	1
Albania_ALB_	2
Algeria_DZA_	3
American_Samoa_ASM_	4
Andorra_AND_	5
Angola_AGO_	6
Anguilla_AIA_	7
Antarctica_ATA_	8
Antigua_and_Barbuda_ATG_	9
Argentina_ARG_	10
Aruba_ABW_	11
Ashmore_and_Cartier_Islands_Australia_ RPRnoteEnumerations13 RPRnoteEnumerationsDeprecated	12
Australia_AUS_	13

Enumerator	Value
Austria_AUT_	14
Bahamas_BHS_	15
Bahrain_BHR_	16
Baker_Island_United_States_ RPRnoteEnumerations14 RPRnoteEnumerationsDeprecated	17
Bangladesh_BGD_	18
Barbados_BRB_	19
Bassas_da_India_France_ RPRnoteEnumerations15 RPRnoteEnumerationsDeprecated	20
Belgium_BEL_	21
Belize_BLZ_	22
Benin_BEN_	23
Bermuda_BMU_	24
Bhutan_BTN_	25
Bolivia_Plurinational_State_of_BOL_	26
Botswana_BWA_	27
Bouvet_Island_BVT_	28
Brazil_BRA_	29
British_Indian_Ocean_Territory_IOT_	30
Virgin_Islands_British_VGB_	31
Brunei_Darussalam_BRN_	32
Bulgaria_BGR_	33
Burkina_Faso_BFA_	34
Myanmar_MMR_	35
Burundi_BDI_	36
Cambodia_KHM_	37
Cameroon_CMR_	38
Canada_CAN_	39
Cabo_Verde_CPV_	40
Cayman_Islands_CYM_	41

Enumerator	Value
Central_African_Republic_CAF_	42
Chad_TCD_	43
Chile_CHL_	44
China_People_s_Republic_of_CHN_	45
Christmas_Island_CXR_	46
Cocos_Keeling_Islands_CCK_	47
Colombia_COL_	48
Comoros_COM_	49
Congo_COG_ RPRnoteEnumerations16	50
Cook_Islands_COK_	51
Coral_Sea_Islands_Australia_ RPRnoteEnumerations13 RPRnoteEnumerationsDeprecated	52
Costa_Rica_CRI_	53
Cuba_CUB_	54
Cyprus_CYP_	55
Czechoslovakia_CSK_ RPRnoteEnumerations17 RPRnoteEnumerationsDeprecated	56
Denmark_DNK_	57
Djibouti_DJI_	58
Dominica_DMA_	59
Dominican_Republic_DOM_	60
Ecuador_ECU_	61
Egypt_EGY_	62
El_Salvador_SLV_	63
Equatorial_Guinea_GNQ_	64
Ethiopia_ETH_	65
Europa_Island_France_ RPRnoteEnumerations15 RPRnoteEnumerationsDeprecated	66
Falkland_Islands_Malvinas_FLK_	67
Faroe_Islands_FRO_	68
Fiji_FJI_	69

Enumerator	Value
Finland_FIN_	70
France_FRA_	71
French_Guiana_GUF_	72
French_Polynesia_PYF_	73
French_Southern_Territories_ATF_	74
Gabon_GAB_	75
Gambia_The_GMB_	76
Gaza_Strip_Israel_ RPRnoteEnumerations18 RPRnoteEnumerationsDeprecated	77
Germany_DEU_	78
Ghana_GHA_	79
Gibraltar_GIB_	80
Glorioso_Islands_France_ RPRnoteEnumerations15 RPRnoteEnumerationsDeprecated	81
Greece_GRC_	82
Greenland_GRL_	83
Grenada_GRD_	84
Guadeloupe_GLP_	85
Guam_GUM_	86
Guatemala_GTM_	87
Guernsey_GGY_	88
Guinea_GIN_	89
Guinea_Bissau_GNB_	90
Guyana_GUY_	91
Haiti HTI_	92
Heard_Island_and_McDonald_Islands_HMD_	93
Honduras_HND_	94
Hong_Kong_HKG_	95
Howland_Island_United_States_ RPRnoteEnumerations14 RPRnoteEnumerationsDeprecated	96
Hungary_HUN_	97

Enumerator	Value
Iceland_ISL_	98
India_IND_	99
Indonesia_IDN_	100
Iran_Islamic_Republic_of_IRN_	101
Iraq_IRQ_	102
Ireland_IRL_	104
Israel_ISR_	105
Italy_ITA_	106
Cote_d_Ivoire_CIV_	107
Jamaica_JAM_	108
Jan_Mayen_Norway_ RPRnoteEnumerations19 RPRnoteEnumerationsDeprecated	109
Japan_JPN_	110
Jarvis_Island_United_States_ RPRnoteEnumerations14 RPRnoteEnumerationsDeprecated	111
Jersey_JEY_	112
Johnston_Atoll_United_States_ RPRnoteEnumerations20 RPRnoteEnumerationsDeprecated	113
Jordan_JOR_	114
Juan_de_Nova_Island RPRnoteEnumerations15 RPRnoteEnumerationsDeprecated	115
Kenya_KEN_	116
Kingman_Reef_United_States_ RPRnoteEnumerations14 RPRnoteEnumerationsDeprecated	117
Kiribati_KIR_	118
Korea_Democratic_People_s_Republic_of_PRK_	119
Korea_Republic_of_KOR_	120
Kuwait_KWT_	121
Lao_People_s_Democratic_Republic_LAO_	122
Lebanon_LBN_	123
Lesotho_LSO_	124
Liberia_LBR_	125
Libya_LBY_	126

Enumerator	Value
Liechtenstein_LIE_	127
Luxembourg_LUX_	128
Madagascar_MDG_	129
Macao_MAC_	130
Malawi_MWI_	131
Malaysia_MYS_	132
Maldives_MDV_	133
Mali_MLI_	134
Malta_MLT_	135
Isle_of_Man_IMN_	136
Marshall_Islands_MHL_	137
Martinique_MTQ_	138
Mauritania_MRT_	139
Mauritius_MUS_	140
Mayotte_MYT_	141
Mexico_MEX_	142
Micronesia_Federated_States_of_FSM_	143
Monaco_MCO_	144
Mongolia_MNG_	145
Montserrat_MSR_	146
Morocco_MAR_	147
Mozambique_MOZ_	148
Namibia_NAM_	149
Nauru_NRU_	150
Navassa_Island_United_States_ RPRnoteEnumerations14 RPRnoteEnumerationsDeprecated	151
Nepal_NPL_	152
Netherlands_NLD_	153
Netherlands_Antilles_Curacao_Bonaire_Saba_Sint_Maarten_Sint_Eustatius_ RPRnoteEnumerations21 RPRnoteEnumerationsDeprecated	154

Enumerator	Value
New_Caledonia_NCL_	155
New_Zealand_NZL_	156
Nicaragua_NIC_	157
Niger_NER_	158
Nigeria_NGA_	159
Niue_NIU_	160
Norfolk_Island_NFK_	161
Northern_Mariana_Islands_MNP_	162
Norway_NOR_	163
Oman_OMN_	164
Pakistan_PAK_	165
Palmyra_Atoll_United_States_ RPRnoteEnumerations14 RPRnoteEnumerationsDeprecated	166
Panama_PAN_	168
Papua_New_Guinea_PNG_	169
Paracel_Islands_International_Occupied_by_China_also_claimed_by_Taiwan_and_Vietnam_ RPRnoteEnumerationsDeprecated	170
Paraguay_PRY_	171
Peru_PER_	172
Philippines_PHL_	173
Pitcairn_PCN_	174
Poland_POL_	175
Portugal_PRT_	176
Puerto_Rico_PRI_	177
Qatar_QAT_	178
Reunion_REU_	179
Romania_ROU_	180
Rwanda_RWA_	181
Saint_Kitts_and_Nevis_KNA_	182
Saint_Helena_Ascension_and_Tristan_da_Cunha_SHN_	183

Enumerator	Value
Saint_Lucia_LCA_	184
Saint_Pierre_and_Miquelon_SPM_	185
Saint_Vincent_and_the_Grenadines_VCT_	186
San_Marino_SMR_	187
Sao_Tome_and_Principe_STP_	188
Saudi_Arabia_SAU_	189
Senegal_SEN_	190
Seychelles_SYC_	191
Sierra_Leone_SLE_	192
Singapore_SGP_	193
Solomon_Islands_SLB_	194
Somalia_SOM_	195
South_Georgia_and_the_South_Sandwich_Islands_SGS_	196
South_Africa_ZAF_	197
Spain_ESP_	198
Spratly_Islands_International_parts_occupied_and_claimed_by_China_Malaysia_Philippines_Taiwan_Vietnam_ RPRnoteEnumerationsDeprecated	199
Sri_Lanka_LKA_	200
Sudan_SDN_	201
Suriname_SUR_	202
Svalbard_Norway_ RPRnoteEnumerations19 RPRnoteEnumerationsDeprecated	203
Eswatini_SWZ_	204
Sweden_SWE_	205
Switzerland_CHE_	206
Syrian_Arab_Republic_SYR_	207
Taiwan_Province_of_China_TWN_	208
Tanzania_United_Republic_of_TZA_	209
Thailand_THA_	210
Togo_TGO_	211

Enumerator	Value
Tokelau_TKL_	212
Tonga_TON_	213
Trinidad_and_Tobago_TTO_	214
Tromelin_Island_France_ RPRnoteEnumerations15 RPRnoteEnumerationsDeprecated	215
Palau_PLW_	216
Tunisia_TUN_	217
Turkiye_Republic_of_TUR_ RPRnoteEnumerations22	218
Turks_and_Caicos_Islands_TCA_	219
Tuvalu_TUV_	220
Uganda_UGA_	221
Russia_RUS_ RPRnoteEnumerations23	222
United_Arab_Emirates_ARE_	223
United_Kingdom_of_Great_Britain_and_Northern_Ireland_GBR_	224
United_States_of_America_USA_	225
Uruguay_URY_	226
Vanuatu_VUT_	227
Holy_See_VAT_	228
Venezuela_Bolivarian_Republic_of_VEN_	229
Viet_Nam_VNM_	230
Virgin_Islands_U_S_VIR_	231
Wake_Island_United_States_ RPRnoteEnumerations14 RPRnoteEnumerationsDeprecated	232
Wallis_and_Futuna_WLF_	233
Western_Sahara_ESH_	234
West_Bank_Israel_ RPRnoteEnumerations18 RPRnoteEnumerationsDeprecated	235
Samoa_WSM_	236
Yemen_YEM_	237
Serbia_and_Montenegro RPRnoteEnumerationsDeprecated	240
Zaire RPRnoteEnumerations24 RPRnoteEnumerationsDeprecated	241

Enumerator	Value
Zambia_ZMB_	242
Zimbabwe_ZWE_	243
Armenia_ARM_	244
Azerbaijan_AZE_	245
Belarus_BLR_	246
Bosnia_and_Herzegovina_BIH_	247
Clipperton_Island_France_ RPRnoteEnumerations15 RPRnoteEnumerationsDeprecated	248
Croatia_HRV_	249
Estonia_EST_	250
Georgia_GEO_	251
Kazakhstan_KAZ_	252
Kyrgyzstan_KGZ_	253
Latvia_LVA_	254
Lithuania_LTU_	255
North_Macedonia_MKD_	256
Midway_Islands_United_States_ RPRnoteEnumerations14 RPRnoteEnumerationsDeprecated	257
Moldova_Republic_of_MDA_	258
Montenegro_MNE_	259
Russia RPRnoteEnumerations25 RPRnoteEnumerationsDeprecated	260
Serbia_and_Montenegro_Montenegro_to_separate_ RPRnoteEnumerationsDeprecated	261
Slovenia_SVN_	262
Tajikistan_TJK_	263
Turkmenistan_TKM_	264
Ukraine_UKR_	265
Uzbekistan_UZB_	266
Czech_Republic_CZE_	267
Slovakia_SVK_	268
Aaland_Islands_ALA_	269

Enumerator	Value
Bonaire_Sint_Eustatius_and_Saba_BES_	270
Congo_Democratic_Republic_of_the_COD_ RPRnoteEnumerations26	271
Curacao_CUW_	272
Eritrea_ERI_	273
Saint_Barthelemy_BLM_	274
Saint_Martin_French_Part_MAF_	275
Serbia_SRB_	276
Sint_Maarten_Dutch_part_SXM_	277
South_Sudan SSD_	278
Svalbard_and_Jan_Mayen_SJM_	279
Timor_Leste_TLS_	280
United_States_Minor_Outlying_Islands_UMI_	281
Palestine_State_of_PSE_	282

CoverShroudStatusEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Cover/Shroud Status [UID 426]*

Enumerator	Value
Closed	0
Opening	1
Cover_Shroud_Blowndetached	2
Open_Attached	3

CryptoKeyIDCryptoModeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Crypto Key ID-Crypto Mode [UID 449]*

Enumerator	Value
Baseband	0
Diphase	1

DEDamageDescriptionComponentDamageStatusEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *DE Damage Description-Component Damage Status [UID 315]*

Enumerator	Value
No_Damage	0
Minor_Damage	1
Medium_Damage	2
Major_Damage	3
Destroyed	4

DEDamageDescriptionComponentVisualSmokeColorEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *DE Damage Description-Component Visual Smoke Color [UID 316]*

Enumerator	Value
No_Smoke	0
White	1
Gray	2
Black	3

DEFireFlagsStateUpdateFlagEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *DE Fire-Flags-State/Update Flag [UID 374]*

Enumerator	Value
Update_Due_to_Heartbeat_Timer	0
State_Change	1

DEFirePulseShapeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *DE Fire-Pulse Shape [UID 312]*

Enumerator	Value
Other	0
Square_Wave	1
Continuous_Wave	2
Gaussian	3

DEPrecisionAimpointBeamSpotTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *DE Precision Aimpoint-Beam Spot Type [UID 311]*

Enumerator	Value
Other	0
Gaussian	1
Top_Hat	2

DRParametersTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *DR Parameters Type [UID 296]*

Enumerator	Value
None	0
Local_Euler_Angles_Yaw_Pitch_Roll_	1
World_Orientation_Quaternion	2

DamageAreaEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize03](#)

Semantics: *Damage Area [UID 889]*

Enumerator	Value
Damage_Area_1_default_is_Side_1_Front_Side_or_Not_Applicable_if_Damage_Areas_are_not_defined_	0
Damage_Area_2_default_is_Side_2_Right_Side_	1
Damage_Area_3_default_is_Side_3_Back_Side_	2
Damage_Area_4_default_is_Side_4_Left_Side_	3

Enumerator	Value
Damage_Area_5_default_is_Corner_1_Front_Right_Corner_	4
Damage_Area_6_default_is_Corner_2_Back_Right_Corner_	5
Damage_Area_7_default_is_Corner_3_Back_Left_Corner_	6
Damage_Area_8_default_is_Corner_4_Front_Left_Corner_	7

DataCategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Data Category [UID 369]*

Enumerator	Value
No_Statement	0
Functional_Data	1
Transponder_Interrogator_Data_Link_Messages	2

DeadReckoningAlgorithmEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Dead Reckoning Algorithm [UID 44]
Enumerated names for the Dead Reckoning Model (DRM) are formed by a combination of letter codes. Following is a description of the letter codes:
F - Fixed orientation; angular velocity not used
R - Rotating; orientation extrapolated using angular velocity
P - Position extrapolated using linear velocity; acceleration not used
V - Velocity and position extrapolated using velocity and acceleration
W - Linear velocity and acceleration provided in world coordinates
B - Linear velocity and acceleration provided in body coordinates; angular velocity required for all extrapolation*

Enumerator	Value
Other	0
Static	1
DRM_FPW	2
DRM_RPW	3
DRM_RVW	4

Enumerator	Value
DRM_FVW	5
DRM_FPB	6
DRM_RPB	7
DRM_RVB	8
DRM_FVB	9

DeltaMode5SAltitudePositiveNegativeIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Delta Mode 5/S Altitude Positive/Negative Indicator [UID 362]*

Enumerator	Value
Positive	0
Negative	1

DesignatorDesignatorCodeEnum16 [RPRnoteEnumerations27](#) [RPRnoteEnumerationsDeprecated](#)

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Designator-Designator Code [UID 81]*

Enumerator	Value
Other RPRnoteEnumerationsDeprecated	0

DesignatorSystemNameEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Designator System Name [UID 80]*

Enumerator	Value
Not_Specified	0
AN_AAQ_4	1000
AN_AAQ_7	1100
AN_AAQ_8	1200
AN_AAQ_14_LANTIRN	1300
AN_AAQ_19	1400

Enumerator	Value
AN_AAQ_22A_SAFIRE	1500
AN_AAQ_22B_SAFIRE_LP	1600
AN_AAQ_22C_Star_SAFIRE_I	1700
AN_AAQ_22D_BRITE_Star	1800
AN_AAQ_24_V_DIRCM_Nemesis	1900
AN_AAQ_25_LTS	2000
AN_AAQ_28_V_LITENING_II	2100
AN_AAQ_30	2200
AN_AAQ_32	2300
AN_AAQ_33_Sniper	2400
AN_AAQ_37	2500
AN_AAQ_38	2600
AN_AAQ_40	2650
AN_AAS_32	2700
AN_AAS_35V	2800
AN_AAS_37	2900
AN_AAS_38	3000
AN_AAS_44_V_	3100
AN_AAS_46	3200
AN_AAS_49	3300
AN_AAS_51	3400
AN_AAS_52_MTS_A	3500
AN_ALQ_10	3600
AN_ASQ_228	3700
AN_AVQ_25	4400
AN_AVQ_26	4500
AN_GVS_5	4600
AN_PED_1_LLDR	4700

Enumerator	Value
TADS_LRF_D	4800
MMS_LRF_D	4900
AH_1_C_NITE	5000
MATES	5100
TCV_115	5200
TIM	5300
TMS_303	5400
TMY_303	5500
ALRAD	5600
RFTDL	5700
VVLR	5800
P0705_HELL	6000
P0708_PULSE	6100
HELD	6200
TYPE_105	6300
TYPE_118	6400
TYPE_121	6500
TYPE_126	6600
TYPE_629	6700
CLDS	6800
TAV_38	6900
TMV_630	7000
ALTM_1020	7100
ALATS	7200
Dark_Star_LAMPS	7300
GLTD_II	7400
MBT_ELRF	7500
Mark_VII	7600

Enumerator	Value
SIRE_V	7700
AN_AAQ_16B	7800
AN_AAQ_16D_AESOP	7900
AN_AAQ_21_Star_SAFIRE_III	8000
AN_AAQ_22E_BRITE_Star	8100
AN_AAQ_36_Star_SAFIRE_II	8200
AN_AAS_38A_Nite_Hawk	8300
AN_AAS_38B_Nite_Hawk	8400
AN_AAS_44C_V_	8500
AN_AAS_53_CSP	8600
AN_ASQ_28_ATFLIR	8700
AN_DAS_1_MTS_B	8800
AN_PAQ_1_LTD	8900
AN_PAQ_3_MULE	9000
AN_PEQ_1_SOFLAM	9090
AN_PEQ_3	9100
AN_PEQ_15_ATPIAL	9140
AN_PEQ_18_IZLID_1000P	9150
AN_TVQ_2_G_VLLD	9200
AN_ZSQ_2_V_1_EOS	9300
AN_ZSQ_2_V_2_EOS	9400
CIRCM	9500
Guardian	9600
IZLID_200P	9700
IZLID_1000P_W	9800
MMS	9900
M_TADS_PNVS_Arrowhead	10000
RBS_70	10100

Enumerator	Value
RBS_90	10200
TADS_PNVS	10300
COLIBRI	10400
Damocles	10500
I_251_Shkval	10600
KPS_53AV_EOTS	10700
Star_SAFIRE_380	10800
JANUS_T_EOS	10900
LOTHAR_EOS	11000
MK46_MOD_1_EOS	11100
MTK_201ME_EOS	11200
Thales_Mirador_Mk2_EOS	11300
TPN_1M_49_23_EOS	11400

DetonationResultEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Detonation Result [UID 62]*

Enumerator	Value
Other	0
Entity_Impact	1
Entity_Proximate_Detonation	2
Ground_Impact	3
Ground_Proximate_Detonation	4
Detonation	5
None_or_No_Detonation_Dud_	6
HE_hit_small	7
HE_hit_medium	8
HE_hit_large	9

Enumerator	Value
Armor_piercing_hit	10
Dirt_blast_small	11
Dirt_blast_medium	12
Dirt_blast_large	13
Water_blast_small	14
Water_blast_medium	15
Water_blast_large	16
Air_hit	17
Building_hit_small	18
Building_hit_medium	19
Building_hit_large	20
Mine_clearing_line_charge	21
Environment_object_impact	22
Environment_object_proximate_detonation	23
Water_Impact	24
Air_Burst	25
Kill_with_fragment_type_1	26
Kill_with_fragment_type_2	27
Kill_with_fragment_type_3	28
Kill_with_fragment_type_1_after_fly_out_failure	29
Kill_with_fragment_type_2_after_fly_out_failure	30
Miss_due_to_fly_out_failure	31
Miss_due_to_end_game_failure	32
Miss_due_to_fly_out_and_end_game_failure	33

ElectromagneticEmissionBeamFunctionEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Electromagnetic Emission-Beam Function [UID 78]*

Enumerator	Value
Other	0
Search	1
Height_Finding	2
Acquisition	3
Tracking	4
Acquisition_and_tracking	5
Command_guidance	6
Illumination	7
Ranging	8
Missile_beacon	9
Missile_Fusing	10
Active_radar_missile_seeker	11
Jamming	12
IFF	13
Navigation_Weather	14
Meteorological	15
Data_transmission	16
Navigational_directional_beacon	17
Time_Shared_Search	20
Time_Shared_Acquisition	21
Time_Shared_Track	22
Time_Shared_Command_Guidance	23
Time_Shared_Illumination	24
Time_Shared_Jamming	25

ElectromagneticEmissionStateUpdateIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Electromagnetic Emission-State Update Indicator [UID 77]*

Enumerator	Value
Heartbeat_Update	0
Changed_Data_Update	1

EmitterNameEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Emitter Name [UID 75]*

Enumerator	Value
_1245_6X	2
_1L117	3
_1L121E	4
_1L250	5
_1L220_U	6
_1L122_1E	7
_1RL257	9
_1RL138	10
_1RL257_Krasukha_4_Jammer	11
_5N20	12
_5H62B	13
Puma_5P_10	14
Puma_5P_10E	15
Puma_5P_10_01	16
Puma_5P_10_01E	17
Puma_5P_10_02	18
Puma_5P_10_02E	19
Laska_5P_10_03	20
Laska_5P_10_03E	21
_5P_10E_MOD	22
_621A_3	25

Enumerator	Value
_860F_1_AL_101	40
_9B_1103M2	42
_1226_DECCA_MIL	45
_9B_1348	46
_3KM6	47
_9KR400	48
Grave_Yard_MFR_SA_28_50N6A	49
_55G6_1	50
_59N6	55
_5N69	57
_67N6	60
_79K6_Pelican	62
_76T6	63
_77T6_ABM	64
_80K6	65
_91N6A_M_	66
_80K6M	67
_96L6E	70
_96L6_TsP	75
_9C18M3	76
_9C36M	77
_9GR400	80
_9_GR_400A	81
_9GR600	90
_9GR606	91
_9_LV_100	125
_9LV_200_TA	135
_9LV_200_TV	180

Enumerator	Value
_9LV_200_TT	181
_9LV200_MK_III	183
_9LV326	185
_9M96E2_Seeker	190
_9M96M_MH	191
_9S15M2	195
_9S19M2	196
_9S19ME	197
_9S32M	198
_9S32ME	199
_9S36E	200
_9S112	215
A310Z	225
A325A	270
A346Z	315
A353B	360
A372A	405
A372B	450
A372C	495
A377A	540
A377B	585
A380Z	630
A381Z	675
A398Z	720
A403Z	765
A409A	810
A418A	855
A419Z	900

Enumerator	Value
A429Z	945
A432Z	990
A434Z	1035
AA_6C_Acrid_R_40_	1070
AA_7C_Apex_R_24R_	1073
A401A	1080
AA_10A_R_27R_	1081
AA_10C_Alamo_C	1082
AA_13_Arrow_Seeker	1085
AAM_4B_MH	1090
AA_300	1094
AA_12_Seeker	1095
AD4A	1096
ADES	1097
ADS_4_LRSR	1098
ACR_430	1099
Agave	1100
ACSOPRI_E	1101
ABD_2000	1102
ADAC_MK_1	1110
ADAC_MK_2	1111
ADAR	1113
ADOUR	1115
AGAT_9B_1348	1117
Adros_KT_01AV	1118
Agat_9E420	1120
AGM_158_JASSM_SAR	1122
AGM_88_HARM_MMW	1123

Enumerator	Value
AGRION_15	1125
AHV_7	1130
AHV_17	1150
AI_MK_23	1170
AIDA_II	1215
AIM_120A	1216
AIM_7M_Sparrow	1218
AISTENOK_1L271	1230
ALA_51	1240
Albatros_MK2	1260
ALT_50	1263
ALTAIR	1264
AM_APS_717	1265
AMES_13_MK_1	1268
WGU_16_B	1270
Box_Spring_1L13_3_55G6_	1280
Box_Spring_B_1L13_3_55G6_	1282
Amber_Wedge_Jammer	1285
AMDR_3D	1288
ANA_SPS_502	1305
Anemone	1306
ANRITSU_Electric_AR_30A	1350
Antilope_V	1395
AN_AAQ_24	1397
AN_ADM_160	1398
AN_ALE_50	1400
AN_ALQ_76	1410
AN_ALQ_99	1440

Enumerator	Value
AN_ALQ_99_Band_4	1441
AN_ALQ_99_LBT	1442
AN_ALQ_100	1485
AN_ALQ_101	1530
AN_ALQ_119	1575
AN_ALQ_122	1585
AN_ALQ_126A	1620
AN_ALQ_128	1621
AN_ALQ_126B	1622
AN_ALQ_131	1626
AN_ALQ_131_Black_II	1627
AN_ALQ_135C_D	1628
AN_ALQ_144A_V_3	1630
AN_ALQ_153	1632
AN_ALQ_157Jammer	1633
AN_ALQ_155	1634
AN_ALQ_156	1635
AN_ALQ_161_A	1636
AN_ALQ_161	1637
AN_ALQ_162	1638
AN_ALQ_164	1639
AN_ALQ_165	1640
AN_ALQ_187_Jammer	1641
AN_ALQ_167	1642
AN_ALQ_172_V_1	1643
AN_ALQ_172_V_2	1644
AN_ALQ_172_V_3	1645
AN_ALQ_176	1646

Enumerator	Value
AN_ALQ_178	1647
AN_ALQ_184	1648
AN_ALQ_184_V_9	1649
AN_ALQ_188	1650
AN_ALQ_214	1651
AN_ALR_56	1652
AN_ALQ_221	1653
AN_ALR_69	1654
SIRFC_AN_ALQ_211_V_	1655
AN_ALT_16A	1656
AN_ALQ_173	1657
AN_ALT_28	1658
AN_ALR_66B_Jammer_1659	1659
AN_ALT_32A	1660
AN_ALQ_196	1661
AN_ALQ_249_V_1	1662
AN_ALQ_240_V_1_Jammer	1663
AN_ALR_66B_Jammer_1664 RPRnoteEnumerationsDeprecated	1664
AN_APD_10	1665
AN_ALQ_249_V_2	1666
AN_ALQ_213	1670
ALQ_214A_V_4_5_Jammer	1672
AN_ALQ_218	1680
AN_ALQ_250_EPAWSS	1685
AN_APG_50	1700
AN_APG_53	1710
AN_APG_59	1755
AN_APG_63	1800

Enumerator	Value
AN_APG_63_V_1	1805
AN_APG_63_V_2	1807
AN_APG_63_V_3	1809
AN_APG_65	1845
AN_APG_66	1870
AN_APG_66_V_	1871
AN_APG_66_V_2	1872
AN_APG_67	1880
AN_APG_68	1890
AN_APG_68_v_9	1895
AN_APG_70	1935
AN_APG_71	1940
AN_APG_73	1945
AN_APG_77	1960
AN_APG_78	1970
AN_APG_79	1971
AN_APG_80	1972
AN_APG_81	1974
AN_APG_82_V_1	1975
AN_APG_83	1976
AN_APG_502	1980
AN_APN_1	2025
AN_APN_22	2070
AN_APN_59	2115
AN_APN_69	2160
AN_APN_81	2205
AN_APN_102	2220
AN_APN_117	2250

Enumerator	Value
AN_APN_118	2295
AN_APN_122	2320
AN_APN_130	2340
AN_APN_131	2385
AN_APN_133	2430
AN_APN_134	2475
AN_APN_141_V_	2476
AN_APN_147	2520
AN_APN_150	2565
AN_APN_153	2610
AN_APN_154	2655
AN_APN_155	2700
AN_APN_159	2745
AN_APN_177	2746
AN_APN_179	2747
AN_APN_169	2748
AN_APN_182	2790
AN_APN_187	2835
AN_APN_190	2880
AN_APN_194	2925
AN_APN_195	2970
AN_APN_198	3015
AN_APN_200	3060
AN_APN_202	3105
AN_APN_205	3106
AN_APN_209	3120
AN_APN_209D	3121
AN_APN_209A	3122

Enumerator	Value
AN_APN_215	3148
AN_APN_217	3150
AN_APN_218	3152
AN_APN_224	3153
AN_APN_227	3154
AN_APN_230	3155
AN_APN_232	3156
AN_APN_237A	3157
AN_APN_234	3158
AN_APN_235	3159
AN_APN_238	3160
AN_APN_222	3161
AN_APN_239	3162
AN_APN_241	3164
AN_APN_242	3166
AN_APN_243	3170
AN_APN_506	3195
AN_APQ_72	3240
AN_APQ_99	3285
AN_APQ_100	3330
AN_APQ_102	3375
AN_APQ_107	3376
AN_APQ_109	3420
AN_APQ_113	3465
AN_APQ_120	3510
AN_APQ_122	3512
AN_APQ_126	3555
AN_APQ_128	3600

Enumerator	Value
AN_APQ_129	3645
AN_APQ_148	3690
AN_APQ_150A	3700
AN_APQ_153	3735
AN_APQ_155	3770
AN_APQ_159	3780
AN_APQ_164	3785
AN_APQ_166	3788
AN_APQ_170	3790
AN_APQ_174	3791
AN_APQ_180	3794
AN_APQ_181	3795
AN_APQ_186	3800
AN_APS_15J	3810
AN_APS_16_V_2	3813
AN_APS_31	3820
AN_APS_42	3825
AN_APS_80	3870
AN_APS_88	3915
AN_APS_88A	3916
AN_APS_94	3920
AN_APS_96	3922
AN_APS_113	3958
AN_APS_115	3960
AN_APS_116	4005
AN_APS_120	4050
AN_APS_121	4095
AN_APS_124	4140

Enumerator	Value
AN_APS_125	4185
AN_APS_127	4190
AN_APS_128	4230
AN_APS_130	4275
AN_APS_133	4320
AN_APS_134	4365
AN_APS_137	4410
AN_APS_137_V_5	4413
AN_APS_137B	4415
AN_APS_137B_V_5	4420
AN_APS_137D_V_5_Elta	4425
AN_APS_138	4455
AN_APS_139	4460
AN_APS_143	4464
AN_APS_143_V_1	4465
AN_APS_143B	4466
AN_APS_143_V_3	4467
AN_APS_143B_V_3	4468
AN_APS_153	4475
AN_APS_154	4476
AN_APS_150	4480
AN_APS_145	4482
AN_APS_147	4485
AN_APS_149	4486
AN_APS_503	4489
AN_APS_504	4490
AN_APS_705	4491
AN_APW_22	4500

Enumerator	Value
AN_APW_23	4545
AN_APX_6	4590
AN_APX_7	4635
AN_APX_39	4680
AN_APX_64_V_	4681
AN_APX_72	4725
AN_APX_76	4770
AN_APX_78	4815
AN_APX_100	4816
AN_APX_101	4860
AN_APX_113_AIFF	4870
AN_APY_1	4900
AN_APY_2	4905
AN_APY_3	4950
AN_APY_7	4952
LYNX_tm_AN_APY_8	4953
AN_APY_9	4954
AN_APY_10	4955
AN_ARN_21	4995
AN_ARN_52	5040
AN_ARN_84	5085
AN_ARN_118	5130
AN_ARN_153_V_	5131
AN_ARN_153	5165
AN_ARW_73	5175
AN_ASB_1	5220
AN_ASG_21	5265
AN_ASN_137	5266

Enumerator	Value
AN_ASN_128	5270
AN_ASQ_108	5280
AN_ASQ_239	5285
AN_AST_502	5290
AN_AVQ_55	5300
AN_AWG_9	5310
AN_BRN_1	5320
AN_BPS_5	5325
AN_BPS_9	5355
AN_BPS_15	5400
AN_BPS_15_H	5401
AN_BPS_15J	5402
AN_BPS_16	5405
AN_BPS_16_V_2	5406
AN_CPN_4	5410
AN_CPN_18	5415
AN_CRM_30	5420
AN_DPW_23	5430
AN_DSQ_26_Phoenix_MH	5445
AN_DSQ_28_Harpoon_MH	5490
AN_FPN_1	5491
AN_FPN_28	5493
AN_FPN_33	5494
AN_FPN_40	5495
AN_FPN_62	5500
AN_FPN_66	5502
AN_FPS_8	5503
AN_FPN_67	5504

Enumerator	Value
AN_FPS_16	5505
AN_FPS_5	5506
AN_FPS_18	5507
AN_FPS_89	5508
AN_FPS_49	5509
AN_FPS_117	5510
AN_FPS_85	5511
AN_FPS_88	5512
AN_FPS_113	5513
AN_FPS_115	5514
AN_FPS_20R	5515
AN_FPS_132	5516
AN_FPS_77	5520
AN_FPS_41	5521
AN_FPS_100A	5522
AN_FPS_103	5525
AN_FPS_108	5526
AN_GPN_12	5527
AN_FPS_124_V_	5528
AN_FPS_129	5529
AN_GPX_6	5530
AN_GPX_8	5535
AN_GRN_12	5537
AN_MPN_14K	5538
AN_MPN_14	5539
AN_MPQ_10	5540
AN_MPN_17	5541
AN_MPQ_33_39_46_57_61_HPIR_ILL	5545

Enumerator	Value
AN_MPQ_34_48_55_62_CWAR_TA	5550
AN_MPQ_49	5551
AN_MPQ_35_50_PAR_TA	5555
AN_MPQ_50C	5556
AN_MPQ_37_51_ROR_TT	5560
AN_MPQ_43	5565
AN_MPQ_50	5567
AN_MPQ_53	5570
AN_MPQ_63	5571
AN_MPQ_64	5575
AN_SLQ_32	5576
AN_MPQ_65	5577
AN_SLQ_32_V_4	5578
AN_SLQ_32A	5579
AN_SPG_34	5580
AN_MSQ_104	5582
AN_MPS_36	5583
AN_SLQ_503	5584
AN_SPG_48_MK_25_MOD_3	5620
AN_SPG_50	5625
AN_SPG_51	5670
AN_PPQ_2	5690
AN_PPS_15	5700
AN_PPS_5	5705
AN_PPS_5D	5710
AN_SPG_51_CWI_TI	5715
AN_SPG_51_FC	5760
AN_SPG_51C_D	5761

Enumerator	Value
AN_SPG_52	5805
AN_SPG_53	5850
AN_SPG_55B	5895
AN_SPG_60	5940
AN_SPG_62	5985
AN_SPG_503	5995
AN_SPN_4	6015
AN_SPN_11	6025
AN_SPN_35	6030
AN_SPN_41	6050
AN_SPN_43	6075
AN_SPN_43A	6076
AN_SPN_43C	6078
AN_SPN_46	6085
AN_SPQ_2	6120
AN_SPQ_5A	6155
AN_SPQ_9A	6165
AN_SPQ_9B	6166
AN_SPQ_34	6190
AN_SPS_4	6210
AN_SPS_5	6255
AN_SPS_5C	6300
AN_SPS_6	6345
AN_SPS_10	6390
AN_SPS_21	6435
AN_SPS_28	6480
AN_SPS_37	6525
AN_SPS_39A	6570

Enumerator	Value
AN_SPS_40	6615
AN_SPS_41	6660
AN_SPS_48	6705
AN_SPS_48C	6750
AN_SPS_48E	6752
AN_SPS_49	6795
AN_SPS_49_V_1	6796
AN_SPS_49_V_2	6797
AN_SPS_49_V_3	6798
AN_SPS_49_V_4	6799
AN_SPS_49_V_5	6800
AN_SPS_49_V_6	6801
AN_SPS_49_V_7	6802
AN_SPS_49_V_8	6803
AN_SPS_49A_V_1	6804
AN_SPS_52	6840
AN_SPS_53	6885
AN_SPS_55	6930
AN_SPS_52C	6945
AN_SPS_55_CS	6970
AN_SPS_55_SS	6975
AN_SPS_58	7020
AN_SPS_58C	7025
AN_SPS_59	7065
AN_SPS_64	7110
AN_SPS_64_V_9	7119
SPS64_V_12	7120
AN_SPS_65	7155

Enumerator	Value
AN_SPS_66	7175
AN_SPS_67	7200
AN_SPS_73_I_	7201
AN_SPS_69	7210
AN_SPS_73	7215
AN_SPS_74	7216
AN_SPS_88	7225
AN_SPS_501	7226
AN_SPS_505	7230
AN_SPY_1	7245
AN_SPY_1A	7250
AN_SPY_1B	7252
AN_SPY_1B_V_	7253
AN_SPY_1D	7260
AN_SPY_1D_V_	7261
AN_SPY_1F	7265
AN_SPY_3	7266
AN_TPN_12	7267
AN_SPY_4	7268
AN_TLQ_32_ARM_Decoy	7269
AN_TPN_17	7270
AN_TPN_8	7271
AN_TPN_22	7272
AN_TLQ_17A	7273
AN_TMS_1	7274
AN_TPN_24	7275
AN_TPN_25	7276
AN_TMS_2	7277

Enumerator	Value
AN_TPN_19	7278
AN_TPN_31	7279
AN_TPQ_18	7280
AN_SPY_6_V_	7281
AN_SPY_7_V1_	7282
AN_TPQ_36	7295
AN_TPQ_37	7300
AN_TPQ_38_V8_	7301
AN_TPQ_39_V_	7302
AN_TPQ_47	7303
AN_TPS_43	7305
AN_TPS_43E	7310
AN_TPQ_48	7311
AN_TPQ_49	7312
AN_TPQ_46A	7313
AN_TPS_34	7314
AN_TPS_59	7315
AN_TPS_44	7316
AN_TPQ_50	7317
AN_TPS_63	7320
AN_TPS_65	7321
AN_TPS_70_V_1	7322
AN_TPS_63SS	7323
AN_TPS_73	7324
AN_TPS_75	7325
AN_TPS_77	7326
AN_TPS_78	7327
AN_TPS_79	7328

Enumerator	Value
AN_TPS_703	7329
AN_TPX_46_V_7	7330
AN_TPS_80	7331
AN_TPY_2	7333
AN_TSQ_288	7334
AN_ULQ_6A	7335
AN_ULQ_19	7340
AN_ULQ_21	7345
AN_UPN_25	7380
AN_UPS_1	7425
AN_UPS_2	7426
AN_UPS_3	7427
AN_UPX_1	7470
AN_UPX_5	7515
AN_UPX_11	7560
AN_UPX_12	7605
AN_UPX_17	7650
AN_UPX_23	7695
AN_USQ_113_V_3	7700
AN_VPS_2	7740
AN_PLM_3	7750
AN_PLM_3A	7751
AN_PLM_4	7752
AN_ZPY1	7753
AN_ZPY_2_MP_RTIP	7754
AN_ZPY_3	7755
AN_ZPY_8	7760
AORL_1AS	7761

Enumerator	Value
AORL_85K_TK_MTA	7762
APAR	7765
Aparna	7770
APECS_II	7780
Apelco_AD_7_7	7785
APG_71	7830
APN_148	7875
APN_227	7920
APQ_113 RPRnoteEnumerationsDeprecated	7965
APQ_120 RPRnoteEnumerationsDeprecated	8010
APQ_148 RPRnoteEnumerationsDeprecated	8055
APS_504_V3	8100
AQUITAINE_II	8102
AR_1	8103
AR_3D	8105
Plessey_AR_5	8112
AR_15	8113
AR_15_2	8114
AR_320	8115
AR_325	8118
AR_327	8120
Arbalet_52	8121
ARBB_31	8122
ARBB_33	8123
ARIES	8126
Aries_Nav	8127
Aries_CS	8128
ARGS_14E	8134

Enumerator	Value
ARGS_31	8135
ARGUS	8140
AR_M31	8145
ARECIBO	8150
ARED	8160
ARI_5954	8190
ARI_5955	8235
ARI_5979	8280
ARGSN_31	8281
ARGOS_10	8282
ARGOS_800	8283
ARI_5983	8284
ARI_5991	8285
ARI_5995	8290
ARINC_564_BNDX_KING_RDR_1E	8325
ARINC_700_BNDX_KING_RDR_1E	8370
ARK_1	8375
ARL_M_MTI	8378
ARMOR	8379
ARSR_3	8380
ARS_400	8381
ARSR_1	8382
ARSR_4	8384
ARSR_18	8390
ARTHUR	8395
ARTHUR_MOD_B	8400
ARTHUR_MOD_C	8405
ARTISAN_3D	8410

Enumerator	Value
AS_2_Kipper	8415
AS_2_Kipper_MH	8460
AS_3_YJ_83K_mmW_MH RPRnoteEnumerationsDeprecated	8470
AS_34_Kormoran_Seeker	8480
AS_4_Kitchen	8505
AS_4_Kitchen_MH	8550
AS_5_Kelt_MH	8595
AS_6_Kingfish_MH	8640
AS_7_Kerry	8685
AS_7_Kerry_MG	8730
AS_15_KENT_altimeter	8735
AS_17A_Krypton_Seeker	8736
AS_17B_Krypton_Seeker	8737
AS901	8750
AS901A	8751
ASARS2	8755
ASDE_KDD	8756
ASLESHA	8757
A_SMGCS	8758
ASMI_18X	8759
Aspide_AAM_SAM_ILL	8760
ASMI_3	8761
Aselsan_MAR	8762
ASR_2000	8771
ASR_4	8772
ASR_4D	8773
ASR_O	8775
ASR_12_8776	8776

Enumerator	Value
ASR_22AL	8778
ASR_3	8779
ASR_5	8780
ASR_7	8782
ASR_8	8785
ASR_9	8790
ASR_9000	8791
ASTI	8792
ASR_11_DASR	8793
ASR_12_8795 RPRnoteEnumerationsDeprecated	8795
Raytheon_ASR_10SS	8812
ASR_23SS	8816
Arabel	8818
ASTRE	8819
AT_2_Swatter_MG	8820
AT_6_Spiral_MG_9K114_Shturm_MG	8824
ASTOR	8825
ASTRA_RCI	8826
ATCR_22	8830
ATCR_22_M	8831
ATCR_2T	8832
ATCR_33	8840
ATCR_33_K_M	8845
ATCR_33S	8846
ATCR_3T	8847
ATCR_44	8848
ATCR_44_K	8849
Argos_73	8850

Enumerator	Value
ATCR_44_M_S	8851
ATCR_4T	8852
Atlas_Elektronk_TRS_N	8865
ATLAS_8600X	8866
Atlas_9600M	8867
ATLAS_9600X	8868
ATLAS_9600S	8869
ATLAS_9740_VTS	8870
ATLASS	8871
ATR_500C	8880
AVG_65	8910
AVH_7	8955
AVIA_CM	8980
AVIA_D	8985
Aviaconversia	8990
Aviaconversia_II	8993
Aviaconversia_III	8995
AVQ_20	9000
AVQ_21	9005
AVQ30X	9045
AVQ_50_RCA_	9075
AVQ_70	9090
AWS_5	9135
AWS_6	9180
AWS_6B_300	9185
B597Z	9200
B636Z	9205
Back_Board	9215

Enumerator	Value
Back_Net_A_B	9225
Back_Trap	9270
BAE_Systems_RT_1805_APN	9280
BAES_DASS_2000_Jammer	9281
Balance_Beam	9285
BALTIKA_B	9300
BALTYK	9310
Ball_End	9315
Ball_Gun	9360
BALL_POINT	9370
Band_Stand	9405
Band_Stand_3	9406
Bar_Lock_P_35_37_A_P_50_B_	9450
BARAX	9475
BASIR_110D	9485
Bass_Tilt	9495
Badger	9505
Barracuda_Jammer	9510
Bavar_373_TTR	9511
Bavar_373_TAR	9512
Bavar_373_TELAR_TER	9520
Baykal_Countermeasures_Suite	9530
Beacon	9540
Bean_Sticks	9585
Bee_Hind	9630
Bell_Nip_Jammer	9638
Bell_Push_Jammer	9639
Bell_Crown_A	9640

Enumerator	Value
Bell_Crown_B	9642
Bell_Squat	9643
BIG_BACK	9645
Big_Bird_A_B_C	9659
Big_Bird_D	9660
Big_Bird_D_Mod	9661
Big_Bird_E_91N6E	9662
Big_Bulge	9675
Big_Bulge_A	9720
Big_Bulge_B	9765
BIG_EYE	9775
Big_Fred_SNAR_10	9780
BIG_HEAD_B	9781
Big_Mesh	9810
Big_Net	9855
Big_Rest_B	9875
Bill_Board_9S15MT	9885
Bill_Fold	9900
BLIGHTER_400	9903
Blowpipe_MG	9905
BLR	9920
Sea_Harrier_FRS_Mk_1_5_Blue_Fox	9930
Blue_Kestrel	9933
Sea_Harrier_F_A_Mk_2_Blue_Vixen	9935
Blue_Silk	9945
Blue_Parrot	9990
Blue_Orchid	10035
BM_DJG_8715	10057

Enumerator	Value
Boat_Sail	10080
BOR_A_550	10090
Bofors_Electronic_9LV_331	10125
Bofors_Ericsson_Sea_Giraffe_50_HC	10170
Bowl_Mesh	10215
Box_Brick	10260
Box_Tail	10305
BM_KG300G_Jamming_Pod	10308
BM_KG600_Jamming_Pod	10310
BM_KG800_Jamming_Pod	10312
BM_KG_8601_8605_8606	10315
BPS_11A	10350
BPS_14	10395
BPS_15A	10440
BR_3440CA_X57	10450
BR_15_Tokyo_KEIKI	10485
BrahMos	10500
BridgeMaster	10510
BridgeMaster_E_EPA	10511
BridgeMaster_E_ATA_and_ARPA	10512
BridgeMaster_E_naval	10513
Brimstone_mmW_MH	10520
Bread_Bin	10530
Brick_Board_Asr	10540
BT_271	10575
BU_304	10595
BX_732	10620
BUK_MB	10630

Enumerator	Value
Buran_D	10642
BUREVISNYK_1	10650
Buzz_Stand	10665
C_5A_Multi_Mode_Radar	10710
C_802_AL	10711
CAESAR	10740
Caiman	10755
Cake_Stand	10800
Calypso_C61	10845
Calypso_C63	10846
Calypso_Ii	10890
Calypso_III	10891
Calypso_IV	10892
Cardion_Coastal	10895
Castor_Ii	10935
Castor_2J_TT_Crotale_NG_	10940
Cat_House	10980
CDR_431	10985
CEAFAR	10987
CEAMOUNT	10988
CEAFAR2_L	10989
CEROS_200	10990
CEROS_200_CWI	10991
CEATAC	10992
CEAOPS	10993
Cerberus_III	10994
CH_SS_N_6	10995
Cerberus_IV	10996

Enumerator	Value
Chair_Back_TT	11000
Chair_Back_ILL	11010
Cheese_Board_LEMZ_96L6	11020
Cheese_Brick	11025
Cheese_Cake	11030
Chrome_Dome	11060
Clam_Pipe_Leninetz_Obzor_MS	11070
Clamshell	11115
CLC_1_TER	11117
CLC_2_TAR	11118
CLC_3_TAR	11119
CLR_155	11120
COAST_WATCHER_100	11123
Coastal_Giraffe	11125
COBRA	11130
Cobra_Shoe	11133
Colibri	11137
Collins_WXR_300	11155
Collins_WXR_700X	11160
Collins_WXR_840_Collins_TWR_850	11165
Collins_DN_101	11205
COMET_1	11230
CONDOR_MK_2	11235
Consilium_Selesmar_RTM_25_XIM	11240
Contraves_Sea_Hunter_MK_4	11250
Corn_Can	11260
COSMO_SKYMED_1	11265
CR_105_RMCA	11270

Enumerator	Value
CREW_Duke_2	11280
CREW_Duke_3	11290
Cross_Bird	11295
Cross_Dome	11340
Cross_Legs	11385
Cross_Out	11430
Cross_Slot	11475
Cross_Sword	11520
Cross_Up	11565
Cross_Sword_FC	11610
THD_5000_Crotale_Acquisition_TA	11655
Griffon_Crotale_NG_TA	11660
Crotale_TT	11665
Crotale_MGMissile_System	11700
CSA_4A_TAR	11705
CSA_4A_TLAR	11706
CSA_4B_TAR	11707
CSA_4B_TLAR	11708
CSA_15A_TAR	11709
CSA_15A_TER	11710
CSA_15B_TAR	11711
CSA_15B_TER	11712
CS_10_TA	11715
Spark_Storm_TER_CSA_20_	11717
Rice_Guard_TAR_CSA_20_CSA_27_	11718
CSA_23_FDV_MG	11719
CSA_23_Missile_MH	11720
CSA_26_TELAR	11724

Enumerator	Value
CSF_Varán	11725
CSS_N_4_MH	11735
CSS_C_3C_CAS_1M1_M2_MH	11745
CSS_C_3_Seersucker_MH_HY_2B_MH	11748
CSS_C_2B_HY_1A_MH	11790
CSS_N_4_Sardine RPRnoteEnumerationsDeprecated	11800
CSS_N_8_MH	11810
Curl_Stone_B	11825
CWS_1	11830
CWS_2	11835
CWS_3	11840
Cygnus	11860
Cylinder_Head	11880
Cymbeline	11902
Cyrano_II	11925
Cyrano_IV	11970
Cyrano_IV_M	11975
DA_01_00	12010
DA_05_00	12015
DA_05_2	12016
DA_08	12018
Dawn	12060
DCR	12090
Dead_Duck	12105
DECCA_20_V90_9	12110
DECCA_20_V90S	12111
DECCA_45	12150
DECCA_50	12195

Enumerator	Value
DECCA_71	12196
Decca_72	12197
DECCA_110	12240
DECCA_170	12285
DECCA_HF_2	12292
DECCA_202	12330
DECCA_D202	12375
DECCA_303	12420
DECCA_535	12430
DECCA_626	12465
DECCA_629	12510
DECCA_914	12555
DECCA_916	12600
DECCA_926	12610
DECCA_1070A	12615
Decca_1008	12616
DECCA_1226_Commercial	12645
DECCA_1290	12655
DECCA_1626	12690
DECCA_2070	12691
Decca_1630	12694
DECCA_2459	12735
DECCA_AWS_1	12780
DECCA_AWS_2	12782
DECCA_AWS_4	12785
DECCA_AWS_4_2_	12787
DECCA_MAR	12800
DECCA_RM_326	12805

Enumerator	Value
DECCA_RM_416	12825
DECCA_RM_970BT	12850
DECCA_RM_914	12870
DF_21D_Seeker	12875
DECCA_RM_1690	12915
DECCA_1690	12916
DECCA_Super_101_MK_3	12960
DISS_1	13005
DISS_7	13006
DISS_013	13007
DISS_15D	13015
DLD_100A	13020
DN_181_Rapier_TTDN_181	13050
BLINDFIRE_FSC_TT_Rapier_2000_TT	13055
Dog_Ear	13095
Dog_House	13140
DM3	13141
DM_3B	13142
DM_5	13143
Don_2	13185
Don_A_B_2_Kay	13230
Donets	13275
Doppler_90_Series	13280
Down_Beat	13320
DR_582	13360
DRAA_2A	13365
DRAA_2B	13410
DRAA_9A	13415

Enumerator	Value
DRAA_11A	13420
DRAC_37B	13450
DRAC_38	13452
DRAC_39	13455
DRAC_39A	13456
DRAC_43A	13460
DRAC_44A	13465
Dragon_Eye	13477
Dragon_Eye_2	13480
Dragon_Eye_3	13481
Dragon_Eye_4	13485
DRBC_30B	13500
DRBC_31A	13545
DRBC_31D	13546
DRBC_32	13585
DRBC_32A	13590
DRBC_32D	13635
DRBC_33A	13680
DRBI_10	13725
DRBI_23	13770
DRBJ_11B	13815
DRBN_30	13860
DRBN_32	13905
DRBN_34	13915
DRBR_51	13950
DRBV_20A	13994
DRBV_20B	13995
DRBV_21_Mars_05	14020

Enumerator	Value
DRBV_22	14040
DRBV_23	14041
DRBV_26C	14085
DRBV_26D	14086
DRBV_30	14130
DRBV_31	14131
DRBV_50	14175
DRBV_51	14220
DRBV_51A	14265
DRBV_51B	14310
DRBV_51C	14355
Drop_Kick	14400
DRUA_31	14445
Drum_Tilt	14490
Drum_Tilt_A	14535
Drum_Tilt_B	14545
DRUN_30A	14560
Dumbo	14580
DWSR_92	14583
DWSR_93S	14585
EAGLE	14586
EAGLE_Mk_1	14587
EAJP_Jamming_Pod	14588
EKCO_E390	14590
CAPTOR_ECR_90	14600
ECR_90_Jammer	14601
Egg_Cup_A_B	14625
EISCAT	14640

Enumerator	Value
EKCO_E120	14660
EKCO_190	14670
Ekran_1	14677
EL_L_8222	14710
EL_L_8240	14713
EL_M_2001B	14715
EL_M_2022	14725
EL_M_2032	14726
EL_M_2052	14727
EL_M_2055	14728
EL_M_2060	14730
EL_M_2075	14735
EL_M_2022U_V_3	14736
EL_M_2080	14737
EL_M_2080S	14738
EL_M_2085	14739
EL_M_2106	14740
EL_M_2106NG	14741
EL_M_2125	14742
EL_M_2129	14743
EL_M_2150	14744
EL_M_2083	14745
EL_M_2084	14746
EL_M_2160_V1	14747
EL_M_2084_MMR	14748
EL_M_2112	14749
EL_M_2200	14750
EL_M_2133	14751

Enumerator	Value
EL_M_2205	14755
EL_M_2207	14760
EL_M_2215	14765
EL_M_2216_V_	14770
EL_M_2216XH	14772
EL_M_2218S	14775
ELT_361	14776
EL_M_2258	14777
ELT_553	14779
ELT_558	14780
ELT_572	14785
ELT_715	14790
Elta_ELM_2022A	14800
ELTA_EL_M_2221_GM_STGR	14805
EL_M_2228S_3D	14806
EL_M_2705	14807
EL_M_2226	14808
EL_M_2228X	14809
ELTA_SIS	14810
EL_M_2238	14811
EL_M_2248	14815
EL_M_2288	14820
EL_M_2311	14821
ELM_2026	14822
ELNA_4007	14830
ELT_318	14831
ELW_2085	14832
ELT_521	14833

Enumerator	Value
ELW_2090	14835
Enhanced_Meteor_Detection_Radar_EMDR	14845
EMD_2900	14850
EMPAR	14851
End_Tray	14895
EQ_36	14896
Ericsson_SLAR	14897
Erieye	14898
ES_3701S_ESM	14899
ESR_1	14900
ESR_220	14901
ESR380	14902
ESTEREL	14903
ET_316	14905
Exocet_Type	14935
Exocet_AL	14936
Exocet_1	14940
Exocet_1_MH	14985
Exocet_2	15030
Eye_Bowl	15075
Eye_Shield	15120
F332Z	15140
Falcon_Claw_TI	15155
Falcon_Claw_TT	15156
FALCON	15160
FALCON_G	15161
Falcon_Eye	15163
Fan_Song_A	15165

Enumerator	Value
Fan_Song_B_F_TA	15200
Fan_Song_B_F_TT	15210
Fan_Song_C_E_TA	15220
Fan_Song_C_E_TT	15230
Fan_Song_C_E_MG	15240
Fan_Song_B_FF_MG	15255
Fan_Tail	15300
FAR_2117	15301
FAR_2827	15302
FAR_2837S	15303
FAR_3000	15304
FB_7_Radar	15305
FCR_1401	15310
FCS_2_12E	15312
FCS_2_12G	15313
FCS_2_21A	15315
FCS_2_21C	15317
FCS_2_22	15318
FCS_2_31	15319
FCS_3	15320
Fin_Curve	15345
Fire_Can	15390
Fire_Dish	15435
Fire_Dome_TA	15470
Fire_Dome_TT	15475
Fire_Dome_TI	15480
Fire_Iron	15525
Fire_Wheel	15570

Enumerator	Value
Fish_Bowl	15615
FK_3	15620
FLAIR	15650
Flap_Lid	15660
Flap_Lid_B_30N6E	15661
Flap_Truck	15705
Flap_Wheel	15750
Flash_Dance	15795
Flash_Dance_M	15800
Flat_Face_A_B_C_D_P_15	15840
Flat_Face_E_35N6	15842
Flat_Screen	15885
Flat_Spin	15930
Flat_Track_Jammer	15970
Flat_Twin	15975
FL_400	15980
FL_1800	15985
FL_1800U RPRnoteEnumerations28	15990
FL_1800S RPRnoteEnumerations28	16000
Fledermaus	16020
FLYCATCHER	16030
FLYCATCHER_MK_2	16035
Fly_Screen	16065
Fly_Screen_A_B	16110
Fly_Trap_B	16155
FM_90	16160
Fog_Lamp_MG	16200
Fog_Lamp_TT	16245

Enumerator	Value
Foil_Two	16290
Football	16300
Fox_Hunter	16335
Fox_Fire_AL	16380
Fox_Fire_ILL	16390
FR_151A	16400
Furuno_FR_1500_FR_1600	16405
FR_1505_DA	16410
FR_1510DS	16412
FR_2000	16420
Furuno_2855W	16421
Fregat_MAE	16422
Fregat_N1	16423
Fregat_N2	16424
Front_Dome	16425
Fregat_MAE_5	16426
Front_Door	16470
Front_Piece	16515
Furby_mmW_MH	16520
Furke	16550
Furke_2_Furke_E_Positiv_ME1_	16552
Furke_4	16554
Furuno	16560
Furuno_1721	16561
Furuno_1934C	16564
Furuno_1715	16565
Furuno_1730	16580
Furuno_1731_Mark_3	16581

Enumerator	Value
Furuno_1832	16585
Furuno_1835	16587
Furuno_1932	16590
Furuno_1935	16596
Furuno_701	16605
Furuno_1940	16606
Furuno_711_2	16650
Furuno_FR_2130S	16652
Furuno_FAR_2137S	16654
Furuno_FAR_28X7	16655
Furuno_FAR_3230S	16658
FR_2110	16660
FR_2115	16662
FR_8062	16663
Furuno_2125	16670
Furuno_240	16690
Furuno_2400	16695
Furuno_FR_810D_FR_801D	16725
Furuno_8051	16730
Furuno_DRS_2D	16732
Furuno_DRS_4D	16733
Furuno_DRS_4A	16734
G030A_APD_31_	16735
Furuno_DRS6A_X_Class	16736
DRS4W	16737
GA_01_00	16740
Gabbiano	16750
Gage	16785

Enumerator	Value
Gaofen_3	16787
GAOFEN_12	16789
GAP_GATE	16790
Gardenia	16800
Garmin_GMR_1224	16815
Garmin_Fantom_24	16820
Garmin_GWX_68_Weather_Radar	16825
Garpin	16830
Gate_Guard	16833
Garpun_Bal_E	16835
Gazetchik	16837
GBS1	16840
GCA_2000	16850
Furuno_1832_1921_Series	16858
GEMINI_DB	16870
GEM_Olympus	16871
GEM_Sentinel	16872
GEM_BX_132	16875
GEM_SC_2050X	16876
GEM_Sea_Eagle_200_N	16877
Generic_Internal_Jammer	16879
Gepard_TA_MPDR_12	16880
GEN_X	16881
Gepard_TT	16884
GERAN_F	16888
GERFAUT	16890
GFE_L_1	16895
GIRAFFE	16900

Enumerator	Value
GIRAFFE_1X	16903
Giraffe_40	16905
Giraffe_50_AT	16908
Giraffe_75	16912
Gin_Sling_TA	16915
Gin_Sling_TT_Gin_Sling	16920
Gin_Sling_MG	16925
Goal_Keeper	16930
Golden_Dome	16935
Golden_Heart	16940
Golden_Rock	16942
Gory_MH	16943
GPN_22	16945
GPSJ_10	16946
GPSJ_25	16947
GPSJ_40	16948
GPSJ_50	16949
GRN_9	16950
GRAN_K	16951
Grackle_MH	16953
Grave_Stone	16960
GRAVES	16963
Green_Stain	16965
Grid_Bow	17010
Grifo_F	17016
Grill_Pan_TT_9S32	17025
GRILL_SCREEN	17027
Grom_2	17029

Enumerator	Value
GROUND_MASTER_400	17030
GT_4	17031
GRS_440	17032
Gruesome_MH	17034
GUARDIAN	17050
Guardsman	17055
Gun_Dish_ZSU_23_4_RPK_2	17070
Gusher_MH	17072
H_RJZ_726_4A_Jammer	17075
H025_NO25E_	17079
HADR	17080
Hair_Net	17100
Half_Plate_A	17145
Half_Plate_B	17190
HARD	17220
Harpoon	17225
Hat_Box	17230
Hawk_Screech	17235
Hay_Pole	17250
Hay_Rick	17255
Head_Light_A	17280
Head_Lights	17325
Head_Lights_C	17370
Head_Lights_MG_A	17415
Head_Lights_MG_B	17460
Head_Lights_TT	17505
Head_Net	17550
Heart_Ache_B	17572

Enumerator	Value
Hellfire_mmW_MH	17590
Hen_Egg	17595
Hen_House	17640
Hen_Nest	17685
Hen_Roost	17730
Herakles	17732
HF_2_MG	17735
HGR_105	17745
Herz_9_TAR	17750
Herz_9_TTR	17751
Herz_9_MG	17752
High_Brick	17775
High_Fix	17820
High_Guard_YLC_2V	17842
High_Lark_TI	17865
High_Lark_1	17910
High_Lark_2	17955
High_Lark_4	18000
High_Lune	18045
High_Pole_A_B	18090
High_Scoop	18135
High_Screen_9S19MT	18150
High_Sieve	18180
Hill_Block	18185
Himalayas_Countermeasures_Suite	18189
HG_9550	18190
HJ_6374	18193
HLD_RADAR_900	18194

Enumerator	Value
HLJQ_520	18195
HN_503	18200
HN_C03_M	18201
Home_Brew_TER_CSA_27_	18223
Home_Talk	18225
Horn_Spoon	18270
Hot_Brick	18280
Hot_Flash	18315
Hot_Flash_2	18316
IHS_6	18318
Hot_Shot_TA_IRL144M	18320
Hot_Shot_TT_IRL144M	18325
Hot_Shot_MG_IRL144M	18330
HPS_106	18331
HPS_104	18332
HQ_9_MH	18339
HSR_1128	18340
HQ_9A_TER	18342
HQ_9B_TER	18344
HT_233	18348
HQ_61	18350
HRJS	18351
I_Derby_ER	18352
IBIS_80	18353
Egg_Drop_Radar_CSA_16_CSA_23_IBIS_150	18355
IBIS_200	18357
Egg_Roll_TER_CSA_16_HQ_16_TER	18359
IFF_MK_XII_AIMS_UPX_29	18360

Enumerator	Value
Janet	18400
IFF_MK_XV	18405
IFF_INT	18406
Jack_Knife	18407
IFF_TRSP	18408
J_MUSIC_Elbit_Systems_Jammer	18409
Javelin_MG	18410
Igla_1_SLAR	18411
_1L267_Moskva_1_Jammer	18412
IR_SA_2B_TER	18415
IR_SA_4_TER	18417
IR_SA_5_TELAR_TER	18419
J_10B_PESA	18420
J_10C_AESA	18421
J_11D_AESA	18422
J_15D_Jammer	18430
J_16D_Jammer	18431
JL_10MP	18443
J_ALQ_8	18445
J_FPS_7	18449
Jay_Bird	18450
J_FPS_3	18451
Mix_Bowl_JH_10	18452
J_MPQ_P7	18453
JL_7	18454
JL_10B	18455
JMA_1576	18456
JRC_JMA_9252_6CA	18457

Enumerator	Value
JLP_40	18458
JRC_JMR_9200_Series_X	18459
JRC_NMD_401	18460
JRC_JRM_310_MK2	18461
JMA_1596	18462
JN_1104	18463
JMA_7000	18464
JRC_JMA_7700	18465
JMA_5320	18466
JRC_JMR_9210_6XC	18467
JERS_1	18468
JINDALEE	18469
JRC_JMA_9900_series	18470
JLP_40D	18471
JRC_JMA_5300_series	18475
Jupiter	18495
Jupiter_II	18540
JY_8	18550
JY_8A	18551
JY_9	18555
JY_9_Modified	18556
JY_11_EW	18557
JY_14	18560
JY_14A	18561
JY_16	18565
JY_24	18570
J_APG_1	18571
J_APG_2	18572

Enumerator	Value
JY_29	18575
JYL_1	18578
JYL_6	18580
JYL_6A	18582
JZ_QF_612	18583
K376Z	18585
K77M	18586
Kaige	18600
KALKAN	18610
KBP_Afganit	18611
KALKAN_II	18615
Kelvin_Hughes_2A	18630
Kelvin_Hughes_14_9	18675
Karpaty	18700
Kashtan_3_Jamming_System	18710
Kelvin_Hughes_type_1006	18720
Kelvin_Hughes_type_1007	18765
Kelvin_Hughes_Type_1007_F_Band	18766
Kelvin_Hughes_2007_F_Band	18767
Kelvin_Hughes_2007_I_Band	18768
KH_Nucleus_5000	18770
KH_MANTA	18774
KH_NUCLEUS_2_6000	18775
KH_NUCLEUS_3_5000	18776
KH_NUCLEUS_3_6000A	18777
KH_Family	18780
Kelvin_Hughes_6000A	18781
Kh_38MAE_MH	18782

Enumerator	Value
KG8605A	18784
KH_902M	18785
KHROM_K	18786
KHIBINY	18787
KG300E	18789
KH_SharpEye	18790
KH_Sharp_Eye_B	18791
KH_Sharp_Eye_C	18792
KH_1700	18795
King_Pin	18797
_3rd_Khordad_TELAR_TIR	18800
_3rd_Khordad_TAR	18801
_15th_Khordad_TER	18803
KG_300	18805
Kite_Screech	18810
Kite_Screech_A	18855
Kite_Screech_B	18900
KLC_3B	18930
KJ_500_Nanjing_Radar	18944
Kivach	18945
KJ_500_Jammer	18946
KLC_1	18947
KLJ_1	18948
KLJ_3_Type_1473_	18950
KLJ_4	18951
KLJ_4B	18952
KLJ_5	18955
KLJ_7	18960

Enumerator	Value
KLJ_7B	18961
KLJ_7A	18962
Knife_Rest	18990
Knife_Rest_B_P_10	19035
KNIFE_REST_C	19037
Koden_MD_3730	19039
KJ_2000	19040
KODEN_MDC_900	19041
Koopol	19042
KOPYO_I	19045
KR_75	19050
KRONOS	19051
KREDO_1E	19052
Krasukha_2	19053
KRONOS_GRAND_NAVAL	19054
KRM_66E	19060
KRTZ_125_2M	19065
KSA_SRN	19080
KSA_TSR	19125
KS_1A_PHASED_ARRAY	19127
KS418	19129
KS418E	19130
KZ100	19131
KZ900	19132
KZ800_Airborne_ELINT_System	19133
L175V	19140
L370_5_President_S_Jammer	19142
L_415	19143

Enumerator	Value
L_88	19145
LAADS	19150
Land_Fall	19170
Land_Roll_MG	19215
Land_Roll_TA	19260
Land_Roll_TT	19305
LAZUR	19306
Last_Straw_Model_791_A	19307
LAP_3000	19309
LC_150	19310
LEER_3	19320
Leg_Drive	19330
Leninets_V_004 RPRnoteEnumerations29	19340
Leningraf	19350
LIANA	19370
Light_Bulb	19395
LIRA_A10	19396
LIROD_8	19397
LIROD_MK_II	19398
LLX05K	19399
LMT_NRAI_6A	19400
LN_55	19440
Ln_66	19485
Liman	19500
Liman_2	19505
Lockheed_Vigilance	19520
Long_Bow	19530
Long_Brick	19575

Enumerator	Value
Long_Bull	19620
Long_Eye	19665
Long_Head	19710
Long_Talk	19755
Long_Track	19800
Long_Trough	19845
Look_Two	19890
LOPAR	19920
LORAN	19935
Low_Blow_TA	19950
Low_Blow_TT	19955
Low_Blow_MG	19960
Low_Jack_TT	19970
Low_Jack_MG	19971
Low_Sieve	19980
Low_Trough	20025
LR_66_TER_LD_2000_	20029
LRA_900	20030
LP_23_TRS_2050	20040
LW_01	20060
LW_08 RPRnoteEnumerations30 RPRnoteEnumerationsDeprecated	20070
M_1983_FCR	20090
M22_40	20115
M44	20160
M401Z	20205
M585Z	20250
M588Z	20295
MA_1_IFF_Portion	20340

Enumerator	Value
MAD_HACK	20350
MARELD	20360
MA_Type_909_	20385
MARC_S_152	20420
Marconi_1810	20430
Marconi_Canada_HC_75	20475
Marconi_S_713	20495
Marconi_S_1802	20520
Marconi_S247	20530
Marconi_S_810	20565
Marconi_SA_10	20585
MARCONI_ST801	20589
Marconi_ST805	20590
Marconi_type_967	20610
Marconi_type_968	20655
Marconi_type_992	20700
Marconi_signaal_type_1022	20745
Marconi_signaal_type_910	20790
Marconi_signaal_type_911	20835
Marconi_signaal_type_992R	20880
MARTELLO_743D	20890
MARTELLO_S_723A	20895
MASTER_A	20897
MBDA_FLAADS_M_Sea_Ceptor_Jammer	20898
Mean_Back_TER_SA_27_	20900
MELCO_3	20915
MELODI	20917
MERLIN	20918

Enumerator	Value
Meraj_4_Ascension_	20919
Northrop_Grumman_MESA	20920
Mesh_Brick	20925
METEOR_1500S	20927
METEOR_200	20929
METEOR_50DX	20930
METEOR_300	20931
Meteor_BVRAAM	20933
MFR	20935
MFSR_2100_45	20940
MICA_MH	20942
MICA_RF	20943
Mineral_ME	20945
Mirage_ILL	20950
Miysis_Jammer	20955
MK_15_Phalanx_BLK_0_	20969
MK_15_CIWS	20970
MK_23	21015
MK_23_TAS	21060
MK_25	21105
Mk_25_Mod_3	21110
Mk_25_Mod_7	21130
MK_35_M2	21150
MK_92	21195
MK_92_CAS	21240
MK_92_STIR	21285
MK_95	21330
MKS_818	21332

Enumerator	Value
MLA_1	21340
MM_APQ_706	21359
MM_950	21360
MM_APS_705	21375
MM_APS_784	21390
MM_SPG_73_RTN_12X_	21419
MM_SPG_74	21420
MM_SPG_75	21465
MM_SPN_703	21490
MM_SPN_730	21492
MM_SPN_753B	21495
MM_SPQ_3	21500
MM_SPS_702	21510
MM_SPS_768	21555
MM_SPS_774	21600
MM_SPS_791_RAN_30X_	21610
MM_SPS_794_RAN_21S_	21615
MM_SPS_798_RAN_40L_	21620
MMSR	21623
Model_17C	21625
Moon_4	21645
Moon_Pie	21646
MOON_CONE	21647
Moon_Rack	21648
MOON_FACE	21649
MMRS	21650
Moon_Shade	21651
Model_360	21655

Enumerator	Value
Model_378	21660
Model_970	21661
Model_974	21665
MONOLIT_B	21672
Monument	21675
Mouse	21680
MP_411_ESM	21682
MPDR_18_S	21685
MPDR_18_X	21690
MPDR_45_E	21692
MR_231_1	21693
MR_231_3	21694
MPR	21695
MR_231_4	21696
MPS_1	21697
MR_36B	21698
MR_231_MOD	21699
MR_1600	21700
MRR	21701
MR35	21702
MR36	21703
MRL_1	21704
MRL_4	21705
MRL_5	21706
MSAM	21707
MR_36A	21708
MSTAR	21709
MT_305X	21710

Enumerator	Value
MR_10M1E	21711
MR_90	21712
MRK_411	21715
MR_320M_Topaz_V	21716
MSP_418K	21720
Muff_Cob	21735
Mushroom	21780
Mushroom_1	21825
Mushroom_2	21870
Mushroom_3	21871
N_23	21872
N_011M_Bars	21873
N_011M_Bars_B	21874
N_011M_Bars_C	21875
N_011M_Bars_R	21876
N035_Irbis_E	21877
N036_Byelka	21878
N_25	21879
N920Z	21880
N001V	21881
N001VE	21882
N001VEP	21883
NACOS_RADARPILOT_Platinum	21884
Nampo_B	21885
NAGIRA	21886
Nanjing_B	21890
Nanjing_C	21895
Nayada	21915

Enumerator	Value
NAYADA_5M	21917
NAYADA_5PV	21918
NEBO_M	21919
Nebo_SVU	21920
Neptun	21960
Nettuno_4100	21965
NIKE_HERCULES_MTR	21970
NIKE_TT	21980
Northrop_Grumman_MFEW_Jammer	21981
NORINCO_3D	21982
NJ_81E	21983
Normandie	21984
NRJ_6A	21985
NOSTRADAMUS	21986
NPG_1240	21987
NPG_1460	21988
NPG_434	21989
NPG_630	21990
NPM_510	21991
Nut_Can	21992
NP_Vega_Liana	21995
Novella_NV1_70	22000
Novella_P_38	22001
NRBA_50	22005
NRBA_51	22050
NRBF_20A	22095
NRJ_5	22110
NS_9005	22115

Enumerator	Value
NS_100_Series	22125
NUR_31	22127
NWS_3	22130
Nysa_B	22140
O524A	22185
O580B	22230
O625Z	22275
O626Z	22320
Ocean_Master	22335
Ocean_Master_400	22340
Odd_Group	22345
Odd_Lot	22365
Odd_Pair	22410
Odd_Rods	22411
Oka	22455
OFOGH	22460
OFOGH_3	22463
OKEAN	22500
OKEAN_A	22505
OKINXE_12C	22545
OKO	22560
OMEGA	22590
Omera_ORB32	22635
OMUL	22640
One_Eye	22680
OP_28	22690
OPRL_4	22695
OPRM_71	22696

Enumerator	Value
OPS_9	22697
OPS_11_B_C	22700
OPS_12	22701
OPS_14B	22705
OPS_14C	22706
OPS_16B	22725
OPS_18	22730
OPS_19	22732
OPS_20	22735
OPS_22	22736
OPS_24	22737
OPS_28	22740
OPS_28C	22745
OPS_39	22750
OPTIMA_3_2	22760
OR_2	22770
ORB_31D	22800
ORB_31S	22810
ORB_32	22815
ORB_42	22830
Orion_Rtn_10X	22860
Surface_Wave_Over_The_Horizon_	22890
Otomat_MK_1	22900
Otomat_MK_II_Teseo	22905
Otomat_Series_AL	22906
Owl_Screech	22950
P360Z	22955
P_14	22956

Enumerator	Value
P_180U	22957
P_18_2	22959
PA_1660	22960
P_18M	22961
P_190U	22962
P_30	22963
P_18_MOD	22964
P_35M	22965
PAGE	22970
Paint_Box	22977
Palm_Frond	22995
Modified_Paint_Box	22998
Palm_Frond_AB	23040
Pandora	23041
PALSAR_2	23042
Pantsir_SM_TAR	23043
PAR_2	23045
Pantsir_S1_2RL80_TAR	23046
Pantsir_S1_1RS2_1_TT	23047
PAR_2000	23050
PAR_2090C	23053
PAR_80	23055
Pat_Hand_TT	23085
Pat_Hand_MG	23095
PATRIOT	23100
Patty_Cake	23130
Pawn_Cake	23175
PBR_4_Rubin	23220

Enumerator	Value
PCS_514	23240
Pea_Sticks	23265
Pechora_SC	23295
Peel_Cone	23310
Peel_Group	23355
Peel_Group_A	23400
Peel_Group_B	23445
Peel_Group_MG	23450
Peel_Pair	23490
Pelena	23500
PGZ_07_TER	23515
Phalanx	23525
Phazotron_Gukol_4	23529
Phazotron_Zhuk_A_AE	23530
Philips_9LV_200	23535
Philips_9LV_331	23580
Philips_LV_223	23625
Philips_Sea_Giraffe_50_HC	23670
Phimat_Jammer	23675
PICOSAR	23680
PILOT_MK2	23685
Pin_Jib	23690
Pin_Tip	23695
PL02_Surveillance	23698
PL_11	23700
PL_12	23701
CH_AA_10_Abaddon_MH_PL_15	23704
CH_AA_12_Auger_MH_PL_17	23705

Enumerator	Value
Plank_Shad	23710
Plank_Shave	23715
Plank_Shave_A	23760
Plank_Shave_B	23805
Plate_Steer	23850
Plessey_AWS_1	23895
Plessey_AWS_2	23925
Plessey_AWS_4	23940
Plessey_AWS_6	23985
Plessey_RJ	23990
Plessey_Type_904	24020
Plessey_type_996	24030
Plessey_AWS_9	24035
Plinth_Net	24075
Pluto	24095
PNA_B_Rubin_Down_Beat	24098
POHJANPALO	24100
Pole_21E_Jammer_L1	24108
Pole_21E_Jammer_L2	24109
Poliment_K	24110
POLLUX	24120
Polozhennya_2	24130
Pop_Group	24165
Pop_Group_MG	24210
Pop_Group_TA	24255
Pop_Group_TT	24300
Pork_Fist	24320
Pork_Trough	24345

Enumerator	Value
Pozitiv_ME1_5P_26	24385
Positive_ME1_2	24386
Pozitiv_MK	24387
Post_Bow	24390
Post_Lamp	24435
Pot_Drum	24480
Pot_Head	24525
Pot_Shot	24535
Praetorian_Countermeasures_Suite	24540
PRIMUS_30A	24569
PRIMUS_40_WXD	24570
Primus_400	24614
PRIMUS_300SL	24615
Primus_500	24616
Primus_650	24617
Primus_700	24618
PRIMUS_800	24619
Primus_3000	24620
Primus_870	24622
PRORA	24630
PRS_2	24631
PRS_3_Argon_2	24633
PRORA_PA_1660	24635
PS_15	24640
PS_05A	24650
PS_46_A	24660
PS_70_R	24705
PS_171_R	24706

Enumerator	Value
PS_860	24707
PS_870	24709
PS_890	24710
PSM_33	24720
Puff_Ball	24750
Quadradar_VI	24755
QW_1A	24757
Phazotron_1RS2_1E	24758
PVS_200	24760
PVS_2000	24761
R_325UMV_Jammer	24767
R_330ZH	24768
R_045	24769
R_76	24770
R_934B	24771
RA_20	24772
RA723	24774
R41XXX	24775
RAC_3D	24776
RAC_30	24780
R_423AM	24781
Raad_1_TER	24785
Raad_2_TER	24787
Racal_1229	24795
DECCA_1230	24800
Racal_AC_2690_BT	24840
Racal_Decca_1216	24885
Racal_DECCA_20V90_9	24890

Enumerator	Value
Racal_Decca_360	24930
Racal_Decca_AC_1290	24975
Racal_Decca_TM_1229	25020
Racal_Decca_TM_1626	25065
Racal_DRBN_34A	25110
RADA_MHR	25150
Radar_24	25155
RADARPILOT_1000	25170
RADARPILOT_1100	25171
RAJENDRA	25180
RAN_7S	25200
RAN_10S	25205
RAN_11_LX	25245
Rani	25250
RAPHAEL_TH	25259
Rapier_TA	25260
Dagger_Rapier_2000_TA	25265
Rapier_MG	25270
RASCAR_3400C	25273
Rashmi	25275
Rasit	25276
Rasit_3190B	25277
RAT_31_DL_M	25278
RAT_31_DL	25279
RAT_31S	25280
RAT_8_S	25281
RAT_31_SL	25282
Raven_ES_05	25283

Enumerator	Value
RATAC_LCT_	25285
RAWL	25286
Rattler	25287
RAWS	25288
RAWL_02	25289
Raytheon_1220	25290
RAWS_03	25291
Raytheon_1210xx	25292
Raytheon_1302	25300
Raytheon_1500	25335
Raytheon_1645	25380
Raytheon_1650	25425
Raytheon_1900	25470
Raytheon_2502	25515
Raytheon_Anschutz_NautoScan_NX	25530
Raytheon_R41	25540
Raytheon_RM_1025_6X	25545
Raytheon_SL_72	25550
Raytheon_TM_1650_6X	25560
Raytheon_TM_1660_12S	25605
RAY_1220XR	25630
RAY_1401	25635
Ray_2900	25650
Raymarine_RD218	25694
Raypath	25695
Raytheon_Pathfinder_ST_mk2	25698
RB_109A_Bylina_Jammer	25710
RB_313A_Jammer	25715

Enumerator	Value
RB_337A_Jammer	25719
RB_341A_Leer_3_Jammer	25721
RB_343A_Jammer	25723
RB_334A_Jammer	25724
RB_336A_Jammer	25725
RB_636AM2_SVET_KU_Jammer	25730
RBE2	25735
RBE2_AA	25736
RCT_180	25739
RDM	25740
RDM_3	25745
RDI	25750
RDY	25760
RDY_3	25762
RDS_86	25770
RDN_72	25785
RDR_1A	25830
RDR_1E	25835
RDR_4A	25840
RDR_150	25845
RDR_160XD	25850
RDR_230_HP	25853
RDR_1100	25855
RDR_1150	25860
RDR_1200	25875
RDR_1400	25885
RDR_1400_C	25890
RDR_4000_25891 RPRnoteEnumerationsDeprecated	25891

Enumerator	Value
RDR_4000_25892	25892
RDR_1500	25895
Rice_Cake	25896
RDR_1600	25897
RDR_2000	25898
RDR_1700B	25899
Remora	25900
Rice_Field	25901
REC_1A	25902
REC_1B	25903
REC_1C	25904
Resolve_EAS	25906
Rice_Cup_C	25907
REL_6E	25908
REC_1	25909
Rice_Bowl	25910
Improved_Reporter	25911
Rice_Bug	25912
Rice_Cup	25915
Rice_Lamp	25920
Rice_Bird	25921
Retia_ReVisor	25930
REVATHI	25940
REZONANS	25950
RGM_UGM_109B	25955
RGM_UGM_109E_Homing_Radar	25958
Rice_Pad	25965
RKL_526	25966

Enumerator	Value
RKZ_764	25967
RKZ_766	25968
RKL_165	25969
RKL_609	25970
RKL_800	25971
RKZ_761	25972
RKZ_2000	25973
RIS_4C_A	25974
RL_2000	25975
RL_41	25976
RIR_778	25977
RISAT	25978
RLM_S	25979
Rim_Hat_ESM_ECM_Suite	25980
Rice_Scoop_REL_86TM	26008
Rice_Screen	26010
DECCA_RM_1070A	26011
RM370BT	26015
Rockwell_Collins_FMR_200X	26020
RM_231_2	26040
RM_231_3	26041
RMT_0100A	26043
RN_222	26045
ROLAND_2	26053
ROLAND_BN	26055
ROLAND_MG	26100
ROLAND_TA	26145
ROLAND_TT	26190

Enumerator	Value
RODODOME	26210
Round_Ball	26235
RP_379D_Tirada_D	26236
RP_3	26237
RP_4G	26238
RP_377L_Lorandit_DF_Jammer	26240
RP_377UVM3_Jammer	26241
Round_House	26280
Round_House_B	26325
RPR_117	26326
RS_02_50	26327
RSR_210N	26328
RT_02_50	26330
RTA_4100	26340
RTN_1A	26350
RTN_25X	26353
RTS_6400	26354
Ruby_Rake	26355
Rum_Sling	26360
Rum_Sling_RO	26361
Rum_Sling_TT	26362
RV2	26370
RV3	26415
RV5	26460
RV10	26505
RV_15M	26506
RV17	26550
RV18	26595

Enumerator	Value
RV_21	26596
RV_21B	26597
RV_25	26600
RV_377	26610
RV_UM	26640
RWD_8	26650
RXN_2_60	26660
Rye_House	26665
S_1810CD	26670
Sahab	26672
Salamandre	26673
Samyung_SMR_7200	26674
S1850M	26675
S_511	26676
S_512	26677
S_600	26678
S_604	26679
S_763_LANZA_3D	26680
S_613	26681
S_631	26682
S_654	26683
S_669	26684
SA_2_Guideline	26685
S_244	26686
S_711	26687
SA_3_Goa	26730
SA_8_Gecko_DT	26775
Gladiator_TELAR_TIR_SA_12A_SA_23A_	26795

Enumerator	Value
Giant_TELAR_TIR_SA_12B_SA_23B_	26797
SABER_M60	26799
Samovar	26805
Sampson	26810
SA_N_7_Gadfly_TI	26820
SA_N_11_Cads_1_UN	26865
Saccade_MH	26900
Salt_Pot_A_B	26910
SAP_14	26920
SAP_518	26925
SAP_518M	26926
Sand_Bar	26930
SA_26_Pechora_2M_TTR	26935
SAR_on_UAVs_	26945
SATRAPE	26950
SATURNE_II	26955
Sayyad_2_TER	26957
Scan_Can	27000
Scan_Fix	27045
Scan_Odd	27090
SCANTER_1002	27095
SCANTER_2001	27100
SCANTER_2002	27101
SCANTER_2100	27102
SCANTER_4002	27109
SCANTER_4100	27110
SCANTER_5102	27111
SCANTER_5502	27113

Enumerator	Value
SCANTER_6000	27115
SCANTER_6002	27116
Scanter_Mil_009	27125
Scan_Three	27135
SCANTER_MIL_S	27137
Scanter_SMR	27139
SCANTER_CSR_	27140
SCORADS	27141
Scimitar	27142
STAR_2000	27143
SCOREBOARD	27150
Scoop_Pair	27175
Scoup_Plate	27180
SCOUT	27183
SCR_584	27190
Sea_Archer_2	27225
Sea_Based_X_Band	27230
Sea_Dragon	27235
Sea_Eagle_Type_381_	27239
Sea_Eagle_S_C_Type_382_	27240
SEA_FALCON	27245
Sea_Giraffe_AMB	27248
Seaguard	27251
Sea_Hawk_SHN_X12	27260
Sea_Hunter_4_MG	27270
Sea_Hunter_4_TA	27315
Sea_Hunter_4_TT	27360
Sea_Gull	27405

Enumerator	Value
Sea_Master_400	27430
Sea_Net	27450
Sea_Sparrow RPRnoteEnumerationsDeprecated	27451
Sea_Spray	27495
Sea_Tiger	27540
Sea_Tiger_M	27550
Sea_Watcher_100_Seastar	27560
Searchwater	27570
Searchwater_2000	27575
SEASONDE	27580
SEASPRAY_7000E	27582
SeaVue	27583
Seaspray_Mk3	27584
Selenia_Orion_7	27585
Selenia_type_912	27630
Selennia_RAN_12_L_X	27675
Selenia_RAN_20S	27680
Selennia_RTN_10X	27720
Selinia_ARP_1645	27765
SENTIR_M20	27770
SERDAR	27771
SERHAT	27773
Series_10_Compact_Submarine_Radar_CSR_	27775
SERIES_52	27780
SERIES_320	27790
SG	27800
SGJ_02	27802
SGJ_03	27803

Enumerator	Value
SGR_102_00	27810
SGR_103_02	27855
SGR_104	27870
Shahed_129_SAR	27873
SHAHINE	27875
Sheet_Bend	27900
Sheet_Curve	27945
SHIKRA	27980
Ship_Globe	27990
Ship_Wheel	28035
SGR_114	28080
Shore_Walk_A	28125
Short_Horn	28170
Shot_Dome	28215
Side_Globe_JN	28260
Side_Net_PRV_11	28280
Side_Walk_A	28305
Signaal_Bharat	28340
Signaal_DA_02	28350
Signaal_DA_05	28395
Signaal_DA_08	28440
Signaal_DA_08_2LS	28445
Signaal_LW_04	28480
Signaal_LW_08	28485
Signaal_LWOR	28530
Signaal_M45	28575
Signaal_MW_08	28620
Signaal_SMART	28665

Enumerator	Value
Signaal_STING	28710
Signaal_STIR	28755
Signaal_STIR_1_8M	28760
Signaal_STIR_24M	28770
Signaal_WM_20_2	28800
Signaal_WM_25	28845
Signaal_WM_27	28890
Signaal_WM_28	28935
Signaal_ZW_01	28980
Signaal_ZW_06	29025
Signaal_ZW_07	29030
Signaal_ZW_0800	29035
SIMRAD_3G	29043
SIMRAD_4G	29045
Simrad_CA54	29050
SIMRAD_Halo_6	29060
Ski_Pole	29070
Skin_Head	29115
Skip_Spin	29160
SKYFENDER	29172
Sky_Wave_Over_The_Horizon_	29175
Skyguard_B	29180
UAR_1021_SKYGUARD_TA	29185
UAR_1021_SKYGUARD_TT	29190
Skyguard_LR	29191
Skymaster	29200
Sky_Watch	29205
Sky_Ranger	29210

Enumerator	Value
SKYSHADOW	29215
SKYSHIELD_TA	29220
SL	29250
SL_ALQ_234	29270
Slap_Shot	29295
Slap_Shot_G	29297
SLC_2	29300
SLC_2E	29301
SLC_4	29305
Slim_Net	29340
Slot_Back_A	29385
Slot_Back_ILL	29400
Slot_Back_B	29430
Slot_Back_IV	29431
Slot_Back_B_Topaz	29432
Slot_Back_E	29433
Slot_Back_G	29434
Slot_Back_VI	29435
Slot_Rest	29440
SM_674A_UPM	29450
SMA_3_RM	29475
SMA_3_RM_20	29520
SMA_3RM_20A_SMG	29565
SMA_BPS_704	29610
SMA_SPIN_749_V_2	29655
SMA_SPN_703	29700
SMA_SPN_751	29745
SMA_SPOS_748	29790

Enumerator	Value
SMA_SPQ_2	29835
SMA_SPQ_2D	29880
SMA_SPQ_701	29925
SMA_SPS_702_UPX	29970
SMA_ST_2_OTOMAT_II_MH	30015
SR_47A	30016
SMA_718_Beacon	30060
Small_Fred	30065
SMART_S	30068
SMART_S_Mk2	30069
SMART_L	30070
SM_932	30072
Smog_Lamp	30075
Snap_Shot	30080
Snoop_Drift	30105
Snoop_Half	30140
Snoop_Head	30150
Snoop_Pair	30195
Snoop_Ping_30200 RPRnoteEnumerationsDeprecated	30200
Snoop_Plate	30240
Snoop_Ping_30255	30255
Snoop_Slab	30285
Snoop_Tray	30330
Snoop_Tray_1	30375
Snoop_Tray_2	30420
SNOOP_TRAY_3	30421
Snoop_Watch	30465
Snow_Drift_9S18M1	30470

Enumerator	Value
Snow_Drift_9S18M1E	30471
SPB_7	30475
Snow_Drop	30480
SNW_10	30490
SO_1	30510
SO_12	30520
SO_A_Communist	30555
SO_69	30580
Sock_Eye	30600
SOM_64	30645
Sopka_Hill_	30650
Sorbsiya	30660
Sorbtsiya_L005	30661
Sorbtsiya_L005S	30662
SPADA_SIR	30665
SPADA_TT	30670
Sparrow_AIM_RIM_7_ILL	30690
SPERRY_RASCAR	30691
SPECTRA	30692
SPEAR3_MMW	30696
Sperry_M_3	30700
SPERRY_VISIONMASTER_FT	30701
SPEXER_2000	30710
SPG_53F	30735
SPG_70_RTN_10X_	30780
SPG_74_RTN_20X_	30825
SPG_75_RTN_30X_	30870
SPG_76_RTN_30X_	30915

Enumerator	Value
Spin_Scan_A	30960
Spin_Scan_B	31005
Spin_Trough	31050
SPINO_D_ADDA_WTR	31070
SPJ_40	31080
Splash_Drop	31095
SPN_2	31096
SPN_4	31097
SPN_30	31100
SPN_35A	31140
SPN_41	31185
SPN_42	31230
SPN_43A	31275
SPN_43B	31320
SPN_44	31365
SPN_46	31410
SPN_703	31455
P_2080_SPN_720	31475
SPN_728_V_1	31500
SPN_748	31545
SPN_750	31590
SPO_8	31592
SPN_753G	31593
Sponge_Cake	31635
Spoon_Rest_P_12	31680
Spoon_Rest_A_P_18	31681
Spoon_Rest_B_P_18	31682
Spoon_Rest_D_P_18	31684

Enumerator	Value
P_18MH2	31685
P_18MA	31686
P_18MU	31687
P_18_Malachite	31688
P_18OU	31689
Spork_Rest	31700
SPQ_712_RAN_12_L_X_	31725
SPR_2	31730
SPR_51	31740
SPS_5_FASOL	31765
SPS_6	31766
SPS_6C	31770
SPS_10F	31815
SPS_12	31860
SPS_22N_BUKET	31870
SPS_33N_BUKET	31875
SPS_44N_BUKET	31880
SPS_55N_BUKET	31890
SPS_58 RPRnoteEnumerationsDeprecated	31905
SPS_62	31925
SPS_100K	31935
SPS_64 RPRnoteEnumerationsDeprecated	31950
SPS_141	31951
SPS_142	31952
SPS_143	31953
SPS_151	31955
SPS_152	31956
SPS_153	31957

Enumerator	Value
SPS_160_Geran	31959
SPS_161	31960
SPS_95K	31970
SPS_171_Jammer	31971
SPS_172_Jammer	31972
SPS_768_RAN_EL_	31995
SPS_540K	32010
SPS_550K_MF	32020
SPS_774_RAN_10S_	32040
SPY_790	32085
Square_Head	32130
Square_Pair	32175
Square_Slot	32220
Square_Tie	32265
Squash_Dome_Shmel RPRnoteEnumerations31	32310
Squat_Eye_P_15M	32330
Squint_Eye	32355
SQUIRE	32365
SR2410C	32373
SR47B_G	32375
SR_64_TAR_LD_2000_	32376
SRE_M5	32385
SRN_6	32400
SRN_15	32445
SRN_206	32455
SRN_745	32490
SRO_1	32535
SRO_2	32580

Enumerator	Value
SS_C_2B_Samlet_MG	32625
SS_N_2A_B_CSSC	32670
SS_N_2A_B_CSSC_2A_3A2_MH	32715
SS_N_2C_Seeker	32760
SS_N_2C_D_Styx	32805
SS_N_2C_D_Styx_C_D_MH	32850
SS_N_2C_Styx_AL	32851
SS_N_2D_Styx_AL	32852
SS_N_3_SSC_SS_C_18_BN	32895
SS_N_3B_Sepal_AL	32940
SS_N_3B_Sepal_MH	32985
SS_N_7_Starbright	33025
SS_N_9_Siren	33030
SS_N_9_Siren_AL	33075
SS_N_9_Siren_MH	33120
SS_N_10A_FL_10_mmW_MH	33125
SS_N_11_Nasr_1_mmW_MH	33140
SS_N_12_Sandbox_AL	33165
SS_N_12_YJ_83J_mmW_MH	33166
SS_N_12_Sandbox_MH	33210
SS_NX_13_Shredder	33230
SS_N_14B_Silex_AL	33231
SS_N_19_Shipwreck	33255
SS_N_19_Shipwreck_AL	33300
SS_N_19_Shipwreck_MH	33345
SS_N_21_AL	33390
SS_N_22_Sunburn	33435
SS_N_22_Sunburn_MH	33480

Enumerator	Value
SS_N_22_Sunburn_AL	33481
SS_N_25_Switchblade_MH	33483
SS_N_26_Strobile_MMW_MH	33484
SS_N_27_Sizzler_MH	33485
SS_N_27A_Sizzler_AL	33486
STING_EO_Mk2	33505
STIR_1_2_EO_Mk2	33510
STIR_2_4_HP_Mk2	33511
Stone_Cake	33525
STR_41	33570
ST_858	33580
START_1M	33582
STENTOR	33584
Storm_Shadow_AHR	33585
STRAIGHT_FLUSH	33586
Straight_Flush_TA	33590
Straight_Flush_TT	33595
Straight_Flush_ILL	33600
Strike_Out	33615
Strut_Curve	33660
Strut_Pair	33705
Strut_Pair_1	33750
Strut_Pair_2	33795
Sun_Visor	33840
SUPERDARN	33850
Superfledermaus	33860
Supersearcher	33870
Swift_Rod_1	33885

Enumerator	Value
Swift_Rod_2	33930
SYMPHONY	33933
SYNOPSIS_Mk2	33935
SY80	33950
T1166	33975
T1171	34020
T1202	34040
T6004	34065
T6031	34110
T8067	34155
T8068	34200
T8124	34245
T8408	34290
T8911	34335
T8937	34380
T8944	34425
T8987	34470
TA_10K	34480
Tall_Board_JY_11B	34500
TACAN_SURF	34505
Tall_King_P_14	34515
TALL_KING_B	34516
TALL_KING_C	34517
Tall_Mike	34560
Tall_Path	34605
TDR_94_MODE_S_	34607
Tea_Spoon	34610
Team_Play	34620

Enumerator	Value
TALISMAN	34624
Team_Work	34625
T1135	34626
TANCAN_SURF	34627
TECSAR	34628
TERRASAR_X	34629
TESAR	34630
THAAD_GBR	34640
Thales_APAR_Block2	34643
Thales_RDY_2	34644
Thales_Nederland_Signaal_APAR	34645
Thales_Scorpion_Jammer	34646
Thales_Variant	34647
Thales_ICMS_Jammer	34648
Thales_IMEWS_Jammer	34649
THD_225	34650
THD_1012	34655
THD_1098	34660
THD_1213	34665
Picador_THD_1940	34670
THD_1955_Palmier	34680
THD_5500	34695
Third_of_Khordad	34700
Thin_Path	34740
Thin_Skin_PRV_9	34785
Thin_Skin_B_PRV_16	34786
Thompson_CSF_TA_10	34795
Thompson_CSF_TH_D_1040_Neptune	34830

Enumerator	Value
Thompson_CSF_Calypso	34875
Thompson_CSF_CASTOR	34920
Thompson_CSF_Castor_II	34965
Thomson_CSF_Domino_30	34966
Thompson_CSF_DRBC_32A	35010
Thompson_CSF_DRBJ_11_D_E	35055
Thompson_CSF_DRBV_15A	35100
Thompson_CSF_DRBV_15C	35145
Thompson_CSF_DRBV_22D	35190
Thompson_CSF_DRBV_23B	35235
Thompson_CSF_DRUA_33	35280
Thompson_CSF_Mars_DRBV_21A	35325
Thompson_CSF_Sea_Tiger	35370
Thompson_CSF_Triton	35415
Thompson_CSF_Vega_with_DRBC_32E	35460
Thomson_ENR_European_Navy_Radar_	35470
Thomson_RDI	35475
Tier_II_Plus	35477
TPS_755	35478
TPS_830K	35479
TIGER_G_TRS_2105	35480
TR_23K	35481
TR_23MR	35482
TRAC_2100	35483
TRAC_2300	35484
Tiger_Paw_HT_223	35485
TRADEX	35486
TRAIL_XI	35487

Enumerator	Value
TRD_1211	35488
TRD_1235	35489
TIGER_S_TRS_2100	35490
TRAC_NG	35491
Tie_Rods	35505
Tin_Shield_36D6	35550
Tin_Trap	35570
TIRSPONDER	35580
TK_25E_5	35583
TMK_Mk2	35585
TMX_Mk2	35586
Toad_Stool_1	35595
Toad_Stool_2	35640
Toad_Stool_3	35685
Toad_Stool_4	35730
Toad_Stool_5	35775
Token_B	35785
Tomb_Stone	35800
Tonson	35810
Top_Bow	35820
Top_Dome	35865
Top_Knot	35910
Top_Mesh	35955
Top_Pair	36000
Top_Plate	36045
Top_Plate_B	36046
Top_Sail	36090
TYPE_208 RPRnoteEnumerationsDeprecated	36120

Enumerator	Value
Top_Steer	36135
Top_Trough	36180
Tornado_GMR	36200
Tornado_TFR	36201
Scrum_Half_TA	36220
Scrum_Half_TT	36225
Scrum_Half_C_TOR_M2_TER	36226
Scrum_Half_MG	36230
Track_Dish	36270
TR_47C	36300
TORSO_M	36315
TQN_2	36320
Trap_Door	36360
TRD_1500	36365
Trick_Shot_TAR	36370
Trick_Shot_TER	36371
TRISPONDE	36380
TRML	36381
TRS_2215	36382
TRML_3D	36383
TRM_S	36384
TRS_2056	36385
TRS_3010	36386
TRS_2060	36387
TRS_2245	36388
TRS_2310	36389
Triton_G	36390
TRS_22XX	36391

Enumerator	Value
TRS_3030	36400
TRS_3033	36405
TRS_3203	36417
TRS_3405	36420
TRS_3410	36425
TRS_3415	36430
TRS_3D	36440
TRS_3D_16	36441
TRS_3D_16_ES	36442
TRS_3D_32	36443
TRS_4D	36446
TRS_C	36447
TRS_N	36450
TRML_4D	36455
TS_4478A	36460
TSE_5000	36495
TSR_333	36540
TSR_793	36550
Tub_Brick	36563
Tube_Arm	36585
TW_1374	36590
TW_1378	36595
TW_1446	36600
Twin_Eyes	36630
Twin_Pill	36675
Twin_Scan	36720
Twin_Scan_Ro	36765
Two_Spot	36810

Enumerator	Value
Type_071_LPD	36821
Type_2_12_J_A	36827
Type_2_21_J_A	36830
Type_2_23	36835
Type_80_ASM_1	36836
Type_120	36838
Type_208	36840
Type_222	36843
Type_226	36846
Type_232H	36850
TYPE_245	36853
TYPE_262	36855
TYPE_275	36900
TYPE_278	36905
TYPE_293	36945
Type_341	36946
TYPE_313	36947
Type_305A	36948
Type_334	36960
Type_342	36985
TYPE_343_SUN_VISOR_B	36990
Type_344	36992
Type_345	37010
Type_346	37011
Type_349A	37033
TYPE_347B	37035
Type_347G	37038
Type_359	37039

Enumerator	Value
Type_352	37040
Type_360	37041
Type_362_ESR_1_SR_47B	37043
Type_354	37045
Type_366	37047
Type_363	37048
Seagull_C_Type_364	37049
Type_404A_CH_	37050
Type_405	37052
TYPE_405J	37053
Moon_Stool_B_Type_408D	37058
Type_517B	37059
Type_518_Hai_Ying_God_Eye_REL_2_	37060
Type_589	37070
TYPE_651	37073
Type_753_37075 RPRnoteEnumerationsDeprecated	37075
Type_702	37077
Type_704M_BL_904_	37078
Type_753_37079	37079
Type_756	37080
TYPE_713	37081
TYPE_714	37082
TYPE_702_D	37083
TYPE_760 RPRnoteEnumerationsDeprecated	37084
Type_760	37086
Type_815	37090
Type_793	37095
Type_8A_813	37100

Enumerator	Value
TYPE_901M	37105
Type_902_FCR	37110
Type_902B	37124
TYPE_903	37125
TYPE_909_TI	37170
TYPE_909_TT	37215
TYPE_910	37260
TYPE_931_CH_	37265
TYPE_965	37305
TYPE_967	37350
TYPE_968	37395
TYPE_974	37440
TYPE_975	37485
TYPE_978	37530
Type_981	37534
Type_981_3	37535
TYPE_982	37540
Type_984	37543
Type_985	37544
TYPE_992	37575
TYPE_993	37620
TYPE_994	37665
Type_996	37670
Type_997_Artisan	37675
TYPE_1006_1_	37710
TYPE_1006_2_	37755
TYPE_1022	37800
Type_1047	37810

Enumerator	Value
Type_1048	37815
Type_1474	37825
Type_1493	37828
ULTRA	37840
UK_MK_10	37845
UPS_220C	37850
UPX_1_10	37890
UPX_27	37935
URN_20	37980
UTES_A	37985
UTES_T	37990
URN_25	38025
VIGILANT	38035
Vitebsk_L370_Jammer	38038
VOLEX_III_IV	38045
VOLGA	38046
VORONEZH_DM	38047
VOSTOK	38048
VOSTOK_E	38049
VSR	38050
VOSTOK_3D	38051
VSTAR_PT	38055
W_160	38058
W1028	38060
W8818	38070
W8838	38115
W8852	38120
WALL_BOARD	38140

Enumerator	Value
Wall_Rust	38150
WAS_74S	38160
Wasp_Head	38205
WATCHDOG	38210
Watch_Guard	38250
Watchman	38260
WAVESTORM	38270
WATCHMAN_S	38275
WATCHMAN_T	38276
WEATHER_SCOUT_2	38280
Western_Electric_MK_10	38295
Westinghouse_ADR_4_LRSR	38320
Westinghouse_Electric_SPG_50	38340
Westinghouse_Electric_W_120	38385
Westinghouse_SPS_29C	38430
Westinghouse_SPS_37	38475
Wet_Eye	38520
Wet_Eye_2	38525
Wet_Eye_Mod	38565
WF44S	38568
WGU_41_B	38570
WGU_44_B	38572
Whiff	38610
Whiff_Brick	38655
Whiff_Fire	38700
WHITE_HOUSE	38715
Wide_Mat	38730
Wine_Glass_Jammer	38735

Enumerator	Value
Wild_Card	38745
WILDCAT	38748
Witch_Eight	38790
Witch_Five	38835
WLR	38840
WM2X_Series	38880
WM2X_Series_CAS	38925
WR_10X	38930
WR_2100	38935
WSR_74C	38950
WSR_74S	38955
WSR_81	38957
WXR_700C	38960
WXR_2100_MSTT	38965
WXR_2100MSTT RPRnoteEnumerationsDeprecated	38966
Wood_Gage	38970
X_TAR25	38990
X_TAR3D	38995
YAOGAN_3	39000
Yaogan_29	39014
Yard_Rake	39015
YH_96	39050
Yew_Loop	39060
YITIAN_ADS	39061
YD_3	39062
YJ_12_MH	39063
YJ_62_MH	39065
YJ_82_MH	39066

Enumerator	Value
YJ_83_MH RPRnoteEnumerationsDeprecated	39067
YJ_63	39068
YLC_2	39070
YLC_2A	39071
YLC_4	39073
YLC_6	39074
YLC_6M	39075
YLC_8	39080
YLC_8B	39081
Tall_Page_YLC_18	39085
Yo_Yo	39105
Zaslon_A	39110
Zaslon_Multi_purpose_X_and_S_band_	39112
ZBD_08_Recon_Surveillance	39115
Zoo_Park_1	39125
ZPS_6	39126
ZOOPARK_3	39127
ZOOPARK_1M	39128
ZD_12	39131
ZW_06	39150
AN_ALQ_136_V_1	39200
AN_ALQ_136_V_2	39201
AN_ALQ_136_V_3	39202
AN_ALQ_136_V_4	39203
AN_ALQ_136_V_5	39204
AN_ALQ_162_V_2 RPRnoteEnumerations32	39210
AN_ALQ_162_V_3	39211
AN_ALQ_162_V_4	39212

Enumerator	Value
Zhuk_M	45300
ZHUK_MAE	45303
ZHUK_ME	45304
ZHUK_MME	45305
Zhuk_MSE	45307

EmitterSystemFunctionEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Emitter System Function [UID 76]*

Enumerator	Value
Other	0
Multi_function	1
Early_Warning_Surveillance	2
Height_Finder	3
Fire_Control	4
Acquisition_Detection	5
Tracker	6
Guidance_Illumination	7
Firing_point_launch_point_location	8
Range_Only	9
Radar_Altimeter	10
Imaging	11
Motion_Detection	12
Navigation	13
Weather_Meteorological	14
Instrumentation_15	15
Identification_Classification_including_IFF_	16
AAA_Anti_Aircraft_Artillery_Fire_Control	17

Enumerator	Value
Air_Search_Bomb	18
Air_Intercept	19
Altimeter	20
Air_Mapping	21
Air_Traffic_Control	22
Beacon	23
Battlefield_Surveillance	24
Ground_Control_Approach	25
Ground_Control_Intercept	26
Coastal_Surveillance	27
Decoy_Mimic	28
Data_Transmission	29
Earth_Surveillance	30
Gun_Lay_Beacon	31
Ground_Mapping	32
Harbor_Surveillance	33
IFF_Identify_Friend_or_Foe_ RPRnoteEnumerationsDeprecated	34
ILS_Instrument_Landing_System_	35
Ionospheric_Sound	36
Interrogator	37
Barrage_Jamming RPRnoteEnumerationsDeprecated	38
Click_Jamming RPRnoteEnumerationsDeprecated	39
Deceptive_Jamming RPRnoteEnumerationsDeprecated	40
Frequency_Swept_Jamming RPRnoteEnumerationsDeprecated	41
Jammer	42
Noise_Jamming RPRnoteEnumerationsDeprecated	43
Pulsed_Jamming RPRnoteEnumerationsDeprecated	44
Repeater_Jamming RPRnoteEnumerationsDeprecated	45

Enumerator	Value
Spot_Noise_Jamming RPRnoteEnumerationsDeprecated	46
Missile_Acquisition	47
Missile_Downlink	48
Meteorological RPRnoteEnumerationsDeprecated	49
Space	50
Surface_Search	51
Shell_Tracking	52
Television	56
Unknown	57
Video_Remoting	58
Experimental_or_Training	59
Missile_Guidance	60
Missile_Homing	61
Missile_Tracking	62
Jamming_noise RPRnoteEnumerationsDeprecated	64
Jamming_deception RPRnoteEnumerationsDeprecated	65
Decoy RPRnoteEnumerationsDeprecated	66
Navigation_Distance_Measuring_Equipment	71
Terrain_Following	72
Weather_Avoidance	73
Proximity_Fuse	74
Instrumentation_75 RPRnoteEnumerationsDeprecated	75
Radiosonde	76
Sonobuoy	77
Bathothermal_Sensor	78
Towed_Counter_Measure	79
Dipping_Sonar	80
Towed_Acoustic_Sensor	81

Enumerator	Value
Weapon_non_lethal	96
Weapon_lethal	97
Test_Equipment	98
Acquisition_Track	99
Track_Guidance	100
Guidance_Illumination_Track_Acquisition	101
Search_Acquisition	102
Dropsonde	103

EntityAssociationAssociationStatusEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Entity Association-Association Status [UID 319]*

Enumerator	Value
Not_Specified	0
Physical_Association_General_Object_1_	1
Functional_Association_General_	2
Association_Broken	3
Physical_Association_Object_2_	4
Functional_Association_Object_1_	5
Functional_Association_Object_2_	6

EntityAssociationAssociationTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Entity Association-Association Type [UID 330]*

Enumerator	Value
Other	0
Target_Entity	1
Target_Location	2
Home_Base_Location	3

Enumerator	Value
CAP_Point_Location	4
Flight_Leader	5
Flight_Member	6
IP_Point	7
Rendezvous_Point	8
On_Station_Location	9
Landing_Zone_Location	10
Downed_Pilot	11
Tanker_Entity_that_Is_Currently_Refueling_the_Transferred_Entity	12
Tanker_Entity_the_Transferred_Entity_Is_Headed_towards_to_Refuel	13
Entity_Headed_towards_to_Join_Up_With	14

EntityAssociationGroupMemberTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Entity Association-Group Member Type [UID 321]*

Enumerator	Value
Not_Part_of_a_Group	0
Group_Leader	1
Group_Member	2
Formation_Leader	3
Formation_Member	4
Convoy_Leader	5
Convoy_Member	6

EntityAssociationPhysicalAssociationTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Entity Association-Physical Association Type [UID 323]*

Enumerator	Value
Not_Specified	0

Enumerator	Value
Towed_in_Air_Single_Hook_Not_Specified_	1
Towed_on_Land	2
Towed_on_Water_Surface	3
Towed_Underwater	4
Mounted_Attached	5
Mounted_Unattached_and_Unsupported	6
Mounted_Unattached_and_Supported	7
Towed_in_Air_Center_Hook_	8
Towed_in_Air_Forward_Hook_	9
Towed_in_Air_Aft_Hook_	10
Towed_in_Air_Tandem_Hook_Fore_and_Aft_	11
Towed_in_Air_Mismanaged_Tandem_Fore_and_Center_	12
Towed_in_Air_Mismanaged_Tandem_Center_and_Aft_	13
Towed_in_Air_All_Hooks_	14
Hoisted	15
Restrained_to_a_Life_Form	30
Restrained_to_a_Platform	31
Restrained_to_an_Object	32
Refueling_Operation	61
Search_and_Rescue_Basket	62
Search_and_Rescue_Rescue_Collar	63
Engagement_Object_2_is_Being_Engaged	64
Return_To_Base_Object_2_is_the_Destination_Object	65
Line_between_Communication_Towers	90
Line_Between_Power_Towers	91
Indoors	92
Top_Surface	93

EntityAssociationPhysicalConnectionTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Entity Association-Physical Connection Type [UID 324]*

Enumerator	Value
Not_Specified	0
Attached_Directly_to_Surface	1
Cable_Wire	2
Rope	3
Chain	4
Power_Line	5
Telephone_Line	6
Cable_Line	7
Refueling_Drogue	8
Refueling_Boom	9
Handcuffs	10
In_Contact_With	11
Fast_Rope	12

EntityCapabilitiesEnum32

Representation: [RPRunsignedInteger32BE](#)

Semantics: *Entity Capabilities [UID 55]*

Enumerator	Value
Land_Platform_Entity_Capabilities	0
Air_Platform_Entity_Capabilities	1
Surface_Platform_Entity_Capabilities	2
Subsurface_Platform_Entity_Capabilities	3
Space_Platform_Entity_Capabilities	4
Munition_Entity_Capabilities	5
Life_Forms_Entity_Capabilities	6

Enumerator	Value
Environmental_Entity_Capabilities	7
Cultural_Feature_Entity_Capabilities	8
Supply_Entity_Capabilities	9
Radio_Entity_Capabilities	10
Expendable_Entity_Capabilities	11
Sensor_Emitter_Entity_Capabilities	12

EntityDamageStatusComponentIdentificationEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Entity Damage Status-Component Identification [UID 314]*

Enumerator	Value
Entity_Center_No_Specific_Component_	0
Entity_Structure	1
Control_System	2
Control_Surface	3
Engine_Propulsion_System	4
Crew_Member	5
Fuse	6
Acquisition_Sensor	7
Tracking_Sensor	8
Fuel_Tank_Solid_Rocket_Motor	9

EntityKindEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Entity Kind [UID 7]*

Enumerator	Value
Other	0
Platform	1
Munition	2

Enumerator	Value
Life_form	3
Environmental	4
Cultural_feature	5
Supply	6
Radio	7
Expendable	8
Sensor_Emitter	9

EntityMarkingCharacterSetEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Entity Marking-Character Set [UID 45]*

Enumerator	Value
Unused	0
ASCII	1
U_S_Army_Marking	2
Digit_Chevron	3

EntityVPRecordChangeIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Entity VP Record-Change Indicator [UID 320]*

Enumerator	Value
Initial_Report_or_No_Change_Since_Last_Issuance	0
Change_Since_Last_Issuance	1

EnvironmentRecordTypeGroupsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Environment Record Type Groups [UID 273]*

Enumerator	Value
State	0

Enumerator	Value
Geometry	1

EnvironmentalProcessModelTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Environmental Process-Model Type [UID 248]*

Enumerator	Value
No_Statement	0

EnvironmentalProcessRecordTypeEnum32

Representation: [RPRunsignedInteger32BE](#)

Semantics: *Environmental Process-Record Type [UID 250]*

Enumerator	Value
COMBIC_State	256
Flare_State	259
Bounding_Sphere_Record	65536
Uniform_Geometry_Record	327680
Point_Record_1	655360
Line_Record_1	786432
Sphere_Record_1	851968
Ellipsoid_Record_1	1048576
Cone_Record_1	3145728
Rectangular_Volume_Record_1	5242880
Rectangular_Volume_Record_3	83886080
Point_Record_2	167772160
Line_Record_2	201326592
Sphere_Record_2	218103808
Ellipsoid_Record_2	268435456
Cone_Record_2	805306368
Rectangular_Volume_Record_2	1342177280

Enumerator	Value
Gaussian_Plume_Record	1610612736
Gaussian_Puff_Record	1879048192

EnvironmentalSubcategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Environmental Subcategory [UID 21]*

Enumerator	Value
Other	0
Very_Small	20
Small	40
Medium	60
Large	80
Very_Large	100

EventReportEventTypeEnum32

Representation: [RPRunsignedInteger32BE](#)

Semantics: *Event Report-Event Type [UID 73]*

Enumerator	Value
Other	0
Ran_Out_of_Ammunition	2
Killed_in_Action_KIA_	3
Damage	4
Mobility_Disabled	5
Fire_Disabled	6
Ran_Out_of_Fuel	7
Entity_Initialization	8
Request_for_Indirect_Fire_or_CAS_Mission	9
Indirect_Fire_or_CAS_Fire	10
Minefield_Entry	11

Enumerator	Value
Minefield_Detonation	12
Vehicle_Master_Power_On	13
Vehicle_Master_Power_Off	14
Aggregate_State_Change_Requested	15
Prevent_Collision_Detonation	16
Ownership_Report	17
Radar_Perception	18
Detect	19

ExpendableAirCategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Expendable-Air Category [UID 25]*

Enumerator	Value
Other	0
Chaff	1
Flare	2
Combined_Chaff_and_Flare	3
Active_Emitter	4
Passive_Decoys	5
Winged_Decoys	6
Signal_Illumination_Flare	7
Smoke_Generator	8
Combined_Flare_and_Smoke_Generator	12
SAR_Night_Light_SRNL_	13
SAR_Buoy	14

ExpendableSubsurfaceCategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Expendable-Subsurface Category [UID 27]*

Enumerator	Value
Other	0
Active_emitter	4
Passive_decoy	5
Signal	7
Noise_Maker_Decoys	9
Bubble_Maker_Decoys	10
Multi_Mode_Decoys	11

ExpendableSurfaceCategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Expendable-Surface Category [UID 26]*

Enumerator	Value
Other	0
Flare	2
Active_Emitter	4
Passive_Decoys	5
Smoke_Generator	8
Combined_Flare_and_Smoke_Generator	12
SAR_Buoy	14

ExplosiveMaterialCategoriesEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Explosive Material Categories [UID 310]*

Enumerator	Value
No_Statement	0
AVGAS_Aviation_Gas_	10
Jet_Fuel_Unspecified_	11
JP_4_F_40_JET_B_	12
JP_5_F_44_JET_A_	13

Enumerator	Value
JP_7	14
JP_8_F_34_JET_A_1_	15
JP_10_Missile_Fuel	16
JPTS	17
Jet_A	18
Jet_A_1	19
Jet_B	20
Jet_Biofuel	21
Gasoline_Petrol_Unspecified_Octane_	151
Diesel_Fuel_Unspecified_Grade_	152
Ethanol	153
E85_Ethanol	154
Fuel_Oil	155
Kerosene	156
Crude_Oil_Unspecified_	157
Light_Crude_Oil	158
Liquid_Petroleum_Gas_LPG_	159
RP_1_Rocket_Fuel	160
LH_2_Rocket_Fuel	161
LOX_Rocket_Fuel	162
Alcohol	164
Hydrogen_Liquid_	166
Nitroglycerin_NG_	301
ANFO	302
Dynamite	451
TNT	452
RDX	453
PETN	454

Enumerator	Value
HMX	455
C_4	456
Composition_C_4	457
Natural_Gas_NG_	601
Butane	602
Propane	603
Helium	604
Hydrogen_Gaseous_	605
Dust_Unspecified_Type_	801
Grain_Dust	802
Flour_Dust	803
Sugar_Dust	804

ExplosiveMaterialGroupsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Explosive Material Groups [UID 309]*

Enumerator	Value
General	0
Liquid_Aviation_Missile_Fuels	1
Liquid_Other_Fuels	2
Liquid_Explosive_Material	3
Solid	4
Gaseous	5
Dust_Material	6

FlareSmokeColorEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize03](#)

Semantics: *Flare/Smoke Color [UID 402]*

Enumerator	Value
White	0
Red	1
Green	2
IR	3

FlareSmokeStatusEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Flare/Smoke Status [UID 403]*

Enumerator	Value
Not_Ignited	0
Burning	1
Burned_Out	2

ForceIDEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Force ID [UID 6]*

Enumerator	Value
Other	0
Friendly	1
Opposing	2
Neutral	3
Friendly_2	4
Opposing_2	5
Neutral_2	6
Friendly_3	7
Opposing_3	8
Neutral_3	9
Friendly_4	10
Opposing_4	11

Enumerator	Value
Neutral_4	12
Friendly_5	13
Opposing_5	14
Neutral_5	15
Friendly_6	16
Opposing_6	17
Neutral_6	18
Friendly_7	19
Opposing_7	20
Neutral_7	21
Friendly_8	22
Opposing_8	23
Neutral_8	24
Friendly_9	25
Opposing_9	26
Neutral_9	27
Friendly_10	28
Opposing_10	29
Neutral_10	30

FormatTypeEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize05

Semantics: *Format Type [UID 363]*

Enumerator	Value
No_Data	0
Identity_Format	4
Surface_Format_5_meter_RNP	5
Surface_Format_100_meter_RNP	6

Enumerator	Value
Airborne_Format_5_meter_RNP_25_foot_Barometric_Altitude	7
Airborne_Format_100_meter_RNP_25_foot_Barometric_Altitude	8
Airborne_Format_0_25_nmi_RNP_25_foot_Barometric_Altitude	9
Airborne_Format_1_0_nmi_RNP_25_foot_Barometric_Altitude	10
Airborne_Format_5_meter_RNP_100_foot_Barometric_Altitude	11
Airborne_Format_100_meter_RNP_100_foot_Barometric_Altitude	12
Airborne_Format_0_25_nmi_RNP_100_foot_Barometric_Altitude	13
Airborne_Format_1_0_nmi_RNP_100_foot_Barometric_Altitude	14
Airborne_Format_5_meter_RNP_GPS_Height	15
Airborne_Format_100_meter_RNP_GPS_Height	16

FuelLocationEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Fuel Location [UID 329]*

Enumerator	Value
Other	0

FuelMeasurementUnitsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Fuel Measurement Units [UID 328]*

Enumerator	Value
Other	0
Liter	1
Kilogram	2

GridAxisDescriptorAxisTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Grid Axis Descriptor-Axis Type [UID 377]*

Enumerator	Value
Regular_Axis	0
Irregular_Axis	1

GriddedDataConstantGridEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Gridded Data-Constant Grid [UID 245]*

Enumerator	Value
Constant_grid	0
Updated_grid	1

GriddedDataCoordinateSystemEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Gridded Data-Coordinate System [UID 244]*

Enumerator	Value
Right_handed_Cartesian_local_topographic_projection_east_north_up_	0
Left_handed_Cartesian_local_topographic_projection_east_north_down_	1
Latitude_Longitude_Height	2
Latitude_Longitude_Depth	3

GriddedDataDataRepresentationEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Gridded Data-Data Representation [UID 247]*

Enumerator	Value
Type_0	0
Type_1	1
Type_2	2

GriddedDataSampleTypeEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Gridded Data-Sample Type [UID 246]*

Enumerator	Value
Not_Specified	0

HighDensityTrackJamEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *High Density Track/Jam [UID 79]*

Enumerator	Value
Not_Selected	0
Selected	1

HighFidelityHAVEQUICKTODTransmitIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *High Fidelity HAVE QUICK-TOD Transmit Indicator [UID 297]*

Enumerator	Value
No_TOD_Is_Being_Transmitted	0
TOD_Transmission_in_Progress	1

IFFAlternateMode4ChallengeReplyEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *IFF-Alternate Mode 4 Challenge/Reply [UID 96]*

Enumerator	Value
No_Statement	0
Valid	1
Invalid	2
No_response	3
Unable_to_Verify	4

IFFApplicableModesEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *IFF Applicable Modes [UID 339]*

Enumerator	Value
No_Applicable_Modes_Data	0
All_Modes	1

IFFDataRecordAvailableEnum8Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)Semantics: *IFF Data Record Available [UID 360]*

Enumerator	Value
Compute_Locally	0
IFF_Data_Record_Available	1

IFFLayerSpecificInformationEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *IFF-Layer Specific Information [UID 87]*

Enumerator	Value
No_Layer_Specific_Information_Is_Present	0

IFFSimulationModeEnum8Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)Semantics: *IFF Simulation Mode [UID 338]*

Enumerator	Value
Regeneration	0
Interactive	1

IFFSystemModeEnum8

Representation: [RPRunsignedInteger8BE](#)Semantics: *IFF-System Mode [UID 84]*

Enumerator	Value
No_Statement	0
Off	1
Standby	2
Normal	3
Emergency	4
Low_or_Low_Sensitivity	5

IFFSystemNameEnum16Representation: [RPRunsignedInteger16BE](#)Semantics: *IFF-System Name [UID 83]*

Enumerator	Value
Not_Used_Invalid_Value_	0
Generic_Mark_X	1
Generic_Mark_XII	2
Generic_ATCRBS	3
Generic_Soviet	4
Generic_Mode_S	5
Generic_Mark_X_XII_ATCRBS	6
Generic_Mark_X_XII_ATCRBS_Mode_S	7
ARI_5954_RRB_	8
ARI_5983_RRB_	9
Generic_RRB	10
Generic_Mark_XIIA	11
Generic_Mode_5	12
Generic_Mark_XIIA_Combined_Interrogator_Transponder_CIT_	13
Generic_Mark_XII_Combined_Interrogator_Transponder_CIT_	14

Enumerator	Value
Generic_TCAS_I_ACAS_I_Transceiver	15
Generic_TCAS_II_ACAS_II_Transceiver	16
Generic_Mark_X_A_	17
Generic_Mark_X_SIF_	18

IFFSystemType1OperationalParameter1Enum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *IFF-System Type 1-Operational Parameter 1 [UID 97]*

Enumerator	Value
No_Operational_Parameter_1_Data	0

IFFSystemType1OperationalParameter2Enum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *IFF-System Type 1-Operational Parameter 2 [UID 98]*

Enumerator	Value
No_Operational_Parameter_2_Data	0

IFFSystemTypeEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *IFF-System Type [UID 82]*

Enumerator	Value
Not_Used_Invalid_Value_	0
Mark_X_XII_ATCRBS_Transponder	1
Mark_X_XII_ATCRBS_Interrogator	2
Soviet_Transponder	3
Soviet_Interrogator	4
RRB_Transponder	5
Mark_XIIA_Interrogator	6
Mode_5_Interrogator	7

Enumerator	Value
Mode_S_Interrogator	8
Mark_XIIA_Transponder	9
Mode_5_Transponder	10
Mode_S_Transponder	11
Mark_XIIA_Combined_Interrogator_Transponder_CIT_	12
Mark_XII_Combined_Interrogator_Transponder_CIT_	13
TCAS_ACAS_Transceiver	14

IOActionIOActionPhaseEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *IO Action-IO Action Phase [UID 288]*

Enumerator	Value
No_Statement	0
Start_Attack_Profile	1
End_Attack_Profile	2
Continue_Attack_Profile_with_Changes	3
Start_Attack_Effects	4
End_Attacked_Effects	5
Continue_Attack_Effects_with_Changes	6

IOActionIOActionTypeEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *IO Action-IO Action Type [UID 287]*

Enumerator	Value
No_Statement	0
IO_Attack_Profile_Data_Parametrics_	1
IO_Attack_Computed_Effects	2
Intent_Based_EW	3
Intent_Based_EW_Computed_Effects	4

IOActionIOSimulationSourceEnum16 [RPRnoteEnumerations33](#)Representation: [RPRunsignedInteger16BE](#)Semantics: *IO Action-IO Simulation Source [UID 286]*

Enumerator	Value
No_Statement	0

IOActionIOWarfareTypeEnum16Representation: [RPRunsignedInteger16BE](#)Semantics: *IO Action-IO Warfare Type [UID 285]*

Enumerator	Value
No_Statement	0
Electronic_Warfare_EW_	1
Computer_Network_Operations_CNO_	2
Psychological_Operations_PSYOPS_	3
Military_Deception_MILDEC_	4
Operations_Security_OPSEC_	5
Physical_Attack	6

IOCommsNodeRecordCommsNodeTypeEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *IO Comms Node Record-Comms Node Type [UID 294]*

Enumerator	Value
No_Statement	0
Sender_Node_ID	1
Receiver_Node_ID	2
Sender_Receiver_Node_ID	3

IOEffectsRecordIOEffectEnum8

Representation: [RPRunsignedInteger8BE](#)Semantics: *IO Effects Record-IO Effect [UID 292]*

Enumerator	Value
No_Statement	0
Denial	1
Degradation	2
Disruption	3
Terminate_Effect	255

IOEffectsRecordIOLinkTypeEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *IO Effects Record-IO Link Type [UID 291]*

Enumerator	Value
No_Statement	0
Logical_Link	1
Physical_Node	2
Physical_Link	3

IOEffectsRecordIOProcessEnum16Representation: [RPRunsignedInteger16BE](#)Semantics: *IO Effects Record-IO Process [UID 293]*

Enumerator	Value
No_Statement	0

IOEffectsRecordIOStatusEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *IO Effects Record-IO Status [UID 290]*

Enumerator	Value
No_Statement	0

Enumerator	Value
Effect_on_Sender	1
Effect_on_Receiver	2
Effect_on_Sender_and_Receiver	3
Effect_on_Message	4
Effect_on_Sender_and_Message	5
Effect_on_Receiver_and_Message	6
Effect_on_Sender_Receiver_and_Message	7

IOReportIOReportTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *IO Report-IO Report Type [UID 289]*

Enumerator	Value
No_Statement	0
Initial_Report	1
Update_Report	2
Final_Report	3

ISLSAntennaTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *ISLS Antenna Type [UID 349]*

Enumerator	Value
No_Statement	0
Monopulse_Antenna	1

IntercomControlCommandEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Intercom Control-Command [UID 182]*

Enumerator	Value
No_Command	0

Enumerator	Value
Status	1
Connect	2
Disconnect	3
Reset	4
On	5
Off	6

IntercomControlCommunicationsClassEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Intercom Control-Communications Class [UID 416]*

Enumerator	Value
Simulated_Communications_Channel	0
Simulation_Support_Communications_Channel	1

IntercomControlCommunicationsTypeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize07](#)

Semantics: *Intercom Control-Communications Type [UID 181]*

Enumerator	Value
Reserved	0
Connection_FDX	1
Connection_HDX_Destination_is_Receive_Only	2
Connection_HDX_Destination_is_Transmit_Only	3
Connection_HDX	4

IntercomControlControlTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Intercom Control-Control Type [UID 180]*

Enumerator	Value
Reserved	0

Enumerator	Value
Status	1
Request_Acknowledge_Required	2
Request_No_Acknowledge	3
Ack_Request_Granted	4
Nack_Request_Denied	5

IntercomControlDestinationLineStateCommandEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Intercom Control-Destination Line State Command [UID 184]*

Enumerator	Value
None	0
Set_Line_State_Transmitting	1
Set_Line_State_Not_Transmitting	2
Return_to_Local_Line_State_Control	3

IntercomControlRecordTypeEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Intercom Control-Record Type [UID 185]*

Enumerator	Value
Specific_Destination_record	1
Group_Destination_record	2
Group_Assignment_record	3

IntercomControlTransmitLineStateEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Intercom Control-Transmit Line State [UID 183]*

Enumerator	Value
Transmit_Line_State_not_applicable	0
Not_Transmitting	1

Enumerator	Value
Transmitting	2

IsGroupOfGroupedEntityCategoryEnum8

Representation: [RPRunsigndInteger8BE](#)

Semantics: *IsGroupOf-Grouped Entity Category [UID 213]*

Enumerator	Value
Undefined	0
Basic_Ground_Combat_Vehicle	1
Enhanced_Ground_Combat_Vehicle	2
Basic_Ground_Combat_Soldier	3
Enhanced_Ground_Combat_Soldier	4
Basic_Rotor_Wing_Aircraft	5
Enhanced_Rotor_Wing_Aircraft	6
Basic_Fixed_Wing_Aircraft	7
Enhanced_Fixed_Wing_Aircraft	8
Ground_Logistics_Vehicle	9

IsGroupOfRestStatusEnum8

Representation: [RPRunsigndInteger8BE](#)

Semantics: *IsGroupOf-Rest Status [UID 214]*

Enumerator	Value
Not_rested_Has_not_slept_in_the_last_three_days_	0
Has_slept_an_average_of_1_hour_per_day_in_the_last_three_days_	1
Has_slept_an_average_of_2_hours_per_day_in_the_last_three_days_	2
Has_slept_an_average_of_3_hours_per_day_in_the_last_three_days_	3
Has_slept_an_average_of_4_hours_per_day_in_the_last_three_days_	4
Has_slept_an_average_of_5_hours_per_day_in_the_last_three_days_	5
Has_slept_an_average_of_6_hours_per_day_in_the_last_three_days_	6
Has_slept_an_average_of_7_hours_per_day_in_the_last_three_days_	7

Enumerator	Value
Fully_rested_Has_slept_an_average_of_8_hours_per_day_in_the_last_three_days_	8

IsPartOfNatureEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *IsPartOf-Nature [UID 210]*

Enumerator	Value
Other	0
Host_fireable_munition	1
Munition_carried_as_cargo	2
Fuel_carried_as_cargo	3
Gunmount_attached_to_host	4
Computer_generated_forces_carried_as_cargo	5
Vehicle_carried_as_cargo	6
Emitter_mounted_on_host	7
Mobile_command_and_control_entity_carried_aboard_host	8
Entity_stationed_at_position_with_respect_to_host	9
Team_member_in_formation_with	10

IsPartOfPositionEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *IsPartOf-Position [UID 211]*

Enumerator	Value
Other	0
On_top_of	1
Inside_of	2

IsPartOfStationNameEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *IsPartOf-Station Name [UID 212]*

Enumerator	Value
Other	0
Aircraft_Wingstation	1
Ship_s_Forward_Gunmount_Starboard_	2
Ship_s_Forward_Gunmount_Port_	3
Ship_s_Forward_Gunmount_Centerline_	4
Ship_s_Aft_Gunmount_Starboard_	5
Ship_s_Aft_Gunmount_Port_	6
Ship_s_Aft_Gunmount_Centerline_	7
Forward_Torpedo_Tube	8
Aft_Torpedo_Tube	9
Bomb_Bay	10
Cargo_Bay	11
Truck_Bed	12
Trailer_Bed	13
Well_Deck	14
On_Station_Range_and_Bearing	15
On_Station_xyz	16
Air_to_Air_Refueling_Boom	17
Aerial_Refueling_Receptacle	18
Port_Side_Refueling_Drogue	19
Starboard_Side_Refueling_Drogue	20
Center_Refueling_Drogue	21
Air_Refueling_Probe	22

IslandSubcategoryEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Island Subcategory [UID 715]*

Enumerator	Value
Other	0
Islands_1_000_2_499_km2_386_965_sq_mi_	1
Islands_2_500_4_999_km2_965_1_930_sq_mi_	2
Islands_5_000_9_999_km2_1_931_3_861_sq_mi_	3
Islands_10_000_24_999_km2_3_861_9_652_sq_mi_	4
Islands_25_000_99_999_km2_9_653_38_610_sq_mi_	5
Islands_100_000_km2_39_000_sq_mi_and_Greater	6

JTIDSPrimaryModeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *The primary mode of a JTIDS system [UID 173]*

Enumerator	Value
NTR	1
JTIDS_Unit_Participant	2

JTIDSSecondaryModeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *The JTIDS secondary mode of operation [UID 174]*

Enumerator	Value
None	0
Net_Position_Reference	1
Primary_Navigation_Controller	2
Secondary_Navigation_Controller	3

JTIDSSynchronizationStateEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Describes the state of synchronization that the JTIDS system has achieved [UID 175]*

Enumerator	Value
No_Statement	0

Enumerator	Value
Initial_Net_Entry	1
Coarse_Synchronization	2
Fine_Synchronization	3
Synchronization_Maintenance RPRnoteEnumerationsDeprecated	4

LVCIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *LVC Indicator [UID 302]*

Enumerator	Value
No_Statement	0
Live	1
Virtual	2
Constructive	3

LeafCoverageEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Leaf Coverage [UID 509]*

Enumerator	Value
Normal	0
Bare	1

Level2SquitterStatusEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Level 2 Squitter Status [UID 353]*

Enumerator	Value
Disabled	0
Enabled	1

LifeFormAirCategoriesEnum8

Representation: [RPRunsignedInteger8BE](#)Semantics: *Life Form Air Categories [UID 478]*

Enumerator	Value
Bird	200
Insect	201
Mammal	202

LifeFormCategoriesAfghanistanEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Life Form Categories-Afghanistan [UID 501]*

Enumerator	Value
Afghan_National_Army_ANA_	11
Afghan_Air_Force_AAF_	13
National_Directorate_of_Security_NDS_	51
Afghan_National_Police_ANP_	52
Afghan_Border_Police_ABP_	53
Afghan_National_Civil_Order_Police_ANCOP_	54
Afghan_Public_Protection_Force_APPF_	55
Non_Military_National_Government_Agencies	70
Terrorist_Combatant	101
Humanitarian_Organizations	120
Red_Crescent	121
Civilian	130
Press	133

LifeFormCategoriesUSEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Life Form Categories-U.S. [UID 476]*

Enumerator	Value
U_S_Army_USA_	11
U_S_Marine_Corps_USMC_	12
U_S_Air_Force_USAF_	13
U_S_Navy_USN_	14
U_S_Coast_Guard_USCG_	15
Special_Forces	30
Rangers	31
ARSOF	32
Force_Reconnaissance	33
Navy_SEAL	34
AFSOF	35
Delta_Force	36
Federal_Bureau_of_Investigation_FBI_	51
Central_Intelligence_Agency_CIA_	52
Department_of_Homeland_Security_DHS_	53
Bureau_of_Alcohol_Tobacco_Firearms_and_Explosives_ATF_	54
U_S_Secret_Service_USSS_	55
U_S_Marshal	56
Law_Enforcement_Non_military_	70
State_Police_Highway_Patrol_	71
County_Sheriff_Police	72
Municipal_Police	73
Red_Cross	124

LifeFormExtraPersonalDataEnum8 [RPRnoteEnumerations34](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form Extra-Personal Data [UID 477]*

Enumerator	Value
Not_Specified_Male_	0
Asian_Male_	1
Pacific_Islander_Male_	2
Black_Male_	3
East_Asian_Male_	4
Hispanic_Male_	5
White_Male_	6
Arab_Male_	7
Homogenous_Country_Code_Male_	8
Indigenous_Country_Code_Male_	9
Infant_0_months_1_year_Male_	10
Toddler_1_3_years_Male_	20
Child_3_10_years_Male_	30
Adolescent_10_12_years_Male_	40
Teenager_13_16_years_Male_	50
Young_Adult_17_25_years_Male_	60
Adult_25_55_years_Male_	70
Senior_Adult_55_70_years_Male_	80
Elderly_71_years_Male_	90
Female	100
Asian_Female_	101
Pacific_Islander_Female_	102
Black_Female_	103
East_Asian_Female_	104
Hispanic_Female_	105
White_Female_	106
Arab_Female_	107
Homogenous_Country_Code_Female_	108

Enumerator	Value
Indigenous_Country_Code_Female_	109
Infant_0_months_1_year_Female_	110
Toddler_1_3_years_Female_	120
Child_3_10_years_Female_	130
Adolescent_10_12_years_Female_	140
Teenager_13_16_years_Female_	150
Young_Adult_17_25_years_Female_	160
Adult_25_55_years_Female_	170
Senior_Adult_55_70_years_Female_	180
Elderly_71_years_Female_	190

LifeFormHumanSpecificAnimalCompanionsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Animal Companions [UID 531]*

Enumerator	Value
Other	0
Military_Working_Dog	1
Military_Working_Dog_Mine_Detection	2
Military_Working_Dog_Drug_Detection	3

LifeFormHumanSpecificAntiMaterielRiflesEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Anti-Materiel Rifles [UID 510]*

Enumerator	Value
Other	0
_12_7mm_AW50	10
_12_7mm_AW50F	11
_12_7mm_TASKO_7ET3	12

LifeFormHumanSpecificAntiTankMissilesEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Life Form-Human-Specific-Anti Tank Missiles [UID 519]*

Enumerator	Value
Other	0
_120mm_Type_64_MAT_KAM_3	30
_153mm_Type_79_Jyu_MAT_KAM_9	31
_120mm_Type_87_Chū_MAT	32
_140mm_Type_01_LMAT	33
_140mm_M47_Dragon	58
_140mm_Saeghe_1_2	59
_127mm_FGM_148_Javelin	60
_139mm_FGM_172_SRAW	63
_139mm_FGM_172B_SRAW_MPV	64
_152mm_BGM_71_TOW	68
_152mm_Orev_TOW_II	69
_120mm_Vickers_Vigilant_Clevite	75
_110mm_Bantam_Rb_53_	80
_150mm_RBS_56_BILL_1	81
_150mm_RBS_56_2_BILL_2	82
_150mm_RB_57_NLAW_MBT_LAW_	83
_130mm_Spike_SR	85
_130mm_Spike_MR_CLU_	86
_130mm_Spike_LR_CLU_	87
_60mm_Mosquito	95
_160mm_SS_10	98
_103mm_MILAN	100
_115mm_MILAN_2	101
_115mm_MILAN_2T	102

Enumerator	Value
_115mm_MILAN_3	103
_115mm_MILAN_ER	104
_136mm_ERYX	105
_152mm_Entac	107
_125mm_RAAD	110
_125mm_I_RAAD_T	111
_152mm_Toophan	112
_152mm_Toophan_2	113
_152mm_Toophan_5	114
_136mm_Bumbar	120
_130mm_Shershen_PK_2	125
_152mm_Shershen_Q_P_2B	126
_130mm_Mectron_MSS_1_2	130
_120mm_HJ_8	140
_120mm_HJ_8A	141
_120mm_HJ_8B	142
_120mm_HJ_8C	143
_120mm_HJ_8D	144
_120mm_HJ_8E	145
_120mm_HJ_8F	146
_120mm_HJ_8FAE	147
_120mm_HJ_8L	148
_120mm_HJ_8H	149
_120mm_HJ_8S	150
_120mm_Baktar_Shikan	151
_120mm_HJ_11_AFT_11_	152
_152mm_HJ_9A	153
_135mm_HJ_12_Red_Arrow	154

Enumerator	Value
_125mm_HJ_73_MCLOS	155
_125mm_HJ_73B_SACLOS	156
_125mm_HJ_73C_SACLOS_ERA	157
_125mm_AT_3_Sagger_A_9M14_Malyutka	170
_125mm_AT_3B_Sagger_B_9M14M_Malyutka_M	171
_125mm_AT_3C_Sagger_C_9M14P_Malyutka_P	172
_125mm_AT_3D_Sagger_D_9M14_2_Malyutka_2	173
_125mm_Susong_Po	174
_125mm_AT_3C_POLK	175
_125mm_Kun_Wu_1	176
_125mm_Maliutka_M2T	177
_120mm_AT_4A_Spigot_A_9M111_Fagot	178
_120mm_AT_4B_Spigot_B_9M111_2_Fagot	179
_120mm_AT_4C_Spigot_C_9M111M_Faktoriya	180
_135mm_AT_5A_Spandrel_9M113_Kronkurs	181
_135mm_AT_5B_Spandrel_9M113M_Kronkurs_M	182
_135mm_Tosan	183
_94mm_AT_7_Saxhorn_9K115_Metis	184
_130mm_AT_13_Saxhorn_2_9K115_2_Metis_M	185
_152mm_AT_14_Spriggan_9M133_Kornet	186
_152mm_Dehlavie	187
_102mm_Mathogo	200

LifeFormHumanSpecificAntiTankRocketsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Anti Tank Rockets [UID 518]*

Enumerator	Value
Other	0

Enumerator	Value
_82mm_B_300	10
_82mm_Shipon	11
_83mm_MK153_Mod_0_SMAW	12
_66mm_M72_LAW	20
_66mm_M72A1_LAW	21
_66mm_M72A2_LAW	22
_66mm_M72A3_LAW	23
_66mm_M72A4_LAW	24
_66mm_M72A5_LAW	25
_66mm_M72A6_LAW	26
_66mm_M72A7_LAW	27
_66mm_M72E8_LAW	28
_66mm_M72E9_LAW	29
_66mm_M72E10_LAW	30
_66mm_M72AS_LAW	31
_94mm_LAW_80	35
_60mm_M1_Bazooka	40
_60mm_M1A1_Bazooka	41
_60mm_M9_Bazooka	42
_60mm_M9A1_Bazooka	43
_89mm_M20_Super_Bazooka	44
_89mm_M20A1_Super_Bazooka	45
_89mm_M20B1_Super_Bazooka	46
_89mm_M20A1B1_Super_Bazooka	47
_89mm_M25_Three_Shot_Bazooka	48
_89mm_Instalaza_M65	49
_90mm_Instalaza_C90	50
_90mm_C90_CR_M3_	51

Enumerator	Value
_90mm_C90_CR_AM_M3_	52
_90mm_C90_CR_BK_M3_	53
_90mm_C90_CR_IN_M3_	54
_60mm_PzF_3	60
_60mm_PzF_3_IT	61
_60mm_PzF_3_Bunkerfaust	62
_44mm_PzF_44	65
_30mm_Panzerfaust_30	70
_50mm_Panzerfaust_60	71
_60mm_Panzerfaust_100	72
_60mm_Panzerfaust_150	73
_88mm_Panzerschreck_RPzB	75
_83mm_RL_83_Blindicide	80
_100mm_RL_100_Blindicide	81
_90mm_M79_Osa_Wasp_	85
_64mm_M80_Zolja_Wasp_	86
_67mm_Armburst_Crossbow	90
_40mm_Type_69_RPG	93
_89mm_PIAT	95
_40mm_RPG_2	100
_64mm_RPG_18_Mukha	101
_72_5mm_RPG_22_Netto	102
_72_5mm_RPG_26_Aglen	103
_105mm_RPG_29_Vampir	104
_105mm_RPG_30_Kryuk	105
_105mm_RPG_32_Nashshab	106
_105mm_RPG_27_Tavolga	107
_125mm_RPG_28_Klyukva	108

Enumerator	Value
_40mm_RPG_7	110
_40mm_PSRL_1_RPG_7USA_	111
_40mm_GS_777_PSRL_2	112
_40mm_RPG_7V2	113
_40mm_RPG_7D3	114
_68mm_RPG_76_Komar_Mosquito_	120
_120mm_SEP_Dard_120	125
_58mm_WASP_58	128
_73mm_LRAC_73_50	130
_89mm_LRAC_89_F1_STRIM	131
_90mm_MATADOR_Man_Portable_Anti_Tank_Anti_DOoR_	135
_90mm_MATADOR_MP	136
_90mm_MATADOR_WB	137
_90mm_MATADOR_AS	138
_78mm_MARA_Anti_Tank_Rocket_Launcher	140
_120mm_Type_98_Pf98	145

LifeFormHumanSpecificAssaultRiflesEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Assault Rifles [UID 474]*

Enumerator	Value
Other	0
_4_5mm_Interdynamics_MKR	1
_5_45mm_AK_74	10
_5_45mm_AKS_74	11
_5_45mm_AK_74M	12
_5_45mm_Kbk_wz_1988_Tantal	13
_5_45mm_Fort_221	14

Enumerator	Value
_5_45mm_5_56mm_7_62mm_IPI_Malyuk	20
_5_56mm_AK_101	30
_5_56mm_Diemaco_C7 RPRnoteEnumerations35	31
_5_56mm_Colt_Canada_C8_Carbine	32
_5_56mm_GIAT_FAMAS_G2	33
_5_56mm_FN_FNC	34
_5_56mm_HK_G36	35
_5_56mm_IMI_Galil	36
_5_56mm_INSAS	37
_5_56mm_Daewoo_K1	38
_5_56mm_Daewoo_K2	39
_5_56mm_M16A1	40
_5_56mm_M16A2_A3_A4	41
_5_56mm_Colt_M4	42
_5_56mm_Colt_M4_Special_Operations_Peculiar_Modification_SOPMOD_	43
_5_56mm_Ruger_Mini_14	44
_5_56mm_Enfield_SA_80A2	45
_5_56mm_Pindad_SS1_V1	46
_5_56mm_Pindad_SS1_V2	47
_5_56mm_Pindad_SS1_V3	48
_5_56mm_Steyr_AUG_A1	49
_5_56mm_T65	50
_5_56mm_T91	51
_5_56mm_Tavor_TAR_21	52
_5_56mm_Type_CQ_M311	53
_5_56mm_Daewoo_K11	54
_5_56mm_Austeyr_F88	55
_5_56mm_Austeyr_F88_GLA	56

Enumerator	Value
_5_56mm_Austeyr_F88_S_A1	57
_5_56mm_Austeyr_F88_S_A2	58
_5_56mm_Austeyr_F88_C	59
_5_56mm_Austeyr_F88_S_A1C	60
_5_56mm_Austeyr_F88_S_A1_LTR	61
_5_56mm_Austeyr_EF88	62
_5_56mm_Bushmaster_XM15	63
_5_56mm_HK416	64
_5_56mm_F90	65
_5_56mm_F90_G_	66
_5_56mm_F90M	67
_5_56mm_F90M_G_	68
_5_56mm_F90CQB	69
_5_56mm_MK17_SCAR_L	70
_5_56mm_CZ_805_BREN	71
_5_56mm_Fort_224	72
_5_56mm_Fort_227	73
_5_56mm_Zbroyar_UAR_15	74
_5_56mm_Zbroyar_Z_15	75
_5_56mm_7_62mm_CZ_BREN_2	76
_5_8mm_QBZ_95_Type_95_	100
_7_62mm_AK_103	110
_7_62mm_AK_104	111
_7_62mm_AK_47	112
_7_62mm_AKM	113
_7_62mm_AKS_47	114
_7_62mm_HK_G3A3	115
_7_62mm_IMI_Galil	116

Enumerator	Value
_7_62mm_KLS	117
_7_62mm_SKS	118
_7_62mm_Type_56	119
_7_62mm_Type_63_68	120
_7_62mm_Type_81	121
_7_62mm_MK17_SCAR_H	122
_7_62mm_Fort_228	123
_7_62mm_Fort_229	124
_7_62mm_Zbroyar_Z_008_non_sniper_variant_	125
_7_62mm_Zbroyar_Z_10	126
_7_62mm_Mayak_MZ_10	127
_7_62mm_Stiletto_Systems_STL_016_300_Winchester	128
_7_62mm_CZ_BREN_2_BR	129
_7_62mm_Sa_vz_58_P	130
_7_62mm_Sa_vz_58_V	131
_6_8mm_M7_Next_Generation_Squad_Weapon_NGSW_	150
_8mm_Lebel_M16	240

LifeFormHumanSpecificDroneGunsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Drone Guns [UID 524]*

Enumerator	Value
Other	0
DroneGun_Tactical	15
DroneGun_MKII	16

LifeFormHumanSpecificEquipmentClassEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Equipment Class [UID 505]*

Enumerator	Value
Signal_Smoke	1
Flash_Light	2
Signal_Mirror	3
IR_Strobe	4
IR_Illuminator	5
Spotlight	6

LifeFormHumanSpecificEquipmentNonSpecificEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Equipment, Non-specific [UID 529]*

Enumerator	Value
Other	0
Toolbox	1

LifeFormHumanSpecificFlameRocketsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Flame Rockets [UID 522]*

Enumerator	Value
Other	0
_66mm_M202_Flash	20
_62mm_FHJ_84	30
_90mm_C90_CR_FIM_M3_	40
_93mm_RPO_A_Shmel	50
_93mm_RPO_Z_Shmel	51
_93mm_RPO_D_Shmel	52

LifeFormHumanSpecificFlameThrowersEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Flame Throwers [UID 523]*

Enumerator	Value
Other	0
Handflammpatrone	10
FmW_41	11
M1A1	20
M2A1_7	21
M9A1_7	22
LPO_50	30
K_Pattern	35
Portable_No_2_Ack_Pack	36
Marsden	37
Harvey	38
ROKS_2	45
ROKS_3	46
Type_93	50
Type_100	51

LifeFormHumanSpecificGrenadeLaunchersEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Grenade Launchers [UID 515]*

Enumerator	Value
Other	0
_40x46mm_Arsenal_UGGL_M1	1
_40x46mm_Arsenal_MSGL	2
_40mm_VOG_Arsenal_MSGL	3
_40x46mm_Arsenal_UBGL_M16	4
_40x46mm_Arsenal_UBGL_M8	5
_40x46mm_Arsenal_UBGL_M7	6
_30mm_BS_1_Tishina	10

Enumerator	Value
_40mm_BTS_203	11
_40mm_Indumil_IMC_40	12
_40mm_VOG_BG_15	20
_40mm_VOG_GP_25_Kostoyor	21
_40mm_VOG_GP_30_Obuvska	22
_40mm_VOG_GP_34	23
_40mm_VOG_RGM_40_Kastet	24
_40mm_VOG_RG_6	25
_40x46mm_CZ_805_G1_UBGL	26
_40x46mm_M79	30
_40x46mm_M203	31
_40x36mm_M320	32
_40x46mm_CIS_40_GL	35
_40x46mm_EAGLE_GL	36
_40x46mm_HK_AG36	37
_40x46mm_HK_AG_C_GLM	38
_40x46mm_HK69A1	39
_40x46mm_Beretta_G LX_160	40
_40x46mm_ARDE_UBGL	41
_40x46mm_XML148	42
_40x46mm_China_Lake_GL	43
_40x46mm_Hawk_MM_1	44
_25x40mm_XM25_CDTE	50
_37mm_Milkor37_38_LL_Stopper	60
_40mm_Milkor40_GL	61
_40mm_Milkor_MGL	62
_40x47mm_Pallad_wz1974	65
_40x47mm_Pallad_wz1983	66

Enumerator	Value
UGL_200_Canister_RWGL_3	70
_20x30mm_ST_Daewoo_K11	80
_30mm_RGSh_30_GL	85
_35mm_Type_91_BreechLoad_GL	90
_40x53mm_CZW_40	95
_40x53mm_UAG_40_AGL	96
_45mm_DP_64	100
_20x42mm_Neopup_PAW_20	105
_68mm_RPG_75_MP	110

LifeFormHumanSpecificGrenadeLaunchingMachineGunEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Grenade Launching Machine Gun [UID 517]*

Enumerator	Value
Other	0
_40x53mm_HK_GMG	20
_40x53mm_Mk47_Striker	25
_40mm_M75	26
_40mm_M129	27
_40x46mm_XM_174	28
_40x46mm_Mk18_Mod_0	29
_40x53mm_Mk19	30
_40x46mm_Mk20_Mod_0	31
_30x29mm_RAG_30_SAG_30	40
_30x29mm_AGS_17_Plamya	41
_30x29mm_AGS_30_Atlant	42
_40mm_VOG_AGS_40_Balkan	43
_40x53mm_SB_LAG_40	44

Enumerator	Value
_40x53mm_Vektor_Y3	50
_40x53mm_CIS_40	55
_40x56mm_Howa_Type_96	60
_40x53mm_Daewoo_Precision_Industries_K4	65
_25x59mm_XM307_Advanced_Crew_Served_Weapon	70
_35x32mm_QLZ87	80

LifeFormHumanSpecificHandGunsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Hand Guns [UID 513]*

Enumerator	Value
Other	0
_5_45mm_PSM	1
_5_8mm_Norinco_QSZ_92	10
_9mm_MK3_SLP	30
_9mm_Beretta_92S_92FS_M9_	31
_9mm_H_K_USP	32
_9mm_Stechkin_APS	33
_9mm_Makarov_PM	34
_9mm_Smith_and_Wesson_SD_Sigma_	35
_9mm_Glock_17	36
_9mm_SIG_Sauer_M17	37
_9mm_SIG_Pro	38
_9mm_Smith_and_Wesson_SW1911	39
_9mm_Smith_and_Wesson_5900_series	40
_45_Cal_M1911	41
_9mm_CZ_75	42
_9mm_CZ_82	43

Enumerator	Value
_9mm_Norinco_QSZ_92_CF_98_	44
_9mm_CZ_P_10_C	45
_9_07mm_Ruger_GP_100	50
_10mm_Glock_20	60
_9mm_Fort_12	70
_9mm_Fort_14TP	71
_9mm_Fort_17	72
_9mm_Fort_21	73
_12_7mm_Norinco_NP762_PX_3_	80

LifeFormHumanSpecificHighPowerRiflesEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-High Power Rifles [UID 475]*

Enumerator	Value
Other	0
_7_62mm_M14	10
_7_62mm_Remington_700	11
_7_62mm_SIG_Sauer_SSG_2000	12
_7_62mm_Stoner_SR_25	13
_7_62mm_Mosin_Nagant_Model_1891_30	14
_7_62mm_HK417	15
_7_62mm_HK417_16_Recce	16
_7_65mm_BAR_M1918	50
_7_65mm_M1_Garand	51

LifeFormHumanSpecificLasersClassEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Lasers Class [UID 527]*

Enumerator	Value
Generic_Laser_Designator	1
Generic_Laser_Pointer	2

LifeFormHumanSpecificLogisticsEQClassEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Logistics EQ Class [UID 525]*

Enumerator	Value
Sling_Load_Pendant	1

LifeFormHumanSpecificMachineGunsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Machine Guns [UID 516]*

Enumerator	Value
Other	0
_5_56x45mm_XM214_Microgun_Six_Pak	10
_7_62x51mm_M134_XM196_Minigun	11
_5_56x45mm_M249_FN_Minimi_SAW_LMG	20
_5_56x45mm_FN_Minimi_Mk3_LMG	21
_7_62x51mm_FN_Minimi_7_62_Mk3_GPMG	22
_7_62x63mm_M1941_Johnson	25
_7_62x63mm_M1918_BAR	26
_7_62x51mm_M1919A4_Mk_21_Mod_0_Browning_MMG	27
_7_62x63mm_M1919A6_Browning_MMG	28
_7_62x51mm_M37_Browning_MMG	29
_5_56x45mm_Ares_Shrike_5_56_LMG	30
_5_56x45mm_LSAT_LMG	31
_5_56x45mm_CMG_1_LMG	32
_5_56x45mm_CMG_2_LMG	33
_5_56x45mm_Stoner_63A_LMG	34

Enumerator	Value
_5_56x45mm_Ultimax_100_LMG	35
_5_56x54mm_Beretta_AS70_90_LMG	36
_5_56x45mm_CETME_Ameli_LMG	37
_5_56x45mm_IMI_Negev_LMG	38
_5_56x45mm_INSAS_LMG	39
_5_56x45mm_AUG_LMG	40
_5_56x45mm_AUG_HBAR_LMG	41
_5_56x45mm_HK_MG4_LMG	43
_5_56x45mm_HK23_GR_9_LMG	44
_5_56x45mm_M27_IAR_SAW	46
_5_56x45mm_L86_LSW	47
_5_56x45mm_Daewoo_K3_LMG	48
_5_56x45mm_Vector_Mini_SS_GPMG	49
_7_62x51mm_M60_GPMG	50
_7_62x51mm_M60E3_GPMG	51
_7_62x51mm_M60E4_GPMG	52
_7_62x51mm_M60E6_GPMG	53
_7_62x51mm_Mark_48_GMPG	55
_7_62x51mm_M240_FN_MAG_58_GPMG	58
_7_62x51mm_M240E4_M240B_GPMG	59
_7_62x51mm_M240E1_M240D_GPMG	60
_7_62x51mm_M240G_GPMG	61
_7_62x51mm_M240E5_M240H_GPMG	62
_7_62x51mm_M240L_GPMG	63
_7_62x39mm_Kk_62_LMG	65
_7_62x51mm_Vector_SS_77_GPMG	70
_7_62x51mm_SIG_MG_710_3_GPMG	71
_7_62x51mm_Sterling_7_62_GMPG	72

Enumerator	Value
_7_62x51mm_Sumitomo_Type_62_GPMG	73
_7_62x51mm_Daewoo_K12_GPMG	74
_7_62x51mm_MG_51_GPMG	75
_7_62x51mm_Rheinmetall_MG_3	76
_7_62x51mm_Rheinmetall_MG_3KWS	77
_7_62x51mm_MG5_HK121_GPMG	80
_7_62x51mm_HK21_GPMG	81
_7_62x51mm_AA_52_GPMP	85
_7_62x51mm_UKM_2000_GPMG	86
_7_62x54mm_Uk_vz_59_GPMG	88
_7_92x57mm_MG_42_GPMG	89
_12_7x99mm_M2A1_Browning_HMG	100
_12_7x99mm_M2HB_Browning_HMG	101
_12_7x99mm_M2HB_QCB_Browning_HMG	102
_12_7x99mm_M85C_HMG	105
_12_7x99mm_Rheinmetall_RMG_50_HMG	108
_12_7x99mm_HK25_HMG	110
_12_7x99mm_CIS_50MG	112
_5_45x39mm_IP_2_LMG	120
_5_45x39mm_Nikonov_LMG	121
_5_45x39mm_M74_RPK	122
_7_62x39mm_M43_RPK	125
_7_62x39mm_RPD_SAW	126
_7_62x39mm_Zastava_M72	127
_7_62x39mm_Type_81_LMG	128
_7_62x51mm_Zastava_M77	135
_7_62x54mm_PK_GPMG	140
_7_62x54mm_AEK_999_GPMP	141

Enumerator	Value
_7_62x54mm_Pecheneg_GPMG	142
_7_62x54mm_Zastava_M84	143
_7_62x54mm_Type_67_GPMG	144
_7_62x54mm_Type_80_GPMG	145
_7_62x54mm_PKP_GPMG	146
_12_7x108mm_NSV_HMG	150
_12_7x108mm_Kord_HMG	151
_12_7x108mm_KPD_12_7_HMG	152
_12_7x108mm_Zastava_M02_Coyotoe_HMG	153
_12_7x108mm_Zastava_M87	154
_12_7x108mm_Type_77_HMG	155
_12_7x108mm_W85_HMG	156
_12_7x108mm_Type_90_HMG	157
_5_8x42mm_QJY_88_LMG	164
_5_8x42mm_QBB_95_DBP87_LMG	165
_5_56x45mm_QBB_95_1_LMG	166
_6_8mm_M250_Next_Generation_Squad_Weapon_NGSW_	170

LifeFormHumanSpecificManPortableAirDefenseSystemEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Man-Portable Air Defense System [UID 520]*

Enumerator	Value
Other	0
_70mm_FIM_43_Redeye	1
_70mm_FIM_92_Stinger	2
_76mm_Blowpipe	10
_76mm_Starburst_Javelin_S_15_	11
_130mm_Starstreak_HVM	12

Enumerator	Value
_90mm_Mistral	15
_72mm_9K32M_Strela_2_SA_7_	20
_72mm_9K36_Strela_3_SA_14_	21
_72mm_9K38_Igla_SA_18_	22
_72mm_9K310_Igla_M_SA_16_	23
_72mm_9K333_Verba_SA_29_	24
_72mm_9K338_Igla_S_SA_24_Grinch_	25
_72mm_9K32M_Strela_2M_SA_7B_	26
_72mm_PPZR_Piorun_Grom_M_	27
_72mm_HN_5_Hong_Ying_5	30
_72mm_CSA_7_QW_1_Vanguard_	31
_72mm_CSA_8_QW_2_	32
_90mm_QW_3	33
_72mm_CSA_10_FN_6_	34
_72mm_CSA_11_QW_18_	35
_72mm_CSA_14_FN_16_	36
_71mm_Misagh_1	45
_71mm_Misagh_2	46
_80mm_Type_91_Kin_SAM	50
_80mm_KP_SAM_Shun_Gung_Chiron_	55
_106mm_RBS_70	60

LifeFormHumanSpecificMedicalEquipmentEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Medical Equipment [UID 530]*

Enumerator	Value
Other	0
Field_Medical_Kit	1

Enumerator	Value
Surgical_Kit	2
First_Aid_Kit	3

LifeFormHumanSpecificMortarsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Mortars [UID 512]*

Enumerator	Value
Others	0
_60mm_M224	30
_60mm_ANTOS	31
_60mm_KBA_118	32
_60mm_KBA_48M	33
_81mm_F2	50
_81mm_L16	51
_81mm_M252	52
_81mm_MX2_KM	53
_82mm_UPIK_82	54
_120mm_Soltam_K6_M120	60
_120mm_M121	61
_120mm_M120_15_Molot	62

LifeFormHumanSpecificPersonalElectronicsClassEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Personal Electronics Class [UID 526]*

Enumerator	Value
Cell_Phone	1
Radio_Telephone_Pack	2
UAV_UAS_Remote_Controller	3

LifeFormHumanSpecificRecoillessRiflesEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Life Form-Human-Specific-Recoilless Rifles [UID 521]*

Enumerator	Value
Other	0
_84mm_M136_AT_4_CS	15
_57mm_M18_RR	20
_75mm_M20_RR	21
_120mm_M_28_Davy_Crockett	22
_155mm_M_29_Davy_Crockett	23
_106mm_M40_Recoilless_Rifle	24
_82mm_M60_RR	25
_90mm_M67_RR	26
_84mm_M1_Carl_Gustav	30
_84mm_M2_Carl_Gustav	31
_84mm_M3_Carl_Gustav	32
_84mm_M4_Carl_Gustav	33
_74mm_Pansarskott_m_68_Miniman	35
_84mm_ALAC	40
_82mm_B_10_RR	45
_107mm_B_11_RR	46
_80mm_Breda_Folgore	50
_120mm_BAT_RR	55
_73mm_SPG_9_Kopye	60
_88mm_RCL_3_45in	65
_90mm_Pvpj_110	70
_50mm_Jagdfaust	75
_30mm_Rheinmetall_RMK30	80
_90mm_Matador_MP_RGW_90_	85

Enumerator	Value
_90mm_Matador_WB	86
_90mm_Matador_AS	87
_60mm_RGW_60	88
_88mm_55_S_55_Raikka	90
_95mm_95_S_58_61	91
_73mm_LG40	95
_105mm_LG40	96
_105mm_LG42	97

LifeFormHumanSpecificRocketLaunchersEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Rocket Launchers [UID 528]*

Enumerator	Value
_90mm_Enforcer	1

LifeFormHumanSpecificShotGunsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Shot Guns [UID 511]*

Enumerator	Value
Other	0
Browning_Superposed_O_U	20
Browning_Cynergy	21
Browning_Auto_5	22
_18_5mm_Browning_Citori_O_U_12_Gauge	23
_16_8mm_Browning_Citori_O_U_16_Gauge	24
_15_6mm_Browning_Citori_O_U_20_Gauge	25
_14mm_Browning_Citori_O_U_28_Gauge	26
_10_4mm_Browning_Citori_O_U_410_Bore	27
_18_5mm_Browning_Double_Automatic_12_Gauge	28

Enumerator	Value
_18_5mm_Ithaca_37_12_Gauge	29
_16_8mm_Ithaca_37_16_Gauge	30
_15_6mm_Ithaca_37_20_Gauge	31
_14mm_Ithaca_37_28_Gauge	32
_19_7mm_Ithaca_Mag_10_SA_10_Gauge	33
_19_7mm_Marlin_Model_55_10_Gauge	34
_18_5mm_Marlin_Model_55_12_Gauge	35
_16_8mm_Marlin_Model_55_16_Gauge	36
_15_6mm_Marlin_Model_55_20_Gauge	37
_18_5mm_Mossberg_500_12_Gauge	38
_15_6mm_Mossberg_500_20_Gauge	39
_10_4mm_Mossberg_500_410_Bore	40
_18_5mm_Mossberg_590_12_Gauge	41
_15_6mm_Mossberg_590_20_Gauge	42
_10_4mm_Mossberg_590_410_Bore	43
_18_5mm_Mossberg_930_SA_12_Gauge	44
Remington_Model_11_SA	45
Remington_Model_10_12_Gauge	46
_15_6mm_Remington_Model_17_20_Gauge	47
Remington_Model_31	48
Remington_Model_11_48_SA	49
_18_5mm_Remington_870_12_Gauge	50
_16_8mm_Remington_870_16_Gauge	51
_15_6mm_Remington_870_20_Gauge	52
_14mm_Remington_870_28_Gauge	53
_10_4mm_Remington_870_410_Bore	54
Remington_Model_58_SA	55
_18_5mm_Remington_878_SA_12_Gauge	56

Enumerator	Value
_18_5mm_Remington_Model_1100_SA_12_Gauge	57
_16_8mm_Remington_Model_1100_SA_16_Gauge	58
_15_6mm_Remington_Model_1100_SA_20_Gauge	59
_14mm_Remington_Model_1100_SA_28_Gauge	60
_10_4mm_Remington_Model_1100_SA_410_Bore	61
_18_5mm_Remington_11_87_SA_12_Gauge	62
_15_6mm_Remington_11_87_SA_20_Gauge	63
_19_7mm_Remington_Model_SP_10_SA_10_Gauge	64
_18_5mm_Remington_887_12_Gauge	65
_18_5mm_Remington_Sparta_100_SxS_12_Gauge	70
_15_6mm_Remington_Sparta_100_SxS_20_Gauge	71
_10_4mm_Remington_Sparta_100_SxS_410_Bore	72
_18_5mm_Remington_Spartan_310_O_U_12_Gauge	73
_15_6mm_Remington_Spartan_310_O_U_20_Gauge	74
_14mm_Remington_Spartan_310_O_U_28_Gauge	75
_10_4mm_Remington_Spartan_310_O_U_410_Bore	76
_18_5mm_Remington_Spartan_453_SA_12_Gauge	77
_18_5mm_Winchester_Model_1200_12_Gauge	80
_16_8mm_Winchester_Model_1200_16_Gauge	81
_15_6mm_Winchester_Model_1200_20_Gauge	82
Winchester_Model_1887_1901	83
Winchester_Model_1897	84
Winchester_Model_1912	85
Winchester_Model_21_SxS	86
Winchester_Model_37_SxS	87
_18_5mm_HR_Ultraslug_SxS_12_Gauge	88
_15_6mm_HR_Ultraslug_SxS_20_Gauge	89
_18_5mm_Ciener_Ultimate_O_U_12_Gauge	90

Enumerator	Value
_18_5mm_Coach_Gun_SxS_Double_Barrel_12_Gauge	91
_18_5mm_Ruger_Gold_Label_SxS_12_Gauge	92
_18_5mm_High_Standard_Model_10_SA_12_Gauge	93
_18_5mm_Kel_Tex_KSG_12_Gauge	94
_18_5_KAC_Masterkey_12_Gauge	95
_18_5mm_M26_M_A_S_S_12_Gauge	96
_18_5mm_SRM_Arms_M1216_SA_12_Gauge	97
_18_5mm_AA_12_FA_Atchisson_Assault	98
_18_5mm_Pancor_Jackhammer_FA_12_Gauge	99
_18_5mm_USAS_12_FA_12_Gauge	110
_18_5mm_MAUL_SA_12_Gauge	111
_18_5mm_FN_SLP_SA_12_Gauge	112
_18_5mm_FN_TPS_12_Gauge	113
_18_5mm_ENARM_Pentagun_SA_12_Gauge	115
Stevens_Model_520_620	116
Stoeger_Coach_Gun_SxS	117
Stoeger_Condor_O_U	118
_18_5mm_Armscor_Model_30_SA_12_Gauge	120
Weatherby_SA_08_SA	121
_18_5mm_Fabarm_SDASS_Tactical_12_Gauge	122
_18_5mm_MAG_7_12_Gauge	123
_18_5mm_Neostead_12_Gauge	124
_18_5mm_Armsel_Striker_SA_12_Gauge	125
_18_5mm_Parker_Hale_Rogun_SA_12_Gauge	127
_26mm_RGA_86_Revolver	130
_18_5mm_Sjorgren_SA_12_Gauge	131
_18_5mm_Akdal_MKA_1919_SA_12_Gauge	132
_18_5mm_Retay_Masai_Mara_SA_12_Gauge	133

Enumerator	Value
_18_5mm_Safir_T_14_SA_12_Gauge	134
_18_5mm_Benelli_M1_Super_90_SA_12_Gauge	150
_15_6mm_Benelli_M1_Super_90_SA_20_Gauge	151
_18_5mm_Benelli_M3_Super_90_SA_12_Gauge	152
_15_6mm_Benelli_M3_Super_90_SA_20_Gauge	153
_18_5mm_Benelli_M4_Super_90_SA_12_Gauge	154
_18_5mm_Benelli_Nova_12_Gauge	155
_15_6mm_Benelli_Nove_20_Gauge	156
_18_5mm_Benelli_Raffaello_SA_12_Gauge	157
_18_5mm_Benelli_Supernova_12_Gauge	158
_18_5mm_Benelli_Vinci_SA_12_Gauge	159
_18_5mm_Beretta_1201FP_SA_12_Gauge	160
_18_5mm_Beretta_682_O_U_12_Gauge	161
_15_6mm_Beretta_682_O_U_20_Gauge	162
_14mm_Beretta_682_O_U_28_Gauge	163
_10_4mm_Beretta_682_O_U_410_Bore	164
_18_5mm_Beretta_A303_SA_12_Gauge	165
_18_5mm_Beretta_AL391_SA_12_Gauge	166
_15_6mm_Beretta_AL391_SA_20_Gauge	167
_18_5mm_Beretta_DT_10_O_U_12_Gauge	168
Beretta_Silver_Pigeon_O_U	169
_18_5mm_Beretta_Xtrema_2_SA_12_Gauge	170
_15_6mm_Franchi_AL_48_SA_20_Gauge	171
_14mm_Franchi_AL_48_SA_28_Gauge	172
_10_4mm_Franchi_mod_410_FA_410_Bore	173
_18_5mm_Franchi_SPAS_12_SA_12_Gauge	174
_18_5mm_Franchi_SPAS_15_SA_12_Gauge	175
_18_5mm_Valtro_PM_5_PM_5_350_12_Gauge	176

Enumerator	Value
Blazer_F3_O_U	180
_18_5mm_HK_FABARM_FP6_12_Gauge	181
_18_5mm_HK_CAWS_FA_12_Gauge	182
_18_5mm_Baikal_MP_153_SA_12_Gauge	200
_18_5mm_Bandayevsky_RB_12_12_Gauge	201
_18_5mm_Molot_Bekas_M_12_Gauge	202
_16_8mm_Molot_Bekas_M_16_Gauge	203
_18_5mm_TOZ_194_12_Gauge	204
_23mm_KS_23	205
MTs_255_Revolver_12_Gauge	206
_18_5mm_RMB_93_12_Gauge	207
_18_5mm_Saiga_12_SA_12_Gauge	208
_15_6mm_Saiga_12_SA_20_Gauge	209
_10_4mm_Saiga_12_SA_410_Bore	210
_18_5mm_Vepr_12_SA_12_Gauge	211
_18_5mm_Fort_500_12_Gauge	212
_18_5mm_Norinco_HP9_1_12_Gauge	220

LifeFormHumanSpecificSniperEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Sniper [UID 481]*

Enumerator	Value
Other	0
_5_8mm_QBU_88_Type_88_	1
_7_62mm_C3	30
_7_62mm_FR_F2	31
_7_62mm_AWM_F_G22_	32
_7_62mm_G3_SG_1	33

Enumerator	Value
_7_62mm_Galil_Sniper	34
_7_62mm_L96A1	35
_7_62mm_M14_DMR	36
_7_62mm_M24_Sniper_Weapon_System_SWS_	37
_7_62mm_M40A1_A3	38
_7_62mm_Steyr_SSG_69	39
_7_62mm_SVD_Dragunov_	40
_7_62mm_TYPE_79	41
_7_62mm_SR_25_MK11	42
_7_62mm_AW_SR_98	43
_7_62mm_Blaser_R93	44
_7_62mm_M2010_Enhanced_Sniper_Rifle_ESR_	45
_7_62mm_M110_Semi_Automatic_Sniper_System_SASS_	46
_7_62mm_L129A1	47
_7_62mm_Sako_TRG_22	48
_7_62mm_Colt_Canada_C20_DMR	49
_7_62mm_BREN_2_PPS	50
_7_62mm_Zbroyar_VPR_308_Sniper_version_of_Z_008_	51
_7_62mm_Zbroyar_UAR_10	52
_7_62mm_Fort_301	53
_7_62mm_AWM_300_Winchester	54
_7_62mm_CZ_750	55
_7_7mm_TYPE_99	100
_8_58mm_Blaser_R93_Tactical_2	105
_8_6mm_Sako_TRG_42	106
_8_6mm_AWSM_338_Lapua_Magnum	107
_9mm_VSS_Vintorez	110
_9_5mm_10_36mm_CheyTac_M200_375_cal_408_cal_	120

Enumerator	Value
_12_7mm_Steyr_HS_50	170
_12_7mm_M82A1A_Special_Applications_Scoped_Rifle_SASR_	171
_12_7mm_NSV	172
_12_7mm_OSV_96	173
_12_7mm_Rangemaster_50	174
_12_7mm_V94	175
_12_7mm_M107	176
_12_7mm_ZVI_Falcon_OP99	177
_12_7mm_Sniper_Rhino_Hunter	178
_12_7mm_Sniper_Alligator	179
_12_7mm_Desert_Tech_HTI_375_cal_408_cal_416_cal_50_cal_	180
_12_7mm_AW50_50_cal_BMG_	181
_14_5mm_Sniper_T_Rex	190
_20mm_Denel_NTW_20	200

LifeFormHumanSpecificSubMachineGunEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Sub Machine Gun [UID 482]*

Enumerator	Value
Other	0
_5_45mm_AKS_74U_AKSU_74_	10
_5_56mm_Daewoo_K1A	20
_7_65mm_Sa_vz_61_Skorpion	40
_9mm_Daewoo_K7	60
_9mm_MAC_10	61
_9mm_Madsen_MK_II	62
_9mm_Mini_Uzi	63
_9mm_Model_83_Skorpion_SMG	64

Enumerator	Value
_9mm_MP5A2	65
_9mm_MP5_N	66
_9mm_Sterling_SMG	67
_9mm_Type_CF_05	68
_9mm_Uzi	69
_9mm_CZ_Scorpion_Evo	70
_9mm_MP7A2	71
_9mm_UMP9	72

LifeFormHumanSpecificWeaponNonSpecificEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Specific-Weapon Non-specific [UID 514]*

Enumerator	Value
Other	0
Knife	10
Machete	50
Explosive_Vest	100
M18A1_Claymore	150
Mine_Clearing_Line_Charge	160

LifeFormHumanSubcategoryEquipmentClassEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form-Human-Subcategory-Equipment Class [UID 473]*

Enumerator	Value
None	0
Weapon_Non_specific	1
Assault_Rifles	5
High_Power_Rifles	10
Sniper_Rifles	15

Enumerator	Value
Anti_Materiel_Rifle_AMR_	17
Sub_Machine_Guns	20
Shot_Guns	25
Grenade_Launchers	30
Machine_Guns	35
Grenade_Launching_Machine_Gun	40
Anti_Tank_Rockets	45
Anti_Tank_Missiles	50
Anti_Tank_Guns	55
Flame_Rockets	60
Flame_Throwers	65
Rocket_Launchers	70
Mortars	75
Hand_Guns	80
Man_Portable_Air_Defense_System_MANPADS_	85
Recoilless_Rifles	90
Drone_Guns	95
Equipment_Non_specific RPRnoteEnumerations36	150
Sensors RPRnoteEnumerations37	151
Signal_Sensor	152
Lasers	153
Animal_Companion RPRnoteEnumerations38	160
Personal_Electronics	171
Logistics_Equipment	172
Medical_Equipment	173

LifeFormLandCategoriesEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Form Land Categories [UID 472]*

Enumerator	Value
Conventional_Armed_Forces RPRnoteEnumerations39	10
Army	11
Naval_Infantry_Marines_	12
Air_Force	13
Navy	14
Coast_Guard	15
United_Nations	16
Special_Operations_Forces_SOF_ RPRnoteEnumerations40	30
Law_Enforcement RPRnoteEnumerations41	50
Non_Military_National_Government_Agencies RPRnoteEnumerations42	70
Regional_Local_Forces RPRnoteEnumerations43	90
Irregular_Forces RPRnoteEnumerations44	100
Terrorist_Combatant	101
Insurgent	102
Paramilitary_Forces RPRnoteEnumerations45	110
Humanitarian_Organizations RPRnoteEnumerations46	120
Civilian RPRnoteEnumerations47	130
Emergency_Medical_Technician_EMT_	131
Firefighter	132
Press	133
Mammal	200
Reptile	201
Amphibian	202
Insect	203
Arachnid	204
Mollusk	205
Marsupial	206

LifeFormSubsurfaceCategoriesEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Life Form Subsurface Categories [UID 479]*

Enumerator	Value
Fish	200
Mammal	201
Mollusk	202
Crustacean	203
Insect	204

LifeFormsSubcategoryCISWeaponsEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Life Forms-Subcategory-C.I.S. Weapons [UID 17]*

Enumerator	Value
Automatic_APS_9_mm_Stechkin	201
PSM_5_45_mm	202
Self_loading_PM_9_mm_Makarov	203
TT_33_7_62_mm_Tokarev	204
Assault_rifle_AK_and_AKM_7_62_mm	205
Assault_rifle_AK_74_and_AKS_74_5_45_mm	206
Self_loading_rifle_SKS_7_62_mm_Simonov	207
Sniper_rifle_SVD_7_62_mm_Dragunov	208
AKSU_74_5_45_mm	209
PPS_43_7_62_mm	210
PPSh_41_7_62_mm	211
General_purpose_PK_7_62_mm	212
Heavy_DShK_38_and_Model_38_46_12_7_mm_Degtyarev	213
Heavy_NSV_12_7_mm	214
Light_RPD_7_62_mm	215

Enumerator	Value
Light_RPK_7_62_mm	216
Light_RPK_74_5_45_mm	217
Hand_grenade_M75	218
Hand_grenade_RGD_5	219
AP_hand_grenade_F1	220
AT_hand_grenade_RKG_3	221
AT_hand_grenade_RKG_3M	222
AT_hand_grenade_RKG_3T	223
Fragmentation_hand_grenade_RGN	224
Fragmentation_hand_grenade_RGO	225
Smoke_hand_grenade_RDG_1	226
Plamya_launcher_30_mm_AGS_17	227
Rifle_mounted_launcher_BG_15_40_mm	228
LPO_50	229
ROKS_3	230
Cart_mounted_TPO_50	231
Gimlet_SA_16	232
Grail_SA_7	233
Gremlin_SA_14	234
Sagger_AT_3_MCLOS_	235
Saxhorn_AT_7	236
Spigot_A_B_AT_14	237
SA_18	238
SA_19	239
Grad_1P_manportable_tripod_rocket_launcher_122_mm_for_Spesnatz_and_other_specialists_aka_9P132_	240
Light_Anti_Armor_weapon_RPG_18	241
Light_Anti_Tank_weapon_RPG_22	242
MG_RPG	243

Enumerator	Value
Portable_rocket_launcher_RPG_16	244
Recoilless_gun_73_mm_SPG_9	245
VAT_rocket_launcher_RPG_7	246
Mon_50_Anti_Personnel_mine	248
RPG_29_Vampir	249
Laser_Designator	250
AT_4_Spigot	251
SA_24_Igla_S	252
Type_69_RPG	253

LifeFormsSubcategoryFrenchWeaponsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Forms-Subcategory-French Weapons [UID 19]*

Enumerator	Value
ACL_STRIM	1
Mistral_missile	2
Milan_AT_missile	3
LRAC_F1_89_mm_AT_rocket_launcher	4
FA_MAS_rifle	5
AA_52_machine_gun	6
_58_mm_rifle_grenade	7
FR_F1_sniper_rifle	8

LifeFormsSubcategoryGermanWeaponsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Forms-Subcategory-German Weapons [UID 20]*

Enumerator	Value
G3_rifle	1
G11_rifle	2

Enumerator	Value
P1_pistol	3
MG3_machine_gun	4
Milan_missile	5
MP1_Uzi_submachine_gun	6
Panzerfaust_3_Light_Anti_Tank_Weapon	7
DM19_hand_grenade	8
DM29_hand_grenade	9

LifeFormsSubcategoryIranianWeaponsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Forms-Subcategory-Iranian Weapons [UID 471]*

Enumerator	Value
Misagh_2	1
RBS_70	2

LifeFormsSubcategoryUKWeaponsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Forms-Subcategory-U.K. Weapons [UID 18]*

Enumerator	Value
LAW_80	1
Blowpipe	2
Javelin	3
_51_mm_mortar	4
SLR_7_62_mm_rifle	5
Sterling_9_mm_submachine_gun	6
L7A2_general_purpose_MG	7
L6_Wombat_Recoilless_rifle_	8
Carl_Gustav_89_mm_recoilless_rifle	9
SA80_Individual_light_support_weapon	10

Enumerator	Value
Trigat	11
Milan_AT_missile	12

LifeFormsSubcategoryUSWeaponsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life Forms-Subcategory-U.S. Weapons [UID 16]*

Enumerator	Value
Assault_machine_pistol_KF_AMP	1
Automatic_model_1911A1_45	2
Combat_Master_Mark_VI_45_Detronics	3
De_cocker_KP90DC_45	4
De_cocker_KP91DC_40	5
General_officer_s_Model_15_45	6
Nova_9_mm_LaFrance	7
Personal_Defense_Weapon_MP5K_PDW_9_mm	8
Silenced_Colt_45_LaFrance	9
_5900_series_9_mm_Smith_Wesson_S_W_	10
M9	11
Model_1911A1_Springfield_Armory	12
Model_2000_9_mm	13
P_9_9_mm_Springfield_Armory	14
P_12_9_mm	15
P_85_Mark_II_9_mm_Ruger	16
Advanced_Combat_Rifle_5_56_mm_AAI	17
Commando_assault_rifle_Model_733_5_56_mm_Colt	18
Infantry_rifle_Mini_14_20_GB_5_56_mm_Ruger	19
Mini_14_5_56_mm_Ruger	20
Mini_Thirty_7_62_mm_Ruger	21

Enumerator	Value
Semi_automatic_model_82A2_50_Barrett	22
Sniper_Weapon_System_M24_7_62_mm	23
Sniping_rifle_M21_Springfield_Armory	24
Sniping_rifle_M40A1_7_62_mm	25
Sniping_rifle_M600_7_62_mm	26
AR_15_M16_5_56_mm	27
M1_30_28	28
M14_7_62_mm_NATO	29
M14_M1A_M1A1_A1_Springfield_Armory	30
M14K_assault_rifle_LaFrance	31
M16A2_assault_rifle_5_56_mm_Colt	32
M21_7_62_mm_U_S_	33
M77_Mark_II_5_56_mm_Ruger	34
M77V_7_62_mm_Ruger	35
S_16_7_62_x_36_mm_Grendel	36
SAR_8_7_62_mm	37
SAR_4800_7_62_mm	38
Assault_carbine_M16K_LaFrance	39
M1_30_40	40
M4_Model_720_5_56_mm_Colt	41
M_900_9_mm_Calico	42
AC_556F_5_56_mm_Ruger	43
M3_45	44
M11_Cobray	45
M951_9_mm_Calico	46
MP5_10_10_mm	47
_9_mm_Colt	48
Ingram	49

Enumerator	Value
Externally_powered_EPG_7_62_mm_Ares	50
GECAL_50	51
General_purpose_M60_7_62_mm	52
Heavy_M2HB_QCB_50_RAMO	53
Light_assault_M60E3_Enhanced_7_62_mm	54
Light_M16A2_5_56_mm_Colt	55
Light_5_56_mm_Ares	56
Lightweight_M2_50_RAMO	57
Lightweight_assault_M60E3_7_62_mm	58
Minigun_M134_7_62_mm_General_Electric	59
MG_system_MK19_Mod_3_40_mm	60
MG_system_or_kit_M2HB_QCB_50_Saco_Defense	61
M1919A4_30_cal_Browning	62
_50_cal_Browning	63
Colored_smoke_hand_grenade_M18	64
Colored_smoke_grenades_Federal_Laboratories	65
Infrared_smoke_grenade_M76	66
Smoke_hand_grenade_AN_M8_HC	67
Delay_fragmentation_hand_grenade_M61	68
Delay_fragmentation_hand_grenade_M67	69
Impact_fragmentation_hand_grenade_M57	70
Impact_fragmentation_hand_grenade_M68	71
Incendiary_hand_grenade_AN_M14_TH3	72
Launcher_I_M203_40_mm	73
Launcher_M79_40_mm	74
Multiple_grenade_launcher_MM_1_40_mm	75
Multi_shot_portable_flame_weapon_M202A2_66_mm	76
Portable_ABC_M9_7	77

Enumerator	Value
Portable_M2A1_7	78
Portable_M9E1_7	79
Dragon_medium_Anti_Armor_missile_M47_FGM_77A	80
Javelin_AAWS_M	81
Light_Anti_Tank_Weapon_M72_LAW_II_	82
Redeye_FIM_43_General_Dynamics	83
Saber_dual_purpose_missile_system	84
Stinger_FIM_92_General_Dynamics	85
TOW_heavy_Anti_Tank_weapon	86
Bear_Trap_AP_device_Pancor	87
Chain_Gun_automatic_weapon_EX_34_7_62_mm	88
Close_Assault_Weapon_System_CAWS_AAI	89
CAWS_Olin_Heckler_and_Koch	90
Crossfire_SAM_Model_88	91
Dragon_and_M16	92
Firing_port_weapon_M231_5_56_mm_Colt	93
Foxhole_Digger_Explosive_Kit_EXFODA_	94
Infantry_Support_Weapon_ASP_30_RM_30_mm	95
Jackhammer_Mk_3_A2_Pancor	96
Light_Anti_Armor_weapon_M136_AT4_	97
M26A2	98
Master_Key_S	99
Minigun_5_56_mm	100
Multipurpose_Individual_Munition_MPIM_Marquardt	101
Multipurpose_weapon_AT8	102
Recoilless_rifle_M40_M40A2_and_M40A4_106_mm	103
Recoilless_rifle_M67_90_mm	104
Revolver_SP_101	105

Enumerator	Value
Revolver_Super_Redhawk_44_magnum_Ruger	106
RAW_rocket_140_mm_Brunswick	107
Rifle_launcher_Anti_Armor_Munition_RAAM_Olin	108
Rocket_launcher_M_20_3_5_in	109
Rocket_launcher_Enhanced_M72_E_series_HEAT_66_mm	110
Selective_fire_weapon_AC_556_5_56_mm_Ruger	111
Selective_fire_weapon_AC_556F_5_56_mm_Ruger	112
Shotgun_M870_Mk_1_U_S_Marine_Corps_Remington	113
SMAW_Mk_193_83_mm_McDonnell_Douglas	114
SMAW_D_Disposable_SMAW	115
Squad_Automatic_Weapon_SAW_M249_5_56_mm	116
Tactical_Support_Weapon_50_12_50_cal_Peregrine	117
Telescoped_Ammunition_Revolver_Gun_TARG_50_cal_Ares	118
Ultimate_over_under_combination_Ciener	119
M18A1_Claymore_mine	120
Mortar_81_mm	121
Machinegun_M240_7_62mm	134

LifeSavingEquipmentEnum8 [RPRnoteEnumerations48](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Life-Saving Equipment [UID 633]*

Enumerator	Value
Lifeboat	1
Liferaft	2
MOB_Boat	3
Lifebuoy	4

Link1111BEncryptionFlagEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 11/11B Encryption Flag [UID 735]*

Enumerator	Value
No_Encryption_Used	0
Encryption_Used	1

Link111BFidelityLevelEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 11/11B Fidelity Level [UID 467]*

Enumerator	Value
Fidelity_Level_0	0
Fidelity_Level_1	1
Fidelity_Level_2	2

Link11BDataSignalingRateEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 11B Data Signaling Rate [UID 740]*

Enumerator	Value
No_Statement	0
Not_Used_1	1
Not_Used_2	2
_1200_bps	3
_2400_bps	4
_600_bps	5

Link11BLinkStateEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 11B Link State [UID 737]*

Enumerator	Value
No_Statement	0
Inactive	1
Ready	2
Active	3
Operational	4

Link11BMessageSubTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 11B Message Sub Type [UID 739]*

Enumerator	Value
No_Statement	0
Transmission_Frame	1
Standby_Signal	2

Link11BModeOfOperationEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Link 11B Mode of Operation [UID 738]*

Enumerator	Value
No_Statement	0
Full_Transmission_of_Data	1
Limited_Transmission_of_Data	2
Receive_only	3

Link11BModulationStandardEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 11B Modulation Standard [UID 741]*

Enumerator	Value
No_Statement	0
CCITT_V_23	1

Link11DataSignalingRateEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Link 11 Data Signaling Rate [UID 732]*

Enumerator	Value
No_Statement	0
_1364_bps	1
_2250_bps	2
_1200_bps RPRnoteEnumerationsDeprecated	3
_2400_bps RPRnoteEnumerationsDeprecated	4

Link11DataTerminalSetIndicatorEnum8 [RPRnoteEnumerations27](#) [RPRnoteEnumerationsDeprecated](#)Representation: [RPRunsignedInteger8BE](#)Semantics: *Link 11 Data Terminal Set Indicator [UID 469]*

Enumerator	Value
No_Statement RPRnoteEnumerationsDeprecated	0
Transmit RPRnoteEnumerationsDeprecated	1
Receive RPRnoteEnumerationsDeprecated	2
Net_Busy RPRnoteEnumerationsDeprecated	3
Transmit_Data_Error RPRnoteEnumerationsDeprecated	4
Receive_Data_Error RPRnoteEnumerationsDeprecated	5
Code_Error RPRnoteEnumerationsDeprecated	6
Synchronization_Complete RPRnoteEnumerationsDeprecated	7

Link11MessageSubTypeEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Link 11 Message Sub Type [UID 730]*

Enumerator	Value
No_Statement	0
Interrogation	1

Enumerator	Value
Data_Start	2
Data	3
Data_Stop	4

Link11MessageTypeIdentifierEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 11 Message Type Identifier [UID 731]*

Enumerator	Value
No_Statement	0
Net_Test	1
Roll_Call	2
Picket_Reply	3
Short_Broadcast	4
Broadcast	5
Net_Sync	6

Link11ModeOfOperationEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Link 11 Mode of Operation [UID 470]*

Enumerator	Value
No_Statement	0
Net_Sync	1
Net_Test	2
Roll_Call	3
Short_Broadcast	4
Broadcast	5

Link11SignalIntegrationIntervalEnum8 [RPRnoteEnumerations27](#) [RPRnoteEnumerationsDeprecated](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 11 Signal Integration Interval [UID 733]*

Enumerator	Value
No_Statement RPRnoteEnumerationsDeprecated	0
_9_ms RPRnoteEnumerationsDeprecated	1
_18_ms RPRnoteEnumerationsDeprecated	2

Link11SignalWaveformEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 11 Signal Waveform [UID 734]*

Enumerator	Value
NoStatement-CLEWFormat	0
ConventionalLinkElevenWaveform_CLEW_	1
SingleToneLinkElevenWaveform_SLEW_	2

Link11TerminalModeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 11 Terminal Mode [UID 468]*

Enumerator	Value
No_Statement	0
Network_Control_Station	1
Picket	2

Link16VersionEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Link 16 version [UID 800]*

Enumerator	Value
No_Statement	0
MIL_STD_6016C	1
MIL_STD_6016D	2

Enumerator	Value
MIL_STD_6016E	3
MIL_STD_6016F	4
MIL_STD_6016F_C1	5
MIL_STD_6016G	6
MIL_STD_6016H	7
STANAG_5516_Ed_3	103
STANAG_5516_Ed_4	104
STANAG_5516_Ed_5	105
STANAG_5516_Ed_6	106
STANAG_5516_Ed_8	108

MaterialEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize04

Semantics: *Material [UID 466]*

Enumerator	Value
Not_Specified	0
Plastic	1
Rubber	2
Road	3

MessageTypeIdentifierEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Message Type Identifier [UID 176]*

Enumerator	Value
JTIDS_Header_Messages	0
RTT_A_B	1
RTT_Reply	2
JTIDS_Voice_CVSD	3
JTIDS_Voice_LPC10	4

Enumerator	Value
JTIDS_Voice_LPC12	5
JTIDS_LET	6
VMF	7

MinefieldAppearanceActiveStatusEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Minefield Appearance-Active Status [UID 419]*

Enumerator	Value
Active	0
Inactive	1

MinefieldAppearanceLaneEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Minefield Appearance-Lane [UID 420]*

Enumerator	Value
Minefield_Has_Active_Lane	0
Minefield_Has_an_Inactive_Lane	1

MinefieldAppearanceMinefieldTypeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Minefield Appearance-Minefield Type [UID 418]*

Enumerator	Value
Mixed_Anti_Personnel_and_Anti_Tank_Minefield	0
Pure_Anti_Personnel_Minefield	1
Pure_Anti_Tank_Minefield	2

MinefieldAppearanceStateEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Minefield Appearance-State [UID 421]*

Enumerator	Value
Active	0
Deactivated	1

MinefieldFusingFuseTypeEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize07

Semantics: *Minefield Fusing-Fuse Type [UID 422]*

Enumerator	Value
No_Fuse	0
Other	1
Pressure	2
Magnetic	3
Tilt_Rod	4
Command	5
Trip_Wire	6

MinefieldPaintSchemeAlgaeEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize02

Semantics: *Minefield Paint Scheme-Algae [UID 424]*

Enumerator	Value
None	0
Light	1
Moderate	2
Heavy	3

MinefieldPaintSchemePaintSchemeEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize06

Semantics: *Minefield Paint Scheme-Paint Scheme [UID 425]*

Enumerator	Value
Other	0

Enumerator	Value
Standard	1
Camouflage_Desert	2
Camouflage_Jungle	3
Camouflage_Snow	4
Camouflage_Gravel	5
Camouflage_Pavement	6
Camouflage_Sand	7
Natural_Wood	8
Clear	9
Red	10
Blue	11
Green	12
Olive	13
White	14
Tan	15
Black	16
Yellow	17
Brown	18

MinefieldSensorTypesEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize04

Semantics: *Minefield-Sensor Types [UID 193]*

Enumerator	Value
Other	0
Optical	1
FLIR	2
RADAR	3
Magnetic	4

Enumerator	Value
Laser	5
SONAR	6
Physical	7
Multispectral	8

MinefieldSensorTypesFLIREnum16

Representation: [RPRunsignedInteger16BE](#) [RPRnoteEnumerationsBitSize12](#)

Semantics: *Minefield-Sensor Types-FLIR [UID 195]*

Enumerator	Value
Generic_3_5	0
Generic_8_12	1
ASTAMIDS_I	2
ASTAMIDS_II	3
GSTAMIDS_3_5	4
GSTAMIDS_8_12	5
HSTAMIDS_3_5	6
HSTAMIDS_8_12	7
COBRA_3_5	8
COBRA_8_12	9

MinefieldSensorTypesLaserEnum16

Representation: [RPRunsignedInteger16BE](#) [RPRnoteEnumerationsBitSize12](#)

Semantics: *Minefield-Sensor Types-Laser [UID 198]*

Enumerator	Value
Generic	0
ASTAMIDS	1

MinefieldSensorTypesMagneticEnum16

Representation: [RPRunsignedInteger16BE](#) [RPRnoteEnumerationsBitSize12](#)Semantics: *Minefield-Sensor Types-Magnetic [UID 197]*

Enumerator	Value
Generic	0
AN_PSS_11	1
AN_PSS_12	2
GSTAMIDS	3

MinefieldSensorTypesMultispectralEnum16Representation: [RPRunsignedInteger16BE](#) [RPRnoteEnumerationsBitSize12](#)Semantics: *Minefield-Sensor Types-Multispectral [UID 201]*

Enumerator	Value
Generic	0

MinefieldSensorTypesOpticalEnum16Representation: [RPRunsignedInteger16BE](#) [RPRnoteEnumerationsBitSize12](#)Semantics: *Minefield-Sensor Types-Optical [UID 194]*

Enumerator	Value
Unaided_Eye_Actively_Searching	0
Unaided_Eye_Not_Actively_Searching	1
Binoculars	2
Image_Intensifier	3
HMMWV_occupant_Actively_Searching	4
HMMWV_occupant_Not_Actively_Searching	5
Truck_occupant_Actively_Searching	6
Truck_occupant_Not_Actively_Searching	7
Tracked_vehicle_occupant_closed_hatch_Actively_Searching	8
Tracked_vehicle_occupant_closed_hatch_Not_Actively_Searching	9
Tracked_vehicle_occupant_open_hatch_Actively_Searching	10

Enumerator	Value
Tracked_vehicle_occupant_open_hatch_Not_Actively_Searching	11

MinefieldSensorTypesPhysicalEnum16

Representation: [RPRunsignedInteger16BE](#) RPRnoteEnumerationsBitSize12

Semantics: *Minefield-Sensor Types-Physical [UID 200]*

Enumerator	Value
Generic_Probe	0
Probe_metal_content	1
Probe_no_metal_content	2

MinefieldSensorTypesRADAREnum16

Representation: [RPRunsignedInteger16BE](#) RPRnoteEnumerationsBitSize12

Semantics: *Minefield-Sensor Types-RADAR [UID 196]*

Enumerator	Value
Generic	0
Generic_GPR	1
GSTAMIDS_I	2
GSTAMIDS_II	3
HSTAMIDS_I	4
HSTAMIDS_II	5

MinefieldSensorTypesSONAREnum16

Representation: [RPRunsignedInteger16BE](#) RPRnoteEnumerationsBitSize12

Semantics: *Minefield-Sensor Types-SONAR [UID 199]*

Enumerator	Value
Generic	0

MinefieldStateProtocolModeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Minefield State-Protocol Mode [UID 336]*

Enumerator	Value
Heartbeat_Mode	0
QRP_Mode	1

Mode5IFFMissionEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize03](#)

Semantics: *Mode 5 IFF Mission [UID 346]*

Enumerator	Value
No_Statement	0
Surveillance_SHORAD	1
SHORAD_associated_with_a_Weapons_System	2
Weapon_System	3
Airborne_and_Surface_Ship_Surveillance_Platforms	4
Airborne_and_Surface_Ship_Weapons_Platforms	5
Ground_to_Ground	6

Mode5LevelSelectionEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Mode 5 Level Selection [UID 412]*

Enumerator	Value
Mode_5_Level_1	0
Mode_5_Level_2	1

Mode5LocationErrorsEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Mode 5 Location Errors [UID 423]*

Enumerator	Value
No_Location_Errors	0
IFF_Data_Record_Present	1

Mode5MessageFormatsStatusEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Mode 5 Message Formats Status [UID 380]*

Enumerator	Value
Capability	0
Active_Interrogation	1

Mode5PlatformTypeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Mode 5 Platform Type [UID 396]*

Enumerator	Value
Ground_Vehicle	0
Air_Vehicle	1

Mode5ReplyEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize03](#)

Semantics: *Mode 5 Reply [UID 350]*

Enumerator	Value
No_Response	0
Valid	1
Invalid	2
Unable_to_Verify	3

Mode5SAltitudeResolutionEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Mode 5/S Altitude Resolution [UID 361]*

Enumerator	Value
_100_foot	0
_25_foot	1

Mode5SquitterTypeEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize03

Semantics: *Mode 5 Squitter Type [UID 352]*

Enumerator	Value
Not_Capable	0
Short	1
Extended	2

ModeCAltitudeIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize01

Semantics: *Mode C Altitude Indicator [UID 340]*

Enumerator	Value
Positive_Altitude_Above_MSL	0
Negative_Altitude_Below_MSL_Use_Alternate_Mode_C_Altitude	1

ModeSInterrogatorIdentifierICTypeEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize01

Semantics: *Mode S Interrogator Identifier IC Type [UID 348]*

Enumerator	Value
II	0
SI	1

ModeSInterrogatorStatusTransmitStateEnum8

Representation: [RPRunsignedInteger8BE](#) RPRnoteEnumerationsBitSize03

Semantics: *Mode S Interrogator Status Transmit State [UID 347]*

Enumerator	Value
No_Statement	0
Roll_Call	1
All_Call	2
Lockout_Override	3
Temporary_Lockout	4
Intermittent_Lockout	5

ModeSSquitterRecordSourceEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Mode S Squitter Record Source [UID 355]*

Enumerator	Value
Layer_4_IFF_Data_Records	0
Layer_5_GICB_IFF_Data_Records	1

ModeSSquitterTypeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize03](#)

Semantics: *Mode S Squitter Type [UID 354]*

Enumerator	Value
Not_Capable	0
Acquisition	1
Extended	2
Short	3

MunitionCategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Munition Category [UID 15]*

Enumerator	Value
Other	0
Guided	1

Enumerator	Value
Ballistic	2
Fixed	3

MunitionDescriptorFuseEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Munition Descriptor-Fuse [UID 61]*

Enumerator	Value
Other	0000
Intelligent_Influence	0010
Sensor	0020
Self_destruct	0030
Ultra_Quick	0040
Body	0050
Deep_Intrusion	0060
Multifunction	0100
Point_Detonation_PD_	0200
Base_Detonation_BD_	0300
Contact	1000
Contact_Instant_Impact_	1100
Contact_Delayed	1200
Contact_Delayed_10_ms_Delay	1201
Contact_Delayed_20_ms_Delay	1202
Contact_Delayed_50_ms_Delay	1205
Contact_Delayed_60_ms_Delay	1206
Contact_Delayed_100_ms_Delay	1210
Contact_Delayed_125_ms_Delay	1212
Contact_Delayed_250_ms_Delay	1225
Contact_Delayed_5_ms_Delay	1250

Enumerator	Value
Contact_Delayed_15_ms_Delay	1251
Contact_Delayed_25_ms_Delay	1252
Contact_Delayed_30_ms_Delay	1253
Contact_Delayed_35_ms_Delay	1254
Contact_Delayed_40_ms_Delay	1255
Contact_Delayed_45_ms_Delay	1256
Contact_Delayed_90_ms_Delay	1257
Contact_Delayed_120_ms_Delay	1258
Contact_Delayed_180_ms_Delay	1259
Contact_Delayed_240_ms_Delay	1260
Contact_Electronic_Oblique_Contact_	1300
Contact_Graze	1400
Contact_Crush	1500
Contact_Hydrostatic	1600
Contact_Mechanical	1700
Contact_Chemical	1800
Contact_Piezoelectric	1900
Contact_Point_Initiating	1910
Contact_Point_Initiating_Base_Detonating	1920
Contact_Base_Detonating	1930
Contact_Ballistic_Cap_and_Base	1940
Contact_Base	1950
Contact_Nose	1960
Contact_Fitted_in_Standoff_Probe	1970
Contact_Non_aligned	1980
Timed	2000
Timed_Programmable	2100
Timed_Burnout	2200

Enumerator	Value
Timed_Pyrotechnic	2300
Timed_Electronic	2400
Timed_Base_Delay	2500
Timed_Reinforced_Nose_Impact_Delay	2600
Timed_Short_Delay_Impact	2700
Timed_Short_Delay_Impact_10_ms_Delay	2701
Timed_Short_Delay_Impact_20_ms_Delay	2702
Timed_Short_Delay_Impact_50_ms_Delay	2705
Timed_Short_Delay_Impact_60_ms_Delay	2706
Timed_Short_Delay_Impact_100_ms_Delay	2710
Timed_Short_Delay_Impact_125_ms_Delay	2712
Timed_Short_Delay_Impact_250_ms_Delay	2725
Timed_Nose_Mounted_Variable_Delay	2800
Timed_Long_Delay_Side	2900
Timed_Selectable_Delay	2910
Timed_Impact	2920
Timed_Sequence	2930
Proximity	3000
Proximity_Active_Laser	3100
Proximity_Magnetic_Magpolarity_	3200
Proximity_Active_Radar_Doppler_Radar_	3300
Proximity_Radio_Frequency_RF_	3400
Proximity_Programmable	3500
Proximity_Programmable_Prefragmented	3600
Proximity_Infrared	3700
Command	4000
Command_Electronic_Remotely_Set	4100
Altitude	5000

Enumerator	Value
Altitude_Radio_Altimeter	5100
Altitude_Air_Burst	5200
Depth	6000
Acoustic	7000
Pressure	8000
Pressure_Delay	8010
Inert	8100
Dummy	8110
Practice	8120
Plug_Representing	8130
Training	8150
Pyrotechnic	9000
Pyrotechnic_Delay	9010
Electro_optical	9100
Electromechanical	9110
Electromechanical_Nose	9120
Strikerless	9200
Strikerless_Nose_Impact	9210
Strikerless_Compression_Ignition	9220
Compression_Ignition	9300
Compression_Ignition_Strikerless_Nose_Impact	9310
Percussion	9400
Percussion_Instantaneous	9410
Electronic	9500
Electronic_Internally_Mounted	9510
Electronic_Range_Setting	9520
Electronic_Programmed	9530
Mechanical	9600

Enumerator	Value
Mechanical_Nose	9610
Mechanical_Tail	9620

MunitionDescriptorWarheadEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Munition Descriptor-Warhead [UID 60]*

Enumerator	Value
Other	0000
Cargo_Variable_Submunitions_	0010
Fuel_Air_Explosive	0020
Glass_Beads	0030
_1_um	0031
_5_um	0032
_10_um	0033
High_Explosive_HE_	1000
HE_Plastic	1100
HE_Incendiary	1200
HE_Fragmentation	1300
HE_Anti_Tank	1400
HE_Bomblets	1500
HE_Shaped_Charge	1600
HE_Continuous_Rod	1610
HE_Tungsten_Ball	1615
HE_Blast_Fragmentation	1620
HE_Steerable_Darts_with_HE	1625
HE_Darts	1630
HE_Flechettes	1635
HE_Directed_Fragmentation	1640

Enumerator	Value
HE_Semi_Armor_Piercing_SAP_	1645
HE_Shaped_Charge_Fragmentation	1650
HE_Semi_Armor_Piercing_Fragmentation	1655
HE_Hollow_Charge	1660
HE_Double_Hollow_Charge	1665
HE_General_Purpose	1670
HE_Blast_Penetrator	1675
HE_Rod_Penetrator	1680
HE_Anti_Personnel	1685
HE_Shaped_Charge_Fragmentation_Incendiary	1690
HE_Penetrator_Blast_Fragmentation	1695
Smoke	2000
WP_White_Phosphorus_	2005
FOGO_Fog_Oil_	2010
HC_HexaChloroEthane_	2015
Illumination	3000
Practice	4000
Blank	4001
Dummy	4002
Kinetic	5000
Mines	6000
Nuclear	7000
Nuclear_IMT	7010
Nuclear_yield_1 RPRnoteEnumerations49	7011
Nuclear_yield_2 RPRnoteEnumerations49	7012
Nuclear_yield_3 RPRnoteEnumerations49	7013
Nuclear_yield_4 RPRnoteEnumerations49	7014
Nuclear_yield_5 RPRnoteEnumerations49	7015

Enumerator	Value
Nuclear_yield_6 RPRnoteEnumerations49	7016
Nuclear_yield_7 RPRnoteEnumerations49	7017
Nuclear_yield_8 RPRnoteEnumerations49	7018
Nuclear_yield_9 RPRnoteEnumerations49	7019
Nuclear_yield_10 RPRnoteEnumerations49	7020
Nuclear_yield_11 RPRnoteEnumerations49	7021
Nuclear_yield_12 RPRnoteEnumerations49	7022
Nuclear_yield_13 RPRnoteEnumerations49	7023
Nuclear_yield_14 RPRnoteEnumerations49	7024
Nuclear_yield_15 RPRnoteEnumerations49	7025
Nuclear_yield_16 RPRnoteEnumerations49	7026
Nuclear_yield_17 RPRnoteEnumerations49	7027
Nuclear_yield_18 RPRnoteEnumerations49	7028
Nuclear_yield_19 RPRnoteEnumerations49	7029
Nuclear_yield_20 RPRnoteEnumerations49	7030
Nuclear_yield_21 RPRnoteEnumerations49	7031
Nuclear_yield_22 RPRnoteEnumerations49	7032
Nuclear_yield_23 RPRnoteEnumerations49	7033
Nuclear_yield_24 RPRnoteEnumerations49	7034
Nuclear_yield_25 RPRnoteEnumerations49	7035
Nuclear_yield_26 RPRnoteEnumerations49	7036
Nuclear_yield_27 RPRnoteEnumerations49	7037
Nuclear_yield_28 RPRnoteEnumerations49	7038
Nuclear_yield_29 RPRnoteEnumerations49	7039
Nuclear_yield_30 RPRnoteEnumerations49	7040
Nuclear_yield_31 RPRnoteEnumerations49	7041
Nuclear_yield_32 RPRnoteEnumerations49	7042
Nuclear_yield_33 RPRnoteEnumerations49	7043

Enumerator	Value
Nuclear_yield_34 RPRnoteEnumerations49	7044
Nuclear_yield_35 RPRnoteEnumerations49	7045
Nuclear_yield_36 RPRnoteEnumerations49	7046
Nuclear_yield_37 RPRnoteEnumerations49	7047
Nuclear_yield_38 RPRnoteEnumerations49	7048
Nuclear_yield_39 RPRnoteEnumerations49	7049
Nuclear_yield_40 RPRnoteEnumerations49	7050
Nuclear_yield_41 RPRnoteEnumerations49	7051
Nuclear_yield_42 RPRnoteEnumerations49	7052
Nuclear_yield_43 RPRnoteEnumerations49	7053
Nuclear_yield_44 RPRnoteEnumerations49	7054
Nuclear_yield_45 RPRnoteEnumerations49	7055
Nuclear_yield_46 RPRnoteEnumerations49	7056
Nuclear_yield_47 RPRnoteEnumerations49	7057
Nuclear_yield_48 RPRnoteEnumerations49	7058
Nuclear_yield_49 RPRnoteEnumerations49	7059
Nuclear_yield_50 RPRnoteEnumerations49	7060
Nuclear_yield_51 RPRnoteEnumerations49	7061
Nuclear_yield_52 RPRnoteEnumerations49	7062
Nuclear_yield_53 RPRnoteEnumerations49	7063
Nuclear_yield_54 RPRnoteEnumerations49	7064
Nuclear_yield_55 RPRnoteEnumerations49	7065
Nuclear_yield_56 RPRnoteEnumerations49	7066
Nuclear_yield_57 RPRnoteEnumerations49	7067
Nuclear_yield_58 RPRnoteEnumerations49	7068
Nuclear_yield_59 RPRnoteEnumerations49	7069
Nuclear_yield_60 RPRnoteEnumerations49	7070
Nuclear_yield_61 RPRnoteEnumerations49	7071

Enumerator	Value
Nuclear_yield_62 RPRnoteEnumerations49	7072
Nuclear_yield_63 RPRnoteEnumerations49	7073
Nuclear_yield_64 RPRnoteEnumerations49	7074
Nuclear_yield_65 RPRnoteEnumerations49	7075
Nuclear_yield_66 RPRnoteEnumerations49	7076
Nuclear_yield_67 RPRnoteEnumerations49	7077
Nuclear_yield_68 RPRnoteEnumerations49	7078
Nuclear_yield_69 RPRnoteEnumerations49	7079
Nuclear_yield_70 RPRnoteEnumerations49	7080
Nuclear_yield_71 RPRnoteEnumerations49	7081
Nuclear_yield_72 RPRnoteEnumerations49	7082
Nuclear_yield_73 RPRnoteEnumerations49	7083
Nuclear_yield_74 RPRnoteEnumerations49	7084
Nuclear_yield_75 RPRnoteEnumerations49	7085
Nuclear_yield_76 RPRnoteEnumerations49	7086
Nuclear_yield_77 RPRnoteEnumerations49	7087
Nuclear_yield_78 RPRnoteEnumerations49	7088
Nuclear_yield_79 RPRnoteEnumerations49	7089
Nuclear_yield_80 RPRnoteEnumerations49	7090
Nuclear_yield_81 RPRnoteEnumerations49	7091
Nuclear_yield_82 RPRnoteEnumerations49	7092
Nuclear_yield_83 RPRnoteEnumerations49	7093
Nuclear_yield_84 RPRnoteEnumerations49	7094
Nuclear_yield_85 RPRnoteEnumerations49	7095
Nuclear_yield_86 RPRnoteEnumerations49	7096
Nuclear_yield_87 RPRnoteEnumerations49	7097
Nuclear_yield_88 RPRnoteEnumerations49	7098
Nuclear_yield_89 RPRnoteEnumerations49	7099

Enumerator	Value
Nuclear_yield_90 RPRnoteEnumerations49	7100
Nuclear_yield_91 RPRnoteEnumerations49	7101
Nuclear_yield_92 RPRnoteEnumerations49	7102
Nuclear_yield_93 RPRnoteEnumerations49	7103
Nuclear_yield_94 RPRnoteEnumerations49	7104
Nuclear_yield_95 RPRnoteEnumerations49	7105
Nuclear_yield_96 RPRnoteEnumerations49	7106
Nuclear_yield_97 RPRnoteEnumerations49	7107
Nuclear_yield_98 RPRnoteEnumerations49	7108
Nuclear_yield_99 RPRnoteEnumerations49	7109
Nuclear_yield_100 RPRnoteEnumerations49	7110
Nuclear_yield_101 RPRnoteEnumerations49	7111
Nuclear_yield_102 RPRnoteEnumerations49	7112
Nuclear_yield_103 RPRnoteEnumerations49	7113
Nuclear_yield_104 RPRnoteEnumerations49	7114
Nuclear_yield_105 RPRnoteEnumerations49	7115
Nuclear_yield_106 RPRnoteEnumerations49	7116
Nuclear_yield_107 RPRnoteEnumerations49	7117
Nuclear_yield_108 RPRnoteEnumerations49	7118
Nuclear_yield_109 RPRnoteEnumerations49	7119
Nuclear_yield_110 RPRnoteEnumerations49	7120
Nuclear_yield_111 RPRnoteEnumerations49	7121
Nuclear_yield_112 RPRnoteEnumerations49	7122
Nuclear_yield_113 RPRnoteEnumerations49	7123
Nuclear_yield_114 RPRnoteEnumerations49	7124
Nuclear_yield_115 RPRnoteEnumerations49	7125
Nuclear_yield_116 RPRnoteEnumerations49	7126
Nuclear_yield_117 RPRnoteEnumerations49	7127

Enumerator	Value
Nuclear_yield_118 RPRnoteEnumerations49	7128
Nuclear_yield_119 RPRnoteEnumerations49	7129
Nuclear_yield_120 RPRnoteEnumerations49	7130
Nuclear_yield_121 RPRnoteEnumerations49	7131
Nuclear_yield_122 RPRnoteEnumerations49	7132
Nuclear_yield_123 RPRnoteEnumerations49	7133
Nuclear_yield_124 RPRnoteEnumerations49	7134
Nuclear_yield_125 RPRnoteEnumerations49	7135
Nuclear_yield_126 RPRnoteEnumerations49	7136
Nuclear_yield_127 RPRnoteEnumerations49	7137
Nuclear_yield_128 RPRnoteEnumerations49	7138
Nuclear_yield_129 RPRnoteEnumerations49	7139
Nuclear_yield_130 RPRnoteEnumerations49	7140
Nuclear_yield_131 RPRnoteEnumerations49	7141
Nuclear_yield_132 RPRnoteEnumerations49	7142
Nuclear_yield_133 RPRnoteEnumerations49	7143
Nuclear_yield_134 RPRnoteEnumerations49	7144
Nuclear_yield_135 RPRnoteEnumerations49	7145
Nuclear_yield_136 RPRnoteEnumerations49	7146
Nuclear_yield_137 RPRnoteEnumerations49	7147
Nuclear_yield_138 RPRnoteEnumerations49	7148
Nuclear_yield_139 RPRnoteEnumerations49	7149
Nuclear_yield_140 RPRnoteEnumerations49	7150
Nuclear_yield_141 RPRnoteEnumerations49	7151
Nuclear_yield_142 RPRnoteEnumerations49	7152
Nuclear_yield_143 RPRnoteEnumerations49	7153
Nuclear_yield_144 RPRnoteEnumerations49	7154
Nuclear_yield_145 RPRnoteEnumerations49	7155

Enumerator	Value
Nuclear_yield_146 RPRnoteEnumerations49	7156
Nuclear_yield_147 RPRnoteEnumerations49	7157
Nuclear_yield_148 RPRnoteEnumerations49	7158
Nuclear_yield_149 RPRnoteEnumerations49	7159
Nuclear_yield_150 RPRnoteEnumerations49	7160
Nuclear_yield_151 RPRnoteEnumerations49	7161
Nuclear_yield_152 RPRnoteEnumerations49	7162
Nuclear_yield_153 RPRnoteEnumerations49	7163
Nuclear_yield_154 RPRnoteEnumerations49	7164
Nuclear_yield_155 RPRnoteEnumerations49	7165
Nuclear_yield_156 RPRnoteEnumerations49	7166
Nuclear_yield_157 RPRnoteEnumerations49	7167
Nuclear_yield_158 RPRnoteEnumerations49	7168
Nuclear_yield_159 RPRnoteEnumerations49	7169
Nuclear_yield_160 RPRnoteEnumerations49	7170
Nuclear_yield_161 RPRnoteEnumerations49	7171
Nuclear_yield_162 RPRnoteEnumerations49	7172
Nuclear_yield_163 RPRnoteEnumerations49	7173
Nuclear_yield_164 RPRnoteEnumerations49	7174
Nuclear_yield_165 RPRnoteEnumerations49	7175
Nuclear_yield_166 RPRnoteEnumerations49	7176
Nuclear_yield_167 RPRnoteEnumerations49	7177
Nuclear_yield_168 RPRnoteEnumerations49	7178
Nuclear_yield_169 RPRnoteEnumerations49	7179
Nuclear_yield_170 RPRnoteEnumerations49	7180
Nuclear_yield_171 RPRnoteEnumerations49	7181
Nuclear_yield_172 RPRnoteEnumerations49	7182
Nuclear_yield_173 RPRnoteEnumerations49	7183

Enumerator	Value
Nuclear_yield_174 RPRnoteEnumerations49	7184
Nuclear_yield_175 RPRnoteEnumerations49	7185
Nuclear_yield_176 RPRnoteEnumerations49	7186
Nuclear_yield_177 RPRnoteEnumerations49	7187
Nuclear_yield_178 RPRnoteEnumerations49	7188
Nuclear_yield_179 RPRnoteEnumerations49	7189
Nuclear_yield_180 RPRnoteEnumerations49	7190
Nuclear_yield_181 RPRnoteEnumerations49	7191
Nuclear_yield_182 RPRnoteEnumerations49	7192
Nuclear_yield_183 RPRnoteEnumerations49	7193
Nuclear_yield_184 RPRnoteEnumerations49	7194
Nuclear_yield_185 RPRnoteEnumerations49	7195
Nuclear_yield_186 RPRnoteEnumerations49	7196
Nuclear_yield_187 RPRnoteEnumerations49	7197
Nuclear_yield_188 RPRnoteEnumerations49	7198
Nuclear_yield_189 RPRnoteEnumerations49	7199
Nuclear_yield_190 RPRnoteEnumerations49	7200
Nuclear_yield_191 RPRnoteEnumerations49	7201
Nuclear_yield_192 RPRnoteEnumerations49	7202
Nuclear_yield_193 RPRnoteEnumerations49	7203
Nuclear_yield_194 RPRnoteEnumerations49	7204
Nuclear_yield_195 RPRnoteEnumerations49	7205
Nuclear_yield_196 RPRnoteEnumerations49	7206
Nuclear_yield_197 RPRnoteEnumerations49	7207
Nuclear_yield_198 RPRnoteEnumerations49	7208
Nuclear_yield_199 RPRnoteEnumerations49	7209
Nuclear_yield_200 RPRnoteEnumerations49	7210
Nuclear_yield_201 RPRnoteEnumerations49	7211

Enumerator	Value
Nuclear_yield_202 RPRnoteEnumerations49	7212
Nuclear_yield_203 RPRnoteEnumerations49	7213
Nuclear_yield_204 RPRnoteEnumerations49	7214
Nuclear_yield_205 RPRnoteEnumerations49	7215
Nuclear_yield_206 RPRnoteEnumerations49	7216
Nuclear_yield_207 RPRnoteEnumerations49	7217
Nuclear_yield_208 RPRnoteEnumerations49	7218
Nuclear_yield_209 RPRnoteEnumerations49	7219
Nuclear_yield_210 RPRnoteEnumerations49	7220
Nuclear_yield_211 RPRnoteEnumerations49	7221
Nuclear_yield_212 RPRnoteEnumerations49	7222
Nuclear_yield_213 RPRnoteEnumerations49	7223
Nuclear_yield_214 RPRnoteEnumerations49	7224
Nuclear_yield_215 RPRnoteEnumerations49	7225
Nuclear_yield_216 RPRnoteEnumerations49	7226
Nuclear_yield_217 RPRnoteEnumerations49	7227
Nuclear_yield_218 RPRnoteEnumerations49	7228
Nuclear_yield_219 RPRnoteEnumerations49	7229
Nuclear_yield_220 RPRnoteEnumerations49	7230
Nuclear_yield_221 RPRnoteEnumerations49	7231
Nuclear_yield_222 RPRnoteEnumerations49	7232
Nuclear_yield_223 RPRnoteEnumerations49	7233
Nuclear_yield_224 RPRnoteEnumerations49	7234
Nuclear_yield_225 RPRnoteEnumerations49	7235
Nuclear_yield_226 RPRnoteEnumerations49	7236
Nuclear_yield_227 RPRnoteEnumerations49	7237
Nuclear_yield_228 RPRnoteEnumerations49	7238
Nuclear_yield_229 RPRnoteEnumerations49	7239

Enumerator	Value
Nuclear_yield_230 RPRnoteEnumerations49	7240
Nuclear_yield_231 RPRnoteEnumerations49	7241
Nuclear_yield_232 RPRnoteEnumerations49	7242
Nuclear_yield_233 RPRnoteEnumerations49	7243
Nuclear_yield_234 RPRnoteEnumerations49	7244
Nuclear_yield_235 RPRnoteEnumerations49	7245
Nuclear_yield_236 RPRnoteEnumerations49	7246
Nuclear_yield_237 RPRnoteEnumerations49	7247
Nuclear_yield_238 RPRnoteEnumerations49	7248
Nuclear_yield_239 RPRnoteEnumerations49	7249
Nuclear_yield_240 RPRnoteEnumerations49	7250
Nuclear_yield_241 RPRnoteEnumerations49	7251
Nuclear_yield_242 RPRnoteEnumerations49	7252
Nuclear_yield_243 RPRnoteEnumerations49	7253
Nuclear_yield_244 RPRnoteEnumerations49	7254
Nuclear_yield_245 RPRnoteEnumerations49	7255
Nuclear_yield_246 RPRnoteEnumerations49	7256
Nuclear_yield_247 RPRnoteEnumerations49	7257
Nuclear_yield_248 RPRnoteEnumerations49	7258
Nuclear_yield_249 RPRnoteEnumerations49	7259
Nuclear_yield_250 RPRnoteEnumerations49	7260
Nuclear_yield_251 RPRnoteEnumerations49	7261
Nuclear_yield_252 RPRnoteEnumerations49	7262
Nuclear_yield_253 RPRnoteEnumerations49	7263
Nuclear_yield_254 RPRnoteEnumerations49	7264
Nuclear_yield_255 RPRnoteEnumerations49	7265
Nuclear_yield_256 RPRnoteEnumerations49	7266
Nuclear_yield_257 RPRnoteEnumerations49	7267

Enumerator	Value
Nuclear_yield_258 RPRnoteEnumerations49	7268
Nuclear_yield_259 RPRnoteEnumerations49	7269
Nuclear_yield_260 RPRnoteEnumerations49	7270
Nuclear_yield_261 RPRnoteEnumerations49	7271
Nuclear_yield_262 RPRnoteEnumerations49	7272
Nuclear_yield_263 RPRnoteEnumerations49	7273
Nuclear_yield_264 RPRnoteEnumerations49	7274
Nuclear_yield_265 RPRnoteEnumerations49	7275
Nuclear_yield_266 RPRnoteEnumerations49	7276
Nuclear_yield_267 RPRnoteEnumerations49	7277
Nuclear_yield_268 RPRnoteEnumerations49	7278
Nuclear_yield_269 RPRnoteEnumerations49	7279
Nuclear_yield_270 RPRnoteEnumerations49	7280
Nuclear_yield_271 RPRnoteEnumerations49	7281
Nuclear_yield_272 RPRnoteEnumerations49	7282
Nuclear_yield_273 RPRnoteEnumerations49	7283
Nuclear_yield_274 RPRnoteEnumerations49	7284
Nuclear_yield_275 RPRnoteEnumerations49	7285
Nuclear_yield_276 RPRnoteEnumerations49	7286
Nuclear_yield_277 RPRnoteEnumerations49	7287
Nuclear_yield_278 RPRnoteEnumerations49	7288
Nuclear_yield_279 RPRnoteEnumerations49	7289
Nuclear_yield_280 RPRnoteEnumerations49	7290
Nuclear_yield_281 RPRnoteEnumerations49	7291
Nuclear_yield_282 RPRnoteEnumerations49	7292
Nuclear_yield_283 RPRnoteEnumerations49	7293
Nuclear_yield_284 RPRnoteEnumerations49	7294
Nuclear_yield_285 RPRnoteEnumerations49	7295

Enumerator	Value
Nuclear_yield_286 RPRnoteEnumerations49	7296
Nuclear_yield_287 RPRnoteEnumerations49	7297
Nuclear_yield_288 RPRnoteEnumerations49	7298
Nuclear_yield_289 RPRnoteEnumerations49	7299
Nuclear_yield_290 RPRnoteEnumerations49	7300
Nuclear_yield_291 RPRnoteEnumerations49	7301
Nuclear_yield_292 RPRnoteEnumerations49	7302
Nuclear_yield_293 RPRnoteEnumerations49	7303
Nuclear_yield_294 RPRnoteEnumerations49	7304
Nuclear_yield_295 RPRnoteEnumerations49	7305
Nuclear_yield_296 RPRnoteEnumerations49	7306
Nuclear_yield_297 RPRnoteEnumerations49	7307
Nuclear_yield_298 RPRnoteEnumerations49	7308
Nuclear_yield_299 RPRnoteEnumerations49	7309
Nuclear_yield_300 RPRnoteEnumerations49	7310
Nuclear_yield_301 RPRnoteEnumerations49	7311
Nuclear_yield_302 RPRnoteEnumerations49	7312
Nuclear_yield_303 RPRnoteEnumerations49	7313
Nuclear_yield_304 RPRnoteEnumerations49	7314
Nuclear_yield_305 RPRnoteEnumerations49	7315
Nuclear_yield_306 RPRnoteEnumerations49	7316
Nuclear_yield_307 RPRnoteEnumerations49	7317
Nuclear_yield_308 RPRnoteEnumerations49	7318
Nuclear_yield_309 RPRnoteEnumerations49	7319
Nuclear_yield_310 RPRnoteEnumerations49	7320
Nuclear_yield_311 RPRnoteEnumerations49	7321
Nuclear_yield_312 RPRnoteEnumerations49	7322
Nuclear_yield_313 RPRnoteEnumerations49	7323

Enumerator	Value
Nuclear_yield_314 RPRnoteEnumerations49	7324
Nuclear_yield_315 RPRnoteEnumerations49	7325
Nuclear_yield_316 RPRnoteEnumerations49	7326
Nuclear_yield_317 RPRnoteEnumerations49	7327
Nuclear_yield_318 RPRnoteEnumerations49	7328
Nuclear_yield_319 RPRnoteEnumerations49	7329
Nuclear_yield_320 RPRnoteEnumerations49	7330
Nuclear_yield_321 RPRnoteEnumerations49	7331
Nuclear_yield_322 RPRnoteEnumerations49	7332
Nuclear_yield_323 RPRnoteEnumerations49	7333
Nuclear_yield_324 RPRnoteEnumerations49	7334
Nuclear_yield_325 RPRnoteEnumerations49	7335
Nuclear_yield_326 RPRnoteEnumerations49	7336
Nuclear_yield_327 RPRnoteEnumerations49	7337
Nuclear_yield_328 RPRnoteEnumerations49	7338
Nuclear_yield_329 RPRnoteEnumerations49	7339
Nuclear_yield_330 RPRnoteEnumerations49	7340
Nuclear_yield_331 RPRnoteEnumerations49	7341
Nuclear_yield_332 RPRnoteEnumerations49	7342
Nuclear_yield_333 RPRnoteEnumerations49	7343
Nuclear_yield_334 RPRnoteEnumerations49	7344
Nuclear_yield_335 RPRnoteEnumerations49	7345
Nuclear_yield_336 RPRnoteEnumerations49	7346
Nuclear_yield_337 RPRnoteEnumerations49	7347
Nuclear_yield_338 RPRnoteEnumerations49	7348
Nuclear_yield_339 RPRnoteEnumerations49	7349
Nuclear_yield_340 RPRnoteEnumerations49	7350
Nuclear_yield_341 RPRnoteEnumerations49	7351

Enumerator	Value
Nuclear_yield_342 RPRnoteEnumerations49	7352
Nuclear_yield_343 RPRnoteEnumerations49	7353
Nuclear_yield_344 RPRnoteEnumerations49	7354
Nuclear_yield_345 RPRnoteEnumerations49	7355
Nuclear_yield_346 RPRnoteEnumerations49	7356
Nuclear_yield_347 RPRnoteEnumerations49	7357
Nuclear_yield_348 RPRnoteEnumerations49	7358
Nuclear_yield_349 RPRnoteEnumerations49	7359
Nuclear_yield_350 RPRnoteEnumerations49	7360
Nuclear_yield_351 RPRnoteEnumerations49	7361
Nuclear_yield_352 RPRnoteEnumerations49	7362
Nuclear_yield_353 RPRnoteEnumerations49	7363
Nuclear_yield_354 RPRnoteEnumerations49	7364
Nuclear_yield_355 RPRnoteEnumerations49	7365
Nuclear_yield_356 RPRnoteEnumerations49	7366
Nuclear_yield_357 RPRnoteEnumerations49	7367
Nuclear_yield_358 RPRnoteEnumerations49	7368
Nuclear_yield_359 RPRnoteEnumerations49	7369
Nuclear_yield_360 RPRnoteEnumerations49	7370
Nuclear_yield_361 RPRnoteEnumerations49	7371
Nuclear_yield_362 RPRnoteEnumerations49	7372
Nuclear_yield_363 RPRnoteEnumerations49	7373
Nuclear_yield_364 RPRnoteEnumerations49	7374
Nuclear_yield_365 RPRnoteEnumerations49	7375
Nuclear_yield_366 RPRnoteEnumerations49	7376
Nuclear_yield_367 RPRnoteEnumerations49	7377
Nuclear_yield_368 RPRnoteEnumerations49	7378
Nuclear_yield_369 RPRnoteEnumerations49	7379

Enumerator	Value
Nuclear_yield_370 RPRnoteEnumerations49	7380
Nuclear_yield_371 RPRnoteEnumerations49	7381
Nuclear_yield_372 RPRnoteEnumerations49	7382
Nuclear_yield_373 RPRnoteEnumerations49	7383
Nuclear_yield_374 RPRnoteEnumerations49	7384
Nuclear_yield_375 RPRnoteEnumerations49	7385
Nuclear_yield_376 RPRnoteEnumerations49	7386
Nuclear_yield_377 RPRnoteEnumerations49	7387
Nuclear_yield_378 RPRnoteEnumerations49	7388
Nuclear_yield_379 RPRnoteEnumerations49	7389
Nuclear_yield_380 RPRnoteEnumerations49	7390
Nuclear_yield_381 RPRnoteEnumerations49	7391
Nuclear_yield_382 RPRnoteEnumerations49	7392
Nuclear_yield_383 RPRnoteEnumerations49	7393
Nuclear_yield_384 RPRnoteEnumerations49	7394
Nuclear_yield_385 RPRnoteEnumerations49	7395
Nuclear_yield_386 RPRnoteEnumerations49	7396
Nuclear_yield_387 RPRnoteEnumerations49	7397
Nuclear_yield_388 RPRnoteEnumerations49	7398
Nuclear_yield_389 RPRnoteEnumerations49	7399
Nuclear_yield_390 RPRnoteEnumerations49	7400
Nuclear_yield_391 RPRnoteEnumerations49	7401
Nuclear_yield_392 RPRnoteEnumerations49	7402
Nuclear_yield_393 RPRnoteEnumerations49	7403
Nuclear_yield_394 RPRnoteEnumerations49	7404
Nuclear_yield_395 RPRnoteEnumerations49	7405
Nuclear_yield_396 RPRnoteEnumerations49	7406
Nuclear_yield_397 RPRnoteEnumerations49	7407

Enumerator	Value
Nuclear_yield_398 RPRnoteEnumerations49	7408
Nuclear_yield_399 RPRnoteEnumerations49	7409
Nuclear_yield_400 RPRnoteEnumerations49	7410
Nuclear_yield_401 RPRnoteEnumerations49	7411
Nuclear_yield_402 RPRnoteEnumerations49	7412
Nuclear_yield_403 RPRnoteEnumerations49	7413
Nuclear_yield_404 RPRnoteEnumerations49	7414
Nuclear_yield_405 RPRnoteEnumerations49	7415
Nuclear_yield_406 RPRnoteEnumerations49	7416
Nuclear_yield_407 RPRnoteEnumerations49	7417
Nuclear_yield_408 RPRnoteEnumerations49	7418
Nuclear_yield_409 RPRnoteEnumerations49	7419
Nuclear_yield_410 RPRnoteEnumerations49	7420
Nuclear_yield_411 RPRnoteEnumerations49	7421
Nuclear_yield_412 RPRnoteEnumerations49	7422
Nuclear_yield_413 RPRnoteEnumerations49	7423
Nuclear_yield_414 RPRnoteEnumerations49	7424
Nuclear_yield_415 RPRnoteEnumerations49	7425
Nuclear_yield_416 RPRnoteEnumerations49	7426
Nuclear_yield_417 RPRnoteEnumerations49	7427
Nuclear_yield_418 RPRnoteEnumerations49	7428
Nuclear_yield_419 RPRnoteEnumerations49	7429
Nuclear_yield_420 RPRnoteEnumerations49	7430
Nuclear_yield_421 RPRnoteEnumerations49	7431
Nuclear_yield_422 RPRnoteEnumerations49	7432
Nuclear_yield_423 RPRnoteEnumerations49	7433
Nuclear_yield_424 RPRnoteEnumerations49	7434
Nuclear_yield_425 RPRnoteEnumerations49	7435

Enumerator	Value
Nuclear_yield_426 RPRnoteEnumerations49	7436
Nuclear_yield_427 RPRnoteEnumerations49	7437
Nuclear_yield_428 RPRnoteEnumerations49	7438
Nuclear_yield_429 RPRnoteEnumerations49	7439
Nuclear_yield_430 RPRnoteEnumerations49	7440
Nuclear_yield_431 RPRnoteEnumerations49	7441
Nuclear_yield_432 RPRnoteEnumerations49	7442
Nuclear_yield_433 RPRnoteEnumerations49	7443
Nuclear_yield_434 RPRnoteEnumerations49	7444
Nuclear_yield_435 RPRnoteEnumerations49	7445
Nuclear_yield_436 RPRnoteEnumerations49	7446
Nuclear_yield_437 RPRnoteEnumerations49	7447
Nuclear_yield_438 RPRnoteEnumerations49	7448
Nuclear_yield_439 RPRnoteEnumerations49	7449
Nuclear_yield_440 RPRnoteEnumerations49	7450
Nuclear_yield_441 RPRnoteEnumerations49	7451
Nuclear_yield_442 RPRnoteEnumerations49	7452
Nuclear_yield_443 RPRnoteEnumerations49	7453
Nuclear_yield_444 RPRnoteEnumerations49	7454
Nuclear_yield_445 RPRnoteEnumerations49	7455
Nuclear_yield_446 RPRnoteEnumerations49	7456
Nuclear_yield_447 RPRnoteEnumerations49	7457
Nuclear_yield_448 RPRnoteEnumerations49	7458
Nuclear_yield_449 RPRnoteEnumerations49	7459
Nuclear_yield_450 RPRnoteEnumerations49	7460
Nuclear_yield_451 RPRnoteEnumerations49	7461
Nuclear_yield_452 RPRnoteEnumerations49	7462
Nuclear_yield_453 RPRnoteEnumerations49	7463

Enumerator	Value
Nuclear_yield_454 RPRnoteEnumerations49	7464
Nuclear_yield_455 RPRnoteEnumerations49	7465
Nuclear_yield_456 RPRnoteEnumerations49	7466
Nuclear_yield_457 RPRnoteEnumerations49	7467
Nuclear_yield_458 RPRnoteEnumerations49	7468
Nuclear_yield_459 RPRnoteEnumerations49	7469
Nuclear_yield_460 RPRnoteEnumerations49	7470
Nuclear_yield_461 RPRnoteEnumerations49	7471
Nuclear_yield_462 RPRnoteEnumerations49	7472
Nuclear_yield_463 RPRnoteEnumerations49	7473
Nuclear_yield_464 RPRnoteEnumerations49	7474
Nuclear_yield_465 RPRnoteEnumerations49	7475
Nuclear_yield_466 RPRnoteEnumerations49	7476
Nuclear_yield_467 RPRnoteEnumerations49	7477
Nuclear_yield_468 RPRnoteEnumerations49	7478
Nuclear_yield_469 RPRnoteEnumerations49	7479
Nuclear_yield_470 RPRnoteEnumerations49	7480
Nuclear_yield_471 RPRnoteEnumerations49	7481
Nuclear_yield_472 RPRnoteEnumerations49	7482
Nuclear_yield_473 RPRnoteEnumerations49	7483
Nuclear_yield_474 RPRnoteEnumerations49	7484
Nuclear_yield_475 RPRnoteEnumerations49	7485
Nuclear_yield_476 RPRnoteEnumerations49	7486
Nuclear_yield_477 RPRnoteEnumerations49	7487
Nuclear_yield_478 RPRnoteEnumerations49	7488
Nuclear_yield_479 RPRnoteEnumerations49	7489
Nuclear_yield_480 RPRnoteEnumerations49	7490
Nuclear_yield_481 RPRnoteEnumerations49	7491

Enumerator	Value
Nuclear_yield_482 RPRnoteEnumerations49	7492
Nuclear_yield_483 RPRnoteEnumerations49	7493
Nuclear_yield_484 RPRnoteEnumerations49	7494
Nuclear_yield_485 RPRnoteEnumerations49	7495
Nuclear_yield_486 RPRnoteEnumerations49	7496
Nuclear_yield_487 RPRnoteEnumerations49	7497
Nuclear_yield_488 RPRnoteEnumerations49	7498
Nuclear_yield_489 RPRnoteEnumerations49	7499
Nuclear_yield_490 RPRnoteEnumerations49	7500
Nuclear_yield_491 RPRnoteEnumerations49	7501
Nuclear_yield_492 RPRnoteEnumerations49	7502
Nuclear_yield_493 RPRnoteEnumerations49	7503
Nuclear_yield_494 RPRnoteEnumerations49	7504
Nuclear_yield_495 RPRnoteEnumerations49	7505
Nuclear_yield_496 RPRnoteEnumerations49	7506
Nuclear_yield_497 RPRnoteEnumerations49	7507
Nuclear_yield_498 RPRnoteEnumerations49	7508
Nuclear_yield_499 RPRnoteEnumerations49	7509
Nuclear_yield_500 RPRnoteEnumerations49	7510
Nuclear_yield_501 RPRnoteEnumerations49	7511
Nuclear_yield_502 RPRnoteEnumerations49	7512
Nuclear_yield_503 RPRnoteEnumerations49	7513
Nuclear_yield_504 RPRnoteEnumerations49	7514
Nuclear_yield_505 RPRnoteEnumerations49	7515
Nuclear_yield_506 RPRnoteEnumerations49	7516
Nuclear_yield_507 RPRnoteEnumerations49	7517
Nuclear_yield_508 RPRnoteEnumerations49	7518
Nuclear_yield_509 RPRnoteEnumerations49	7519

Enumerator	Value
Nuclear_yield_510 RPRnoteEnumerations49	7520
Nuclear_yield_511 RPRnoteEnumerations49	7521
Nuclear_yield_512 RPRnoteEnumerations49	7522
Nuclear_yield_513 RPRnoteEnumerations49	7523
Nuclear_yield_514 RPRnoteEnumerations49	7524
Nuclear_yield_515 RPRnoteEnumerations49	7525
Nuclear_yield_516 RPRnoteEnumerations49	7526
Nuclear_yield_517 RPRnoteEnumerations49	7527
Nuclear_yield_518 RPRnoteEnumerations49	7528
Nuclear_yield_519 RPRnoteEnumerations49	7529
Nuclear_yield_520 RPRnoteEnumerations49	7530
Nuclear_yield_521 RPRnoteEnumerations49	7531
Nuclear_yield_522 RPRnoteEnumerations49	7532
Nuclear_yield_523 RPRnoteEnumerations49	7533
Nuclear_yield_524 RPRnoteEnumerations49	7534
Nuclear_yield_525 RPRnoteEnumerations49	7535
Nuclear_yield_526 RPRnoteEnumerations49	7536
Nuclear_yield_527 RPRnoteEnumerations49	7537
Nuclear_yield_528 RPRnoteEnumerations49	7538
Nuclear_yield_529 RPRnoteEnumerations49	7539
Nuclear_yield_530 RPRnoteEnumerations49	7540
Nuclear_yield_531 RPRnoteEnumerations49	7541
Nuclear_yield_532 RPRnoteEnumerations49	7542
Nuclear_yield_533 RPRnoteEnumerations49	7543
Nuclear_yield_534 RPRnoteEnumerations49	7544
Nuclear_yield_535 RPRnoteEnumerations49	7545
Nuclear_yield_536 RPRnoteEnumerations49	7546
Nuclear_yield_537 RPRnoteEnumerations49	7547

Enumerator	Value
Nuclear_yield_538 RPRnoteEnumerations49	7548
Nuclear_yield_539 RPRnoteEnumerations49	7549
Nuclear_yield_540 RPRnoteEnumerations49	7550
Nuclear_yield_541 RPRnoteEnumerations49	7551
Nuclear_yield_542 RPRnoteEnumerations49	7552
Nuclear_yield_543 RPRnoteEnumerations49	7553
Nuclear_yield_544 RPRnoteEnumerations49	7554
Nuclear_yield_545 RPRnoteEnumerations49	7555
Nuclear_yield_546 RPRnoteEnumerations49	7556
Nuclear_yield_547 RPRnoteEnumerations49	7557
Nuclear_yield_548 RPRnoteEnumerations49	7558
Nuclear_yield_549 RPRnoteEnumerations49	7559
Nuclear_yield_550 RPRnoteEnumerations49	7560
Nuclear_yield_551 RPRnoteEnumerations49	7561
Nuclear_yield_552 RPRnoteEnumerations49	7562
Nuclear_yield_553 RPRnoteEnumerations49	7563
Nuclear_yield_554 RPRnoteEnumerations49	7564
Nuclear_yield_555 RPRnoteEnumerations49	7565
Nuclear_yield_556 RPRnoteEnumerations49	7566
Nuclear_yield_557 RPRnoteEnumerations49	7567
Nuclear_yield_558 RPRnoteEnumerations49	7568
Nuclear_yield_559 RPRnoteEnumerations49	7569
Nuclear_yield_560 RPRnoteEnumerations49	7570
Nuclear_yield_561 RPRnoteEnumerations49	7571
Nuclear_yield_562 RPRnoteEnumerations49	7572
Nuclear_yield_563 RPRnoteEnumerations49	7573
Nuclear_yield_564 RPRnoteEnumerations49	7574
Nuclear_yield_565 RPRnoteEnumerations49	7575

Enumerator	Value
Nuclear_yield_566 RPRnoteEnumerations49	7576
Nuclear_yield_567 RPRnoteEnumerations49	7577
Nuclear_yield_568 RPRnoteEnumerations49	7578
Nuclear_yield_569 RPRnoteEnumerations49	7579
Nuclear_yield_570 RPRnoteEnumerations49	7580
Nuclear_yield_571 RPRnoteEnumerations49	7581
Nuclear_yield_572 RPRnoteEnumerations49	7582
Nuclear_yield_573 RPRnoteEnumerations49	7583
Nuclear_yield_574 RPRnoteEnumerations49	7584
Nuclear_yield_575 RPRnoteEnumerations49	7585
Nuclear_yield_576 RPRnoteEnumerations49	7586
Nuclear_yield_577 RPRnoteEnumerations49	7587
Nuclear_yield_578 RPRnoteEnumerations49	7588
Nuclear_yield_579 RPRnoteEnumerations49	7589
Nuclear_yield_580 RPRnoteEnumerations49	7590
Nuclear_yield_581 RPRnoteEnumerations49	7591
Nuclear_yield_582 RPRnoteEnumerations49	7592
Nuclear_yield_583 RPRnoteEnumerations49	7593
Nuclear_yield_584 RPRnoteEnumerations49	7594
Nuclear_yield_585 RPRnoteEnumerations49	7595
Nuclear_yield_586 RPRnoteEnumerations49	7596
Nuclear_yield_587 RPRnoteEnumerations49	7597
Nuclear_yield_588 RPRnoteEnumerations49	7598
Nuclear_yield_589 RPRnoteEnumerations49	7599
Nuclear_yield_590 RPRnoteEnumerations49	7600
Nuclear_yield_591 RPRnoteEnumerations49	7601
Nuclear_yield_592 RPRnoteEnumerations49	7602
Nuclear_yield_593 RPRnoteEnumerations49	7603

Enumerator	Value
Nuclear_yield_594 RPRnoteEnumerations49	7604
Nuclear_yield_595 RPRnoteEnumerations49	7605
Nuclear_yield_596 RPRnoteEnumerations49	7606
Nuclear_yield_597 RPRnoteEnumerations49	7607
Nuclear_yield_598 RPRnoteEnumerations49	7608
Nuclear_yield_599 RPRnoteEnumerations49	7609
Nuclear_yield_600 RPRnoteEnumerations49	7610
Nuclear_yield_601 RPRnoteEnumerations49	7611
Nuclear_yield_602 RPRnoteEnumerations49	7612
Nuclear_yield_603 RPRnoteEnumerations49	7613
Nuclear_yield_604 RPRnoteEnumerations49	7614
Nuclear_yield_605 RPRnoteEnumerations49	7615
Nuclear_yield_606 RPRnoteEnumerations49	7616
Nuclear_yield_607 RPRnoteEnumerations49	7617
Nuclear_yield_608 RPRnoteEnumerations49	7618
Nuclear_yield_609 RPRnoteEnumerations49	7619
Nuclear_yield_610 RPRnoteEnumerations49	7620
Nuclear_yield_611 RPRnoteEnumerations49	7621
Nuclear_yield_612 RPRnoteEnumerations49	7622
Nuclear_yield_613 RPRnoteEnumerations49	7623
Nuclear_yield_614 RPRnoteEnumerations49	7624
Nuclear_yield_615 RPRnoteEnumerations49	7625
Nuclear_yield_616 RPRnoteEnumerations49	7626
Nuclear_yield_617 RPRnoteEnumerations49	7627
Nuclear_yield_618 RPRnoteEnumerations49	7628
Nuclear_yield_619 RPRnoteEnumerations49	7629
Nuclear_yield_620 RPRnoteEnumerations49	7630
Nuclear_yield_621 RPRnoteEnumerations49	7631

Enumerator	Value
Nuclear_yield_622 RPRnoteEnumerations49	7632
Nuclear_yield_623 RPRnoteEnumerations49	7633
Nuclear_yield_624 RPRnoteEnumerations49	7634
Nuclear_yield_625 RPRnoteEnumerations49	7635
Nuclear_yield_626 RPRnoteEnumerations49	7636
Nuclear_yield_627 RPRnoteEnumerations49	7637
Nuclear_yield_628 RPRnoteEnumerations49	7638
Nuclear_yield_629 RPRnoteEnumerations49	7639
Nuclear_yield_630 RPRnoteEnumerations49	7640
Nuclear_yield_631 RPRnoteEnumerations49	7641
Nuclear_yield_632 RPRnoteEnumerations49	7642
Nuclear_yield_633 RPRnoteEnumerations49	7643
Nuclear_yield_634 RPRnoteEnumerations49	7644
Nuclear_yield_635 RPRnoteEnumerations49	7645
Nuclear_yield_636 RPRnoteEnumerations49	7646
Nuclear_yield_637 RPRnoteEnumerations49	7647
Nuclear_yield_638 RPRnoteEnumerations49	7648
Nuclear_yield_639 RPRnoteEnumerations49	7649
Nuclear_yield_640 RPRnoteEnumerations49	7650
Nuclear_yield_641 RPRnoteEnumerations49	7651
Nuclear_yield_642 RPRnoteEnumerations49	7652
Nuclear_yield_643 RPRnoteEnumerations49	7653
Nuclear_yield_644 RPRnoteEnumerations49	7654
Nuclear_yield_645 RPRnoteEnumerations49	7655
Nuclear_yield_646 RPRnoteEnumerations49	7656
Nuclear_yield_647 RPRnoteEnumerations49	7657
Nuclear_yield_648 RPRnoteEnumerations49	7658
Nuclear_yield_649 RPRnoteEnumerations49	7659

Enumerator	Value
Nuclear_yield_650 RPRnoteEnumerations49	7660
Nuclear_yield_651 RPRnoteEnumerations49	7661
Nuclear_yield_652 RPRnoteEnumerations49	7662
Nuclear_yield_653 RPRnoteEnumerations49	7663
Nuclear_yield_654 RPRnoteEnumerations49	7664
Nuclear_yield_655 RPRnoteEnumerations49	7665
Nuclear_yield_656 RPRnoteEnumerations49	7666
Nuclear_yield_657 RPRnoteEnumerations49	7667
Nuclear_yield_658 RPRnoteEnumerations49	7668
Nuclear_yield_659 RPRnoteEnumerations49	7669
Nuclear_yield_660 RPRnoteEnumerations49	7670
Nuclear_yield_661 RPRnoteEnumerations49	7671
Nuclear_yield_662 RPRnoteEnumerations49	7672
Nuclear_yield_663 RPRnoteEnumerations49	7673
Nuclear_yield_664 RPRnoteEnumerations49	7674
Nuclear_yield_665 RPRnoteEnumerations49	7675
Nuclear_yield_666 RPRnoteEnumerations49	7676
Nuclear_yield_667 RPRnoteEnumerations49	7677
Nuclear_yield_668 RPRnoteEnumerations49	7678
Nuclear_yield_669 RPRnoteEnumerations49	7679
Nuclear_yield_670 RPRnoteEnumerations49	7680
Nuclear_yield_671 RPRnoteEnumerations49	7681
Nuclear_yield_672 RPRnoteEnumerations49	7682
Nuclear_yield_673 RPRnoteEnumerations49	7683
Nuclear_yield_674 RPRnoteEnumerations49	7684
Nuclear_yield_675 RPRnoteEnumerations49	7685
Nuclear_yield_676 RPRnoteEnumerations49	7686
Nuclear_yield_677 RPRnoteEnumerations49	7687

Enumerator	Value
Nuclear_yield_678 RPRnoteEnumerations49	7688
Nuclear_yield_679 RPRnoteEnumerations49	7689
Nuclear_yield_680 RPRnoteEnumerations49	7690
Nuclear_yield_681 RPRnoteEnumerations49	7691
Nuclear_yield_682 RPRnoteEnumerations49	7692
Nuclear_yield_683 RPRnoteEnumerations49	7693
Nuclear_yield_684 RPRnoteEnumerations49	7694
Nuclear_yield_685 RPRnoteEnumerations49	7695
Nuclear_yield_686 RPRnoteEnumerations49	7696
Nuclear_yield_687 RPRnoteEnumerations49	7697
Nuclear_yield_688 RPRnoteEnumerations49	7698
Nuclear_yield_689 RPRnoteEnumerations49	7699
Nuclear_yield_690 RPRnoteEnumerations49	7700
Nuclear_yield_691 RPRnoteEnumerations49	7701
Nuclear_yield_692 RPRnoteEnumerations49	7702
Nuclear_yield_693 RPRnoteEnumerations49	7703
Nuclear_yield_694 RPRnoteEnumerations49	7704
Nuclear_yield_695 RPRnoteEnumerations49	7705
Nuclear_yield_696 RPRnoteEnumerations49	7706
Nuclear_yield_697 RPRnoteEnumerations49	7707
Nuclear_yield_698 RPRnoteEnumerations49	7708
Nuclear_yield_699 RPRnoteEnumerations49	7709
Nuclear_yield_700 RPRnoteEnumerations49	7710
Nuclear_yield_701 RPRnoteEnumerations49	7711
Nuclear_yield_702 RPRnoteEnumerations49	7712
Nuclear_yield_703 RPRnoteEnumerations49	7713
Nuclear_yield_704 RPRnoteEnumerations49	7714
Nuclear_yield_705 RPRnoteEnumerations49	7715

Enumerator	Value
Nuclear_yield_706 RPRnoteEnumerations49	7716
Nuclear_yield_707 RPRnoteEnumerations49	7717
Nuclear_yield_708 RPRnoteEnumerations49	7718
Nuclear_yield_709 RPRnoteEnumerations49	7719
Nuclear_yield_710 RPRnoteEnumerations49	7720
Nuclear_yield_711 RPRnoteEnumerations49	7721
Nuclear_yield_712 RPRnoteEnumerations49	7722
Nuclear_yield_713 RPRnoteEnumerations49	7723
Nuclear_yield_714 RPRnoteEnumerations49	7724
Nuclear_yield_715 RPRnoteEnumerations49	7725
Nuclear_yield_716 RPRnoteEnumerations49	7726
Nuclear_yield_717 RPRnoteEnumerations49	7727
Nuclear_yield_718 RPRnoteEnumerations49	7728
Nuclear_yield_719 RPRnoteEnumerations49	7729
Nuclear_yield_720 RPRnoteEnumerations49	7730
Nuclear_yield_721 RPRnoteEnumerations49	7731
Nuclear_yield_722 RPRnoteEnumerations49	7732
Nuclear_yield_723 RPRnoteEnumerations49	7733
Nuclear_yield_724 RPRnoteEnumerations49	7734
Nuclear_yield_725 RPRnoteEnumerations49	7735
Nuclear_yield_726 RPRnoteEnumerations49	7736
Nuclear_yield_727 RPRnoteEnumerations49	7737
Nuclear_yield_728 RPRnoteEnumerations49	7738
Nuclear_yield_729 RPRnoteEnumerations49	7739
Nuclear_yield_730 RPRnoteEnumerations49	7740
Nuclear_yield_731 RPRnoteEnumerations49	7741
Nuclear_yield_732 RPRnoteEnumerations49	7742
Nuclear_yield_733 RPRnoteEnumerations49	7743

Enumerator	Value
Nuclear_yield_734 RPRnoteEnumerations49	7744
Nuclear_yield_735 RPRnoteEnumerations49	7745
Nuclear_yield_736 RPRnoteEnumerations49	7746
Nuclear_yield_737 RPRnoteEnumerations49	7747
Nuclear_yield_738 RPRnoteEnumerations49	7748
Nuclear_yield_739 RPRnoteEnumerations49	7749
Nuclear_yield_740 RPRnoteEnumerations49	7750
Nuclear_yield_741 RPRnoteEnumerations49	7751
Nuclear_yield_742 RPRnoteEnumerations49	7752
Nuclear_yield_743 RPRnoteEnumerations49	7753
Nuclear_yield_744 RPRnoteEnumerations49	7754
Nuclear_yield_745 RPRnoteEnumerations49	7755
Nuclear_yield_746 RPRnoteEnumerations49	7756
Nuclear_yield_747 RPRnoteEnumerations49	7757
Nuclear_yield_748 RPRnoteEnumerations49	7758
Nuclear_yield_749 RPRnoteEnumerations49	7759
Nuclear_yield_750 RPRnoteEnumerations49	7760
Nuclear_yield_751 RPRnoteEnumerations49	7761
Nuclear_yield_752 RPRnoteEnumerations49	7762
Nuclear_yield_753 RPRnoteEnumerations49	7763
Nuclear_yield_754 RPRnoteEnumerations49	7764
Nuclear_yield_755 RPRnoteEnumerations49	7765
Nuclear_yield_756 RPRnoteEnumerations49	7766
Nuclear_yield_757 RPRnoteEnumerations49	7767
Nuclear_yield_758 RPRnoteEnumerations49	7768
Nuclear_yield_759 RPRnoteEnumerations49	7769
Nuclear_yield_760 RPRnoteEnumerations49	7770
Nuclear_yield_761 RPRnoteEnumerations49	7771

Enumerator	Value
Nuclear_yield_762 RPRnoteEnumerations49	7772
Nuclear_yield_763 RPRnoteEnumerations49	7773
Nuclear_yield_764 RPRnoteEnumerations49	7774
Nuclear_yield_765 RPRnoteEnumerations49	7775
Nuclear_yield_766 RPRnoteEnumerations49	7776
Nuclear_yield_767 RPRnoteEnumerations49	7777
Nuclear_yield_768 RPRnoteEnumerations49	7778
Nuclear_yield_769 RPRnoteEnumerations49	7779
Nuclear_yield_770 RPRnoteEnumerations49	7780
Nuclear_yield_771 RPRnoteEnumerations49	7781
Nuclear_yield_772 RPRnoteEnumerations49	7782
Nuclear_yield_773 RPRnoteEnumerations49	7783
Nuclear_yield_774 RPRnoteEnumerations49	7784
Nuclear_yield_775 RPRnoteEnumerations49	7785
Nuclear_yield_776 RPRnoteEnumerations49	7786
Nuclear_yield_777 RPRnoteEnumerations49	7787
Nuclear_yield_778 RPRnoteEnumerations49	7788
Nuclear_yield_779 RPRnoteEnumerations49	7789
Nuclear_yield_780 RPRnoteEnumerations49	7790
Nuclear_yield_781 RPRnoteEnumerations49	7791
Nuclear_yield_782 RPRnoteEnumerations49	7792
Nuclear_yield_783 RPRnoteEnumerations49	7793
Nuclear_yield_784 RPRnoteEnumerations49	7794
Nuclear_yield_785 RPRnoteEnumerations49	7795
Nuclear_yield_786 RPRnoteEnumerations49	7796
Nuclear_yield_787 RPRnoteEnumerations49	7797
Nuclear_yield_788 RPRnoteEnumerations49	7798
Nuclear_yield_789 RPRnoteEnumerations49	7799

Enumerator	Value
Nuclear_yield_790 RPRnoteEnumerations49	7800
Nuclear_yield_791 RPRnoteEnumerations49	7801
Nuclear_yield_792 RPRnoteEnumerations49	7802
Nuclear_yield_793 RPRnoteEnumerations49	7803
Nuclear_yield_794 RPRnoteEnumerations49	7804
Nuclear_yield_795 RPRnoteEnumerations49	7805
Nuclear_yield_796 RPRnoteEnumerations49	7806
Nuclear_yield_797 RPRnoteEnumerations49	7807
Nuclear_yield_798 RPRnoteEnumerations49	7808
Nuclear_yield_799 RPRnoteEnumerations49	7809
Nuclear_yield_800 RPRnoteEnumerations49	7810
Nuclear_yield_801 RPRnoteEnumerations49	7811
Nuclear_yield_802 RPRnoteEnumerations49	7812
Nuclear_yield_803 RPRnoteEnumerations49	7813
Nuclear_yield_804 RPRnoteEnumerations49	7814
Nuclear_yield_805 RPRnoteEnumerations49	7815
Nuclear_yield_806 RPRnoteEnumerations49	7816
Nuclear_yield_807 RPRnoteEnumerations49	7817
Nuclear_yield_808 RPRnoteEnumerations49	7818
Nuclear_yield_809 RPRnoteEnumerations49	7819
Nuclear_yield_810 RPRnoteEnumerations49	7820
Nuclear_yield_811 RPRnoteEnumerations49	7821
Nuclear_yield_812 RPRnoteEnumerations49	7822
Nuclear_yield_813 RPRnoteEnumerations49	7823
Nuclear_yield_814 RPRnoteEnumerations49	7824
Nuclear_yield_815 RPRnoteEnumerations49	7825
Nuclear_yield_816 RPRnoteEnumerations49	7826
Nuclear_yield_817 RPRnoteEnumerations49	7827

Enumerator	Value
Nuclear_yield_818 RPRnoteEnumerations49	7828
Nuclear_yield_819 RPRnoteEnumerations49	7829
Nuclear_yield_820 RPRnoteEnumerations49	7830
Nuclear_yield_821 RPRnoteEnumerations49	7831
Nuclear_yield_822 RPRnoteEnumerations49	7832
Nuclear_yield_823 RPRnoteEnumerations49	7833
Nuclear_yield_824 RPRnoteEnumerations49	7834
Nuclear_yield_825 RPRnoteEnumerations49	7835
Nuclear_yield_826 RPRnoteEnumerations49	7836
Nuclear_yield_827 RPRnoteEnumerations49	7837
Nuclear_yield_828 RPRnoteEnumerations49	7838
Nuclear_yield_829 RPRnoteEnumerations49	7839
Nuclear_yield_830 RPRnoteEnumerations49	7840
Nuclear_yield_831 RPRnoteEnumerations49	7841
Nuclear_yield_832 RPRnoteEnumerations49	7842
Nuclear_yield_833 RPRnoteEnumerations49	7843
Nuclear_yield_834 RPRnoteEnumerations49	7844
Nuclear_yield_835 RPRnoteEnumerations49	7845
Nuclear_yield_836 RPRnoteEnumerations49	7846
Nuclear_yield_837 RPRnoteEnumerations49	7847
Nuclear_yield_838 RPRnoteEnumerations49	7848
Nuclear_yield_839 RPRnoteEnumerations49	7849
Nuclear_yield_840 RPRnoteEnumerations49	7850
Nuclear_yield_841 RPRnoteEnumerations49	7851
Nuclear_yield_842 RPRnoteEnumerations49	7852
Nuclear_yield_843 RPRnoteEnumerations49	7853
Nuclear_yield_844 RPRnoteEnumerations49	7854
Nuclear_yield_845 RPRnoteEnumerations49	7855

Enumerator	Value
Nuclear_yield_846 RPRnoteEnumerations49	7856
Nuclear_yield_847 RPRnoteEnumerations49	7857
Nuclear_yield_848 RPRnoteEnumerations49	7858
Nuclear_yield_849 RPRnoteEnumerations49	7859
Nuclear_yield_850 RPRnoteEnumerations49	7860
Nuclear_yield_851 RPRnoteEnumerations49	7861
Nuclear_yield_852 RPRnoteEnumerations49	7862
Nuclear_yield_853 RPRnoteEnumerations49	7863
Nuclear_yield_854 RPRnoteEnumerations49	7864
Nuclear_yield_855 RPRnoteEnumerations49	7865
Nuclear_yield_856 RPRnoteEnumerations49	7866
Nuclear_yield_857 RPRnoteEnumerations49	7867
Nuclear_yield_858 RPRnoteEnumerations49	7868
Nuclear_yield_859 RPRnoteEnumerations49	7869
Nuclear_yield_860 RPRnoteEnumerations49	7870
Nuclear_yield_861 RPRnoteEnumerations49	7871
Nuclear_yield_862 RPRnoteEnumerations49	7872
Nuclear_yield_863 RPRnoteEnumerations49	7873
Nuclear_yield_864 RPRnoteEnumerations49	7874
Nuclear_yield_865 RPRnoteEnumerations49	7875
Nuclear_yield_866 RPRnoteEnumerations49	7876
Nuclear_yield_867 RPRnoteEnumerations49	7877
Nuclear_yield_868 RPRnoteEnumerations49	7878
Nuclear_yield_869 RPRnoteEnumerations49	7879
Nuclear_yield_870 RPRnoteEnumerations49	7880
Nuclear_yield_871 RPRnoteEnumerations49	7881
Nuclear_yield_872 RPRnoteEnumerations49	7882
Nuclear_yield_873 RPRnoteEnumerations49	7883

Enumerator	Value
Nuclear_yield_874 RPRnoteEnumerations49	7884
Nuclear_yield_875 RPRnoteEnumerations49	7885
Nuclear_yield_876 RPRnoteEnumerations49	7886
Nuclear_yield_877 RPRnoteEnumerations49	7887
Nuclear_yield_878 RPRnoteEnumerations49	7888
Nuclear_yield_879 RPRnoteEnumerations49	7889
Nuclear_yield_880 RPRnoteEnumerations49	7890
Nuclear_yield_881 RPRnoteEnumerations49	7891
Nuclear_yield_882 RPRnoteEnumerations49	7892
Nuclear_yield_883 RPRnoteEnumerations49	7893
Nuclear_yield_884 RPRnoteEnumerations49	7894
Nuclear_yield_885 RPRnoteEnumerations49	7895
Nuclear_yield_886 RPRnoteEnumerations49	7896
Nuclear_yield_887 RPRnoteEnumerations49	7897
Nuclear_yield_888 RPRnoteEnumerations49	7898
Nuclear_yield_889 RPRnoteEnumerations49	7899
Nuclear_yield_890 RPRnoteEnumerations49	7900
Nuclear_yield_891 RPRnoteEnumerations49	7901
Nuclear_yield_892 RPRnoteEnumerations49	7902
Nuclear_yield_893 RPRnoteEnumerations49	7903
Nuclear_yield_894 RPRnoteEnumerations49	7904
Nuclear_yield_895 RPRnoteEnumerations49	7905
Nuclear_yield_896 RPRnoteEnumerations49	7906
Nuclear_yield_897 RPRnoteEnumerations49	7907
Nuclear_yield_898 RPRnoteEnumerations49	7908
Nuclear_yield_899 RPRnoteEnumerations49	7909
Nuclear_yield_900 RPRnoteEnumerations49	7910
Nuclear_yield_901 RPRnoteEnumerations49	7911

Enumerator	Value
Nuclear_yield_902 RPRnoteEnumerations49	7912
Nuclear_yield_903 RPRnoteEnumerations49	7913
Nuclear_yield_904 RPRnoteEnumerations49	7914
Nuclear_yield_905 RPRnoteEnumerations49	7915
Nuclear_yield_906 RPRnoteEnumerations49	7916
Nuclear_yield_907 RPRnoteEnumerations49	7917
Nuclear_yield_908 RPRnoteEnumerations49	7918
Nuclear_yield_909 RPRnoteEnumerations49	7919
Nuclear_yield_910 RPRnoteEnumerations49	7920
Nuclear_yield_911 RPRnoteEnumerations49	7921
Nuclear_yield_912 RPRnoteEnumerations49	7922
Nuclear_yield_913 RPRnoteEnumerations49	7923
Nuclear_yield_914 RPRnoteEnumerations49	7924
Nuclear_yield_915 RPRnoteEnumerations49	7925
Nuclear_yield_916 RPRnoteEnumerations49	7926
Nuclear_yield_917 RPRnoteEnumerations49	7927
Nuclear_yield_918 RPRnoteEnumerations49	7928
Nuclear_yield_919 RPRnoteEnumerations49	7929
Nuclear_yield_920 RPRnoteEnumerations49	7930
Nuclear_yield_921 RPRnoteEnumerations49	7931
Nuclear_yield_922 RPRnoteEnumerations49	7932
Nuclear_yield_923 RPRnoteEnumerations49	7933
Nuclear_yield_924 RPRnoteEnumerations49	7934
Nuclear_yield_925 RPRnoteEnumerations49	7935
Nuclear_yield_926 RPRnoteEnumerations49	7936
Nuclear_yield_927 RPRnoteEnumerations49	7937
Nuclear_yield_928 RPRnoteEnumerations49	7938
Nuclear_yield_929 RPRnoteEnumerations49	7939

Enumerator	Value
Nuclear_yield_930 RPRnoteEnumerations49	7940
Nuclear_yield_931 RPRnoteEnumerations49	7941
Nuclear_yield_932 RPRnoteEnumerations49	7942
Nuclear_yield_933 RPRnoteEnumerations49	7943
Nuclear_yield_934 RPRnoteEnumerations49	7944
Nuclear_yield_935 RPRnoteEnumerations49	7945
Nuclear_yield_936 RPRnoteEnumerations49	7946
Nuclear_yield_937 RPRnoteEnumerations49	7947
Nuclear_yield_938 RPRnoteEnumerations49	7948
Nuclear_yield_939 RPRnoteEnumerations49	7949
Nuclear_yield_940 RPRnoteEnumerations49	7950
Nuclear_yield_941 RPRnoteEnumerations49	7951
Nuclear_yield_942 RPRnoteEnumerations49	7952
Nuclear_yield_943 RPRnoteEnumerations49	7953
Nuclear_yield_944 RPRnoteEnumerations49	7954
Nuclear_yield_945 RPRnoteEnumerations49	7955
Nuclear_yield_946 RPRnoteEnumerations49	7956
Nuclear_yield_947 RPRnoteEnumerations49	7957
Nuclear_yield_948 RPRnoteEnumerations49	7958
Nuclear_yield_949 RPRnoteEnumerations49	7959
Nuclear_yield_950 RPRnoteEnumerations49	7960
Nuclear_yield_951 RPRnoteEnumerations49	7961
Nuclear_yield_952 RPRnoteEnumerations49	7962
Nuclear_yield_953 RPRnoteEnumerations49	7963
Nuclear_yield_954 RPRnoteEnumerations49	7964
Nuclear_yield_955 RPRnoteEnumerations49	7965
Nuclear_yield_956 RPRnoteEnumerations49	7966
Nuclear_yield_957 RPRnoteEnumerations49	7967

Enumerator	Value
Nuclear_yield_958 RPRnoteEnumerations49	7968
Nuclear_yield_959 RPRnoteEnumerations49	7969
Nuclear_yield_960 RPRnoteEnumerations49	7970
Nuclear_yield_961 RPRnoteEnumerations49	7971
Nuclear_yield_962 RPRnoteEnumerations49	7972
Nuclear_yield_963 RPRnoteEnumerations49	7973
Nuclear_yield_964 RPRnoteEnumerations49	7974
Nuclear_yield_965 RPRnoteEnumerations49	7975
Nuclear_yield_966 RPRnoteEnumerations49	7976
Nuclear_yield_967 RPRnoteEnumerations49	7977
Nuclear_yield_968 RPRnoteEnumerations49	7978
Nuclear_yield_969 RPRnoteEnumerations49	7979
Nuclear_yield_970 RPRnoteEnumerations49	7980
Nuclear_yield_971 RPRnoteEnumerations49	7981
Nuclear_yield_972 RPRnoteEnumerations49	7982
Nuclear_yield_973 RPRnoteEnumerations49	7983
Nuclear_yield_974 RPRnoteEnumerations49	7984
Nuclear_yield_975 RPRnoteEnumerations49	7985
Nuclear_yield_976 RPRnoteEnumerations49	7986
Nuclear_yield_977 RPRnoteEnumerations49	7987
Nuclear_yield_978 RPRnoteEnumerations49	7988
Nuclear_yield_979 RPRnoteEnumerations49	7989
Nuclear_yield_980 RPRnoteEnumerations49	7990
Nuclear_yield_981 RPRnoteEnumerations49	7991
Nuclear_yield_982 RPRnoteEnumerations49	7992
Nuclear_yield_983 RPRnoteEnumerations49	7993
Nuclear_yield_984 RPRnoteEnumerations49	7994
Nuclear_yield_985 RPRnoteEnumerations49	7995

Enumerator	Value
Nuclear_yield_986 RPRnoteEnumerations49	7996
Nuclear_yield_987 RPRnoteEnumerations49	7997
Nuclear_yield_988 RPRnoteEnumerations49	7998
Nuclear_yield_989 RPRnoteEnumerations49	7999
Chemical_General	8000
Chemical_Blister_Agent	8100
HD_Mustard_	8110
Thickened_HD_Mustard_	8115
Dusty_HD_Mustard_	8120
L_Lewisite_	8125
HN3_Nitrogen_Mustard_	8130
HL_Mustard_Lewisite_	8135
CX_Phosgene_Oxime_	8140
DMMP_Phosphate_Dimethyl_Hydrogen_	8145
DMHP_Phosphite_	8150
DMA_Dimethyl_Acrylate_	8155
DEM	8160
PX_P_xlene_	8165
Chemical_Blood_Agent	8200
AC_HCN_	8210
CK_CNCl_	8215
CG_Phosgene_	8220
Chemical_Nerve_Agent	8300
VX	8310
Thickened_VX	8315
Dusty_VX	8320
GA_Tabun_	8325
Thickened_GA_Tabun_	8330

Enumerator	Value
Dusty_GA_Tabun_	8335
GB_Sarin_	8340
Thickened_GB_Sarin_	8345
Dusty_GB_Sarin_	8350
GD_Soman_	8355
Thickened_GD_Soman_	8360
Dusty_GD_Soman_	8365
GF	8370
Thickened_GF	8375
Dusty_GF	8380
SVX_Soviet_VX_	8385
BIS	8410
TCP	8415
MS_Methyl_Salicylate_	8425
TEP	8430
H2O_Water_	8445
TO1_Toxic_Organic_1_	8450
TO2_Toxic_Organic_2_	8455
TO3_Toxic_Organic_3_	8460
Sulfur_Hexafluoride	8465
AA_Acetic_Acid_	8470
HF_Hydrofluoric_Acid_	8475
Biological	9000
Biological_Virus	9100
Biological_Bacteria	9200
Biological_Rickettsia	9300
Biological_Genetically_Modified_Micro_organisms	9400
Biological_Toxin	9500

MunitionDomainEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Munition Domain [UID 14]*

Enumerator	Value
Other	0
Anti_Air	1
Anti_Armor	2
Anti_Guided_Weapon	3
Anti_Radar	4
Anti_Satellite	5
Anti_Ship	6
Anti_Submarine	7
Anti_Personnel	8
Battlefield_Support	9
Strategic	10
Tactical	11
Directed_Energy_DE_Weapon	12

MunitionExpendableStatusEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Munition/Expendable Status [UID 327]*

Enumerator	Value
Other	0
Ready	1
Inventory	2

NETIDRecordFrequencyTableEnum8Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)Semantics: *NET ID Record-Frequency Table [UID 299]*

Enumerator	Value
HQ_I_Operations	0
HQII_NATO_Europe_Area_Operations	1
HQII_Non_NATO_Europe_Area_Operations	2
SATURN_Operations RPRnoteEnumerations50	3

NETIDRecordModeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *NET ID Record-Mode [UID 298]*

Enumerator	Value
A_HAVE_QUICK_I_or_HAVE_QUICK_II_COMBAT	1
B_SATURN_COMBAT RPRnoteEnumerations50	2
T_TRAINING	3

NavigationSourceEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Navigation Source [UID 359]*

Enumerator	Value
No_Statement	0
GPS	1
INS	2
INS_GPS	3

ObjectKindEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Object Kind [UID 225]*

Enumerator	Value
Other	0
Obstacle	1
Prepared_Position	2

Enumerator	Value
Cultural_Feature	3
Passageway	4
Tactical_Smoke	5
Obstacle_Marker	6
Obstacle_Breach	7
Environmental_Object	8

OwnershipStatusEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Ownership Status [UID 332]*

Enumerator	Value
Other	0
New_Owner	1
Ownership_Query_Response	2
Ownership_Conflict	3
Local_Entity_Cancelled_Auto_Resolve_Conflict	4
Local_Entity_Cancelled_Manual_Resolve_Conflict	5
Local_Entity_Cancelled_Remove_Entity_TCR_Received	6

ParachuteEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Parachute [UID 401]*

Enumerator	Value
None	0
Deployed	1
Collapsed	2
Malfunction_Streamers	3

PhysicalAssociationTypeGroupsEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Physical Association Type Groups [UID 322]*

Enumerator	Value
Not_Specified	0
Towed_Mounted_Sling_Load	1
Restrained	2
Mission	3
Other_Connections	4

PlatformAirCategoryEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Platform-Air Category [UID 10]*

Enumerator	Value
Other	0
Fighter_Air_Defense	1
Attack_Strike	2
Bomber	3
Cargo_Tanker	4
ASW_Patrol_Observation	5
Electronic_Warfare_EW_	6
Reconnaissance	7
Surveillance_C2_Airborne_Early_Warning_	8
Air_Sea_Rescue_ASR_	9
Attack_Helicopter	20
Utility_Helicopter	21
Anti_Submarine_Warfare_Patrol_Helicopter	22
Cargo_Helicopter	23
Observation_Helicopter	24

Enumerator	Value
Special_Operations_Helicopter	25
Training_Helicopter	26
Trainer	40
Unmanned	50
Non_Combatant_Commercial_Aircraft RPRnoteEnumerationsDeprecated	57
Civilian_Ultralight_Aircraft_Non_rigid_Wing	80
Civilian_Ultralight_Aircraft_Rigid_Wing	81
Civilian_Fixed_Wing_Aircraft_Glider	83
Civilian_Fixed_Wing_Aircraft_Light_Sport_up_to_1320_lbs_600_kg_	84
Civilian_Fixed_Wing_Aircraft_Small_up_to_12_500_lbs_5_670_kg_	85
Civilian_Fixed_Wing_Aircraft_Medium_up_to_41_000_lbs_18_597_kg_	86
Civilian_Fixed_Wing_Aircraft_Large_up_to_255_000_lbs_115_666_kg_	87
Civilian_Fixed_Wing_Aircraft_Heavy_above_255_000_lbs_115_666_kg_	88
Civilian_Helicopter_Small_up_to_7_000_lbs_3_175_kg_	90
Civilian_Helicopter_Medium_up_to_20_000_lbs_9_072_kg_	91
Civilian_Helicopter_Large_above_20_000_lbs_9_072_kg_	92
Civilian_Autogyro	93
Civilian_Lighter_than_Air_Balloon	100
Civilian_Lighter_than_Air_Airship	101

PlatformAirCivilianFixedWingAircraftSubcategoriesEnum8 [RPRnoteEnumerations51](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Air Civilian Fixed Wing Aircraft Subcategories [UID 277]*

Enumerator	Value
Single_Piston_Engine	11
Twin_Piston_Engine	12
Single_Engine_Turboprop	21
Twin_Engine_Turboprop	22

Enumerator	Value
Four_Engine_Turboprop	24
Twin_Jet	32
Tri_Jet	33
Four_Engine_Jet	34
Twin_Engine_Tilt_Rotor	40
Tri_Engine_Tilt_Rotor	41
Four_Engine_Tilt_Rotor	42
Six_Engine_Tilt_Rotor	43

PlatformAirCivilianGliderSubcategoriesEnum8 [RPRnoteEnumerations52](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Air Civilian Glider Subcategories [UID 276]*

Enumerator	Value
Sail_Plane	1
Motor_Glider	2

PlatformAirCivilianHelicopterSubcategoriesEnum8 [RPRnoteEnumerations53](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Air Civilian Helicopter Subcategories [UID 278]*

Enumerator	Value
Single_Rotor_Piston_Engine	11
Single_Rotor_Turboshaft_Engine_Conventional_Tail_Rotor	12
Single_Rotor_Turboshaft_Engine_Shrouded_Tail_Rotor	13
Single_Rotor_Turboshaft_Engine_No_Tail_Rotor	14
Tandem_Rotor	21
Coaxial_Rotor	22
Intermeshing_Rotor	23

PlatformAirCivilianLighterThanAirAirshipSubcategoriesEnum8 [RPRnoteEnumerations54](#)Representation: [RPRunsignedInteger8BE](#)Semantics: *Platform-Air Civilian Lighter than Air, Airship Subcategories [UID 280]*

Enumerator	Value
Non_rigid_blimp_	1
Semi_rigid	2
Rigid	3
Hybrid	4

PlatformAirCivilianLighterThanAirBalloonSubcategoriesEnum8 [RPRnoteEnumerations55](#)Representation: [RPRunsignedInteger8BE](#)Semantics: *Platform-Air Civilian Lighter than Air, Balloon Subcategories [UID 279]*

Enumerator	Value
Gas_filled_free	1
Gas_filled_tethered	2
Hot_Air	3
Roziere_Balloon	4
Helikite	5

PlatformAirCivilianUltralightNonRigidWingAircraftSubcategoriesEnum8 [RPRnoteEnumerations56](#)Representation: [RPRunsignedInteger8BE](#)Semantics: *Platform-Air Civilian Ultralight Non-rigid Wing Aircraft Subcategories [UID 274]*

Enumerator	Value
Hang_Glider_Unpowered	1
Hang_Glider_Powered	2
Paraglider_Unpowered	3
Paraglider_Powered	4
Powered_Parachute	5

PlatformAirCivilianUltralightRigidWingAircraftSubcategoriesEnum8 [RPRnoteEnumerations57](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Air Civilian Ultralight Rigid Wing Aircraft Subcategories [UID 275]*

Enumerator	Value
Weight_shift_control	1
Control_surface_elevator_rudder_aileron_control	2

PlatformDomainEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform Domain [UID 8]*

Enumerator	Value
Other	0
Land	1
Air	2
Surface	3
Subsurface	4
Space	5

PlatformLandBusSubcategoriesEnum8 [RPRnoteEnumerations58](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Bus Subcategories [UID 429]*

Enumerator	Value
Other	0
Commuter_Flat_Nose_	1
Commuter_Snout_Nose_	2
Shuttle	3
Double_Decker	4
Guided	5
Kneeling	6

Enumerator	Value
Midibus	7
Minibus	8
Mini_Wheelchair	9
Motorcoach	10
Prison_Bus	11
Schoolbus	12
School_Wheelchair	13
Tour	14
Tram_Parking_Lot	15
Trolley	16
Airport_Transport	17
Articulated_Multi_Unit_	18

PlatformLandCarSubcategoriesEnum8 [RPRnoteEnumerations59](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Car Subcategories [UID 428]*

Enumerator	Value
Other	0
Generic RPRnoteEnumerations60	10
Generic_Mini_Microcar_Tiny_ RPRnoteEnumerations61	11
Generic_Economy_Compact_Small_ RPRnoteEnumerations62	12
Generic_Intermediate_Standard_Medium_ RPRnoteEnumerations63	13
Generic_Full_Premium_Luxury_Large_ RPRnoteEnumerations64	14
Generic_Oversize RPRnoteEnumerations65	15
_2_Door_Coupe_ RPRnoteEnumerations60	20
_2_Door_Coupe_Mini_Microcar_Tiny_	21
_2_Door_Coupe_Economy_Compact_Small_	22
_2_Door_Coupe_Intermediate_Standard_Medium_	23

Enumerator	Value
_3_Door_Hatchback RPRnoteEnumerations60	30
_3_Door_Hatchback_Mini_Microcar_Tiny_	31
_3_Door_Hatchback_Economy_Compact_Small_	32
_4_Door_Sedan RPRnoteEnumerations60	40
_4_Door_Sedan_Mini_Microcar_Tiny_	41
_4_Door_Sedan_Economy_Compact_Small_	42
_4_Door_Sedan_Intermediate_Standard_Medium_	43
_4_Door_Sedan_Full_Premium_Luxury_Large_	44
_4_Door_Sedan_Oversize	45
_5_Door_Hatchback RPRnoteEnumerations60	50
_5_Door_Hatchback_Mini_Microcar_Tiny_	51
_5_Door_Hatchback_Economy_Compact_Small_	52
_5_Door_Hatchback_Intermediate_Standard_Medium_	53
_5_Door_Hatchback_Full_Premium_Luxury_Large_	54
Wagon RPRnoteEnumerations60	60
Wagon_Economy_Compact_Small_	62
Wagon_Intermediate_Standard_Medium_	63
Wagon_Full_Premium_Luxury_Large_	64
Minivan RPRnoteEnumerations60	70
Limousine RPRnoteEnumerations60	80
Limousine_Full_Premium_Luxury_Large_	84
Limousine_Oversize	85
Sports_High_Performance_	90
Convertible RPRnoteEnumerations60	100
Convertible_Mini_Microcar_Tiny_	101
Convertible_Economy_Compact_Small_	102
Convertible_Intermediate_Standard_Medium_	103
Convertible_Full_Premium_Luxury_Large_	104

Enumerator	Value
Sports_Utility_Vehicle_SUV_ RPRnoteEnumerations60	110
Sports_Utility_Vehicle_SUV_Economy_Compact_Small_	112
Sports_Utility_Vehicle_SUV_Intermediate_Standard_Medium_	113
Sports_Utility_Vehicle_SUV_Full_Premium_Luxury_Large_	114
Sports_Utility_Vehicle_SUV_Oversize	115

PlatformLandCategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land Category [UID 9]*

Enumerator	Value
Other	0
Tank	1
Armored_Fighting_Vehicle RPRnoteEnumerations66	2
Armored_Utility_Vehicle RPRnoteEnumerations67	3
Self_Propelled_Artillery	4
Towed_Artillery RPRnoteEnumerations68	5
Small_Wheeled_Utility_Vehicle RPRnoteEnumerations69	6
Large_Wheeled_Utility_Vehicle RPRnoteEnumerations70	7
Small_Tracked_Utility_Vehicle RPRnoteEnumerations71	8
Large_Tracked_Utility_Vehicle RPRnoteEnumerations72	9
Mortar	10
Mine_Plow	11
Mine_Rake	12
Mine_Roller	13
Cargo_Trailer	14
Fuel_Trailer	15
Generator_Trailer	16
Water_Trailer	17

Enumerator	Value
Engineer_Equipment	18
Heavy_Equipment_Transport_Trailer	19
Maintenance_Equipment_Trailer	20
Limber	21
Chemical_Decontamination_Trailer	22
Warning_System	23
Train_Engine	24
Train_Car	25
Train_Caboose	26
Civilian_Vehicle RPRnoteEnumerations73 RPRnoteEnumerationsDeprecated	27
Air_Defense_Missile_Defense_Unit_Equipment	28
Command_Control_Communications_and_Intelligence_C3I_System	29
Operations_Facility	30
Intelligence_Facility	31
Surveillance_Facility	32
Communications_Facility	33
Command_Facility	34
C4I_Facility	35
Control_Facility	36
Fire_Control_Facility	37
Missile_Defense_Facility	38
Field_Command_Post	39
Observation_Post	40
Mine_Flail	41
Fixed_Barrel_Turret_Gun RPRnoteEnumerations74	42
Fixed_Missile_Launcher RPRnoteEnumerations75	43
Unmanned	50
Motorcycle	80

Enumerator	Value
Car	81
Bus	82
Single_Unit_Cargo_Truck	83
Single_Unit_Utility_Emergency_Truck	84
Multiple_Unit_Cargo_Truck	85
Multiple_Unit_Utility_Emergency_Truck	86
Construction_Specialty_Vehicle	87
Farm_Specialty_Vehicle	88
Trailer	89
Recreational	90
Non_motorized	91
Trains	92
Utility_Emergency_Car	93

PlatformLandConstructionSpecialtyVehicleSubcategoriesEnum8 [RPRnoteEnumerations76](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Construction Specialty Vehicle Subcategories [UID 434]*

Enumerator	Value
Other	0
Tug	1
Forklift	2
Loader	3
Loader_Backhoe	4
Crane_Tractor_Mounted	5
Crane_Wheeled	6
Grader	7
Road_Roller_Other	8
Road_Roller_Double_Drum_Smooth	9

Enumerator	Value
Road_Roller_Single_Drum_Smooth	10
Road_Roller_Double_Drum_Sheeps	11
Road_Roller_Single_Drum_Sheeps	12
Road_Roller_Pneumatic_Tired	13
Excavator_Other	14
Excavator_Dragline	15
Excavator_Long_Reach	16
Excavator_Mobile_Tire	17
Mini_Excavator	18
Excavator_Giant	19
Bulldozer_Tractor_Mounted	20
Bulldozer_Tracked	21
Scraper	22
Skid_Steer	23
Dump_Truck_Other	24
Dump_Truck_Articulated	25
Dump_Truck_Transfer	26
Dump_Truck_Super	27
Dump_Truck_Off_Road	28
Paver	29
Drilling_Machine	30
Concrete_Mixer_Other	31
Concrete_Mixer_Rear_Discharge	32
Concrete_Mixer_Front_Discharge	33
Concrete_Mixer_Six_Axle	34
Concrete_Mixer_Long_Reach_Boom	35
Concrete_Mixer_Volumetric	36
Trencher_Chain	37

Enumerator	Value
Trencher_Rockwheel	38
Snowcat	39
Crane_Tracked	40
Crane_Shovel	41
Sweeper_Rotary	42
Roller_Vibratory_Compactor	43
Fork_Lift_Truck	44
Fork_Lift_Rought_Terrain	45
Transloader	46
Truck_Water_Construction	47
Truck_Fuel_Delivery	48
Truck_Sawmill	49
Truck_Line_Marking_Construction	50
Tractor_Industrial	51
Compactor_High_Speed	52
Truck_Drilling	53
Truck_Drilling_Support	54
Crane_Construction	55

PlatformLandFarmSpecialtyVehicleSubcategoriesEnum8 [RPRnoteEnumerations77](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Farm Specialty Vehicle Subcategories [UID 435]*

Enumerator	Value
Other	0
Tractor	1
Harvester_Reaper	2
Skidder	3
Forwarder	4

Enumerator	Value
Lawn_Mower_Other	5
Lawn_Mower_Riding	6
Lawn_Mower_Standing	7
Lawn_Mower_Push	8

PlatformLandMotorcycleSubcategoriesEnum8 [RPRnoteEnumerations78](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Motorcycle Subcategories [UID 427]*

Enumerator	Value
Other	0
Scooter_Small_	1
Sport_Street_Mid_Size_	2
Cruiser_Large_	3
Dirt_Bike	4
Standard	5
Touring	6
Dual_Purpose	7

PlatformLandMultipleUnitCargoTruckSubcategoriesEnum8 [RPRnoteEnumerations79](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Multiple Unit Cargo Truck Subcategories [UID 432]*

Enumerator	Value
Other	0
Tractor_Trailer	1
Tanker	2

PlatformLandMultipleUnitUtilityEmergencyTruckSubcategoriesEnum8 [RPRnoteEnumerations80](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Multiple Unit Utility Emergency Truck Subcategories [UID 433]*

Enumerator	Value
Other	0
Fire_Engine_Hook_And_Ladder	1

PlatformLandNonMotorizedSubcategoriesEnum8 [RPRnoteEnumerations81](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Non-motorized Subcategories [UID 438]*

Enumerator	Value
Other	0
Unicycle	1
Bicycle	2
Bicycle_Mountain	3
Bicycle_Racing	4
Tricycle	5
Quadricycle	6
Rickshaw_Two_Person	7
Rickshaw_One_Person	8
Tandem_Bicycle	9
Cycle_Trailer	10
Cycle_Sidecar	11
Sled	12
Skis	13
Snowboard	14
Skateboard	15
Skates	16
Skates_In_Line	17
Wagon_Cart	18
Dolly	19
Handtruck	20

Enumerator	Value
Push_Cart	21
Wheelbarrow	22
Kick_Scooter	23
Wheelchair	24

PlatformLandRecreationalSubcategoriesEnum8 [RPRnoteEnumerations82](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Recreational Subcategories [UID 437]*

Enumerator	Value
Other	0
ATV_2X4	1
ATV_4X4	2
ATV_6X6	3
ATV_3_wheeled	4
Toy_Other	5
Toy_Car	6
Toy_ATV	7
Golf_Cart	8
Snowmobile	9
Recreational_Vehicle	10
Recreational_Vehicle_Type_A_Motorhome	11
Recreational_Vehicle_Type_B_Motorhome	12
Recreational_Vehicle_Type_C_Motorhome	13
Conversion_Van	14

PlatformLandSingleUnitCargoTruckSubcategoriesEnum8 [RPRnoteEnumerations83](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Single Unit Cargo Truck Subcategories [UID 430]*

Enumerator	Value
Other	0
Pickup_Truck_Mini	1
Pickup_Truck_Mid_Size	2
Pickup_Truck_Full_Size	3
Pickup_Truck_Crew_Cab	4
Pickup_Truck_Extended_Cab	5
Pickup_Truck_Long_Bed	6
Pickup_Truck_Cab_Forward	7
Cargo_Truck	10
Cargo_Truck_up_to_2_5_Ton	11
Cargo_Truck_up_to_5_Ton	12
Cargo_Truck_up_to_7_5_Ton	13
Cargo_Truck_up_to_10_Ton	14
Cargo_Truck_over_10_Ton	15
Tanker	20
Semi_Trailer_Cab_w_o_Trailer_	30
Van	70
Van_Extended	71
Van_Compact	72
Van_Wheelchair	73
Van_Delivery	74
Delivery_Truck	100
Delivery_Truck_Box	101
Delivery_Truck_Flatbed	102
Delivery_Truck_Stake_Bed	103
Mess_Truck	104
Truck_Palletised_Load_System_PLS_ RPRnoteEnumerations84	105
Truck_Petroleum_Oil_and_Lubricants_POL_Palletised_Load_System_PLS_	106

Enumerator	Value
Truck_Petroleum_Oil_and_Lubricants_POL_Surveillance	107
Refrigerated_Truck_Small	108
Refrigerated_Truck_Medium	109
Refrigerated_Truck_Large	110

PlatformLandSingleUnitUtilityEmergencyTruckSubcategoriesEnum8 [RPRnoteEnumerations85](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Single Unit Utility Emergency Truck Subcategories [UID 431]*

Enumerator	Value
Other	0
Ambulance_Truck	1
Fire_Paramedic_Truck	2
Ambulance_Advanced_Life_Support	3
Ambulance_Pickup_Truck	4
Fire_Engine	10
Aerial_Ladder_Fire_Engine	11
Airport_Fire_Engine	12
Wildland_Fire_Engine	13
Fire_Chief	14
Police_Paddy_Wagon	20
Police_SWAT	21
Police_Bomb_Squad	22
Police_Pickup_Truck	23
Hazmat	30
Wrecker_Normal_Hook_and_Chain	40
Wrecker_Normal_Boom	41
Wrecker_Normal_Wheel_Lift	42
Wrecker_Normal_Flatbed	43

Enumerator	Value
Wrecker_Normal_Integrated	44
Wrecker_Heavy_Hook_and_Chain	45
Wrecker_Heavy_Boom	46
Wrecker_Heavy_Wheel_Lift	47
Wrecker_Heavy_Flatbed	48
Wrecker_Heavy_Integrated	49
Postal_Truck	60
Street_Sweeper	70
Street_Sweeper_Three_Wheeled	71
Waste_Collection_Other	80
Waste_Collection_Front_Loader	81
Waste_Collection_Rear_Loader	82
Waste_Collection_Automated_Side_Loader	83
Waste_Collection_Pneumatic_Collection	84
Waste_Collection_Grapple	85
Utility_Truck	90
Utility_Truck_w_Boom	91
Aerial_Work_Platform_Other	100
Aerial_Work_Platform_Scissor_Lift	101
Aerial_Work_Platform_Telescoping	102
Maintenance_Truck	120
Decontamination_Truck	121
Water_Cannon_Truck	122
Water_Purification_Truck	123
Smoke_Generator_Truck	124
Auto_Rickshaw	150

PlatformLandTrailerSubcategoriesEnum8 [RPRnoteEnumerations86](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Trailer Subcategories [UID 436]*

Enumerator	Value
Other	0
Trailer_Flatbed	1
Trailer_Container	2
Trailer_Container_Refrigerated	3
Trailer_Double	4
Trailer_Auto_Transport	5
Trailer_Articulated	6
Trailer_Tanker	7
Trailer_Tanker_Small	8
Trailer_Tanker_Large	9
Trailer_Tanker_Gasoline	10
Trailer_Tanker_Milk	11
Trailer_Tanker_Water	12
Trailer_Tanker_Septic	13
Trailer_Boat	14
Trailer_Boat_Small	15
Trailer_Boat_Large	16
Trailer_Recreational	17
Trailer_Recreational_Conventional	18
Trailer_Recreational_Travel_Expandable	19
Trailer_Recreational_Fifth_Wheel_Travel	20
Trailer_Recreational_Folding_Camping	21
Trailer_Recreational_Truck_Camper	22
Trailer_Aerostat_Mooring_Platform	23
Trailer_Household	24
Trailer_Kitchen	25
Trailer_UltraLight_Aircraft	26

Enumerator	Value
Trailer_Heavy_Equipment	27

PlatformLandTrainsSubcategoriesEnum8 [RPRnoteEnumerations87](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Trains Subcategories [UID 439]*

Enumerator	Value
Other	0
Engine_Locomotive_	1
Box_Car	2
Tanker	3
Flatcar	4
Caboose	5
Passenger_Car	6
Hopper	7

PlatformLandUtilityEmergencyCarSubcategoriesEnum8 [RPRnoteEnumerations88](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Land-Utility Emergency Car Subcategories [UID 440]*

Enumerator	Value
Other	0
Ambulance_Car	1
Police_Car	2
Police_Chief	3
Hearse	4
Taxi	5

PlatformSpaceCategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Space Category [UID 13]*

Enumerator	Value
Other	0
Manned_Spacecraft	1
Unmanned RPRnoteEnumerationsDeprecated	2
Booster	3
Debris	10
Satellite_Unknown_Unspecified_Mission	11
Satellite_Communication	12
Satellite_Navigation	13
Satellite_Science_Experimental_Demonstration	14
Satellite_Inert_Target_Reflector_Calibration_	15
Satellite_Earth_Observation	16
Satellite_Space_Surveillance	17
Satellite_Astronomy	18

PlatformSubsurfaceCategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Subsurface Category [UID 12]*

Enumerator	Value
Other	0
SSBN_Nuclear_Ballistic_Missile_	1
SSGN_Nuclear_Guided_Missile_	2
SSN_Nuclear_Attack_Torpedo_	3
SSG_Conventional_Guided_Missile_	4
SS_Conventional_Attack_Torpedo_Patrol_	5
SSAN_Nuclear_Auxiliary_	6
SSA_Conventional_Auxiliary_	7
Unmanned_Underwater_Vehicle_UUV_	8
SSB_Submarine_Ballistic_Ballistic_Missile_Submarine_	9

Enumerator	Value
SSC_Coastal_Submarine_over_150_tons_	10
SSP_Attack_Submarine_Diesel_Air_Independent_Propulsion_	11
SSM_Midget_Submarine_under_150_tons_	12
SSNR_Special_Attack_Submarine_	13
SST_Training_Submarine_	14
AGSS_Auxiliary_Submarine_	15
Semi_Submersible_Boats	16
Civilian_Submarines	80
Civilian_Submersibles	81
Civilian_Semi_Submersible_Boats	82

PlatformSubsurfaceCivilianSemiSubmersiblesSubcategoriesEnum8 [RPRnoteEnumerations89](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Subsurface-Civilian Semi-Submersibles-Subcategories [UID 508]*

Enumerator	Value
Narco_Submarine	1

PlatformSurfaceCategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Surface Category [UID 11]*

Enumerator	Value
Other	0
Carrier	1
Command_Ship_Cruiser	2
Guided_Missile_Cruiser	3
Guided_Missile_Destroyer_DDG_	4
Destroyer_DD_	5
Guided_Missile_Frigate_FFG_	6
Light_Patrol_Craft	7

Enumerator	Value
Mine_Countermeasure_Ship_Craft	8
Dock_Landing_Ship	9
Tank_Landing_Ship	10
Landing_Craft	11
Light_Carrier	12
Cruiser_Helicopter_Carrier	13
Hydrofoil	14
Air_Cushion_Surface_Effect	15
Auxiliary	16
Auxiliary_Merchant_Marine	17
Utility	18
Unmanned_Surface_Vehicle_USV_	19
Littoral_Combat_Ships_LCS_	20
Surveillance_Ship	21
Frigate_including_Corvette_	50
Battleship	51
Heavy_Cruiser	52
Destroyer_Tender	53
Amphibious_Assault_Ship	54
Amphibious_Cargo_Ship	55
Amphibious_Transport_Dock	56
Ammunition_Ship	57
Combat_Stores_Ship	58
Surveillance_Towed_Array_Sonar_System_SURTASS_	59
Fast_Combat_Support_Ship	60
Non_Combatant_Ship RPRnoteEnumerationsDeprecated	61
Coast_Guard_Cutters	62
Coast_Guard_Boats	63

Enumerator	Value
Fast_Attack_Craft	64
Inflatable_Boat RPRnoteEnumerations90	65
Passenger_Vessel_Group_1_Merchant_	80
Dry_Cargo_Ship_Group_2_Merchant_	81
Tanker_Group_3_Merchant_	82
Offshore_Support_Vessel	83
Private_Motorboat	84
Private_Sailboat	85
Fishing_Vessel	86
Other_Vessels	87
Human_Powered_Watercraft	88
Search_and_Rescue_Vessels	100
Life_Saving_Equipment	101

PlatformSurfaceDryCargoShipSubcategoriesEnum8 [RPRnoteEnumerations91](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Surface-Dry Cargo Ship Subcategories [UID 442]*

Enumerator	Value
Common_Dry_Cargo_Ship	1
Dry_Bulk_Cargo_Ship	2
Container_Ship	3
Reefer_Ship_Refrigerator_Ship_	4
Ro_Ro_Ship_Roll_on_Roll_off_Ship_	5
Barge	6
Heavy_Lift_Ship	7

PlatformSurfaceFishingVesselSubcategoriesEnum8 [RPRnoteEnumerations92](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Surface Fishing Vessel Subcategories [UID 447]*

Enumerator	Value
Small_Fishing_Vessel_up_to_26ft_7_9m_	1
Medium_Fishing_Vessel_up_to_65ft_19_8m_	2
Large_Fishing_Vessel_greater_than_65ft_19_8m_	3
Fish_Processing_Vessel	4
Masted_Fishing_Vessel	5

PlatformSurfaceHumanPoweredWatercraftSubcategoriesEnum8 [RPRnoteEnumerations93](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Surface Human-Powered Watercraft Subcategories [UID 897]*

Enumerator	Value
Other	0
Rowboat	1
Kayak	2
Canoe	3
Stand_Up_Paddle_Board	4

PlatformSurfaceOtherVesselsSubcategoriesEnum8 [RPRnoteEnumerations94](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Surface Other Vessels Subcategories [UID 448]*

Enumerator	Value
Go_Fast_Boat	1
Research_Vessel	2
Hydrofoil_Vessel	3
Cable_Layer_Vessel	4
Dredger_Vessel	5
Junk_Dhow_Vessel	6
Catamaran	7
Pontoon	8
Personal_Water_Craft	9

Enumerator	Value
Refugee_Raft	10
Lightship	11
Law_Enforcement_Vessel	12
Pilot_Vessel	13

PlatformSurfacePassengerVesselSubcategoriesEnum8 [RPRnoteEnumerations95](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Surface-Passenger Vessel Subcategories [UID 441]*

Enumerator	Value
Cruise_Ship	1
Cruise_Ferry	2
High_Speed_Ferry	3
Ferry	4
Ocean_Liner	5

PlatformSurfacePrivateMotorboatSubcategoriesEnum8 [RPRnoteEnumerations96](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Surface Private Motorboat Subcategories [UID 445]*

Enumerator	Value
Small_Motorboat_up_to_26ft_7_9m_	1
Medium_Motorboat_up_to_39ft_11_9m_	2
Large_Motorboat_up_to_65ft_19_8m_	3
Very_Large_Motorboat_greater_than_65ft_19_8m_	4

PlatformSurfacePrivateSailboatSubcategoriesEnum8 [RPRnoteEnumerations97](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Surface Private Sailboat Subcategories [UID 446]*

Enumerator	Value
Small_Sailboat_up_to_26ft_7_9m_	1

Enumerator	Value
Medium_Sailboat_up_to_39ft_11_9m_	2
Large_Sailboat_up_to_65ft_19_8m_	3
Very_Large_Sailboat_greater_than_65ft_19_8m_	4

PlatformSurfaceSupportVesselSubcategoriesEnum8 [RPRnoteEnumerations98](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Surface Support Vessel Subcategories [UID 444]*

Enumerator	Value
Platform_Supply_Vessel	1
Tender_Vessel	2
Tugboat	3
Dive_Support_Vessel	4
Fireboat	5
Well_Stimulation_Vessel_WSV_	6
Anchor_Handling_Tug_Supply_Vessel_AHTS_	7
Offshore_Construction_Vessel_OCV_	8
Emergency_Response_and_Rescue_Vessel_ERRV_	9

PlatformSurfaceTankerSubcategoriesEnum8 [RPRnoteEnumerations99](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Platform-Surface Tanker Subcategories [UID 443]*

Enumerator	Value
Liquid_Petroleum_Gas_LPG_Tanker	1
Chemical_Tanker	2
Liquid_Natural_Gas_LNG_Tanker	3
Coastal_Trading_Vessel_Coaster_	4
Crude_Oil_Tanker_up_to_159_999_DWT_	5
Liquid_Bulk_Tanker	6
Very_Large_Crude_Carrier_160_000_319_999_DWT_	7

Enumerator	Value
Ultra_Large_Crude_Carrier_320_000_549_999_DWT_	8
Condensate_Storage_Tanker	9

RadioAttachedIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Radio Attached Indicator [UID 306]*

Enumerator	Value
No_Statement	0
Unattached	1
Attached	2

RadioCategoryEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Radio Category [UID 22]*

Enumerator	Value
Other	0
Voice_Transmission_Reception	1
Data_Link_Transmission_Reception	2
Voice_and_Data_Link_Transmission_Reception	3
Instrumented_Landing_System_ILS_Glideslope_Transmitter	4
Instrumented_Landing_System_ILS_Localizer_Transmitter	5
Instrumented_Landing_System_ILS_Outer_Marker_Beacon	6
Instrumented_Landing_System_ILS_Middle_Marker_Beacon	7
Instrumented_Landing_System_ILS_Inner_Marker_Beacon	8
Instrumented_Landing_System_ILS_Receiver_Platform_Radio_	9
Tactical_Air_Navigation_TACAN_Transmitter_Ground_Fixed_Equipment_	10
Tactical_Air_Navigation_TACAN_Receiver_Moving_Platform_Equipment_	11
Tactical_Air_Navigation_TACAN_Transmitter_Receiver_Moving_Platform_Equipment_	12
Variable_Omni_Ranging_VOR_Transmitter_Ground_Fixed_Equipment_	13

Enumerator	Value
Variable_Omni_Ranging_VOR_with_Distance_Measuring_Equipment_DME_Transmitter_Ground_Fixed_Equipmen t_	14
Combined_VOR_ILS_Receiver_Moving_Platform_Equipment_	15
Combined_VOR_TACAN_VORTAC_Transmitter	16
Non_Directional_Beacon_NDB_Transmitter	17
Non_Directional_Beacon_NDB_Receiver	18
Non_Directional_Beacon_NDB_with_Distance_Measuring_Equipment_DME_Transmitter	19
Distance_Measuring_Equipment_DME_	20
Link_16_Terminal	21
Link_11_Terminal	22
Link_11B_Terminal	23
EPLRS_SADL_Terminal	24
F_22_Intra_Flight_Data_Link_IFDL_	25
F_35_Multifunction_Advanced_Data_Link_MADL_	26
SINCGARS_Terminal	27
L_Band_SATCOM_Terminal	28
IBS_Terminal	29
GPS	30
Tactical_Video	31
Air_to_Air_Missile_Datalink	32
Link_16_Surrogate_for_Non_NATO_TDL_Terminal RPRnoteEnumerations100	33
MQ_1_9_C_Band_LOS_Datalink	34
MQ_1_9_Ku_Band_SATCOM_Datalink	35
Air_to_Ground_Weapon_Datalink	36
Automatic_Identification_System_AIS_	37
JPALS_Data_Link	38
Combat_Search_and_Rescue_CSAR_Radio	40
Counter_Unmanned_Aircraft_System_C_UAS_Radio	41
Emergency_Position_Indicating_Radio_Beacons_EPIRB_	42

Enumerator	Value
Electronic_Attack_Systems	50
Tactical_Targeting_Network_Technology_TTNT_	51

RadioSubcategoryEnum8 [RPRnoteEnumerations101](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Radio Subcategory [UID 23]*

Enumerator	Value
Other	0
Joint_Electronics_Type_Designation_System_JETDS_Non_specific_Series	1
Manufacturer_Designation	2
National_Designation	3
JETDS_ARC_Set_1	11
JETDS_ARC_Set_2	12
JETDS_ARC_Set_3	13
JETDS_ARC_Set_4	14
JETDS_BRC_Set_1	15
JETDS_BRC_Set_2	16
JETDS_BRC_Set_3	17
JETDS_BRC_Set_4	18
JETDS_CRC_Set_1	19
JETDS_CRC_Set_2	20
JETDS_CRC_Set_3	21
JETDS_CRC_Set_4	22
JETDS_DRC_Set_1	23
JETDS_DRC_Set_2	24
JETDS_DRC_Set_3	25
JETDS_DRC_Set_4	26
JETDS_FRC_Set_1	27

Enumerator	Value
JETDS_FRC_Set_2	28
JETDS_FRC_Set_3	29
JETDS_FRC_Set_4	30
JETDS_GRC_Set_1	31
JETDS_GRC_Set_2	32
JETDS_GRC_Set_3	33
JETDS_GRC_Set_4	34
JETDS_KRC_Set_1	35
JETDS_KRC_Set_2	36
JETDS_KRC_Set_3	37
JETDS_KRC_Set_4	38
JETDS_MRC_Set_1	39
JETDS_MRC_Set_2	40
JETDS_MRC_Set_3	41
JETDS_MRC_Set_4	42
JETDS_PRC_Set_1	43
JETDS_PRC_Set_2	44
JETDS_PRC_Set_3	45
JETDS_PRC_Set_4	46
JETDS_SRC_Set_1	47
JETDS_SRC_Set_2	48
JETDS_SRC_Set_3	49
JETDS_SRC_Set_4	50
JETDS_TRC_Set_1	51
JETDS_TRC_Set_2	52
JETDS_TRC_Set_3	53
JETDS_TRC_Set_4	54
JETDS_VRC_Set_1	55

Enumerator	Value
JETDS_VRC_Set_2	56
JETDS_VRC_Set_3	57
JETDS_VRC_Set_4	58
JETDS_WRC_Set_1	59
JETDS_WRC_Set_2	60
JETDS_WRC_Set_3	61
JETDS_WRC_Set_4	62
JETDS_ZRC_Set_1	63
JETDS_ZRC_Set_2	64
JETDS_ZRC_Set_3	65
JETDS_ZRC_Set_4	66

ReceiverReceiverStateEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Receiver-Receiver State [UID 179]*

Enumerator	Value
Off	0
On_but_not_receiving	1
On_and_receiving	2

RecordQueryREventTypeEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Record Query-R-Event Type [UID 334]*

Enumerator	Value
Periodic	0
Internal_Entity_State_Data	1

RecordREventTypeEnum16

Representation: [RPRunsignedInteger16BE](#)Semantics: *Record-R-Event Type [UID 333]*

Enumerator	Value
Other	0

RepairCompleteRepairEnum16Representation: [RPRunsignedInteger16BE](#)Semantics: *Repair Complete-Repair [UID 64]*

Enumerator	Value
no_repairs_performed	0
all_requested_repairs_performed	1
motor_engine	10
starter	20
alternator	30
generator	40
battery	50
engine_coolant_leak	60
fuel_filter	70
transmission_oil_leak	80
engine_oil_leak	90
pumps	100
filters_110	110
transmission	120
brakes	130
suspension_system	140
oil_filter	150
hull	1000
airframe	1010
truck_body	1020

Enumerator	Value
tank_body	1030
trailer_body	1040
turret	1050
propeller	1500
filters_1520	1520
wheels	1540
tire	1550
track	1560
gun_elevation_drive	2000
gun_stabilization_system	2010
gunner_s_primary_sight_GPS_	2020
commander_s_extension_to_the_GPS	2030
loading_mechanism	2040
gunner_s_auxiliary_sight	2050
gunner_s_control_panel	2060
gunner_s_control_assembly_handle_s_	2070
commander_s_control_handles_assembly	2090
commander_s_weapon_station	2100
commander_s_independent_thermal_viewer_CITV_	2110
general_weapons	2120
fuel_transfer_pump	4000
fuel_lines	4010
gauges	4020
general_fuel_system	4030
electronic_warfare_systems	4500
detection_systems	4600
detection_systems_radio_frequency	4610
detection_systems_microwave	4620

Enumerator	Value
detection_systems_infrared	4630
detection_systems_laser	4640
range_finders	4700
range_only_radar	4710
laser_range_finder	4720
electronic_systems	4800
electronics_systems_radio_frequency	4810
electronics_systems_microwave	4820
electronics_systems_infrared	4830
electronics_systems_laser	4840
radios	5000
communication_systems	5010
intercoms	5100
encoders	5200
encryption_devices	5250
decoders	5300
decryption_devices	5350
computers	5500
navigation_and_control_systems	6000
fire_control_systems	6500
air_supply	8000
filters_8010	8010
water_supply_8020	8020
refrigeration_system	8030
chemical_biological_and_radiological_protection	8040
water_wash_down_systems	8050
decontamination_systems	8060
water_supply_9000	9000

Enumerator	Value
cooling_system	9010
winches	9020
catapults	9030
cranes	9040
launchers	9050
life_boats	10000
landing_craft	10010
ejection_seats	10020

RepairGroupsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Repair Groups [UID 272]*

Enumerator	Value
General_Repair_Codes	0
Drive_Train	1
Hull_Airframe_Body	2
Interfaces_with_Environment	3
Weapons	4
Fuel_Systems	5
Electronics	6
Life_Support_Systems	7
Hydraulic_Systems_and_Actuators	8
Auxiliary_Craft	9

RepairResponseRepairResultEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Repair Response-Repair Result [UID 65]*

Enumerator	Value
other	0

Enumerator	Value
repair_ended	1
invalid_repair	2
repair_interrupted	3
service_canceled_by_the_supplier	4

ReplyAmplificationEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Reply Amplification [UID 373]*

Enumerator	Value
No_Statement	0
Complete	1
Limited	2
Unable_to_Respond	3

RequiredReliabilityServiceEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Required Reliability Service [UID 74]*

Enumerator	Value
Acknowledged	0
Unacknowledged	1

SISOSTD002VersionEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *SISO-STD-002 version [UID 736]*

Enumerator	Value
SISO_STD_002_2006	0
SISO_STD_002_2021	1

SeamarkCategoryAndColorEnum8

Representation: [RPRunsignedInteger8BE](#)Semantics: *Seamark Category and Color [UID 892]*

Enumerator	Value
Other	0
Lateral_Mark_Red_	1
Lateral_Mark_Green_	2
Preferred_Channel_Mark_Red_Green_Red_	3
Preferred_Channel_Mark_Green_Red_Green_	4
North_Cardinal_Mark_Black_Yellow_	5
East_Cardinal_Mark_Black_Yellow_Black_	6
South_Cardinal_Mark_Yellow_Back_	7
West_Cardinal_Mark_Yellow_Black_Yellow_	8
Isolated_Danger_Mark_Black_Red_Black_	9
Safe_Water_Mark_Red_White_Stripes_	10
Special_Purpose_Mark_Yellow_	11
Emergency_Wreck_Mark_Yellow_Blue_Stripes_	12

SeamarkTopmarkEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Seamark Topmark [UID 893]*

Enumerator	Value
None	0
Cone_point_up	1
Cone_point_down	2
Sphere	3
_2_Spheres	4
Cylinder_can_	5
Board	6
X_shape_St_Andrew_s_cross_	7

Enumerator	Value
Upright_cross_St_George_s_cross_	8
Cube_point_up	9
_2_cones_point_to_point	10
_2_cones_base_to_base	11
Rhombus_diamond_	12
_2_cones_points_upward_	13
_2_cones_points_downward_	14
Besom_point_up_broom_or_perch_	15
Besom_point_down_broom_or_perch_	16
Flag	17
Sphere_over_rhombus_Circle_over_rhombus_	18
Square	19
Rectangle_horizontal	20
Rectangle_vertical	21
Trapezium_up	22
Trapezium_down	23
Triangle_point_up	24
Triangle_point_down	25
Circle	26
Two_upright_crosses_one_over_the_other_	27
T_shape	28
Triangle_pointing_up_over_a_circle	29
Upright_cross_over_a_circle	30
Rhombus_over_a_circle	31
Circle_over_a_triangle_pointing_up	32
Other_shape	33

SeamarkTypeEnum8

Representation: [RPRunsignedInteger8BE](#)Semantics: *Seamark Type [UID 891]*

Enumerator	Value
Other	0
Conical_Buoy_Nun_Ogival_	1
Can_Buoy_Cylindrical_	2
Spherical_Buoy	3
Pillar_Buoy	4
Spar_Buoy_Spindle_	5
Barrel_Buoy_Tun_	6
Superbuoy	7
Stake_Pole_Post	8
Perch	9
Withy	10
Beacon	11
Beacon_Tower	12
Lattice_Beacon	13
Pile_Beacon	14
Cairn	15
Buoyant_Beacon	16
Fishing_Buoy	17

SeasonEnum8Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)Semantics: *Season [UID 465]*

Enumerator	Value
Summer	0
Winter	1
Spring	2

Enumerator	Value
Autumn	3

SensorEmitterCategoryEnum8

Representation: [RPRunsigndInteger8BE](#)

Semantics: *Sensor/Emitter Category [UID 28]*

Enumerator	Value
Other	0
Multi_spectral	1
RF_Active	2
RF_Passive_intercept_and_DF_	3
Optical_direct_viewing_with_or_without_optics_	4
Electro_Optical	5
Seismic	6
Chemical_point_detector	7
Chemical_standoff	8
Thermal_temperature_sensing_	9
Acoustic_Active	10
Acoustic_Passive	11
Contact_Pressure_physical_hydrostatic_barometric_	12
Electro_Magnetic_Radiation_gamma_radiation_	13
Particle_Radiation_Neutrons_alpha_beta_particles_	14
Magnetic	15
Gravitational	16

SensorOnOffStatusEnum8

Representation: [RPRunsigndInteger8BE](#)

Semantics: *Sensor On/Off Status [UID 331]*

Enumerator	Value
Off	0

Enumerator	Value
On	1

SensorRecordSensorTypeOtherActiveSensorsEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Sensor Record-Sensor Type-Other Active Sensors [UID 325]*

Enumerator	Value
Undefined	0

SensorRecordSensorTypePassiveSensorsEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Sensor Record-Sensor Type-Passive Sensors [UID 326]*

Enumerator	Value
ALR_400	60000
AN_AAR_47	60001
AN_AAR_50	60002
AN_AAR_54	60003
AN_AAR_56	60004
AN_AAR_57	60005
AN_ALQ_142	60006
AN_ALR_45	60007
AN_ALR_46	60008
AN_ALR_56	60009
AN_ALR_59	60010
AN_ALR_64	60011
AN_ALR_66	60012
AN_ALR_67	60013
AN_ALR_69	60014
AN_ALR_73	60015
AN_ALR_76	60016

Enumerator	Value
AN_ALR_91	60017
AN_ALR_93	60018
AN_ALR_94	60019
AN_ALR_801	60020
AN_APR_39	60021
AN_AYR_2	60022
ARI_18223	60023
BOW_21	60024
Chaparral_IRST	60025
Flanker_IRST	60026
Foxbat_IRST	60027
Foxhound_IRST	60028
Fulcrum_IRST	60029
Havoc_IRST	60030
Hind_IRST	60031
KJ_200	60032
KJ_8602	60033
L_150_Pastel	60034
Serval	60035
Sherloc	60036
Sherlocvf	60037
Sirena_2	60038
Sirena_3	60039
Sirena_3M	60040
Sky_Guardian	60041
SPO_15	60042
SPS_200	60043
Tarang	60044

Enumerator	Value
AN_AAQ_29A	60045
_101KS_U_MAW	60046
Abrams_2GF_FLIR	60047
Abrams_3GF_FLIR	60048
AN_AAQ_13_LANTIRN_FLIR	60049
AN_ALR_74	60050
AN_ALR_90	60051
AN_APR_48	60052
ELT_156X_V_	60053
Felon_IRST_101KS_V	60054
Flogger_IRST_TP_23ML	60055
Generic_FLIR	60056
Generic_IRST	60057
Generic_MAWS	60058
Generic_RWR	60059
L_136_Mak	60060
Leonardo_SASS_IRST	60061
OSF_IRST	60062
Pirate_IRST	60063
RECCELITE	60064
Thales_Nederland_Sirius_IRST	60065
Tornado_RWR	60066
TOES_521_FLIR	60067
Safran_Vampir_MB	60068

SensorTypeSourceEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Sensor Type Source [UID 414]*

Enumerator	Value
Other_Active_Sensors	0
Electromagnetic	1
Passive_Sensors	2
Minefield_Sensors	3
Underwater_Acoustics	4
Lasers	5

SeparationVPPreEntityIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Separation VP-Pre-Entity Indicator [UID 283]*

Enumerator	Value
No_Statement	0
Entity_ID_Existed_Prior_to_Separation_without_Entity_State_PDU	1
Entity_ID_Existed_Prior_to_Separation_with_Entity_State_PDU_Issued	2
Entity_Initially_Created_at_Separation_Event	3

SeparationVPReasonForSeparationEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Separation VP-Reason for Separation [UID 282]*

Enumerator	Value
No_Statement	0
Attached_Part_Separation	1
Submunition_Separation	2

ServiceRequestServiceTypeRequestedEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Service Request-Service Type Requested [UID 63]*

Enumerator	Value
Other	0

Enumerator	Value
Resupply	1
Repair	2
Aerial_Refueling_High_Fidelity	3
Aerial_Refueling_Low_Fidelity	4

SignalEncodingClassEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Signal-Encoding Class [UID 270]*

Enumerator	Value
Encoded_audio	0
Raw_Binary_Data	1
Application_Specific_Data	2
Database_index	3

SignalEncodingTypeEnum16

Representation: [RPRunsignedInteger16BE](#) [RPRnoteEnumerationsBitSize14](#)

Semantics: *Signal-Encoding Type [UID 271]*

Enumerator	Value
_8_bit_mu_law_ITU_T_G_711_	1
CVSD_MIL_STD_188_113_ RPRnoteEnumerations102	2
ADPCM_ITU_T_G_726_	3
_16_bit_Linear_PCM_2s_Complement_Big_Endian	4
_8_bit_Linear_PCM_Unsigned	5
VQ_Vector_Quantization_ RPRnoteEnumerationsDeprecated	6
_unavailable_for_use__7	7
GSM_Full_Rate_ETSI_06_10_	8
GSM_Half_Rate_ETSI_06_20_	9
Speex_Narrow_Band RPRnoteEnumerations103	10
Opus RPRnoteEnumerations104	11

Enumerator	Value
LPC_10_FIPS_PUB_137_	12
_16_bit_Linear_PCM_2s_Complement_Little_Endian	100
_unavailable_for_use__255	255

SignalTDLTypeEnum16

Representation: [RPRunsigedInteger16BE](#)

Semantics: *Signal-TDL Type [UID 178]*

Enumerator	Value
Other	0
PADIL	1
NATO_Link_1	2
ATDL_1	3
Link_11B_TADIL_B_	4
Situational_Awareness_Data_Link_SADL_	5
Link_16_Legacy_Format_JTIDS_TADIL_J_	6
Link_16_Legacy_Format_JTIDS_FDL_TADIL_J_	7
Link_11_TADIL_A_	8
IJMS	9
Link_4A_TADIL_C_	10
Link_4C	11
TIBS	12
ATL	13
Constant_Source	14
Abbreviated_Command_and_Control_15	15
MILSTAR	16
ATHS	17
OTHGOLD	18
TACELINT	19

Enumerator	Value
Weapons_Data_Link_AWW_13_	20
Abbreviated_Command_and_Control_21 RPRnoteEnumerationsDeprecated	21
Enhanced_Position_Location_Reporting_System_EPLRS_	22
Position_Location_Reporting_System_PLRS_	23
SINCGARS	24
HAVE_QUICK_I	25
HAVE_QUICK_II	26
SATURN RPRnoteEnumerations105	27
Intra_Flight_Data_Link_1	28
Intra_Flight_Data_Link_2	29
Improved_Data_Modem_IDM_	30
Air_Force_Application_Program_Development_AFAPD_	31
Cooperative_Engagement_Capability_CEC_	32
Forward_Area_Air_Defense_FAAD_Data_Link_FDL_	33
Ground_Based_Data_Link_GBDL_	34
Intra_Vehicular_Info_System_IVIS_	35
Marine_Tactical_System_MTS_	36
Tactical_Fire_Direction_System_TACFIRE_	37
Integrated_Broadcast_Service_IBS_	38
Airborne_Information_Transfer_ABIT_	39
Advanced_Tactical_Airborne_Reconnaissance_System_ATARS_Data_Link	40
Battle_Group_Passive_Horizon_Extension_System_BGPHERS_Data_Link	41
Common_High_Bandwidth_Data_Link_CHBDL_	42
Guardrail_Interoperable_Data_Link_IDL_	43
Guardrail_Common_Sensor_System_One_CSS1_Data_Link	44
Guardrail_Common_Sensor_System_Two_CSS2_Data_Link	45
Guardrail_CSS2_Multi_Role_Data_Link_MRDL_	46
Guardrail_CSS2_Direct_Air_to_Satellite_Relay_DASR_Data_Link	47

Enumerator	Value
Line_of_Sight_LOS_Data_Link_Implementation_LOS_tether_	48
Lightweight_CDL_LWCDL_	49
L_52M_SR_71_	50
Rivet_Reach_Rivet_Owl_Data_Link	51
Senior_Span	52
Senior_Spur	53
Senior_Stretch_	54
Senior_Year_Interoperable_Data_Link_IDL_	55
Space_CDL	56
TR_1_mode_MIST_Airborne_Data_Link	57
Ku_band_SATCOM_Data_Link_Implementation_UAV_	58
Mission_Equipment_Control_Data_link_MECDL_	59
Radar_Data_Transmitting_Set_Data_Link	60
Surveillance_and_Control_Data_Link_SCDL_	61
Tactical_UAV_Video	62
UHF_SATCOM_Data_Link_Implementation_UAV_	63
Tactical_Common_Data_Link_TCDDL_	64
Low_Level_Air_Picture_Interface_LLAPI_	65
Weapons_Data_Link_AGM_130_	66
Automatic_Identification_System_AIS_	67
Weapons_Data_Link_AIM_120_	68
Weapons_Data_Link_AIM_9_	69
Weapons_Data_Link_CAMM_	70
GC3	99
Link_16_Standardized_Format_JTIDS_MIDS_TADIL_J_	100
Link_16_Enhanced_Data_Rate_EDR_JTIDS_MIDS_TADIL_J_	101
JTIDS_MIDS_Net_Data_Load_TIMS_TOMS_	102
Link_22	103

Enumerator	Value
AFIWC_IADS_Communications_Links	104
F_22_Intra_Flight_Data_Link_IFDL_	105
L_Band_SATCOM	106
TSAF_Communications_Link	107
Enhanced_SINCGARS_7_3	108
F_35_Multifunction_Advanced_Data_Link_MADL_	109
Cursor_on_Target	110
All_Purpose_Structured_Eurocontrol_Surveillance_Information_Exchange_ASTERIX_	111
Variable_Message_Format_VMF_over_Combat_Net_Radio_VMF_over_CNR_	112
Link_16_Surrogate_for_Non_NATO_TDL RPRnoteEnumerations100	113
MQ_1_9_C_Band_LOS_Uplink	114
MQ_1_9_C_Band_LOS_Downlink	115
MQ_1_9_Ku_Band_SATCOM_Uplink	116
MQ_1_9_Ku_Band_SATCOM_Downlink	117
Weapons_Datalink_SDB_II_	118
JTAC_SA_Uplink	119
Common_Interactive_Broadcast_CIB_	120
Joint_Range_Extension_Application_Protocol_A_JREAP_A_	121
JPALS_Data_Link	125
OneSAF_IADS_Communications_Link	126
Tactical_Targeting_Network_Technology_TTNT_Application	127

SignalUserProtocolIdentificationNumberEnum32

Representation: [RPRunsignedInteger32BE](#)

Semantics: *Signal-User Protocol Identification Number [UID 177]*

Enumerator	Value
CCSIL	1
A2ATD_SINCGARS_ERF	5

Enumerator	Value
A2ATD_CAC2	6
Battle_Command	20
AFIWC_IADS_Track_Report	30
AFIWC_IADS_Comm_C2_Message	31
AFIWC_IADS_Ground_Control_Interceptor_GCI_Command	32
AFIWC_Voice_Text_Message	35
ModSAF_Text_Radio	177
CCTT_SINGARS_ERF_LOCKOUT	200
CCTT_SINGARS_ERF_HOPSET	201
CCTT_SINGARS_OTAR	202
CCTT_SINGARS_DATA	203
ModSAF_FWA_Forward_Air_Controller	546
ModSAF_Threat_ADA_C3	832
F_16_MTC_AFAPD_Protocol	1000
F_16_MTC_IDL_Protocol	1100
Automatic_Identification_System_AIS_ RPRnoteEnumerationsDeprecated	1371
ModSAF_Artillery_Fire_Control	4570
AGTS	5361
GC3	6000
WNCP_data	6010
Spoken_text_message	6020
Longbow_IDM_message	6661
Comanche_IDM_message	6662
Longbow_Airborne_TACFIRE_Message	6663
Longbow_Ground_TACFIRE_Message	6664
Longbow_AFAPD_Message	6665
Longbow_ERF_message	6666
VMF_IDM	7000

Enumerator	Value
CSAR_Radio_Survivor_Message	7010
CSAR_Radio_Interrogator_Message	7020
Image_File_Transfer_Message	7030
Geotag_Data_Message	7040
Tactical_Video_Regeneration_Data	7050
Call_Message	8000

SpecificDimensionEnumerationsForAirAreaSizeEnum8 [RPRnoteEnumerations1](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Specific Dimension Enumerations for Air Area Size [UID 132]*

Enumerator	Value
Small_Flock_Swarm_5km_long_x_5m_wide_1km_x_5m_	222
Small_Flock_Swarm_Dense_5km_long_x_5m_wide_1km_x_5m_	223
Medium_Flock_Swarm_1km_long_x_10m_wide_2km_x_10m_	224
Medium_Flock_Swarm_Dense_1km_long_x_10m_wide_2km_x_10m_	225
Large_Flock_Swarm_10km_long_x_100m_wide_20km_x_100m_	226
Large_Flock_Swarm_Dense_10km_long_x_100m_wide_20km_x_100m_	227

SpecificDimensionEnumerationsForLandAreaSizeEnum8 [RPRnoteEnumerations1](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Specific Dimension Enumerations for Land Area Size [UID 131]*

Enumerator	Value
Small_Area_2_000m2_4_000m2_5_to_1_sq_acre_	222
Small_Area_Dense_2_000m2_4_000m2_5_to1_sq_acre_	223
Medium_Area_20_000m2_41_000m2_10_to_20_sq_acres_	224
Medium_Area_Dense_20_000m2_41_000m2_10_to_20_sq_acres_	225
Large_Area_40_000m2_81_000m2_20_to_40_sq_acres_	226
Large_Area_Dense_40_000m2_81_000m2_20_to_40_sq_acres_	227

SpotChaffStatusEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Spot Chaff Status [UID 404]*

Enumerator	Value
None	0
Deployed	1
Malfunction	2

StateIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *State Indicator [UID 300]*

Enumerator	Value
Heartbeat_Update	0
Changed_Data	1
Has_Ceased	2

StopFreezeReasonEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Stop/Freeze-Reason [UID 67]*

Enumerator	Value
Other	0
Recess	1
Termination	2
System_Failure	3
Security_Violation	4
Entity_Reconstitution	5
Stop_for_reset	6
Stop_for_restart	7
Abort_Training_Return_to_Tactical_Operations	8

SubcategoriesForAirCategory200BirdEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Subcategories for Air Category 200 - Bird [UID 110]*

Enumerator	Value
Penguin RPRnoteEnumerations106	1
Seagull	2
Pelican	3
Albatross	4
Swan	5
Cormorant	6
Heron	7
Crane	8
Osprey	9
Loon	10
Stork	11
Flamingo	12
Duck	13
Ostrich RPRnoteEnumerations107	20
Emu	21
Chicken	22
Black_Bird RPRnoteEnumerations108	30
Starling	31
Budgerigar_Parakeet_	32
Canadian_Goose RPRnoteEnumerations109	40
Crow	41
Eagle RPRnoteEnumerations110	50
Vulture	55
Falcon	60
Hawk	65

Enumerator	Value
Owl	70
Kite	80

SubcategoriesForAirCategory201InsectEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Air Category 201 - Insect [UID 111]*

Enumerator	Value
Moth RPRnoteEnumerations111	1
Butterfly	2
Fly RPRnoteEnumerations112	20
Mosquito RPRnoteEnumerations113	30
Wasp RPRnoteEnumerations114	40
Bee RPRnoteEnumerations115	50
Beetle RPRnoteEnumerations116	60
Dragonfly RPRnoteEnumerations117	70
Locust RPRnoteEnumerations118	80

SubcategoriesForAirCategory202MammalEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Air Category 202 - Mammal [UID 112]*

Enumerator	Value
Bat	1
Flying_Squirrel	10
Gliding_Possum	20

SubcategoriesForLandCategory200MammalEnum8 [RPRnoteEnumerations119](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Land Category 200 - Mammal [UID 100]*

Enumerator	Value
Small_Dog RPRnoteEnumerations120	1
Chihuahua	2
Medium_Dog RPRnoteEnumerations121	10
Australian_Cattle_Dog	11
Large_Dog RPRnoteEnumerations122	20
German_Shepherd	21
Very_Large_Dog RPRnoteEnumerations123	30
Giant_Turkish_Kangal	31
Sheep RPRnoteEnumerations124	40
Goat	41
Pig RPRnoteEnumerations125	50
Cow RPRnoteEnumerations126	60
Ox	61
Ox_With_Cart RPRnoteEnumerations127	70
Horse RPRnoteEnumerations128	80
Donkey	81
Mule	82
Horse_With_Rider RPRnoteEnumerations129	90
Horse_With_Cargo	91
Donkey_With_Rider	92
Donkey_With_Cargo	93
Mule_With_Rider	94
Mule_With_Cargo	95
Camel RPRnoteEnumerations130	100
Dromedary_Camel_One_Hump_	101
Bactrian_Camel_Two_Humps_	102
Dromedary_Camel_With_Rider RPRnoteEnumerations131	110
Dromedary_Camel_With_Cargo	111

Enumerator	Value
Rat RPRnoteEnumerations132	200

SubcategoriesForLandCategory201ReptileEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Land Category 201 - Reptile [UID 101]*

Enumerator	Value
New_Zealand_Northern_Tuatara RPRnoteEnumerations133	1
Monitor	3
Gecko	8
Iguana	13
Chameleon	17
Non_Venomous_Snake RPRnoteEnumerations134	30
Boa	31
Python	35
Bullsnake	39
Kingsnake	43
Venomous_Snake RPRnoteEnumerations135	60
Rattlesnake	61
Copperhead	62
Cottonmouth	63
Taipan	64
Viper	65
Cobra	66
Australian_Brown_Snake	67
Tortoise RPRnoteEnumerations136	90
Turtle	100
American_Alligator RPRnoteEnumerations137	120
Crocodile	121

Enumerator	Value
Australian_Freshwater_Crocodile	122

SubcategoriesForLandCategory202AmphibianEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Land Category 202 - Amphibian [UID 102]*

Enumerator	Value
Frog RPRnoteEnumerations138	1
Toad	2
Salamander RPRnoteEnumerations139	170
Caecilian RPRnoteEnumerations140	230

SubcategoriesForLandCategory203InsectEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Land Category 203 - Insect [UID 103]*

Enumerator	Value
Beetle RPRnoteEnumerations141	1
Mantis RPRnoteEnumerations142	60
Cockroach RPRnoteEnumerations143	70
Army_Ant RPRnoteEnumerations144	80
Fire_Ant	81
Grasshopper RPRnoteEnumerations145	90
Centipede RPRnoteEnumerations146	100

SubcategoriesForLandCategory204ArachnidEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Land Category 204 - Arachnid [UID 104]*

Enumerator	Value
Spider RPRnoteEnumerations147	1
Tick RPRnoteEnumerations148	20

Enumerator	Value
Scorpion RPRnoteEnumerations149	30
Harvestmen RPRnoteEnumerations150	40
Mite RPRnoteEnumerations151	50

SubcategoriesForLandCategory205MolluskEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Land Category 205 - Mollusk [UID 105]*

Enumerator	Value
Snail RPRnoteEnumerations152	1
Slug RPRnoteEnumerations153	50

SubcategoriesForLandCategory206MarsupialEnum8 [RPRnoteEnumerations154](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Land Category 206 - Marsupial [UID 106]*

Enumerator	Value
Brown_Four_Eyed_Opossum RPRnoteEnumerations155	1
Bushy_Tailed_Opossum	2
Tate_s_Shrew_Opossum RPRnoteEnumerations156	90
Greater_Bilby RPRnoteEnumerations157	100
Tasmanian_Devil RPRnoteEnumerations158	110
Brush_Tailed_Rock_Wallaby RPRnoteEnumerations159	150
Eastern_Wallaroo	160
Red_Kangaroo	170
Queensland_Koala RPRnoteEnumerations160	200
Southern_Hairy_Nosed_Wombat RPRnoteEnumerations161	205
Brushtail_Possum RPRnoteEnumerations162	210
Sugar_Glider	211

SubcategoriesForSubsurfaceCategory200FishEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Subcategories for Subsurface Category 200 - Fish [UID 120]*

Enumerator	Value
Forage_Fish_Small_Schooling RPRnoteEnumerations163	1
Herring	2
Sardines	3
Krill	4
Squid	5
Medium_Schooling_Fish RPRnoteEnumerations164	30
Hake	31
Cod	32
Haddock	33
Mackerel	34
Large_Schooling_Fish RPRnoteEnumerations165	60
Tuna	61
Small_Shark RPRnoteEnumerations166	90
Dogfish_Shark	91
Medium_Shark RPRnoteEnumerations167	120
Mako_Shark	121
Hammerhead_Shark	122
Large_Shark RPRnoteEnumerations168	150
Great_White_Shark	151
Tiger_Shark	152
Blue_Shark	153
Whale_Shark	154
Skate RPRnoteEnumerations169	180
Stingray	181
Eel RPRnoteEnumerations170	190

Enumerator	Value
Marlin RPRnoteEnumerations171	200
Swordfish	201

SubcategoriesForSubsurfaceCategory201MammalEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Subsurface Category 201 - Mammal [UID 121]*

Enumerator	Value
Whale RPRnoteEnumerations172	1
Beaked_Whale	2
Beluga_Whale	3
Blue_Whale	4
Bottlenose_Whale	5
Northern_Bottlenose_Whale	6
Southern_Bottlenose_Whale	7
Bowhead_Whale	8
Bryde_s_Whale	9
Dwarf_Sperm_Whale	10
Finback_Whale	11
Gray_Whale	12
Humpback_Whale	13
Long_Finned_Pilot_Whale	14
Minke_Whale	15
Northern_Minke_Whale	16
Southern_Minke_Whale	17
Narwhal_Whale	18
Orca_Whale	19
Pygmy_Sperm_Whale	20
Right_Whale	21

Enumerator	Value
North_Atlantic_Right_Whale	22
North_Pacific_Right_Whale	23
Southern_Right_Whale	24
Sei_Whale	25
Short_Finned_Pilot_Whale	26
Sperm_Whale	27
Dolphin RPRnoteEnumerations173	50
Bottlenose_Dolphin	51
Bottlenose_Indo_Pacific_Dolphin	52
Bottlenose_Burrunan_Dolphin	53
Atlantic_Spotted_Dolphin	54
Australian_Snubfin_Dolphin	55
Chilean_Black_Dolphin	56
Chinese_White_Dolphin	57
Clymene_Dolphin	58
Porpoise RPRnoteEnumerations174	100
Harbour_Porpoise	101
Californian_Porpoise	102
Dall_s_Porpoise	103
Burmeister_s_Porpoise	104
Seal RPRnoteEnumerations175	120
Bearded_Seal	121
Harbor_Seal	122
Fur_Seal	123
Weddell_Seal	124
Elephant_Seal	125
Sea_Lion RPRnoteEnumerations176	130
Australian_Sea_Lion	131

Enumerator	Value
California_Sea_Lion	132
Walrus RPRnoteEnumerations177	140
Atlantic_Walrus	141
Pacific_Walrus	142
Otter RPRnoteEnumerations178	150
Sea_Otter	151
Manatee RPRnoteEnumerations179	160
Florida_Manatee	161
Dugongs	162
Polar_Bear RPRnoteEnumerations180	200

SubcategoriesForSubsurfaceCategory202MolluskEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Subsurface Category 202 - Mollusk [UID 122]*

Enumerator	Value
Snail	1
Slug	10
Octopus	20
Squid	30
Cuttlefish	40
Clam	50
Muscle	60
Oyster	70
Scallop	80

SubcategoriesForSubsurfaceCategory203CrustaceanEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Subsurface Category 203 - Crustacean [UID 123]*

Enumerator	Value
Shrimp	1
Snapping_Shrimp	2
Crayfish	10
Lobster	20
Crab	30

SubcategoriesForSubsurfaceCategory204InsectEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Subcategories for Subsurface Category 204 - Insect [UID 124]*

Enumerator	Value
Sea_Skater	1
Water_Beetle	2

SupplyDomainEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Supply Domain [UID 600]*

Enumerator	Value
Not_Used RPRnoteEnumerationsDeprecated	0
Class_1_Subistence	1
Class_2_Clothing_Individual_Equipment_Tools_Admin_Supplies	2
Class_3_Petroleum_Oils_Lubricants	3
Class_4_Construction_Materials	4
Class_5_Ammunition RPRnoteEnumerations181 RPRnoteEnumerationsDeprecated	5
Class_6_Personnel_Demand_Items	6
Class_7_Major_Items RPRnoteEnumerations182	7
Class_8_Medical_Material	8
Class_9_Repair_Parts_and_Components	9
Class_10_Material_to_Support_Non_Military_Programs	10
Class_11_Supplies_Non_Doctrinal_	11

Enumerator	Value
Class_12_Sling_Loads_Non_Doctrinal_	12

SupplyFuelTypeEnum8 [RPRnoteEnumerations183](#)

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Supply-Fuel Type [UID 413]*

Enumerator	Value
Other	0
Gasoline	1
Diesel_Fuel_F_54_DF_2_	2
JP_4_F_40_JET_B_	3
Fuel_Oil	4
JP_8_F_34_JET_A_1_	5
Fog_Oil	6
Multi_Spectral_Fog_Oil	7
JP_5_F_44_JET_A_	8
JPTS	9
TS_1_Russia_RUS_	10

SurfaceCategoriesEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Surface Categories [UID 890]*

Enumerator	Value
Other	0
Oil_Platform	1
Buoy	2
Pier	3
Marker	4
Electrical_Power_Installation	7
Target	8

Enumerator	Value
Container	9
Seamark	10
Floating_Brdge	11

SurveillanceStatusEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Surveillance/Status [UID 365]*

Enumerator	Value
No_Information	0
Emergency_Loss_of_Communications	1
SPI	2
ATCRBS_Code_Change	3

TCASACASBasicAdvancedIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *TCAS/ACAS Basic/Advanced Indicator [UID 341]*

Enumerator	Value
Basic	0
Advanced	1

TCASACASIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *TCAS/ACAS Indicator [UID 342]*

Enumerator	Value
TCAS	0
ACAS	1

TCASACASSoftwareVersionEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize03](#)

Semantics: *TCAS/ACAS Software Version [UID 343]*

Enumerator	Value
No_Statement	0
_6_0_2	1
_7_0	2

TCASACASTypeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize03](#)

Semantics: *TCAS/ACAS Type [UID 344]*

Enumerator	Value
No_Statement	0
ACAS_I	1
ACAS_II	2

TCASIIITypeEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *TCAS I/II Type [UID 345]*

Enumerator	Value
TCAS_I_or_No_Statement_	0
TCAS_II	1

TILinkTypeEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *T/I Link Type [UID 370]*

Enumerator	Value
Not_Used	0
Ground_Initiated_Communications_B_GICB_	1
Automatic_Dependent_Surveillance_ADS_	2

Enumerator	Value
Global_Navigation_Satellite_System_GNSS_	3
Data_Link_Initiation_Capability_DLIC_	4
Aircraft_Communications_Addressing_and_Reporting_System_ACARS_	5
ATC_Communications_Management_ACM_	6
VHF_Digital_Link_VDL_	7
Aeronautical_Telecommunication_Network_ATN_	8
Mode_Select_S_	9
Airborne_Collision_Avoidance_Systems_ACAS_	10
Traffic_Collision_Avoidance_Systems_TCAS_	11
Automatic_Dependent_Surveillance_B_ADS_B_	12

TimeSlotAllocationLevelEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Time Slot Allocation (TSA) level of fidelity. See SISO-STD-002 for detailed descriptions and requirements [UID 172]*

Enumerator	Value
Low_Fidelity_Level_0	0
Low_Fidelity_Level_1	1
Medium_Fidelity_Level_2	2
Medium_Fidelity_Level_3	3
High_Fidelity_Level_4	4

TimeTypeSourceEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Time Type Source [UID 367]*

Enumerator	Value
Compute_Locally	0
Even_Second	1
Odd_Second	2

TransferControlTransferTypeEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Transfer Control-Transfer Type [UID 224]*

Enumerator	Value
Other	0
Push_Transfer_Entity	1
Automatic_Pull_Transfer_Entity	2
Not_Used_3	3
Push_Transfer_Environmental_Process	4
Automatic_Pull_Transfer_Environmental_Process	5
Not_Used_6	6
Cancel_Transfer	7
Manual_Pull_Transfer_Entity	8
Manual_Pull_Transfer_Environmental_Process	9
Remove_Entity	10

TransmissionIndicatorEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Transmission Indicator [UID 372]*

Enumerator	Value
No_Statement	0
Original_Interrogation	1
Interrogation_Reply	2
Squitter_Transmission	3

TransmitterAntennaPatternReferenceSystemEnum8Representation: [RPRunsignedInteger8BE](#)Semantics: *Transmitter-Antenna Pattern-Reference System [UID 168]*

Enumerator	Value
World_Coordinates	1
Entity_Coordinates	2

TransmitterAntennaPatternTypeEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Transmitter-Antenna Pattern Type [UID 167]*

Enumerator	Value
Isotropic_Spherical_Radiation_Pattern_ RPRnoteEnumerations184	0
Beam	1
Spherical_harmonic RPRnoteEnumerationsDeprecated	2
Transmitter_Radiation_Volume	4
Beam_and_Transmitter_Radiation_Volume	5
Omnidirectional_Toroidal_Radiation_Pattern_	6

TransmitterCryptoSystemEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Transmitter-Crypto System [UID 166]*

Enumerator	Value
No_Encryption_Device	0
KY_28	1
KY_58	2
Narrow_Spectrum_Secure_Voice_NSVE_	3
Wide_Spectrum_Secure_Voice_WSVE_	4
SINCGARS_ICOM	5
KY_75	6
KY_100	7
KY_57	8
KYV_5	9
Link_11_KG_40A_P_NTDS_	10

Enumerator	Value
Link_11B_KG_40A_S	11
Link_11_KG_40AR	12
KGV_135A	13
Tactical_Secure_Voice_TSV_	14

TransmitterDetailAmplitudeAndAngleModulationEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Transmitter-Detail-Amplitude and Angle Modulation [UID 157]*

Enumerator	Value
Other	0
Amplitude_and_Angle	1

TransmitterDetailAmplitudeModulationEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Transmitter-Detail-Amplitude Modulation [UID 156]*

Enumerator	Value
Other	0
AFSK_Audio_Frequency_Shift_Keying_	1
AM_Amplitude_Modulation_	2
CW_Continuous_Wave_Modulation_	3
DSB_Double_Sideband_	4
ISB_Independent_Sideband_	5
LSB_Single_Band_Suppressed_Carrier_Lower_Sideband_Mode_	6
SSB_Full_Single_Sideband_Full_Carrier_	7
SSB_Reduc_Single_Band_Reduced_Carrier_	8
USB_Single_Band_Suppressed_Carrier_Upper_Sideband_Mode_	9
VSB_Vestigial_Sideband_	10

TransmitterDetailAngleModulationEnum16Representation: [RPRunsignedInteger16BE](#)Semantics: *Transmitter-Detail-Angle modulation [UID 158]*

Enumerator	Value
Other	0
FM_Frequency_Modulation_	1
FSK_Frequency_Shift_Keying_	2
PM_Phase_Modulation_	3
MSK_Minimum_Shift_Keying_	4

TransmitterDetailCarrierPhaseShiftModulationEnum16Representation: [RPRunsignedInteger16BE](#)Semantics: *Transmitter-Detail-Carrier Phase Shift Modulation [UID 162]*

Enumerator	Value
Other	0

TransmitterDetailCombinationModulationEnum16Representation: [RPRunsignedInteger16BE](#)Semantics: *Transmitter-Detail-Combination Modulation [UID 159]*

Enumerator	Value
Other	0
Amplitude_Angle_Pulse	1

TransmitterDetailPulseModulationEnum16Representation: [RPRunsignedInteger16BE](#)Semantics: *Transmitter-Detail-Pulse Modulation [UID 160]*

Enumerator	Value
Other	0
Pulse	1

Enumerator	Value
X_Band_TACAN_Pulse	2
Y_Band_TACAN_Pulse	3

TransmitterDetailSATCOMModulationEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Transmitter-Detail-SATCOM-Modulation [UID 589]*

Enumerator	Value
Other	0
No_Delay RPRnoteEnumerations185	1

TransmitterDetailUnmodulatedModulationEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Transmitter-Detail-Unmodulated Modulation [UID 161]*

Enumerator	Value
Other	0
Continuous_Wave_emission_of_an_unmodulated_carrier	1

TransmitterInputSourceEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Transmitter-Input Source [UID 165]*

Enumerator	Value
Other	0
Pilot	1
Copilot	2
First_Officer	3
Driver	4
Loader	5
Gunner	6
Commander	7

Enumerator	Value
Digital_Data_Device	8
Intercom	9
Audio_Jammer	10
Data_Jammer	11
GPS_Jammer	12
GPS_Meaconer	13
SATCOM_Uplink_Jammer	14
Crew_Observer_1	15
Crew_Observer_2	16
Aerial_Refueling_Officer_ARO_	17
Aerial_Refueling_Officer_Instructor_AROI_	18

TransmitterMajorModulationEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Transmitter-Major Modulation [UID 155]*

Enumerator	Value
No_Statement	0
Amplitude	1
Amplitude_and_Angle	2
Angle RPRnoteEnumerations186	3
Combination	4
Pulse	5
Unmodulated	6
Carrier_Phase_Shift_Modulation_CPSM_	7
SATCOM RPRnoteEnumerations185	8

TransmitterModulationTypeSystemEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *Transmitter-Modulation Type System [UID 163]*

Enumerator	Value
Other	0
Generic_Radio_or_Simple_Intercom	1
HAVE_QUICK_I	2
HAVE_QUICK_II	3
SATURN RPRnoteEnumerations105	4
SINCGARS	5
CCTT_SINCGARS	6
EPLRS_Enhanced_Position_Location_Reporting_System_	7
JTIDS_MIDS RPRnoteEnumerations187	8
Link_11 RPRnoteEnumerations188	9
Link_11B RPRnoteEnumerations188	10
L_Band_SATCOM	11
Enhanced_SINCGARS_7_3	12
Navigation_Aid	13
MUOS	14

TransmitterTransmitStateEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Transmitter-Transmit State [UID 164]*

Enumerator	Value
Off	0
On_but_not_transmitting	1
On_and_transmitting	2

TransponderInterrogatorIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize01](#)

Semantics: *Transponder/Interrogator Indicator [UID 337]*

Enumerator	Value
Transponder	0

Enumerator	Value
Interrogator	1

TurnRateSourceEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize02](#)

Semantics: *Turn Rate Source [UID 366]*

Enumerator	Value
Compute_Locally	0
Less_Than_1_Degree_Turn_or_Not_Turning	1
_1_Degree_or_Greater_Turn_Rate	2

UAAcousticEmitterSystemFunctionEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *UA-Acoustic Emitter System Function [UID 145]*

Enumerator	Value
Other	0
Platform_search_detect_track	1
Navigation	2
Mine_hunting	3
Weapon_search_detect_track_detect	4

UAAcousticSystemNameEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *UA-Acoustic System Name [UID 144]*

Enumerator	Value
Other	0
AN_BQQ_5	1
AN_SSQ_62	2
AN_SQS_23	3
AN_SQS_26	4

Enumerator	Value
AN_SQS_53	5
ALFS	6
LFA	7
AN_AQS_901	8
AN_AQS_902	9

UAActiveEmissionParameterIndexEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *UA-Active Emission Parameter Index [UID 146]*

Enumerator	Value
Other	0

UAAdditionalPassiveActivityParameterIndexEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *UA-Additional Passive Activity Parameter Index [UID 150]*

Enumerator	Value
Other	0

UAPassiveParameterIndexEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *UA-Passive Parameter Index [UID 148]*

Enumerator	Value
Other	0

UAPropulsionPlantConfigurationConfigurationEnum8

Representation: [RPRunsignedInteger8BE](#) [RPRnoteEnumerationsBitSize07](#)

Semantics: *UA Propulsion Plant Configuration-Configuration [UID 335]*

Enumerator	Value
Other	0

Enumerator	Value
Diesel_Electric	1
Diesel	2
Battery	3
Turbine_Reduction	4
Steam	6
Gas_Turbine	7

UAScanPatternEnum16

Representation: [RPRunsignedInteger16BE](#)

Semantics: *UA-Scan Pattern [UID 147]*

Enumerator	Value
Scan_pattern_not_used	0
Conical	1
Helical	2
Raster	3
Sector_search	4
Continuous_search	5

UAStateChangeUpdateIndicatorEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *UA-State/Change Update Indicator [UID 143]*

Enumerator	Value
State_Update	0
Changed_Data_Update	1

VariableParameterRecordTypeEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Variable Parameter Record Type [UID 56]*

Enumerator	Value
Articulated_Part	0
Attached_Part	1
Separation	2
Entity_Type	3
Entity_Association	4

VariableRecordTypesEnum32

Representation: [RPRunsignedInteger32BE](#)

Semantics: *Variable Record Types [UID 66]*

Enumerator	Value
Not_Used_Invalid_Value_	0
Entity_ID_List	1
DDCP_Join_Transaction_Join_Request_Message	1001
DDCP_Set_Playback_Window_Transaction_Set_Playback_Window_Request_Message	1002
DDCP_Load_Mission_Recording_Transaction_Load_Mission_Recording_Request_Message	1003
DDCP_Cue_Transaction_Cue_Request_Message	1004
DDCP_Play_Transaction_Play_Request_Message	1005
DDCP_Stop_Transaction_Stop_Request_Message	1006
DDCP_Pause_Transaction_Pause_Request_Message	1007
DDCP_End_Transaction_End_Request_Message	1009
DDCP_Join_Response_Message	1051
DDCP_Request_Receipt_Message	1052
DDCP_Playback_Window_Confirmed_Message	1053
DDCP_Mission_Recording_Loaded_Message	1054
DDCP_Cue_Confirmed_Message	1055
DDCP_Time_to_Complete_Message	1056
DDCP_Play_Commenced_Message	1057
DDCP_Stop_Confirmed_Message	1058

Enumerator	Value
DDCP_Pause_Confirmed_Message	1059
DDCP_End_Response_Message	1061
DDCP_Master_Announce_Message	1111
DDCP_Device_Announce_Message	1112
DDCP_Device_Exit_Message	1114
DDCP_Device_Heartbeat_Message	1115
DDCP_Master_Time_Sync_Message	1116
DDCP_Error_Message	1118
DDCP_Master_Stop_Sync_Message	1119
DDCP_Master_Transition_Message	1120
Mission_Time	1200
High_Fidelity_HAVE_QUICK_SATURN_Radio RPRnoteEnumerations50	3000
Blanking_Sector_attribute_record	3500
Angle_Deception_attribute_record	3501
False_Targets_attribute_record	3502
DE_Precision_Aimpoint_record	4000
DE_Area_Aimpoint_record	4001
Directed_Energy_Damage_Description_record	4500
Crypto_Control	5000
Mode_5_S_Transponder_Location	5001
Mode_5_S_Transponder_Location_Error	5002
Squitter_Airborne_Position_Report	5003
Squitter_Airborne_Velocity_Report	5004
Squitter_Surface_Position_Report	5005
Squitter_Identification_Report	5006
GICB	5007
Squitter_Event_Driven_Report	5008
Antenna_Location	5009

Enumerator	Value
Basic_Interactive	5010
Interactive_Mode_4_Reply	5011
Interactive_Mode_5_Reply	5012
Interactive_Basic_Mode_5	5013
Interactive_Basic_Mode_S	5014
IO_Effect	5500
IO_Communications_Node	5501
Identification_10000	10000
Trainer_Initial_Conditions_Filename	10010
Increment_3_1_Mission_Data_Load_Name	10020
Increment_2_Mission_Data_Load_Name	10030
Set_Markpoint_Command	10110
Markpoint_ID	10115
Reaction_Level	10140
Weapon_Reload	10150
CES_Entity_Set_Clear_Status	10157
Activate_Entity	10160
Disengage_Reengage_10170	10170
Fuel_Freeze	10190
Fire_Launch_Dispende	10250
Target_Assignment	10254
CIC_Enable	10256
Shoot_Inhibit	10258
Posture	10259
Jammer_State	10262
Jammer_Type	10263
Dynamic_Targeting	10264
Manual_Jamming_On_Override	10267

Enumerator	Value
SOJ_Axis	10268
Emitter_Override_10280	10280
Shields_10290	10290
Crash_Override	10300
Stop_Buzzer	10306
Target_Lasing_On_Off	10307
Target_Lasing_Laser_Code	10308
Power_Plant	10310
Tactical_Lighting_On_Off_Control_Light_Control	10311
Tactical_Lighting_Blinker_Control_Blinker_Value	10312
Tactical_Lighting_On_Off_Control_Light_Control_Type	10313
Park_Vehicle	10314
Signaling_On_Off	10315
Signaling_Device	10316
Ownship_ID	10400
State_Change	10600
Entity_Type_11000	11000
Concatenated	11100
Kind_11110	11110
Domain_11120	11120
Country_11130	11130
Category_11140	11140
Subcategory_11150	11150
Specific_11160	11160
Extra_11170	11170
Force_ID_11180	11180
Force_ID_11200	11200
Description_11300	11300

Enumerator	Value
Tanker_Boom_Control	11500
Airport_Lights	11501
Weather_Post	11502
Localizer_and_GlideSlope	11503
TACAN_NavAids	11504
Alternative_Entity_Type_12000	12000
Kind_12110	12110
Domain_12120	12120
Country_12130	12130
Category_12140	12140
Subcategory_12150	12150
Specific_12160	12160
Extra_12170	12170
Description_12300	12300
Entity_Marking	13000
Entity_Marking_Characters	13100
Crew_ID	13200
Task_Organization	14000
Regiment_Name	14200
Battalion_Name	14300
Company_Name	14400
Platoon_Name	14500
Squad_Name	14520
Team_Name	14540
Bumper_Number	14600
Vehicle_Number	14700
Unit_Number	14800
DIS_Identity	15000

Enumerator	Value
DIS_Site_ID	15100
DIS_Host_ID	15200
DIS_Entity_ID	15300
Mount_Intent	15400
Tether_Untether_Command_ID	15500
Teleport_Entity_Data_Record	15510
DIS_Aggregate_ID_Set_if_communication_to_aggregate_	15600
Ownership_Status	15800
Reconstitute	19177
Loads	20000
Crew_Members	21000
Crew_Member_ID	21100
Health	21200
Job_Assignment	21300
Fuel	23000
Quantity_23100	23100
Quantity_23105	23105
Ammunition	24000
_120_mm_HEAT_quantity	24001
_120_mm_SABOT_quantity	24002
_12_7_mm_M8_quantity	24003
_12_7_mm_M20_quantity	24004
_7_62_mm_M62_quantity	24005
M250_UKL8A1_quantity	24006
M250_UKL8A3_quantity	24007
_7_62_mm_M80_quantity	24008
_12_7_mm_quantity	24009
_7_62_mm_quantity	24010

Enumerator	Value
Mines_quantity	24060
Type	24100
Kind_24110	24110
Domain_24120	24120
Country_24130	24130
Category_24140	24140
Subcategory_24150	24150
Extra_24160	24160
Description_24300	24300
Cargo	25000
Vehicle_Mass	26000
Supply_Quantity	27000
Armament	28000
Status	30000
Activate_entity	30010
Subscription_State	30100
Round_trip_time_delay	30300
TADIL_J_message_count_label_0_	30400
TADIL_J_message_count_label_1_	30401
TADIL_J_message_count_label_2_	30402
TADIL_J_message_count_label_3_	30403
TADIL_J_message_count_label_4_	30404
TADIL_J_message_count_label_5_	30405
TADIL_J_message_count_label_6_	30406
TADIL_J_message_count_label_7_	30407
TADIL_J_message_count_label_8_	30408
TADIL_J_message_count_label_9_	30409
TADIL_J_message_count_label_10_	30410

Enumerator	Value
TADIL_J_message_count_label_11_	30411
TADIL_J_message_count_label_12_	30412
TADIL_J_message_count_label_13_	30413
TADIL_J_message_count_label_14_	30414
TADIL_J_message_count_label_15_	30415
TADIL_J_message_count_label_16_	30416
TADIL_J_message_count_label_17_	30417
TADIL_J_message_count_label_18_	30418
TADIL_J_message_count_label_19_	30419
TADIL_J_message_count_label_20_	30420
TADIL_J_message_count_label_21_	30421
TADIL_J_message_count_label_22_	30422
TADIL_J_message_count_label_23_	30423
TADIL_J_message_count_label_24_	30424
TADIL_J_message_count_label_25_	30425
TADIL_J_message_count_label_26_	30426
TADIL_J_message_count_label_27_	30427
TADIL_J_message_count_label_28_	30428
TADIL_J_message_count_label_29_	30429
TADIL_J_message_count_label_30_	30430
TADIL_J_message_count_label_31_	30431
Position_31000	31000
Route_Waypoint_type	31010
MilGrid10	31100
Geocentric_Coordinates	31200
X_31210	31210
Y_31220	31220
Z_31230	31230

Enumerator	Value
Latitude_31300	31300
Longitude_31400	31400
Line_of_Sight	31500
X_31510	31510
Y_31520	31520
Z_31530	31530
Altitude_31600	31600
Destination_Latitude	31700
Destination_Longitude	31800
Destination_Altitude	31900
Orientation	32000
Hull_Heading_Angle	32100
Hull_Pitch_Angle	32200
Roll_Angle	32300
X_32500	32500
Y_32600	32600
Z_32700	32700
Appearance	33000
Ambient_Lighting	33100
Lights_33101	33101
Paint_Scheme	33200
Smoke	33300
Trailing_Effects	33400
Flaming	33500
Marking	33600
Mine_Plows_Attached	33710
Mine_Rollers_Attached	33720
Tank_Turret_Azimuth	33730

Enumerator	Value
Failures_and_Malfunctions	34000
Age	34100
Kilometers	34110
Damage	35000
Cause	35050
Mobility_Kill_35100	35100
Fire_Power_Kill	35200
Personnel_Casualties	35300
Velocity	36000
X_velocity	36100
Y_velocity	36200
Z_velocity	36300
Speed_36400	36400
Acceleration	37000
X_acceleration	37100
Y_acceleration	37200
Z_acceleration	37300
Engine_Status	38100
Primary_Target_Line_PTL_	39000
Exercise	40000
Exercise_State	40010
Restart_Refresh	40015
AFATDS_File_Name	40020
Terrain_Database	41000
Missions	42000
Mission_ID	42100
Mission_Type	42200
Mission_Request_Time_Stamp	42300

Enumerator	Value
Exercise_Description	43000
Name	43100
Entities	43200
Version	43300
Guise_Mode	43410
Simulation_Application_Active_Status	43420
Simulation_Application_Role_Record	43430
Simulation_Application_State	43440
Visual_Output_Mode	44000
Simulation_Manager_Role	44100
Simulation_Manager_Site_ID	44110
Simulation_Manager_Applic_ID	44120
Simulation_Manager_Entity_ID	44130
Simulation_Manager_Active_Status	44140
After_Active_Review_Role	44200
After_Active_Review_Site_ID	44210
After_Active_Applic_ID	44220
After_Active_Review_Entity_ID	44230
After_Active_Review_Active_Status	44240
Exercise_Logger_Role	44300
Exercise_Logger_Site_ID	44310
Exercise_Logger_Applic_ID	44320
Exercise_Entity_ID	44330
Exercise_Logger_Active_Status	44340
Synthetic_Environment_Manager_Role	44400
Synthetic_Environment_Manager_Site_ID	44410
Synthetic_Environment_Manager_Applic_ID	44420
Synthetic_Environment_Manager_Entity_ID	44430

Enumerator	Value
Synthetic_Environment_Manager_Active_Status	44440
SIMNET_DIS_Translator_Role	44500
SIMNET_DIS_Translator_Site_ID	44510
SIMNET_DIS_Translator_Applic_ID	44520
SIMNET_DIS_Translator_Entity_ID	44530
SIMNET_DIS_Translator_Active_Status	44540
Application_Rate	45000
Application_Time	45005
Application_Timestep	45010
Feedback_Time	45020
Simulation_Rate	45030
Simulation_Time	45040
Simulation_Timestep	45050
Time_Interval	45060
Time_Latency	45070
Time_Scheme	45080
Exercise_Elapsed_Time	46000
Elapsed_Time	46010
Environment	50000
Scenario_Date	50103
Time_Date_Valid	50106
Scenario_Time	50118
Snow_Enable_Disable	50120
Weather_Attributes_Request	50124
MET_Heartbeat_Message	50126
Contrails_Enable	50600
Contrail_Altitudes	50700
Weather_51000	51000

Enumerator	Value
Weather_Condition	51010
Thermal_Condition	51100
Thermal_Visibility_51110	51110
Thermal_Visibility_51111	51111
Time_52000	52000
Time_52001	52001
Time_of_Day_Discrete	52100
Time_of_Day_Continuous	52200
Time_Mode	52300
Time_Scene	52305
Current_Hour	52310
Current_Minute	52320
Current_Second	52330
Azimuth_52340	52340
Maximum_Elevation	52350
Time_Zone	52360
Time_Rate	52370
The_number_of_simulation_seconds_since_the_start_of_the_exercise_simulation_time_	52380
Time_Sunrise_Enabled	52400
Sunrise_Hour	52410
Sunrise_Minute	52420
Sunrise_Second	52430
Sunrise_Azimuth	52440
Time_Sunset_Enabled	52500
Sunset_Hour_52510	52510
Sunset_Hour_52511	52511
Sunset_Minute	52520
Sunset_Second	52530

Enumerator	Value
Date	52600
Date_European_	52601
Date_US_	52602
Month	52610
Day	52620
Year	52630
Clouds	53000
Cloud_Layer_Enable	53050
Cloud_Layer_Selection	53060
Visibility_53100	53100
Base_Altitude_53200	53200
Base_Altitude_53250	53250
Ceiling_53300	53300
Ceiling_53350	53350
Characteristics	53400
Concentration_Length	53410
Transmittance	53420
Radiance	53430
Precipitation	54000
Rain	54100
Fog	55000
Visibility_55100	55100
Visibility_55101	55101
Visibility_55105	55105
Density_55200	55200
Base	55300
View_Layer_from_above_	55401
Transition_Range	55410

Enumerator	Value
Bottom_55420	55420
Bottom_55425	55425
Ceiling_55430	55430
Ceiling_55435	55435
Heavenly_Bodies	56000
Sun	56100
Sun_Visible	56105
Position_56110	56110
Sun_Position_Elevation_Degrees	56111
Position_Azimuth_56120	56120
Sun_Position_Azimuth_Degrees	56121
Position_Elevation_56130	56130
Position_Intensity_56140	56140
Moon	56200
Moon_Visible	56205
Position_56210	56210
Position_Azimuth_56220	56220
Moon_Position_Azimuth_Degrees	56221
Position_Elevation_56230	56230
Moon_Position_Elevation_Degrees	56231
Position_Intensity_56240	56240
Horizon	56310
Horizon_Azimuth	56320
Horizon_Elevation	56330
Horizon_Heading	56340
Horizon_Intensity	56350
Humidity	57200
Visibility_57300	57300

Enumerator	Value
Winds	57400
Speed_57410	57410
Wind_Speed_Knots	57411
Wind_Direction_57420	57420
Wind_Direction_Degrees	57421
Rainsoak	57500
Tide_Speed	57610
Tide_Speed_Knots	57611
Tide_Direction	57620
Tide_Direction_Degrees	57621
Haze	58000
Visibility_58100	58100
Visibility_58105	58105
Density_58200	58200
Ceiling_58430	58430
Ceiling_58435	58435
Contaminants_and_Obscurants	59000
Contaminant_Obscurant_Type	59100
Persistence	59110
Chemical_Dosage	59115
Chemical_Air_Concentration	59120
Chemical_Ground_Deposition	59125
Chemical_Maximum_Ground_Deposition	59130
Chemical_Dosage_Threshold	59135
Biological_Dosage	59140
Biological_Air_Concentration	59145
Biological_Dosage_Threshold	59150
Biological_Binned_Particle_Count	59155

Enumerator	Value
Radiological_Dosage	59160
Communications	60000
Fire_Bottle_Reload	61005
Channel_Type_61100	61100
Channel_Type_61101	61101
Channel_Identification	61200
Alpha_Identification	61300
Radio_Identification	61400
Land_Line_Identification	61500
Intercom_Identification	61600
Group_Network_Channel_Number	61700
Radio_Communications_Status	62100
Boom_Interphone	62101
Stationary_Radio_Transmitters_Default_Time	62200
Moving_Radio_Transmitters_Default_Time	62300
Stationary_Radio_Signals_Default_Time	62400
Moving_Radio_Signal_Default_Time	62500
Radio_Initialization_Transsec_Security_Key	63101
Radio_Initialization_Internal_Noise_Level	63102
Radio_Initialization_Squelch_Threshold	63103
Radio_Initialization_Antenna_Location	63104
Radio_Initialization_Antenna_Pattern_Type	63105
Radio_Initialization_Antenna_Pattern_Length	63106
Radio_Initialization_Beam_Definition	63107
Radio_Initialization_Transmit_Heartbeat_Time	63108
Radio_Initialization_Transmit_Distance_Threshold_Variable_Record	63109
Radio_Channel_Initialization_Lockout_ID	63110
Radio_Channel_Initialization_Hopset_ID	63111

Enumerator	Value
Radio_Channel_Initialization_Preset_Frequency	63112
Radio_Channel_Initialization_Frequency_Sync_Time	63113
Radio_Channel_Initialization_Comsec_Key	63114
Radio_Channel_Initialization_Alpha	63115
Algorithm_Parameters	70000
Dead_Reckoning_Algorithm_DRA_	71000
DRA_Location_Threshold	71100
DRA_Orientation_Threshold	71200
DRA_Time_Threshold	71300
Simulation_Management_Parameters	72000
Checkpoint_Interval	72100
Transmitter_Time_Threshold	72600
Receiver_Time_Threshold	72700
Interoperability_Mode	73000
SIMNET_Data_Collection RPRnoteEnumerations189	74000
Event_ID	75000
Source_Site_ID	75100
Source_Host_ID	75200
Articulated_Parts	90000
Part_ID	90050
Index	90070
Position_90100	90100
Position_Rate	90200
Extension	90300
Extension_Rate	90400
X_90500	90500
X_rate	90600
Y_90700	90700

Enumerator	Value
Y_rate	90800
Z_90900	90900
Z_rate	91000
Azimuth_91100	91100
Azimuth_Rate	91200
Elevation	91300
Elevation_Rate	91400
Rotation_91500	91500
Rotation_Rate	91600
DRA_Angular_X_Velocity	100001
DRA_Angular_Y_Velocity	100002
DRA_Angular_Z_Velocity	100003
Appearance_Trailing_Effects	100004
Appearance_Hatch	100005
Appearance_Character_Set	100008
Capability_Ammunition_Supplier	100010
Capability_Miscellaneous_Supplier	100011
Capability_Repair_Provider	100012
Articulation_Parameter	100014
Articulation_Parameter_Type_100047	100047
Articulation_Parameter_Value_100048	100048
Time_of_Day_Scene	100058
Latitude_North_Location_of_weather_cell_	100061
Longitude_East_Location_of_weather_cell_	100063
Tactical_Driver_Status	100068
Sonar_System_Status	100100
Accomplished_accept	100160
Upper_latitude	100161

Enumerator	Value
Latitude_South_Location_of_weather_cell_	100162
Western_longitude	100163
Longitude_West_location_of_weather_cell_	100164
CD_ROM_Number_Disk_ID_for_terrain_	100165
DTED_disk_ID	100166
Altitude_100167	100167
Tactical_System_Status	100169
JTIDS_Status	100170
TADIL_J_Status	100171
DSDD_Status	100172
Weapon_System_Status	100200
Subsystem_status	100205
Number_of_interceptors_fired	100206
Number_of_interceptor_detonations	100207
Number_of_message_buffers_dropped	100208
Satellite_sensor_background_year_day_	100213
Satellite_sensor_background_hour_minute_	100214
Script_Number	100218
Entity_Track_Update_Data	100300
Local_Force_Training	100400
Entity_Track_Identity_Data	100500
Entity_for_Track_Event	100510
IFF_Friend_Foe_status	100520
Engagement_Data	100600
Target_Latitude	100610
Target_Longitude	100620
Area_of_Interest_Ground_Impact_Circle_Center_Latitude	100631
Area_of_Interest_Ground_Impact_Circle_Center_Longitude	100632

Enumerator	Value
Area_of_Interest_Ground_Impact_Circle_Radius	100633
Area_of_Interest_Type	100634
Target_Aggregate_ID	100640
GIC_Identification_Number	100650
Estimated_Time_of_Flight_to_TBM_Impact	100660
Estimated_Intercept_Time	100661
Estimated_Time_of_Flight_to_Next_Waypoint	100662
Entity_Track_Equipment_Data	100700
Emission_EW_Data	100800
Appearance_Data	100900
Command_Order_Data	101000
Environmental_Data	101100
Significant_Event_Data	101200
Operator_Action_Data	101300
ADA_Engagement_Mode	101310
ADA_Shooting_Status	101320
ADA_Mode	101321
ADA_Radar_Status	101330
Shoot_Command	101340
ADA_Weapon_Status	101350
ADA_Firing_Disciple	101360
Order_Status	101370
Time_Synchronization	101400
Tomahawk_Data	101500
Number_of_Detonations	102100
Number_of_Intercepts	102200
OBT_Control_MT_201	200201
Sensor_Data_MT_202	200202

Enumerator	Value
Environmental_Data_MT_203	200203
Ownship_Data_MT_204	200204
Acoustic_Contact_Data_MT_205	200205
Sonobuoy_Data_MT_207	200207
Sonobuoy_Contact_Data_MT_210	200210
Helo_Control_MT_211	200211
ESM_Control_Data	200213
ESM_Contact_Data_MT_214	200214
ESM_Emitter_Data_MT_215	200215
Weapon_Definition_Data_MT_217	200216
Weapon_Preset_Data_MT_217	200217
OBT_Control_MT_301	200301
Sensor_Data_MT_302	200302
Environmental_Data_MT_303m	200303
Ownship_Data_MT_304	200304
Acoustic_Contact_Data_MT_305	200305
Sonobuoy_Data_MT_307	200307
Sonobuoy_Contact_Data_MT_310	200310
Helo_Scenario_Equipment_Status	200311
ESM_Control_Data_MT_313	200313
ESM_Contact_Data_MT_314	200314
ESM_Emitter_Data_MT_315	200315
Weapon_Definition_Data_MT_316	200316
Weapon_Preset_Data_MT_317	200317
Pairing_Association_eMT_56_	200400
Pointer_eMT_57_	200401
Reporting_Responsibility_eMT_58_	200402
Track_Number_eMT_59_	200403

Enumerator	Value
ID_for_Link_11_Reporting_eMT_60_	200404
Remote_Track_eMT_62_	200405
Link_11_Error_Rate_eMT_63_	200406
Track_Quality_eMT_64_	200407
Gridlock_eMT_65_	200408
Kill_eMT_66_	200409
Track_ID_Change_Resolution_eMT_68_	200410
Weapons_Status_eMT_69_	200411
Link_11_Operator_eMT_70_	200412
Force_Training_Transmit_eMT_71_	200413
Force_Training_Receive_eMT_72_	200414
Interceptor_Amplification_eMT_75_	200415
Consumables_eMT_78_	200416
Link_11_Local_Track_Quality_eMT_95_	200417
DLRP_eMT_19_	200418
Force_Order_eMT_52_	200419
Wilco_Cantco_eMT_53_	200420
EMC_Bearing_eMT_54_	200421
Change_Track_Eligibility_eMT_55_	200422
Land_Mass_Reference_Point	200423
System_Reference_Point	200424
PU_Amplification	200425
Set_Drift	200426
Begin_Initialization_MT_1_	200427
Status_and_Control_MT_3_	200428
Scintillation_Change_MT_39_	200429
Link_11_ID_Control_MT_61_	200430
PU_Guard_List	200431

Enumerator	Value
Winds_Aloft_MT_14_	200432
Surface_Winds_MT_15_	200433
Sea_State_MT_17_	200434
Magnetic_Variation_MT_37_	200435
Track_Eligibility_MT_29_	200436
Training_Track_Notification	200437
Tacan_Data_MT_32_	200501
Interceptor_Amplification_MT_75_	200502
Tacan_Assignment_MT_76_	200503
Autopilot_Status_MT_77_	200504
Consumables_MT_78_	200505
Downlink_MT_79_	200506
TIN_Report_MT_80_	200507
Special_Point_Control_MT_81_	200508
Control_Discretes_MT_82_	200509
Request_Target_Discretes_MT_83_	200510
Target_Discretes_MT_84_	200511
Reply_Discretes_MT_85_	200512
Command_Maneuvers_MT_86_	200513
Target_Data_MT_87_	200514
Target_Pointer_MT_88_	200515
Intercept_Data_MT_89_	200516
Decrement_Missile_Inventory_MT_90_	200517
Link_4A_Alert_MT_91_	200518
Strike_Control_MT_92_	200519
Speed_Change_MT_25_	200521
Course_Change_MT_26_	200522
Altitude_Change_MT_27_	200523

Enumerator	Value
ACLS_AN_SPN_46_Status	200524
ACLS_Aircraft_Report	200525
SPS_67_Radar_Operator_Functions	200600
SPS_55_Radar_Operator_Functions	200601
SPQ_9A_Radar_Operator_Functions	200602
SPS_49_Radar_Operator_Functions	200603
MK_23_Radar_Operator_Functions	200604
SPS_48_Radar_Operator_Functions	200605
SPS_40_Radar_Operator_Functions	200606
MK_95_Radar_Operator_Functions	200607
Kill_No_Kill	200608
CMT_pc	200609
CMC4AirGlobalData	200610
CMC4GlobalData	200611
LINKSIM_COMMENT_PDU	200612
NSST_Ownship_Control	200613
Other	240000
Mass_Of_The_Vehicle	240001
Force_ID_240002	240002
Entity_Type_Kind	240003
Entity_Type_Domain	240004
Entity_Type_Country	240005
Entity_Type_Category	240006
Entity_Type_Sub_Category	240007
Entity_Type_Specific	240008
Entity_Type_Extra	240009
Alternative_Entity_Type_Kind	240010
Alternative_Entity_Type_Domain	240011

Enumerator	Value
Alternative_Entity_Type_Country	240012
Alternative_Entity_Type_Category	240013
Alternative_Entity_Type_Sub_Category	240014
Alternative_Entity_Type_Specific	240015
Alternative_Entity_Type_Extra	240016
Entity_Location_X	240017
Entity_Location_Y	240018
Entity_Location_Z	240019
Entity_Linear_Velocity_X	240020
Entity_Linear_Velocity_Y	240021
Entity_Linear_Velocity_Z	240022
Entity_Orientation_Psi	240023
Entity_Orientation_Theta	240024
Entity_Orientation_Phi	240025
Dead_Reckoning_Algorithm	240026
Dead_Reckoning_Linear_Acceleration_X	240027
Dead_Reckoning_Linear_Acceleration_Y	240028
Dead_Reckoning_Linear_Acceleration_Z	240029
Dead_Reckoning_Angular_Velocity_X	240030
Dead_Reckoning_Angular_Velocity_Y	240031
Dead_Reckoning_Angular_Velocity_Z	240032
Entity_Appearance	240033
Entity_Marking_Character_Set	240034
Entity_Marking_11_Bytes	240035
Capability	240036
Number_Articulation_Parameters	240037
Articulation_Parameter_ID	240038
Articulation_Parameter_Type_240039	240039

Enumerator	Value
Articulation_Parameter_Value_240040	240040
Type_Of_Stores	240041
Quantity_Of_Stores	240042
Fuel_Quantity	240043
Radar_System_Status	240044
Radio_Communication_System_Status	240045
Default_Time_For_Radio_Transmission_For_Stationary_Transmitters	240046
Default_Time_For_Radio_Transmission_For_Moving_Transmitters	240047
Body_Part_Damaged_Ratio	240048
Name_Of_The_Terrain_Database_File	240049
Name_Of_Local_File	240050
Aimpoint_Bearing	240051
Aimpoint_Elevation	240052
Aimpoint_Range	240053
Air_Speed	240054
Altitude_240055	240055
Application_Status	240056
Auto_Iff	240057
Beacon_Delay	240058
Bingo_Fuel_Setting	240059
Cloud_Bottom	240060
Cloud_Top	240061
Direction	240062
End_Action	240063
Frequency	240064
Freeze	240065
Heading	240066
Identification_240067	240067

Enumerator	Value
Initial_Point_Data	240068
Latitude_240069	240069
Lights_240070	240070
Linear	240071
Longitude_240072	240072
Low_Altitude	240073
Mfd_Formats	240074
Nctr	240075
Number_Projectiles	240076
Operation_Code	240077
Pitch	240078
Profiles	240079
Quantity_240080	240080
Radar_Modes	240081
Radar_Search_Volume	240082
Roll	240083
Rotation_240084	240084
Scale_Factor_X	240085
Scale_Factor_Y	240086
Shields_240087	240087
Steerpoint	240088
Spare1	240089
Spare2	240090
Team	240091
Text	240092
Time_Of_Day	240093
Trail_Flag	240094
Trail_Size	240095

Enumerator	Value
Type_Of_Projectile	240096
Type_Of_Target	240097
Type_Of_Threat	240098
Uhf_Frequency	240099
Utm_Altitude	240100
Utm_Latitude	240101
Utm_Longitude	240102
Vhf_Frequency	240103
Visibility_Range	240104
Void_Aaa_Hit	240105
Void_Collision	240106
Void_Earth_Hit	240107
Void_Friendly	240108
Void_Gun_Hit	240109
Void_Rocket_Hit	240110
Void_Sam_Hit	240111
Weapon_Data	240112
Weapon_Type	240113
Weather_240114	240114
Wind_Direction_240115	240115
Wind_Speed	240116
Wing_Station	240117
Yaw	240118
Memory_Offset	240119
Memory_Data	240120
VASI	240121
Beacon	240122
Strobe	240123

Enumerator	Value
Culture	240124
Approach	240125
Runway_End	240126
Obstruction	240127
Runway_Edge	240128
Ramp_Taxiway	240129
Laser_Bomb_Code	240130
Rack_Type	240131
HUD	240132
RoleFileName	240133
PilotName	240134
PilotDesignation	240135
Model_Type	240136
DIS_Type	240137
Class	240138
Channel	240139
Entity_Type_240140	240140
Alternative_Entity_Type_240141	240141
Entity_Location	240142
Entity_Linear_Velocity	240143
Entity_Orientation	240144
Dead_Reckoning	240145
Failure_Symptom	240146
Max_Fuel	240147
Refueling_Boom_Connect	240148
Altitude_AGL	240149
Calibrated_Airspeed	240150
TACAN_Channel_240151	240151

Enumerator	Value
TACAN_Band_240152	240152
TACAN_Mode_240153	240153
Fuel_Flow_Rate_kg_min_	270115
Fuel_Temperature_degC_	270116
Fuel_Pressure_Pa_	270117
SKE_Slot	270150
SKE_Lead	270151
SKE_Frequency	270152
FCI_Cmd	270153
FCI_Num	270154
SKE_Bit_Field	270155
Formation_Position	270156
Formation_Number	270157
FFS_Mode_Active	270158
FFS_Role	270159
FFS_VCAS	270160
FFS_Bit_Field	270161
FFS_Call_Sign	270162
FFS_Guidance_Data	270163
FFS_Text_Data	270164
FFS_Airdrop_Request_Data	270165
FFS_Airdrop_Data	270166
Horizontal_Circular_Error_Probable_m_	300000
Horizontal_Position_Error_m_	300001
Vertical_Position_Error_m_	300002
Horizontal_Velocity_Error_m_s_	300003
Vertical_Velocity_Error_m_s_	300004
_4th_Lowest_Jammer_to_Signal_Ratio_for_P_Y_L1_dB_	300005

Enumerator	Value
_4th_Lowest_Jammer_to_Signal_Ratio_for_P_Y_L2_dB_	300006
GPS_Figure_of_Merit	300007
Weapon_Transfer_GPS_State	300008
Weapon_Transfer_Horizontal_Position_Error_m_	300009
Weapon_Transfer_Vertical_Position_Error_m_300010	300010
Weapon_Transfer_Vertical_Position_Error_m_300011	300011
Weapon_Transfer_Horizontal_Velocity_Error_m_s_	300012
Time_Transfer_Error_sec_	300013
Age_of_Ephemeris_sec_	300014
Non_Flyout_Munition_Entity_Request_DIS_Type_Enumeration	300016
Non_Flyout_Munition_Entity_Request_Launch_Point_X_m_	300017
Non_Flyout_Munition_Entity_Request_Launch_Point_Y_m_	300018
Non_Flyout_Munition_Entity_Request_Launch_Point_Z_m_	300019
Non_Flyout_Munition_Entity_Request_Maximum_Altitude_m_MSL_	300020
Non_Flyout_Munition_Entity_Request_Flight_Path	300021
Non_Flyout_Munition_Entity_Request_Impact_Point_X_m_	300022
Non_Flyout_Munition_Entity_Request_Impact_Point_Y_m_	300023
Non_Flyout_Munition_Entity_Request_Impact_Point_Z_m_	300024
Non_Flyout_Munition_Entity_Request_Elapsed_Flight_Time_sec_	300025
Non_Flyout_Munition_Entity_Request_Launch_Time_sec_	300026
Time_Error_sec_	300027
Link_16_Command_Variety_1	301100
Push	301130
Rolex	301140
Terminate_Intercept	301150
Heal_Damage	301151
Destroy	301152
Transfer_Control_Management	301160

Enumerator	Value
Link_16_Controls_PPLI_Enable	301170
Link_16_Controls_Command_Control_Enable	301171
Link_16_Reference_Point_Message_Initiation	301174
Assign_External_Entity_Link_16_Track_Number	301175
Link_16_Intelligence_Info	301176
Link_16_Track_Management	301177
Link_16_Controls_CES_Global_PPLI_Publish	301178
Link_16_Controls_CES_Global_Surveillance_Publish	301179
Request_Global_Link_16_Configuration	301180
Link_16_Controls_Surveillance_Enable	301181
Link_16_Pointer	301182
Link_16_Vector	301183
Link_16_Control_Unit_Change	301184
Link_16_Text	301185
Request_Link_16_Objects	301186
Link_16_Ref_Object_Name_List	301187
Total_Number_of_PDUs_in_Link_16_Ref_Objects_Response	301189
PDU_Number_in_Link_16_Ref_Objects_Response	301190
Total_Number_of_Link_16_Ref_Objects	301191
Link_16_Controls_F2F_A_Enable	301197
Link_16_Controls_F2F_B_Enable	301198
STN_of_Formation_Leader	301199
Formation_Name	301200
Formation_Role	301201
Surveillance_Contributor_Sensor_Based_Detection	301202
F2F_A_NPG	301220
Link_16_Controls_F2F_A_Net	301221
F2F_B_NPG	301222

Enumerator	Value
Link_16_Controls_F2F_B_Net	301223
Surveillance_Enabled_NPB	301224
Surveillance_Enabled_Net	301225
Control_Unit_Enabled	301226
Control_Unit_Enabled_NPG	301227
Control_Unit_Enabled_Net	301228
Voice_Frequency	301229
Link_16_JTIDS_Voice_Callsign	301234
Entity_ID_of_Control_Unit	301237
STN_of_Control_Unit	301238
NTR_Participation_Level	301239
Link_16_Controls_CES_Global_PPLI_Subscribe	301240
Link_16_Controls_CES_Global_Surveillance_Subscribe	301241
NTR_in_Mission	301242
NTR_Marking	301243
NTR_Receipt_Compliance	301244
Formation_F2F_NPG	301255
Formation_F2F_Channel	301256
JLVC_JSPA_LogReport	400008
JLVC_JSPA_SupplyAdjust	400009
JLVC_JSPA_EntityControl	400010
JLVC_JSPA_HealthUpdate	400011
JLVC_JSPA_RepairComplete	400012
JLVC_JSPA_UnitActivation	400013
JLVC_JSPA_BattleDamageRepair	400014
JLVC_JSPA_Minefield	400015
JLVC_JSPA_Wire	400016
JLVC_JSPA_Abatis	400017

Enumerator	Value
JLVC_JSPA_Crater	400018
JLVC_JSPA_Ditch	400019
JLVC_JSPA_Lanes	400020
JLVC_JSPA_IED	400021
JLVC_JSPA_Rubble	400022
JLVC_JSPA_SubmergedBarrier	400023
JLVC_JSPA_FloatingBarrier	400024
JLVC_JSPA_Foxhole	400025
JLVC_JSPA_VehicleHole	400026
JLVC_JSPA_VehicleFortification	400027
JLVC_JSPA_Sandbag	400028
JLVC_JSPA_Checkpoint	400029
JLVC_JSPA_ContamCloud2D	400030
JLVC_JSPA_PopulationEffect	400031
JLVC_JSPA_Mine	400032
JLVC_JSPA_SeaMinefield	400033
JLVC_ReasonForEffect	400034
JLVC_PhysicalEffect	400035
JLVC_FunctionalEffect	400036
JLVC_SourceLVCID	400037
JLVC_TargetLVCID	400038
JLVC_EventUUID	400039
JLVC_EffectGeometry	400040
JLVC_EffectRequestAcknowledgement	400041
JLVC_EffectAcknowledgeResponse	400042
JLVC_EffectLifecycle	400043
JLVC_RelatedEventID	400044
JLVC_SystemFunctionalType	400045

Enumerator	Value
JLVC_CapabilityType	400046
JLVC_NetworkFunctionalType	400047
JLVC_TargetEntityType	400048
JLVC_TargetEmitterType	400049
Munition	500001
Engine_Fuel	500002
Storage_Fuel	500003
Not_Used	500004
Expendable	500005
Total_Record_Sets	500006
Launched_Munition	500007
Association	500008
Sensor	500009
Munition_Reload	500010
Engine_Fuel_Reload	500011
Storage_Fuel_Reload	500012
Expendable_Reload	500013
IFF_Change_Control_Mode_1_Code	500014
IFF_Change_Control_Mode_2_Code	500015
IFF_Change_Control_Mode_3_Code	500016
IFF_Change_Control_Mode_4_Code	500017
IFF_Change_Control_Mode_5_Code	500018
IFF_Change_Control_Mode_6_Code	500019
Link_16_Data	500021
ARM_Alert	500022
IFF_Change_Control_Mode_On_Off	500023
Weapon_Status_Data	500024
Expendable_Status_Data	500025

Enumerator	Value
Tactic_Status_Data	500026
Emitter_Sensor_Data	500027
IOS_Control_Data	500028
Static_Status_Data	500029
Request_Inactive_Entities	500200
Inactive_Entity_Quantity	500201
Inactive_Entity_ID	500202
Inactive_Entity_Type	500203
Activation_Trigger_Type	500204
Activation_Trigger_Value	500205
Air_to_Air_Missile_Qty	551001
AIM_7_Missile_Qty	551002
AIM_9_Missile_Qty	551003
AIM_120_Missile_Qty	551004
Air_to_Ground_Missile_Qty	551005
Surface_to_Air_Missile_Qty	551006
Bullet_Qty_551007	551007
Chaff_Qty	552001
Flare_Qty	552002
Fuel_Level	553001
Route_Type	553002
Threat_Mode	553003
Target_Occluded	553004
Terrain_Height	553005
Entity_Status	553006
Marshal_Status	553007
Power_Plant_Status	553008
Nav_Light_Status	553009

Enumerator	Value
Interior_Light_Status	553010
Landing_Light_Status	553011
Formation_Light_Status	553012
Anti_Collision_Light_Status	553013
Nav_Formation_Flash_Rate	553014
Anti_Col_On_Duration	553015
Anti_Col_Off_Duration	553016
Intercept_Status	553017
LifeForm_Signaling_Device_Type	553018
LifeForm_Movement_Type	553019
LifeForm_In_Vehicle	553020
Mobility_Kill_553021	553021
Firepower_Kill	553022
Tanker_Enabled_Disabled	553028
Threat_Status_Tactic_OK_to_Shoot_Down_Weapons	553029
TACAN_Channel_554001	554001
TACAN_Band_554002	554002
TACAN_Mode_554003	554003
RWR_Status	554004
UHF_Radio_Frequency	554005
Emit_Jamming_Status	554006
Emit_Jamming_Type	554007
Receive_Jamming_Status	554008
RADAR_Mode	554009
Available_RADAR_Modes	554010
Jammer_Pod_Enumeration	554100
Jammer_Pod_Behavior	554101
Jammer_Pod_Programs	554102

Enumerator	Value
Jammer_Pod_Receiver_Sensitivity	554103
Jammer_Pod_Receiver_Frequency_Minimum	554104
Jammer_Pod_Receiver_Frequency_Maximum	554105
Jammer_Pod_Power	554106
Jammer_Pod_Variability	554107
Jammer_Pod_Number_of_False_Targets	554108
Jammer_Pod_Jammer_Knob	554109
Jammer_Pod_Missile_Jamming	554110
Emitter_Override_555001	555001
Jammer_Override	555002
Disengage_Reengage_555003	555003
Heading_Override	555004
Altitude_Override	555005
Speed_Override	555006
Verbose_Override	555007
Occlusion_Override	555008
Commit_Range	556001
Current_Scenario_IFF_Mode_4A_Code_for_This_Threat_s_Affiliation	556007
Current_Scenario_IFF_Mode_4B_Code_for_This_Threat_s_Affiliation	556008
Ok_to_Engage_Waypoint_Number	556016
Max_Speed_at_Sea_Level	556017
Max_Speed	556018
Current_Waypoint_Number	556019
Route_Information	556020
Threat_Status_Static_Multi_Target_Track	556029
Air_Air_IR_Missile_Qty	557001
Air_Air_Radar_Missile_Qty	557002
Air_Ground_IR_Missile_Qty	557003

Enumerator	Value
Air_Ground_Radar_Missile_Qty	557004
Air_Ground_Anti_Radiation_Missile_Qty	557005
Air_Ground_Bomb_Qty	557006
Air_Ground_Rocket_Qty	557007
Surface_Air_IR_Missile_Qty	557008
Surface_Air_Radar_Missile_Qty	557009
Bullet_Qty_557010	557010
PPLI_Publish_Enabled	559001
Surveillance_PublishEnabled	559002
NPG	559003
NPG_Channel	559004
JTIDS_Track_Number	559005
Link_16_Controls_Surveillance_Reportable	559006
Link_16_Controls_Surveillance_Track_Quality	559007
Link_16_Controls_Target_Position_Quality	559008
Link_16_Controls_Quality_Error_Type	559009
Link_16_Controls_Affiliation_Determination_Rule	559010
Link_16_Controls_Reset_Entity_Affiliation	559011
Link_16_Controls_Reset_All_Affiliation	559012
End_of_Messages	559999
Malfunction_Activate_Deactivate_Set	600001
Malfunction_Status	600002
Request_JTIDS_Track_Numbers	600210
Track_Numbers_vs_EID	600212
Total_Number_of_JTIDS_Track_Numbers	600214
PDU_Number_in_JTIDS_Track_Number_Response	600215
Total_Number_of_PDUs_in_JTIDS_Track_Number_Response	600216
Air_to_Air_Refueler_Entities_Request	600218

Enumerator	Value
Air_to_Air_Refueling_Count	600219
Air_To_Air_Refueler_Entity	600220
Formation_Library_Request	600300
Total_Number_Formation_Library_PDUs_600301	600301
PDU_Number_in_Formation_Library_Response_600302	600302
Total_Number_Formation_Library_Items_in_PDU	600303
Formation_Library_Variable	600304
Create_Runtime_Formation	600305
Formation_Request_Header	600306
Formation_Position_Absolute	600307
Formation_Position_Relative	600308
Expendables_Reload	610006
Position_Freeze	610007
Activate_Ownship	610008
Chocks	610009
Warm_up_Cool_down_Override	610010
Ground_Power	610011
Scramble_Start	610012
Ownship_as_a_Threat	610013
Fuel_External	610015
Fuel_Internal	610016
Fuel_Tank_Temp	610017
Gross_Weight	610025
Angle_Of_Attack	610026
G_Load	610027
Weight_On_Wheels	610029
Stored_Energy_System_Reload	610032
Kill_Override	610035

Enumerator	Value
Expendables_Freeze	610036
GPS_Satellites_Enable_Disable	610037
Ownship_Message_Display	610040
Weapon_Quantity_Freeze	610042
Global_Control_Freeze_Weapons_Quantity_On_All_Ownships	610043
Global_Control_Freeze_Fuel_Quantity_On_All_Ownships	610044
Global_Control_Freeze_Kill_Override_On_All_Ownships	610045
Global_Control_Freeze_Crash_Override_On_All_Ownships	610046
Ownship_OFP_Block_Number	610047
Waypoint_Information_Query	610048
Waypoint_Information	610049
Ownship_Subsystem_Status_Data	610050
Cockpit_Switch_Status	613002
Integrated_Control_Panel_Messages	613003
Throttle_Positions	613004
Current_Critical_Switch_Position	613005
Correct_Critical_Switch_Position	613006
Current_Critical_Switch_Data	613007
Correct_Critical_Switch_Data	613008
Mission_Initial_Conditions_Set	613013
Global_Control_Malfunction_Active_on_All_Ownships	613016
Global_Control_Malfunction_Clear_On_All_Ownships	613017
Validated_Critical_Switch_Report	613020
SAR_Map_Pathname	613021
Validated_Critical_Switch_Ownship_ID	613022
Lower_Boom_Event_Report	613027
Raise_Boom_Event_Report	613028
Breakaway_Event_Report	613029

Enumerator	Value
Complete_Event_Report	613030
Aux_Comm_Panel_Frequency_Display	613031
Network_Station_Information	615000
Global_Control_Select_Network_Station	615001
Network_Station_Under_Global_Control	615002
Global_Control_Still_Controlling	615003
Global_Control_Release_Control_of_Network_Station	615004
Global_Control_Freeze_Weapon_Quantity	615005
Global_Control_Freeze_Fuel_Quantity	615006
Global_Control_Freeze_Kill_Override	615007
Global_Control_Freeze_Crash_Override	615008
Global_Control_Malfunction_Active	615009
Global_Control_Malfunction_Clear	615010
Global_Control_Start_Devices	615011
Global_Control_Freeze_Devices	615012
Global_Control_JTIDS_Command	615013
Network_Station_IC_Set_Information	615015
Global_Control_Reset_IC_Set	615017
Number_of_Controlling_Units	615018
Network_Station_JTIDS_Controlling_Units	615019
Network_Station_JTIDS_Objective_Tracks	615020
Number_of_Reference_Objects	615021
Network_Station_JTIDS_Reference_Objects	615022
Networked_Station_Still_Under_Control	615023
Global_Control_Delete_Threat_Entities	615024
Network_Station_Ownship_Callsigns	615025
Global_Control_Request_Formation_Library_Data	615026
Total_Number_Formation_Library_PDUs_615027	615027

Enumerator	Value
PDU_Number_in_Formation_Library_Response_615028	615028
Total_Number_Formation_Library_Items_in_PDUs	615029
Network_Station_Formation_Library_Item	615030
Global_Control_Add_Relative_Formation	615031
Network_Station_TIC_Filename	615032
Global_Control_Freeze_Warm_up_Override	615033
Global_Control_Reload_SES	615034
Global_Control_Reload_Weapons	615035
Global_Control_Reload_Expendables	615036
Global_Control_Reload_Fuel	615037
Global_Control_Reload_Firebottle	615038
Test_Pattern_DORT_	700000
Audio_Test_DORT_	700001
Audio_Tone_DORT_	700002
Calibrate_Throttles_DORT_	700003
Operational_Limits_Event_Report	700004
Operational_Limits	700005
Event_Marker_Message	1000620
Receiver_Aircraft_Aero_Model_Data	2000000
Tanker_Aircraft_Aero_Model_Data	2000010
Boom_Aircraft_Aero_Model_Data	2000020
Access_to_Image_Generator_Data_2000030	2000030
Access_to_Image_Generator_Data_2000031	2000031
Access_to_Image_Generator_Data_2000032	2000032
Access_to_Image_Generator_Data_2000033	2000033
Access_to_Image_Generator_Data_2000034	2000034
Access_to_Image_Generator_Data_2000035	2000035
Access_to_Image_Generator_Data_2000036	2000036

Enumerator	Value
Access_to_Image_Generator_Data_2000037	2000037
Access_to_Image_Generator_Data_2000038	2000038
Access_to_Image_Generator_Data_2000039	2000039
Access_to_Image_Generator_Data_2000040	2000040
Access_to_Image_Generator_Data_2000041	2000041
Access_to_Image_Generator_Data_2000042	2000042
Access_to_Image_Generator_Data_2000043	2000043
Access_to_Image_Generator_Data_2000044	2000044
Access_to_Image_Generator_Data_2000045	2000045
Host_Load_Number	2000050
Extended_Fire_Event_Reports	5005001
Battle_Damage_Assessment_BDA_Event_Report	5005002
Extended_Fire_Event_Launcher	5005003
Extended_Fire_Event_Missile	5005006
Extended_Fire_Event_MRM_Weapon	5005008
Extended_Fire_Event_Gun_Fire_Control	5005009
Extended_Fire_Event_Bomb	5005010
Extended_Fire_Event_Expendable	5005011
Battle_Damage_Assessment	5005012
Extended_Fire_Pickle_Event	5005014
Radar_Track_Report	5005055
Jammer_Report	5005060
Jammer_False_Targets_Report	5005061
Detect_Event_Report	5005063
MALD_Beam_Report	5005070
Transmitter_Radiation_Volume RPRnoteEnumerationsDeprecated	5005080
Transmitter_Radiation_Volume_v2	5005081
Call_State	5005100

Enumerator	Value
Call_Endpoint_Data	5005101
Physical_Network_Definition	5007010
Network_Channel_Definition	5007020
Logical_Network_Definition	5007030
Logical_Network_Entity_Definition	5007040
Physical_Network_Entity_Definition	5007050
C2_Message	5008010
Candidate_Object	5008020
Set_of_Candidate_Objects	5008030
Bounded_Region	5008040
Angular_Region	5008050
RoE_Object	5008060
Track_Object	5008070
Set_of_Track_Objects	5008080
Logical_Entity_Definition	5009010
Logical_Entity_Relationship_Definition	5009020
Intent_Based_EW_Message	5507010

VariantsForAirCategory200BirdEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Variants for Air Category 200 - Bird [UID 136]*

Enumerator	Value
Bird_with_Fish	1
V_Pattern_Flock_Shape	2
Circular_Flock_Shape	3
Irregular_Flock_Shape	4

VariantsForLandCategoriesReptilesAmphibiansInsectsAndArachnidsEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Variants for Land Categories Reptiles, Amphibians, Insects, and Arachnids [UID 135]*

Enumerator	Value
Black	1
Green	2
Spotted	3
Red	4
Brown	5

VariantsForSubsurfaceCategory201MammalEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *Variants for Subsurface Category 201 - Mammal [UID 139]*

Enumerator	Value
Singing	1
Spouting	2

2.2. Notes

RPRnoteEnumerations1

Semantics: *The gaps in enumeration values are intentional and are reserved for future additions.*

RPRnoteEnumerations2

Semantics: *Surface includes a water platform surface such as a ship deck*

RPRnoteEnumerations3

Semantics: *Nose and Left/Right Main move in unison; intended for low-fidelity simulations and not to be used if the others are used.*

RPRnoteEnumerations4

Semantics: *Point defense systems like Phalanx guns, or visible chaff, flare, or smoke dispensers.*

RPRnoteEnumerations5

Semantics: *Any radar dish or movable antenna or sensor.*

RPRnoteEnumerations6

Semantics: *Masts are numbered from forward to aft, masts that are side by side are numbered from port to starboard*

RPRnoteEnumerations7

Semantics: *May be a sprocket, idler, or road wheel*

RPRnoteEnumerations8

Semantics: *Any repair part for a modelled function of a platform, not related to Mobility, Firepower, Sensor, or Communications, e.g. Bulldozer Blade, Crane, Winch, Mine Plow, etc.*

RPRnoteEnumerations9

Semantics: *Called Pilots Clearview window in C-17 flight manual*

RPRnoteEnumerations10

Semantics: *Called Copilots Clearview window in C-17 flight manual*

RPRnoteEnumerations11

Semantics: *Collision Type 55 indicates that during an aerial refueling exercise the boom nozzle has cleared the receiver's refueling receptacle.*

RPRnoteEnumerations12

Semantics: *The color values 1 to 15 are defined in the HTML 3.0 specification; color values 20 to 159 are defined in the SVG 1.0 specification*

RPRnoteEnumerations13

Semantics: *Use Australia*

RPRnoteEnumerations14

Semantics: *Use United States Minor Outlying Islands*

RPRnoteEnumerations15

Semantics: *Use French Southern Territories*

RPRnoteEnumerations16

Semantics: *Not to be confused with the neighbouring Democratic Republic of the Congo*

RPRnoteEnumerations17

Semantics: *Now dissolved; succeeded by the Czech Republic and Slovakia*

RPRnoteEnumerations18

Semantics: *Use Israel instead*

RPRnoteEnumerations19

Semantics: *Use Svalbard and Jan Mayen*

RPRnoteEnumerations20

Semantics: *Island no longer occupied, removed from ISO 3166 country list*

RPRnoteEnumerations21

Semantics: *Use Netherlands*

RPRnoteEnumerations22

Semantics: *United Nations recognized the usage of Turkiye in May, 2022.*

RPRnoteEnumerations23

Semantics: *Includes CIS and USSR era types*

RPRnoteEnumerations24

Semantics: *Now Democratic Republic of the Congo*

RPRnoteEnumerations25

Semantics: *Replaced by country code 222*

RPRnoteEnumerations26

Semantics: *Not to be confused with the neighbouring Republic of the Congo*

RPRnoteEnumerations27

Semantics: *UID table only for historical purposes. Do not add new entries*

RPRnoteEnumerations28

Semantics: *There is a requirement for this variant to have an ID separate from the FL 1800.*

RPRnoteEnumerations29

Semantics: *Sometimes known as Leninets B-004*

RPRnoteEnumerations30

Semantics: *Use 28485 value.*

RPRnoteEnumerations31

Semantics: *Add SHMEL National Nomenclature for clarity*

RPRnoteEnumerations32

Semantics: *See also 1638*

RPRnoteEnumerations33

Semantics: *Values 1-999 are reserved for United States IO Simulation Sources – See applicable agreement of the organizers of the event (training, exercise, etc.) in which information operations is included.*

RPRnoteEnumerations34

Semantics: *The 1's digit is reserved for Race/Ethnicity and the enumerations are taken from the U.S. OMB. The 10's digit is reserved for general age group. The 100's digit is reserved for gender, where 0 is Male and 1 is Female.*

RPRnoteEnumerations35

Semantics: *Diemaco is now Colt Canada.*

RPRnoteEnumerations36

Semantics: *Subcategories 150-199 are restricted to non-weapon equipment classes.*

RPRnoteEnumerations37

Semantics: *Subcategories 151-159 are restricted to sensor equipment classes.*

RPRnoteEnumerations38

Semantics: *Subcategories 160-169 are restricted to animal companion classes.*

RPRnoteEnumerations39

Semantics: *Values from 10-29 are restricted to Conventional Armed Forces and may be further clarified by country code-specific values.*

RPRnoteEnumerations40

Semantics: *Values from 30-49 are restricted to Special Operations Forces and may be further clarified by country code-specific values.*

RPRnoteEnumerations41

Semantics: *Values from 50-69 are restricted to Law Enforcement and may be further clarified by country code-specific values.*

RPRnoteEnumerations42

Semantics: *Values from 70-89 are restricted to Non-Military National Government Agencies and may be further clarified by country code-specific values.*

RPRnoteEnumerations43

Semantics: *Values from 90-99 are restricted to Regional / Local Forces (e.g., state guard) and may be further clarified by country code-specific values.*

RPRnoteEnumerations44

Semantics: *Values from 100-109 are restricted to Irregular Forces and may be further clarified by country code-specific values.*

RPRnoteEnumerations45

Semantics: *Values from 110-119 are restricted to Paramilitary Forces and may be further clarified by country code-specific values.*

RPRnoteEnumerations46

Semantics: *Values from 120-129 are restricted to non-governmental Humanitarian Organizations and may be further clarified by country code-specific values.*

RPRnoteEnumerations47

Semantics: *Values from 130-159 are restricted to Civilians and may be further clarified by country code-specific values.*

RPRnoteEnumerations48

Semantics: *Subcategories for Surface Platform Category 101. For Subcategory 1-3. Use the Extra field to specify the capacity 1-255 persons.*

RPRnoteEnumerations49

Semantics: *Where $7_{xy} = xx * 10^y$ equivalent tons of TNT, example: $7126 = 12 * 10^6$ tons of TNT or 12 Megatons of TNT.*

RPRnoteEnumerations50

Semantics: *SATURN was formerly HAVE QUICK IIA (HQIIA)*

RPRnoteEnumerations51

Semantics: *Subcategories for Air Platform Categories 84-88. Subcategory values have the type of engine in the 10's digit and the number of engines in the 1's digit.*

RPRnoteEnumerations52

Semantics: *Subcategories for Air Platform Category 83*

RPRnoteEnumerations53

Semantics: *Subcategories for Air Platform Categories 90-92. Subcategory values have the number of rotors in the 10's digit and the rotor configuration in the 1's digit.*

RPRnoteEnumerations54

Semantics: *Subcategories for Air Platform Category 101*

RPRnoteEnumerations55

Semantics: *Subcategories for Air Platform Category 100*

RPRnoteEnumerations56

Semantics: *Subcategories for Air Platform Category 80*

RPRnoteEnumerations57

Semantics: *Subcategories for Air Platform Category 81*

RPRnoteEnumerations58

Semantics: *Subcategories for Land Platform Category 82*

RPRnoteEnumerations59

Semantics: *Subcategories for Land Platform Category 81. Subcategory values have the general form/style in the 10's/100's digits and the size in the 1's digit. A zero in the 1's digit indicates an unspecified size.*

RPRnoteEnumerations60

Semantics: *No specific enumerations may be placed in this subcategory*

RPRnoteEnumerations61

Semantics: *ACRIS M, N*

RPRnoteEnumerations62

Semantics: *ACRIS E, H, C, D*

RPRnoteEnumerations63

Semantics: *ACRIS I, J, S, R*

RPRnoteEnumerations64

Semantics: *ACRIS F, G, P, U, L, W*

RPRnoteEnumerations65

Semantics: *ACRIS O*

RPRnoteEnumerations66

Semantics: *FV, APC, SP mortars, armored cars, chemical reconnaissance, Anti-Tank guided missile launchers, etc.*

RPRnoteEnumerations67

Semantics: *Engineering vehicle, tracked load carriers, towing vehicles, recovery vehicles, AVLB, etc.*

RPRnoteEnumerations68

Semantics: *Anti-Tank guns, guns and howitzers*

RPRnoteEnumerations69

Semantics: *0-3 tons*

RPRnoteEnumerations70

Semantics: *Greater than 3 tons*

RPRnoteEnumerations71

Semantics: *0-4999 kg weight load*

RPRnoteEnumerations72

Semantics: *Greater than 4999 kg weight load*

RPRnoteEnumerations73

Semantics: *Civilian land vehicles were moved to a group of Category values starting at 80*

RPRnoteEnumerations74

Semantics: *Howtizers and Air Defense guns that aren't mounted on any vehicle or trailer.*

RPRnoteEnumerations75

Semantics: *Missile launchers that aren't mounted on any vehicle or trailer.*

RPRnoteEnumerations76

Semantics: *Subcategories for Land Platform Category 87*

RPRnoteEnumerations77

Semantics: *Subcategories for Land Platform Category 88*

RPRnoteEnumerations78

Semantics: *Subcategories for Land Platform Category 80*

RPRnoteEnumerations79

Semantics: *Subcategories for Land Platform Category 85*

RPRnoteEnumerations80

Semantics: *Subcategories for Land Platform Category 86*

RPRnoteEnumerations81

Semantics: *Subcategories for Land Platform Category 91*

RPRnoteEnumerations82

Semantics: *Subcategories for Land Platform Category 90*

RPRnoteEnumerations83

Semantics: *Subcategories for Land Platform Category 83*

RPRnoteEnumerations84

Semantics: *https://en.wikipedia.org/wiki/Palletized_load_system*

RPRnoteEnumerations85

Semantics: *Subcategories for Land Platform Category 84*

RPRnoteEnumerations86

Semantics: *Subcategories for Land Platform Category 89*

RPRnoteEnumerations87

Semantics: *Subcategories for Land Platform Category 92*

RPRnoteEnumerations88

Semantics: *Subcategories for Land Platform Category 93*

RPRnoteEnumerations89

Semantics: *Subcategories for Subsurface Platform Category 82*

RPRnoteEnumerations90

Semantics: *Includes Rigid Inflatable Boats*

RPRnoteEnumerations91

Semantics: *Subcategories for Surface Platform Category 81.*

RPRnoteEnumerations92

Semantics: *Subcategories for Surface Platform Category 86.*

RPRnoteEnumerations93

Semantics: *Subcategories for Surface Platform Category 88.*

RPRnoteEnumerations94

Semantics: *Subcategories for Surface Platform Category 87.*

RPRnoteEnumerations95

Semantics: *Subcategories for Surface Platform Category 80.*

RPRnoteEnumerations96

Semantics: *Subcategories for Surface Platform Category 84.*

RPRnoteEnumerations97

Semantics: *Subcategories for Surface Platform Category 85.*

RPRnoteEnumerations98

Semantics: *Subcategories for Surface Platform Category 83*

RPRnoteEnumerations99

Semantics: *Subcategories for Surface Platform Category 82.*

RPRnoteEnumerations100

Semantics: *Using Link 16 and SISO-STD-002 to simulate a non-NATO TDL system*

RPRnoteEnumerations101

Semantics: *JETDS Specific Series are reserved in the range 11-100. Each set allows for 255 radios.*

RPRnoteEnumerations102

Semantics: *There are variants of CVSD encoding types that exist in the DIS community. These are not compatible with each other. At least two additional types of CVSD, known as CCTT CVSD and CECOM CVSD, also use Enumeration value 2. Make sure that if you are using CVSD that your system is compatible with other systems playing in an exercise.*

RPRnoteEnumerations103

Semantics: *<http://www.speex.org/>*

RPRnoteEnumerations104

Semantics: *<https://tools.ietf.org/html/rfc6716>*

RPRnoteEnumerations105

Semantics: *Formerly HAVE QUICK IIA (HQIIA)*

RPRnoteEnumerations106

Semantics: *1-49 Marine Birds*

RPRnoteEnumerations107

Semantics: *20-29 Flightless Land Birds*

RPRnoteEnumerations108

Semantics: *30-39 Small birds that fly in large flocks*

RPRnoteEnumerations109

Semantics: *40-49 Large birds that fly in large flocks*

RPRnoteEnumerations110

Semantics: *50-99 Birds of prey*

RPRnoteEnumerations111

Semantics: *1-19 Moth/Butterfly*

RPRnoteEnumerations112

Semantics: *20-29 Fly/Midge*

RPRnoteEnumerations113

Semantics: *30-39 Mosquito/Gnat*

RPRnoteEnumerations114

Semantics: *40-49 Wasp/Hornet*

RPRnoteEnumerations115

Semantics: *50-59 Bee*

RPRnoteEnumerations116

Semantics: *60-69 Beetle*

RPRnoteEnumerations117

Semantics: *70-79 Dragonfly/Damselfly*

RPRnoteEnumerations118

Semantics: *80-89 Locust*

RPRnoteEnumerations119

Semantics: *1-199 Domesticated animals, 200-255 Wild animals*

RPRnoteEnumerations120

Semantics: *1-9 Small dog (22cm/4kg)*

RPRnoteEnumerations121

Semantics: *10-19 Medium dog (50cm/16kg)*

RPRnoteEnumerations122

Semantics: *20-29 Large dog (65cm/40kg)*

RPRnoteEnumerations123

Semantics: *30-39 Very large dog(76cm/61kg)*

RPRnoteEnumerations124

Semantics: *40-49 Small Livestock (sheep/ goat)*

RPRnoteEnumerations125

Semantics: *50-59 Pig*

RPRnoteEnumerations126

Semantics: *60-69 Medium Livestock (cow/steer/bull/oxen)*

RPRnoteEnumerations127

Semantics: *70-79 Medium Livestock harnessed (cart/plow)*

RPRnoteEnumerations128

Semantics: *80-89 Horse/mule/donkey*

RPRnoteEnumerations129

Semantics: *90-99 Horse/mule/donkey with rider or cargo*

RPRnoteEnumerations130

Semantics: *100-109 Camel*

RPRnoteEnumerations131

Semantics: *110-119 Camel with rider or cargo*

RPRnoteEnumerations132

Semantics: *200-209 Rodent*

RPRnoteEnumerations133

Semantics: *1-29 Lizard/tuatara*

RPRnoteEnumerations134

Semantics: *30-59 Non-venomous snake*

RPRnoteEnumerations135

Semantics: *60-89 Venomous snake*

RPRnoteEnumerations136

Semantics: *90-119 Turtle/Tortoise*

RPRnoteEnumerations137

Semantics: *120-149 Crocodilian*

RPRnoteEnumerations138

Semantics: *1-169 Frog/Toad*

RPRnoteEnumerations139

Semantics: *170-229 Salamander/Newt*

RPRnoteEnumerations140

Semantics: *230-255 Caecilian*

RPRnoteEnumerations141

Semantics: *1-59 Beetle/Weevil*

RPRnoteEnumerations142

Semantics: *60-69 Mantis*

RPRnoteEnumerations143

Semantics: *70-79 Cockroach*

RPRnoteEnumerations144

Semantics: *80-89 Ant*

RPRnoteEnumerations145

Semantics: *90-99 Grasshopper/Cricket/Katydid*

RPRnoteEnumerations146

Semantics: *100-109 Centipede/Millipede*

RPRnoteEnumerations147

Semantics: *1-19 Spider*

RPRnoteEnumerations148

Semantics: *20-29 Tick*

RPRnoteEnumerations149

Semantics: *30-39 Scorpion*

RPRnoteEnumerations150

Semantics: *40-49 Harvestmen*

RPRnoteEnumerations151

Semantics: 50-59 *Mite*

RPRnoteEnumerations152

Semantics: 1-49 *Snail*

RPRnoteEnumerations153

Semantics: 50-99 *Slug*

RPRnoteEnumerations154

Semantics: 1-99 *Ameridelphia*, 100-250 *Australidelphia*

RPRnoteEnumerations155

Semantics: 1-89 *Opossum*

RPRnoteEnumerations156

Semantics: 90-99 *Shrew Opossum*

RPRnoteEnumerations157

Semantics: 100-109 *Bandicoots and Bilbies*

RPRnoteEnumerations158

Semantics: 110-149 *Carnivorous Marsupials*

RPRnoteEnumerations159

Semantics: 150-199 *Wallabies, Wallaroos and Kangaroos*

RPRnoteEnumerations160

Semantics: 200-204 *Koalas*

RPRnoteEnumerations161

Semantics: 205-209 *Wombats*

RPRnoteEnumerations162

Semantics: *210-249 Possums, Gliders, and Cuscus*

RPRnoteEnumerations163

Semantics: *1-29 Small size schooling (forage) fish (herring-like, sardines, squid, krill)*

RPRnoteEnumerations164

Semantics: *30-59 Medium size schooling fish (cod-like, hake, haddock)*

RPRnoteEnumerations165

Semantics: *60-89 Large size schooling fish (tuna)*

RPRnoteEnumerations166

Semantics: *90-119 Small shark*

RPRnoteEnumerations167

Semantics: *120-149 Medium shark*

RPRnoteEnumerations168

Semantics: *150-179 Large to vary large shark*

RPRnoteEnumerations169

Semantics: *180-189 Skate/Ray*

RPRnoteEnumerations170

Semantics: *190-199 Eel*

RPRnoteEnumerations171

Semantics: *200-255 Other fish*

RPRnoteEnumerations172

Semantics: *1-49 Whale*

RPRnoteEnumerations173

Semantics: *50-99 Dolphin*

RPRnoteEnumerations174

Semantics: *100-119 Porpoise*

RPRnoteEnumerations175

Semantics: *120-129 Seal*

RPRnoteEnumerations176

Semantics: *130-139 Sea Lion*

RPRnoteEnumerations177

Semantics: *140-149 Walrus*

RPRnoteEnumerations178

Semantics: *150-159 Otter*

RPRnoteEnumerations179

Semantics: *160-164 Manatee*

RPRnoteEnumerations180

Semantics: *200-255 Other Mammals*

RPRnoteEnumerations181

Semantics: *Not used, as it is described in Munitions.*

RPRnoteEnumerations182

Semantics: *Only used for add on equipment like bomb racks.*

RPRnoteEnumerations183

Semantics: *Specific values for Supply Subcategory Fuels (1).*

RPRnoteEnumerations184

Semantics: *Except for underwater acoustics, this was called 'omni-directional', 'omnidirectional' and 'omni directional' in DIS 7, RPR 2 and earlier versions of the DIS and RPR standards, but the intended meaning was a spherical radiation pattern.*

RPRnoteEnumerations185

Semantics: *A low-fidelity representation of SATCOM modulation*

RPRnoteEnumerations186

Semantics: *Angle refers to the modulation of the frequency or phase of the carrier signal*

RPRnoteEnumerations187

Semantics: *See SISO-STD-002, Standard for: Link 16 Simulations*

RPRnoteEnumerations188

Semantics: *See SISO-STD-005 DRAFT, Standard for: Link 11/11B Simulation*

RPRnoteEnumerations189

Semantics: *The variable-datum-value field consists of:32-bit unsigned integer, Data Collection Type; 32-bit unused; and multiple of 64 bits, SIMNET Data Collection Variant.*

RPRnoteEnumerationsDeprecated

Semantics: *Deprecated in SISO-REF-010.*

RPRnoteEnumerationsBitSize01

Semantics: *WARNING: The enumeration is defined in SISO-REF-010 with a size of 1 bit. If interoperability with DIS-based simulation applications is important, limit the value range to 0-1 when modifying this enumeration.*

RPRnoteEnumerationsBitSize02

Semantics: *WARNING: The enumeration is defined in SISO-REF-010 with a size of 2 bits. If interoperability with DIS-based simulation applications is important, limit the value range to 0-3 when modifying this enumeration.*

RPRnoteEnumerationsBitSize03

Semantics: *WARNING: The enumeration is defined in SISO-REF-010 with a size of 3 bits. If interoperability with DIS-based simulation applications is important, limit the value range to 0-7 when modifying this enumeration.*

RPRnoteEnumerationsBitSize04

Semantics: *WARNING: The enumeration is defined in SISO-REF-010 with a size of 4 bits. If interoperability with DIS-based simulation applications is important, limit the value range to 0-15 when modifying this enumeration.*

RPRnoteEnumerationsBitSize05

Semantics: *WARNING: The enumeration is defined in SISO-REF-010 with a size of 5 bits. If interoperability with DIS-based simulation applications is important, limit the value range to 0-31 when modifying this enumeration.*

RPRnoteEnumerationsBitSize06

Semantics: *WARNING: The enumeration is defined in SISO-REF-010 with a size of 6 bits. If interoperability with DIS-based simulation applications is important, limit the value range to 0-63 when modifying this enumeration.*

RPRnoteEnumerationsBitSize07

Semantics: *WARNING: The enumeration is defined in SISO-REF-010 with a size of 7 bits. If interoperability with DIS-based simulation applications is important, limit the value range to 0-127 when modifying this enumeration.*

RPRnoteEnumerationsBitSize12

Semantics: *WARNING: The enumeration is defined in SISO-REF-010 with a size of 12 bits. If interoperability with DIS-based simulation applications is important, limit the value range to 0-4095 when modifying this enumeration.*

RPRnoteEnumerationsBitSize14

Semantics: *WARNING: The enumeration is defined in SISO-REF-010 with a size of 14 bits. If interoperability with DIS-based simulation applications is important, limit the value range to 0-16383 when modifying this enumeration.*

3. Module Base



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	This module provides a common base for RPR based FOM modules. It contains common datatypes and the BaseEntity and EmbeddedSystem object class definitions.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

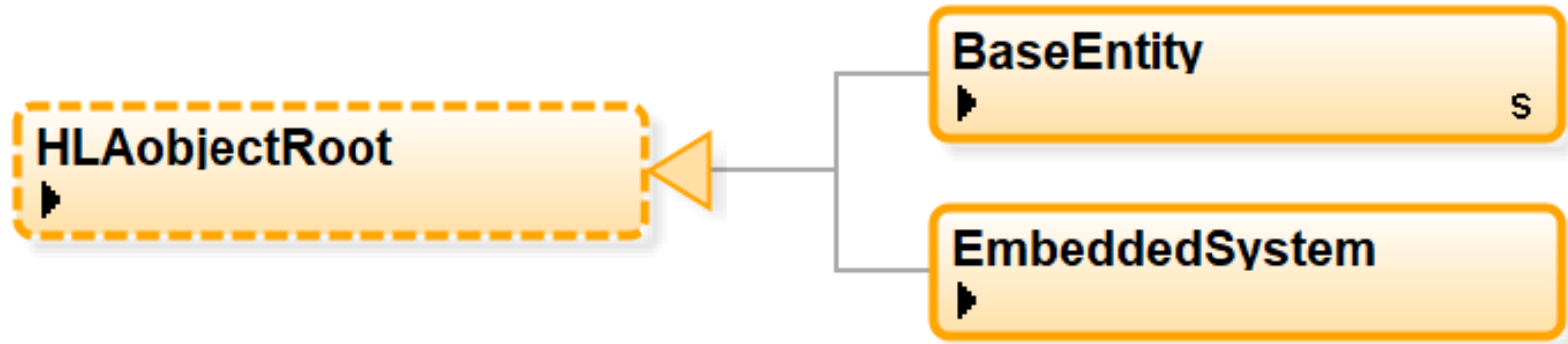
References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Enumerations
Foundation

3.1. Object Classes



3.1.1. BaseEntity [RPRnoteBase1](#)

Full Name: HLAObjectRoot.BaseEntity

Sharing: Subscribe

Semantics: *A base class of aggregate and discrete scenario domain participants. The BaseEntity class is characterized by being located at a particular location in space and independently movable, if capable of movement at all. It specifically excludes elements normally considered to be a component of another element. The BaseEntity class is intended to be a container for common attributes for entities of this type. Since it lacks sufficient class specific attributes that are required for simulation purposes, federates cannot publish objects of this class. Certain simulation management federates, e.g. viewers, may subscribe to this class. Simulation federates will normally subscribe to one of the subclasses, to gain the extra information required to properly simulate the entity.*

Attributes:

EntityType RPRnoteBase2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The category of the entity.					
EntityIdentifier RPRnoteBase2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The unique identifier for the entity instance.					
IsPartOf RPRnoteBase3	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.					
Spatial RPRnoteBase2 RPRnoteBase4	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the world coordinate system.					

RelativeSpatial RPRnoteBase3RPRnoteBase4	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>					

HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

3.1.2. EmbeddedSystem

Full Name: HLAobjectRoot.EmbeddedSystem

Sharing:

Semantics: A base class used to associate components such as sensor and emitting systems with their parent entity object.

Attributes:

EntityIdentifier <u>RPRnoteBase2</u>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The EntityIdentifier of the object which this embedded system is a part of.					
HostObjectIdentifier <u>RPRnoteBase2</u> <u>RPRnoteBase6</u>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The RTI object instance ID of the object of which this embedded system is part of.					

RelativePosition RPRnoteBase2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The position of the embedded system, relative to the host object's position.					
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

3.2. Datatypes

3.2.1. Simple Datatypes

AccelerationMeterPerSecondSquaredFloat32

Representation: HLAAfloat32BE

Units: meter per second squared ($\text{m}/(\text{s}^2)$)

Resolution: NA

Accuracy: NA

Semantics: Linear acceleration vector composed of SI base units. Based on the Linear Acceleration Vector record as specified in IEEE Std 1278.1-2012 Section 6.2.96 item b.

AngleDegreeFloat32

Representation: HLAAfloat32BE

Units: degree (deg)

Resolution: NA

Accuracy: NA

Semantics: Angle, based on unit degree (of arc), unit symbol °.

AngleRadianFloat32

Representation: HLAAfloat32BE

Units: radian (rad)

Resolution: NA

Accuracy: NA

Semantics: Angle, based on SI derived unit radian, unit symbol rad.

AngularVelocityRadianPerSecondFloat32

Representation: HLAAfloat32BE

Units: radian per second (rad/s)

Resolution: NA

Accuracy: perfect

Semantics: Angular velocity vector composed of SI base units. Based on the Angular Velocity Vector record as specified in IEEE Std 1278.1-2012 Section 6.2.7.

BitsUnsignedInteger16

Representation: [RPRunsignedInteger16BE](#)

Units: bit

Resolution: 1

Accuracy: perfect

Semantics: Number of valid data bits.

BitsUnsignedInteger32

Representation: [RPRunsignedInteger32BE](#)

Units: bit

Resolution: 1

Accuracy: perfect

Semantics: Number of valid data bits.

ClockTimeHourInteger32

Representation: HLAinteger32BE

Units: hour

Resolution: 1

Accuracy: perfect

Semantics: Time past on the clock in full hours since a specified point in time.

DepthMeterFloat32

Representation: HLAfloat32BE

Units: meter (m)

Resolution: NA

Accuracy: NA

Semantics: Depth, based on SI base unit meter, unit symbol m.

Float32

Representation: HLAfloat32BE

Units: NA

Resolution: NA

Accuracy: NA

Semantics: Single-precision floating point number.

Float64

Representation: HLAfloat64BE

Units: NA

Resolution: NA

Accuracy: NA

Semantics: Double-precision floating point number.

FrequencyHertzFloat32

Representation: HLAfloat32BE

Units: hertz (Hz)

Resolution: NA

Accuracy: NA

Semantics: Frequency, based on SI derived unit hertz, unit symbol Hz.

Integer16

Representation: HLAinteger16BE

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Integer in the range $[-2^{15}, 2^{15}-1]$.

Integer32

Representation: HLAinteger32BE

Units: NA

Resolution: 1
Accuracy: perfect
Semantics: Integer in the range $[-2^{31}, 2^{31}-1]$.

InterrogationsPerSecondFloat32

Representation: HLAfloat32BE
Units: interrogations/second
Resolution: NA
Accuracy: perfect
Semantics: Number of interrogations per second.

LengthMeterFloat32

Representation: HLAfloat32BE
Units: meter (m)
Resolution: NA
Accuracy: NA
Semantics: Length, based on SI base unit meter, unit symbol m.

MassKilogramFloat32

Representation: HLAfloat32BE
Units: kilogram (kg)
Resolution: NA
Accuracy: NA
Semantics: Mass, based on SI base unit kilogram, unit symbol kg.

MeterFloat32

Representation: HLAfloat32BE
Units: meter (m)
Resolution: NA
Accuracy: perfect
Semantics: Datatype based on SI base unit meter, unit symbol m.

MeterFloat64

Representation: HLAfloat64BE

Units: meter (m)

Resolution: NA

Accuracy: perfect

Semantics: Datatype based on SI base unit meter, unit symbol m.

Octet

Representation: HLAoctet

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Uninterpreted 8-bit value.

PercentFloat32

Representation: HLAfloat32BE

Units: percent (%)

Resolution: NA

Accuracy: NA

Semantics: Percentage

PercentUnsignedInteger8

Representation: [RPRunsignedInteger8BE](#)

Units: percent (%)

Resolution: 1

Accuracy: perfect

Semantics: Percentage

PowerRatioDecibelMilliwattFloat32

Representation: HLAfloat32BE

Units: decibel milliwatt (dBm)

Resolution: NA
Accuracy: perfect
Semantics: Power ratio in decibels (dB) of a measured power referenced to 1 milliwatt (mW).

RevolutionsPerMinuteInteger16

Representation: HLAinteger16BE
Units: revolutions per minute (RPM)
Resolution: 1
Accuracy: NA
Semantics: Frequency of rotation, expressed in revolutions per minute.

TemperatureDegreeCelsiusFloat32

Representation: HLAfloat32BE
Units: degree Celsius (C)
Resolution: NA
Accuracy: NA
Semantics: Temperature, based on SI derived unit degree Celsius, unit symbol °C.

TimeMicrosecondFloat32

Representation: HLAfloat32BE
Units: microsecond
Resolution: NA
Accuracy: NA
Semantics: Time, based on SI base unit second, expressed in microsecond, unit symbol s.

TimeMillisecondUnsignedInteger32

Representation: [RPRunsignedInteger32BE](#)
Units: millisecond (ms)
Resolution: NA
Accuracy: NA
Semantics: Time, based on SI base unit second, expressed in millisecond, unit symbol ms.

TimeSecondFloat32

Representation: HLAfloat32BE

Units: second

Resolution: NA

Accuracy: NA

Semantics: Time in SI base unit second, unit symbol s.

TimeSecondInteger32

Representation: HLAinteger32BE

Units: second (s)

Resolution: 1

Accuracy: perfect

Semantics: Time, based on SI base unit second, unit symbol s.

TimestampUnsignedInteger32

Representation: [RPRunsignedInteger32BE](#)

Units: $3600/(2^{31})$ second

Resolution: 1

Accuracy: perfect

Semantics: The time past the hour, scaled so that value 0 represents the start of the hour and value $2^{31} - 1$ represents one time unit before the start of the next hour, thereby resulting in each time unit representing exactly $3600/(2^{31})$ s, which is approximately 1.67638063 microsecond.

UnsignedInteger16

Representation: [RPRunsignedInteger16BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Integer in the range $[0, 2^{16}-1]$.

UnsignedInteger32

Representation: [RPRunsignedInteger32BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Integer in the range $[0, 2^{32}-1]$.

UnsignedInteger64

Representation: [RPRunsignedInteger64BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Integer in the range $[0, 2^{64}-1]$.

UnsignedInteger8

Representation: [RPRunsignedInteger8BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Integer in the range $[0, 2^8-1]$.

VelocityMeterPerSecondFloat32

Representation: HLAfloat32BE

Units: meter per second (m/s)

Resolution: NA

Accuracy: perfect

Semantics: Speed/Velocity in meter per second.

WavelengthMicronFloat32

Representation: HLAfloat32BE

Units: micron

Resolution: NA

Accuracy: perfect

Semantics: Wavelength expressed in micrometer.

3.2.2. Array Datatypes

ClockTimeStructArray

Element [ClockTimeStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Dynamic array of ClockTimeStruct elements, may also contain no elements.*

EntityTypeStructArray

Element [EntityTypeStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Dynamic array of EntityTypeStruct elements, may also contain no elements.*

Float32Array1Plus

Element [Float32](#)

Type:

Cardinality: [1..2147483647]

Encoding: HLAVariableArray

Semantics: *Generic dynamic array of Float32 elements, containing at least one element.*

LocationStructArray

Element [LocationStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Dynamic array of LocationStruct elements, may also contain no elements.*

OctetArray

Element [Octet](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Generic dynamic array of Octet elements, may also contain no elements.*

OctetArray1Plus

Element [Octet](#)
Type:
Cardinality: [1..2147483647]
Encoding: HLAvariableArray
Semantics: *Generic dynamic array of Octet elements, containing at least one element.*

OctetArray3

Element [Octet](#)
Type:
Cardinality: 3
Encoding: HLAfixedArray
Semantics: *Generic array of three Octet elements.*

OctetArray7

Element [Octet](#)
Type:
Cardinality: 7
Encoding: HLAfixedArray
Semantics: *Generic array of seven Octet elements.*

OpaqueDatum32

Element [Octet](#)
Type:
Cardinality: 4
Encoding: HLAfixedArray

Semantics: *Structure to hold an opaque datum of up to 32 bits. The datum bits are followed by padding bits to align to a 32-bit boundary.*

OpaqueDatum64

Element [Octet](#)

Type:

Cardinality: 8

Encoding: HLAfixedArray

Semantics: *Structure to hold an opaque datum, or part thereof, of up to 64 bits. For larger datums, the datatype OpaqueDatum64Array1Plus may be used. The datum bits are followed by padding bits to align to a 64-bit boundary.*

OpaqueDatum64Array1Plus

Element [OpaqueDatum64](#)

Type:

Cardinality: [1..2147483647]

Encoding: HLAvariableArray

Semantics: *Structure to hold an opaque datum of any number of bits, padded to a 64-bit boundary.*

OrientationStructArray

Element [OrientationStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAvariableArray

Semantics: *Dynamic array of OrientationStruct elements, may also contain no elements.*

UnsignedInteger16Array1Plus

Element [UnsignedInteger16](#)

Type:

Cardinality: [1..2147483647]

Encoding: HLAvariableArray

Semantics: *Generic dynamic array of UnsignedInteger16 elements, containing at least one element.*

UnsignedInteger32Array

Element [UnsignedInteger32](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Generic dynamic array of UnsignedInteger32 elements, may also contain no elements.*

UnsignedInteger8Array

Element [UnsignedInteger8](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Generic dynamic array of UnsignedInteger8 elements, may also contain no elements.*

VariableDatumStructArray

Element [VariableDatumStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Array of VariableDatumStruct.*

VariableParameterVariantStructArray

Element [VariableParameterVariantStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Dynamic array of VariableParameterVariantStruct elements, may also contain no elements.*

3.2.3. Fixed Record Datatypes

AccelerationVectorStruct

Encoding: HLAfixedRecord

Semantics: *The magnitude of the change in linear velocity over time.*

Name	Type	Semantic
XAcceleration	AccelerationMeterPerSecondSquaredFloat32	Acceleration component along the X axis.
YAcceleration	AccelerationMeterPerSecondSquaredFloat32	Acceleration component along the Y axis.
ZAcceleration	AccelerationMeterPerSecondSquaredFloat32	Acceleration component along the Z axis.

AngularVelocityVectorStruct

Encoding: HLAfixedRecord

Semantics: *The rate at which the orientation is changing over time, in body coordinates.*

Name	Type	Semantic
XAngularVelocity	AngularVelocityRadianPerSecondFloat32	Acceleration component about the X axis.
YAngularVelocity	AngularVelocityRadianPerSecondFloat32	Acceleration component about the Y axis.
ZAngularVelocity	AngularVelocityRadianPerSecondFloat32	Acceleration component about the Z axis.

ArticulatedPartVPStruct

Encoding: HLAfixedRecord

Semantics: *Structure to specify a movable part or a combination of a moveable/attached part of an entity. Based on the Articulated Part VP record as specified in IEEE Std 1278.1-2012 Section 6.2.94.2.*

Name	Type	Semantic
ChangeIndicator	UnsignedInteger8	Indicator of a change to the part. This field shall be set to zero for each exercise and sequentially incremented by one for each change in articulation parameters. In the case where all possible values are exhausted, the numbers shall be reused beginning at zero.
PartAttachedTo	UnsignedInteger16	Identification of the articulated part to which this articulation parameter is attached. This field shall contain the value zero if the articulated part is attached directly to the entity.

Name	Type	Semantic
ParameterTypeClass	ArticulatedPartsTypeClassEnum32	<i>The type class uniquely identifies a particular articulated part on a given entity type. Guidance for uniquely assigning type classes to an entity's articulated parts is given in Section 4.8 of the enumeration document (SISO-REF-010).</i>
ParameterTypeMetric	ArticulatedPartsTypeMetricEnum8	<i>The type metric uniquely identifies the transformation to be applied to the articulated part.</i>
Value RPRnoteBase7	Float32	<i>Value of the transformation to be applied to the articulated part.</i>

AttachedPartVPStruct

Encoding: HLAfixedRecord

Semantics: *Structure to specify a removable part that may be attached to an entity. Based on the Attached Part VP record as specified in IEEE Std 1278.1-2012 Section 6.2.94.3.*

Name	Type	Semantic
DetachedIndicator	AttachedPartDetachedIndicatorEnum8	<i>Indicates whether an attached part is attached or detached.</i>
PartAttachedTo	UnsignedInteger16	<i>Identification of the articulated or attached part to which this attached part is attached. This field shall contain the value zero if the attached part is attached directly to the entity.</i>
ParameterType	AttachedPartsEnum32	<i>Identification of the location (or station) to which the part is attached.</i>
AttachedPartType	EntityTypeStruct	<i>The entity type of the attached part.</i>

BeamAntennaStruct

Encoding: HLAfixedRecord

Semantics: *Specifies the direction, pattern, and polarization of radiation from a radio transmitter's antenna.*

Name	Type	Semantic
BeamOrientation	OrientationStruct	<i>The rotation that transforms the reference coordinate system into the beam coordinate system.</i>
BeamAzimuthBeamwidth	AngleRadianFloat32	<i>The full width of the beam to the -3 dB power density points in the x-y plane of the beam coordinate system.</i>
BeamElevationBeamwidth	AngleRadianFloat32	<i>The full width of the beam to the -3 dB power density points in the x-z plane of the beam coordinate system.</i>

Name	Type	Semantic
ReferenceSystem	TransmitterAntennaPatternReferenceSystemEnum8	The reference coordinate system with respect to which beam direction is specified.
Ez	Float32	The magnitude of the Z-component (in beam coordinates) of the Electrical field at some arbitrary single point in the main beam and in the far field of the antenna.
Ex	Float32	The magnitude of the X-component (in beam coordinates) of the Electrical field at some arbitrary single point in the main beam and in the far field of the antenna.
BeamPhaseAngle	AngleRadianFloat32	The phase angle between Ez and Ex in radians.

ClockTimeStruct

Encoding: HLAfixedRecord

Semantics: Specification of the point in time of an occurrence. Based on the Clock Time record as specified in IEEE Std 1278.1-2012 Section 6.2.14.

Name	Type	Semantic
Hours	ClockTimeHourInteger32	The number of hours since 0000 hours, January 1, 1970 UTC.
TimePastTheHour	TimestampUnsignedInteger32	The time past the hour indicated in the Hours field.

ConstituentPartRelationshipStruct

Encoding: HLAfixedRecord

Semantics: The relationship of the constituent part object to its host object. Based on the Relationship record as specified in IEEE Std 1278.1-2012 Section 6.2.74.

Name	Type	Semantic
Nature	IsPartOfNatureEnum16	The nature or purpose for the joining of the constituent part object to the host object.
Position	IsPartOfPositionEnum16	The position of the constituent part object with respect to the host object.

DimensionStruct

Encoding: HLAfixedRecord

Semantics: Bounding box in X,Y,Z axis.

Name	Type	Semantic
XAxisLength	MeterFloat32	Length in meters along X axis.
YAxisLength	MeterFloat32	Length in meters along Y axis.
ZAxisLength	MeterFloat32	Length in meters along Z axis.

EntityAssociationVPStruct

Encoding: HLAfixedRecord

Semantics: *Structure to specify an association or disassociation of two entities, or an entity with another object. Based on the Entity Association VP record as specified in IEEE Std 1278.1-2012 Section 6.2.94.4.*

Name	Type	Semantic
ChangeIndicator	EntityVPRecordChangeIndicatorEnum8	Indicates whether the (dis)association changed since the last issuance.
AssociationStatus	EntityAssociationAssociationStatusEnum8	The association status between two entities, or an entity with another object.
AssociationType	EntityAssociationPhysicalAssociationTypeEnum8	The type of association that exists between two entities, or an entity with another object.
ObjectID	RTIobjectId	The object ID of the entity or other object associated with this entity.
OwnStationLocation	IsPartOfStationNameEnum16	The station location on one's own entity where a physical connection is attached, if known.
PhysicalConnectionType	EntityAssociationPhysicalConnectionTypeEnum8	The type of physical connection, if any, between this entity and another entity or object.
GroupMemberType	EntityAssociationGroupMemberTypeEnum8	The type of member the entity is within the group.
GroupNumber	UnsignedInteger16	The group, if any, to which an entity belongs.

EntityIdentifierStruct

Encoding: HLAfixedRecord

Semantics: *Unique, exercise-wide identification of the entity, or a symbolic group address referencing multiple entities or a simulation application. Based on the Entity Identifier record as specified in IEEE Std 1278.1-2012 Section 6.2.28. Also refer to IEEE Std 1278.1-2012 Section 4.2.5.2 for general requirements for all object identifiers.*

Name	Type	Semantic
FederateIdentifier	FederateIdentifierStruct	Simulation application (federate) identifier.

Name	Type	Semantic
EntityNumber	UnsignedInteger16	<i>The reference number of the entity identifier.</i>

EntityTypeStruct RPRnoteBase8

Encoding: HLAfixedRecord

Semantics: *Type of entity. Based on the Entity Type and Aggregate Type record as specified in IEEE Std 1278.1-2012 Sections 6.2.30 and 6.2.5, respectively.*

Name	Type	Semantic
EntityKind	UnsignedInteger8	<i>Kind of entity.</i>
Domain	UnsignedInteger8	<i>Domain in which the entity operates.</i>
CountryCode	CountryEnum16	<i>Country to which the design of the entity is attributed.</i>
Category	UnsignedInteger8	<i>Main category that describes the entity.</i>
Subcategory	UnsignedInteger8	<i>Subcategory to which an entity belongs based on the Category field.</i>
Specific	UnsignedInteger8	<i>Specific information about an entity based on the Subcategory field.</i>
Extra	UnsignedInteger8	<i>Extra information required to describe a particular entity.</i>

EntityTypeVPStruct

Encoding: HLAfixedRecord

Semantics: *Structure to specify the Entity Type for an entity when it is different than the entity type value contained in the EntityType attribute. Based on the Entity Type VP record as specified in IEEE Std 1278.1-2012 Section 6.2.94.5.*

Name	Type	Semantic
ChangeIndicator	EntityVPRecordChangeIndicatorEnum8	<i>Indicates whether the Entity Type changed since the last issuance.</i>
EntityType	EntityTypeStruct	<i>Entity Type associated with an entity.</i>

EventIdentifierStruct

Encoding: HLAfixedRecord

Semantics: *Identification of an event. Based on the Event Identifier record as specified in IEEE Std 1278.1-2012 Section 6.2.33.*

Name	Type	Semantic
EventCount	UnsignedInteger16	The event number. Uniquely assigned by the simulation application (federate) that initiates the sequence of events. It shall be set to one for each exercise and incremented by one for each event. In the case where all possible values are exhausted, the numbers may be reused beginning again at one.
IssuingObjectIdentifier	RTObjectId	Identification of the object issuing the event.

FederateIdentifierStruct

Encoding: HLAfixedRecord

Semantics: *Unique identification of the simulation application (federate) in an exercise, or a symbolic group address referencing multiple simulation applications. Based on the Simulation Address record as specified in IEEE Std 1278.1-2012 Section 6.2.80.*

Name	Type	Semantic
SiteID	UnsignedInteger16	Each site participating in an event (e.g., training exercise, experiment, and test) shall be assigned a unique SiteID that is different from any other site that is part of the same event. The range of valid SiteID values that may be assigned to a specific site is from 1 to 65 534. The manner in which SiteID values are assigned is part of federation agreements. A SiteID value of NO_SITE (fixed value: 0) indicates no specific site. A SiteID value of ALL_SITES (fixed value: FFFF (H)) indicates all sites.
ApplicationID	UnsignedInteger16	Each application (federate) participating in an event (e.g., training exercise) shall be assigned a unique ApplicationID value for the site with which the application is associated. An application may be assigned multiple ApplicationID values, see IEEE Std 1278.1-2012 Section 4.2.3.3. The range of valid ApplicationID values that may be assigned to a specific application is from 1 to 65 534. The manner in which ApplicationID values are assigned is part of federation agreements. An ApplicationID value of NO_APPLIC (fixed value: 0) indicates no specific application. An ApplicationID value of ALL_APPLIC (fixed value: FFFF (H)) indicates all applications.

IsPartOfStruct

Encoding: HLAfixedRecord

Semantics: *Defines the spatial relationship between two objects.*

Name	Type	Semantic
HostEntityIdentifier	EntityIdentifierStruct	<i>The identifier of the entity of which the object is a constituent part.</i>
HostRTIObjectIdentifier	RTIobjectId	<i>The RTI instance identifier of the object of which this object is a constituent part.</i>
Relationship	ConstituentPartRelationshipStruct	<i>The relationship of the constituent part object to its host object.</i>
NamedLocation	NamedLocationStruct	<i>The discrete positional relationship of the constituent part object with respect to its host object.</i>

LinearSegmentStruct

Encoding: HLAfixedRecord

Semantics: *Specifies linear object segment characteristics.*

Name	Type	Semantic
SegmentNumber	UnsignedInteger8	<i>Identifies the individual segment.</i>
PercentComplete	PercentUnsignedInteger8	<i>Specifies the percent completion of the segment.</i>
IsLocationModified	RPRboolean	<i>Whether the location of the linear segment has been modified since the last update.</i>
IsOrientationModified	RPRboolean	<i>Whether the orientation of the linear segment has been modified since the last update.</i>
Location	LocationStruct	<i>Specifies the location of the segment in the world coordinate system.</i>
Orientation	OrientationStruct	<i>Specifies the orientation of the segment.</i>
Length	MeterFloat32	<i>Specifies the length of the segment, in meters, extending in the positive X direction.</i>
Width	MeterFloat32	<i>Specifies the total width of the segment, in meters; one-half of the width shall extend in the positive Y direction, and one-half of the width shall extend in the negative Y direction.</i>
Height	MeterFloat32	<i>Specifies the height of the segment, in meters, above ground.</i>
Depth	MeterFloat32	<i>Specifies the depth of the segment, in meters, below ground level.</i>
Damage	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the segment.</i>

Name	Type	Semantic
State	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the segment has been deactivated (it has ceased to exist in the synthetic environment).</i>
IsFlaming	RPRboolean	<i>Specifies whether the segment is burning and flames are visible.</i>
Predistributed	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the segment was created before the start of the exercise.</i>
IsSmoking	RPRboolean	<i>Specifies whether or not the segment is smoking (creating a smoke plume).</i>
IEDPresent	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the segment has or contains an IED.</i>

LocalEulerAnglesStruct

Encoding: HLAfixedRecord

Semantics: *The orientation of an object in the local coordinate system, as specified in IEEE Std 1278.1-2012 Section E.8.2.2.*

Name	Type	Semantic
LocalYaw	AngleRadianFloat32	<i>Heading from true North, positive to the right, in radians.</i>
LocalPitch	AngleRadianFloat32	<i>Elevation angle above or below the local horizon, positive up, in radians. Limited to between -pi/2 and +pi/2, inclusive.</i>
LocalRoll	AngleRadianFloat32	<i>Bank angle from the local horizontal, positive tilt to the right (also called right wing down), in radians.</i>

LocationStruct

Encoding: HLAfixedRecord

Semantics: *The location of a point in space. Unless specified otherwise for the attribute, parameter, or datatype field using this datatype, the location is in the world coordinate system, as specified in IEEE Std 1278.1-2012 Section 1.6.3.*

Name	Type	Semantic
X	MeterFloat64	<i>Distance from the origin along the X axis.</i>
Y	MeterFloat64	<i>Distance from the origin along the Y axis.</i>
Z	MeterFloat64	<i>Distance from the origin along the Z axis.</i>

NamedLocationIdentificationStruct

Encoding: HLAfixedRecord

Semantics: *Structure to specify the discrete positional relationship of the part entity with respect to its host entity. Based on the Named Location Identification record as specified in IEEE Std 1278.1-2012 Section 6.2.62.*

Name	Type	Semantic
StationName	IsPartOfStationNameEnum16	The station name within the host at which the part entity is located.
StationNumber	UnsignedInteger16	The number of the particular wing station, cargo hold, and so on, at which the part is attached.

NamedLocationStruct

Encoding: HLAfixedRecord

Semantics: *The discrete positional relationship of the constituent part object with respect to its host object. Based on the specifications in IEEE Std 1278.1-2012 of the IsPartOf PDU Part Location (Section 7.8.5 item e) and Named Location ID (Section 7.8.5 item f) fields.*

Name	Type	Semantic
StationNumber	UnsignedInteger16	The number of the particular station at which the constituent part is attached.
StationName	StationNameLocationVariantStruct	The name of the station where the constituent part is located.

OrientationStruct

Encoding: HLAfixedRecord

Semantics: *The orientation of an object in space. Unless specified otherwise for the attribute, parameter, or datatype field using this datatype, the orientation is in the world coordinate system, as specified in IEEE Std 1278.1-2012 Section 1.6.3*

Name	Type	Semantic
Psi	AngleRadianFloat32	Rotation about the Z axis.
Theta	AngleRadianFloat32	Rotation about the Y axis.
Phi	AngleRadianFloat32	Rotation about the X axis.

RelativePositionStruct

Encoding: HLAfixedRecord

Semantics: *Relative position in right-handed Cartesian coordinates.*

Name	Type	Semantic
BodyXDistance	MeterFloat32	<i>The distance from the reference location along the X axis.</i>
BodyYDistance	MeterFloat32	<i>The distance from the reference location along the Y axis.</i>
BodyZDistance	MeterFloat32	<i>The distance from the reference location along the Z axis.</i>

RelativeRangeBearingStruct

Encoding: HLAfixedRecord

Semantics: *Relative position in polar coordinates.*

Name	Type	Semantic
Range	LengthMeterFloat32	<i>The range from the reference location.</i>
Bearing	AngleRadianFloat32	<i>The bearing from the reference location.</i>

RPRUserDefinedTag

Encoding: HLAfixedRecord

Semantics: *This User Defined Tag enables federations to exchange timestamps for updates and interactions for real-time execution.
Federations that wish to exchange additional information in the User Defined Tag may create a custom version of this fixed record that contains additional fields at the end.*

Name	Type	Semantic
Timestamp	HLAinteger64Time	<i>Time for which the update or interaction is valid. See the Time Representation table for semantics.</i>
IsAbsoluteTime	RPRboolean	<i>Set to true to indicate if the timestamp is Absolute time, i.e. synchronized with the reference time.</i>

ScaledWorldOrientationQuaternionStruct

Encoding: HLAfixedRecord

Semantics: *The orientation of an object in the world coordinate system, described by a quaternion, as specified in IEEE Std 1278.1-2012 Section E.8.2.3.*

Name	Type	Semantic
Scaled0	UnsignedInteger16	Scalar component of the quaternion, scaled to a 16-bit unsigned integer.
X	Float32	X component of the quaternion.
Y	Float32	Y component of the quaternion.
Z	Float32	Z component of the quaternion.

SeparationVPStruct

Encoding: HLAfixedRecord

Semantics: Structure to specify the physical separation of an entity from another entity. Based on the Separation VP record as specified in IEEE Std 1278.1-2012 Section 6.2.94.6.

Name	Type	Semantic
SeparationReason	SeparationVPReasonForSeparationEnum8	The reason for the separation.
PreEntityIndicator	SeparationVPPreEntityIndicatorEnum8	Indicates whether the entity existed prior to the separation and, if so, in what manner.
ParentObjectID	RTObjectId	The object ID of the parent entity/object.
StationLocation	NamedLocationIdentificationStruct	The station that this entity was attached to prior to separation if known.

SpatialFPStruct RPRnoteBase9

Encoding: HLAfixedRecord

Semantics: Spatial structure for Dead Reckoning Algorithm FPW (2) and FPB (6).

Name	Type	Semantic
Location	LocationStruct	Location of the object, in reference system coordinates.
IsFrozen RPRnoteBase10	RPRboolean	Whether the object is frozen or not.
Orientation	OrientationStruct	The angles of rotation around the coordinate axes between the object's attitude and the reference coordinate system axes (calculated as the Tait-Bryan Euler angles specifying the successive rotations needed to transform from the reference coordinate system to the entity coordinate system).
VelocityVector	VelocityVectorStruct	The rate at which an object's position is changing over time, in reference or own body coordinates depending on the Dead Reckoning Algorithm.

Name	Type	Semantic
OtherParameters	DeadReckoningParametersVariantStruct	Optional additional parameters required for more complex dead reckoning algorithms.

SpatialFVStruct [RPRnoteBase9](#)

Encoding: HLAfixedRecord

Semantics: *Spatial structure for Dead Reckoning Algorithm FVW (5) and FVB (9).*

Name	Type	Semantic
Location	LocationStruct	Location of the object, in reference system coordinates.
IsFrozen RPRnoteBase10	RPRboolean	Whether the object is frozen or not.
Orientation	OrientationStruct	The angles of rotation around the coordinate axes between the object's attitude and the reference coordinate system axes (calculated as the Tait-Bryan Euler angles specifying the successive rotations needed to transform from the reference coordinate system to the entity coordinate system).
VelocityVector	VelocityVectorStruct	The rate at which an object's position is changing over time, in reference or own body coordinates depending on the Dead Reckoning Algorithm.
AccelerationVector	AccelerationVectorStruct	The magnitude of the change in linear velocity of an object over time, in reference or own body coordinates depending on the Dead Reckoning Algorithm.
OtherParameters	DeadReckoningParametersVariantStruct	Optional additional parameters required for more complex dead reckoning algorithms.

SpatialRPStruct [RPRnoteBase9](#)

Encoding: HLAfixedRecord

Semantics: *Spatial structure for Dead Reckoning Algorithm RPW (3) and RPB (7).*

Name	Type	Semantic
Location	LocationStruct	Location of the object, in reference system coordinates.
IsFrozen RPRnoteBase10	RPRboolean	Whether the object is frozen or not.

Name	Type	Semantic
Orientation	OrientationStruct	The angles of rotation around the coordinate axes between the object's attitude and the reference coordinate system axes (calculated as the Tait-Bryan Euler angles specifying the successive rotations needed to transform from the reference coordinate system to the entity coordinate system).
VelocityVector	VelocityVectorStruct	The rate at which an object's position is changing over time, in reference or own body coordinates depending on the Dead Reckoning Algorithm.
AngularVelocity	AngularVelocityVectorStruct	The rate at which an object's orientation is changing over time, in body coordinates.
OtherParameters	DeadReckoningParametersVariantStruct	Optional additional parameters required for more complex dead reckoning algorithms.

SpatialRVStruct [RPRnoteBase9](#)

Encoding: HLAfixedRecord

Semantics: *Spatial structure for Dead Reckoning Algorithm RVW (4) and RVB (8).*

Name	Type	Semantic
Location	LocationStruct	Location of the object, in reference system coordinates.
IsFrozen RPRnoteBase10	RPRboolean	Whether the object is frozen or not.
Orientation	OrientationStruct	The angles of rotation around the coordinate axes between the object's attitude and the reference coordinate system axes (calculated as the Tait-Bryan Euler angles specifying the successive rotations needed to transform from the reference coordinate system to the entity coordinate system).
VelocityVector	VelocityVectorStruct	The rate at which an object's position is changing over time, in reference or own body coordinates depending on the Dead Reckoning Algorithm.
AccelerationVector	AccelerationVectorStruct	The magnitude of the change in linear velocity of an object over time, in reference or own body coordinates depending on the Dead Reckoning Algorithm.
AngularVelocity	AngularVelocityVectorStruct	The rate at which an object's orientation is changing over time, in body coordinates.
OtherParameters	DeadReckoningParametersVariantStruct	Optional additional parameters required for more complex dead reckoning algorithms.

SpatialStaticStruct [RPRnoteBase9](#)

Encoding: HLAfixedRecord

Semantics: *Spatial structure for Dead Reckoning Algorithm Static (1).*

Name	Type	Semantic
Location	LocationStruct	Location of the object, in reference system coordinates.
IsFrozen RPRnoteBase10	RPRboolean	Whether the object is frozen or not.
Orientation	OrientationStruct	The angles of rotation around the coordinate axes between the object's attitude and the reference coordinate system axes (calculated as the Tait-Bryan Euler angles specifying the successive rotations needed to transform from the reference coordinate system to the entity coordinate system).
OtherParameters	DeadReckoningParametersVariantStruct	Optional additional parameters required for more complex dead reckoning algorithms.

VariableDatumStruct [RPRnoteBase11](#) [RPRnoteBase12](#)

Encoding: HLAfixedRecord

Semantics: *These fields shall specify the types of variable datum, their length, and their value.*

Name	Type	Semantic
DatumID	VariableRecordTypesEnum32	The fixed datum id represented by a 32-bit enumeration.
DatumLength	BitsUnsignedInteger32	This field shall specify the length of the variable datum in bits.
DatumValue	OpaqueDatum64Array1Plus	The value of the variable datum. The field format depends on the type of datum as specified by the DatumID. If necessary, this field includes padding to a 64-bit boundary.

VelocityVectorStruct

Encoding: HLAfixedRecord

Semantics: *The rate at which the position is changing over time.*

Name	Type	Semantic
XVelocity	VelocityMeterPerSecondFloat32	Velocity component along the X axis.
YVelocity	VelocityMeterPerSecondFloat32	Velocity component along the Y axis.

Name	Type	Semantic
ZVelocity	VelocityMeterPerSecondFloat32	<i>Velocity component along the Z axis.</i>

3.2.4. Variant Record Datatypes

DeadReckoningParametersVariantStruct

Encoding: HLAvariantRecord

Discriminant name: DRParametersType

Discriminant type: [DRParametersTypeEnum8](#)

Semantics: *Variant record for additional dead reckoning parameters.*

Name	Enumerator	Type	Semantics
None	None	RPRnoData	<i>No additional dead reckoning parameter data. See IEEE Std 1278.1-2012 Section E.8.2.1.</i>
LocalEulerAngles	Local_Euler_Angles_Yaw_Pitch_Roll_	LocalEulerAnglesStruct	<i>Alternative for local coordinate Yaw-Pitch-Roll Euler angles.</i>
WorldOrientationQuaternion	World_Orientation_Quaternion	ScaledWorldOrientationQuaternionStruct	<i>Alternative for world coordinate quaternion.</i>

MinefieldSensorTypeVariantStruct

Encoding: RPRextendedVariantRecord

Discriminant name: SensorTypeCategory

Discriminant type: [MinefieldSensorTypesEnum8](#)

Semantics: *Information about a particular mine sensor. Based on the Minefield Sensor Type record as specified in IEEE Std 1278.1-2012 Section 6.2.57.*

Name	Enumerator	Type	Semantics
Optical	Optical	MinefieldSensorTypesOpticalEnum16	<i>Optical subcategory.</i>
FLIR	FLIR	MinefieldSensorTypesFLIREnum16	<i>FLIR subcategory.</i>
RADAR	RADAR	MinefieldSensorTypesRADAREnum16	<i>RADAR subcategory.</i>
Magnetic	Magnetic	MinefieldSensorTypesMagneticEnum16	<i>Magnetic subcategory.</i>
Laser	Laser	MinefieldSensorTypesLaserEnum16	<i>Laser subcategory.</i>
SONAR	SONAR	MinefieldSensorTypesSONAREnum16	<i>SONAR subcategory.</i>
Physical	Physical	MinefieldSensorTypesPhysicalEnum16	<i>Physical subcategory.</i>

Name	Enumerator	Type	Semantics
Multispectral	Multispectral	MinefieldSensorTypesMultispectralEnum16	<i>Multispectral subcategory.</i>

SpatialVariantStruct

Encoding: HLAvariantRecord

Discriminant name: DeadReckoningAlgorithm

Discriminant type: [DeadReckoningAlgorithmEnum8](#)

Semantics: *Variant Record for a single spatial attribute.*

Name	Enumerator	Type	Semantics
SpatialStatic	Static	SpatialStaticStruct	<i>Variant for representing a static object.</i>
SpatialFPW	DRM_FPW	SpatialFPStruct	<i>Variant for representing an object with a constant velocity (or low acceleration) linear motion. With the linear velocity provided in reference system coordinates (world or host entity's coordinate system).</i>
SpatialRPW	DRM_RPW	SpatialRPStruct	<i>Variant for representing an object with a constant velocity (or low acceleration) linear motion, including rotation information. With the linear velocity provided in reference system coordinates (world or host entity's coordinate system).</i>
SpatialRVW	DRM_RVW	SpatialRVStruct	<i>Variant for representing an object with high speed or maneuvering at any speed, including rotation information. With the linear velocity and acceleration provided in reference system coordinates (world or host entity's coordinate system).</i>
SpatialFVW	DRM_FVW	SpatialFVStruct	<i>Variant for representing an object with high speed or maneuvering at any speed. With the linear velocity and acceleration provided in reference system coordinates (world or host entity's coordinate system).</i>
SpatialFPB	DRM_FPB	SpatialFPStruct	<i>Variant for representing an object with a constant velocity (or low acceleration) linear motion. With the linear velocity provided in own body axis coordinates.</i>

Name	Enumerator	Type	Semantics
SpatialRPB	DRM_RPB	SpatialRPStruct	Variant for representing an object with a constant velocity (or low acceleration) linear motion, including rotation information. With the linear velocity provided in own body axis coordinates.
SpatialRVB	DRM_RVB	SpatialRVStruct	Variant for representing an object with high speed or maneuvering at any speed, including rotation information. With the linear velocity and acceleration provided in own body axis coordinates.
SpatialFVB	DRM_FVB	SpatialFVStruct	Variant for representing an object with high speed or maneuvering at any speed. With the linear velocity and acceleration provided in own body axis coordinates.

StationNameLocationVariantStruct [RPRnoteBase13](#)

Encoding: HLAvariantRecord

Discriminant name: StationName

Discriminant type: [IsPartOfStationNameEnum16](#)

Semantics: The station name at which the constituent part is located. In case of 'On Station', the alternative specifies its location relative to the host object.

Name	Enumerator	Type	Semantics
RelativeLocation	On_Station_xyz	RelativePositionStruct	The location of the constituent part object relative to the host object entity coordinate system.
RelativeRangeAndBearing	On_Station_Range_and_Bearing	RelativeRangeBearingStruct	The location of the constituent part object relative to the host object in polar coordinates.

VariableParameterVariantStruct

Encoding: RPRextendedVariantRecord

Discriminant name: RecordType

Discriminant type: [VariableParameterRecordTypeEnum8](#)

Semantics: *Variant record specifying information on articulated parts, attached parts, and other data associated with an entity or detonation. Based on the Variable Parameter record as specified in IEEE Std 1278.1-2012 Section 6.2.94.*

Note that usage of this datatype for the PhysicalEntity object class and MunitionDetonation interaction class attribute/parameter VariableParametersArray shall be in accordance with IEEE Std 1278.1-2012 Annex I.

Name	Enumerator	Type	Semantics
ArticulatedPart	Articulated_Part	ArticulatedPartVPStruct	<i>Alternative for an articulated part.</i>
AttachedPart	Attached_Part	AttachedPartVPStruct	<i>Alternative for an attached part.</i>
Separation	Separation	SeparationVPStruct	<i>Alternative for a separation.</i>
EntityType	Entity_Type	EntityTypeVPStruct	<i>Alternative for an entity type.</i>
EntityAssociation	Entity_Association	EntityAssociationVPStruct	<i>Alternative for an entity association.</i>

3.3. User Supplied Tags

Update/Reflect

Datatype: [RPRUserDefinedTag](#)

Semantics: *User-supplied tag provided with each update/reflect of object instance attribute values. Contains (at least) the time at which the provided data is valid.*

Send/Receive

Datatype: [RPRUserDefinedTag](#)

Semantics: *User-supplied tag provided with each send/receive of an interaction. Contains (at least) the time at which the provided data is valid.*

Delete/Remove

Datatype: NA

Semantics: NA

Divestiture Request

Datatype: NA

Semantics: NA

Divestiture Completion

Datatype: NA

Semantics: NA

Acquisition Request

Datatype: NA

Semantics: NA

Request Update

Datatype: NA

Semantics: NA

3.4. Time Representation

Timestamp

Datatype: HLAinteger64Time

Semantics: *Number of microseconds since the Epoch. The default Epoch is 00:00 Jan 1, 1970 in the UTC+0 time zone, but other Epochs may be agreed by a federation. Leap days are included in the calculation, but leap seconds are not.*

Lookahead

Datatype: HLAinteger64Time

Semantics: *Number or microseconds.*

3.5. Synchronization Points

RPRobjectsRegistered

Tag Datatype: NA

Capability: Not Applicable

Semantics: *Indicates that all object instances have been registered.*

RPRattributesUpdated

Tag Datatype: NA

Capability: Not Applicable

Semantics: *Indicates that all required updates with initial values have been provided for all registered instances.*

3.6. Notes

RPRnoteBase1

Semantics: *Federates shall send the time at which the data is valid in the user-supplied tag with every attribute values update and interaction; see datatype RPRUserDefinedTag for the encoding. All federates shall transmit this field, even if they do not use it themselves, so that other federates can use its value to compensate for network transport delays.*

RPRnoteBase2

Semantics: *Not optional*

RPRnoteBase3

Semantics: *Default value: all zeros*

RPRnoteBase4

Semantics: *If the entity is a constituent part of another entity (denoted by the IsPartOf attribute being set appropriately) then the Spatial attribute may be ignored by a receiving federate. Instead, the receiving federate can calculate spatial attribute values by adding the offsets provided in the RelativeSpatial attribute to the values provided in the host entity's Spatial attribute. Even if a federate is updating RelativeSpatial, it should still update Spatial for the benefit of federates who do not subscribe to the optional RelativeSpatial and IsPartOf attributes.*

RPRnoteBase5

Semantics: *The update condition shall be evaluated as follows:
The owner of a base entity object shall maintain two state models of the object in support of the dead reckoning process. One model shall be the internal model used by the simulation application to represent that object. The other shall be a dead reckoning model of the object. Certain thresholds shall be established as criteria for determining if the object's actual TSPI data has varied by an allowable amount from the dead reckoned TSPI data. The update condition is TRUE when either:
a) the object's actual position differs from the dead reckoned position by more than DRA_POS_THRSH (default value: 1 m), or
b) the object's actual orientation differs from the dead reckoned orientation by more than DRA_ORIENT_THRSH (default value: 3 degree).
See IEEE Std 1278.1-2012 Sections 4.6.6 and 5.3.2.3 and Annex E for more details on dead reckoning.*

RPRnoteBase6

Semantics: *This must reference a valid Object instance.*

RPRnoteBase7

Semantics: *The units of the Value field depends on the value of the ParameterTypeMetric field. The units are defined in IEEE Std 1278.1-2012 Section I.2.2.5.*

RPRnoteBase8

Semantics: *All fields in the entity type struct are enumerations. The values for the individual fields are to be derived from the federation agreements, which could refer to SISO-REF-010. The values used in this structure should comply with the requirements specified in IEEE Std 1278.1-2012 Section 6.2.30 (for platform and environmental entities) and Section 6.2.5 (for aggregate entities).*

RPRnoteBase9

Semantics: *Frozen entities should not be dead-reckoned, i.e. should be displayed as fixed at the current location even if non-zero velocity, acceleration or rotation data received from the frozen entity.*

RPRnoteBase10

Semantics: *If the entity is a constituent part of another entity (denoted by the IsPartOf attribute being set appropriately) then the IsFrozen attribute is no longer updated. The frozen status of the entity is the same as the frozen status of the host entity.*

RPRnoteBase11

Semantics: *The DatumLength equals the length in bits of the DatumValue only. The total size of a VariableDatumStruct record must account for the padding length.*

RPRnoteBase12

Semantics: *The type of the DatumValue field is determined by the value of the DatumID field. The types and associated units, etc., for each of the DatumID enumeration values are to be derived from the federation agreements, which could refer to SISO-REF-010. The DatumValue element type is defined as a UnsignedInteger64 (64 bits) to ensure the correct byte alignment for types that include 64-bit elements.*

RPRnoteBase13

Semantics: *This note applies when this datatype is used within the BaseEntity.IsPartOf attribute. The following StationName enumerations - On station RNG/BRG (15) and On station - x,y,z (16), are optional to transmit along with associated RelativeLocation or RelativeRangeAndBearing information. If these enumeration values (15) and (16) are received, they shall be ignored. The RelativeSpatial attribute shall be used in all cases. (Note: The RelativeLocation field uses the same Location contained in the RelativeSpatial attribute. In both these cases, the values represent the relative location of a part entity to the host entity in the referenced entity coordinate system. The RelativeRangeAndBearing field does not provide full relative spatial data and, therefore, cannot be substituted for the RelativeSpatial attribute.)*

4. Module Physical



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Physical FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	This module provides object class definitions for representing physical entities including aircraft, ground vehicles, ships, life forms, ammunition, etc. In addition, it provides interaction classes to signal collisions between physical entities.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

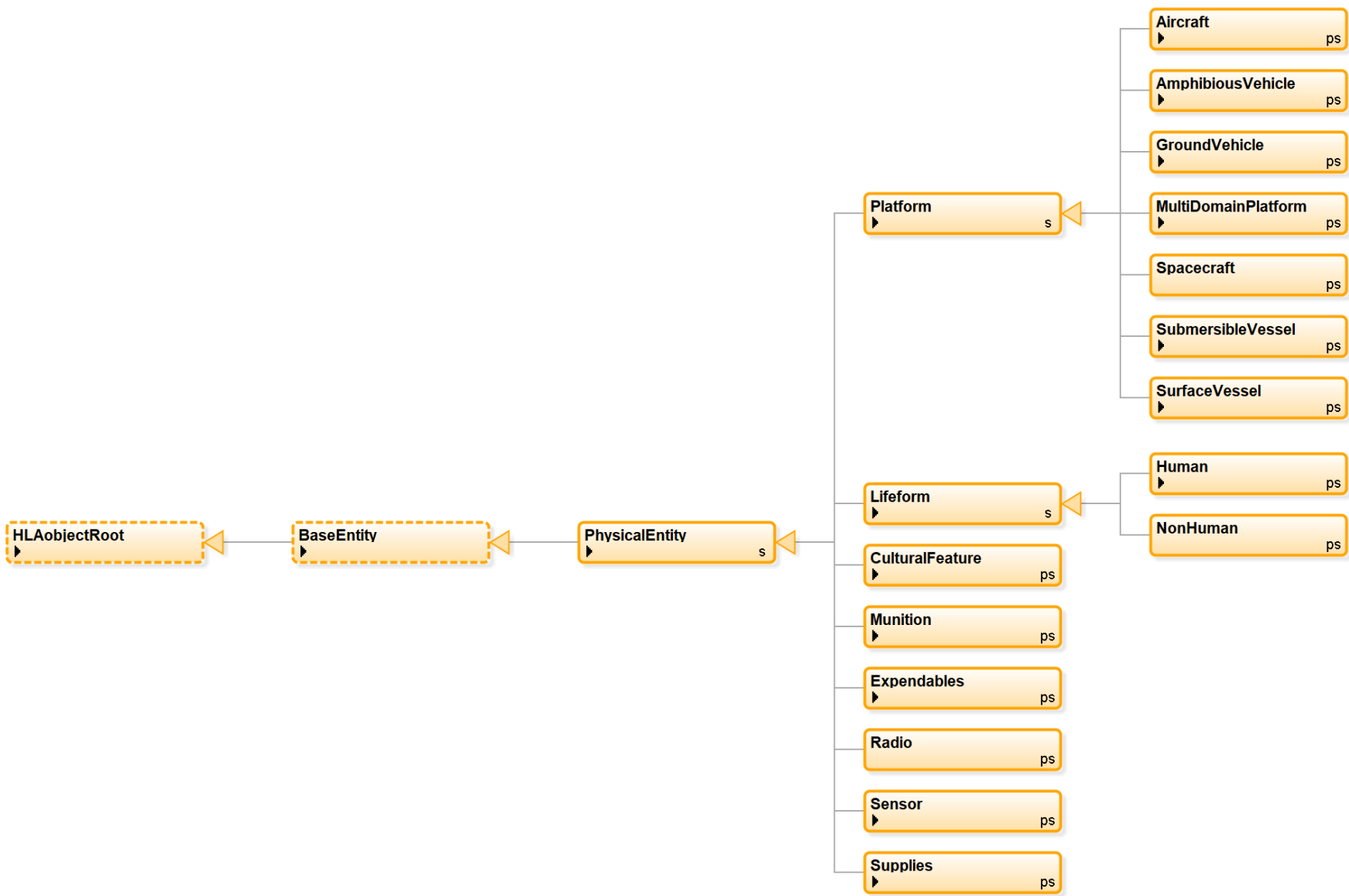
References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Base
Foundation
Enumerations

4.1. Object Classes



4.1.1. PhysicalEntity

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity

Sharing: Subscribe

Semantics: *A base class of all discrete platform scenario domain participants.*

Attributes:

LVCIndicator <u>RPRnotePhysical1</u>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.					
AcousticSignatureIndex <u>RPRnotePhysical2</u>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Index used to obtain the acoustics (sound through air) signature state of the entity.					
AlternateEntityType <u>RPRnotePhysical3</u>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The category of entity to be used when viewed by entities on the 'opposite' side.					
ForceIdentifier <u>RPRnotePhysical4</u>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The identification of the force that the entity belongs to.					

InfraredSignatureIndex RPRnotePhysical2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed RPRnotePhysical2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					
Marking RPRnotePhysical5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					
PropulsionSystemsData RPRnotePhysical5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					
RadarCrossSectionSignatureIndex RPRnotePhysical2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the radar cross section signature state of the entity.</i>					

VectoringNozzleSystemData RPRnotePhysical5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The basic operational data for the vectoring nozzle systems aboard the entity.				
ExtendedDamageStatus RPRnotePhysical5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Detailed information on damage sustained by the entity.				
VariableParametersArray RPRnotePhysical5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Array of information on articulated parts, attached parts, and other data associated with an entity.				
EntityType RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The category of the entity.				
EntityIdentifier RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The unique identifier for the entity instance.				

IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>					
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>					
RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>					
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.2. Platform

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Platform

Sharing: Subscribe

Semantics: *A physical object under the control of armed forces upon which sensor, communication, or weapon systems may be mounted.*

Attributes:

PaintScheme RPRnotePhysical6 RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the visual paint design.</i>					
MobilityKilled RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 RPRnotePhysical13	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether it is capable of moving on its own power.</i>					
Damage RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical14	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the damaged appearance.</i>					
IsSmokeEmanating RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not smoke is emanating from the entity.</i>					
IsEngineEmittingSmoke RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not the engine is emitting smoke.</i>					

IsFlaming RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is burning and flames are visible.</i>					
PowerPlantOn RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the power plant is on or off.</i>					
AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.</i>					
FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.</i>					
RecoveryCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.</i>					

RepairCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether the entity is able to supply certain repair services in response to an appropriate service request.</i>				
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>				
AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>				
AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>				
ForceIdentifier RPRnotePhysical4 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The identification of the force that the entity belongs to.</i>				

InfraredSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the infra-red signature state of the entity.</i>				
LiveEntityMeasuredSpeed RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>				
Marking RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>				
PropulsionSystemsData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operating data of the propulsion systems aboard the entity.</i>				
RadarCrossSectionSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the radar cross section signature state of the entity.</i>				

VectoringNozzleSystemData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>				
ExtendedDamageStatus RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Detailed information on damage sustained by the entity.</i>				
VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>				
EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The category of the entity.</i>				
EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The unique identifier for the entity instance.</i>				

IsPartOf RPRnoteBase3 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.					
Spatial RPRnoteBase2 RPRnoteBase4 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the world coordinate system.					
RelativeSpatial RPRnoteBase3 RPRnoteBase4 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.					
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.3. Aircraft

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Platform.Aircraft

Sharing: Publish/Subscribe

Semantics: A platform entity that operates mainly in the air, such as aircraft, balloons, etc. This includes the entities when they are on the ground.

Attributes:

NVGMode RPRnotePhysical8 RPRnotePhysical20	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceNVGModeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether air platform lighting is in covert or overt mode.					
TrailingEffects RPRnotePhysical8 RPRnotePhysical21	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceTrailingEffectsEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the size of the contrails or ionization trailing effects.					
CanopyTroopDoor RPRnotePhysical8 RPRnotePhysical22	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceCanopyEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the state of the canopy/troop door.					
LandingLightsOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether landing lights are on or off.					
NavigationLightsOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether navigation lights are on or off.					

AntiCollisionLightsOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether Anti-Collision lights are on or off.</i>				
AfterburnerOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes if the air platform is in afterburner.</i>				
LowerAntiCollisionLightOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether the lower Anti-Collision light is on or off.</i>				
UpperAntiCollisionLightOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether the upper Anti-Collision light is on or off.</i>				
AntiCollisionLightDayNight RPRnotePhysical8 RPRnotePhysical23	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceAntiCollisionDayNightEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the day/night status of the Anti-Collision lights.</i>				

IsBlinking RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether any air platform lights are blinking or not.					

FormationLightsOn RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether formation lights are on or off.					

LandingGearExtended RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the landing gear is wholly retracted or extended.					

CargoDoorsOpened RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the cargo doors (main door) are closed or open.					

NavigationPositionBrightness RPRnotePhysical8RPRnotePhysical24	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceNavigationPositionBrightnessEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the brightness of the navigation/position lights.					

SpotSearchLight_1On RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether spot/search light #1 is on or off.					
InteriorLightsOn RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether interior lights are on or off.					
ReverseThrustEngaged RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the air platform has engaged reverse thrust.					
WeightOnWheels RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the air platform has weight on its main landing gear.					
ADSBCapability RPRnotePhysical12RPRnotePhysical16	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is equipped with Automatic Dependent Surveillance - Broadcast (ADS-B).					

SlingLoadCarrierCapability RPRnotePhysical12 RPRnotePhysical16	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity is able to carry a payload in a sling load. The extended appearance record (if available) will identify the current sling load status and entity association and/or entity offset records (if available) will provide additional sling load details (such as payload).					
SlingLoadableCapability RPRnotePhysical12 RPRnotePhysical16	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity is able to be carried as a sling load payload. The extended appearance record (if available) will identify if it is currently sling loaded and entity association and/or entity offset records (if available) will provide additional sling load details (such as carrier).					
IEDPresenceIndicator RPRnotePhysical12 RPRnotePhysical16	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.					
TaskOrganizableCapability RPRnotePhysical12 RPRnotePhysical16	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity (normally a virtual manned module) can be task organized into an existing mixed mode unit (where mixed mode is intended to comprise a combination of computer-generated forces and virtual or even live forces).					

LAIRCMCapability RPRnotePhysical12 RPRnotePhysical16	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether the entity is equipped with Large Aircraft Infrared Countermeasures (LAIRCM).</i>				
PaintScheme RPRnotePhysical6 RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the visual paint design.</i>				
MobilityKilled RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 RPRnotePhysical13 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether it is capable of moving on its own power.</i>				
Damage RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical14 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the damaged appearance.</i>				
IsSmokeEmanating RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether or not smoke is emanating from the entity.</i>				

IsEngineEmittingSmoke RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not the engine is emitting smoke.</i>					
IsFlaming RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is burning and flames are visible.</i>					
PowerPlantOn RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the power plant is on or off.</i>					
AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.</i>					
FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.</i>					

RecoveryCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.</i>						
RepairCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Describes whether the entity is able to supply certain repair services in response to an appropriate service request.</i>						
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>						
AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>						
AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Static	<i>NA</i>					
	Semantics						
	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>						

ForceIdentifier <i>RPRnotePhysical4</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					
Marking <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					
PropulsionSystemsData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					

RadarCrossSectionSignatureIndex RPRnotePhysical2 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Index used to obtain the radar cross section signature state of the entity.				
VectoringNozzleSystemData RPRnotePhysical5 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The basic operational data for the vectoring nozzle systems aboard the entity.				
ExtendedDamageStatus RPRnotePhysical5 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Detailed information on damage sustained by the entity.				
VariableParametersArray RPRnotePhysical5 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Array of information on articulated parts, attached parts, and other data associated with an entity.				
EntityType RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The category of the entity.				

EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The unique identifier for the entity instance.</i>					
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>					
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>					
RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>					

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.4. AmphibiousVehicle

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Platform.AmphibiousVehicle

Sharing: Publish/Subscribe

Semantics: *A platform entity that can operate both on the land and the sea.*

Attributes:

FirePowerKilled RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether it can fire weapons.</i>				
TrailingEffects RPRnotePhysical7 RPRnotePhysical9 RPRnotePhysical21 RPRnotePhysical25	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceTrailingEffectsEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the size of the dust cloud or wake trailing effect.</i>				
PrimaryHatch RPRnotePhysical7 RPRnotePhysical26	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceHatchEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether the primary hatch is open or closed and whether someone is visible.</i>				

HeadLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether head lights are on or off.				
TailLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether tail lights are on or off.				
BrakeLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether brake lights are on or off.				
LauncherOperational RPRnotePhysical7 RPRnotePhysical27	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLauncherOperationalEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the status of the mechanism required for the platform to be operational, such as the elevated status of the primary launcher.				
CamouflageType RPRnotePhysical7 RPRnotePhysical28	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceCamouflageTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the camouflage color.				

ConcealedPosition RPRnotePhysical7 RPRnotePhysical29	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceConcealedPositionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the type of concealment.				
TentExtended RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the tent is extended.				
RampExtended RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the ramp is extended.				
BlackoutLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether blackout lights are on or off.				
BlackoutBrakeLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether blackout brake lights are on or off.				

SpotSearchLight_1On RPRnotePhysical7 RPRnotePhysical9 RPRnotePhysical12 RPRnotePhysical30	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether spot/search light #1 is on or off.					

InteriorLightsOn RPRnotePhysical7 RPRnotePhysical9 RPRnotePhysical12 RPRnotePhysical31	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether interior lights (forward) are on or off.					

OccupantsSurrendered RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the vehicle occupants have surrendered.					

MaskedCloaked RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the entity is masked or cloaked.					

RunningLightsOn RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether running lights are on or off.					

IsAccommodationLadderLowered RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes if the accommodation ladder is lowered or not.				
IsFenceRaised RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the safety fence around a helicopter landing deck is raised.				
IsFlagRaised RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the national identification flag is raised.				
TowedArraySonar RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the TAS is deployed or not.				
SlingLoadableCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical17	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The Entity is able to be carried as a sling load payload. The extended appearance record (if available) will identify if it is currently sling loaded and entity association and/or entity offset records (if available) will provide additional sling load details (such as carrier).				

IEDPresenceIndicator RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical17	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.</i>					
TaskOrganizableCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical17	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The Entity (normally a virtual manned module) can be task organized into an existing mixed mode unit (where mixed mode is intended to comprise a combination of computer-generated forces and virtual or even live forces).</i>					
PaintScheme RPRnotePhysical6 RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the visual paint design.</i>					
MobilityKilled RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 RPRnotePhysical13 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether it is capable of moving on its own power.</i>					
Damage RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical14 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the damaged appearance.</i>					

IsSmokeEmanating RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not smoke is emanating from the entity.</i>					
IsEngineEmittingSmoke RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not the engine is emitting smoke.</i>					
IsFlaming RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is burning and flames are visible.</i>					
PowerPlantOn RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the power plant is on or off.</i>					
AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.</i>					

FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.</i>					
RecoveryCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.</i>					
RepairCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply certain repair services in response to an appropriate service request.</i>					
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>					
AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>					

AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The category of entity to be used when viewed by entities on the 'opposite' side.					
ForceIdentifier RPRnotePhysical4 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The identification of the force that the entity belongs to.					
InfraredSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Index used to obtain the infra-red signature state of the entity.					
LiveEntityMeasuredSpeed RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The entity's own measurement of speed (e.g. air speed for aircraft).					
Marking RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	A unique marking or combination of characters used to distinguish the entity from other entities.					

PropulsionSystemsData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					
RadarCrossSectionSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the radar cross section signature state of the entity.</i>					
VectoringNozzleSystemData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>					
ExtendedDamageStatus <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Detailed information on damage sustained by the entity.</i>					
VariableParametersArray <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>					

EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The category of the entity.</i>					
EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The unique identifier for the entity instance.</i>					
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>					
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>					

RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.5. GroundVehicle

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Platform.GroundVehicle

Sharing: Publish/Subscribe

Semantics: *A platform entity that operates wholly on the surface of the earth.*

Attributes:

FirePowerKilled RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether it can fire weapons.</i>				
TrailingDustCloud RPRnotePhysical7 RPRnotePhysical21	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceTrailingEffectsEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the size of the dust cloud trailing effect.</i>				

PrimaryHatch RPRnotePhysical7 RPRnotePhysical26	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceHatchEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the primary hatch is open or closed and whether someone is visible.					
HeadLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether head lights are on or off.					
TailLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether tail lights are on or off.					
BrakeLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether brake lights are on or off.					
LauncherOperational RPRnotePhysical7 RPRnotePhysical27	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLauncherOperationalEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the status of the mechanism required for the platform to be operational, such as the elevated status of the primary launcher.					

CamouflageType RPRnotePhysical7 RPRnotePhysical28	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceCamouflageTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the camouflage color.				
ConcealedPosition RPRnotePhysical7 RPRnotePhysical29	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceConcealedPositionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the type of concealment.				
TentExtended RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the tent is extended.				
RampExtended RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the ramp is extended.				
BlackoutLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether blackout lights are on or off.				

BlackoutBrakeLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether blackout brake lights are on or off.					
SpotSearchLight_1On RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether spot/search light #1 is on or off.					
InteriorLightsForwardOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether interior lights (forward) are on or off.					
OccupantsSurrendered RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the vehicle occupants have surrendered.					
MaskedCloaked RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the entity is masked or cloaked.					

SlingLoadableCapability RPRnotePhysical12 RPRnotePhysical15	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The Entity is able to be carried as a sling load payload. The extended appearance record (if available) will identify if it is currently sling loaded and entity association and/or entity offset records (if available) will provide additional sling load details (such as carrier).</i>					
IEDPresenceIndicator RPRnotePhysical12 RPRnotePhysical15	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.</i>					
TaskOrganizableCapability RPRnotePhysical12 RPRnotePhysical15	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The Entity (normally a virtual manned module) can be task organized into an existing mixed mode unit (where mixed mode is intended to comprise a combination of computer-generated forces and virtual or even live forces).</i>					
PaintScheme RPRnotePhysical6 RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the visual paint design.</i>					
MobilityKilled RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 RPRnotePhysical13 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether it is capable of moving on its own power.</i>					

Damage RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical14 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the damaged appearance.</i>					
IsSmokeEmanating RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not smoke is emanating from the entity.</i>					
IsEngineEmittingSmoke RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not the engine is emitting smoke.</i>					
IsFlaming RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is burning and flames are visible.</i>					
PowerPlantOn RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the power plant is on or off.</i>					

AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.</i>					
FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.</i>					
RecoveryCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.</i>					
RepairCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply certain repair services in response to an appropriate service request.</i>					
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>					

AcousticSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>					
AlternateEntityType <i>RPRnotePhysical3</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					
	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>					
ForceIdentifier <i>RPRnotePhysical4</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					

Marking <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					
PropulsionSystemsData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					
RadarCrossSectionSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the radar cross section signature state of the entity.</i>					
VectoringNozzleSystemData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>					
ExtendedDamageStatus <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Detailed information on damage sustained by the entity.</i>					

VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>				
EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The category of the entity.</i>				
EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The unique identifier for the entity instance.</i>				
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>				

Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>				
RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.6. MultiDomainPlatform

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Platform.MultiDomainPlatform

Sharing: Publish/Subscribe

Semantics: *A platform entity that operates in more than one domain (excluding those combinations explicitly defined as subclasses of the superclass of this class).*

Attributes:

FirePowerKilled RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether it can fire weapons.</i>				

TrailingEffects <a>RPRnotePhysical7 <a>RPRnotePhysical8 <a>RPRnotePhysical9 <a>RPRnotePhysical21 <a>RPRnotePhysical25	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	<a>AppearanceTrailingEffectsEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the size of the dust cloud, contrails or ionization, or wake trailing effects.					
Hatch <a>RPRnotePhysical7 <a>RPRnotePhysical10 <a>RPRnotePhysical26 <a>RPRnotePhysical32	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	<a>AppearanceHatchEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the (primary) hatch is open or closed.					
HeadLightsOn <a>RPRnotePhysical7 <a>RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	<a>RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether head lights are on or off.					
TailLightsOn <a>RPRnotePhysical7 <a>RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	<a>RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether tail lights are on or off.					
BrakeLightsOn <a>RPRnotePhysical7 <a>RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	<a>RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether brake lights are on or off.					

LauncherOperational RPRnotePhysical7 RPRnotePhysical27	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLauncherOperationalEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the status of the mechanism required for the platform to be operational, such as the elevated status of the primary launcher.				
CamouflageType RPRnotePhysical7 RPRnotePhysical28	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceCamouflageTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the camouflage color.				
ConcealedPosition RPRnotePhysical7 RPRnotePhysical29	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceConcealedPositionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the type of concealment.				
TentExtended RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the tent is extended.				
RampExtended RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the ramp is extended.				

BlackoutLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether blackout lights are on or off.					
BlackoutBrakeLightsOn RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether blackout brake lights are on or off.					
SpotSearchLight_1On RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical12 RPRnotePhysical30	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether spot/search light #1 is on or off.					
InteriorLightsOn RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical12 RPRnotePhysical31	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether interior lights (forward) are on or off.					
OccupantsSurrendered RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the vehicle occupants have surrendered.					

MaskedCloaked RPRnotePhysical7 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the entity is masked or cloaked.				
RunningLightsOn RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether running lights are on or off.				
IsAccommodationLadderLowered RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes if the accommodation ladder is lowered or not.				
IsFenceRaised RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the safety fence around a helicopter landing deck is raised.				
IsFlagRaised RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the national identification flag is raised.				

TowedArraySonar RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the TAS is deployed or not.				
Periscope RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the periscope is extended or not.				
Snorkel RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the snorkel is extended or not.				
RadarMast RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the radar mast is extended or not.				
CommsMast RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the communication mast is extended or not.				

ESMMast RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the ESM mast is extended or not.				
TorpedoTubes RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the torpedo tubes are open or not.				
MissileHatches RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the missile hatches are open or not.				
NVGMode RPRnotePhysical8 RPRnotePhysical20	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceNVGModeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether air platform lighting is in covert or overt mode.				
CanopyTroopDoor RPRnotePhysical8 RPRnotePhysical22	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceCanopyEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the state of the canopy/troop door.				

LandingLightsOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether landing lights are on or off.				
NavigationLightsOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether navigation lights are on or off.				
AntiCollisionLightsOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether Anti-Collision lights are on or off.				
AfterburnerOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes if the air platform is in afterburner.				
LowerAntiCollisionLightOn RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the lower Anti-Collision light is on or off.				

UpperAntiCollisionLightOn RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the upper Anti-Collision light is on or off.					

AntiCollisionLightDayNight RPRnotePhysical8RPRnotePhysical23	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceAntiCollisionDayNightEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the day/night status of the Anti-Collision lights.					

IsBlinking RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether any air platform lights are blinking or not.					

FormationLightsOn RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether formation lights are on or off.					

LandingGearExtended RPRnotePhysical8RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the landing gear is wholly retracted or extended.					

CargoDoorsOpened RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the cargo doors (main door) are closed or open.					
NavigationPositionBrightness RPRnotePhysical8 RPRnotePhysical24	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceNavigationPositionBrightnessEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the brightness of the navigation/position lights.					
ReverseThrustEngaged RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the air platform has engaged reverse thrust.					
WeightOnWheels RPRnotePhysical8 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the air platform has weight on its main landing gear.					
ADSBCapability RPRnotePhysical12 RPRnotePhysical16	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is equipped with Automatic Dependent Surveillance - Broadcast (ADS-B).					

SlingLoadCarrierCapability RPRnotePhysical12 RPRnotePhysical16	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity is able to carry a payload in a sling load. The extended appearance record (if available) will identify the current sling load status and entity association and/or entity offset records (if available) will provide additional sling load details (such as payload).					
SlingLoadableCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity is able to be carried as a sling load payload. The extended appearance record (if available) will identify if it is currently sling loaded and entity association and/or entity offset records (if available) will provide additional sling load details (such as carrier).					
IEDPresenceIndicator RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.					
TaskOrganizableCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity (normally a virtual manned module) can be task organized into an existing mixed mode unit (where mixed mode is intended to comprise a combination of computer-generated forces and virtual or even live forces).					

LAIRCMCapability RPRnotePhysical12 RPRnotePhysical16	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether the entity is equipped with Large Aircraft Infrared Countermeasures (LAIRCM).</i>				
PaintScheme RPRnotePhysical6 RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the visual paint design.</i>				
MobilityKilled RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 RPRnotePhysical13 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether it is capable of moving on its own power.</i>				
Damage RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical14 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the damaged appearance.</i>				
IsSmokeEmanating RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether or not smoke is emanating from the entity.</i>				

IsEngineEmittingSmoke RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not the engine is emitting smoke.</i>					
IsFlaming RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is burning and flames are visible.</i>					
PowerPlantOn RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the power plant is on or off.</i>					
AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.</i>					
FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.</i>					

RecoveryCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.</i>						
RepairCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Describes whether the entity is able to supply certain repair services in response to an appropriate service request.</i>						
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>						
AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>						
AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Static	<i>NA</i>					
	Semantics						
	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>						

ForceIdentifier <i>RPRnotePhysical4</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					
Marking <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					
PropulsionSystemsData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					

RadarCrossSectionSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the radar cross section signature state of the entity.</i>				
VectoringNozzleSystemData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>				
ExtendedDamageStatus RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Detailed information on damage sustained by the entity.</i>				
VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>				
EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The category of the entity.</i>				

EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The unique identifier for the entity instance.</i>				
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjIdentifier shall be set to the empty string.</i>				
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>				
RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.7. Spacecraft

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Platform.Spacecraft

Sharing: Publish/Subscribe

Semantics: *A platform entity that operates mainly in space.*

Attributes:

PaintScheme RPRnotePhysical6 RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the visual paint design.</i>				
MobilityKilled RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 RPRnotePhysical13 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether it is capable of moving on its own power.</i>				
Damage RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical14 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the damaged appearance.</i>				

IsSmokeEmanating RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not smoke is emanating from the entity.</i>					
IsEngineEmittingSmoke RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not the engine is emitting smoke.</i>					
IsFlaming RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is burning and flames are visible.</i>					
PowerPlantOn RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the power plant is on or off.</i>					
AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.</i>					

FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.</i>						
RecoveryCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.</i>						
RepairCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Describes whether the entity is able to supply certain repair services in response to an appropriate service request.</i>						
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>						
AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>						

AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The category of entity to be used when viewed by entities on the 'opposite' side.					

ForceIdentifier RPRnotePhysical4 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The identification of the force that the entity belongs to.					

InfraredSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Index used to obtain the infra-red signature state of the entity.					

LiveEntityMeasuredSpeed RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The entity's own measurement of speed (e.g. air speed for aircraft).					

Marking RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	A unique marking or combination of characters used to distinguish the entity from other entities.					

PropulsionSystemsData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operating data of the propulsion systems aboard the entity.</i>				
RadarCrossSectionSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the radar cross section signature state of the entity.</i>				
VectoringNozzleSystemData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>				
ExtendedDamageStatus RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Detailed information on damage sustained by the entity.</i>				
VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>				

EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The category of the entity.</i>					
EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The unique identifier for the entity instance.</i>					
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>					
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>					

RelativeSpatial RPRnoteBase3 RPRnoteBase4 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.					

HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.8. SubmersibleVessel

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Platform.SubmersibleVessel

Sharing: Publish/Subscribe

Semantics: *A platform entity that operates either on the surface of the sea, or beneath it.*

Attributes:

Hatch RPRnotePhysical10 RPRnotePhysical26	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceSubsurfaceHatchEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the hatch is open or closed.					

RunningLightsOn RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether running lights are on or off.					

Periscope RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the periscope is extended or not.					
Snorkel RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the snorkel is extended or not.					
RadarMast RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the radar mast is extended or not.					
CommsMast RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the communication mast is extended or not.					
ESMMast RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the ESM mast is extended or not.					

TorpedoTubes RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the torpedo tubes are open or not.					
MissileHatches RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the missile hatches are open or not.					
TowedArraySonar RPRnotePhysical10 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the TAS is deployed or not.					
SlingLoadableCapability RPRnotePhysical12 RPRnotePhysical18	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity is able to be carried as a sling load payload. The extended appearance record (if available) will identify if it is currently sling loaded and entity association and/or entity offset records (if available) will provide additional sling load details (such as carrier).					
IEDPresenceIndicator RPRnotePhysical12 RPRnotePhysical18	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.					

TaskOrganizableCapability RPRnotePhysical12 RPRnotePhysical18	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The Entity (normally a virtual manned module) can be task organized into an existing mixed mode unit (where mixed mode is intended to comprise a combination of computer-generated forces and virtual or even live forces).</i>					
PaintScheme RPRnotePhysical6 RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the visual paint design.</i>					
MobilityKilled RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 RPRnotePhysical13 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether it is capable of moving on its own power.</i>					
Damage RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical14 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the damaged appearance.</i>					
IsSmokeEmanating RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not smoke is emanating from the entity.</i>					

IsEngineEmittingSmoke RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not the engine is emitting smoke.</i>					
IsFlaming RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is burning and flames are visible.</i>					
PowerPlantOn RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the power plant is on or off.</i>					
AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.</i>					
FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.</i>					

RecoveryCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.</i>						
RepairCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Describes whether the entity is able to supply certain repair services in response to an appropriate service request.</i>						
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>						
AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Conditional	<i>On change</i>					
	Semantics						
	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>						
AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype		Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct		PS	DA	RO	HLAbestEffort	
	Update type	Update Condition					
	Static	<i>NA</i>					
	Semantics						
	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>						

ForceIdentifier <i>RPRnotePhysical4</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					
Marking <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					
PropulsionSystemsData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					

RadarCrossSectionSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the radar cross section signature state of the entity.</i>				
VectoringNozzleSystemData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>				
ExtendedDamageStatus RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Detailed information on damage sustained by the entity.</i>				
VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>				
EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The category of the entity.</i>				

EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The unique identifier for the entity instance.</i>				
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjIdentifier shall be set to the empty string.</i>				
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>				
RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.9. SurfaceVessel

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Platform.SurfaceVessel

Sharing: Publish/Subscribe

Semantics: *A platform entity that operates wholly on the surface of the sea.*

Attributes:

WakeSize RPRnotePhysical9 RPRnotePhysical21	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceTrailingEffectsEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the size of the wake trailing effect.					
RunningLightsOn RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether running lights are on or off.					
IsAccommodationLadderLowered RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes if the accommodation ladder is lowered or not.					

IsFenceRaised RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the safety fence around a helicopter landing deck is raised.					
IsFlagRaised RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the national identification flag is raised.					
TowedArraySonar RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the TAS is deployed or not.					
SpotLightsOn RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether spot lights are on or off.					
InteriorLightsOn RPRnotePhysical9 RPRnotePhysical12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether interior lights are on or off.					

SlingLoadableCapability RPRnotePhysical12 RPRnotePhysical17	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The Entity is able to be carried as a sling load payload. The extended appearance record (if available) will identify if it is currently sling loaded and entity association and/or entity offset records (if available) will provide additional sling load details (such as carrier).</i>					
IEDPresenceIndicator RPRnotePhysical12 RPRnotePhysical17	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.</i>					
TaskOrganizableCapability RPRnotePhysical12 RPRnotePhysical17	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The Entity (normally a virtual manned module) can be task organized into an existing mixed mode unit (where mixed mode is intended to comprise a combination of computer-generated forces and virtual or even live forces).</i>					
PaintScheme RPRnotePhysical6 RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the visual paint design.</i>					
MobilityKilled RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 RPRnotePhysical13 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether it is capable of moving on its own power.</i>					

Damage RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical14 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the damaged appearance.</i>					
IsSmokeEmanating RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not smoke is emanating from the entity.</i>					
IsEngineEmittingSmoke RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not the engine is emitting smoke.</i>					
IsFlaming RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is burning and flames are visible.</i>					
PowerPlantOn RPRnotePhysical7 RPRnotePhysical8 RPRnotePhysical9 RPRnotePhysical10 RPRnotePhysical11 RPRnotePhysical12 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the power plant is on or off.</i>					

AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.</i>					
FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.</i>					
RecoveryCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.</i>					
RepairCapability RPRnotePhysical12 RPRnotePhysical15 RPRnotePhysical16 RPRnotePhysical17 RPRnotePhysical18 RPRnotePhysical19 <i>Inherited from Platform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply certain repair services in response to an appropriate service request.</i>					
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>					

AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>					
AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					
	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>					
ForceIdentifier RPRnotePhysical4 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					

Marking <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					
PropulsionSystemsData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					
RadarCrossSectionSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the radar cross section signature state of the entity.</i>					
VectoringNozzleSystemData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>					
ExtendedDamageStatus <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Detailed information on damage sustained by the entity.</i>					

VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>				
EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The category of the entity.</i>				
EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The unique identifier for the entity instance.</i>				
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>				

Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>				
RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.10. Lifeform

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Lifeform

Sharing: Subscribe

Semantics: *A living military platform (human or not).*

Attributes:

Health RPRnotePhysical33 RPRnotePhysical34 RPRnotePhysical35	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLifeFormHealthEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the visual appearance of the severity of any injury.</i>				

MountedHoistedStatus RPRnotePhysical12 RPRnotePhysical34 RPRnotePhysical35	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the life form is mounted/hoisted on another platform, such as a troop transport or helicopter hoist.					
AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical36	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.					
FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical36	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.					
RecoveryCapability RPRnotePhysical12 RPRnotePhysical36	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.					
RepairCapability RPRnotePhysical12 RPRnotePhysical36	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is able to supply certain repair services in response to an appropriate service request.					

LVCIndicator <i>RPRnotePhysical1</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>					
AcousticSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>					
AlternateEntityType <i>RPRnotePhysical3</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>					
ForceIdentifier <i>RPRnotePhysical4</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					

LiveEntityMeasuredSpeed RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>				
Marking RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>				
PropulsionSystemsData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operating data of the propulsion systems aboard the entity.</i>				
RadarCrossSectionSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the radar cross section signature state of the entity.</i>				
VectoringNozzleSystemData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>				

ExtendedDamageStatus RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Detailed information on damage sustained by the entity.</i>				
VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>				
EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The category of the entity.</i>				
EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The unique identifier for the entity instance.</i>				

IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>					
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>					
RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>					
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.11. Human

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Lifeform.Human

Sharing: Publish/Subscribe

Semantics: A human lifeform.

Attributes:

PaintScheme RPRnotePhysical6 RPRnotePhysical34	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the visual paint design.					

ComplianceStatus RPRnotePhysical34 RPRnotePhysical37	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLifeFormComplianceStatusEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the manner and degree to which the life form is complying.					

ClothingIRSignature RPRnotePhysical34 RPRnotePhysical38	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ClothingIRSignatureEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the general nature of the IR Signature due to clothing or covering.					

SignalSmokeInUse RPRnotePhysical12 RPRnotePhysical34	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether signal smoke is being used or not.					

FlashLightsOn RPRnotePhysical12 RPRnotePhysical34	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether flash lights are on or off.					

SignalMirrorInUse RPRnotePhysical12 RPRnotePhysical34	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether a signal mirror is being used or not.					

IRStrobeOn RPRnotePhysical12 RPRnotePhysical34	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether an IR strobe is on or off.					

IRIlluminatorOn RPRnotePhysical12 RPRnotePhysical34	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether an IR illuminator (flare) is on or off.					

LifeFormPosture RPRnotePhysical34 RPRnotePhysical39	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLifeFormPostureEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the posture (position) of the life form.					

IsSmokingCigarette RPRnotePhysical12 RPRnotePhysical34	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the life form has a lit cigarette or not. The primary purpose is for IR signature generation, so it could be a cigar or other item.					

WeaponImplement1 RPRnotePhysical34 RPRnotePhysical40	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLifeFormWeaponImplementEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the position of the life form's primary weapon/implement.</i>				
WeaponImplement2 RPRnotePhysical34 RPRnotePhysical40	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLifeFormWeaponImplementEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the position of the life form's secondary weapon/implement.</i>				
CamouflageType RPRnotePhysical28 RPRnotePhysical34	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceCamouflageTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the camouflage color.</i>				
ConcealedStationary RPRnotePhysical29 RPRnotePhysical34	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceConcealedPositionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether or not the life form is in a prepared concealed position.</i>				

ConcealedMovement RPRnotePhysical34 RPRnotePhysical41	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceConcealedMovementEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the life form uses concealment during movement.				
Health RPRnotePhysical33 RPRnotePhysical34 RPRnotePhysical35 Inherited from Lifeform in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLifeFormHealthEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the visual appearance of the severity of any injury.				
MountedHoistedStatus RPRnotePhysical12 RPRnotePhysical34 RPRnotePhysical35 Inherited from Lifeform in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the life form is mounted/hoisted on another platform, such as a troop transport or helicopter hoist.				
AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical36 Inherited from Lifeform in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.				
FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical36 Inherited from Lifeform in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.				

RecoveryCapability RPRnotePhysical12 RPRnotePhysical36 <i>Inherited from Lifeform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.</i>					
RepairCapability RPRnotePhysical12 RPRnotePhysical36 <i>Inherited from Lifeform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply certain repair services in response to an appropriate service request.</i>					
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>					
AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>					
AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					
	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>					

ForceIdentifier <i>RPRnotePhysical4</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					
Marking <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					
PropulsionSystemsData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					

RadarCrossSectionSignatureIndex RPRnotePhysical2 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Index used to obtain the radar cross section signature state of the entity.				
VectoringNozzleSystemData RPRnotePhysical5 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The basic operational data for the vectoring nozzle systems aboard the entity.				
ExtendedDamageStatus RPRnotePhysical5 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Detailed information on damage sustained by the entity.				
VariableParametersArray RPRnotePhysical5 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Array of information on articulated parts, attached parts, and other data associated with an entity.				
EntityType RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The category of the entity.				

EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The unique identifier for the entity instance.</i>				
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjIdentifier shall be set to the empty string.</i>				
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>				
RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.12. NonHuman

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Lifeform.NonHuman

Sharing: Publish/Subscribe

Semantics: *An animal or other non-human lifeform.*

Attributes:

Health RPRnotePhysical33 RPRnotePhysical34 RPRnotePhysical35 <i>Inherited from Lifeform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLifeFormHealthEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the visual appearance of the severity of any injury.</i>				
MountedHoistedStatus RPRnotePhysical12 RPRnotePhysical34 RPRnotePhysical35 <i>Inherited from Lifeform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether or not the life form is mounted/hoisted on another platform, such as a troop transport or helicopter hoist.</i>				
AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical36 <i>Inherited from Lifeform in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.</i>				

FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical36 Inherited from Lifeform in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.					
RecoveryCapability RPRnotePhysical12 RPRnotePhysical36 Inherited from Lifeform in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is able to provide recovery (e.g., towing) services in response to an appropriate service request.					
RepairCapability RPRnotePhysical12 RPRnotePhysical36 Inherited from Lifeform in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is able to supply certain repair services in response to an appropriate service request.					
LVCIndicator RPRnotePhysical1 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.					
AcousticSignatureIndex RPRnotePhysical2 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Index used to obtain the acoustics (sound through air) signature state of the entity.					

AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>					
ForceIdentifier RPRnotePhysical4 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					
Marking RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					

PropulsionSystemsData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					
RadarCrossSectionSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the radar cross section signature state of the entity.</i>					
VectoringNozzleSystemData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>					
ExtendedDamageStatus RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Detailed information on damage sustained by the entity.</i>					
VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>					

EntityType RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The category of the entity.					

EntityIdentifier RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The unique identifier for the entity instance.					

IsPartOf RPRnoteBase3 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.					

Spatial RPRnoteBase2 RPRnoteBase4 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the world coordinate system.					

RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.13. CulturalFeature

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.CulturalFeature

Sharing: Publish/Subscribe

Semantics: *Engineering and natural effects such as craters, bridges, vehicle tracks, etc.*

Attributes:

DamageArea RPRnotePhysical42 RPRnotePhysical43	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageAreaEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the damaged area.</i>				
Damage RPRnotePhysical14 RPRnotePhysical43	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the damaged appearance.</i>				

IsSmokeEmanating RPRnotePhysical12 RPRnotePhysical43	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not smoke is emanating from the entity.					
IsFlaming RPRnotePhysical12 RPRnotePhysical43	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is burning and flames are visible.					
InternalHeatOn RPRnotePhysical12 RPRnotePhysical43	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the internal heat is on or off.					
ExteriorLightsOn RPRnotePhysical12 RPRnotePhysical43	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the exterior lights are on or off.					
InteriorLightsOn RPRnotePhysical12 RPRnotePhysical43	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the interior lights are on or off.					

MaskedCloaked RPRnotePhysical12 RPRnotePhysical43	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the entity is masked or cloaked.				
SlingLoadableCapability RPRnotePhysical12 RPRnotePhysical44	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The Entity is able to be carried as a sling load payload. The extended appearance record (if available) will identify if it is currently sling loaded and entity association and/or entity offset records (if available) will provide additional sling load details (such as carrier).				
IEDPresenceIndicator RPRnotePhysical12 RPRnotePhysical44	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.				
LVCIndicator RPRnotePhysical1 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.				
AcousticSignatureIndex RPRnotePhysical2 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Index used to obtain the acoustics (sound through air) signature state of the entity.				

AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The category of entity to be used when viewed by entities on the 'opposite' side.					

ForceIdentifier RPRnotePhysical4 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The identification of the force that the entity belongs to.					

InfraredSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Index used to obtain the infra-red signature state of the entity.					

LiveEntityMeasuredSpeed RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The entity's own measurement of speed (e.g. air speed for aircraft).					

Marking RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	A unique marking or combination of characters used to distinguish the entity from other entities.					

PropulsionSystemsData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operating data of the propulsion systems aboard the entity.</i>				
RadarCrossSectionSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the radar cross section signature state of the entity.</i>				
VectoringNozzleSystemData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>				
ExtendedDamageStatus RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Detailed information on damage sustained by the entity.</i>				
VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>				

EntityType RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The category of the entity.					

EntityIdentifier RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The unique identifier for the entity instance.					

IsPartOf RPRnoteBase3 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.					

Spatial RPRnoteBase2 RPRnoteBase4 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the world coordinate system.					

RelativeSpatial RPRnoteBase3 RPRnoteBase4 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.					

HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.14. Munition

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Munition

Sharing: Publish/Subscribe

Semantics: *A complete device charged with explosives, propellants, pyrotechnics, initiating composition, or nuclear, biological or chemical material for use in military operations, including demolitions.*

Attributes:

Damage RPRnotePhysical14 RPRnotePhysical45	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the damaged appearance.					
IsSmokeVaporEmanating RPRnotePhysical12 RPRnotePhysical45	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not smoke or vapor is emanating from the entity.					

IsEngineEmittingSmoke RPRnotePhysical12 RPRnotePhysical45	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the engine is emitting smoke.					
VaporTrailSize RPRnotePhysical21 RPRnotePhysical45	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceTrailingEffectsEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the size of the vapor trail.					
IsFlaming RPRnotePhysical12 RPRnotePhysical45	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is burning and flames are visible.					
LaunchFlashPresent RPRnotePhysical12 RPRnotePhysical45	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the guided munition's launch flash is present.					
PowerPlantOn RPRnotePhysical12 RPRnotePhysical45	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the power plant is on or off.					

CoverShroudStatus RPRnotePhysical45 RPRnotePhysical46	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	CoverShroudStatusEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the status of the cover or shroud.					
MaskedCloaked RPRnotePhysical12 RPRnotePhysical45	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the entity is masked or cloaked.					
IEDPresenceIndicator RPRnotePhysical12 RPRnotePhysical47	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.					
LVCIndicator RPRnotePhysical1 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.					
AcousticSignatureIndex RPRnotePhysical2 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Index used to obtain the acoustics (sound through air) signature state of the entity.					

AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>					
ForceIdentifier RPRnotePhysical4 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					
Marking RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					

PropulsionSystemsData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					
RadarCrossSectionSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the radar cross section signature state of the entity.</i>					
VectoringNozzleSystemData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>					
ExtendedDamageStatus <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Detailed information on damage sustained by the entity.</i>					
VariableParametersArray <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>					

EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The category of the entity.</i>					
EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The unique identifier for the entity instance.</i>					
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>					
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>					

RelativeSpatial RPRnoteBase3 RPRnoteBase4 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.					

HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.15. Expendables

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Expendables

Sharing: Publish/Subscribe

Semantics: *Countermeasures devices that are dispensed from another entity. The devices may be active emitters or passive reflectors of energy.*

Attributes:

Damage RPRnotePhysical14 RPRnotePhysical48	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the damaged appearance.					
IsSmokeEmanating RPRnotePhysical12 RPRnotePhysical48	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not smoke is emanating from the entity.					

ParachuteStatus RPRnotePhysical48 RPRnotePhysical49	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ParachuteEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the status of a flare's parachute.					

FlareSmokeColor RPRnotePhysical48 RPRnotePhysical50	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FlareSmokeColorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the color of a flare's light output or the color of smoke emanating from a smoke expendable.					

IsFlaming RPRnotePhysical12 RPRnotePhysical48	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the entity is burning and flames are visible.					

LaunchFlashPresent RPRnotePhysical12 RPRnotePhysical48	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not launch flash is present.					

FlareSmokeStatus RPRnotePhysical48 RPRnotePhysical51	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FlareSmokeStatusEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the status of a flare or smoke expendable.					

PowerPlantOn RPRnotePhysical12 RPRnotePhysical48	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the power plant is on or off.					
SpotChaffStatus RPRnotePhysical48 RPRnotePhysical52	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpotChaffStatusEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the status of spot chaff.					
MaskedCloaked RPRnotePhysical12 RPRnotePhysical48	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the entity is masked or cloaked.					
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.					
AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Index used to obtain the acoustics (sound through air) signature state of the entity.					

AlternateEntityType <i>RPRnotePhysical3</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The category of entity to be used when viewed by entities on the 'opposite' side.					
ForceIdentifier <i>RPRnotePhysical4</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The identification of the force that the entity belongs to.					
InfraredSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Index used to obtain the infra-red signature state of the entity.					
LiveEntityMeasuredSpeed <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The entity's own measurement of speed (e.g. air speed for aircraft).					
Marking <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	A unique marking or combination of characters used to distinguish the entity from other entities.					

PropulsionSystemsData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					
RadarCrossSectionSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the radar cross section signature state of the entity.</i>					
VectoringNozzleSystemData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>					
ExtendedDamageStatus RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Detailed information on damage sustained by the entity.</i>					
VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>					

EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The category of the entity.</i>					
EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The unique identifier for the entity instance.</i>					
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>					
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>					

RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.16. Radio

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Radio

Sharing: Publish/Subscribe

Semantics: *Electronic devices for the communication of both audio and data, operated by entities belonging to armed forces.*

Attributes:

LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>				
AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>				

AlternateEntityType <i>RPRnotePhysical3</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>					
ForceIdentifier <i>RPRnotePhysical4</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					
Marking <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					

PropulsionSystemsData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operating data of the propulsion systems aboard the entity.</i>				
RadarCrossSectionSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the radar cross section signature state of the entity.</i>				
VectoringNozzleSystemData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>				
ExtendedDamageStatus RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Detailed information on damage sustained by the entity.</i>				
VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>				

EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The category of the entity.</i>					
EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The unique identifier for the entity instance.</i>					
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>					
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>					

RelativeSpatial RPRnoteBase3 RPRnoteBase4 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.					

HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.17. Sensor

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Sensor

Sharing: Publish/Subscribe

Semantics: *Sensors and emitters, such as stand-alone radars, jammers, and detection systems, that are not part of another platform or system described by another Physical Entity, and are operated by armed forces.*

Attributes:

PaintScheme RPRnotePhysical6 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the visual paint design.					
MobilityKilled RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether it is capable of moving on its own power.					

MissionKilled RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether it is capable of carrying out its mission (e.g., damaged antenna).					
Damage RPRnotePhysical14 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the damaged appearance.					
IsSmokeEmanating RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not smoke is emanating from the entity.					
IsEngineEmittingSmoke RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether or not the engine is emitting smoke.					
TrailingEffects RPRnotePhysical21 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceTrailingEffectsEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the size of the trailing effects.					

LightsOn RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the lights are on or off.</i>					
IsFlaming RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is burning and flames are visible.</i>					
AntennaRaised RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the antenna is raised or not.</i>					
CamouflageType RPRnotePhysical28 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceCamouflageTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the camouflage color.</i>					
ConcealedPosition RPRnotePhysical29 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceConcealedPositionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the type of concealment.</i>					

PowerPlantOn RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether the power plant is on or off.				
TentExtended RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the tent is extended.				
BlackoutLightsOn RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether blackout lights are on or off.				
InteriorLightsOn RPRnotePhysical12 RPRnotePhysical53	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether interior lights are on or off.				
SlingLoadableCapability RPRnotePhysical12 RPRnotePhysical54	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The Entity is able to be carried as a sling load payload. The extended appearance record (if available) will identify if it is currently sling loaded and entity association and/or entity offset records (if available) will provide additional sling load details (such as carrier).				

IEDPresenceIndicator RPRnotePhysical12 RPRnotePhysical54	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	<i>The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.</i>				
TaskOrganizableCapability RPRnotePhysical12 RPRnotePhysical54	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	<i>The Entity (normally a virtual manned module) can be task organized into an existing mixed mode unit (where mixed mode is intended to comprise a combination of computer-generated forces and virtual or even live forces).</i>				
LVCIndicator RPRnotePhysical1 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>				
AcousticSignatureIndex RPRnotePhysical2 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>				
AlternateEntityType RPRnotePhysical3 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>				

ForceIdentifier <i>RPRnotePhysical4</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The identification of the force that the entity belongs to.					
InfraredSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Index used to obtain the infra-red signature state of the entity.					
LiveEntityMeasuredSpeed <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The entity's own measurement of speed (e.g. air speed for aircraft).					
Marking <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	A unique marking or combination of characters used to distinguish the entity from other entities.					
PropulsionSystemsData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The basic operating data of the propulsion systems aboard the entity.					

RadarCrossSectionSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Index used to obtain the radar cross section signature state of the entity.</i>				
VectoringNozzleSystemData RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The basic operational data for the vectoring nozzle systems aboard the entity.</i>				
ExtendedDamageStatus RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Detailed information on damage sustained by the entity.</i>				
VariableParametersArray RPRnotePhysical5 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Array of information on articulated parts, attached parts, and other data associated with an entity.</i>				
EntityType RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The category of the entity.</i>				

EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The unique identifier for the entity instance.</i>				
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjIdentifier shall be set to the empty string.</i>				
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>				
RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.1.18. Supplies

Full Name: HLAobjectRoot.BaseEntity.PhysicalEntity.Supplies

Sharing: Publish/Subscribe

Semantics: *Supplies other than munitions, such as fuel, food and personnel.*

Attributes:

PaintScheme <i>RPRnotePhysical6 RPRnotePhysical55</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearancePaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the visual paint design.</i>				
Damage <i>RPRnotePhysical14 RPRnotePhysical55</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the damaged appearance.</i>				
ParachuteStatus <i>RPRnotePhysical49 RPRnotePhysical55</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ParachuteEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the status of a supply's parachute.</i>				

IsFlaming RPRnotePhysical12 RPRnotePhysical55	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is burning and flames are visible.</i>					
DeployedStatus RPRnotePhysical55 RPRnotePhysical56	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceSupplyDeployedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes the deployed status.</i>					
MaskedCloaked RPRnotePhysical12 RPRnotePhysical55	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether or not the entity is masked or cloaked.</i>					
AmmunitionSupplyCapability RPRnotePhysical12 RPRnotePhysical57	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of ammunition in response to an appropriate service request.</i>					
FuelSupplyCapability RPRnotePhysical12 RPRnotePhysical57	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Describes whether the entity is able to supply some type of fuel in response to an appropriate service request.</i>					

SlingLoadableCapability RPRnotePhysical12 RPRnotePhysical57	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
<i>The Entity is able to be carried as a sling load payload. The extended appearance record (if available) will identify if it is currently sling loaded and entity association and/or entity offset records (if available) will provide additional sling load details (such as carrier).</i>						
IEDPresenceIndicator RPRnotePhysical12 RPRnotePhysical57	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
<i>The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.</i>						
LVCIndicator RPRnotePhysical1 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LVCIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
<i>Indicates whether the data contained in this object instance is related to a Live, Virtual, or Constructive entity.</i>						
AcousticSignatureIndex RPRnotePhysical2 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
<i>Index used to obtain the acoustics (sound through air) signature state of the entity.</i>						
AlternateEntityType RPRnotePhysical3 <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
<i>The category of entity to be used when viewed by entities on the 'opposite' side.</i>						

ForceIdentifier <i>RPRnotePhysical4</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The identification of the force that the entity belongs to.</i>					
InfraredSignatureIndex <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Index used to obtain the infra-red signature state of the entity.</i>					
LiveEntityMeasuredSpeed <i>RPRnotePhysical2</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VelocityDecimeterPerSecondInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The entity's own measurement of speed (e.g. air speed for aircraft).</i>					
Marking <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>A unique marking or combination of characters used to distinguish the entity from other entities.</i>					
PropulsionSystemsData <i>RPRnotePhysical5</i> <i>Inherited from PhysicalEntity in Physical</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PropulsionSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The basic operating data of the propulsion systems aboard the entity.</i>					

RadarCrossSectionSignatureIndex RPRnotePhysical2 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Index used to obtain the radar cross section signature state of the entity.				
VectoringNozzleSystemData RPRnotePhysical5 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VectoringNozzleSystemDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The basic operational data for the vectoring nozzle systems aboard the entity.				
ExtendedDamageStatus RPRnotePhysical5 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DamageDescriptionVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Detailed information on damage sustained by the entity.				
VariableParametersArray RPRnotePhysical5 Inherited from PhysicalEntity in Physical	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Array of information on articulated parts, attached parts, and other data associated with an entity.				
EntityType RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The category of the entity.				

EntityIdentifier RPRnoteBase2 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The unique identifier for the entity instance.</i>				
IsPartOf RPRnoteBase3 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjIdentifier shall be set to the empty string.</i>				
Spatial RPRnoteBase2 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>				
RelativeSpatial RPRnoteBase3 RPRnoteBase4 <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteBase5				
	Semantics	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

4.2. Interaction Classes



4.2.1. HLAIinteractionRoot

Full Name: HLAIinteractionRoot
Sharing: Publish/Subscribe
Transportation type: HLAreliable
Order: Receive
Dimensions:
Semantics:
Parameters: -

4.2.2. Collision

Full Name: HLAIinteractionRoot.Collision
Sharing: Publish/Subscribe
Transportation type: HLABestEffort
Order: Receive
Dimensions:
Semantics: *The act or instance of coming together with solid impact.*
Parameters:

Name	Datatype	Semantics
CollidingObjectIdentifier RPRnotePhysical58	RTObjectId	<i>The object instance ID of the object that the issuing object has collided with.</i>

Name	Datatype	Semantics
IssuingObjectMass	MassKilogramFloat32	<i>The mass of the issuing object.</i>
IssuingObjectVelocityVector	VelocityVectorStruct	<i>The velocity vector of the issuing object at the moment of impact.</i>
CollisionType	CollisionTypeEnum8	<i>The type of collision.</i>
CollisionLocation	RelativePositionStruct	<i>The location of the collision relative to the object that the issuing object has collided with.</i>
EventIdentifier	EventIdentifierStruct	<i>An ID assigned by the issuing object to associate related collision events.</i>
IssuingObjectIdentifier RPRnotePhysical59	RTIobjectId	<i>The object instance ID of the object that has detected the collision and issued the collision interaction.</i>

4.2.3. CollisionElastic

Full Name: HLAIinteractionRoot.Collision.CollisionElastic

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *The act or instance of coming together with solid impact in an elastic manner. An elastic collision allows a higher fidelity collision to be modeled, taking into account linear and rotational momentum transfer, variable elasticity, and momentum transfer that is dependent on surface orientation.*

Parameters:

Name	Datatype	Semantics
CoefficientOfRestitution	Float32	<i>The degree that energy is conserved in a collision.</i>
IntermediateResultXX	Float32	<i>X-X Component of the positive semi-definite Collision Intermediate Result matrix.</i>
IntermediateResultXY	Float32	<i>X-Y Component of the positive semi-definite Collision Intermediate Result matrix.</i>

Name	Datatype	Semantics
IntermediateResultXZ	Float32	<i>X-Z Component of the positive semi-definite Collision Intermediate Result matrix.</i>
IntermediateResultYY	Float32	<i>Y-Y Component of the positive semi-definite Collision Intermediate Result matrix.</i>
IntermediateResultYZ	Float32	<i>Y-Z Component of the positive semi-definite Collision Intermediate Result matrix.</i>
IntermediateResultZZ	Float32	<i>Z-Z Component of the positive semi-definite Collision Intermediate Result matrix.</i>
UnitSurfaceNormal	EntityCoordinateVectorStruct	<i>The normal vector to the surface at the point of collision detection.</i>
CollidingObjectIdentifier RPRnotePhysical58 Inherited from Collision in Physical	RTObjectId	<i>The object instance ID of the object that the issuing object has collided with.</i>
IssuingObjectMass Inherited from Collision in Physical	MassKilogramFloat32	<i>The mass of the issuing object.</i>
IssuingObjectVelocityVector Inherited from Collision in Physical	VelocityVectorStruct	<i>The velocity vector of the issuing object at the moment of impact.</i>
CollisionType Inherited from Collision in Physical	CollisionTypeEnum8	<i>The type of collision.</i>
CollisionLocation Inherited from Collision in Physical	RelativePositionStruct	<i>The location of the collision relative to the object that the issuing object has collided with.</i>
EventIdentifier Inherited from Collision in Physical	EventIdentifierStruct	<i>An ID assigned by the issuing object to associate related collision events.</i>
IssuingObjectIdentifier RPRnotePhysical59 Inherited from Collision in Physical	RTObjectId	<i>The object instance ID of the object that has detected the collision and issued the collision interaction.</i>

4.3. Datatypes

4.3.1. Simple Datatypes

PowerSettingFloat32

Representation: HLAfloat32BE

Units: NA

Resolution: NA

Accuracy: NA

Semantics: A normalized value indicating the operator's power controller setting (e.g., the throttle position). In normal operation, the value shall be set in the range of POWER_IDLE (fixed value: 0.0) to POWER_MILITARY (maximum without afterburner; fixed value: 50.0) inclusive, proportional to the controller position. If an afterburner is engaged, the setting shall be in the range of POWER_MIN_AFTERBURNER (fixed value: 51.0) to POWER_MAX_AFTERBURNER (fixed value: 100.0) inclusive. If the propulsion system is off, the setting shall be set to POWER_ENGINE_OFF (fixed value: -100.0).

RevolutionsPerMinuteFloat32

Representation: HLAfloat32BE

Units: RPM

Resolution: NA

Accuracy: perfect

Semantics: Rotation speed expressed in revolutions per minute.

VelocityDecimeterPerSecondInteger16

Representation: [RPRunsignedInteger16BE](#)

Units: decimeter per second (dm/s)

Resolution: 1

Accuracy: perfect

Semantics: Velocity/Speed measured in decimeter per second.

4.3.2. Array Datatypes

DamageDescriptionVariantStructArray

Element [DamageDescriptionVariantStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Array with detailed damage information.*

MarkingArray11

Element [Octet](#)
Type:
Cardinality: 11
Encoding: HLAfixedArray
Semantics: *String of characters represented by an 11 element character string.*

PropulsionSystemDataStructArray

Element [PropulsionSystemDataStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *A set of Propulsion System Data descriptions.*

VectoringNozzleSystemDataStructArray

Element [VectoringNozzleSystemDataStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *A set of Vectoring Nozzle System Data description.*

4.3.3. Fixed Record Datatypes

DEDamageDescriptionComponentVisualDamageStatusDataStruct

Encoding: HLAfixedRecord

Semantics: *DE Damage Description-Component Visual Damage Status.*

Name	Type	Semantic
IsFirePresent	RPRboolean	<i>Describes presence of fire at the damage site.</i>
Smoke	ComponentVisualDamageStatusSmokeEnum8	<i>Describes presence of smoke emanating from the damage site.</i>
SurfaceDamage	ComponentVisualDamageStatusSurfaceDamageEnum8	<i>Describes general surface appearance at the damage site.</i>

DirectedEnergyDamageDescriptionStruct

Encoding: HLAfixedRecord

Semantics: *Detailed damage sustained by an entity due to directed energy.*

Name	Type	Semantic
DamageLocation	RelativePositionStruct	<i>The location of the centroid of the damage location in the entity's coordinate frame.</i>
DamageDiameter	MeterFloat32	<i>The size of the damaged location.</i>
Temperature RPRnotePhysical60	TemperatureDegreeCelsiusFloat32	<i>The average temperature of the damaged region.</i>
ComponentIdentification	EntityDamageStatusComponentIdentificationEnum8	<i>Component Identification.</i>
ComponentDamageStatus	DEDamageDescriptionComponentDamageStatusEnum8	<i>Component Damage Status.</i>
ComponentVisualDamageStatus	DEDamageDescriptionComponentVisualDamageStatusDataStruct	<i>Component Visual Damage Status.</i>
ComponentVisualSmokeColor	DEDamageDescriptionComponentVisualSmokeColorEnum8	<i>Component Visual Smoke Color.</i>
EventIdentifier RPRnotePhysical61	EventIdentifierStruct	<i>Association to related (DE) fire and detonation events.</i>

EntityCoordinateVectorStruct

Encoding: HLAfixedRecord

Semantics: *Vector in the entity coordinate system. Based on the Entity Coordinate Vector record as specified in IEEE Std 1278.1-2012 Section 6.2.96 item a.*

Name	Type	Semantic
XComponent	MeterFloat32	Component along the X axis.
YComponent	MeterFloat32	Component along the Y axis.
ZComponent	MeterFloat32	Component along the Z axis.

MarkingStruct [RPRnotePhysical62](#)

Encoding: HLAfixedRecord

Semantics: *Character set used in the marking and the string of characters to be interpreted for display.*

Name	Type	Semantic
MarkingEncodingType	EntityMarkingCharacterSetEnum8	Character set representation.
MarkingData	MarkingArray11	11 element character string.

PropulsionSystemDataStruct

Encoding: HLAfixedRecord

Semantics: *Information describing a propulsion system in terms of power settings and current RPM.*

Name	Type	Semantic
PowerSetting	PowerSettingFloat32	The power setting of the propulsion system, after any response lags have been incorporated.
EngineRPM	RevolutionsPerMinuteFloat32	The current engine speed.

VectoringNozzleSystemDataStruct

Encoding: HLAfixedRecord

Semantics: *Operational data for describing the vectoring nozzle systems being simulated.*

Name	Type	Semantic
HorizontalDeflectionAngle	AngleDegreeFloat32	The nozzle deflection angle in the horizontal body axis of the entity.
VerticalDeflectionAngle	AngleDegreeFloat32	The nozzle deflection angle in the vertical body axis of the entity.

4.3.4. Variant Record Datatypes

DamageDescriptionVariantStruct

Encoding: RPRextendedVariantRecord
Discriminant name: DamageDescriptionType
Discriminant type: [VariableRecordTypesEnum32](#)
Semantics: Variant record for specifying detailed information on the entity damage state.

Name	Enumerator	Type	Semantics
DirectedEnergyDamageDescription	Directed_Energy_Damage_Description_rec ord	DirectedEnergyDamageDescriptionStruct	Alternative for DE damage description.

4.4. Notes

RPRnotePhysical1

Semantics: *Default value: No_Statement*

RPRnotePhysical2

Semantics: *Default value: 0*

RPRnotePhysical3

Semantics: *Default value: BaseEntity.EntityType*

RPRnotePhysical4

Semantics: *Default value: Other*

RPRnotePhysical5

Semantics: *Default value: empty*

RPRnotePhysical6

Semantics: *Default value: Uniform_Color*

RPRnotePhysical7

Semantics: *Source: SISO-REF-010 [UID 31]*

RPRnotePhysical8

Semantics: *Source: SISO-REF-010 [UID 32]*

RPRnotePhysical9

Semantics: *Source: SISO-REF-010 [UID 33]*

RPRnotePhysical10

Semantics: *Source: SISO-REF-010 [UID 34]*

RPRnotePhysical11

Semantics: *Source: SISO-REF-010 [UID 35]*

RPRnotePhysical12

Semantics: *Default value: False*

RPRnotePhysical13

Semantics: *For the Air Domain named 'Propulsion Killed' in SISO-REF-010.*

RPRnotePhysical14

Semantics: *Default value: No_Damage*

RPRnotePhysical15

Semantics: *Source: SISO-REF-010 [UID 450]*

RPRnotePhysical16

Semantics: *Source: SISO-REF-010 [UID 451]*

RPRnotePhysical17

Semantics: *Source: SISO-REF-010 [UID 452]*

RPRnotePhysical18

Semantics: *Source: SISO-REF-010 [UID 453]*

RPRnotePhysical19

Semantics: *Source: SISO-REF-010 [UID 454]*

RPRnotePhysical20

Semantics: *Default value: Overt_Lighting*

RPRnotePhysical21

Semantics: *Default value: None*

RPRnotePhysical22

Semantics: *Default value: Not_Applicable*

RPRnotePhysical23

Semantics: *Default value: Day*

RPRnotePhysical24

Semantics: *Default value: Bright*

RPRnotePhysical25

Semantics: *For the Land Domain named 'Trailing Dust Cloud' and for the Surface Domain named 'Wake Size' in SISO-REF-010.*

RPRnotePhysical26

Semantics: *Default value: Not_Applicable*

RPRnotePhysical27

Semantics: *Default value: Not_Raised_Not_Operational*

RPRnotePhysical28

Semantics: *Default value: Other*

RPRnotePhysical29

Semantics: *Default value: Not_Concealed*

RPRnotePhysical30

Semantics: *For the Surface Domain named 'Spot Lights On' in SISO-REF-010.*

RPRnotePhysical31

Semantics: *For the Land Domain named 'Interior Lights (Forward) On' in SISO-REF-010.*

RPRnotePhysical32

Semantics: *For the Land Domain named 'Primary Hatch' in SISO-REF-010.
If the entity can operate beneath the surface of the sea, the valid values may need to be limited to 0, 1, and 4, in accordance with AppearanceSubsurfaceHatchEnum8.*

RPRnotePhysical33

Semantics: *Default value: No_Injury*

RPRnotePhysical34

Semantics: *Source: SISO-REF-010 [UID 37]*

RPRnotePhysical35

Semantics: *Source: SISO-REF-010 [UID 480]*

RPRnotePhysical36

Semantics: *Source: SISO-REF-010 [UID 456]*

RPRnotePhysical37

Semantics: *Default value: Not_Specified*

RPRnotePhysical38

Semantics: *Default value: Standard_Clothing*

RPRnotePhysical39

Semantics: *Default value: Not_Specified*

RPRnotePhysical40

Semantics: *Default value: Not_Present*

RPRnotePhysical41

Semantics: *Default value: Open_Movement*

RPRnotePhysical42

Semantics: *Default value:*
Damage_Area_1_default_is_Side_1_Front_Side_or_Not_Applicable_if_Damage_Areas_are_not_defined
—

RPRnotePhysical43

Semantics: *Source: SISO-REF-010 [UID 39]*

RPRnotePhysical44

Semantics: *Source: SISO-REF-010 [UID 458]*

RPRnotePhysical45

Semantics: *Source: SISO-REF-010 [UID 36]*

RPRnotePhysical46

Semantics: *Default value: Cover_Shroud_Blown_Detached*

RPRnotePhysical47

Semantics: *Source: SISO-REF-010 [UID 455]*

RPRnotePhysical48

Semantics: *Source: SISO-REF-010 [UID 42]*

RPRnotePhysical49

Semantics: *Default value: None*

RPRnotePhysical50

Semantics: *Default value: White*

RPRnotePhysical51

Semantics: *Default value: Burning*

RPRnotePhysical52

Semantics: *Default value: Deployed*

RPRnotePhysical53

Semantics: *Source: SISO-REF-010 [UID 43]*

RPRnotePhysical54

Semantics: *Source: SISO-REF-010 [UID 462]*

RPRnotePhysical55

Semantics: *Source: SISO-REF-010 [UID 40]*

RPRnotePhysical56

Semantics: *Default value: Not_Applicable*

RPRnotePhysical57

Semantics: *Source: SISO-REF-010 [UID 459]*

RPRnotePhysical58

Semantics: *If there is no object instance associated with the parameter, then this should be set to the empty string (no characters). Refer to SISO-STD-001 Section 7.8.6 for handling empty strings.*

RPRnotePhysical59

Semantics: *This must reference a valid object instance.*

RPRnotePhysical60

Semantics: *If the entity does not model or otherwise cannot provide this information, a value of -273.15 °C shall be used.*

RPRnotePhysical61

Semantics: *If no fire or detonation event is applicable, this field shall be set to all zeroes.*

RPRnotePhysical62

Semantics: *The units and semantics for the MarkingData array elements are specified by the value of the MarkingEncodingType.*

5. Module Aggregate



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Aggregate FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	This module provides the object class definition for representing aggregates of entities.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

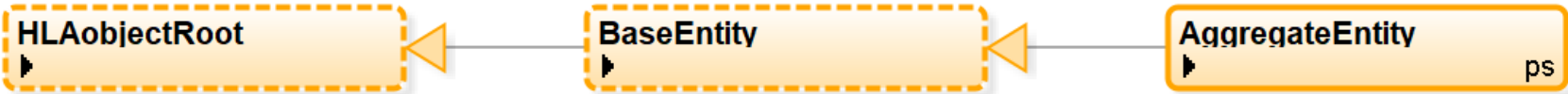
References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Base
Foundation
Enumerations

5.1. Object Classes



5.1.1. AggregateEntity

Full Name: HLAobjectRoot.BaseEntity.AggregateEntity
Sharing: Publish/Subscribe
Semantics: *A group of one or more separate objects that operate together as part of an organization. These objects may be discrete, may be other aggregate objects, or may be a mixture of both.*

Attributes:

AggregateMarking RPRnoteAggregate1	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AggregateMarkingStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	A unique marking or combination of characters used to distinguish the aggregate from other aggregates.					
AggregateState RPRnoteAggregate2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AggregateStateAggregateStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	An indicator of the extent of association of objects form an operating group.					

Dimensions RPRnoteAggregate2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DimensionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The size of the area covered by the units in the aggregate.					
EntityIdentifiers RPRnoteAggregate1	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectIdArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The identification of entities that are contained within the aggregate.					
ForceIdentifier RPRnoteAggregate3	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The identification of the force that the aggregate belongs to.					
Formation RPRnoteAggregate3	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AggregateStateFormationEnum32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The category of positional arrangement of the entities within the aggregate.					
SilentAggregates RPRnoteAggregate1	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SilentAggregateStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The numbers and types, of silent aggregates contained in the aggregate. Silent aggregates are sub-aggregates that are in the aggregate, but that are not separately represented in the virtual world.					

SilentEntities <u>RPRnoteAggregate1</u>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SilentEntityStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The numbers and types, of silent entities in the aggregate. Silent entities are entities that are in the aggregate, but that are not separately represented in the virtual world.</i>					
SubAggregateIdentifiers <u>RPRnoteAggregate1</u>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectIdArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The identifications of aggregates represented in the virtual world that are contained in the aggregate.</i>					
VariableDatums <u>RPRnoteAggregate1</u>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	VariableDatumStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Extra data that describes the aggregate.</i>					
EntityType <u>RPRnoteBase2</u> <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					
	<i>The category of the entity.</i>					
EntityIdentifier <u>RPRnoteBase2</u> <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					
	<i>The unique identifier for the entity instance.</i>					

IsPartOf <i>RPRnoteBase3</i> <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.</i>					
Spatial <i>RPRnoteBase2 RPRnoteBase4</i> <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> <i>RPRnoteBase5</i>				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the world coordinate system.</i>					
RelativeSpatial <i>RPRnoteBase3 RPRnoteBase4</i> <i>Inherited from BaseEntity in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> <i>RPRnoteBase5</i>				
	Semantics					
	<i>Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.</i>					
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

5.2. Datatypes

5.2.1. Array Datatypes

MarkingArray31

Element [Octet](#)
Type:
Cardinality: 31
Encoding: HLAfixedArray
Semantics: *String of characters represented by a 31 element character string.*

SilentAggregateStructArray

Element [SilentAggregateStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Set of silent aggregates (aggregates not registered in the federation).*

SilentEntityStructArray

Element [SilentEntityStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Set of silent entities (entities not registered in the federation).*

5.2.2. Fixed Record Datatypes

AggregateMarkingStruct RPRnoteAggregate4

Encoding: HLAfixedRecord

Semantics: *Unique marking associated with an aggregate.*

Name	Type	Semantic
MarkingEncodingType	EntityMarkingCharacterSetEnum8	<i>The type of marking.</i>
MarkingData	MarkingArray31	<i>The marking itself.</i>

SilentAggregateStruct

Encoding: HLAfixedRecord

Semantics: *These fields contain information about subaggregates not registered in the federation.*

Name	Type	Semantic
AggregateType	EntityTypeStruct	<i>This field shall specify the aggregates common to this system list.</i>
NumberOfAggregatesOfThisType	UnsignedInteger16	<i>This field shall specify the number of aggregates that have the type specified in AggregateType field.</i>

SilentEntityStruct

Encoding: HLAfixedRecord

Semantics: *These fields contain information about entities not registered in the federation.*

Name	Type	Semantic
NumberOfEntitiesOfThisType	UnsignedInteger16	<i>This field shall specify the number of entities that have the type specified in the field EntityType.</i>
EntityType	EntityTypeStruct	<i>This field shall specify the entity types common to the entities in this system list.</i>
EntityAppearance RPRnoteAggregate5	UnsignedInteger32Array	<i>This field shall specify the entity appearances of entities in the aggregate that deviate from the default. The number of array elements shall be between zero and the number of entities of this type. Simulation applications representing the aggregate that do not model entity appearances shall set this field to zero elements. Other simulations can safely assume that any entity appearances not specified are default appearances.</i>

5.3. Notes

RPRnoteAggregate1

Semantics: *Default value: empty*

RPRnoteAggregate2

Semantics: *Not optional*

RPRnoteAggregate3

Semantics: *Default value: Other*

RPRnoteAggregate4

Semantics: *The units and semantics for the MarkingData array elements are specified by the value of the MarkingEncodingType.*

RPRnoteAggregate5

Semantics: *The interpretation of the 32 bits defining the entity appearance is to be derived from the federation agreements, which could refer to the entity appearance record as defined in SISO-REF-010. The reason that this has not been split out into separate fields (as has been done for the subclasses of BaseEntity) is the difficulty of providing an efficient manner of defining an array of such appearance fields.*

6. Module Synthetic Environment



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Synthetic Environment FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	The Synthetic Environment FOM module relates to the simulation of environmental information both under the form of (point, linear, areal) objects and processes.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Base
Foundation
Enumerations

6.1. Object Classes

6.1.1. EnvironmentalEntity

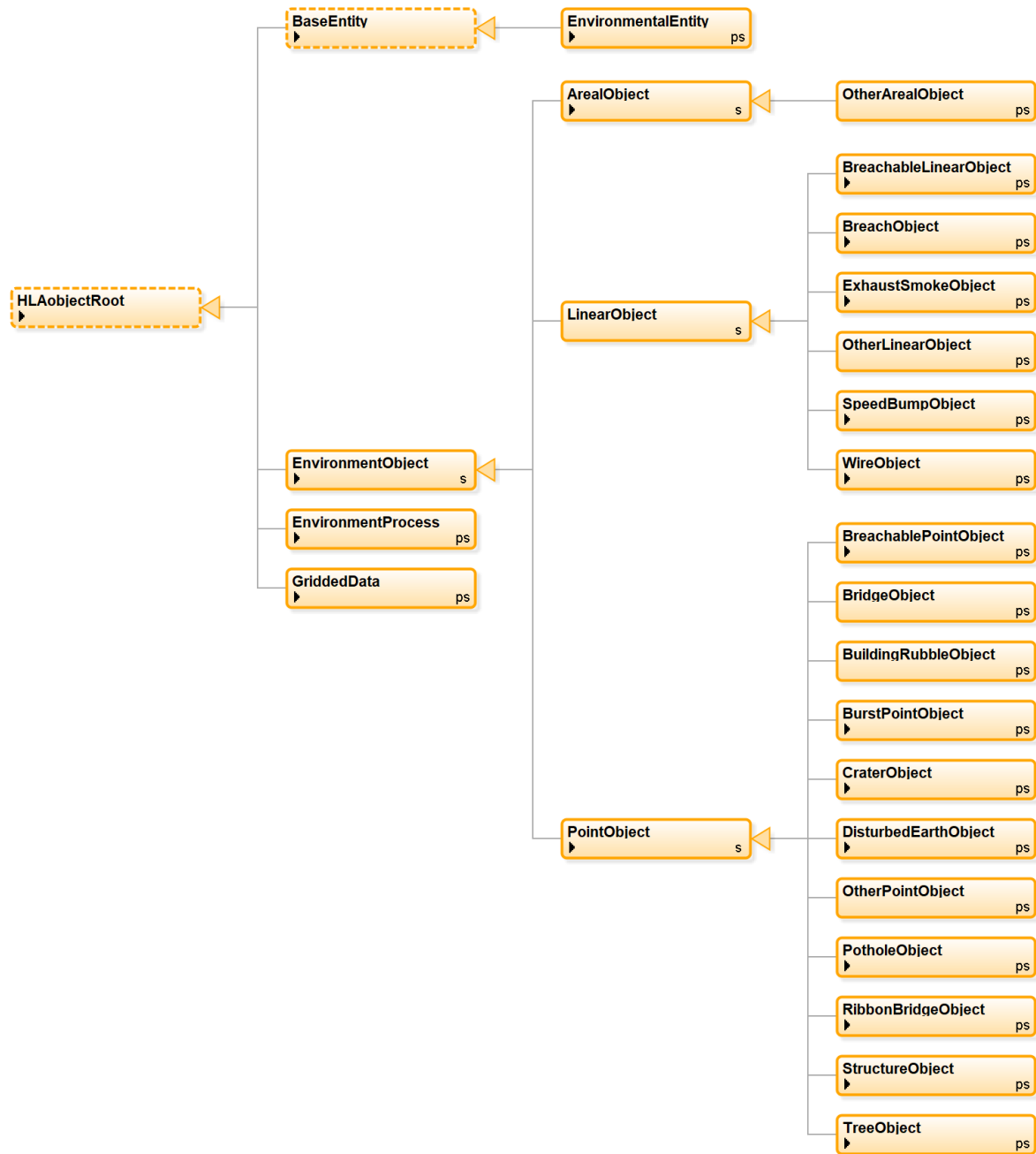
Full Name: HLAObjectRoot.BaseEntity.EnvironmentalEntity

Sharing: Publish/Subscribe

Semantics: *An object which has physical extent but not necessarily fixed size and shape, such as meteorological effects such as clouds or smokes.*

Attributes:

Density RPRnoteSE1 RPRnoteSE2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEnvironmentalDensityEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes the density.				
MaskedCloaked RPRnoteSE2 RPRnoteSE3	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Describes whether or not the entity is masked or cloaked.				
IEDPresenceIndicator RPRnoteSE3 RPRnoteSE4	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The Entity is an IED or contains an IED. The extended appearance record (if available) will identify how well hidden the IED is on the Entity. An Attached Part (if applicable, for instance a jury-rigged munition does not apply here) will identify the IED explicitly.				



EntityType RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The category of the entity.					

EntityIdentifier RPRnoteBase2 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The unique identifier for the entity instance.					

IsPartOf RPRnoteBase3 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IsPartOfStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Defines if the entity is a constituent part of another entity (denoted the host entity). If the entity is a constituent part of another entity then the HostEntityIdentifier shall be set to the EntityIdentifier of the host entity and the HostRTIObjectIdentifier shall be set to the RTI object instance ID of the host entity. If the entity is not a constituent part of another entity then the HostEntityIdentifier shall be set to 0.0.0 and the HostRTIObjectIdentifier shall be set to the empty string.					

Spatial RPRnoteBase2 RPRnoteBase4 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the world coordinate system.					

RelativeSpatial RPRnoteBase3 RPRnoteBase4 Inherited from BaseEntity in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpatialVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteBase5				
	Semantics					
	Spatial state of the entity's coordinate system with respect to the host entity's coordinate system. References to the host entity are provided in the IsPartOf attribute.					

HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.2. EnvironmentObject

Full Name: HLAobjectRoot.EnvironmentObject

Sharing: Subscribe

Semantics: *A base class of environment point, linear, or areal object classes.*

Attributes:

ObjectIdentifier RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Identifies this EnvironmentObject instance (point, linear or areal).					
ReferencedObjectIdentifier RPRnoteSE6	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.					

ForceIdentifier RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.3. ArealObject

Full Name: HLAobjectRoot.EnvironmentObject.ArealObject

Sharing: Subscribe

Semantics: A synthetic environment object that is geometrically anchored to the terrain with a set of three or more points, which comes to a closure.

Attributes:

PointsData RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the physical location (a collection of points, in the world coordinate system) of the object.				

PercentComplete RPRnoteSE7 RPRnoteSE8	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the percent completion of the object.				
Damage RPRnoteSE8 RPRnoteSE9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the damaged appearance of the object.				
Predistributed RPRnoteSE8 RPRnoteSE10	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether or not the object was created before the start of the exercise.				
State RPRnoteSE8 RPRnoteSE11	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).				
IsSmoking RPRnoteSE3 RPRnoteSE8	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether or not the object is smoking (creating a smoke plume).				

IsFlaming RPRnoteSE3 RPRnoteSE8	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object is burning and flames are visible.				
IEDPresent RPRnoteSE8 RPRnoteSE12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object has or contains an IED.				
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				

ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.4. OtherArealObject

Full Name: HLAobjectRoot.EnvironmentObject.ArealObject.OtherArealObject

Sharing: Publish/Subscribe

Semantics: *Areal objects other than Minefield objects.*

Attributes:

PointsData RPRnoteSE5 Inherited from ArealObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the physical location (a collection of points, in the world coordinate system) of the object.				
PercentComplete RPRnoteSE7 RPRnoteSE8 Inherited from ArealObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the percent completion of the object.				

Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from ArealObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the damaged appearance of the object.</i>				
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from ArealObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object was created before the start of the exercise.</i>				
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from ArealObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>				
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from ArealObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>				
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from ArealObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object is burning and flames are visible.</i>				

IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from ArealObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object has or contains an IED.				
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliaible	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.5. LinearObject

Full Name: HLAobjectRoot.EnvironmentObject.LinearObject

Sharing: Subscribe

Semantics: *A synthetic environment object that has size and an orientation and is geometrically anchored to the terrain with one point.*

Attributes:

ObjectIdentifier <i>RPRnoteSE5</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>				
ReferencedObjectIdentifier <i>RPRnoteSE6</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>				
ForceIdentifier <i>RPRnoteSE5</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>				

ObjectType <i>RPRnoteSES</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>Identifies the type of this EnvironmentObject instance.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.6. BreachableLinearObject

Full Name: HLAobjectRoot.EnvironmentObject.LinearObject.BreachableLinearObject

Sharing: Publish/Subscribe

Semantics: *A linear object that can be breached.*

Attributes:

SegmentRecords <i>RPRnoteSES</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	BreachableSegmentStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies a breachable linear object.</i>				
ObjectIdentifier <i>RPRnoteSES</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>				

ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTLObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.7. BreachObject

Full Name: HLAobjectRoot.EnvironmentObject.LinearObject.BreachObject

Sharing: Publish/Subscribe

Semantics: A breach linear object.

Attributes:

SegmentRecords RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	BreachStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies a breach linear object.					
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Identifies this EnvironmentObject instance (point, linear or areal).					
ReferencedObjectIdentifier RPRnoteSE6 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.					
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Identifies the force that created or modified this EnvironmentObject instance.					
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Identifies the type of this EnvironmentObject instance.					

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliaible	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.8. ExhaustSmokeObject

Full Name: HLAobjectRoot.EnvironmentObject.LinearObject.ExhaustSmokeObject

Sharing: Publish/Subscribe

Semantics: *An exhaust smoke linear object.*

Attributes:

SegmentRecords <i>RPRnoteSE5</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ExhaustSmokeStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies an exhaust smoke linear object.</i>				
ObjectIdentifier <i>RPRnoteSE5</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>				
ReferencedObjectIdentifier <i>RPRnoteSE6</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>				

ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.9. OtherLinearObject

Full Name: HLAobjectRoot.EnvironmentObject.LinearObject.OtherLinearObject

Sharing: Publish/Subscribe

Semantics: Linear objects other than BreachableLinear, Breach, ExhaustSmoke, or MinefieldLaneMarker objects.

Attributes:

ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				

ReferencedObjectIdentifier <i>RPRnoteSE6</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTLObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>				
ForceIdentifier <i>RPRnoteSE5</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>				
ObjectType <i>RPRnoteSE5</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>Identifies the type of this EnvironmentObject instance.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

6.1.10. SpeedBumpObject

Full Name: HLAobjectRoot.EnvironmentObject.LinearObject.SpeedBumpObject

Sharing: Publish/Subscribe

Semantics: *A speed bump object.*

Attributes:

SegmentRecords RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SpeedBumpStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies a speed bump object.				
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliaible	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.11. WireObject

Full Name: HLAobjectRoot.EnvironmentObject.LinearObject.WireObject

Sharing: Publish/Subscribe

Semantics: *A wire object.*

Attributes:

SegmentRecords <i>RPRnoteSE5</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	WireStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies a wire object.</i>				
ObjectIdentifier <i>RPRnoteSE5</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>				
ReferencedObjectIdentifier <i>RPRnoteSE6</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>				

ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.12. PointObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject

Sharing: Subscribe

Semantics: A synthetic environment object that is geometrically anchored to the terrain with a single point.

Attributes:

Location RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the physical location of the object in the world coordinate system.				

Orientation RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>					
PercentComplete RPRnoteSE7 RPRnoteSE8	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the percent completion of the object.</i>					
Damage RPRnoteSE8 RPRnoteSE9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the damaged appearance of the object instance.</i>					
Predistributed RPRnoteSE8 RPRnoteSE10	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies whether or not the object was created before the start of the exercise.</i>					

State RPRnoteSE8 RPRnoteSE11	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).					
IsSmoking RPRnoteSE3 RPRnoteSE8	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not the object is smoking (creating a smoke plume).					
IsFlaming RPRnoteSE3 RPRnoteSE8	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether the object is burning and flames are visible.					
IEDPresent RPRnoteSE8 RPRnoteSE12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether the object has or contains an IED.					
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Identifies this EnvironmentObject instance (point, linear or areal).					

ReferencedObjectIdentifier <i>RPRnoteSE6</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>				
ForceIdentifier <i>RPRnoteSE5</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>				
ObjectType <i>RPRnoteSE5</i> <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>Identifies the type of this EnvironmentObject instance.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

6.1.13. BreachablePointObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.BreachableObject

Sharing: Publish/Subscribe

Semantics: *A point object that can be breached.*

Attributes:

BreachState RPRnoteSE5 RPRnoteSE13	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectSpecificBreachStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the breached appearance of the object.				
Location RPRnoteSE5 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the physical location of the object in the world coordinate system.				
Orientation RPRnoteSE5 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.				
PercentComplete RPRnoteSE7 RPRnoteSE8 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the percent completion of the object.				

Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies the damaged appearance of the object instance.</i>					
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object was created before the start of the exercise.</i>					
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>					
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object is smoking (creating a smoke plume).</i>					
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether the object is burning and flames are visible.</i>					

IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object has or contains an IED.				
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliaible	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.14. BridgeObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.BridgeObject

Sharing: Publish/Subscribe

Semantics: *A stationary bridge, AVLB object.*

Attributes:

Location <i>RPRnoteSE5</i> <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the physical location of the object in the world coordinate system.</i>				
Orientation <i>RPRnoteSE5</i> <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>				
PercentComplete <i>RPRnoteSE7 RPRnoteSE8</i> <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the percent completion of the object.</i>				

Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies the damaged appearance of the object instance.</i>					
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object was created before the start of the exercise.</i>					
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>					
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object is smoking (creating a smoke plume).</i>					
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether the object is burning and flames are visible.</i>					

IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object has or contains an IED.				
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliaible	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.15. BuildingRubbleObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.BuildingRubbleObject

Sharing: Publish/Subscribe

Semantics: *A building rubble object.*

Attributes:

Location <i>RPRnoteSE5</i> <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the physical location of the object in the world coordinate system.</i>				
Orientation <i>RPRnoteSE5</i> <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>				
PercentComplete <i>RPRnoteSE7 RPRnoteSE8</i> <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the percent completion of the object.</i>				

Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the damaged appearance of the object instance.</i>				
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object was created before the start of the exercise.</i>				
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>				
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>				
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object is burning and flames are visible.</i>				

IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object has or contains an IED.				
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliaible	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.16. BurstPointObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.BurstPointObject

Sharing: Publish/Subscribe

Semantics: *A burst point object.*

Attributes:

OpacityPercent RPRnoteSE5 RPRnoteSE14	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLA bestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the opacity of the smoke.</i>					
Size RPRnoteSE5 RPRnoteSE14	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MeterUnsignedInteger8	PS	DA	RO	HLA bestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the radius of the cylinder approximating an individual smoke burst ; for multiple bursts, the center bottom of each cylinder is calculated based on the model used to represent the multiple bursts.</i>					
Height RPRnoteSE5 RPRnoteSE14	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MeterUnsignedInteger8	PS	DA	RO	HLA bestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the height of the cylinder approximating an individual smoke burst ; for multiple bursts, the center bottom of each cylinder is calculated based on the model used to represent the multiple bursts.</i>					

NumberOfBursts RPRnoteSE5 RPRnoteSE14 RPRnoteSE15	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the number of bursts in the instance of tactical smoke.</i>					
ChemicalType RPRnoteSE5 RPRnoteSE14	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectSpecificChemicalTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the chemical content of the smoke.</i>					
Location RPRnoteSE5 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the physical location of the object in the world coordinate system.</i>					
Orientation RPRnoteSE5 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>					
PercentComplete RPRnoteSE7 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the percent completion of the object.</i>					

Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies the damaged appearance of the object instance.</i>					
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object was created before the start of the exercise.</i>					
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>					
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object is smoking (creating a smoke plume).</i>					
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether the object is burning and flames are visible.</i>					

IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object has or contains an IED.				
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.17. CraterObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.CraterObject

Sharing: Publish/Subscribe

Semantics: *A pit, depression, or cavity formed in the surface of the earth by an explosion. The depression's shape can range from saucer to conical, depending largely on the depth of burst.*

Attributes:

Diameter <i>RPRnoteSE5 RPRnoteSE16</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MeterUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the diameter of the crater in meters, where the center of the crater is at the point object location.</i>				
Depth <i>RPRnoteSE5 RPRnoteSE16</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	CentimeterUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the depth of the crater in centimeters.</i>				
Height <i>RPRnoteSE5 RPRnoteSE16</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	CentimeterUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the height of the crater in centimeters.</i>				

BreachState RPRnoteSE5 RPRnoteSE16	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceLinearObjectTankDitchBreachEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the breached appearance of the object.					
Location RPRnoteSE5 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the physical location of the object in the world coordinate system.					
Orientation RPRnoteSE5 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.					
PercentComplete RPRnoteSE7 RPRnoteSE8 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the percent completion of the object.					

Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies the damaged appearance of the object instance.</i>					
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object was created before the start of the exercise.</i>					
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>					
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether or not the object is smoking (creating a smoke plume).</i>					
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Specifies whether the object is burning and flames are visible.</i>					

IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object has or contains an IED.				
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.18. DisturbedEarthObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.DisturbedEarthObject

Sharing: Publish/Subscribe

Semantics: *A disturbed earth / road object.*

Attributes:

Diameter RPRnoteSE5 RPRnoteSE17	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DecimeterUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the diameter in decimeters.</i>				
Height RPRnoteSE5 RPRnoteSE17	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	CentimeterUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the height in centimeters.</i>				
Contrast RPRnoteSE5 RPRnoteSE17	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ContrastLevelUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes indicating 16 levels of contrast (low to high) distinguishing variation of object from surrounding surface.</i>				

Location RPRnoteSE5 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the physical location of the object in the world coordinate system.</i>					
Orientation RPRnoteSE5 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>					
PercentComplete RPRnoteSE7 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the percent completion of the object.</i>					
Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the damaged appearance of the object instance.</i>					

Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object was created before the start of the exercise.</i>				
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>				
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>				
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object is burning and flames are visible.</i>				
IEDPresent RPRnoteSE8 RPRnoteSE12 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object has or contains an IED.</i>				

ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.19. OtherPointObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.OtherPointObject

Sharing: Publish/Subscribe

Semantics: *Point objects other than Breachable, Bridge, BuildingRubble, Burst, Crater, DisturbedEarth, Pothole, RibbonBridge, Structure, Tree objects.*

Attributes:

Location RPRnoteSE5 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the physical location of the object in the world coordinate system.</i>					
Orientation RPRnoteSE5 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>					
PercentComplete RPRnoteSE7 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the percent completion of the object.</i>					
Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the damaged appearance of the object instance.</i>					

Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object was created before the start of the exercise.</i>				
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>				
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>				
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object is burning and flames are visible.</i>				
IEDPresent RPRnoteSE8 RPRnoteSE12 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object has or contains an IED.</i>				

ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTLObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.20. PotholeObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.PotholeObject

Sharing: Publish/Subscribe

Semantics: *A pothole object.*

Attributes:

Diameter RPRnoteSE5RPRnoteSE18	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DecimeterUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the diameter in decimeters.					
Depth RPRnoteSE5RPRnoteSE18	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	CentimeterUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the depth in centimeters.					
Location RPRnoteSE5 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the physical location of the object in the world coordinate system.					
Orientation RPRnoteSE5 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.					

PercentComplete RPRnoteSE7 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the percent completion of the object.</i>				
Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the damaged appearance of the object instance.</i>				
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object was created before the start of the exercise.</i>				
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>				
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>				

IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object is burning and flames are visible.</i>				
IEDPresent RPRnoteSE8 RPRnoteSE12 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object has or contains an IED.</i>				
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>				
ReferencedObjectIdentifier RPRnoteSE6 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>				
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>				

ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.21. RibbonBridgeObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.RibbonBridgeObject

Sharing: Publish/Subscribe

Semantics: A ribbon bridge object.

Attributes:

NumberOfSegments RPRnoteSE5 RPRnoteSE19	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the number of segments composing the ribbon bridge.				
Location RPRnoteSE5 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the physical location of the object in the world coordinate system.				

Orientation <i>RPRnoteSE5</i> <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>					
PercentComplete <i>RPRnoteSE7 RPRnoteSE8</i> <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the percent completion of the object.</i>					
Damage <i>RPRnoteSE8 RPRnoteSE9</i> <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the damaged appearance of the object instance.</i>					
Predistributed <i>RPRnoteSE8 RPRnoteSE10</i> <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies whether or not the object was created before the start of the exercise.</i>					

State RPRnoteSE8 RPRnoteSE11 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).				
IsSmoking RPRnoteSE3 RPRnoteSE8 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether or not the object is smoking (creating a smoke plume).				
IsFlaming RPRnoteSE3 RPRnoteSE8 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object is burning and flames are visible.				
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object has or contains an IED.				
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				

ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTLObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.22. StructureObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.StructureObject

Sharing: Publish/Subscribe

Semantics: A building structure object.

Attributes:

Color RPRnoteSE5 RPRnoteSE20	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ColorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the color of the building or structure.					
PaintScheme RPRnoteSE5 RPRnoteSE20	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	BuildingPaintSchemeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes the visual paint design.					
IsApertureOpen RPRnoteSE5 RPRnoteSE20	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Describes whether the aperture (e.g., tent) is open or closed.					
Location RPRnoteSE5 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the physical location of the object in the world coordinate system.					
Orientation RPRnoteSE5 Inherited from PointObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.					

PercentComplete RPRnoteSE7 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the percent completion of the object.</i>				
Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the damaged appearance of the object instance.</i>				
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object was created before the start of the exercise.</i>				
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>				
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>				

IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object is burning and flames are visible.</i>				
IEDPresent RPRnoteSE8 RPRnoteSE12 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object has or contains an IED.</i>				
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>				
ReferencedObjectIdentifier RPRnoteSE6 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>				
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>				

ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>Identifies the type of this EnvironmentObject instance.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.23. TreeObject

Full Name: HLAobjectRoot.EnvironmentObject.PointObject.TreeObject

Sharing: Publish/Subscribe

Semantics: *A tree object.*

Attributes:

Season RPRnoteSE5 RPRnoteSE21	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SeasonEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the visually represented season.</i>				
LeafCoverage RPRnoteSE5 RPRnoteSE21	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LeafCoverageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Describes the leaf coverage.</i>				

Location RPRnoteSE5 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the physical location of the object in the world coordinate system.</i>					
Orientation RPRnoteSE5 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>					
PercentComplete RPRnoteSE7 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the percent completion of the object.</i>					
Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the damaged appearance of the object instance.</i>					

Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object was created before the start of the exercise.</i>				
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>				
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>				
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object is burning and flames are visible.</i>				
IEDPresent RPRnoteSE8 RPRnoteSE12 <i>Inherited from PointObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object has or contains an IED.</i>				

ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTLObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.24. EnvironmentProcess

Full Name: HLAobjectRoot.EnvironmentProcess

Sharing: Publish/Subscribe

Semantics: *An object class used to communicate information about environmental processes and effects.*

Attributes:

ProcessIdentifier RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Identifies which process issued the environmental process update.					
Type RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Identifies the environmental process type.					
ModelType RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentalProcessModelTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Identifies the model used for generating this environmental process update.					
IsActive RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether the environmental process is active or not.					

SequenceNumber	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Optional, used to support update sequencing. If not used, set to EP_NO_SEQUENCE (fixed value: FFFF (H)); when used, set to zero for each exercise and incremented by one for each update sent.</i>				
EnvironmentRecData RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentRecStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies environment records.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.1.25. GriddedData

Full Name: HLAobjectRoot.GriddedData

Sharing: Publish/Subscribe

Semantics: *An object class used to depict global, spatially varying environmental effects.*

Attributes:

GridIdentifier RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>Identifies the environmental simulation application.</i>				

CoordinateSystem RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	GriddedDataCoordinateSystemEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the coordinate system used to locate the data grid.</i>					
ConstantGrid RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	GriddedDataConstantGridEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies whether the grid axes remain constant for the life of the data grid.</i>					
EnvironmentType RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					
	<i>Identifies the type of environmental entity being described.</i>					
Orientation RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the orientation of the data grid, with Euler angles.</i>					
SampleTime RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ClockTimeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the valid time of the environmental data sample.</i>					

TotalValues RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the number of data values that make up this grid.				
VectorDimension RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the number of data values at each grid point ; VectorDimension shall be one for scalar data, and shall be greater than one when multiple enumerated environmental data values are sent for each grid point (e.g. u, v, w wind components have VectorDimension = 3).				
GridAxisInfo RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	GridAxisStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies information on grid axes used to define the data grid (e.g. three grid axes for an x, y, z coordinate system).				
GridDataInfo RPRnoteSE5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	GridDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies information on grid data representations.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

6.2. Interaction Classes

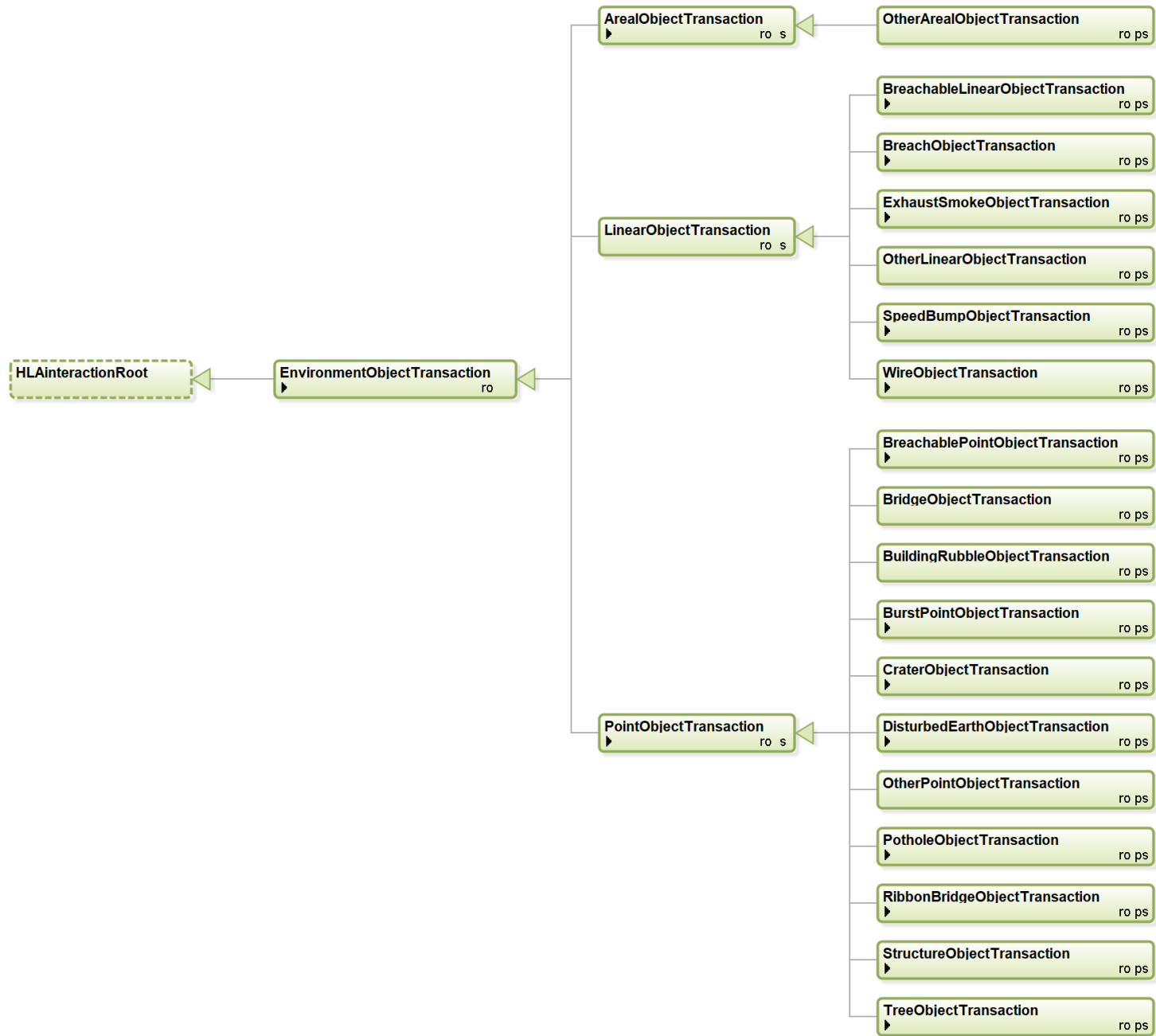
6.2.1. HLAinteractionRoot

Full Name: HLAinteractionRoot
Sharing: Publish/Subscribe
Transportation type: HLAREliable
Order: Receive
Dimensions:
Semantics:
Parameters: -

6.2.2. EnvironmentObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction
Sharing:
Transportation type: HLABestEffort
Order: Receive
Dimensions:
Semantics: A base interaction for modifying instances of point, linear and areal environment object classes.
Parameters:

Name	Datatype	Semantics
ObjectIdentifier RPRnoteSE5	EntityIdentifierStruct	Identifies this EnvironmentObject instance (point, linear or areal).
ReferencedObjectIdentifier RPRnoteSE5	RTIobjectId	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.
ForceIdentifier RPRnoteSE5	ForceIDEnum8	Identifies the force that created or modified this EnvironmentObject instance.
ObjectType RPRnoteSE5	EnvironmentObjectTypeStruct	Identifies the type of this EnvironmentObject instance.



Name	Datatype	Semantics
RequestingIdentifier RPRnoteSE5	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.3. ArealObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.ArealObjectTransaction

Sharing: Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An interaction for modifying instances of the ArealObject class.*

Parameters:

Name	Datatype	Semantics
PointsData RPRnoteSE5	LocationStructArray	<i>Specifies the physical location (a collection of points, in the world coordinate system) that defines the object.</i>
PercentComplete RPRnoteSE7 RPRnoteSE8	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>
IsSmoking RPRnoteSE8 RPRnoteSE3	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE8 RPRnoteSE3	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>
IEDPresent RPRnoteSE8 RPRnoteSE12	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the object has or contains an IED.</i>

Name	Datatype	Semantics
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTLObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.4. OtherArealObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.ArealObjectTransaction.OtherArealObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the OtherArealObject class.*

Parameters:

Name	Datatype	Semantics
PointsData RPRnoteSE5 Inherited from ArealObjectTransaction in Synthetic Environment	LocationStructArray	<i>Specifies the physical location (a collection of points, in the world coordinate system) that defines the object.</i>
PercentComplete RPRnoteSE7 RPRnoteSE8 Inherited from ArealObjectTransaction in Synthetic Environment	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9 Inherited from ArealObjectTransaction in Synthetic Environment	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10 Inherited from ArealObjectTransaction in Synthetic Environment	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11 Inherited from ArealObjectTransaction in Synthetic Environment	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>
IsSmoking RPRnoteSE8 RPRnoteSE3 Inherited from ArealObjectTransaction in Synthetic Environment	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE8 RPRnoteSE3 Inherited from ArealObjectTransaction in Synthetic Environment	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from ArealObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the object has or contains an IED.</i>
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>

Name	Datatype	Semantics
ReferencedObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	RTLObjectId	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	ForceIDEnum8	Identifies the force that created or modified this EnvironmentObject instance.
ObjectType RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EnvironmentObjectTypeStruct	Identifies the type of this EnvironmentObject instance.
RequestingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application sending the EnvironmentObjectTransaction interaction.
ReceivingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.

6.2.5. LinearObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.LinearObjectTransaction

Sharing: Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: An interaction for modifying instances of the LinearObject class.

Parameters:

Name	Datatype	Semantics
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EntityIdentifierStruct	Identifies this EnvironmentObject instance (point, linear or areal).

Name	Datatype	Semantics
ReferencedObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	RTLObjectId	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	ForceIDEnum8	Identifies the force that created or modified this EnvironmentObject instance.
ObjectType RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EnvironmentObjectTypeStruct	Identifies the type of this EnvironmentObject instance.
RequestingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application sending the EnvironmentObjectTransaction interaction.
ReceivingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.

6.2.6. BreachableLinearObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.LinearObjectTransaction.BreachableLinearObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BreachableLinearObject class.

Parameters:

Name	Datatype	Semantics
SegmentRecords RPRnoteSE5	BreachableSegmentStructArray	Specifies a breachable linear object.

Name	Datatype	Semantics
ObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTLObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.7. BreachObjectTransaction

Full Name: HLAIinteractionRoot.EnvironmentObjectTransaction.LinearObjectTransaction.BreachObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BreachObject class.*

Parameters:

Name	Datatype	Semantics
SegmentRecords <u>RPRnoteSE5</u>	BreachStructArray	<i>Specifies a breach linear object.</i>
ObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTIobjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.8. ExhaustSmokeObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.LinearObjectTransaction.ExhaustSmokeObjectTransaction

Sharing: Publish/Subscribe

Transportation: HLAbestEffort

type:

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the ExhaustSmokeObject class.*

Parameters:

Name	Datatype	Semantics
SegmentRecords RPRnoteSE5	ExhaustSmokeStructArray	<i>Specifies an exhaust smoke linear object.</i>
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTIobjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.9. OtherLinearObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.LinearObjectTransaction.OtherLinearObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the OtherLinearObject class.*

Parameters:

Name	Datatype	Semantics
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTIobjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDenum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.10. SpeedBumpObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.LinearObjectTransaction.SpeedBumpObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the SpeedBumpObject class.*

Parameters:

Name	Datatype	Semantics
SegmentRecords RPRnoteSE5	SpeedBumpStructArray	<i>Specifies a speed bump linear object.</i>
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.11. WireObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.LinearObjectTransaction.WireObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the WireObject class.*

Parameters:

Name	Datatype	Semantics
SegmentRecords RPRnoteSE5	WireStructArray	<i>Specifies a wire linear object.</i>
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTIobjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.12. PointObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction
 Sharing: Subscribe
 Transportation type: HLABestEffort
 Order: Receive
 Dimensions:
 Semantics: *An interaction for modifying instances of the PointObject class.*
 Parameters:

Name	Datatype	Semantics
Location RPRnoteSE5	LocationStruct	<i>Specifies the physical location of the object in the world coordinate system.</i>
Orientation RPRnoteSE5	OrientationStruct	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>
PercentComplete RPRnoteSE8 RPRnoteSE7	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>
IsSmoking RPRnoteSE3 RPRnoteSE8	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE3 RPRnoteSE8	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>
IEDPresent RPRnoteSE8 RPRnoteSE12	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the object has or contains an IED.</i>
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>

Name	Datatype	Semantics
ReferencedObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	RTLObjectId	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	ForceIDEnum8	Identifies the force that created or modified this EnvironmentObject instance.
ObjectType RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EnvironmentObjectTypeStruct	Identifies the type of this EnvironmentObject instance.
RequestingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application sending the EnvironmentObjectTransaction interaction.
ReceivingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.

6.2.13. BreachablePointObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.BreachablePointObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BreachablePointObject class.

Parameters:

Name	Datatype	Semantics
BreachState RPRnoteSE5 RPRnoteSE13	AppearanceObjectSpecificBreachStateEnum8	Specifies the breached appearance of the object.

Name	Datatype	Semantics
Location RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	LocationStruct	<i>Specifies the physical location of the object in the world coordinate system.</i>
Orientation RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	OrientationStruct	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>
PercentComplete RPRnoteSE8 RPRnoteSE7 Inherited from PointObjectTransaction in Synthetic Environment	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>
IsSmoking RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the object has or contains an IED.</i>

Name	Datatype	Semantics
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTLObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.14. BridgeObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.BridgeObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BridgeObject class.*

Parameters:

Name	Datatype	Semantics
Location RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	LocationStruct	<i>Specifies the physical location of the object in the world coordinate system.</i>
Orientation RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	OrientationStruct	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>
PercentComplete RPRnoteSE8 RPRnoteSE7 Inherited from PointObjectTransaction in Synthetic Environment	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>
IsSmoking RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the object has or contains an IED.</i>

Name	Datatype	Semantics
ObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.15. BuildingRubbleObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.BuildingRubbleObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BuildingRubbleObject class.*

Parameters:

Name	Datatype	Semantics
Location RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	LocationStruct	Specifies the physical location of the object in the world coordinate system.
Orientation RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	OrientationStruct	Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.
PercentComplete RPRnoteSE8 RPRnoteSE7 Inherited from PointObjectTransaction in Synthetic Environment	PercentUnsignedInteger8	Specifies the percent completion of the object.
Damage RPRnoteSE8 RPRnoteSE9 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralDamageEnum8	Specifies the damaged appearance of the object.
Predistributed RPRnoteSE8 RPRnoteSE10 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralPredistributedEnum8	Specifies whether or not the object was created before the start of the exercise.
State RPRnoteSE8 RPRnoteSE11 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceEntityOrObjectStateEnum8	Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).
IsSmoking RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	Specifies whether or not the object is smoking (creating a smoke plume).
IsFlaming RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	Specifies whether the object is burning and flames are visible.
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	Specifies whether the object has or contains an IED.

Name	Datatype	Semantics
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.16. BurstPointObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.BurstPointObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the BurstPointObject class.*

Parameters:

Name	Datatype	Semantics
OpacityPercent RPRnoteSE5 RPRnoteSE14	PercentUnsignedInteger8	<i>Specifies the opacity of the smoke.</i>
Size RPRnoteSE5 RPRnoteSE14	MeterUnsignedInteger8	<i>Specifies the radius of the cylinder approximating an individual smoke burst ; for multiple bursts, the center bottom of each cylinder is calculated based on the model used to represent the multiple bursts.</i>
Height RPRnoteSE5 RPRnoteSE14	MeterUnsignedInteger8	<i>Specifies the height of the cylinder approximating an individual smoke burst ; for multiple bursts, the center bottom of each cylinder is calculated based on the model used to represent the multiple bursts.</i>
NumberOfBursts RPRnoteSE5 RPRnoteSE14 RPRnoteSE15	UnsignedInteger8	<i>Specifies the number of bursts in the instance of tactical smoke.</i>
ChemicalType RPRnoteSE5 RPRnoteSE14	AppearanceObjectSpecificChemicalTypeEnum8	<i>Specifies the chemical content of the smoke.</i>
Location RPRnoteSE5 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	LocationStruct	<i>Specifies the physical location of the object in the world coordinate system.</i>
Orientation RPRnoteSE5 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	OrientationStruct	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>
PercentComplete RPRnoteSE8 RPRnoteSE7 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>

Name	Datatype	Semantics
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>
IEDPresent RPRnoteSE8 RPRnoteSE12 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the object has or contains an IED.</i>
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.17. CraterObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.CraterObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the CraterObject class.*

Parameters:

Name	Datatype	Semantics
Diameter RPRnoteSE5 RPRnoteSE16	MeterUnsignedInteger8	<i>Specifies the diameter of the crater in meters, where the center of the crater is at the point object location.</i>
Depth RPRnoteSE5 RPRnoteSE16	CentimeterUnsignedInteger8	<i>Specifies the depth of the crater in centimeters.</i>
Height RPRnoteSE5 RPRnoteSE16	CentimeterUnsignedInteger8	<i>Specifies the height of the crater in centimeters.</i>
BreachState RPRnoteSE5 RPRnoteSE16	AppearanceLinearObjectTankDitchBreachEnum8	<i>Specifies the breached appearance of the object.</i>
Location RPRnoteSE5 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	LocationStruct	<i>Specifies the physical location of the object in the world coordinate system.</i>
Orientation RPRnoteSE5 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	OrientationStruct	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>
PercentComplete RPRnoteSE8 RPRnoteSE7 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>

Name	Datatype	Semantics
Damage RPRnoteSE8 RPRnoteSE9 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralDamageEnum8	Specifies the damaged appearance of the object.
Predistributed RPRnoteSE8 RPRnoteSE10 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralPredistributedEnum8	Specifies whether or not the object was created before the start of the exercise.
State RPRnoteSE8 RPRnoteSE11 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceEntityOrObjectStateEnum8	Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).
IsSmoking RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	Specifies whether or not the object is smoking (creating a smoke plume).
IsFlaming RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	Specifies whether the object is burning and flames are visible.
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	Specifies whether the object has or contains an IED.
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EntityIdentifierStruct	Identifies this EnvironmentObject instance (point, linear or areal).
ReferencedObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	RTObjectId	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	ForceIDEnum8	Identifies the force that created or modified this EnvironmentObject instance.

Name	Datatype	Semantics
ObjectType RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EnvironmentObjectTypeStruct	Identifies the type of this EnvironmentObject instance.
RequestingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application sending the EnvironmentObjectTransaction interaction.
ReceivingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.

6.2.18. DisturbedEarthObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.DisturbedEarthObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the DisturbedEarthObject class.

Parameters:

Name	Datatype	Semantics
Diameter RPRnoteSE5 RPRnoteSE17	DecimeterUnsignedInteger8	Specifies the diameter in decimeters.
Height RPRnoteSE5 RPRnoteSE17	CentimeterUnsignedInteger8	Specifies the height in centimeters.
Contrast RPRnoteSE5 RPRnoteSE17	ContrastLevelUnsignedInteger8	Specifies 16 levels of contrast (low to high) distinguishing variation of object from surrounding surface.

Name	Datatype	Semantics
Location RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	LocationStruct	Specifies the physical location of the object in the world coordinate system.
Orientation RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	OrientationStruct	Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.
PercentComplete RPRnoteSE8 RPRnoteSE7 Inherited from PointObjectTransaction in Synthetic Environment	PercentUnsignedInteger8	Specifies the percent completion of the object.
Damage RPRnoteSE8 RPRnoteSE9 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralDamageEnum8	Specifies the damaged appearance of the object.
Predistributed RPRnoteSE8 RPRnoteSE10 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralPredistributedEnum8	Specifies whether or not the object was created before the start of the exercise.
State RPRnoteSE8 RPRnoteSE11 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceEntityOrObjectStateEnum8	Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).
IsSmoking RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	Specifies whether or not the object is smoking (creating a smoke plume).
IsFlaming RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	Specifies whether the object is burning and flames are visible.
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	Specifies whether the object has or contains an IED.

Name	Datatype	Semantics
ObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTLObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.19. OtherPointObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.OtherPointObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the OtherPointObject class.*

Parameters:

Name	Datatype	Semantics
Location RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	LocationStruct	Specifies the physical location of the object in the world coordinate system.
Orientation RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	OrientationStruct	Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.
PercentComplete RPRnoteSE8 RPRnoteSE7 Inherited from PointObjectTransaction in Synthetic Environment	PercentUnsignedInteger8	Specifies the percent completion of the object.
Damage RPRnoteSE8 RPRnoteSE9 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralDamageEnum8	Specifies the damaged appearance of the object.
Predistributed RPRnoteSE8 RPRnoteSE10 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralPredistributedEnum8	Specifies whether or not the object was created before the start of the exercise.
State RPRnoteSE8 RPRnoteSE11 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceEntityOrObjectStateEnum8	Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).
IsSmoking RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	Specifies whether or not the object is smoking (creating a smoke plume).
IsFlaming RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	Specifies whether the object is burning and flames are visible.
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	Specifies whether the object has or contains an IED.

Name	Datatype	Semantics
ObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTLObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.20. PotholeObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.PotholeObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the PotholeObject class.*

Parameters:

Name	Datatype	Semantics
Diameter RPRnoteSE5 RPRnoteSE18	DecimeterUnsignedInteger8	<i>Specifies the diameter in decimeters.</i>
Depth RPRnoteSE5 RPRnoteSE18	CentimeterUnsignedInteger8	<i>Specifies the depth in centimeters.</i>
Location RPRnoteSE5 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	LocationStruct	<i>Specifies the physical location of the object in the world coordinate system.</i>
Orientation RPRnoteSE5 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	OrientationStruct	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>
PercentComplete RPRnoteSE8 RPRnoteSE7 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>

Name	Datatype	Semantics
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	Specifies whether the object has or contains an IED.
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EntityIdentifierStruct	Identifies this EnvironmentObject instance (point, linear or areal).
ReferencedObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	RTIobjectId	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	ForceIDEnum8	Identifies the force that created or modified this EnvironmentObject instance.
ObjectType RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EnvironmentObjectTypeStruct	Identifies the type of this EnvironmentObject instance.
RequestingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application sending the EnvironmentObjectTransaction interaction.
ReceivingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.

6.2.21. RibbonBridgeObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.RibbonBridgeObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the RibbonBridgeObject class.*

Parameters:

Name	Datatype	Semantics
NumberOfSegments RPRnoteSE5 RPRnoteSE19	UnsignedInteger8	<i>Specifies the number of segments composing the ribbon bridge.</i>
Location RPRnoteSE5 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	LocationStruct	<i>Specifies the physical location of the object in the world coordinate system.</i>
Orientation RPRnoteSE5 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	OrientationStruct	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>
PercentComplete RPRnoteSE8 RPRnoteSE7 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>

Name	Datatype	Semantics
IEDPresent RPRnoteSE8 RPRnoteSE12 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the object has or contains an IED.</i>
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTIobjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.22. StructureObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.StructureObjectTransaction
 Sharing: Publish/Subscribe
 Transportation type: HLABestEffort
 Order: Receive
 Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the StructureObject class.*

Parameters:

Name	Datatype	Semantics
Color RPRnoteSE5 RPRnoteSE20	ColorEnum8	<i>Specifies the color of the building or structure.</i>
PaintScheme RPRnoteSE5 RPRnoteSE20	BuildingPaintSchemeEnum8	<i>Specifies the visual paint design.</i>
IsApertureOpen RPRnoteSE5 RPRnoteSE20	RPRboolean	<i>Specifies whether the aperture (e.g., tent) is open or closed.</i>
Location RPRnoteSE5 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	LocationStruct	<i>Specifies the physical location of the object in the world coordinate system.</i>
Orientation RPRnoteSE5 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	OrientationStruct	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>
PercentComplete RPRnoteSE8 RPRnoteSE7 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from PointObjectTransaction in Synthetic Environment</i>	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>

Name	Datatype	Semantics
IsSmoking RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the object has or contains an IED.</i>
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	RTObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.2.23. TreeObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.PointObjectTransaction.TreeObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the TreeObject class.*

Parameters:

Name	Datatype	Semantics
Season RPRnoteSE5 RPRnoteSE21	SeasonEnum8	<i>Specifies the visually represented season.</i>
LeafCoverage RPRnoteSE5 RPRnoteSE21	LeafCoverageEnum8	<i>Specifies the leaf coverage.</i>
Location RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	LocationStruct	<i>Specifies the physical location of the object in the world coordinate system.</i>
Orientation RPRnoteSE5 Inherited from PointObjectTransaction in Synthetic Environment	OrientationStruct	<i>Specifies the angles of rotation around the coordinate axis between the object's attitude and the reference coordinate system axes ; these are calculated as the Tait-Bryan Euler angles, specifying the successive rotations needed to transform from the world coordinate system to the object coordinate system.</i>
PercentComplete RPRnoteSE8 RPRnoteSE7 Inherited from PointObjectTransaction in Synthetic Environment	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>

Name	Datatype	Semantics
State RPRnoteSE8 RPRnoteSE11 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceEntityOrObjectStateEnum8	Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).
IsSmoking RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	Specifies whether or not the object is smoking (creating a smoke plume).
IsFlaming RPRnoteSE3 RPRnoteSE8 Inherited from PointObjectTransaction in Synthetic Environment	RPRboolean	Specifies whether the object is burning and flames are visible.
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from PointObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	Specifies whether the object has or contains an IED.
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EntityIdentifierStruct	Identifies this EnvironmentObject instance (point, linear or areal).
ReferencedObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	RTObjectId	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	ForceIDEnum8	Identifies the force that created or modified this EnvironmentObject instance.
ObjectType RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EnvironmentObjectTypeStruct	Identifies the type of this EnvironmentObject instance.
RequestingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application sending the EnvironmentObjectTransaction interaction.

Name	Datatype	Semantics
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

6.3. Datatypes

6.3.1. Simple Datatypes

CentimeterUnsignedInteger8

Representation: [RPRunsignedInteger8BE](#)

Units: centimeter (cm)

Resolution: 1

Accuracy: perfect

Semantics: Datatype based on SI derived unit centimeter, unit symbol cm.

ContrastLevelUnsignedInteger8

Representation: [RPRunsignedInteger8BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: 16 levels of contrast, low to high (least significant 4 bits).

DecimeterUnsignedInteger8

Representation: [RPRunsignedInteger8BE](#)

Units: decimeter (cm)

Resolution: 1

Accuracy: perfect

Semantics: Datatype based on SI derived unit decimeter, unit symbol dm.

MeterUnsignedInteger8

Representation: [RPRunsignedInteger8BE](#)

Units: meter (m)

Resolution: 1

Accuracy: perfect

Semantics: Datatype based on SI base unit meter, unit symbol m.

6.3.2. Array Datatypes

BreachableSegmentStructArray

Element [BreachableSegmentStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAvariableArray

Semantics: *Specifies a breachable linear object (a collection of segments).*

BreachStructArray

Element [BreachStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAvariableArray

Semantics: *Specifies a breach linear object (a collection of segments).*

EnvironmentRecStructArray

Element [EnvironmentRecStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAvariableArray

Semantics: *Specifies environment records as a collection of geometry and state records.*

ExhaustSmokeStructArray

Element [ExhaustSmokeStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAvariableArray

Semantics: *Specifies an exhaust smoke linear object (a collection of smoke segments).*

GridAxisStructArray

Element [GridAxisStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Specifies detailed information for a collection of grid axes.*

GridDataStructArray

Element [GridDataStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Specifies detailed information for a collection of grid data representations.*

RPRbooleanArray8

Element [RPRboolean](#)

Type:

Cardinality: 8

Encoding: HLAfixedArray

Semantics: *Specifies whether each individual segment portion of length = BreachLength is breached or not.*

SpeedBumpStructArray

Element [SpeedBumpStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Specifies a speed bump object (a collection of segments).*

WireStructArray

Element [WireStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Specifies a wire object (a collection of segments).*

6.3.3. Fixed Record Datatypes

BreachableSegmentStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the characteristics of a breachable linear object segment.*

Name	Type	Semantic
SegmentParameters	LinearSegmentStruct	<i>Specifies the breachable linear object segment characteristics.</i>
BreachLength RPRnoteSE22	MeterUnsignedInteger8	<i>Specifies the fixed length in meters of a breached segment.</i>
BreachState RPRnoteSE22	AppearanceLinearObjectTankDitchBreachEnum8	<i>Specifies the breached appearance of the breachable linear object segment.</i>
SegmentBreached RPRnoteSE5 RPRnoteSE22	RPRbooleanArray8	<i>Specifies whether the segment portion beginning at the segment origin + (i*BreachLength) and extending BreachLength meters is breached or not.</i>

BreachStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the characteristics of a breach linear object segment.*

Name	Type	Semantic
SegmentParameters	LinearSegmentStruct	<i>Specifies the breach linear object segment characteristics.</i>

COMBICStateRecStruct [RPRnoteSE23](#)

Encoding: HLAfixedRecord

Semantics: *Record specifying COMBIC state record.*

Name	Type	Semantic
TimeSinceCreation	UnsignedInteger32	<i>Time since creation.</i>
MunitionSource	EntityTypeStruct	<i>Munition source.</i>
NumberOfSources	Integer32	<i>Number of sources.</i>
GeometryIndex	UnsignedInteger16	<i>Geometry index.</i>
SourceType	UnsignedInteger32	<i>Source type.</i>
BarrageRate	Float32	<i>Barrage rate.</i>

Name	Type	Semantic
BarrageDuration	Float32	<i>Barrage duration.</i>
BarrageCrosswindLength	Float32	<i>Barrage crosswind length.</i>
BarrageDownwindLength	Float32	<i>Barrage downwind length.</i>
DetonationVelocity	VelocityVectorStruct	<i>Detonation velocity.</i>

Cone1GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Cone 1 geometry record.*

Name	Type	Semantic
VertexLocation	LocationStruct	<i>Vertex location X, Y, Z.</i>
Orientation	OrientationStruct	<i>Orientation, specified by Euler angles.</i>
Height	Float32	<i>Height.</i>
PeakAngle	Float32	<i>Peak angle.</i>

Cone2GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Cone 2 geometry record.*

Name	Type	Semantic
VertexLocation	LocationStruct	<i>Vertex location X, Y, Z.</i>
Orientation	OrientationStruct	<i>Orientation, specified by Euler angles.</i>
Velocity	VelocityVectorStruct	<i>Velocity Vx, Vy, Vz.</i>
AngularVelocity	AngularVelocityVectorStruct	<i>Angular velocity Vx, Vy, Vz.</i>
Height	Float32	<i>Height.</i>
HeightRate	Float32	<i>Variation of height.</i>
PeakAngle	Float32	<i>Peak angle.</i>
PeakAngleRate	Float32	<i>Variation of peak angle.</i>

DimensionRateStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying length X, Y, Z rates.*

Name	Type	Semantic
XRate	VelocityMeterPerSecondFloat32	Variation of X axis length.
YRate	VelocityMeterPerSecondFloat32	Variation of Y axis length.
ZRate	VelocityMeterPerSecondFloat32	Variation of Z axis length.

Ellipsoid1GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Ellipsoid 1 geometry record.*

Name	Type	Semantic
CentroidLocation	LocationStruct	Centroid location X, Y, Z.
SigmaValue	DimensionStruct	Sigma dimensions.
Orientation	OrientationStruct	Orientation, specified by Euler angles.

Ellipsoid2GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Ellipsoid 2 geometry record.*

Name	Type	Semantic
CentroidLocation	LocationStruct	Centroid location X, Y, Z.
SigmaValue	DimensionStruct	Sigma dimensions.
SigmaRate	VelocityVectorStruct	Variation of sigma dimensions.
Orientation	OrientationStruct	Orientation, specified by Euler angles.
Velocity	VelocityVectorStruct	Velocity Vx, Vy, Vz.
AngularVelocity	AngularVelocityVectorStruct	Angular velocity Vx, Vy, Vz.

EnvironmentObjectTypeStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the domain, the kind and the specific identification of the environment object.*

Name	Type	Semantic
Domain	PlatformDomainEnum8	<i>Specifies the domain in which the object exists.</i>
Kind	ObjectKindEnum8	<i>Identifies the kind of object.</i>
Category	UnsignedInteger8	<i>Specifies the main category that describes the object.</i>
Subcategory	UnsignedInteger8	<i>Specifies a particular subcategory to which an object belongs based on the Category field.</i>

EnvironmentRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying an environment (geometry or state) record.*

Name	Type	Semantic
Index	UnsignedInteger32	<i>Identifies the sequentially numbered record index.</i>
DataVariant	EnvironmentRecVariantStruct	<i>Specifies geometry and state record alternatives.</i>

ExhaustSmokeStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the characteristics of a linear object smoke segment.*

Name	Type	Semantic
SegmentParameters	LinearSegmentStruct	<i>Specifies the linear object smoke segment characteristics.</i>
OpacityPercent RPRnoteSE24	PercentUnsignedInteger8	<i>Specifies the opacity of the linear object smoke segment.</i>
SmokeIsAttached RPRnoteSE24	RPRboolean	<i>Specifies whether the linear object smoke segment is attached to the vehicle.</i>
ChemicalType RPRnoteSE24	AppearanceObjectSpecificChemicalTypeEnum8	<i>Specifies the chemical content of the linear object smoke segment.</i>

FlareStateRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Flare state record.*

Name	Type	Semantic
TimeSinceCreation	UnsignedInteger32	<i>Time since creation.</i>
Source	EntityTypeStruct	<i>Source.</i>

Name	Type	Semantic
NumberIntensity	Integer32	<i>Number intensity.</i>
NumberOfSources	Integer32	<i>Number of sources.</i>
GeometryIndex	UnsignedInteger16	<i>Geometry index.</i>

GaussPlumeGeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Gaussian Plume geometry record.*

Name	Type	Semantic
SourceLocation	LocationStruct	<i>Source location X, Y, Z.</i>
Orientation	OrientationStruct	<i>Orientation, specified by Euler angles.</i>
PlumeDimension	PlumeDimensionStruct	<i>Plume dimensions.</i>
PlumeDimensionRate	PlumeDimensionRateStruct	<i>Variation of plume dimensions.</i>
LeadingEdge	Float32	<i>Leading edge.</i>
LeadingEdgeVelocity	VelocityVectorStruct	<i>Leading edge velocity.</i>

GaussPuffGeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Gaussian Puff geometry record.*

Name	Type	Semantic
PuffLocation	LocationStruct	<i>Puff location X, Y, Z.</i>
OriginationLocation	LocationStruct	<i>Origination location X, Y, Z.</i>
SigmaValue	DimensionStruct	<i>Sigma dimensions.</i>
SigmaRate	DimensionRateStruct	<i>Variation of sigma dimensions.</i>
Orientation	OrientationStruct	<i>Orientation, specified by Euler angles.</i>
Velocity	VelocityVectorStruct	<i>Velocity Vx, Vy, Vz.</i>
AngularVelocity	AngularVelocityVectorStruct	<i>Angular velocity Vx, Vy, Vz.</i>
CentroidHeight	Float32	<i>Centroid height.</i>

GridAxisStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying grid Xi axis data.*

Name	Type	Semantic
InitialValue	Float64	<i>Specifies the coordinate of the origin (or initial value) for the Xi axis.</i>
FinalValue	Float64	<i>Specifies the coordinate of the endpoint (or final value) for the Xi axis.</i>
TotalNumberOfPoints	UnsignedInteger16	<i>Specifies the total number of grid points along the Xi domain axis.</i>
InterleafFactor	UnsignedInteger8	<i>Specifies the integer valued interleaf factor along a domain (grid) Xi axis ; a value of one shall indicate no sub-sampling (interleaving), while integer values greater than one shall indicate the sampling frequency along an axis.</i>
AxisData	GridAxisDataVariantStruct	<i>Specifies axis data alternatives.</i>

GridDataStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying grid data representation.*

Name	Type	Semantic
SampleType	GriddedDataSampleTypeEnum16	<i>Specifies the environmental data sample.</i>
DataRepresentation	GridDataRepresentationVariantStruct	<i>Specifies grid data representation alternatives.</i>

GridValueType0Struct

Encoding: HLAfixedRecord

Semantics: *Record specifying type 0 data representation.*

Name	Type	Semantic
DataValues	OctetArray1Plus	<i>Specifies the environmental state variable data values. The data shall be represented as a stream of octets, the interpretation of which shall be agreed to prior to the start of the exercise.</i>

GridValueType1Struct

Encoding: HLAfixedRecord

Semantics: *Record specifying type 1 data representation.*

Name	Type	Semantic
Scale	Float32	<i>Specifies the constant scale factor used to scale the environmental state variable data values.</i>
Offset	Float32	<i>Specifies the constant offset used to scale the environmental state variable data values.</i>
DataValues	UnsignedInteger16Array1Plus	<i>Specifies the environmental state variable data values.</i>

GridValueType2Struct

Encoding: HLAfixedRecord

Semantics: *Record specifying type 2 data representation.*

Name	Type	Semantic
DataValues	Float32Array1Plus	<i>Specifies the environmental state variable data values.</i>

IrregularGridAxisStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying irregular (variable spacing) axis data.*

Name	Type	Semantic
NumberOfPoints	UnsignedInteger16	<i>Specifies the number of grid locations along the Xi axis.</i>
InitialIndex	UnsignedInteger16	<i>Specifies the index of the initial grid point along the Xi domain axis.</i>
CoordinateScale	Float64	<i>Specifies the value that linearly scales the coordinates of the grid locations for the Xi axis.</i>
CoordinateOffset	Float64	<i>Specifies the constant offset value that shall be applied to the grid locations for the Xi axis.</i>
GridLocations	UnsignedInteger16Array1Plus	<i>Specifies the coordinate values for the Ni grid locations along the irregular (variable spacing) Xi axis.</i>

Line1GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Line 1 geometry record.*

Name	Type	Semantic
StartPointLocation	LocationStruct	Start point location.
EndPointLocation	LocationStruct	End point location.

Line2GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Line 2 geometry record.*

Name	Type	Semantic
StartPointLocation	LocationStruct	Start point location.
EndPointLocation	LocationStruct	End point location.
StartPointVelocity	VelocityVectorStruct	Start point velocity.
EndPointVelocity	VelocityVectorStruct	End point velocity.

PlumeDimensionRateStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying plume dimension rates.*

Name	Type	Semantic
LengthRate	VelocityMeterPerSecondFloat32	Variation of plume length.
WidthRate	VelocityMeterPerSecondFloat32	Variation of plume width.
HeightRate	VelocityMeterPerSecondFloat32	Variation of plume height.

PlumeDimensionStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying plume dimensions.*

Name	Type	Semantic
Length	MeterFloat32	Plume length.
Width	MeterFloat32	Plume width.

Name	Type	Semantic
Height	MeterFloat32	<i>Plume height.</i>

Point2GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Point 2 geometry record.*

Name	Type	Semantic
Location	LocationStruct	<i>Location X, Y, Z.</i>
Velocity	VelocityVectorStruct	<i>Velocity Vx, Vy, Vz.</i>

RectVol1GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Rectangular Volume 1 geometry record.*

Name	Type	Semantic
CornerLocation	LocationStruct	<i>Corner location X, Y, Z.</i>
Dimensions	DimensionStruct	<i>Dimensions.</i>
Orientation	OrientationStruct	<i>Orientation, specified by Euler angles.</i>

RectVol2GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Rectangular Volume 2 geometry record.*

Name	Type	Semantic
CornerLocation	LocationStruct	<i>Corner location X, Y, Z.</i>
Dimensions	DimensionStruct	<i>Dimensions.</i>
DimensionsRate	DimensionRateStruct	<i>Variation of dimensions.</i>
Orientation	OrientationStruct	<i>Orientation, specified by Euler angles.</i>
Velocity	VelocityVectorStruct	<i>Velocity Vx, Vy, Vz.</i>
AngularVelocity	AngularVelocityVectorStruct	<i>Angular velocity Vx, Vy, Vz.</i>

RectVol3GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Rectangular Volume 3 geometry record.*

Name	Type	Semantic
CenterLocation	LocationStruct	Center location X, Y, Z.
Dimensions	DimensionStruct	Dimensions.
Orientation	OrientationStruct	Orientation, specified by Euler angles.

RegularGridAxisStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying regular (fixed spacing) axis data.*

Name	Type	Semantic
NumberOfPoints	UnsignedInteger16	Specifies the number of grid locations along the Xi axis.
InitialIndex	UnsignedInteger16	Specifies the index of the initial grid point along the Xi domain axis.

SpeedBumpStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the characteristics of a speed bump segment.*

Name	Type	Semantic
SegmentParameters	LinearSegmentStruct	Specifies the generic linear object segment characteristics.
Color RPRnoteSE25	ColorEnum8	Describes the color.
Contrast RPRnoteSE25	ContrastLevelUnsignedInteger8	Describes 16 levels of contrast (low to high) distinguishing variation of object from surrounding surface.
Material RPRnoteSE25	MaterialEnum8	Describes the type of material.

Sphere1GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Bounding Sphere & Sphere 1 geometry record.*

Name	Type	Semantic
CentroidLocation	LocationStruct	Centroid location X, Y, Z.

Name	Type	Semantic
Radius	Float32	<i>Radius.</i>

Sphere2GeomRecStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying Sphere 2 geometry record.*

Name	Type	Semantic
CentroidLocation	LocationStruct	<i>Centroid location X, Y, Z.</i>
Radius	Float32	<i>Radius.</i>
RadiusRate	Float32	<i>Variation of radius.</i>
Velocity	VelocityVectorStruct	<i>Velocity Vx, Vy, Vz.</i>
AngularVelocity	AngularVelocityVectorStruct	<i>Angular velocity Vx, Vy, Vz.</i>

UniformGeomRecStruct [RPRnoteSE26](#)

Encoding: HLAfixedRecord

Semantics: *The Uniform Geometry record represents a geometry that extends in all directions without limit.*

Name	Type	Semantic
Field1	UnsignedInteger8	<i>This field shall be set to zero.</i>

WireStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the characteristics of a wire segment.*

Name	Type	Semantic
SegmentParameters	LinearSegmentStruct	<i>Specifies the generic linear object segment characteristics.</i>
Color RPRnoteSE27	ColorEnum8	<i>Describing the color.</i>
Contrast RPRnoteSE27	ContrastLevelUnsignedInteger8	<i>Describing 16 levels of contrast (low to high) distinguishing variation of object from surrounding surface.</i>
IsBuried RPRnoteSE27	RPRboolean	<i>Describes whether the wire is buried or not.</i>

6.3.4. Variant Record Datatypes

EnvironmentRecVariantStruct

Encoding: RPRextendedVariantRecord

Discriminant
name: Type

Discriminant type: [EnvironmentalProcessRecordTypeEnum32](#)

Semantics: *Record specifying either a geometry or a state record.*

Name	Enumerator	Type	Semantics
Point1GeometryData	Point_Record_1	LocationStruct	<i>Specifies Point 1 geometry record alternative.</i>
Point2GeometryData	Point_Record_2	Point2GeomRecStruct	<i>Specifies Point 2 geometry record alternative.</i>
Line1GeometryData	Line_Record_1	Line1GeomRecStruct	<i>Specifies Line 1 geometry record alternative.</i>
Line2GeometryData	Line_Record_2	Line2GeomRecStruct	<i>Specifies Line 2 geometry record alternative.</i>
BoundingSphereGeometryData	Bounding_Sphere_Record	Sphere1GeomRecStruct	<i>Specifies Bounding Sphere geometry record alternative.</i>
Sphere1GeometryData	Sphere_Record_1	Sphere1GeomRecStruct	<i>Specifies Sphere 1 geometry record alternative.</i>
Sphere2GeometryData	Sphere_Record_2	Sphere2GeomRecStruct	<i>Specifies Sphere 2 geometry record alternative.</i>
Ellipsoid1GeometryData	Ellipsoid_Record_1	Ellipsoid1GeomRecStruct	<i>Specifies Ellipsoid 1 geometry record alternative.</i>
Ellipsoid2GeometryData	Ellipsoid_Record_2	Ellipsoid2GeomRecStruct	<i>Specifies Ellipsoid 2 geometry record alternative.</i>
Cone1GeometryData	Cone_Record_1	Cone1GeomRecStruct	<i>Specifies Cone 1 geometry record alternative.</i>
Cone2GeometryData	Cone_Record_2	Cone2GeomRecStruct	<i>Specifies Cone 2 geometry record alternative.</i>
RectVol1GeometryData	Rectangular_Volume_Record_1	RectVol1GeomRecStruct	<i>Specifies Rectangular Volume 1 geometry record alternative.</i>
RectVol2GeometryData	Rectangular_Volume_Record_2	RectVol2GeomRecStruct	<i>Specifies Rectangular Volume 2 geometry record alternative.</i>

Name	Enumerator	Type	Semantics
GaussPlumeGeometryData	Gaussian_Plume_Record	GaussPlumeGeomRecStruct	<i>Specifies Gaussian Plume geometry record alternative.</i>
GaussPuffGeometryData	Gaussian_Puff_Record	GaussPuffGeomRecStruct	<i>Specifies Gaussian Puff geometry record alternative.</i>
UniformGeometryData	Uniform_Geometry_Record	UniformGeomRecStruct	<i>Specifies Uniform geometry record alternative.</i>
COMBICStateData	COMBIC_State	COMBICStateRecStruct	<i>Specifies COMBIC state record alternative.</i>
FlareStateData	Flare_State	FlareStateRecStruct	<i>Specifies Flare state record alternative.</i>
RectVol3GeometryData	Rectangular_Volume_Record_3	RectVol3GeomRecStruct	<i>Specifies Rectangular Volume 3 geometry record alternative.</i>

GridAxisDataVariantStruct

Encoding: RPRextendedVariantRecord

Discriminant name: AxisType

Discriminant type: [GridAxisDescriptorAxisTypeEnum8](#)

Semantics: *Record specifying either regular (fixed spacing) or irregular (variable spacing) axis data.*

Name	Enumerator	Type	Semantics
RegularGridAxis	Regular_Axis	RegularGridAxisStruct	<i>Specifies regular (fixed spacing) axis data alternative.</i>
IrregularGridAxis	Irregular_Axis	IrregularGridAxisStruct	<i>Specifies irregular (variable spacing) axis data alternative.</i>

GridDataRepresentationVariantStruct

Encoding: RPRextendedVariantRecord

Discriminant name: DataRepresentation

Discriminant type: [GriddedDataDataRepresentationEnum16](#)

Semantics: *Record specifying either type 0 or type 1 or type 2 data representation.*

Name	Enumerator	Type	Semantics
Type0	Type_0	GridValueType0Struct	<i>Specifies type 0 data representation alternative.</i>

Name	Enumerator	Type	Semantics
Type1	Type_1	GridValueType1Struct	<i>Specifies type 1 data representation alternative.</i>
Type2	Type_2	GridValueType2Struct	<i>Specifies type 2 data representation alternative.</i>

6.4. Notes

RPRnoteSE1

Semantics: *Default value: Clear*

RPRnoteSE2

Semantics: *Source: SISO-REF-010 [UID 38]*

RPRnoteSE3

Semantics: *Default value: False*

RPRnoteSE4

Semantics: *Source: SISO-REF-010 [UID 457]*

RPRnoteSE5

Semantics: *Default value: Not optional*

RPRnoteSE6

Semantics: *Default value: Zero*

RPRnoteSE7

Semantics: *Default value: 100% complete*

RPRnoteSE8

Semantics: *Source: SISO-REF-010 [UID 229]*

RPRnoteSE9

Semantics: *Default value: No_Damage*

RPRnoteSE10

Semantics: *Default value: Object_Created_During_the_Exercise*

RPRnoteSE11

Semantics: *Default value: Active*

RPRnoteSE12

Semantics: *Default value: None*

RPRnoteSE13

Semantics: *Source: SISO-REF-010 [UID 231]*

RPRnoteSE14

Semantics: *Source: SISO-REF-010 [UID 232]*

RPRnoteSE15

Semantics: *For interoperability with DIS, according to SISO-REF-010, limit the value range to 0 to 63 (6 bits).*

RPRnoteSE16

Semantics: *Source: SISO-REF-010 [UID 233]*

RPRnoteSE17

Semantics: *Source: SISO-REF-010 [UID 485]*

RPRnoteSE18

Semantics: *Source: SISO-REF-010 [UID 486]*

RPRnoteSE19

Semantics: *Source: SISO-REF-010 [UID 234]*

RPRnoteSE20

Semantics: *Source: SISO-REF-010 [UID 230]*

RPRnoteSE21

Semantics: *Source: SISO-REF-010 [UID 487]*

RPRnoteSE22

Semantics: *Source: SISO-REF-010 [UID 235]*

RPRnoteSE23

Semantics: *The structure of this datatype follows that defined in SISO-REF-010 (Section 12.3.5.1).*

RPRnoteSE24

Semantics: *Source: SISO-REF-010 [UID 236]*

RPRnoteSE25

Semantics: *Source: SISO-REF-010 [UID 489]*

RPRnoteSE26

Semantics: *There is no data associated with a uniform geometry record. However, to enable inclusion as an alternative in the EnvironmentRecVariantStruct, the fixed record has been modeled according to [UID 266].*

RPRnoteSE27

Semantics: *Source: SISO-REF-010 [UID 488]*

7. Module Minefield



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Minefield FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	The Minefield FOM module relates to the simulation of minefields at both an aggregate and individual level simultaneously. In addition, it implements the two modes defined for exchanging the data of individual mines within minefields, depending on the number of mines.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

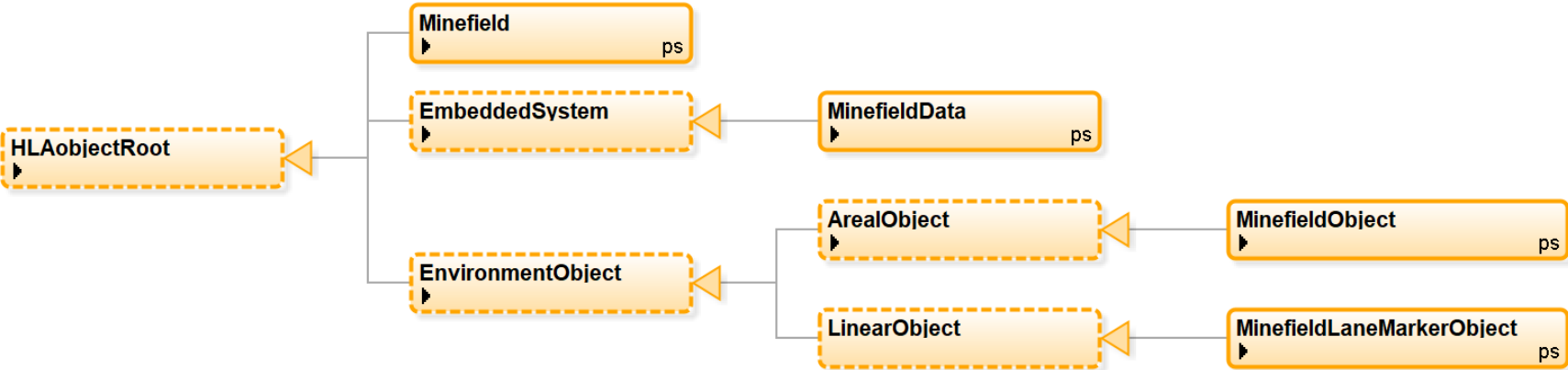
References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Synthetic Environment FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Synthetic Environment
Foundation
Enumerations
Base

7.1. Object Classes



7.1.1. Minefield

Full Name: HLAobjectRoot.Minefield
Sharing: Publish/Subscribe
Semantics: *An area of ground or water containing mines laid with or without a pattern.*
Attributes:

ActiveStatus RPRnoteMinefield1 RPRnoteMinefield2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MinefieldAppearanceActiveStatusEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the active status of the minefield.					

ForceIdentifier RPRnoteMinefield3	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Identifies the force to which the minefield belongs.</i>					
Lane RPRnoteMinefield2 RPRnoteMinefield4	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MinefieldAppearanceLaneEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies whether or not the minefield has an active lane.</i>					
MinefieldAppearanceType RPRnoteMinefield2 RPRnoteMinefield5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MinefieldAppearanceMinefieldTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the appearance information needed for displaying the symbology of the minefield.</i>					
MinefieldIdentifier RPRnoteMinefield6	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					
	<i>Uniquely identifies this minefield instance in association with the federate's site and application.</i>					
MinefieldLocation RPRnoteMinefield6	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the location of the center of the minefield.</i>					

MinefieldOrientation RPRnoteMinefield6	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the orientation of the minefield, with Euler angles.</i>					
MinefieldType RPRnoteMinefield6	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the minefield type.</i>					
MineTypes RPRnoteMinefield6	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the type of each mine contained within the minefield.</i>					
PerimeterPointCoordinates RPRnoteMinefield7	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PerimeterPointStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the location of each perimeter point, relative to the minefield location.</i>					
ProtocolMode RPRnoteMinefield8	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MinefieldStateProtocolModeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the mode (Heartbeat or Query Response Protocol) being used to communicate data about the minefield.</i>					

State RPRnoteMinefield2 RPRnoteMinefield6	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MinefieldAppearanceStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the minefield has been deactivated.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

7.1.2. MinefieldData

Full Name: HLAobjectRoot.EmbeddedSystem.MinefieldData

Sharing: Publish/Subscribe

Semantics: *The MinefieldData object class provides data about collections of mines within a minefield on an individual mine basis.*

Attributes:

GroundBurialDepthOffset RPRnoteMinefield9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DepthMeterFloat32Array	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the offset of the origin of the mine coordinate system with respect to the ground surface.</i>				
Fusing RPRnoteMinefield10	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MineFusingStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the primary and secondary fuse and anti-handling device for each mine in a collection of mines.</i>				

MineEmplacementTime RPRnoteMinefield9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ClockTimeStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the real-world (UTC) emplacement time of each mine in a collection of mines.</i>					
MineEntityIdentifier RPRnoteMinefield6	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MineIdentifierArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Identifies the mine entity identifier; the MineEntityIdentifier in conjunction with the MinefieldIdentifier form the unique identifier for each mine.</i>					
MinefieldIdentifier RPRnoteMinefield6	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Identifies the minefield to which the mines belong.</i>					
MineLocation RPRnoteMinefield10	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the location of the mine relative to the minefield location for each mine in a collection of mines.</i>					
MineOrientation RPRnoteMinefield10	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OrientationStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the orientation of the center axis direction of fire of the mine, relative to the minefield Coordinate System for each mine in a collection of mines.</i>					

MineType RPRnoteMinefield6	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the type of mine for the collection of mines contained within the MinefieldData object.					
NumberTripDetonationWires RPRnoteMinefield11	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8Array	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the number of trip detonation wires that exist for each mine in a collection of mines.					
NumberWireVertices RPRnoteMinefield12	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8Array	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the number of vertices for each trip wire of each mine in a collection of mines.					
PaintScheme RPRnoteMinefield9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MinefieldDataPaintSchemeStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the camouflage scheme/color of the mine.					
Reflectance RPRnoteMinefield9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MineDielectricDifferenceArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the local dielectric difference between the mine and the surrounding soil.					

ScalarDetectionCoefficient RPRnoteMinefield9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8Array	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the coefficient to be utilized to insure proper correlation between detectors located on different simulation platforms.				
SensorTypes RPRnoteMinefield9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MinefieldSensorTypeVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	In QRP mode, specifies the requesting sensor types which were specified in the minefield query whereas in Heartbeat mode, specifies the sensor types that are being served by the minefield.				
SnowBurialDepthOffset RPRnoteMinefield9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DepthMeterFloat32Array	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the offset of the origin of the mine coordinate system with respect to the snow surface.				
ThermalContrast RPRnoteMinefield9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TemperatureDegreeCelsiusFloat32Array	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the temperature difference between the mine and the surround soil in degrees Centigrade.				
WaterBurialDepthOffset RPRnoteMinefield9	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DepthMeterFloat32Array	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies the offset of the origin of the mine coordinate system with respect to the water surface.				

WireVertices RPRnoteMinefield13	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Specifies the locations of vertices in a trip wire.</i>					
EntityIdentifier RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					
	<i>The EntityIdentifier of the object which this embedded system is a part of.</i>					
HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					
	<i>The RTI object instance ID of the object of which this embedded system is part of.</i>					
RelativePosition RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The position of the embedded system, relative to the host object's position.</i>					
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

7.1.3. MinefieldObject

Full Name: HLAobjectRoot.EnvironmentObject.ArealObject.MinefieldObject

Sharing: Publish/Subscribe

Semantics: *A mine, mine weapon, mine row, mine strip, mine lane, mine marker or minefield, defined as an areal environment object.*

Attributes:

BreachState RPRnoteMinefield6 RPRnoteMinefield14	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectSpecificBreachStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the breached appearance of the minefield object.					
MineCount RPRnoteMinefield6 RPRnoteMinefield14	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the number of mines in the minefield.					
PointsData RPRnoteSE5 Inherited from ArealObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the physical location (a collection of points, in the world coordinate system) of the object.					
PercentComplete RPRnoteSE7 RPRnoteSE8 Inherited from ArealObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies the percent completion of the object.					

Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from ArealObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralDamageEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies the damaged appearance of the object.</i>				
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from ArealObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralPredistributedEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object was created before the start of the exercise.</i>				
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from ArealObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceEntityOrObjectStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>				
IsSmoking RPRnoteSE3 RPRnoteSE8 <i>Inherited from ArealObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>				
IsFlaming RPRnoteSE3 RPRnoteSE8 <i>Inherited from ArealObject in Synthetic Environment</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether the object is burning and flames are visible.</i>				

IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from ArealObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AppearanceObjectGeneralIEDPresentEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Specifies whether the object has or contains an IED.				
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies this EnvironmentObject instance (point, linear or areal).				
ReferencedObjectIdentifier RPRnoteSE6 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.				
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Identifies the force that created or modified this EnvironmentObject instance.				
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Identifies the type of this EnvironmentObject instance.				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

7.1.4. MinefieldLaneMarkerObject

Full Name: HLAobjectRoot.EnvironmentObject.LinearObject.MinefieldLaneMarkerObject

Sharing: Publish/Subscribe

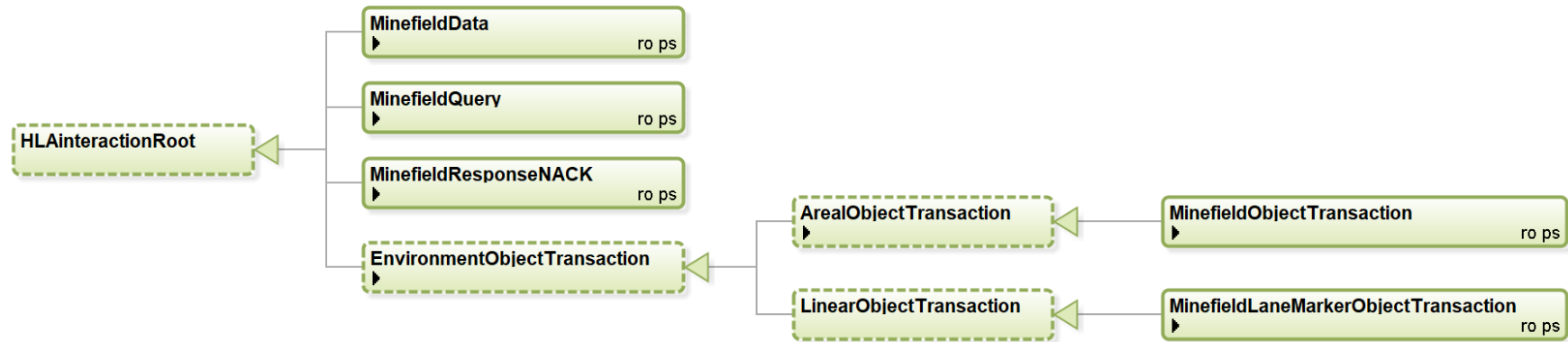
Semantics: *A minefield lane marker showing a cleared lane through a specific minefield, defined as a linear object.*

Attributes:

SegmentRecords <i>RPRnoteMinefield6</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MinefieldLaneMarkerStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies a minefield lane marker showing a cleared lane through a minefield.					
ObjectIdentifier <i>RPRnoteSE5</i> Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Identifies this EnvironmentObject instance (point, linear or areal).					
ReferencedObjectIdentifier <i>RPRnoteSE6</i> Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.					

ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ForceIDEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Identifies the force that created or modified this EnvironmentObject instance.					
ObjectType RPRnoteSE5 Inherited from EnvironmentObject in Synthetic Environment	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EnvironmentObjectTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Identifies the type of this EnvironmentObject instance.					
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

7.2. Interaction Classes



7.2.1. HLAinteractionRoot

Full Name: HLAinteractionRoot
 Sharing: Publish/Subscribe
 Transportation type: HLAREliable
 Order: Receive
 Dimensions:
 Semantics:
 Parameters: -

7.2.2. MinefieldData

Full Name: HLAinteractionRoot.MinefieldData
 Sharing: Publish/Subscribe
 Transportation type: HLAbestEffort
 Order: Receive
 Dimensions:
 Semantics: *Provides information on individual mines contained within a minefield.*

Parameters:

Name	Datatype	Semantics
GroundBurialDepthOffset RPRnoteMinefield9	DepthMeterFloat32Array	<i>Specifies the offset of the origin of the mine coordinate system with respect to the ground surface.</i>
Fusing RPRnoteMinefield10	MineFusingStructArray	<i>Specifies the primary and secondary fuse and anti-handling device for each mine.</i>
MineEmplacementTime RPRnoteMinefield9	ClockTimeStructArray	<i>Specifies the real-world (UTC) emplacement time of the mine.</i>
MineEntityIdentifier RPRnoteMinefield6	MineIdentifierArray	<i>Identifies the mine entity identifier; the MineEntityID in conjunction with the MinefieldID form the unique identifier for each mine.</i>
MinefieldIdentifier RPRnoteMinefield6	RTObjectId	<i>Identifies the minefield to which the mines belong.</i>
MineLocation RPRnoteMinefield6	LocationStructArray	<i>Specifies the location of the mine relative to the minefield location.</i>
MineOrientation RPRnoteMinefield10	OrientationStructArray	<i>Specifies the orientation of the center axis direction of fire of the mine, relative to the minefield Coordinate System.</i>
MineType RPRnoteMinefield6	EntityTypeStruct	<i>Specifies the type of each mine contained within the minefield interaction.</i>
NumberOfRecords RPRnoteMinefield6	UnsignedInteger8	<i>Specifies the total number of minefield records being published in response to a Minefield Query interaction.</i>
NumberTripDetonationWires RPRnoteMinefield11	UnsignedInteger8Array	<i>Specifies the number of trip detonation wires that exist for each mine.</i>
NumberWireVertices RPRnoteMinefield12	UnsignedInteger8Array	<i>Specifies the number of vertices for each trip wire.</i>
PaintScheme RPRnoteMinefield9	MinefieldDataPaintSchemeStructArray	<i>Specifies the camouflage scheme/color of the mine.</i>
RecordSequenceNumber RPRnoteMinefield6	UnsignedInteger8	<i>Specifies the number of the current record in a sequence of minefield records published in response to a Minefield Query interaction.</i>
Reflectance RPRnoteMinefield9	MineDielectricDifferenceArray	<i>Specifies the local dielectric difference between the mine and the surrounding soil.</i>
RequestIdentifier RPRnoteMinefield6	UnsignedInteger8	<i>Identifies the matching response to a request for mine information from the minefield simulation made by means of a Minefield Query interaction.</i>

Name	Datatype	Semantics
RequestingIdentifier RPRnoteMinefield6	FederateIdentifierStruct	<i>Identifies the simulation application that requested the information from the minefield simulation in QRP (Query Response Protocol) mode.</i>
ScalarDetectionCoefficient RPRnoteMinefield9	UnsignedInteger8Array	<i>Specifies the coefficient to be utilized to insure proper correlation between detectors located on different simulation platforms.</i>
SensorTypes RPRnoteMinefield6	MinefieldSensorTypeVariantStructArray	<i>In QRP mode, specifies the requesting sensor types which were specified in the minefield query whereas in Heartbeat mode, specifies the sensor types that are being served by the minefield.</i>
SnowBurialDepthOffset RPRnoteMinefield9	DepthMeterFloat32Array	<i>Specifies the offset of the origin of the mine coordinate system with respect to the snow surface.</i>
ThermalContrast RPRnoteMinefield9	TemperatureDegreeCelsiusFloat32Array	<i>Specifies the temperature difference between the mine and the surrounding soil in degrees Centigrade.</i>
WaterBurialDepthOffset RPRnoteMinefield9	DepthMeterFloat32Array	<i>Specifies the offset of the origin of the mine coordinate system with respect to the water surface.</i>
WireVertices RPRnoteMinefield13	LocationStructArray	<i>Specifies the locations of vertices in a trip wire.</i>

7.2.3. MinefieldQuery

Full Name: HLAinteractionRoot.MinefieldQuery

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *Provides the means by which a federate shall query a minefield simulation for information on the individual mines within the minefield operating in QRP mode.*

Parameters:

Name	Datatype	Semantics
MinefieldIdentifier RPRnoteMinefield6	RTObjectId	<i>Identifies the minefield to which this query is addressed.</i>
PerimeterPoints RPRnoteMinefield6	PerimeterPointStructArray	<i>Specifies the location of each perimeter point in the requested area relative to the minefield location.</i>

Name	Datatype	Semantics
QueryFusing RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not fusing is requested.</i>
QueryMineOrientation RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not orientation is requested.</i>
QueryGroundBurialDepthOffset RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not ground burial depth offset is requested.</i>
QueryMineEmplacementAge RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not emplacement age is requested.</i>
QueryPaintScheme RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not paint scheme is requested.</i>
QueryReflectance RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not reflectance is requested.</i>
QueryScalarDetectionCoefficient RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not scalar detection coefficient is requested.</i>
QuerySnowBurialDepthOffset RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not snow burial depth offset is requested.</i>
QueryThermalContrast RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not thermal contrast is requested.</i>
QueryTripDetonationWire RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not trip detonation wire is requested.</i>
QueryWaterBurialDepthOffset RPRnoteMinefield15	RPRboolean	<i>Specifies whether or not water burial depth offset is requested.</i>
RequestingIdentifier RPRnoteMinefield6	FederateIdentifierStruct	<i>Identifies the simulation application that requests the information from the minefield simulation.</i>
RequestIdentifier RPRnoteMinefield6	UnsignedInteger8	<i>Identifies the minefield query request.</i>
RequestedMineType RPRnoteMinefield6	EntityTypeStruct	<i>Identifies the type of mine being queried by the requesting federate.</i>
SensorTypes RPRnoteMinefield6	MinefieldSensorTypeVariantStructArray	<i>Specifies the types of sensors requesting the data.</i>

7.2.4. MinefieldResponseNACK

Full Name: HLAinteractionRoot.MinefieldResponseNACK

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A response to a MinefieldQuery providing information on individual mines within a minefield.*

Parameters:

Name	Datatype	Semantics
MinefieldIdentifier RPRnoteMinefield6	RTIobjectId	<i>Identifies the minefield to which this query is addressed.</i>
MissingRecordNumbers RPRnoteMinefield6	MissingRecordNumbersArray1Plus	<i>Specifies the record numbers that were not received in a sequence of minefield records.</i>
RequestIdentifier RPRnoteMinefield6	UnsignedInteger8	<i>Identifies the minefield query request.</i>
RequestingIdentifier RPRnoteMinefield6	FederateIdentifierStruct	<i>Identifies the simulation application that requested the information from the minefield simulation.</i>

7.2.5. EnvironmentObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAreliable

Order: Receive

Dimensions:

Semantics:

Parameters:

Name	Datatype	Semantics
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

7.2.6. ArealObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.ArealObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics:

Parameters:

Name	Datatype	Semantics
PointsData RPRnoteSE5 Inherited from ArealObjectTransaction in Synthetic Environment	LocationStructArray	<i>Specifies the physical location (a collection of points, in the world coordinate system) that defines the object.</i>
PercentComplete RPRnoteSE7 RPRnoteSE8 Inherited from ArealObjectTransaction in Synthetic Environment	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9 Inherited from ArealObjectTransaction in Synthetic Environment	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10 Inherited from ArealObjectTransaction in Synthetic Environment	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11 Inherited from ArealObjectTransaction in Synthetic Environment	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>
IsSmoking RPRnoteSE8 RPRnoteSE3 Inherited from ArealObjectTransaction in Synthetic Environment	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE8 RPRnoteSE3 Inherited from ArealObjectTransaction in Synthetic Environment	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>
IEDPresent RPRnoteSE8 RPRnoteSE12 Inherited from ArealObjectTransaction in Synthetic Environment	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the object has or contains an IED.</i>
ObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>

Name	Datatype	Semantics
ReferencedObjectIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	RTLObjectId	Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.
ForceIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	ForceIDEnum8	Identifies the force that created or modified this EnvironmentObject instance.
ObjectType RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	EnvironmentObjectTypeStruct	Identifies the type of this EnvironmentObject instance.
RequestingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application sending the EnvironmentObjectTransaction interaction.
ReceivingIdentifier RPRnoteSE5 Inherited from EnvironmentObjectTransaction in Synthetic Environment	FederateIdentifierStruct	Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.

7.2.7. MinefieldObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.ArealObjectTransaction.MinefieldObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the MinefieldObject class.

Parameters:

Name	Datatype	Semantics
BreachState RPRnoteMinefield6 RPRnoteMinefield14	AppearanceObjectSpecificBreachStateEnum8	Specifies the breached appearance of the minefield object.

Name	Datatype	Semantics
MineCount RPRnoteMinefield6 RPRnoteMinefield14	UnsignedInteger16	<i>Specifies the number of mines in the minefield.</i>
PointsData RPRnoteSE5 <i>Inherited from ArealObjectTransaction in Synthetic Environment</i>	LocationStructArray	<i>Specifies the physical location (a collection of points, in the world coordinate system) that defines the object.</i>
PercentComplete RPRnoteSE7 RPRnoteSE8 <i>Inherited from ArealObjectTransaction in Synthetic Environment</i>	PercentUnsignedInteger8	<i>Specifies the percent completion of the object.</i>
Damage RPRnoteSE8 RPRnoteSE9 <i>Inherited from ArealObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralDamageEnum8	<i>Specifies the damaged appearance of the object.</i>
Predistributed RPRnoteSE8 RPRnoteSE10 <i>Inherited from ArealObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralPredistributedEnum8	<i>Specifies whether or not the object was created before the start of the exercise.</i>
State RPRnoteSE8 RPRnoteSE11 <i>Inherited from ArealObjectTransaction in Synthetic Environment</i>	AppearanceEntityOrObjectStateEnum8	<i>Specifies whether or not the object has been deactivated (it has ceased to exist in the synthetic environment).</i>
IsSmoking RPRnoteSE8 RPRnoteSE3 <i>Inherited from ArealObjectTransaction in Synthetic Environment</i>	RPRboolean	<i>Specifies whether or not the object is smoking (creating a smoke plume).</i>
IsFlaming RPRnoteSE8 RPRnoteSE3 <i>Inherited from ArealObjectTransaction in Synthetic Environment</i>	RPRboolean	<i>Specifies whether the object is burning and flames are visible.</i>
IEDPresent RPRnoteSE8 RPRnoteSE12 <i>Inherited from ArealObjectTransaction in Synthetic Environment</i>	AppearanceObjectGeneralIEDPresentEnum8	<i>Specifies whether the object has or contains an IED.</i>

Name	Datatype	Semantics
ObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier <u>RPRnoteSE5</u> <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

7.2.8. LinearObjectTransaction

Full Name: HLAIinteractionRoot.EnvironmentObjectTransaction.LinearObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics:

Parameters:

Name	Datatype	Semantics
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTObjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDEnum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

7.2.9. MinefieldLaneMarkerObjectTransaction

Full Name: HLAinteractionRoot.EnvironmentObjectTransaction.LinearObjectTransaction.MinefieldLaneMarkerObjectTransaction

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An interaction sent to an environment manager to request the creation, modification, or deletion of instances of the MinefieldLaneMarkerObject class.*

Parameters:

Name	Datatype	Semantics
SegmentRecords RPRnoteMinefield6	MinefieldLaneMarkerStructArray	<i>Specifies a minefield lane marker linear object.</i>
ObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EntityIdentifierStruct	<i>Identifies this EnvironmentObject instance (point, linear or areal).</i>
ReferencedObjectIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	RTIobjectId	<i>Identifies the Synthetic Environment object instance to which this EnvironmentObject instance is associated.</i>
ForceIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	ForceIDenum8	<i>Identifies the force that created or modified this EnvironmentObject instance.</i>
ObjectType RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	EnvironmentObjectTypeStruct	<i>Identifies the type of this EnvironmentObject instance.</i>
RequestingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application sending the EnvironmentObjectTransaction interaction.</i>
ReceivingIdentifier RPRnoteSE5 <i>Inherited from EnvironmentObjectTransaction in Synthetic Environment</i>	FederateIdentifierStruct	<i>Identifies the simulation application receiving the EnvironmentObjectTransaction interaction.</i>

7.3. Datatypes

7.3.1. Simple Datatypes

MineDielectricDifference

Representation: HLAfloat32BE

Units: NA

Resolution: NA

Accuracy: NA

Semantics: Local dielectric difference between the mine and the surrounding soil (reflectance).

MineIdentifier

Representation: [RPRunsignedInteger16BE](#)

Units: NA

Resolution: 1

Accuracy: NA

Semantics: Specifies a mine entity identifier.

7.3.2. Array Datatypes

DepthMeterFloat32Array

Element [DepthMeterFloat32](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Specifies ground - snow - water burial depth offset for each mine in a collection of mines.*

MineDielectricDifferenceArray

Element [MineDielectricDifference](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Specifies local dielectric difference between the mine and the surrounding soil (reflectance) for each mine in a collection of mines.*

MinefieldDataPaintSchemeStructArray

Element [MinefieldDataPaintSchemeStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Specifies the camouflage scheme/color for each mine in a collection of mines.*

MinefieldLaneMarkerStructArray

Element [MinefieldLaneMarkerStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Specifies a minefield lane marker (a collection of segments).*

MinefieldSensorTypeVariantStructArray

Element [MinefieldSensorTypeVariantStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Specifies a collection of minefield sensor types.*

MineFusingStructArray

Element [MineFusingStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Specifies the type of primary fuse, the type of the secondary fuse and the anti-handling device status for a collection of mines.*

MineIdentifierArray

Element [MineIdentifier](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Identifies the mine entity identifier for each mine in a collection of mines.*

MissingRecordNumbersArray1Plus

Element [UnsignedInteger8](#)
Type:
Cardinality: [1..2147483647]
Encoding: HLAVariableArray
Semantics: *Specifies missing record numbers as a collection.*

PerimeterPointStructArray

Element [PerimeterPointStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray

Semantics: *Specifies the location of perimeter points (collection).*

TemperatureDegreeCelsiusFloat32Array

Element [TemperatureDegreeCelsiusFloat32](#)
Type:

Cardinality: Dynamic

Encoding: HLAvariableArray

Semantics: *Specifies thermal contrast for each mine in a collection of mines.*

7.3.3. Fixed Record Datatypes

MinefieldDataPaintSchemeStruct

Encoding: HLAfixedRecord

Semantics: *Minefield Data-Paint Scheme [UID 202].*

Name	Type	Semantic
Algae	MinefieldPaintSchemeAlgaeEnum8	<i>Identifies the algae build-up on the mine.</i>
PaintScheme	MinefieldPaintSchemePaintSchemeEnum8	<i>Identifies the paint scheme of the mine.</i>

MinefieldLaneMarkerStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying a minefield lane marker segment.*

Name	Type	Semantic
SegmentParameters	LinearSegmentStruct	<i>Specifies the minefield lane marker segment characteristics.</i>
VisibleSide RPRnoteMinefield16	AppearanceLinearObjectLaneMarkerVisibleEnum8	<i>Specifies the visible side(s) of the minefield lane marker segment.</i>

MineFusingStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the type of primary fuse, the type of the secondary fuse and the anti-handling device status of a mine.*

Name	Type	Semantic
Primary	MinefieldFusingFuseTypeEnum8	<i>Specifies the type of the primary fuse of a mine.</i>
Secondary	MinefieldFusingFuseTypeEnum8	<i>Specifies the type of the secondary fuse of a mine.</i>
HasAntiHandlingDevice	RPRboolean	<i>Specifies the anti-handling device status of a mine.</i>

PerimeterPointStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the location of a perimeter point.*

Name	Type	Semantic
X	MeterFloat32	<i>Specifies the X coordinate.</i>
Y	MeterFloat32	<i>Specifies the Y coordinate.</i>

7.4. Notes

RPRnoteMinefield1

Semantics: *Default value: Inactive*

RPRnoteMinefield2

Semantics: *Source: SISO-REF-010 [UID 190]*

RPRnoteMinefield3

Semantics: *Default value: Other*

RPRnoteMinefield4

Semantics: *Default value: Minefield_Has_an_Inactive_Lane*

RPRnoteMinefield5

Semantics: *Default value: Mixed_Anti_Personnel_and_Anti_Tank_Minefield*

RPRnoteMinefield6

Semantics: *Not optional*

RPRnoteMinefield7

Semantics: *Default value: Minefield Location (not needed for single mines)*

RPRnoteMinefield8

Semantics: *Default value: Heartbeat_Mode*

RPRnoteMinefield9

Semantics: *Default value: 0*

RPRnoteMinefield10

Semantics: *Default value: {0,0,0}*

RPRnoteMinefield11

Semantics: *Required if tripwires are used, otherwise the default value is Empty.*

RPRnoteMinefield12

Semantics: *Required if the sum of values in NumberTripDetonationWires is greater than 0, otherwise the default value is Empty.*

RPRnoteMinefield13

Semantics: *Required if the sum of values in NumberWireVertices is greater than 0, otherwise the default values is Empty.*

RPRnoteMinefield14

Semantics: *Source: SISO-REF-010 [UID 239]*

RPRnoteMinefield15

Semantics: *Default value: False*

RPRnoteMinefield16

Semantics: *Source: SISO-REF-010 [UID 237]*

8. Module Communication



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Communication FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	The Communication module is used to simulate radio transmitters and receivers as well as the radio signals that are transmitted between them.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

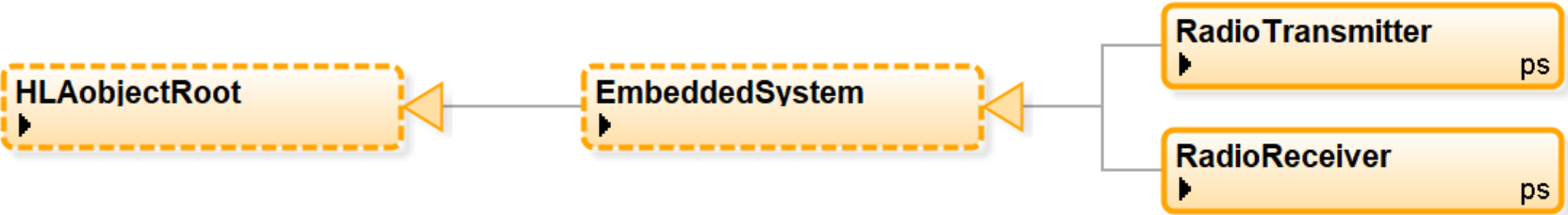
References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Base
Foundation
Enumerations

8.1. Object Classes



8.1.1. RadioTransmitter

Full Name: HLAObjectRoot.EmbeddedSystem.RadioTransmitter
Sharing: Publish/Subscribe
Semantics: *A device that sends out information encoded in electromagnetic waves in the radio frequency range.*
Attributes:

AntennaPatternData	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AntennaPatternVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The radiation pattern of the radio's antenna.				
CryptographicMode	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	CryptoKeyIDCryptoModeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The mode of the cryptographic system.				

CryptoSystem	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TransmitterCryptoSystemEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The type of cryptographic equipment in use.</i>				
EncryptionKeyIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The identification of the key used to encrypt the radio signals being transmitted. The transmitter and receiver should be considered to be using the same key if these numbers match.</i>				
Frequency	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzUnsignedInteger64	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The center frequency of the radio transmissions.</i>				
FrequencyBandwidth	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The bandpass of the radio transmissions.</i>				
FrequencyHopInUse	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Whether the radio is using frequency hopping or not.</i>				

HighFidelityHAVEQUICKData	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HighFidelityHAVEQUICKStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Additional information related to High Fidelity HAVE QUICK and SATURN radios.</i>				
Power	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PowerRatioDecibelMilliwattFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The EIRP when transmitting.</i>				
PseudoNoiseInUse	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Whether the radio is using pseudo noise or not.</i>				
RadioAttachedIndicator	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RadioAttachedIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether the radio transmitter is attached to an entity or object.</i>				
RadioIndex	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>A number that uniquely identifies this radio from others on the host.</i>				

RadioInputSource	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TransmitterInputSourceEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies which position or data port provided the input for the transmission.					
RadioSystemType	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RadioTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The type of radio transmitter.					
RFModulationSystemParameters	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RFModulationSystemParametersVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Classification of the modulation system type and parameters.					
RFModulationType	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RFModulationTypeVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Classification of the modulation type.					
StreamTag	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger64	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	A globally unique identifier for the associated audio stream.					

TimeHopInUse	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Whether the radio is using time hopping or not.</i>				
TransmitterOperationalStatus	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TransmitterTransmitStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The current operational state of the radio transmitter.</i>				
WorldLocation	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> <i>TRANS_POS_THRSH</i> RPRnoteCommunication1				
	Semantics	<i>The location of the antenna in the world coordinate system.</i>				
EntityIdentifier RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The EntityIdentifier of the object which this embedded system is a part of.</i>				
HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The RTI object instance ID of the object of which this embedded system is part of.</i>				

RelativePosition RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The position of the embedded system, relative to the host object's position.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

8.1.2. RadioReceiver

Full Name: HLAobjectRoot.EmbeddedSystem.RadioReceiver

Sharing: Publish/Subscribe

Semantics: *A device that converts incoming electromagnetic waves in the radio frequency range into information.*

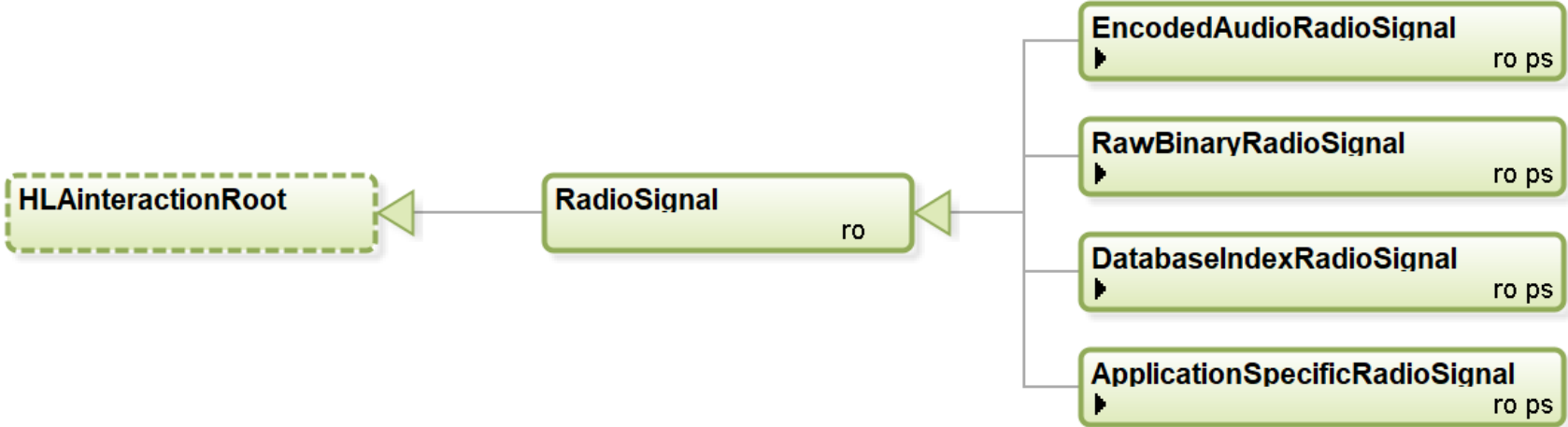
Attributes:

RadioAttachedIndicator	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RadioAttachedIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether the radio receiver is attached to an entity or object.</i>				
RadioIndex	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>A number that uniquely identifies this radio receiver from other receivers on the host object.</i>				

ReceivedPower	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PowerRatioDecibelMilliwattFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The radio frequency power received, after applying any propagation loss and antenna gain.</i>				
ReceivedTransmitterIdentifier RPRnoteCommunication2	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The object instance ID of the transmitter that generated the received radio signal.</i>				
ReceiverOperationalStatus	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ReceiverReceiverStateEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The operational state of the radio receiver.</i>				
EntityIdentifier RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The EntityIdentifier of the object which this embedded system is a part of.</i>				
HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The RTI object instance ID of the object of which this embedded system is part of.</i>				

RelativePosition <u>RPRnoteBase2</u> <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The position of the embedded system, relative to the host object's position.</i>					
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

8.2. Interaction Classes



8.2.1. HLAinteractionRoot

Full Name: HLAinteractionRoot
Sharing: Publish/Subscribe
Transportation type: HLAreliable
Order: Receive
Dimensions:
Semantics:
Parameters: -

8.2.2. RadioSignal

Full Name: HLAinteractionRoot.RadioSignal
Sharing:
Transportation type: HLAbestEffort

Order: Receive
 Dimensions:
 Semantics: *The wireless transmission and reception of audio or digital data by means of electromagnetic waves.*
 Parameters: -

8.2.3. EncodedAudioRadioSignal

Full Name: HLAIinteractionRoot.RadioSignal.EncodedAudioRadioSignal
 Sharing: Publish/Subscribe
 Transportation type: HLABestEffort
 Order: Receive
 Dimensions:
 Semantics: *A radio signal used to transmit voice/audio data encoded according to a standard encoding scheme.*
 Parameters:

Name	Datatype	Semantics
AudioData	AudioDataTypeStruct	<i>The data associated with the encoded audio radio signal.</i>

8.2.4. RawBinaryRadioSignal

Full Name: HLAIinteractionRoot.RadioSignal.RawBinaryRadioSignal
 Sharing: Publish/Subscribe
 Transportation type: HLABestEffort
 Order: Receive
 Dimensions:
 Semantics: *A radio signal used to transmit raw binary data.*
 Parameters:

Name	Datatype	Semantics
HostRadioIndex RPRnoteCommunication2	RTObjectId	<i>The object instance ID of the radio transmitting this signal.</i>

Name	Datatype	Semantics
DataRate	BitRateBitPerSecondUnsignedInteger32	<i>The rate at which the data is being transmitted.</i>
SignalDataLength	BitsUnsignedInteger16	<i>The length of the signal data.</i>
SignalData	SignalDataArray	<i>The signal data.</i>
TacticalDataLinkType	SignalTDLTypeEnum16	<i>The type of tactical data link used to transmit this signal (if any).</i>
TDLMessageCount	UnsignedInteger16	<i>The number of tactical data link messages contained in this signal.</i>

8.2.5. DatabaseIndexRadioSignal

Full Name: HLAinteractionRoot.RadioSignal.DatabaseIndexRadioSignal

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *A radio signal that represents transmission of a pre-determined message by using a pre-defined database index.*

Parameters:

Name	Datatype	Semantics
HostRadioIndex RPRnoteCommunication2	RTObjectId	<i>The object instance ID of the radio transmitting this signal.</i>
DatabaseIndex	UnsignedInteger32	<i>The index into the federation specific database of stored signals.</i>
Duration	TimeMillisecondUnsignedInteger32	<i>The duration of the stored signal to be replayed.</i>
StartOffset	TimeMillisecondUnsignedInteger32	<i>The offset, from the start of the stored signal, that the signal is replayed from.</i>

Name	Datatype	Semantics
TacticalDataLinkType	SignalTDLTypeEnum16	<i>The type of tactical data link used to transmit this signal (if any).</i>
TDLMessageCount	UnsignedInteger16	<i>The number of tactical data link messages contained in this signal.</i>

8.2.6. ApplicationSpecificRadioSignal

Full Name: HLAinteractionRoot.RadioSignal.ApplicationSpecificRadioSignal

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *A radio signal that can be used for an application specific encoding scheme.*

Parameters:

Name	Datatype	Semantics
HostRadioIndex RPRnoteCommunication2	RTObjectId	<i>The object instance ID of the radio transmitting this signal.</i>
DataRate	BitRateBitPerSecondUnsignedInteger32	<i>The rate at which the data is being transmitted.</i>
SignalDataLength	BitsUnsignedInteger16	<i>The length of the signal data.</i>
SignalData	SignalDataArray	<i>The signal data.</i>
TacticalDataLinkType	SignalTDLTypeEnum16	<i>The type of tactical data link used to transmit this signal (if any).</i>
TDLMessageCount	UnsignedInteger16	<i>The number of tactical data link messages contained in this signal.</i>
UserProtocolID	SignalUserProtocolIdentificationNumberEnum32	<i>The ID of the user protocol in use.</i>

8.3. Datatypes

8.3.1. Simple Datatypes

BitRateBitPerSecondUnsignedInteger32

Representation: [RPRunsignedInteger32BE](#)

Units: bit/second

Resolution: 1

Accuracy: perfect

Semantics: Rate of transmission, in bits per second.

FrequencyHertzUnsignedInteger64

Representation: [RPRunsignedInteger64BE](#)

Units: hertz (Hz)

Resolution: NA

Accuracy: NA

Semantics: Frequency of a radio transmission, in hertz.

NetNumberUnsignedInteger16

Representation: [RPRunsignedInteger16BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Net number, ranging from 0 to 999 decimal.

SampleRateHertzUnsignedInteger32

Representation: [RPRunsignedInteger32BE](#)

Units: Hertz

Resolution: 1

Accuracy: perfect

Semantics: Rate of sampling, in samples per second.

TimeHundredsOfMicrosecondInteger32

Representation: HLAinteger32BE

Units: 100 microsecond

Resolution: 1

Accuracy: perfect

Semantics: Time, based on SI base unit second, expressed in hundreds of microseconds.

WordOfDayUnsignedInteger32

Representation: [RPRunsignedInteger32BE](#)

Units: NA

Resolution: NA

Accuracy: NA

Semantics: A Word of Day (WOD) contains six decimal digits. The first digit can only be a 2 or a 3, while the second third and fourth can be 0 through 9. The last two digits can only be 00, 25, 50, or 75. WOD notation has a decimal between the third and fourth digit. Example notation: "300.025".

8.3.2. Array Datatypes

HighFidelityHAVEQUICKStructArray

Element [HighFidelityHAVEQUICKStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Array with High Fidelity HAVE QUICK data.*

SignalDataArray

Element [Octet](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *The audio or digital data conveyed in a radio transmission.*

8.3.3. Fixed Record Datatypes

AudioDataTypeDefStruct

Encoding: HLAfixedRecord

Semantics: *Specifies an encoded audio radio signal.*

Name	Type	Semantic
StreamTag	UnsignedInteger64	Identifier for the audio stream. This must be globally unique.
EncodingType	SignalEncodingTypeEnum16	The type of encoding used for the audio data.
SampleRate	SampleRateHertzUnsignedInteger32	The number of samples per second in the audio data.
DataLength	BitsUnsignedInteger16	The length of the signal data.
SampleCount	UnsignedInteger32	The number of samples in this transmission.
Data	SignalDataArray	The signal data.

BasicHAVEQUICKModulationStruct

Encoding: HLAfixedRecord

Semantics: *Modulation parameters for Basic HAVE QUICK radio system.*

Name	Type	Semantic
NETID	NETIDStruct	The NET ID being used.
MWODIndex	UnsignedInteger16	This is an index into the "MWOD Index - WOD Correlation Table" provided by the exercise agreement. When exercise agreement data is not available or MWOD Index is not known, the default maintenance and training MWOD Index value of 1 shall be used, which corresponds to the WOD values in Table C.2 of IEEE Std 1278.1-2012 Section C.4.2.2.
TimeOfDay	UnsignedInteger32	The Time of Day (TOD) field for the Basic HAVE QUICK MP record is used to specify a representative Time of Day value for a given radio. A value of zero shall be considered a match to any other value received.

CCTTSINCGARSModulationStruct

Encoding: HLAfixedRecord

Semantics: *Modulation parameters for CCTT SINCGARS radio system.*

Name	Type	Semantic
FHNetID	UnsignedInteger16	Frequency hopping network identifier.
HopSetID	UnsignedInteger16	Identification for the set of frequencies used when creating a hopping pattern.
LockoutSetID	UnsignedInteger16	Identification for the set of frequencies that are excluded from the hopping pattern.
StartOfMessage	CCTTSINCGARSStartOfMessageEnum8	Specifies whether the radio is starting or continuing a transmission. A CCTT SINCGARS radio shall not receive any part of an FH transmission if the Start of Message is missed. Start of message may be missed if the radio was transmitting, receiving a different radio, or otherwise not ready to start receiving a transmission.
ClearChannel	CCTTSINCGARSClearChannelEnum8	Specifies that the transmission is not subject to propagation loss, interference, COMSEC, or any other form of signal degradation. Clear channels are used for interoperability with older systems and for control purposes.
FHSynchronizationTimeOffset	TimeSecondInteger32	The offset to exercise time in seconds for the clock in the radio.
TransmissionSecurityKey	UnsignedInteger16	Identifies the transmission security key that is used when generating the hopping pattern.

HighFidelityHAVEQUICKStruct

Encoding: HLAfixedRecord

Semantics: *Specification of a High Fidelity HAVE QUICK record. See IEEE Std 1278.1-2012 Section C.4.2.3 for more information.*

Name	Type	Semantic
NETID	NETIDStruct	Contains the NET ID being used.
TODTransmitIndicator	HighFidelityHAVEQUICKTODTransmitIndicatorEnum8	Indicates whether a TOD is being transmitted. This is also referred to as a Mickey.

Name	Type	Semantic
TODDelta	TimeHundredsOfMicrosecondInteger32	<i>Specifies the difference between the transceiver internal clock and the reference clock, which is the amount of clock drift that has occurred. This field is expressed in units of hundreds of microseconds. The reference clock is an internally simulated clock that simulates a drift or difference from the reference value of zero, with which all simulated HAVE QUICK radios are assumed to be synchronized. This drift shall be placed in the TOD Delta field of the transmitting radio. Receiving radios may then use the TOD Delta field to emulate audio effects (such as clipped or garbled audio) due to HAVE QUICK time mismatches. If the TOD Delta field is zero, the radio is considered to be synchronized with the reference clock for all of the High-Fidelity HAVE QUICK radios participating in the exercise.</i>
WOD1	WordOfDayUnsignedInteger32	<i>Word of Day (WOD) 1. All six WODs collectively are called a Multiple Word of Day (MWOD). May also be specified as segment 1 or CH20.</i>
WOD2	WordOfDayUnsignedInteger32	<i>Word of Day (WOD) 2. All six WODs collectively are called a Multiple Word of Day (MWOD). May also be specified as segment 2 or CH19.</i>
WOD3	WordOfDayUnsignedInteger32	<i>Word of Day (WOD) 3. All six WODs collectively are called a Multiple Word of Day (MWOD). May also be specified as segment 3 or CH18.</i>
WOD4	WordOfDayUnsignedInteger32	<i>Word of Day (WOD) 4. All six WODs collectively are called a Multiple Word of Day (MWOD). May also be specified as segment 4 or CH17.</i>
WOD5	WordOfDayUnsignedInteger32	<i>Word of Day (WOD) 5. All six WODs collectively are called a Multiple Word of Day (MWOD). May also be specified as segment 5 or CH16.</i>
WOD6	WordOfDayUnsignedInteger32	<i>Word of Day (WOD) 6. All six WODs collectively are called a Multiple Word of Day (MWOD). May also be specified as segment 6 or CH15.</i>

NETIDStruct

Encoding: HLAfixedRecord

Semantics: *The NET ID for HAVE QUICK radios.*

Name	Type	Semantic
NetNumber	NetNumberUnsignedInteger16	<i>The net number ranging from 0 to 999 decimal.</i>

Name	Type	Semantic
FrequencyTable	NETIDRecordFrequencyTableEnum8	<i>The Frequency Table portion of the NET ID in use.</i>
Mode	NETIDRecordModeEnum8	<i>The mode.</i>

RadioTypeStruct RPRnoteCommunication3

Encoding: HLAfixedRecord

Semantics: *Type of radio, based on the Radio Type record as specified in IEEE Std 1278.1-2012 Section 6.2.71.*

Name	Type	Semantic
EntityKind	EntityKindEnum8	<i>The kind of the entity, set to Radio (7).</i>
Domain	PlatformDomainEnum8	<i>The domain that the radio normally operates in.</i>
CountryCode	CountryEnum16	<i>The country to which the design of the radio entity is attributed.</i>
Category	RadioCategoryEnum8	<i>The primary type of information that is transmitted or received by the radio when in operation.</i>
Subcategory	RadioSubcategoryEnum8	<i>The Radio Nomenclature System associated with the radio identified in the Specific field (e.g., a JETDS PRC-series radio).</i>
Specific	UnsignedInteger8	<i>The specific radio associated with the Radio Nomenclature System contained in the Subcategory field.</i>
Extra	UnsignedInteger8	<i>The version or variation of the radio.</i>

8.3.4. Variant Record Datatypes

AntennaPatternVariantStruct

Encoding: HLAvariantRecord

Discriminant name: AntennaPatternType

Discriminant type: [TransmitterAntennaPatternTypeEnum16](#)

Semantics: *Specifies the radiation pattern from the antenna, its orientation in space, and the polarization of the radiation.*

Name	Enumerator	Type	Semantics
IsotropicSphericalRadiationPattern	Isotropic_Spherical_Radiation_Pattern_	RPRnoData	<i>No Antenna Pattern Type and Antenna Pattern data. See IEEE Std 1278.1-2012 Section 5.8.3.3 item b5.</i>
BeamAntenna	Beam	BeamAntennaStruct	<i>Specifies the direction, pattern, and polarization of radiation from a radio transmitter's antenna, represented with respect to a beam coordinate system.</i>

RFModulationSystemParametersVariantStruct

Encoding: HLAvariantRecord

Discriminant name: SystemType

Discriminant type: [TransmitterModulationTypeSystemEnum16](#)

Semantics: *Specifies the system type as well as modulation parameters specific to the type.*

Name	Enumerator	Type	Semantics
GenericRadioOrSimpleIntercom	Generic_Radio_or_Simple_Intercom	RPRnoData	<i>No Modulation Parameter data. See IEEE Std 1278.1-2012 Section C.3.2.2.</i>
CCTTSINCGARSModulation	CCTT_SINCGARS	CCTTSINCGARSModulationStruct	<i>Modulation parameters for CCTT SINCGARS radio system.</i>
BasicHAVEQUICKModulation	HAVE_QUICK_I, HAVE_QUICK_II, SATURN	BasicHAVEQUICKModulationStruct	<i>Modulation parameters for HAVE QUICK or SATURN radio system.</i>

RFModulationTypeVariantStruct

Encoding: HLAvariantRecord

Discriminant name: MajorModulationType

Discriminant type: [TransmitterMajorModulationEnum16](#)Semantics: *Specifies the major modulation type as well as certain detailed information specific to the type.*

Name	Enumerator	Type	Semantics
NoStatementModulationType	No_Statement	UnsignedInteger16	<i>Detail for a major modulation type for which no statement is provided. This alternative is also used to implement a Simple Intercom capability using RadioTransmitter objects, see IEEE Std 1278.1 Section C.3. Then the value indicates the intercom priority in the range 0 to 255, with 0 = Exempt - Always heard regardless of the highest priority in use 1 = Highest Priority 255 = Lowest Priority</i>
AmplitudeModulationType	Amplitude	TransmitterDetailAmplitudeModulationEnum16	<i>Detailed modulation type for the amplitude major modulation type.</i>
AmplitudeAngleModulationType	Amplitude_and_Angle	TransmitterDetailAmplitudeAndAngleModulationEnum16	<i>Detailed modulation type for the amplitude and angle major modulation type.</i>
AngleModulationType	Angle	TransmitterDetailAngleModulationEnum16	<i>Detailed modulation type for the angle major modulation type.</i>
CombinationModulationType	Combination	TransmitterDetailCombinationModulationEnum16	<i>Detailed modulation type for the combination major modulation type.</i>
PulseModulationType	Pulse	TransmitterDetailPulseModulationEnum16	<i>Detailed modulation type for the pulse major modulation type.</i>
UnmodulatedType	Unmodulated	TransmitterDetailUnmodulatedModulationEnum16	<i>Detailed modulation type for the unmodulated major modulation type.</i>
CarrierPhaseShiftModulationType	Carrier_Phase_Shift_Modulation_CPSM_	TransmitterDetailCarrierPhaseShiftModulationEnum16	<i>Detailed modulation type for the Carrier Phase Shift Modulation (CPSM) major modulation type.</i>
SATCOMModulationType	SATCOM	TransmitterDetailSATCOMModulationEnum16	<i>Detailed modulation type for the SATCOM major modulation type. Note that this is a low-fidelity representation of SATCOM modulation.</i>

8.4. Notes

RPRnoteCommunication1

Semantics: *This attribute is updated if the current value differs from the previously updated value by more than the value specified by the symbolic name (default value: 500m).*

RPRnoteCommunication2

Semantics: *This must reference a valid Object instance.*

RPRnoteCommunication3

Semantics: *All the fields in the radio type struct are enumerations. The values for the individual fields (that do not use enumeration datatypes from within the RPR FOM) are to be derived from the federation agreements, which could refer to SISO-REF-010.*

9. Module Distributed Emission Regeneration



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Distributed Emission Regeneration FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	Represent electromagnetic emissions using the regeneration principle.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Base
Foundation
Enumerations

9.1. Object Classes

9.1.1. Designator

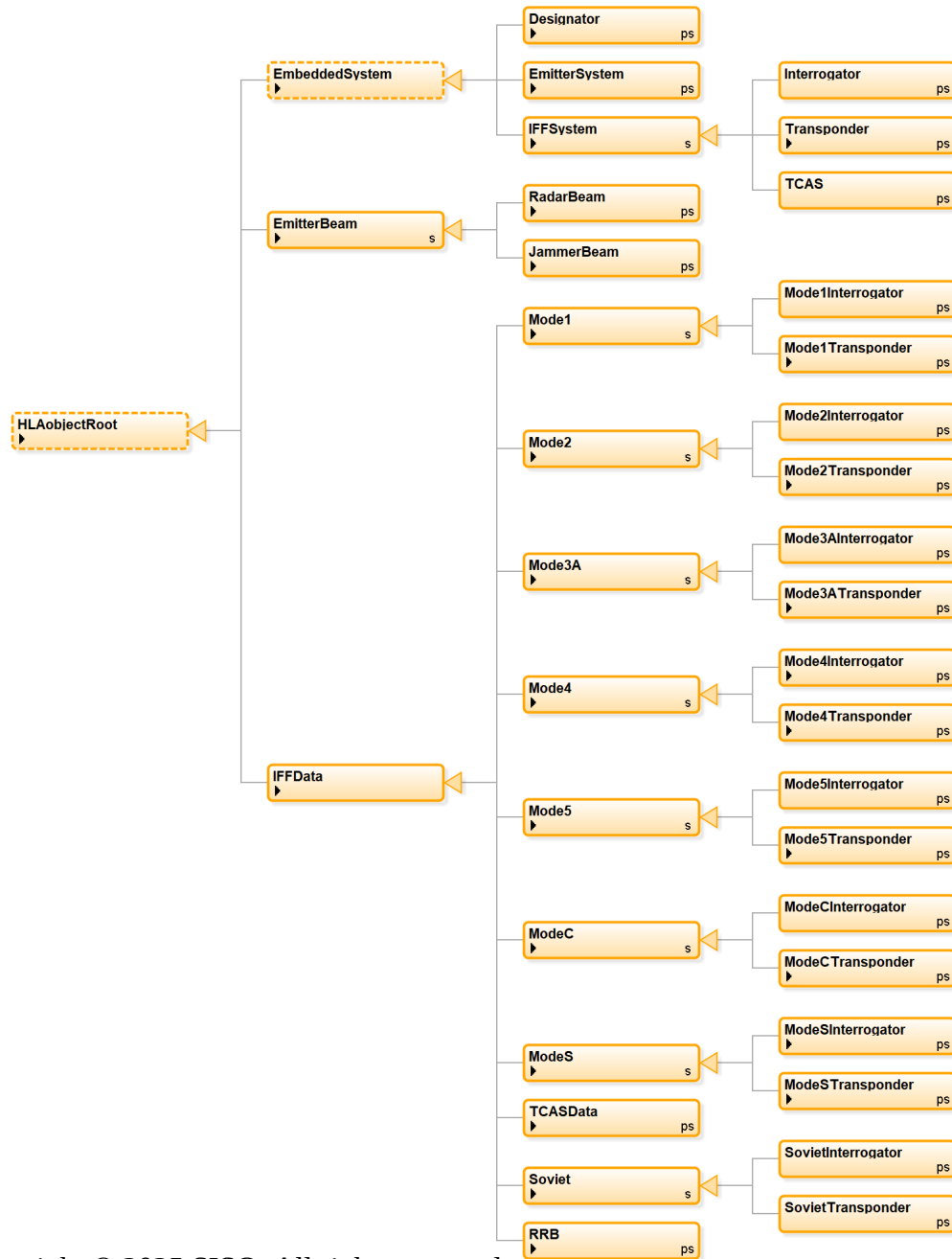
Full Name: HLAobjectRoot.EmbeddedSystem.Designator

Sharing: Publish/Subscribe

Semantics: *A system used to designate or mark a location or object, such as a laser designator which supports a laser-guided weapon engagement.*

Attributes:

CodeName	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DesignatorSystemNameEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The code name of the designator system.					
DesignatedObjectIdentifier RPRnoteDER1	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The object instance ID of the entity that is currently being designated (if any).					
DesignatorCode	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DesignatorDesignatorCodeEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The designator code being used by the designating entity.					



DesignatorEmissionWavelength	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	WavelengthMicronFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The wavelength of the designator system.					
DesignatorOutputPower	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PowerWattFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The output power of the designator system.					
DesignatorSpotLocation	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	LocationStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteDER2 RPRnoteDER3				
	Semantics					
	The location, in the world coordinate system, of the designator spot.					
DeadReckoningAlgorithm	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DeadReckoningAlgorithmEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Dead reckoning algorithm used by the issuing object.					
RelativeSpotLocation	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The location of the designator spot, relative to the object being designated (if any).					

SpotLinearAccelerationVector	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AccelerationVectorStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteDER2 RPRnoteDER4				
	Semantics					
	The rate of change in linear velocity of the designator spot over time.					
EntityIdentifier RPRnoteBase2 Inherited from EmbeddedSystem in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The EntityIdentifier of the object which this embedded system is a part of.					
HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 Inherited from EmbeddedSystem in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The RTI object instance ID of the object of which this embedded system is part of.					
RelativePosition RPRnoteBase2 Inherited from EmbeddedSystem in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The position of the embedded system, relative to the host object's position.					
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.2. EmitterSystem

Full Name: HLAAobjectRoot.EmbeddedSystem.EmitterSystem

Sharing: Publish/Subscribe

Semantics: *A device that radiates an electromagnetic signal, such as a radar or jammer.*

Attributes:

EmitterFunctionCode	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EmitterSystemFunctionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The function of the emitter system.					
EmitterType	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EmitterNameEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The type of the emitter system.					
EmitterIndex	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	A unique number, which uniquely identifies the emitter system from other on the same host entity.					
EventIdentifier RPRnoteDER5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteDER6				
	Semantics					
	The EventIdentifier is used by the generating federate to associate related events. The event number shall start at one at the beginning of the exercise, and be incremented by one for each event.					

EntityIdentifier RPRnoteBase2 Inherited from EmbeddedSystem in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The EntityIdentifier of the object which this embedded system is a part of.				
HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 Inherited from EmbeddedSystem in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The RTI object instance ID of the object of which this embedded system is part of.				
RelativePosition RPRnoteBase2 Inherited from EmbeddedSystem in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The position of the embedded system, relative to the host object's position.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.3. IFFSystem

Full Name: HLAobjectRoot.EmbeddedSystem.IFFSystem

Sharing: Subscribe

Semantics: Cooperative or noncooperative military IFF transponder and interrogator systems, including military Battlefield Target Identification Device (BTID) systems used on military vehicles, and cooperative civilian Air Traffic Control (ATC) radar beacon systems.

Attributes:

SystemType	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemTypeEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	General type of IFF/SSR system in use.					
SystemName	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemNameEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	On change				
	Semantics					
	Particular named type of the IFF/SSR system in use.					
SystemDesignator	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	A unique decimal number assigned to this interrogator or transponder to distinguish it from multiple interrogators or transponders that are associated with the same entity.					
InteractiveModeIsActive	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether or not the IFFSystem is currently operating in the Interactive Mode.					
EventIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Used to associate related events.					

SystemIsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Whether or not the system is on.					
SystemMode	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemModeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Mode of operation.					
OperationalStatus	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OperationalStatusEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Whether the system is operational or whether it has experienced a system failure.					
TestModeIsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether the IFF/SSR system has been placed in the equipment test mode.					
EmissionsDataAvailable	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies if emissions data (DIS IFF PDU layer 2) is available for this IFF/SSR system. When set to True, then the EmissionsDataSpecificInformation, BeamAzimuthCenter, BeamAzimuthSweep, BeamElevationCenter, BeamElevationSweep, BeamSweepSync, and FundamentalParameterData attributes shall be generated.					

EmissionsDataSpecificInformation RPRnoteDER7	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Layer-specific information for emissions data (layer number 2) that varies by SystemType.</i>				
BeamAzimuthCenter RPRnoteDER7	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>> IFF_AZ_THRSH</i> RPRnoteDER8				
	Semantics	<i>The azimuth center of the IFF/SSR beam's scan volume relative to the IFF/SSR system.</i>				
BeamAzimuthSweep RPRnoteDER7	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The azimuth half-angle of the IFF/SSR beam's scan volume relative to the IFF/SSR system.</i>				
BeamElevationCenter RPRnoteDER7	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>> IFF_EL_THRSH</i> RPRnoteDER9				
	Semantics	<i>The elevation center of the IFF/SSR beam's scan volume relative to the IFF/SSR system.</i>				
BeamElevationSweep RPRnoteDER7	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The elevation half-angle of the IFF/SSR beam's scan volume relative to the IFF/SSR system.</i>				

BeamSweepSync RPRnoteDER7	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Periodic	<i>On change</i> RPRnoteDER10				
	Semantics	<i>The percentage of time a scan is through its pattern from its origin.</i>				
FundamentalParameterData RPRnoteDER7	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FundamentalParameterDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The fundamental energy radiation characteristics of the IFF/SSR system.</i>				
AntennaLocationArray	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AntennaLocationIFFDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The location of each auxiliary IFF/SSR antenna. An auxiliary IFF/SSR antenna is one that is in addition to the primary antenna whose location is contained in the EmbeddedSystem attribute RelativePosition.</i>				
DataCommunicationsSpecificInformation	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Layer-specific information for data communications (layer number 5) that varies by SystemType.</i>				

EntityIdentifier <i>RPRnoteBase2</i> <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The EntityIdentifier of the object which this embedded system is a part of.</i>					
HostObjectIdentifier <i>RPRnoteBase2 RPRnoteBase6</i> <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>The RTI object instance ID of the object of which this embedded system is part of.</i>					
RelativePosition <i>RPRnoteBase2</i> <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The position of the embedded system, relative to the host object's position.</i>					
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.4. Interrogator

Full Name: HLAobjectRoot.EmbeddedSystem.IFFSystem.Interrogator

Sharing: Publish/Subscribe

Semantics: A transmitter that emits a signal to trigger a response from an IFF/SSR system.

Attributes:

SystemType <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemTypeEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	<i>General type of IFF/SSR system in use.</i>					
SystemName <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemNameEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>On change</i>				
	Semantics					
	<i>Particular named type of the IFF/SSR system in use.</i>					
SystemDesignator <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>A unique decimal number assigned to this interrogator or transponder to distinguish it from multiple interrogators or transponders that are associated with the same entity.</i>					
InteractiveModeIsActive <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Indicates whether or not the IFFSystem is currently operating in the Interactive Mode.</i>					
EventIdentifier <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Used to associate related events.</i>					

SystemIsOn <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Whether or not the system is on.</i>				
SystemMode <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemModeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode of operation.</i>				
OperationalStatus <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OperationalStatusEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Whether the system is operational or whether it has experienced a system failure.</i>				
TestModeIsOn <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether the IFF/SSR system has been placed in the equipment test mode.</i>				
EmissionsDataAvailable <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies if emissions data (DIS IFF PDU layer 2) is available for this IFF/SSR system. When set to True, then the EmissionsDataSpecificInformation, BeamAzimuthCenter, BeamAzimuthSweep, BeamElevationCenter, BeamElevationSweep, BeamSweepSync, and FundamentalParameterData attributes shall be generated.</i>				

EmissionsDataSpecificInformation RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Layer-specific information for emissions data (layer number 2) that varies by SystemType.				
BeamAzimuthCenter RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> IFF_AZ_THRSH RPRnoteDER8				
	Semantics	The azimuth center of the IFF/SSR beam's scan volume relative to the IFF/SSR system.				
BeamAzimuthSweep RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The azimuth half-angle of the IFF/SSR beam's scan volume relative to the IFF/SSR system.				
BeamElevationCenter RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> IFF_EL_THRSH RPRnoteDER9				
	Semantics	The elevation center of the IFF/SSR beam's scan volume relative to the IFF/SSR system.				
BeamElevationSweep RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The elevation half-angle of the IFF/SSR beam's scan volume relative to the IFF/SSR system.				

BeamSweepSync RPRnoteDER7 <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Periodic	<i>On change</i> RPRnoteDER10				
	Semantics					
	<i>The percentage of time a scan is through its pattern from its origin.</i>					
FundamentalParameterData RPRnoteDER7 <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FundamentalParameterDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The fundamental energy radiation characteristics of the IFF/SSR system.</i>					
AntennaLocationArray <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AntennaLocationIFFDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>The location of each auxiliary IFF/SSR antenna. An auxiliary IFF/SSR antenna is one that is in addition to the primary antenna whose location is contained in the EmbeddedSystem attribute RelativePosition.</i>					
DataCommunicationsSpecificInformation <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics					
	<i>Layer-specific information for data communications (layer number 5) that varies by SystemType.</i>					

EntityIdentifier RPRnoteBase2 Inherited from EmbeddedSystem in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The EntityIdentifier of the object which this embedded system is a part of.				
HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 Inherited from EmbeddedSystem in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The RTI object instance ID of the object of which this embedded system is part of.				
RelativePosition RPRnoteBase2 Inherited from EmbeddedSystem in Base	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The position of the embedded system, relative to the host object's position.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.5. Transponder

Full Name: HLAobjectRoot.EmbeddedSystem.IFFSystem.Transponder

Sharing: Publish/Subscribe

Semantics: A system that produces IFF/SSR responses when it receives an appropriate signal.

Attributes:

EmergencyOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Set to True when the transponder has set Military Emergency to On.</i>				
IdentSquawkFlashOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Set to True when the transponder has set the I/P to On. This is also referred to as the Special Process Indicator (SPI), "Squawk Flash" or "Squawk Ident".</i>				
UnmannedAircraftOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Set to True when the transponder has set the Unmanned Aircraft reply to On.</i>				
SystemType <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemTypeEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>General type of IFF/SSR system in use.</i>				
SystemName <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemNameEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>On change</i>				
	Semantics	<i>Particular named type of the IFF/SSR system in use.</i>				

SystemDesignator <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>A unique decimal number assigned to this interrogator or transponder to distinguish it from multiple interrogators or transponders that are associated with the same entity.</i>				
InteractiveModeIsActive <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether or not the IFFSystem is currently operating in the Interactive Mode.</i>				
EventIdentifier <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
SystemIsOn <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Whether or not the system is on.</i>				
SystemMode <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemModeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode of operation.</i>				

OperationalStatus <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OperationalStatusEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Whether the system is operational or whether it has experienced a system failure.</i>				
TestModeIsOn <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether the IFF/SSR system has been placed in the equipment test mode.</i>				
EmissionsDataAvailable <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies if emissions data (DIS IFF PDU layer 2) is available for this IFF/SSR system. When set to True, then the EmissionsDataSpecificInformation, BeamAzimuthCenter, BeamAzimuthSweep, BeamElevationCenter, BeamElevationSweep, BeamSweepSync, and FundamentalParameterData attributes shall be generated.</i>				
EmissionsDataSpecificInformation <i>RPRnoteDER7</i> <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Layer-specific information for emissions data (layer number 2) that varies by SystemType.</i>				
BeamAzimuthCenter <i>RPRnoteDER7</i> <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>> IFF_AZ_THRSH</i> <i>RPRnoteDER8</i>				
	Semantics	<i>The azimuth center of the IFF/SSR beam's scan volume relative to the IFF/SSR system.</i>				

BeamAzimuthSweep RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The azimuth half-angle of the IFF/SSR beam's scan volume relative to the IFF/SSR system.				
BeamElevationCenter RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> IFF_EL_THRSH RPRnoteDER9				
	Semantics	The elevation center of the IFF/SSR beam's scan volume relative to the IFF/SSR system.				
BeamElevationSweep RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The elevation half-angle of the IFF/SSR beam's scan volume relative to the IFF/SSR system.				
BeamSweepSync RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Periodic	On change RPRnoteDER10				
	Semantics	The percentage of time a scan is through its pattern from its origin.				
FundamentalParameterData RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FundamentalParameterDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The fundamental energy radiation characteristics of the IFF/SSR system.				

AntennaLocationArray <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AntennaLocationIFFDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>The location of each auxiliary IFF/SSR antenna. An auxiliary IFF/SSR antenna is one that is in addition to the primary antenna whose location is contained in the EmbeddedSystem attribute RelativePosition.</i>					
DataCommunicationsSpecificInformation <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>Layer-specific information for data communications (layer number 5) that varies by SystemType.</i>					
EntityIdentifier RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics <i>The EntityIdentifier of the object which this embedded system is a part of.</i>					
HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics <i>The RTI object instance ID of the object of which this embedded system is part of.</i>					
RelativePosition RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>The position of the embedded system, relative to the host object's position.</i>					

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.6. TCAS

Full Name: HLAobjectRoot.EmbeddedSystem.IFFSystem.TCAS

Sharing: Publish/Subscribe

Semantics: *TCAS or ACAS transceiver system.*

Attributes:

SystemType <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemTypeEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>General type of IFF/SSR system in use.</i>				
SystemName <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemNameEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>On change</i>				
	Semantics	<i>Particular named type of the IFF/SSR system in use.</i>				
SystemDesignator <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>A unique decimal number assigned to this interrogator or transponder to distinguish it from multiple interrogators or transponders that are associated with the same entity.</i>				

InteractiveModeIsActive <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether or not the IFFSystem is currently operating in the Interactive Mode.</i>				
EventIdentifier <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
SystemIsOn <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Whether or not the system is on.</i>				
SystemMode <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFSystemModeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode of operation.</i>				
OperationalStatus <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	OperationalStatusEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Whether the system is operational or whether it has experienced a system failure.</i>				

TestModeIsOn <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether the IFF/SSR system has been placed in the equipment test mode.</i>				
EmissionsDataAvailable <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies if emissions data (DIS IFF PDU layer 2) is available for this IFF/SSR system. When set to True, then the EmissionsDataSpecificInformation, BeamAzimuthCenter, BeamAzimuthSweep, BeamElevationCenter, BeamElevationSweep, BeamSweepSync, and FundamentalParameterData attributes shall be generated.</i>				
EmissionsDataSpecificInformation <i>RPRnoteDER7</i> <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Layer-specific information for emissions data (layer number 2) that varies by SystemType.</i>				
BeamAzimuthCenter <i>RPRnoteDER7</i> <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>> IFF_AZ_THRSH</i> <i>RPRnoteDER8</i>				
	Semantics	<i>The azimuth center of the IFF/SSR beam's scan volume relative to the IFF/SSR system.</i>				
BeamAzimuthSweep <i>RPRnoteDER7</i> <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The azimuth half-angle of the IFF/SSR beam's scan volume relative to the IFF/SSR system.</i>				

BeamElevationCenter RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> IFF_EL_THRSH RPRnoteDER9				
	Semantics	The elevation center of the IFF/SSR beam's scan volume relative to the IFF/SSR system.				
BeamElevationSweep RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The elevation half-angle of the IFF/SSR beam's scan volume relative to the IFF/SSR system.				
BeamSweepSync RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Periodic	On change RPRnoteDER10				
	Semantics	The percentage of time a scan is through its pattern from its origin.				
FundamentalParameterData RPRnoteDER7 Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FundamentalParameterDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The fundamental energy radiation characteristics of the IFF/SSR system.				
AntennaLocationArray Inherited from IFFSystem in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AntennaLocationIFFDataStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The location of each auxiliary IFF/SSR antenna. An auxiliary IFF/SSR antenna is one that is in addition to the primary antenna whose location is contained in the EmbeddedSystem attribute RelativePosition.				

DataCommunicationsSpecificInformation <i>Inherited from IFFSystem in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Layer-specific information for data communications (layer number 5) that varies by SystemType.</i>				
EntityIdentifier RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The EntityIdentifier of the object which this embedded system is a part of.</i>				
HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The RTI object instance ID of the object of which this embedded system is part of.</i>				
RelativePosition RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The position of the embedded system, relative to the host object's position.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.7. EmitterBeam

Full Name: HLAobjectRoot.EmitterBeam

Sharing: Subscribe

Semantics: *A sector of concentrated energy from a device that radiates an electromagnetic signal.*

Attributes:

BeamAzimuthCenter	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> EE_AZ_THRSH RPRnoteDER11				
	Semantics					
	The azimuth center of the emitter beam's scan volume relative to the emitter system.					
BeamAzimuthSweep	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> EE_AZ_THRSH RPRnoteDER11				
	Semantics					
	The azimuth half-angle of the emitter beam's scan volume relative to the emitter system.					
BeamElevationCenter	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> EE_EL_THRSH RPRnoteDER12				
	Semantics					
	The elevation center of the emitter beam's scan volume relative to the emitter system.					
BeamElevationSweep	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> EE_EL_THRSH RPRnoteDER12				
	Semantics					
	The elevation half-angle of the emitter beam's scan volume relative to the emitter system.					

BeamFunctionCode	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ElectromagneticEmissionBeamFunctionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The function of the emitter beam.</i>				
BeamIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The identification of the emitter beam (must be unique on the emitter system).</i>				
BeamParameterIndex	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The index, into the federation specific emissions database, of the current operating mode of the emitter beam.</i>				
EffectiveRadiatedPower	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PowerRatioDecibelMilliwattFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The effective radiated power for the emission (in the direction of the strongest beam). For a radar or a noise jammer, this attribute shall represent the peak ERP. RPRnoteDER13</i>				
EmissionFrequency	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The center frequency of the emitter beam.</i>				

EmitterSystemIdentifier RPRnoteDER14	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the emitter system that is generating this emitter beam.</i>				
EventIdentifier RPRnoteDER5	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteDER6				
	Semantics	<i>The EventIdentifier is used by the generating federate to associate related events. The event number shall start at one at the beginning of the exercise, and be incremented by one for each event.</i>				
FrequencyRange	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The bandwidth of the frequencies covered by the emitter beam.</i>				
PulseRepetitionFrequency	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The Pulse Repetition Frequency of the emitter beam.</i>				
PulseWidth	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TimeMicrosecondFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The pulse width of the emitter beam.</i>				

SweepSynch	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Periodic	<i>On change</i> RPRnoteDER10				
	Semantics	<i>The percentage of time a scan is through its pattern from its origin.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.8. RadarBeam

Full Name: HLAobjectRoot.EmitterBeam.RadarBeam

Sharing: Publish/Subscribe

Semantics: *A sector of concentrated energy from a device that radiates an electromagnetic signal.*

Attributes:

HighDensityTrack	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HighDensityTrackJamEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>When set to Selected (1), the receiving simulation can assume that all targets that are in the scan pattern of the radar beam are being tracked.</i>				
TrackObjectIdentifiers RPRnoteDER15	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectIdArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identification of the entities being tracked.</i>				

BlankingSectors	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	BlankingSectorStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i> RPRnoteDER16				
	Semantics					
	<i>Information about the radar beam's blanking sectors.</i>					
BeamAzimuthCenter <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>> EE_AZ_THRSH</i> RPRnoteDER11				
	Semantics					
	<i>The azimuth center of the emitter beam's scan volume relative to the emitter system.</i>					
BeamAzimuthSweep <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>> EE_AZ_THRSH</i> RPRnoteDER11				
	Semantics					
	<i>The azimuth half-angle of the emitter beam's scan volume relative to the emitter system.</i>					
BeamElevationCenter <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>> EE_EL_THRSH</i> RPRnoteDER12				
	Semantics					
	<i>The elevation center of the emitter beam's scan volume relative to the emitter system.</i>					
BeamElevationSweep <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>> EE_EL_THRSH</i> RPRnoteDER12				
	Semantics					
	<i>The elevation half-angle of the emitter beam's scan volume relative to the emitter system.</i>					

BeamFunctionCode <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ElectromagneticEmissionBeamFunctionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The function of the emitter beam.</i>				
BeamIdentifier <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The identification of the emitter beam (must be unique on the emitter system).</i>				
BeamParameterIndex <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The index, into the federation specific emissions database, of the current operating mode of the emitter beam.</i>				
EffectiveRadiatedPower <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PowerRatioDecibelMilliwattFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The effective radiated power for the emission (in the direction of the strongest beam). For a radar or a noise jammer, this attribute shall represent the peak ERP. ^{RPRnoteDER13}</i>				
EmissionFrequency <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The center frequency of the emitter beam.</i>				

EmitterSystemIdentifier RPRnoteDER14 Inherited from EmitterBeam in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The identification of the emitter system that is generating this emitter beam.				
EventIdentifier RPRnoteDER5 Inherited from EmitterBeam in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteDER6				
	Semantics	The EventIdentifier is used by the generating federate to associate related events. The event number shall start at one at the beginning of the exercise, and be incremented by one for each event.				
FrequencyRange Inherited from EmitterBeam in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The bandwidth of the frequencies covered by the emitter beam.				
PulseRepetitionFrequency Inherited from EmitterBeam in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The Pulse Repetition Frequency of the emitter beam.				
PulseWidth Inherited from EmitterBeam in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TimeMicrosecondFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The pulse width of the emitter beam.				

SweepSynch <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Periodic	<i>On change</i> RPRnoteDER10				
	Semantics	<i>The percentage of time a scan is through its pattern from its origin.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.9. JammerBeam

Full Name: HLAobjectRoot.EmitterBeam.JammerBeam

Sharing: Publish/Subscribe

Semantics: *An emitter beam that is designed to jam or otherwise interfere or confuse another emitter system.*

Attributes:

JammingModeSequence	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	JammingTechniqueStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The jamming mode technique or series of techniques being applied.</i>				
JammedObjectIdentifiers RPRnoteDER17	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectIdArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Identification of the emitter beams being jammed.</i>				

HighDensityJam	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HighDensityTrackJamEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	When set to Selected (1), the receiving simulation can assume that all emitter beams that are in the scan pattern of the jammer beam are being jammed.					
AngleDeceptionData	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleDeceptionStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteDER16				
	Semantics					
	Information about angle deception jamming.					
FalseTargetsData	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FalseTargetsStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteDER16				
	Semantics					
	Information about false targets jamming.					
BeamAzimuthCenter	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> EE_AZ_THRSH RPRnoteDER11				
	Semantics					
	The azimuth center of the emitter beam's scan volume relative to the emitter system.					
BeamAzimuthSweep	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> EE_AZ_THRSH RPRnoteDER11				
	Semantics					
	The azimuth half-angle of the emitter beam's scan volume relative to the emitter system.					

BeamElevationCenter <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> EE_EL_THRSH RPRnoteDER12				
	Semantics	The elevation center of the emitter beam's scan volume relative to the emitter system.				
BeamElevationSweep <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	> EE_EL_THRSH RPRnoteDER12				
	Semantics	The elevation half-angle of the emitter beam's scan volume relative to the emitter system.				
BeamFunctionCode <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ElectromagneticEmissionBeamFunctionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The function of the emitter beam.				
BeamIdentifier <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The identification of the emitter beam (must be unique on the emitter system).				
BeamParameterIndex <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The index, into the federation specific emissions database, of the current operating mode of the emitter beam.				

EffectiveRadiatedPower <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PowerRatioDecibelMilliwattFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>The effective radiated power for the emission (in the direction of the strongest beam). For a radar or a noise jammer, this attribute shall represent the peak ERP. RPRnoteDER13</i>					
EmissionFrequency <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>The center frequency of the emitter beam.</i>					
EmitterSystemIdentifier RPRnoteDER14 <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics <i>The identification of the emitter system that is generating this emitter beam.</i>					
EventIdentifier RPRnoteDER5 <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change RPRnoteDER6</i>				
	Semantics <i>The EventIdentifier is used by the generating federate to associate related events. The event number shall start at one at the beginning of the exercise, and be incremented by one for each event.</i>					
FrequencyRange <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics <i>The bandwidth of the frequencies covered by the emitter beam.</i>					

PulseRepetitionFrequency <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The Pulse Repetition Frequency of the emitter beam.</i>				
PulseWidth <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TimeMicrosecondFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The pulse width of the emitter beam.</i>				
SweepSynch <i>Inherited from EmitterBeam in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PercentFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Periodic	<i>On change</i> RPRnoteDER10				
	Semantics	<i>The percentage of time a scan is through its pattern from its origin.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

9.1.10. IFFData

Full Name: HLAobjectRoot.IFFData

Sharing:

Semantics: *Data applicable to distinct IFF/SSR system capabilities.*

Attributes:

IFFSystemIdentifier RPRnoteDER14	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The identification of the IFFSystem object to which the IFF/SSR capability is composed.				
EventIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Used to associate related events.				
ReportingSimulation	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The simulation publishing this IFF data.				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.11. Mode1

Full Name: HLAobjectRoot.IFFData.Mode1

Sharing: Subscribe

Semantics: Mode 1 data applicable to both IFF transponders and IFF interrogators.

Attributes:

IsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 1 is on.					
IsDamaged	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 1 is damaged.					
IsMalfunctioning	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 1 is malfunctioning.					
IFFSystemIdentifier RPRnoteDER14 Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The identification of the IFFSystem object to which the IFF/SSR capability is composed.					
EventIdentifier Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Used to associate related events.					

ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

9.1.12. Mode1Interrogator

Full Name: HLAobjectRoot.IFFData.Mode1.Mode1Interrogator

Sharing: Publish/Subscribe

Semantics: *IFF interrogator Mode 1 capability.*

Attributes:

IsOn <i>Inherited from Mode1 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 1 is on.</i>				
IsDamaged <i>Inherited from Mode1 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 1 is damaged.</i>				

IsMalfunctioning <i>Inherited from Mode1 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 1 is malfunctioning.</i>				
IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.13. Mode1Transponder

Full Name: HLAobjectRoot.IFFData.Mode1.Mode1Transponder

Sharing: Publish/Subscribe

Semantics: *IFF transponder Mode 1 capability.*

Attributes:

Code	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TransponderCode4DigitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Mode 1 code. For Mark X/XII systems, two octal values in the range 00 to 73 with the first digit in the range 0 to 7 and the second digit in the range 0 to 3 reading left to right. The first (left) digit is Code Element 2 (A) and occupies bits 3 to 5, and the second (right) digit is Code Element 1 (B) and occupies bits 0 to 2. Mark XIIA systems may employ extended Mode 1 codes, four octal values in the range 0000 to 7777 (as per Mode 2 and Mode 3/A codes). The attribute ExtendedCodeIsSet indicates whether or not an extended code has been set. If an extended Mode 1 code has been set, Mark X/XII interrogators need to extract bits 9-11 (Code Element A) and bits 6-7 (Code Element B).					
ExtendedCodeIsSet RPRnoteDER18	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mark XIIA Enhanced Mode 1 is on.					
IsOn Inherited from Mode1 in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 1 is on.					
IsDamaged Inherited from Mode1 in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 1 is damaged.					

IsMalfunctioning <i>Inherited from Mode1 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 1 is malfunctioning.</i>				
IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.14. Mode2

Full Name: HLAobjectRoot.IFFData.Mode2

Sharing: Subscribe

Semantics: *Mode 2 data applicable to both IFF transponders and IFF interrogators.*

Attributes:

IsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 2 is on.					
IsDamaged	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 2 is damaged.					
IsMalfunctioning	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 2 is malfunctioning.					
IFFSystemIdentifier RPRnoteDER14 Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The identification of the IFFSystem object to which the IFF/SSR capability is composed.					

EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

9.1.15. Mode2Interrogator

Full Name: HLAobjectRoot.IFFData.Mode2.Mode2Interrogator

Sharing: Publish/Subscribe

Semantics: *IFF interrogator Mode 2 capability.*

Attributes:

IsOn <i>Inherited from Mode2 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 2 is on.</i>				

IsDamaged <i>Inherited from Mode2 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 2 is damaged.</i>				
IsMalfunctioning <i>Inherited from Mode2 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 2 is malfunctioning.</i>				
IFFSystemIdentifier RPRnoteDER14 <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.16. Mode2Transponder

Full Name: HLAobjectRoot.IFFData.Mode2.Mode2Transponder

Sharing: Publish/Subscribe

Semantics: *IFF transponder Mode 2 capability.*

Attributes:

Code	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TransponderCode4DigitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode 2 code, four octal values in the range 0000 to 7777. The first (left-most) digit is Code Element 4 (A) and occupies bits 9 to 11, followed by Code Element 3 (B) in bits 6 to 8, Code Element 2 (C) in bits 3 to 5, and the last (right-most) digit is Code Element 1 (D) and occupies bits 0 to 2.</i>				
IsOn <i>Inherited from Mode2 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 2 is on.</i>				
IsDamaged <i>Inherited from Mode2 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 2 is damaged.</i>				

IsMalfunctioning <i>Inherited from Mode2 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 2 is malfunctioning.</i>				
IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.17. Mode3A

Full Name: HLAobjectRoot.IFFData.Mode3A

Sharing: Subscribe

Semantics: *Mode 3/A data applicable to IFF/SSR transponders and IFF/SSR interrogators.*

Attributes:

IsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 3/A is on.					
IsDamaged	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 3/A is damaged.					
IsMalfunctioning	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 3/A is malfunctioning.					
IFFSystemIdentifier RPRnoteDER14 Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The identification of the IFFSystem object to which the IFF/SSR capability is composed.					

EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

9.1.18. Mode3AInterrogator

Full Name: HLAobjectRoot.IFFData.Mode3A.Mode3AInterrogator

Sharing: Publish/Subscribe

Semantics: *IFF/SSR interrogator Mode 3/A capability.*

Attributes:

IsOn <i>Inherited from Mode3A in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 3/A is on.</i>				

IsDamaged <i>Inherited from Mode3A in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 3/A is damaged.</i>				
IsMalfunctioning <i>Inherited from Mode3A in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 3/A is malfunctioning.</i>				
IFFSystemIdentifier RPRnoteDER14 <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.19. Mode3ATransponder

Full Name: HLAobjectRoot.IFFData.Mode3A.Mode3ATransponder

Sharing: Publish/Subscribe

Semantics: *IFF/SSR transponder Mode 3/A capability.*

Attributes:

Code	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TransponderCode4DigitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode 3/A code, four octal values in the range 0000 to 7777. The first (left-most) digit is Code Element 4 (A) and occupies bits 9 to 11, followed by Code Element 3 (B) in bits 6 to 8, Code Element 2 (C) in bits 3 to 5, and the last (right-most) digit is Code Element 1 (D) and occupies bits 0 to 2.</i>				
IsOn <i>Inherited from Mode3A in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 3/A is on.</i>				
IsDamaged <i>Inherited from Mode3A in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 3/A is damaged.</i>				

IsMalfunctioning <i>Inherited from Mode3A in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 3/A is malfunctioning.</i>				
IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAToken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.20. Mode4

Full Name: HLAobjectRoot.IFFData.Mode4

Sharing: Subscribe

Semantics: *Mode 4 data applicable to both IFF transponders and IFF interrogators.*

Attributes:

IsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 4 is on.					
IsDamaged	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 4 is damaged.					
IsMalfunctioning	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 4 is malfunctioning.					
PseudoCryptoCodeAAvailable	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether a Mode 4 pseudo crypto code (A) value is available in the attribute PseudoCryptoCodeA or that Challenge or Reply is to be used instead. If also a code B is available, the attribute PseudoCryptoCodeActive indicates which code is to be used.					

PseudoCryptoCodeA	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PseudoCode12BitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode 4 pseudo crypto code (A) in the range 0 to 4094 decimal.</i>				
PseudoCryptoCodeBAvailable	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether or not a Mode 4 pseudo crypto code B value is available in the attribute PseudoCryptoCodeB. If available, PseudoCryptoCodeA must also be set to a valid value, PseudoCryptoCodeAAvailable to True, and PseudoCryptoCodeActive to indicate which code is to be used. If not, the PseudoCryptoCodeA is to be used, if available, or the Challenge or Reply attribute.</i>				
PseudoCryptoCodeB	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PseudoCode12BitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode 4 pseudo crypto code B in the range 0 to 4094 decimal.</i>				
PseudoCryptoCodeActive	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFMode4CodeActiveEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether Mode 4 Code A or Code B is active, or neither.</i>				

IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The identification of the IFFSystem object to which the IFF/SSR capability is composed.				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Used to associate related events.				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The simulation publishing this IFF data.				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.21. Mode4Interrogator

Full Name: HLAobjectRoot.IFFData.Mode4.Mode4Interrogator

Sharing: Publish/Subscribe

Semantics: IFF interrogator Mode 4 capability.

Attributes:

Challenge	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFAlternateMode4ChallengeReplyEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates the validity of a Mode 4 interrogation. This attribute shall be set to Valid (1) or Invalid (2).</i>				
IsOn <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 4 is on.</i>				
IsDamaged <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 4 is damaged.</i>				
IsMalfunctioning <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 4 is malfunctioning.</i>				
PseudoCryptoCodeAAvailable <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether a Mode 4 pseudo crypto code (A) value is available in the attribute PseudoCryptoCodeA or that Challenge or Reply is to be used instead. If also a code B is available, the attribute PseudoCryptoCodeActive indicates which code is to be used.</i>				

PseudoCryptoCodeA <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PseudoCode12BitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode 4 pseudo crypto code (A) in the range 0 to 4094 decimal.</i>				
PseudoCryptoCodeBAvailable <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether or not a Mode 4 pseudo crypto code B value is available in the attribute PseudoCryptoCodeB. If available, PseudoCryptoCodeA must also be set to a valid value, PseudoCryptoCodeAAvailable to True, and PseudoCryptoCodeActive to indicate which code is to be used. If not, the PseudoCryptoCodeA is to be used, if available, or the Challenge or Reply attribute.</i>				
PseudoCryptoCodeB <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PseudoCode12BitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode 4 pseudo crypto code B in the range 0 to 4094 decimal.</i>				
PseudoCryptoCodeActive <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFMode4CodeActiveEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether Mode 4 Code A or Code B is active, or neither.</i>				

IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The identification of the IFFSystem object to which the IFF/SSR capability is composed.				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Used to associate related events.				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The simulation publishing this IFF data.				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.22. Mode4Transponder

Full Name: HLAobjectRoot.IFFData.Mode4.Mode4Transponder

Sharing: Publish/Subscribe

Semantics: IFF transponder Mode 4 capability.

Attributes:

Reply	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFAlternateMode4ChallengeReplyEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates the reply that would be received from a Mode 4 transponder by a Mode 4 interrogator. This attribute shall be set to Valid (1), Invalid (2), or No Response (3). The Unable to Verify (4) reply is optional.					
IsOn <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 4 is on.					
IsDamaged <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 4 is damaged.					
IsMalfunctioning <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 4 is malfunctioning.					
PseudoCryptoCodeAAvailable <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether a Mode 4 pseudo crypto code (A) value is available in the attribute PseudoCryptoCodeA or that Challenge or Reply is to be used instead. If also a code B is available, the attribute PseudoCryptoCodeActive indicates which code is to be used.					

PseudoCryptoCodeA <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PseudoCode12BitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode 4 pseudo crypto code (A) in the range 0 to 4094 decimal.</i>				
PseudoCryptoCodeBAvailable <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether or not a Mode 4 pseudo crypto code B value is available in the attribute PseudoCryptoCodeB. If available, PseudoCryptoCodeA must also be set to a valid value, PseudoCryptoCodeAAvailable to True, and PseudoCryptoCodeActive to indicate which code is to be used. If not, the PseudoCryptoCodeA is to be used, if available, or the Challenge or Reply attribute.</i>				
PseudoCryptoCodeB <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PseudoCode12BitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode 4 pseudo crypto code B in the range 0 to 4094 decimal.</i>				
PseudoCryptoCodeActive <i>Inherited from Mode4 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFMode4CodeActiveEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether Mode 4 Code A or Code B is active, or neither.</i>				

IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The identification of the IFFSystem object to which the IFF/SSR capability is composed.				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Used to associate related events.				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The simulation publishing this IFF data.				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.23. Mode5

Full Name: HLAobjectRoot.IFFData.Mode5

Sharing: Subscribe

Semantics: Mode 5 data applicable to both IFF transponders and IFF interrogators.

Attributes:

IsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 5 is on.					
IsDamaged	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 5 is damaged.					
IsMalfunctioning	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode 5 is malfunctioning.					
PseudoCryptoCodeAvailable	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether a Mode 5 pseudo crypto code value is available in the attribute PseudoCryptoCode or that Reply is to be used instead.					
PseudoCryptoCode	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PseudoCode12BitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Mode 5 pseudo crypto code in the range 0 to 4094 decimal.					

MessageFormatsCapable	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5MessageFormatsStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The included message formats that this Mode 5 interrogator or transponder is capable of supporting and could be in a Mode 5 interrogation or in a reply to a Mode 5 interrogation.</i>				
Mission	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5IFFMissionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The IFF mission of this Mode 5 interrogator or transponder. (Some of the enumerators are applicable to either a transponder or an interrogator.)</i>				
SpecificInformation	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specific information for Mode 5 data (DIS IFF PDU layer 3) that varies by System Type.</i>				
IFFSystemIdentifier RPRnoteDER14 Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				

ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

9.1.24. Mode5Interrogator

Full Name: HLAobjectRoot.IFFData.Mode5.Mode5Interrogator

Sharing: Publish/Subscribe

Semantics: *IFF interrogator Mode 5 capability.*

Attributes:

IsOn <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 5 is on.</i>				
IsDamaged <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 5 is damaged.</i>				

IsMalfunctioning <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 5 is malfunctioning.</i>				
PseudoCryptoCodeAvailable <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether a Mode 5 pseudo crypto code value is available in the attribute PseudoCryptoCode or that Reply is to be used instead.</i>				
PseudoCryptoCode <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PseudoCode12BitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode 5 pseudo crypto code in the range 0 to 4094 decimal.</i>				
MessageFormatsCapable <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5MessageFormatsStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The included message formats that this Mode 5 interrogator or transponder is capable of supporting and could be in a Mode 5 interrogation or in a reply to a Mode 5 interrogation.</i>				
Mission <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5IFFMissionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The IFF mission of this Mode 5 interrogator or transponder. (Some of the enumerators are applicable to either a transponder or an interrogator.)</i>				

SpecificInformation <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specific information for Mode 5 data (DIS IFF PDU layer 3) that varies by System Type.</i>				
IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.25. Mode5Transponder

Full Name: HLAobjectRoot.IFFData.Mode5.Mode5Transponder

Sharing: Publish/Subscribe

Semantics: *IFF transponder Mode 5 capability.*

Attributes:

Reply	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5ReplyEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The validity of a reply that would be transmitted by a Mode 5 transponder if interrogated. The following enumerators shall be capable of being set: No Response (0), Valid (1), and Invalid (2). The Unable to Verify (3) value is optional.					
LevelSelection	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5LevelSelectionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether Mode 5 Level 1 (0) or Mode 5 Level 2 (1) is currently selected for this Mode 5 transponder. (The Level 2 selection includes both Level 1 and Level 2 message formats.)					
SquitterIsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether Mode 5 squitter is On (1) or Off (0).					
Level2SquitterStatus	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Level2SquitterStatusEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether Mode 5 Level 2 squitter is Disabled (0) or Enabled (1).					

PlatformType	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5PlatformTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates the type of platform (air or ground) that is associated with this Mode 5 transponder.					
PlatformIdentificationNumber	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Platform Identification Number (PIN) that uniquely identifies the platform carrying this Mode 5 transponder. The PIN shall be a number in the range 00 000 octal to 37 777 octal.					
NationalOrigin	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Enumeration16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The national origin, the country that owns the platform carrying a Mode 5 transponder.					
FigureOfMerit	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Figure of Merit, a number in the range of 0 to 31 decimal.					
AntennaSelection	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AntennaSelectionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicate the present Mode 5 transponder antenna selection and shall be set to either No Statement (0) if the Mode 5 antenna position is not available or to Top (1), Bottom (2), or Diversity (3).					

LineTestIsEnabled	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether a line test is in progress.					
NavigationSource	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	NavigationSourceEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Navigation source.					
LocationDataIsPresent	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether the source of Mode 5 latitude, longitude, and altitude information is present in the TransponderLocation attribute (1), or is to be computed locally (0).					
TransponderLocation	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5STransponderLocationIFFDataStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Real-time location and altitude information transmitted by the transponder.					
LocationErrorsDataPresence	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5LocationErrorsEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether there are location errors associated with Mode 5 latitude, longitude, or altitude present in the TransponderLocationError attribute (1), or No Location Errors are provided (0).					

TransponderLocationError	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5STransponderLocationErrorIFFDataStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The specification of errors in location and altitude information transmitted by the transponder.</i>				
IsOn <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 5 is on.</i>				
IsDamaged <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 5 is damaged.</i>				
IsMalfunctioning <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode 5 is malfunctioning.</i>				
PseudoCryptoCodeAvailable <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates whether a Mode 5 pseudo crypto code value is available in the attribute PseudoCryptoCode or that Reply is to be used instead.</i>				

PseudoCryptoCode <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PseudoCode12BitUnsignedInteger16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode 5 pseudo crypto code in the range 0 to 4094 decimal.</i>				
MessageFormatsCapable <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5MessageFormatsStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The included message formats that this Mode 5 interrogator or transponder is capable of supporting and could be in a Mode 5 interrogation or in a reply to a Mode 5 interrogation.</i>				
Mission <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5IFFMissionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The IFF mission of this Mode 5 interrogator or transponder. (Some of the enumerators are applicable to either a transponder or an interrogator.)</i>				
SpecificInformation <i>Inherited from Mode5 in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specific information for Mode 5 data (DIS IFF PDU layer 3) that varies by System Type.</i>				

IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The identification of the IFFSystem object to which the IFF/SSR capability is composed.				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Used to associate related events.				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The simulation publishing this IFF data.				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.26. ModeC

Full Name: HLAobjectRoot.IFFData.ModeC

Sharing: Subscribe

Semantics: Mode C data applicable to both IFF/SSR transponders and IFF/SSR interrogators.

Attributes:

IsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode C is on.					
IsDamaged	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode C is damaged.					
IsMalfunctioning	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode C is malfunctioning.					
IFFSystemIdentifier RPRnoteDER14 Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The identification of the IFFSystem object to which the IFF/SSR capability is composed.					
EventIdentifier Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Used to associate related events.					

ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

9.1.27. ModeCInterrogator

Full Name: HLAobjectRoot.IFFData.ModeC.ModeCInterrogator

Sharing: Publish/Subscribe

Semantics: *IFF/SSR interrogator Mode C capability.*

Attributes:

IsOn <i>Inherited from ModeC in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode C is on.</i>				
IsDamaged <i>Inherited from ModeC in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode C is damaged.</i>				

IsMalfunctioning <i>Inherited from ModeC in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode C is malfunctioning.</i>				
IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

9.1.28. ModeCTransponder

Full Name: HLAobjectRoot.IFFData.ModeC.ModeCTransponder

Sharing: Publish/Subscribe

Semantics: *IFF/SSR transponder Mode C capability.*

Attributes:

AltitudeAvailable	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether the actual Mode C altitude is available in the attribute Altitude or is to be derived from entity truth data.					
Altitude	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ModeCAltitudeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The actual Mode C altitude.					
IsOn Inherited from ModeC in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode C is on.					
IsDamaged Inherited from ModeC in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode C is damaged.					

IsMalfunctioning <i>Inherited from ModeC in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode C is malfunctioning.</i>				
IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.29. ModeS

Full Name: HLAobjectRoot.IFFData.ModeS

Sharing: Subscribe

Semantics: *Mode S data applicable to both IFF/SSR transponders and IFF/SSR interrogators.*

Attributes:

IsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode S is on.					
IsDamaged	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode S is damaged.					
IsMalfunctioning	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Mode S is malfunctioning.					
LevelsPresent	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ModeSLevelsPresentStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates the Mode S levels present. All levels that would be able to be responded to for a Mode S interrogation shall be set.					

SpecificInformation	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specific information for Mode S data (DIS IFF PDU layer 4) that varies by System Type.</i>				
IFFSystemIdentifier RPRnoteDER14 Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.30. ModeSInterrogator

Full Name: HLAobjectRoot.IFFData.ModeS.ModeSInterrogator

Sharing: Publish/Subscribe

Semantics: *IFF/SSR interrogator Mode S capability.*

Attributes:

InterrogatorIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ModeSInterrogatorIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Mode S Interrogator Code (IC) information.</i>				
TransmitState	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ModeSInterrogatorStatusTransmitStateEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The state of the Mode S interrogator. This field shall be set to No Statement (0) when the Mode S interrogator simulation is operating in the Regeneration Mode. It may be set to other values if operating in the Interactive Mode.</i>				
IsOn <i>Inherited from ModeS in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode S is on.</i>				
IsDamaged <i>Inherited from ModeS in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode S is damaged.</i>				

IsMalfunctioning <i>Inherited from ModeS in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode S is malfunctioning.</i>				
LevelsPresent <i>Inherited from ModeS in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ModeSLevelsPresentStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates the Mode S levels present. All levels that would be able to be responded to for a Mode S interrogation shall be set.</i>				
SpecificInformation <i>Inherited from ModeS in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specific information for Mode S data (DIS IFF PDU layer 4) that varies by System Type.</i>				
IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				

ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

9.1.31. ModeSTransponder

Full Name: HLAobjectRoot.IFFData.ModeS.ModeSTransponder

Sharing: Publish/Subscribe

Semantics: *IFF/SSR transponder Mode S capability.*

Attributes:

AircraftAddress	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>A unique ICAO Mode S Aircraft Address. When a military aircraft has both a permanent and an operational Mode S Aircraft Address, only one address will be active at a time. The active aircraft address shall be included in this attribute.</i>				
AircraftIdentificationType	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AircraftIdentificationTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The aircraft identification type.</i>				

AircraftIdentification	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ASCIICharArray8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The International Civil Aviation Organization (ICAO) flight identity of the aircraft, either the flight ID (i.e., call sign) or the registration number (also referred to as the tail number) of the aircraft. This attribute shall consist of an eight-element American Standard Code for Information Interchange (ASCII) character string consisting of the letters A to Z and the numbers 0 to 9. An aircraft identification may be less than eight characters. This string shall begin with the left-most character located at the lowest index. Unused characters shall be indicated by Null (0) values.					
SquitterIsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether the squitter is On (1) or Off (0).					
SquitterType	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ModeSSquitterTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates the type of Mode S squitter.					
CapabilityReport	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	CapabilityReportEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The Mode S Transponder Capability Report. The numeric values are from Annex 10 to ICAO Vol. IV, para 3.1.2.5.2.2.1, except for No Statement. If no specific capability is indicated, this field shall be set to No Statement (255).					

AircraftPresentDomain	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AircraftPresentDomainEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates the present domain of the aircraft. Surface means on a water platform surface such as a ship deck or on the water surface directly (e.g., pontoon-equipped aircraft). This field is referred to as Flight Status in ICAO Publications.					
AltitudeAvailable	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether the actual Mode S altitude is available in the attribute Altitude or is to be derived from entity truth data.					
Altitude	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TransponderRelativeAltitudeVariantStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The actual Mode S altitude as determined by the Mode S transponder model, represented as a binary coded altitude referenced to -1000 feet. All altitudes greater than 50 175 feet shall be reported with 100-foot resolution. A receiving simulation is not required to process an actual Mode S altitude value by this standard and may always compute it from the entity truth altitude.					
SquitterRecordSource	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ModeSSquitterRecordSourceEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether the squitter records are only present in the Squitter*Report attributes (0) or are also included in the GroundInitiatedCommBData array attribute (1).					

AirbornePositionReportDataIsPresent	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether there is a Squitter Airborne Position Report IFF Data record either Present (1) or Not Present (0) in the SquitterAirbornePositionReport attribute.					
SquitterAirbornePositionReport	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SquitterAirbornePositionReportIFFDataStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	This replicates the functional data contained in a Squitter Airborne Position Report transmitted by a Mode S transponder in accordance with ICAO standards.					
AirborneVelocityReportDataIsPresent	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether there is a Squitter Airborne Velocity Report IFF Data record either Present (1) or Not Present (0) in the SquitterAirborneVelocityReport attribute.					
SquitterAirborneVelocityReport	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SquitterAirborneVelocityReportIFFDataStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	This replicates the functional data contained in a Squitter Airborne Velocity Report transmitted by a Mode S transponder in accordance with ICAO standards.					

SurfacePositionReportDataIsPresent	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether there is a Squitter Surface Position Report IFF Data record either Present (1) or Not Present (0) in the SquitterSurfacePositionReport attribute.					
SquitterSurfacePositionReport	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SquitterSurfacePositionReportIFFDataStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	This replicates the functional data contained in a Squitter Surface Position Report transmitted by a Mode S transponder in accordance with ICAO standards. This report is transmitted when an aircraft is on the ground or aboard ship.					
IdentificationReportDataIsPresent	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether there is a Squitter Identification Report IFF Data record either Present (1) or Not Present (0) in the SquitterIdentificationReport attribute.					
SquitterIdentificationReport	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SquitterIdentificationReportIFFDataStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	This replicates the functional data contained in a Squitter Identification Report transmitted by a Mode S transponder in accordance with ICAO standards.					

EventDrivenReportDataIsPresent	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether there is a Squitter Event-Driven Report IFF Data record either Present (1) or Not Present (0) in the SquitterEventDrivenReport attribute.					
SquitterEventDrivenReport	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	SquitterEventDrivenReportIFFDataStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	This replicates the functional data contained in a Squitter Event-Driven Report transmitted by a Mode S transponder in accordance with ICAO standards.					
TransponderLocation	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5STransponderLocationIFFDataStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Real-time location and altitude information transmitted by the transponder.					
GroundInitiatedCommBData	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	GroundInitiatedCommBIFFDataStructureArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Variable data related to Ground-Initiated Comm-B (GICB) data communications.					

DAPSource	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ModeSDAPSourceStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether IFF Data records are present in the DAPData array attribute that can be used as the source for Mode S Downlink Aircraft Parameters (DAPs).					
DAPData	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ModeSDownlinkAircraftParameterVariantStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	IFF Data records representing Mode S Downlink Aircraft Parameters (DAPs).					
LocationErrorsDataPresence	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5LocationErrorsEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Indicates whether there are location errors associated with Mode S latitude, longitude, or altitude present in the TransponderLocationError attribute (1), or No Location Errors are provided (0).					
TransponderLocationError	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Mode5STransponderLocationErrorIFFDataStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The specification of errors in location and altitude information transmitted by the transponder.					

IsOn <i>Inherited from ModeS in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode S is on.</i>				
IsDamaged <i>Inherited from ModeS in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode S is damaged.</i>				
IsMalfunctioning <i>Inherited from ModeS in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Mode S is malfunctioning.</i>				
LevelsPresent <i>Inherited from ModeS in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ModeSLevelsPresentStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Indicates the Mode S levels present. All levels that would be able to be responded to for a Mode S interrogation shall be set.</i>				
SpecificInformation <i>Inherited from ModeS in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	IFFLayerSpecificInformationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specific information for Mode S data (DIS IFF PDU layer 4) that varies by System Type.</i>				

IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The identification of the IFFSystem object to which the IFF/SSR capability is composed.				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Used to associate related events.				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The simulation publishing this IFF data.				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.32. TCASData

Full Name: HLAobjectRoot.IFFData.TCASData

Sharing: Publish/Subscribe

Semantics: TCAS/ACAS capability, applicable to IFF/SSR transponders, IFF/SSR interrogators, and TCAS/ACAS systems.

Attributes:

IsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not the TCAS/ACAS system is on.					
IsDamaged	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not the TCAS/ACAS system is damaged.					
IsMalfunctioning	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not the TCAS/ACAS system is malfunctioning.					
TCAS_I_II_Type	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TCASIIITypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Indicates whether a TCAS system is a TCAS I (0) or TCAS II (1) system when the TCAS/ACAS Type field does not indicate a TCAS III system. If the TCAS/ACAS Type field contains a nonzero value, then this field shall be set to No Statement (0).					

BasicAdvancedIndicator	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TCASACASBasicAdvancedIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Indicates whether this TCACS/ACAS object contains Basic (0) or Advanced (1) data. Basic data is the minimum required to be set by any TCAS/ACAS capable interrogator or transponder. Advanced data means that the record contains additional data fields for a higher fidelity simulation of TCAS/ACAS capabilities. The attributes TCAS_ACASType, TCAS_ACASIndicator, and SoftwareVersion are not applicable to Basic and required for Advanced data.					
TCAS_ACASType	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TCASACASTypeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Indicates the type of TCAS/ACAS system other than a TCAS I or II system, if known, or shall be set to No Statement (0) if not known.					
TCAS_ACASIndicator	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TCASACASIndicatorEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Indicates whether the record applies to a TCAS (0) or an ACAS (1) system.					
SoftwareVersion	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TCASACASSoftwareVersionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Indicates the software version of the TCAS/ACAS system if known or set to No Statement (0) if not known.					

IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The identification of the IFFSystem object to which the IFF/SSR capability is composed.				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	Used to associate related events.				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The simulation publishing this IFF data.				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.33. Soviet

Full Name: HLAobjectRoot.IFFData.Soviet

Sharing: Subscribe

Semantics: Data applicable to both Soviet IFF transponders and Soviet IFF interrogators.

Attributes:

IsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not the Soviet mode is on.					
IsDamaged	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not the Soviet mode is damaged.					
IsMalfunctioning	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Soviet mode is malfunctioning.					
IFFSystemIdentifier RPRnoteDER14 Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The identification of the IFFSystem object to which the IFF/SSR capability is composed.					
EventIdentifier Inherited from IFFData in Distributed Emission Regeneration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Used to associate related events.					

ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

9.1.34. SovietInterrogator

Full Name: HLAobjectRoot.IFFData.Soviet.SovietInterrogator

Sharing: Publish/Subscribe

Semantics: *IFF interrogator Soviet capability.*

Attributes:

IsOn <i>Inherited from Soviet in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the Soviet mode is on.</i>				
IsDamaged <i>Inherited from Soviet in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the Soviet mode is damaged.</i>				

IsMalfunctioning <i>Inherited from Soviet in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not Soviet mode is malfunctioning.</i>				
IFFSystemIdentifier <i>RPRnoteDER14</i> <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.1.35. SovietTransponder

Full Name: HLAobjectRoot.IFFData.Soviet.SovietTransponder

Sharing: Publish/Subscribe

Semantics: *IFF transponder Soviet capability.*

Attributes:

IsOn <i>Inherited from Soviet in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not the Soviet mode is on.					
IsDamaged <i>Inherited from Soviet in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not the Soviet mode is damaged.					
IsMalfunctioning <i>Inherited from Soviet in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Specifies whether or not Soviet mode is malfunctioning.					
IFFSystemIdentifier RPRnoteDER14 <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The identification of the IFFSystem object to which the IFF/SSR capability is composed.					

EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

9.1.36. RRB [RPRnoteDER19](#)

Full Name: HLAobjectRoot.IFFData.RRB

Sharing: Publish/Subscribe

Semantics: *Reply Receiver 'B' (RRB) transponder capability.*

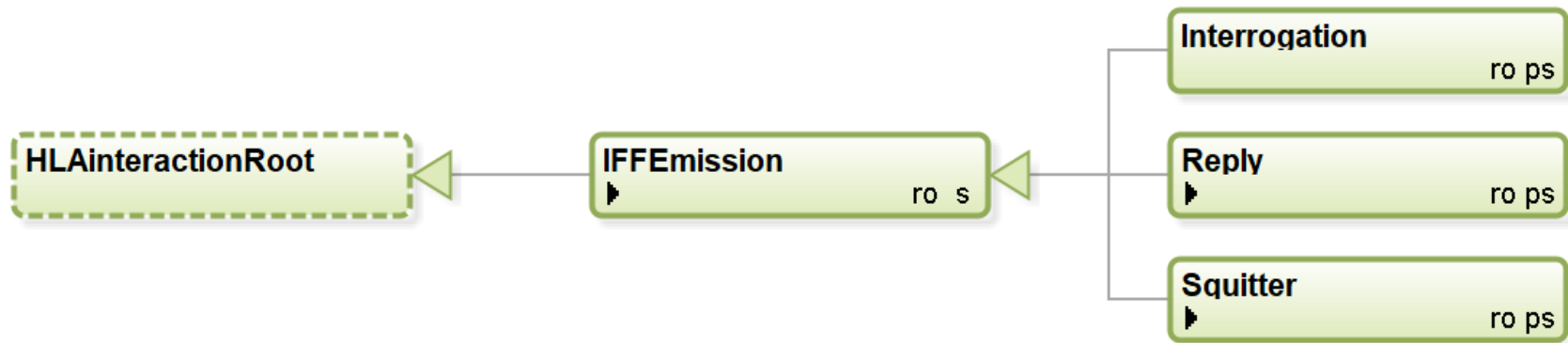
Attributes:

IsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the RRB transponder is set to reply.</i>				

IsDamaged	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the RRB system is damaged.</i>				
IsMalfunctioning	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the RRB system is malfunctioning.</i>				
Code	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RRBCodeUnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>RRB Code (range 0-16).</i>				
PowerReductionIsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not power reduction is on.</i>				
RadarEnhancementIsOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Specifies whether or not the radar enhancement is on.</i>				

IFFSystemIdentifier RPRnoteDER14 <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The identification of the IFFSystem object to which the IFF/SSR capability is composed.</i>				
EventIdentifier <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to associate related events.</i>				
ReportingSimulation <i>Inherited from IFFData in Distributed Emission Regeneration</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FederateIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The simulation publishing this IFF data.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAToken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

9.2. Interaction Classes



9.2.1. HLAinteractionRoot

Full Name: HLAinteractionRoot
 Sharing: Publish/Subscribe
 Transportation type: HLAREliable
 Order: Receive
 Dimensions:
 Semantics:
 Parameters: -

9.2.2. IFFEmission

Full Name: HLAinteractionRoot.IFFEmission
 Sharing: Subscribe
 Transportation type: HLABestEffort
 Order: Receive
 Dimensions:

Semantics: *An IFF electromagnetic signal, used by IFF systems in Interactive Mode.*

Parameters:

Name	Datatype	Semantics
IFFSystemIdentifier RPRnoteDER14	RTObjectId	The identification of the IFFSystem object emitting this IFF electromagnetic signal or composite set of signals.
TransmissionIndicator	TransmissionIndicatorEnum8	Indication whether this is an interrogation, a reply to an interrogation, or a squitter transmission.
InterrogatingObjectIdentifier	RTObjectId	The object that is interrogating another object.
InterrogatedObjectIdentifier	RTObjectId	The object to which the interrogation is being directed.
InteractiveEventIdentifier	EventIdentifierStruct	Used to associate a reply to an interrogation. The Event Identifier is set by an interrogator that issues an interactive IFF emission, and the identical number is included in the reply by an interactive transponder to an interrogation. The first number assigned is one, and then is incremented by one for each succeeding interactive interrogation for the same IFF system. A transponder that has a capability to issue an interactive IFF transmission representing of one or more squitter message(s) shall always set the Event Identifier to zero.
InterrogatedModes	InterrogatedModesStruct	The modes being interrogated.
ReportingSimulation	FederateIdentifierStruct	The simulation publishing this IFF emission.
DataCommunicationsSpecificInformation	IFFLayerSpecificInformationEnum8	Specific information for data communications (DIS IFF PDU layer 5) that varies by System Type.
BasicMode5	InteractiveBasicMode5IFFDataStruct	Basic data for an interactive Mode 5 interrogation or corresponding reply.
BasicModeS	InteractiveBasicModeSIFFDataStruct	Basic data for an interactive Mode S interrogation or corresponding reply.

9.2.3. Interrogation

Full Name: HLAinteractionRoot.IFFEmission.Interrogation

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An IFF interrogation.*

Parameters:

Name	Datatype	Semantics
IFFSystemIdentifier RPRnoteDER14 <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	RTObjectId	<i>The identification of the IFFSystem object emitting this IFF electromagnetic signal or composite set of signals.</i>
TransmissionIndicator <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	TransmissionIndicatorEnum8	<i>Indication whether this is an interrogation, a reply to an interrogation, or a squitter transmission.</i>
InterrogatingObjectIdentifier <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	RTObjectId	<i>The object that is interrogating another object.</i>
InterrogatedObjectIdentifier <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	RTObjectId	<i>The object to which the interrogation is being directed.</i>
InteractiveEventIdentifier <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	EventIdentifierStruct	<i>Used to associate a reply to an interrogation.</i> <i>The Event Identifier is set by an interrogator that issues an interactive IFF emission, and the identical number is included in the reply by an interactive transponder to an interrogation. The first number assigned is one, and then is incremented by one for each succeeding interactive interrogation for the same IFF system. A transponder that has a capability to issue an interactive IFF transmission representing of one or more squitter message(s) shall always set the Event Identifier to zero.</i>
InterrogatedModes <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	InterrogatedModesStruct	<i>The modes being interrogated.</i>
ReportingSimulation <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	FederateIdentifierStruct	<i>The simulation publishing this IFF emission.</i>

Name	Datatype	Semantics
DataCommunicationsSpecificInformation <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	IFFLayerSpecificInformationEnum8	<i>Specific information for data communications (DIS IFF PDU layer 5) that varies by System Type.</i>
BasicMode5 <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	InteractiveBasicMode5IFFDataStruct	<i>Basic data for an interactive Mode 5 interrogation or corresponding reply.</i>
BasicModeS <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	InteractiveBasicModeSIFFDataStruct	<i>Basic data for an interactive Mode S interrogation or corresponding reply.</i>

9.2.4. Reply

Full Name: HLAinteractionRoot.IFFEmission.Reply

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *Reply to an IFF interrogation.*

Parameters:

Name	Datatype	Semantics
ReplyAmplification	ReplyAmplificationEnum8	<i>Amplifying information about the reply.</i>
Mode4Reply	InteractiveMode4ReplyIFFDataStruct	<i>Specification of a Mode 4 reply to an Interactive Mode interrogation.</i>
Mode5Reply	InteractiveMode5ReplyIFFDataStruct	<i>Specification of a Mode 5 reply to an Interactive Mode interrogation.</i>
ModeSDAPSource	ModeSDAPSourceStruct	<i>Indicates whether IFF Data records are present in the ModeSDAPData array attribute that can be used as the source for Mode S Downlink Aircraft Parameters (DAPs).</i>

Name	Datatype	Semantics
ModeSDAPData	ModeSDownlinkAircraftParameterVariantStructArray	IFF Data records representing Mode S Downlink Aircraft Parameters (DAPs).
IFFSystemIdentifier RPRnoteDER14 <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	RTObjectId	The identification of the IFFSystem object emitting this IFF electromagnetic signal or composite set of signals.
TransmissionIndicator <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	TransmissionIndicatorEnum8	Indication whether this is an interrogation, a reply to an interrogation, or a squitter transmission.
InterrogatingObjectIdentifier <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	RTObjectId	The object that is interrogating another object.
InterrogatedObjectIdentifier <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	RTObjectId	The object to which the interrogation is being directed.
InteractiveEventIdentifier <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	EventIdentifierStruct	Used to associate a reply to an interrogation. The Event Identifier is set by an interrogator that issues an interactive IFF emission, and the identical number is included in the reply by an interactive transponder to an interrogation. The first number assigned is one, and then is incremented by one for each succeeding interactive interrogation for the same IFF system. A transponder that has a capability to issue an interactive IFF transmission representing of one or more squitter message(s) shall always set the Event Identifier to zero.
InterrogatedModes <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	InterrogatedModesStruct	The modes being interrogated.
ReportingSimulation <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	FederateIdentifierStruct	The simulation publishing this IFF emission.
DataCommunicationsSpecificInformation <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	IFFLayerSpecificInformationEnum8	Specific information for data communications (DIS IFF PDU layer 5) that varies by System Type.

Name	Datatype	Semantics
BasicMode5 <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	InteractiveBasicMode5IFFDataStruct	<i>Basic data for an interactive Mode 5 interrogation or corresponding reply.</i>
BasicModeS <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	InteractiveBasicModeSIFFDataStruct	<i>Basic data for an interactive Mode S interrogation or corresponding reply.</i>

9.2.5. Squitter

Full Name: HLAinteractionRoot.IFFEmission.Squitter

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *An IFF transponder emission transmitting information without receiving an interrogation.*

Parameters:

Name	Datatype	Semantics
ModeSDAPSource	ModeSDAPSourceStruct	<i>Indicates whether IFF Data records are present in the ModeSDAPData array attribute that can be used as the source for Mode S Downlink Aircraft Parameters (DAPs).</i>
ModeSDAPData	ModeSDownlinkAircraftParameterVariantStructArray	<i>IFF Data records representing Mode S Downlink Aircraft Parameters (DAPs).</i>
IFFSystemIdentifier RPRnoteDER14 <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	RTObjectId	<i>The identification of the IFFSystem object emitting this IFF electromagnetic signal or composite set of signals.</i>
TransmissionIndicator <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	TransmissionIndicatorEnum8	<i>Indication whether this is an interrogation, a reply to an interrogation, or a squitter transmission.</i>
InterrogatingObjectIdentifier <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	RTObjectId	<i>The object that is interrogating another object.</i>

Name	Datatype	Semantics
InterrogatedObjectIdentifier <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	RTObjectId	<i>The object to which the interrogation is being directed.</i>
InteractiveEventIdentifier <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	EventIdentifierStruct	<i>Used to associate a reply to an interrogation. The Event Identifier is set by an interrogator that issues an interactive IFF emission, and the identical number is included in the reply by an interactive transponder to an interrogation. The first number assigned is one, and then is incremented by one for each succeeding interactive interrogation for the same IFF system. A transponder that has a capability to issue an interactive IFF transmission representing of one or more squitter message(s) shall always set the Event Identifier to zero.</i>
InterrogatedModes <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	InterrogatedModesStruct	<i>The modes being interrogated.</i>
ReportingSimulation <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	FederateIdentifierStruct	<i>The simulation publishing this IFF emission.</i>
DataCommunicationsSpecificInformation <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	IFFLayerSpecificInformationEnum8	<i>Specific information for data communications (DIS IFF PDU layer 5) that varies by System Type.</i>
BasicMode5 <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	InteractiveBasicMode5IFFDataStruct	<i>Basic data for an interactive Mode 5 interrogation or corresponding reply.</i>
BasicModeS <i>Inherited from IFFEmission in Distributed Emission Regeneration</i>	InteractiveBasicModeSIFFDataStruct	<i>Basic data for an interactive Mode S interrogation or corresponding reply.</i>

9.3. *Datatypes*

9.3.1. *Simple Datatypes*

AngularAccelerationRadianPerSecondSquaredFloat32

Representation: HLAfloat32BE

Units: radian per second squared ($\text{rad}/(\text{s}^2)$)

Resolution: NA

Accuracy: perfect

Semantics: Angular acceleration composed of SI base units.

Enumeration16

Representation: [RPRunsignedInteger16BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Generic representation of a 16-bit enumeration. The enumerators for this enumeration are defined outside of the RPR FOM.

Enumeration32

Representation: [RPRunsignedInteger32BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Generic representation of a 32-bit enumeration. The enumerators for this enumeration are defined outside of the RPR FOM.

Enumeration8

Representation: [RPRunsignedInteger8BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Generic representation of an 8-bit enumeration. The enumerators for this enumeration are defined outside of the RPR FOM.

PowerWattFloat32

Representation: HLAfloat32BE

Units: watt (W)

Resolution: NA

Accuracy: perfect

Semantics: The unit of power is the watt (W), which is equal to one joule per second.

PseudoCode12BitUnsignedInteger16

Representation: [RPRunsignedInteger16BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Pseudo crypto code in the range 0 to 4094 decimal.

RRBCodeUnsignedInteger8

Representation: [RPRunsignedInteger8BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Decimal RRB code in (the least significant) 5 bits. See RRB documentation for the range of decimal values.

TimeHundredthMicrosecond

Representation: [RPRunsignedInteger8BE](#)

Units: one-hundredth of a microsecond

Resolution: NA

Accuracy: NA

Semantics: Time, based on SI base unit second, expressed in one-hundredths of a microsecond.

TransponderAngle180Degree18Bit

Representation: [RPRunsignedInteger32BE](#)

Units: $180/(2^{17})$ degree (deg)

Resolution: 1

Accuracy: perfect

Semantics: Two's complement coded angle in 18 bits (0-17) with the Most Significant Bit (MSB, 17) equal to -180 degrees.

TransponderAngle90Degree18Bit

Representation: [RPRunsignedInteger32BE](#)

Units: $90/(2^{17})$ degree (deg)

Resolution: 1

Accuracy: perfect

Semantics: Two's complement coded angle in 18 bits (0-17) with the Most Significant Bit (MSB, 17) equal to -90 degrees.

TransponderAngleError180DegreeInteger32

Representation: HLAinteger32BE

Units: $180/(2^{31})$ degree (deg)

Resolution: 1

Accuracy: perfect

Semantics: Delta angle represented as two's complement coded latitude with the Most Significant Bit (MSB) equal to -90 degrees.

TransponderAngleError90DegreeInteger32

Representation: HLAinteger32BE

Units: $90/(2^{31})$ degree (deg)

Resolution: 1

Accuracy: perfect

Semantics: Delta angle represented as two's complement coded latitude with the Most Significant Bit (MSB) equal to -90 degrees.

TransponderCode4DigitUnsignedInteger16

Representation: [RPRunsignedInteger16BE](#)

Units: NA

Resolution: 1

Accuracy: perfect

Semantics: Four-digit octal code in the range 0000 to 7777 (base 8). The first (left-most) digit is Code Element 4 (A) and occupies bits 9 to 11, followed by Code Element 3 (B) in bits 6 to 8, Code Element 2 (C) in bits 3 to 5, and the last (right-most) digit is Code Element 1 (D) and occupies bits 0 to 2.

TransponderDeltaAltitude100Foot

Representation: [RPRunsignedInteger16BE](#)

Units: 100-foot increment

Resolution: 1

Accuracy: perfect

Semantics: Binary coded altitude in 11 bits (0-10) in 100-foot increments (range $0-100 \cdot (2^{11})$).

TransponderDeltaAltitude25Foot

Representation: [RPRunsignedInteger16BE](#)

Units: 25-foot increment

Resolution: 1

Accuracy: perfect

Semantics: Binary coded altitude in 11 bits (0-10) in 25-foot increments (range $0-25 \cdot (2^{11})$).

TransponderModeCAltitude100Foot

Representation: [RPRunsignedInteger16BE](#)

Units: 100-foot increment

Resolution: 1

Accuracy: perfect

Semantics: Binary coded Mode C altitude in 11 bits (0-10) in the range 0-126,000 feet in 100-foot increments.

TransponderRelativeAltitude100Foot

Representation: [RPRunsignedInteger16BE](#)

Units: 100-foot increment

Resolution: 1

Accuracy: perfect

Semantics: Binary coded altitude in 11 bits (0-10) in 100-foot increments referenced to -1000 feet.

TransponderRelativeAltitude25Foot

Representation: [RPRunsignedInteger16BE](#)

Units: 25-foot increment

Resolution: 1

Accuracy: perfect

Semantics: Binary coded altitude in 11 bits (0-10) in 25-foot increments referenced to -1000 feet.

Undefined2Bit

Representation: HLAoctet

Units: NA

Resolution: NA

Accuracy: NA

Semantics: Undefined or undisclosed semantics of a 2-bit field.

Undefined6Bit

Representation: HLAoctet

Units: NA

Resolution: NA

Accuracy: NA

Semantics: Undefined or undisclosed semantics of a 6-bit field.

9.3.2. Enumerated Datatypes

FundamentalParameterSystemSpecificDataTypeEnum32

Representation: HLAinteger32BE

Semantics: *System-specific alternatives for fundamental energy radiation characteristics of a transponder or interrogator system emission.*

Enumerator	Value
None	0
InterrogatorSideLobeSuppression	1

IFFMode4CodeActiveEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *IFF Mode 4 active code.*

Enumerator	Value
None	0
CodeA	1
CodeB	2

OperationalStatusEnum8

Representation: [RPRunsignedInteger8BE](#)

Semantics: *IFF system operational status.*

Enumerator	Value
Operational	0
SystemFailed	1

9.3.3. Array Datatypes

AngleDeceptionStructArray

Element [AngleDeceptionStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Array with angle deception data.*

AntennaLocationIFFDataStructArray

Element [AntennaLocationIFFDataStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Array of Antenna Location IFF Data records.*

ASCIIcharArray8

Element HLAASCIIchar
Type:
Cardinality: 8
Encoding: HLAfixedArray
Semantics: *String of up to 8 ASCII characters. Unused characters shall be indicated by Null (0) values.*

BlankingSectorStructArray

Element [BlankingSectorStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Array with blanking sector data.*

FalseTargetsStructArray

Element [FalseTargetsStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Array with false targets data.*

FundamentalParameterDataStructArray

Element [FundamentalParameterDataStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Array of Fundamental Parameter Data records.*

GroundInitiatedCommBIFFDDataStructArray

Element [GroundInitiatedCommBIFFDDataStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Array of IFF Data records related to Mode S transponder Ground-Initiated Comm-B.*

ModeSDownlinkAircraftParameterVariantStructArray

Element [ModeSDownlinkAircraftParameterVariantStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAvariableArray
Semantics: *Array of IFF Data records related to Mode S Downlink Aircraft Parameters.*

9.3.4. Fixed Record Datatypes

AngleDeceptionAngleDataStruct

Encoding: HLAfixedRecord

Semantics: *Data defining the angle deception jamming in one plane.*

Name	Type	Semantic
Offset RPRnoteDER20	AngleRadianFloat32	Indicates the relative angle at which the deceptive radar returns are centered. This angle is measured with respect to the x-y plane of the victim radar's entity coordinate system. This angle is measured in radians from the victim radar entity's azimuth/elevation for the true jammer position to the center of the range of azimuths/elevations in which deceptive radar returns are perceived. For azimuth offset, positive and negative values indicate that the perceived positions of the jammer are right and left of the true position of the jammer, respectively. The range of permissible values is $-\pi$ to π . For elevation offset, positive and negative values indicate that the perceived positions of the jammer are above and below the true position of the jammer, respectively. The range of permissible values is $-\pi/2$ to $\pi/2$.
Width RPRnoteDER20	AngleRadianFloat32	Indicates the range through which the deceptive radar returns are perceived, centered on the Offset. For azimuth width, the range of permissible values is 0 to 2π radians. For elevation width, the range of permissible values is 0 to π radians.
PullRate RPRnoteDER21	AngularVelocityRadianPerSecondFloat32	Indicates the rate (in radians per second) at which the Offset value is changing. For azimuth pull rate, positive and negative values indicate that the azimuth offset is moving to the right or left, respectively. For elevation pull rate, positive and negative values indicate that the elevation offset is moving up or down, respectively.
PullAcceleration RPRnoteDER22	AngularAccelerationRadianPerSecondSquaredFloat32	Indicates the rate (in radians per second squared) at which the PullRate value is changing. Azimuth Pull Acceleration is defined as positive to the right and negative to the left. Elevation Pull Acceleration is defined as positive to upward and negative downward.

AngleDeceptionStruct

Encoding: HLAfixedRecord

Semantics: *Specification of discrete values associated with angle deception jamming that cannot be referenced to an emitter mode. The values provided in the record shall be considered valid only for the victim radar beams listed in the jamming beam's JammedObjectIdentifiers. See IEEE Std 1278.1-2012 Section 6.2.21.2 for more information, including Figure 46 and Figure 47 illustrating the concept of angle deception jamming.*

Name	Type	Semantic
AzimuthAngleData	AngleDeceptionAngleDataStruct	The azimuth relative angle, width, pull rate, and pull acceleration.
ElevationAngleData	AngleDeceptionAngleDataStruct	The elevation relative angle, width, pull rate, and pull acceleration.

AntennaLocationIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *Specification of the location of an auxiliary IFF antenna for a transponder or interrogator.*

Name	Type	Semantic
SystemName	IFFSystemNameEnum16	The name of the system with which this auxiliary IFF antenna is associated.
AntennaReferenceNumber	UnsignedInteger8	The reference number associated with each auxiliary antenna.
AntennaStatus	AntennaStatusEnum8	The status of the auxiliary antenna.
RelativeAntennaLocation	RelativePositionStruct	The location of the auxiliary antenna relative to the associated emitting entity's coordinate system.

BlankingSectorStruct

Encoding: HLAfixedRecord

Semantics: *Specification of a persistent area within a scan volume where emitter power is reduced to an insignificant value. See IEEE Std 1278.1-2012 Section 6.2.21.1 for more information, including Figure 45 illustrating an example.*

Name	Type	Semantic
LeftAzimuth RPRnoteDER20	AngleRadianFloat32	Indicates the left-most azimuth (clockwise in radians) for which emitted power is reduced. This angle is measured in the x-y plane of the radar's entity coordinate system. The range of permissible values is 0 to 2pi, with zero pointing in the x-direction.
RightAzimuth RPRnoteDER20	AngleRadianFloat32	Indicates the right-most azimuth (clockwise in radians) for which emitted power is reduced. This angle is measured in the x-y plane of the radar's entity coordinate system. The range of permissible values is 0 to 2pi, with zero pointing in the x-direction.
LowerElevation RPRnoteDER20	AngleRadianFloat32	Indicates the lowest elevation (in radians) for which emitted power is reduced. This angle is measured positive upward with respect to the x-y plane of the radar's entity coordinate system. The range of permissible values is -pi/2 to pi/2.
UpperElevation RPRnoteDER20	AngleRadianFloat32	Indicates the highest elevation (in radians) for which emitted power is reduced. This angle is measured positive upward with respect to the x-y plane of the radar's entity coordinate system. The range of permissible values is -pi/2 to pi/2.
ResidualPower	PowerRatioDecibelMilliwattFloat32	Specifies the residual effective radiated power in the blanking sector in dBm.

DeltaBarometricAltitudeStruct

Encoding: HLAfixedRecord

Semantics: *The delta error in barometric altitude of the entity.*

Name	Type	Semantic
DeltaAltitude	TransponderDeltaAltitudeVariantStruct	The delta barometric altitude.
PositiveNegativeIndicator	DeltaMode5SAltitudePositiveNegativeIndicatorEnum8	Indicates whether the delta altitude value is Positive (0) or Negative (1). The Positive indicator means the offset is above the entity's present truth altitude and a Negative indicator means the offset is below the entity's present truth altitude.

DeltaMode5SAltitudeStruct

Encoding: HLAfixedRecord

Semantics: *The delta error in Mode 5 / Mode S Altitude of the entity with a Mode 5 or S transponder.*

Name	Type	Semantic
DeltaAltitude	TransponderDeltaAltitudeVariantStruct	The delta Mode 5 / Mode S altitude.
PositiveNegativeIndicator	DeltaMode5SAltitudePositiveNegativeIndicatorEnum8	Indicates whether the delta altitude value is Positive (0) or Negative (1). The Positive indicator means the offset is above the entity's present truth altitude and a Negative indicator means the offset is below the entity's present truth altitude.

FalseTargetsStruct

Encoding: HLAfixedRecord

Semantics: *Specification of discrete values that are associated with false targets jamming that cannot be referenced to an emitter mode. The values provided in the record shall be considered valid only for the victim radar beams listed in the jamming beam's JammedObjectIdentifiers. See IEEE Std 1278.1-2012 Section 6.2.21.3 for more information, including Figure 48 illustrating the effect.*

Name	Type	Semantic
FalseTargetCount	UnsignedInteger16	Specifies the number of false targets presented to the victim radar.
WalkSpeed RPRnoteDER23	VelocityMeterPerSecondFloat32	Specifies the speed (in meters per second) at which false targets move toward the victim radar. Negative values shall indicate a velocity away from the victim radar.
WalkAcceleration RPRnoteDER24	AccelerationMeterPerSecondSquaredFloat32	Specifies the rate (in meters per second squared) at which false targets accelerate toward the victim radar. Negative values shall indicate an acceleration direction away from the victim radar.
MaximumWalkDistance RPRnoteDER25	MeterFloat32	Specifies the distance (in meters) that a false target is to walk before it pauses in range.
KeepTime RPRnoteDER26	TimeSecondFloat32	Specifies the simulation time (in seconds) that a false target is to be held at the MaximumWalkDistance before it resets to its initial position.
EchoSpacing RPRnoteDER27	MeterFloat32	Specifies the distance between false targets in meters. A positive value for this field shall indicate that the second and subsequent false targets appear at the given spacing relative to the first false target away from the victim radar (i.e., at increasing ranges from the victim). A negative value shall indicate that the second and subsequent false targets appear toward the victim radar relative to the first false target.

Name	Type	Semantic
FirstTargetOffset	MeterFloat32	Sets the position of the first false target relative to the jamming entity in meters. A positive value for this field shall indicate that the first target appears at the given distance relative to the jamming entity away from the victim radar (i.e., at an increasing range from the victim). A negative value shall indicate the first false target appears toward the victim radar relative to the jamming entity. A value of zero shall place the first false target at the same position as the jamming entity.

FundamentalParameterDataStruct

Encoding: HLAfixedRecord

Semantics: *Fundamental Parameter Data record.*

Name	Type	Semantic
EIRP RPRnoteDER28	PowerRatioDecibelMilliwattFloat32	The average peak radiated power of the emission.
Frequency RPRnoteDER29	FrequencyHertzFloat32	The center frequency of the emission.
PgRF	InterrogationsPerSecondFloat32	When applied to originators, this field shall specify the number of interrogations per second emitted. This field shall be set to zero when applied to a responder (i.e. transponder) systems.
PulseWidth	TimeMicrosecondFloat32	The duration of the fundamental pulse of which the interrogation or reply is composed.
BurstLength	UnsignedInteger32	The number of emissions generated in a single burst. This field shall contain zero for continuously emitting systems and shall contain the value 1 for responders.
ApplicableModes	IFFApplicableModesEnum8	Specifies the modes to which the fundamental parameter data applies.
SystemSpecificData	FundamentalParameterSystemSpecificVariantStruct	System specific data associated with the system type.

GroundInitiatedCommBIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *Information contained in a Transponder/Interrogator (T/I) data link message in Ground-Initiated Comm-B (GICB) format.*

Name	Type	Semantic
TILinkType	TILinkTypeEnum16	The T/I link type.

Name	Type	Semantic
MessageDesignator	Enumeration16	The GICB message designator as defined in appropriate ICAO technical publications, represented by a 16-bit enumeration.
RegisterMessageNumber	UnsignedInteger8	The applicable register and message number for this message.
ProcessStatus	Enumeration8	GICB Process Status.
RegisterDataValidity	Enumeration8	GICB Register Data Validity.
Register	Enumeration32	GICB register data.
Message	OctetArray7	56-bit GICB message data.

InteractiveBasicMode5IFFDataStruct

Encoding: HLAfixedRecord

Semantics: Basic data for an interactive Mode 5 interrogation or corresponding reply.

Name	Type	Semantic
MessageFormats	Mode5MessageFormatsStruct	For a Mode 5 interrogator, this field contains the specific Message Formats being requested. These message formats may be different from those contained in the regeneration IFF object for this interrogator. For a Mode 5 transponder, this field contains the specific Message Formats that it actually responded to for this interrogation. This may be different from the Message Formats contained in the interrogation request, as well as the present Message Formats contained in the regeneration IFF object for this transponder.

InteractiveBasicModeSIFFDDataStruct

Encoding: HLAfixedRecord

Semantics: Basic data for an interactive Mode S interrogation or corresponding reply.

Name	Type	Semantic
Levels	ModeSLevelsPresentStruct	For a Mode S interrogator, this field indicates the Mode S levels it is interrogating. For a Mode S transponder, this field indicates the actual levels contained in this reply. The value of this field may or may not be the same as the present value for the ModeSLevelsPresent attribute of the IFF object.

Name	Type	Semantic
TransmitState	ModeSInterrogatorStatusTransmitStateEnum8	The Mode S Transmit State that would be contained in this specific Mode S interactive interrogation. The value of this field may or may not be the same as the present value for the ModeSTransmitState attribute of the IFF object. Not applicable to a transponder and set to zero.
InterrogatorIdentifier	ModeSInterrogatorIdentifierStruct	The Mode S Interrogator Identifier that would be contained in this specific Mode S interactive interrogation. The values of this record may or may not be the same as the present value for the ModeSInterrogatorIdentifier attribute of the IFF object. Not applicable to a transponder and set to zero.

InteractiveMode4ReplyIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *Specification of an interactive Mode 4 reply to an Interactive Mode 4 interrogation.*

Name	Type	Semantic
Reply	IFFAlternateMode4ChallengeReplyEnum8	The Mode 4 reply that would be sent by this transponder to this Interactive Mode 4 interrogation. This may not be the same as the regeneration value.

InteractiveMode5ReplyIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *The specification of an Interactive Mode 5 reply to an Interactive Mode 5 interrogation.*

Name	Type	Semantic
Reply	Mode5ReplyEnum8	The Mode 5 reply that would be sent by this IFF system to this Interactive Mode 5 interrogation.

InterrogatedModesStruct

Encoding: HLAfixedRecord

Semantics: *The specific modes and other basic data types related to an interactive interrogation, reply, or squitter transmission.*

Name	Type	Semantic
Mode1Included	RPRboolean	Set to True when Mode 1 is being interrogated or contained in a transponder reply.

Name	Type	Semantic
Mode2Included	RPRboolean	Set to True when Mode 2 is being interrogated or contained in a transponder reply.
Mode3AIncluded	RPRboolean	Set to True when Mode 3/A is being interrogated or contained in a transponder reply.
Mode4Included	RPRboolean	Set to True when Mode 4 is being interrogated or contained in a transponder reply.
Mode5Included	RPRboolean	Set to True when Mode 5 is being interrogated or contained in a transponder reply.
ModeSIncluded	RPRboolean	Set to True when Mode S is being interrogated or contained in a transponder reply.
ModeCIncluded	RPRboolean	Set to True when Mode C Altitude is being interrogated or contained in a transponder reply.
TCASIncluded	RPRboolean	Set to True when TCAS/ACAS is being interrogated or contained in a transponder reply.
SpecialIPIncluded	RPRboolean	Set to True when this is a Special Identification of Position (IP) transmission.
SquitterIncluded	RPRboolean	Set to True when this is an interactive squitter transmission.

InterrogatorSideLobeSuppressionStruct

Encoding: HLAfixedRecord

Semantics: *Interrogator Side Lobe Suppression (ISLS) inhibits transponder replies to all challenges not radiated in the main beam of the interrogating antenna.*

Name	Type	Semantic
AntennaType	ISLSAntennaTypeEnum8	The type of antenna used by the system.
PulseRiseTime	TimeHundredthMicrosecond	The real-world time required for the leading edge of pulses P1, P2, and P3 to rise from 10 to 90 percent of its maximum voltage amplitude. Pulse Rise Time shall be expressed in one-hundredths of a microsecond where 1 is 0.01 microsecond and 99 is 0.99 microsecond.
PulseFallTime	TimeHundredthMicrosecond	The real-world time required for the leading edge of pulses P1, P2, and P3 to fall from 10 to 90 percent of its maximum voltage amplitude. Pulse Fall Time shall be expressed in one-hundredths of a microsecond where 1 is 0.01 microsecond and 99 is 0.99 microsecond.

JammingTechniqueStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the kind, category, subcategory, and specific indication of a jamming technique.*

Name	Type	Semantic
Kind	UnsignedInteger8	Indicates the broadest category of jamming.
Category	UnsignedInteger8	Permits more detailed definition of techniques defined by the Kind.
Subcategory	UnsignedInteger8	Permits more detailed definition of techniques defined by the Kind and Category.
Specific	UnsignedInteger8	Provides a means for high-fidelity electronic warfare emitters to indicate the most detailed definition of techniques.

Mode5MessageFormatsStruct

Encoding: HLAfixedRecord

Semantics: *The message formats associated with an IFF Mode 5 transponder or interrogator.*
When used for a transponder object in the Regeneration mode, it indicates the message formats that the transponder is capable of supporting and could be in a reply to a Mode 5 interrogation.
When used for a transponder in the Interactive mode, in a Reply or Squitter interaction, it indicates the specific message formats that it actually responded to for the Mode 5 interrogation.
When used for an interrogator object in the Regeneration mode, it indicates the message formats that the interrogator is capable of supporting and could be sent in a Mode 5 interrogation.
When used for an interrogator in the Interactive mode, in an Interrogation interaction, it indicates the specific Mode 5 message formats being requested.

Name	Type	Semantic
MessageFormat0Included	RPRboolean	Set to True when a message format 0 capability exists or is included in an interactive interrogation or reply.
MessageFormat1Included	RPRboolean	Set to True when a message format 1 capability exists or is included in an interactive interrogation or reply.
MessageFormat2Included	RPRboolean	Set to True when a message format 2 capability exists or is included in an interactive interrogation or reply.
MessageFormat3Included	RPRboolean	Set to True when a message format 3 capability exists or is included in an interactive interrogation or reply.
MessageFormat4Included	RPRboolean	Set to True when a message format 4 capability exists or is included in an interactive interrogation or reply.

Name	Type	Semantic
MessageFormat5Included	RPRboolean	Set to True when a message format 5 capability exists or is included in an interactive interrogation or reply.
MessageFormat6Included	RPRboolean	Set to True when a message format 6 capability exists or is included in an interactive interrogation or reply.
MessageFormat7Included	RPRboolean	Set to True when a message format 7 capability exists or is included in an interactive interrogation or reply.
MessageFormat8Included	RPRboolean	Set to True when a message format 8 capability exists or is included in an interactive interrogation or reply.
MessageFormat9Included	RPRboolean	Set to True when a message format 9 capability exists or is included in an interactive interrogation or reply.
MessageFormat10Included	RPRboolean	Set to True when a message format 10 capability exists or is included in an interactive interrogation or reply.
MessageFormat11Included	RPRboolean	Set to True when a message format 11 capability exists or is included in an interactive interrogation or reply.
MessageFormat12Included	RPRboolean	Set to True when a message format 12 capability exists or is included in an interactive interrogation or reply.
MessageFormat13Included	RPRboolean	Set to True when a message format 13 capability exists or is included in an interactive interrogation or reply.
MessageFormat14Included	RPRboolean	Set to True when a message format 14 capability exists or is included in an interactive interrogation or reply.
MessageFormat15Included	RPRboolean	Set to True when a message format 15 capability exists or is included in an interactive interrogation or reply.
MessageFormat16Included	RPRboolean	Set to True when a message format 16 capability exists or is included in an interactive interrogation or reply.
MessageFormat17Included	RPRboolean	Set to True when a message format 17 capability exists or is included in an interactive interrogation or reply.
MessageFormat18Included	RPRboolean	Set to True when a message format 18 capability exists or is included in an interactive interrogation or reply.
MessageFormat19Included	RPRboolean	Set to True when a message format 19 capability exists or is included in an interactive interrogation or reply.
MessageFormat20Included	RPRboolean	Set to True when a message format 20 capability exists or is included in an interactive interrogation or reply.
MessageFormat21Included	RPRboolean	Set to True when a message format 21 capability exists or is included in an interactive interrogation or reply.
MessageFormat22Included	RPRboolean	Set to True when a message format 22 capability exists or is included in an interactive interrogation or reply.

Name	Type	Semantic
MessageFormat23Included	RPRboolean	Set to True when a message format 23 capability exists or is included in an interactive interrogation or reply.
MessageFormat24Included	RPRboolean	Set to True when a message format 24 capability exists or is included in an interactive interrogation or reply.
MessageFormat25Included	RPRboolean	Set to True when a message format 25 capability exists or is included in an interactive interrogation or reply.
MessageFormat26Included	RPRboolean	Set to True when a message format 26 capability exists or is included in an interactive interrogation or reply.
MessageFormat27Included	RPRboolean	Set to True when a message format 27 capability exists or is included in an interactive interrogation or reply.
MessageFormat28Included	RPRboolean	Set to True when a message format 28 capability exists or is included in an interactive interrogation or reply.
MessageFormat29Included	RPRboolean	Set to True when a message format 29 capability exists or is included in an interactive interrogation or reply.
MessageFormat30Included	RPRboolean	Set to True when a message format 30 capability exists or is included in an interactive interrogation or reply.
MessageFormat31Included	RPRboolean	Set to True when a message format 31 capability exists or is included in an interactive interrogation or reply.

Mode5STransponderLocationErrorIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *The specification of errors in location and altitude information transmitted by a Mode 5 or Mode S transponder. Errors are defined as delta values from the truth location of the entity platform. They reflect either human or navigation system errors that might be present and manifest themselves in the data sent in a Mode 5 or Mode S reply or squitter report. If Mode 5 or S location errors are modeled, then this record shall be issued to convey the magnitude of the error for one or more of the location values that could have an error. Fields with no errors shall have all bits of that field set to zero.*

Name	Type	Semantic
DeltaLatitude	TransponderAngleError90DegreeInteger32	The delta error in latitude from the true latitude computed for the entity with a Mode 5 or S transponder. Delta latitude shall be represented as two's complement coded latitude with the Most Significant Bit (MSB) equal to -90 degrees. All bits of this field shall be set to zero if there is no error in latitude.

Name	Type	Semantic
DeltaLongitude	TransponderAngleError180DegreeInteger32	The delta error in longitude from the true longitude computed for the entity. Delta longitude shall be represented as two's complement coded longitude with the Most Significant Bit (MSB) equal to -180 degrees. All bits of this field shall be set to zero if there is no error in longitude.
DeltaMode5SAltitude	DeltaMode5SAltitudeStruct	The delta error in Mode 5 / Mode S Altitude of the entity with a Mode 5 or S transponder. All bits of this field shall be set to zero if there is no error in Mode 5 / Mode S altitude.
DeltaBarometricAltitude	DeltaBarometricAltitudeStruct	The delta error in barometric altitude of the entity. All bits of this field shall be set to zero if there is no error in barometric altitude.

Mode5STransponderLocationIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *Real-time location and altitude information transmitted by a Mode 5 or Mode S transponder.*

Name	Type	Semantic
Latitude	TransponderAngle90Degree18Bit	The latitude of the entity, represented as an 18-bit two's complement coded angle with the Most Significant Bit (MSB) equal to -90 degrees.
Longitude	TransponderAngle180Degree18Bit	The longitude of the entity, represented as an 18-bit two's complement coded angle with the Most Significant Bit (MSB) equal to -180 degrees.
Mode5SAltitude	TransponderRelativeAltitudeVariantStruct	The Mode 5 / Mode S Altitude of the entity.
BarometricAltitude	TransponderRelativeAltitudeVariantStruct	The Barometric Altitude of the entity.

ModeCAltitudeStruct

Encoding: HLAfixedRecord

Semantics: *The actual Mode C altitude.*

Name	Type	Semantic
PositiveNegativeIndicator	ModeCAltitudeIndicatorEnum8	Indicates either a Positive Altitude Above MSL (0) or a Negative Altitude Below MSL (1).

Name	Type	Semantic
ModeCAltitude	TransponderModeCAltitude100Foot	The actual Mode C altitude, in the range 0 feet to 126 000 feet in 100-foot increments based on the PositiveNegativeIndicator field. A value of 2047 indicates to use Alternate Mode C Altitude (i.e., derived from entity truth data).

ModeSDAPSourceStruct

Encoding: HLAfixedRecord

Semantics: Mode S source of Downlink Aircraft Parameters (DAPs).

Name	Type	Semantic
IndicatedAirSpeedDataIsPresent RPRnoteDER30	RPRboolean	Indicates whether there is an IFF Data record either Present (1) or Not Present (0) in the IFFData array attribute that can be used as the source for the Mode S Downlink Aircraft Parameter (DAP) for the Indicated Air Speed (IAS).
MachNumberDataIsPresent RPRnoteDER30	RPRboolean	Indicates whether there is an IFF Data record either Present (1) or Not Present (0) in the IFFData array attribute that can be used as the source for the Mode S Downlink Aircraft Parameter (DAP) for the Mach Number.
GroundSpeedDataIsPresent RPRnoteDER30	RPRboolean	Indicates whether there is an IFF Data record either Present (1) or Not Present (0) in the IFFData array attribute that can be used as the source for the Mode S Downlink Aircraft Parameter (DAP) for the Ground Speed.
MagneticHeadingDataIsPresent RPRnoteDER30	RPRboolean	Indicates whether there is an IFF Data record either Present (1) or Not Present (0) in the IFFData array attribute that can be used as the source for the Mode S Downlink Aircraft Parameter (DAP) for the Magnetic Heading.
TrackAngleRateDataIsPresent RPRnoteDER30	RPRboolean	Indicates whether there is an IFF Data record either Present (1) or Not Present (0) in the IFFData array attribute that can be used as the source for the Mode S Downlink Aircraft Parameter (DAP) for the Track Angle Rate.
TrueTrackAngleDataIsPresent RPRnoteDER30	RPRboolean	Indicates whether there is an IFF Data record either Present (1) or Not Present (0) in the IFFData array attribute that can be used as the source for the Mode S Downlink Aircraft Parameter (DAP) for the True Track Angle.

Name	Type	Semantic
TrueAirspeedDataIsPresent RPRnoteDER30	RPRboolean	Indicates whether there is an IFF Data record either Present (1) or Not Present (0) in the IFFData array attribute that can be used as the source for the Mode S Downlink Aircraft Parameter (DAP) for the True Airspeed.
VerticalRateDataIsPresent RPRnoteDER30	RPRboolean	Indicates whether there is an IFF Data record either Present (1) or Not Present (0) in the IFFData array attribute that can be used as the source for the Mode S Downlink Aircraft Parameter (DAP) for the Vertical Rate.

ModeSInterrogatorIdentifierStruct

Encoding: HLAfixedRecord

Semantics: *Mode S Interrogator Code (IC) information.*

Name	Type	Semantic
PrimaryICCode	ModeSInterrogatorCodeVariantStruct	Primary IC Code.
SecondaryICCode	ModeSInterrogatorCodeVariantStruct	Secondary IC Code.

ModeSLevelsPresentStruct

Encoding: HLAfixedRecord

Semantics: *The Mode S levels the transponder or interrogator is able to support.*

Name	Type	Semantic
Level1IsPresent	RPRboolean	Set to True when Mode S level 1 is present.
Level2ElementarySurveillanceIsPresent	RPRboolean	Set to True when Mode S level 2 (Elementary Surveillance sublevel) is present.
Level2EnhancedSurveillanceIsPresent	RPRboolean	Set to True when Mode S level 2 (Enhanced Surveillance sublevel) is present.
Level3IsPresent	RPRboolean	Set to True when Mode S level 3 is present.
Level4IsPresent	RPRboolean	Set to True when Mode S level 4 is present.

SquitterAirbornePositionReportIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *This replicates the functional data contained in a Squitter Airborne Position Report transmitted by a Mode S transponder in accordance with ICAO standards.
If it is desired to transmit an exact replication of this squitter report, it may be sent using the GroundInitiatedCommBIFFDDataStruct.*

Name	Type	Semantic
AircraftAddressSource	AircraftAddressSourceEnum8	The source of the Aircraft Address.
FormatType	FormatTypeEnum8	The Format Type.
ICAOSpare	Undefined2Bit	Represent the spare field that exists in a Squitter Airborne Position Report.
SurveillanceStatus	SurveillanceStatusEnum8	Indicates the surveillance status associated with this report.
TurnRateSource	TurnRateSourceEnum8	Indicates whether the aircraft is turning or whether the receiving simulation is to make that determination by examining entity truth data.
AltitudeSource	IFFDataRecordAvailableEnum8	Indicates whether the altitude is to be computed locally or is provided in a Mode5STransponderLocationIFFDataStruct.
TimeTypeSource	TimeTypeSourceEnum8	Indicates whether this squitter report was output on an even or odd second or whether the receiving simulation is to determine that based on its Mode S interrogator model.
LatLongSource	IFFDataRecordAvailableEnum8	Indicates whether the lat/long is to be computed locally or is provided in a Mode5STransponderLocationIFFDataStruct.

SquitterAirborneVelocityReportIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *This replicates the functional data contained in a Squitter Airborne Velocity Report transmitted by a Mode S transponder in accordance with ICAO standards.
If it is desired to transmit an exact replication of this squitter report, it may be sent using the GroundInitiatedCommBIFFDDataStruct.*

Name	Type	Semantic
AircraftAddressSource	AircraftAddressSourceEnum8	The source of the Aircraft Address.
FormatType	FormatTypeEnum8	The Format Type.
Subtype	Enumeration8	The subtype associated with this report. (See the appropriate ICAO publication for the subtype values.)

Name	Type	Semantic
IsSynced	RPRboolean	Indicates the sync status as being either Synced (True) or Not Synced (False).
GroundTrackSource	IFFDataRecordAvailableEnum8	Indicates whether the ground track is to be computed locally from truth data or is provided in a GroundInitiatedCommBIFFDDataStruct.
TRNRateSource	IFFDataRecordAvailableEnum8	Indicates whether the Turn (TRN) Rate is to be computed locally from truth data or is provided in a GroundInitiatedCommBIFFDDataStruct.
VRTRateSource	IFFDataRecordAvailableEnum8	Indicates whether the Vertical Rate of Climb (VRT) Rate is to be computed locally from truth data or is provided in a GroundInitiatedCommBIFFDDataStruct.

SquitterEventDrivenReportIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *This replicates the functional data contained in a Squitter Event-Driven Report transmitted by a Mode S transponder in accordance with ICAO standards.
If it is desired to transmit an exact replication of this squitter report, it may be sent using the GroundInitiatedCommBIFFDDataStruct.*

Name	Type	Semantic
SquitterEventIdentification	Enumeration16	The ICAO Downlink Format (DF) number assigned to this type of event data.
SquitterEventData	OctetArray1Plus	The specific event data being sent.

SquitterIdentificationReportIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *This replicates the functional data contained in a Squitter Identification Report transmitted by a Mode S transponder in accordance with ICAO standards.
If it is desired to transmit an exact replication of this squitter report, it may be sent using the GroundInitiatedCommBIFFDDataStruct.*

Name	Type	Semantic
AircraftAddressSource	AircraftAddressSourceEnum8	The source of the Aircraft Address.
FormatType	FormatTypeEnum8	The Format Type.
AircraftTypeWake	AircraftTypeWakeEnum8	Aircraft type and wake data.
AircraftIDSource	AircraftIDSourceEnum8	The data source for the Aircraft ID.

Name	Type	Semantic
ICAOSpare	Undefined6Bit	<i>Represent the spare field that exists in a Squitter Identification Report.</i>

SquitterSurfacePositionReportIFFDataStruct

Encoding: HLAfixedRecord

Semantics: *This replicates the functional data contained in a Squitter Surface Position Report transmitted by a Mode S transponder in accordance with ICAO standards. This report is transmitted when an aircraft is on the ground or aboard ship.
If it is desired to transmit an exact replication of this squitter report, it may be sent using the GroundInitiatedCommBIFFDDataStruct.*

Name	Type	Semantic
AircraftAddressSource	AircraftAddressSourceEnum8	<i>The source of the Aircraft Address.</i>
FormatType	FormatTypeEnum8	<i>The Format Type.</i>
Movement RPRnoteDER31	Enumeration8	<i>Movement data.</i>
HeadingSource	IFFDataRecordAvailableEnum8	<i>Indicates whether the heading is to be computed locally from truth data or is provided in a GroundInitiatedCommBIFFDDataStruct.</i>
VRTRateSource	IFFDataRecordAvailableEnum8	<i>Indicates whether the Vertical Rate of Climb (VRT) Rate is to be computed locally from truth data or is provided in a GroundInitiatedCommBIFFDDataStruct.</i>
ICAOSpare	OctetArray3	<i>Represents the 24-bit ICAO Spare field contained in a Squitter Surface Position Report.</i>

9.3.5. Variant Record Datatypes

FundamentalParameterSystemSpecificVariantStruct

Encoding: RPRextendedVariantRecord
 Discriminant name: FundamentalParameterSystemSpecificDataType
 Discriminant type: [FundamentalParameterSystemSpecificDataTypeEnum32](#)
 Semantics: *System-specific alternatives for fundamental energy radiation characteristics of a transponder or interrogator system emission.*

Name	Enumerator	Type	Semantics
None	None	RPRnoData	<i>No fundamental energy radiation characteristics of a transponder or interrogator system. See IEEE Std 1278.1-2012 Section 6.2.44.</i>
InterrogatorSideLobeSuppression	InterrogatorSideLobeSuppression	InterrogatorSideLobeSuppressionStruct	<i>Interrogator Side Lobe Suppression (ISLS) inhibits transponder replies to all challenges not radiated in the main beam of the interrogating antenna.</i>

ModeSDownlinkAircraftParameterVariantStruct [RPRnoteDER32](#)

Encoding: RPRextendedVariantRecord
 Discriminant name: RecordType
 Discriminant type: [VariableRecordTypesEnum32](#)
 Semantics: *For optional Mode S transponder Downlink Aircraft Parameter data.*

ModeSInterrogatorCodeVariantStruct

Encoding: RPRextendedVariantRecord
 Discriminant name: InterrogatorCodeType
 Discriminant type: [ModeSInterrogatorIdentifierICTypeEnum8](#)
 Semantics: *Mode S Interrogator Code (IC) information.*

Name	Enumerator	Type	Semantics
InterrogatorIdentifierCode	II	UnsignedInteger8	<i>Interrogator Identifier (II) Code: 0 to 16 decimal.</i>
SurveillanceIdentifierCode	SI	UnsignedInteger8	<i>Surveillance Identifier (SI) Code: 0 to 64 decimal.</i>

TransponderDeltaAltitudeVariantStruct

Encoding: RPRextendedVariantRecord

Discriminant
name: Resolution

Discriminant type: [Mode5SAltitudeResolutionEnum8](#)

Semantics: *Delta to the transponder reported altitude represented as a 11-bit binary coded altitude.*

Name	Enumerator	Type	Semantics
DeltaAltitude100Foot	_100_foot	TransponderDeltaAltitude100Foot	<i>Delta altitude in 100-foot increments.</i>
DeltaAltitude25Foot	_25_foot	TransponderDeltaAltitude25Foot	<i>Delta altitude in 25-foot increments.</i>

TransponderRelativeAltitudeVariantStruct

Encoding: RPRextendedVariantRecord

Discriminant
name: Resolution

Discriminant type: [Mode5SAltitudeResolutionEnum8](#)

Semantics: *Transponder reported altitude represented as a 11-bit binary coded altitude referenced to -1000 feet. All altitudes greater than 50175 feet shall be reported with 100-foot resolution. When no altitude is reported, the alternative value shall be set to 2047 (decimal) to indicate to use entity truth altitude to derive the transponder reported altitude.*

Name	Enumerator	Type	Semantics
RelativeAltitude100Foot	_100_foot	TransponderRelativeAltitude100Foot	<i>Relative altitude in 100-foot increments.</i>
RelativeAltitude25Foot	_25_foot	TransponderRelativeAltitude25Foot	<i>Relative altitude in 25-foot increments.</i>

9.4. Notes

RPRnoteDER1

Semantics: *If there is no object instance associated with the attribute, then this should be set to the empty string (no characters). Refer to SISO-STD-001 Section 7.8.6 for handling empty strings.*

RPRnoteDER2

Semantics: *The update condition shall be evaluated as follows:
The owner of a designator object shall maintain two state models of the object in support of the dead reckoning process. One model shall be the internal model used by the simulation application to represent that object. The other shall be a dead reckoning model of the object. Certain thresholds shall be established as criteria for determining if the object's actual spot location has varied by an allowable amount from the dead reckoned spot location. The attribute is updated when the object's actual spot location differs from the dead reckoned spot location by more than DRA_POS_THRSH (default value: 1 m).
See IEEE Std 1278.1-2012 Sections 4.6.6 and 5.3.2.3 and Annex E for more details.*

A TSPI update includes all attributes applicable to the dead reckoning algorithm. A TSPI update attribute may be omitted if its last updated value is within an established epsilon amount from its current value.

RPRnoteDER3

Semantics: *The epsilon value for the location is DRA_POS_EPSILSON (default value: 0.001 m), in any direction.*

RPRnoteDER4

Semantics: *The epsilon value for the linear acceleration is DRA_ACCEL_EPSILSON (default value: 0.001 m/(s²)), in any direction.*

RPRnoteDER5

Semantics: *The EventIdentifier attribute is used to link together updates arising from a common event. For example an emitter function change can be linked to a frequency change in one of its associated emitter beams, by using the same event ID in the attribute updates to the appropriate EmitterSystem and EmitterBeam classes. It is important to note that in order that the EventIdentifier is delivered in the same reflect attribute values call in the receiving federates as the associated attributes, then all the attributes, including the EventIdentifier, must share the same transport type.*

RPRnoteDER6

Semantics: *The federate should change the value of the EventIdentifier when associating changes between two or more emitter beams and/or emitter systems. It is not necessary to update the EventIdentifier field with every attribute update if the change is not associated with changes in other instances of other classes.*

RPRnoteDER7

Semantics: *If emissions data is available (determined by the state of the EmissionsDataAvailable attribute) then this attribute shall be updated by the federate. If emissions data is not available then this attribute shall not be updated by the federate.*

RPRnoteDER8

Semantics: *This attribute is updated if the current value differs from the previously updated value by more than the value specified by the symbolic name (default value: 3 degrees).*

RPRnoteDER9

Semantics: *This attribute is updated if the current value differs from the previously updated value by more than the value specified by the symbolic name (default value: 3 degrees).*

RPRnoteDER10

Semantics: *A change in sweep sync value alone does not require an update to be published. When one of the other beam attributes (center and sweep for azimuth and elevation) requires an update publication, the sweep sync attribute shall be included when its value has changed since the last update.*

RPRnoteDER11

Semantics: *This attribute is updated if the current value differs from the previously updated value by more than the value specified by the symbolic name (default value: 1 degree).*

RPRnoteDER12

Semantics: *This attribute is updated if the current value differs from the previously updated value by more than the value specified by the symbolic name (default value: 1 degree).*

RPRnoteDER13

Semantics: *The term "ERP" is widely used throughout the radar community (including radar databases), but in radar contexts, the intended meaning is actually that of EIRP.*

RPRnoteDER14

Semantics: *This must reference a valid Object instance.*

RPRnoteDER15

Semantics: *The object IDs in this structure shall refer to PhysicalEntity objects (or subclasses of PhysicalEntity).*

RPRnoteDER16

Semantics: *Special update conditions may apply to the datatypes or their fields used for the array elements. Refer to the datatype or field semantics, including any referenced notes, for more details.*

RPRnoteDER17

Semantics: *The object IDs in this structure shall refer to EmitterBeam objects (or subclasses of EmitterBeam).*

RPRnoteDER18

Semantics: *If the IFF system is Mode 5 capable then this attribute shall be updated by the federate. If the IFF system is not Mode 5 capable then this attribute shall either not be published or set to False.*

RPRnoteDER19

Semantics: *The RRB system is a transponder only system.*

RPRnoteDER20

Semantics: *Upon a value change, an update is only required to be published immediately when the change exceeds a threshold value. For azimuth related data, the threshold is defined by the symbolic name EE_AZ_THRSH (default value: 1 degree). For elevation related data, the threshold is defined by the symbolic name EE_EL_THRSH (default value: 1 degree).*

RPRnoteDER21

Semantics: *Upon a value change, an update is only required to be published immediately when the change exceeds a threshold value. The threshold is defined by the symbolic name EE_AD_PULRAT_THRSH (default value: 0.017 rad/s).*

RPRnoteDER22

Semantics: *Upon a value change, an update is only required to be published immediately when the change exceeds a threshold value. The threshold is defined by the symbolic name EE_AD_PULACC_THRSH (default value: 0.017 rad/(s²)).*

RPRnoteDER23

Semantics: *Upon a value change, an update is only required to be published immediately when the change exceeds a threshold value. The threshold is defined by the symbolic name EE_FT_VEL_THRSH (default value: 1.0 m/s).*

RPRnoteDER24

Semantics: *Upon a value change, an update is only required to be published immediately when the change exceeds a threshold value. The threshold is defined by the symbolic name EE_FT_ACC_THRSH (default value: 1.0 m/(s²)).*

RPRnoteDER25

Semantics: *Upon a value change, an update is only required to be published immediately when the change exceeds a threshold value. The threshold is defined by the symbolic name EE_FT_MWD_THRSH (default value: 10000 m).*

RPRnoteDER26

Semantics: *Upon a value change, an update is only required to be published immediately when the change exceeds a threshold value. The threshold is defined by the symbolic name EE_FT_KT_THRSH (default value: 10 s).*

RPRnoteDER27

Semantics: *Upon a value change, an update is only required to be published immediately when the change exceeds a threshold value. The threshold is defined by the symbolic name EE_FT_ESP_THRSH (default value: 10 m).*

RPRnoteDER28

Semantics: *For radar-like IFF interrogators, this may still be known as 'ERP' because EIRP is generally known throughout the radar community as ERP. The field is named "ERP" in IEEE Std 1278.1-2012, but it is believed that EIRP was the intended meaning.*

RPRnoteDER29

Semantics: *Frequency modulation shall be derived from database parameters stored for the particular emitting system and specified mode.*

RPRnoteDER30

Semantics: *IFF Data records have not been defined for specific DAP data. When such records are developed, it is recommended that the DAP field format from the appropriate ICAO publication be used.*

RPRnoteDER31

Semantics: *The size of this field in IEEE Std 1278.1-2012 Section B.2.51 is assumed to be incorrect. According to ICAO documentation, this field in the extended squitter surface position BDS register is 7 bits, not 3. RPR FOM users should be careful in using the full range as per the ICAO specifications. For in heterogeneous exercises with DIS-based simulations this may lead to unexpected results. It is advised to explicitly note the allowed values and the mapping to be applied in gateways in federation/exercise agreements (when using Mode S squitter surface position reports).*

RPRnoteDER32

Semantics: *At the time of publication of the Standard for RPR FOM 3.0, no IFF Data records have been defined for specific DAP data.*

10. Module Underwater Acoustics



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Underwater Acoustics FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	Definitions of the objects and interactions required to express acoustic signatures of surface and subsurface vessels for passive sonar sensors. Includes sounds produced as byproducts of propulsion and other ship board systems and activities as well as sounds produced by acoustic transducers for active sonar sensing and depth measurement.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

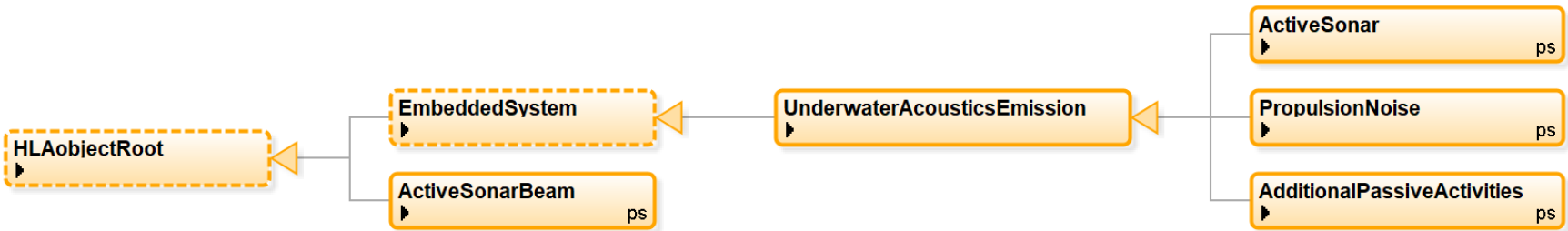
References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Base
Foundation
Enumerations

10.1. Object Classes



10.1.1. UnderwaterAcousticsEmission

Full Name: HLAobjectRoot.EmbeddedSystem.UnderwaterAcousticsEmission

Sharing:

Semantics: *The underwater acoustic classes used to communicate underwater acoustic active, intentional emissions and Passive Signature or Unintentional Emissions information. These emissions are used during undersea warfare scenarios to detect, classify, and track hostile forces when electronic warfare mechanisms are unavailable.*

Attributes:

EventIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The generating federate uses the Event Identifier to associate related events. The event number begins at one at the beginning of the exercise and is incremented by one for each event.					
EntityIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The EntityIdentifier of the object which this embedded system is a part of.					

HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTLObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	The RTI object instance ID of the object of which this embedded system is part of.				
RelativePosition RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics	The position of the embedded system, relative to the host object's position.				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

10.1.2. ActiveSonar

Full Name: HLAobjectRoot.EmbeddedSystem.UnderwaterAcousticsEmission.ActiveSonar

Sharing: Publish/Subscribe

Semantics: *Describes the state of an active sonar system.*

Attributes:

AcousticName	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UAAcousticSystemNameEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	Defines the type of sonar being represented.				

FunctionCode	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UAAcousticEmitterSystemFunctionEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Declares the current primary use of the sonar. It may be used by simulations to infer intent.</i>				
AcousticsIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>Unique identifier for the sonar on an entity, starts with 1.</i>				
EventIdentifier <i>Inherited from UnderwaterAcousticsEmission in Underwater Acoustics</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The generating federate uses the Event Identifier to associate related events. The event number begins at one at the beginning of the exercise and is incremented by one for each event.</i>				
EntityIdentifier RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The EntityIdentifier of the object which this embedded system is a part of.</i>				
HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The RTI object instance ID of the object of which this embedded system is part of.</i>				

RelativePosition RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The position of the embedded system, relative to the host object's position.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

10.1.3. PropulsionNoise

Full Name: HLAobjectRoot.EmbeddedSystem.UnderwaterAcousticsEmission.PropulsionNoise

Sharing: Publish/Subscribe

Semantics: *Describes the steady state component of unintended passive emissions that are normally associated with the power plant.*

Attributes:

HullMaskerOn	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Whether the hull masker silencing system is on.</i>				
PassiveParameterIndex	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UAPassiveParameterIndexEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Database index to look up the entities acoustic signature.</i>				

PropulsionPlantConfiguration	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UAPropulsionPlantConfigurationConfigurationEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The operating mode of the propulsion plant.</i>				
ShaftRateData	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ShaftDataStructArray1Plus	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Information about each of the propulsion shafts associated with the entity. Shafts are ordered from port to starboard, when looking from the stern to the bow.</i>				
EventIdentifier <i>Inherited from UnderwaterAcousticsEmission in Underwater Acoustics</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The generating federate uses the Event Identifier to associate related events. The event number begins at one at the beginning of the exercise and is incremented by one for each event.</i>				
EntityIdentifier RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The EntityIdentifier of the object which this embedded system is a part of.</i>				
HostObjectIdentifier RPRnoteBase2 RPRnoteBase6 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics	<i>The RTI object instance ID of the object of which this embedded system is part of.</i>				

RelativePosition RPRnoteBase2 <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The position of the embedded system, relative to the host object's position.</i>				
HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	<i>NA</i>				
	Semantics					

10.1.4. AdditionalPassiveActivities

Full Name: HLAobjectRoot.EmbeddedSystem.UnderwaterAcousticsEmission.AdditionalPassiveActivities

Sharing: Publish/Subscribe

Semantics: *Describes the steady state components of non-propulsion passive emissions such as those held in the Additional Narrowband Database (ANDB).*

Attributes:

ActivityCode	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UAAdditionalPassiveActivityParameterIndexEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Database index used to indicate the passive signature of the entity.</i>				
ActivityParameter	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	Integer16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The current state of this activity.</i>				

IsSilent	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RPRboolean	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Used to silence an additional passive activity without destroying this object.</i>				
EventIdentifier <i>Inherited from UnderwaterAcousticsEmission in Underwater Acoustics</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The generating federate uses the Event Identifier to associate related events. The event number begins at one at the beginning of the exercise and is incremented by one for each event.</i>				
EntityIdentifier <i>RPRnoteBase2</i> <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The EntityIdentifier of the object which this embedded system is a part of.</i>				
HostObjectIdentifier <i>RPRnoteBase2 RPRnoteBase6</i> <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The RTI object instance ID of the object of which this embedded system is part of.</i>				
RelativePosition <i>RPRnoteBase2</i> <i>Inherited from EmbeddedSystem in Base</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The position of the embedded system, relative to the host object's position.</i>				

HLAprivilegeToDeleteObject <i>Inherited from HLAobjectRoot in MIM</i>	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

10.1.5. ActiveSonarBeam

Full Name: HLAobjectRoot.ActiveSonarBeam

Sharing: Publish/Subscribe

Semantics: *A sector of concentrated acoustic energy from an active sonar device.*

Attributes:

ActiveEmissionParameterIndex	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UAAActiveEmissionParameterIndexEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>An index into the database of active (intentional) underwater acoustics emissions.</i>				
ActiveSonarIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTIobjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The RTI ID of the ActiveSonar emitting this beam.</i>				
AzimuthBeamwidth	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The horizontal width of the main beam (as opposed to any side lobes) measured from beam center to the 3 dB down point. Omnidirectional beams shall have a beam width of 0 radians.</i>				

AzimuthCenter	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The center azimuthal bearing of the main beam (as opposed to side lobes) in relation to the own ship heading, clockwise positive. Omnidirection beams shall have an azimuthal center of 0 radians.					
BeamIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UnsignedInteger8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The identification of the active sonar beam, which must be unique on the active sonar system.					
ElevationBeamwidth	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Vertical angle from beam center to 3db down point. O is omnidirectional.					
ElevationCenter	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AngleRadianFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The angle of axis of the beam center to the horizontal plane. Positive upward.					
EventIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	Used to associate changes to the beam with its ActiveSonar.					

ScanPattern	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	UAScanPatternEnum16	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change				
	Semantics					
	The pattern that describes the angular movement of the sonar beam during its sweep.					
HLAprivilegeToDeleteObject Inherited from HLAobjectRoot in MIM	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					

10.2. Interaction Classes



10.2.1. HLAinteractionRoot

Full Name: HLAinteractionRoot
 Sharing: Publish/Subscribe
 Transportation type: HLAREliable
 Order: Receive
 Dimensions:
 Semantics:
 Parameters: -

10.2.2. AcousticTransient

Full Name: HLAinteractionRoot.AcousticTransient
 Sharing: Publish/Subscribe
 Transportation type: HLABestEffort
 Order: Receive
 Dimensions:
 Semantics: *Specifies the occurrence of a transient acoustic event such as torpedo tube floodings, hatch slams, and wrench drops.*
 Parameters:

Name	Datatype	Semantics
ActivityCode	UAAAdditionalPassiveActivityParameterIndexEnum16	<i>Index into a data base of transient acoustic events.</i>
ActivityParameter	Integer16	<i>The current state of the activity specified by the activity code.</i>
HostObjectIdentifier	RTObjectId	<i>The Object Instance ID of the object emitting the sound.</i>
RelativePosition	RelativePositionStruct	<i>The location of the sound relative to the hosts coordinate system.</i>

10.3. Datatypes

10.3.1. Simple Datatypes

SpeedChangeRateRPMPerSecondInteger32

Representation: HLAinteger32BE

Units: RPM/s

Resolution: 1

Accuracy: perfect

Semantics: Angular acceleration.

10.3.2. Array Datatypes

ShaftDataStructArray1Plus

Element [ShaftDataStruct](#)

Type:

Cardinality: [1..2147483647]

Encoding: HLAvariableArray

Semantics: *Array of propulsion shaft states, one per shaft.*

10.3.3. Fixed Record Datatypes

ShaftDataStruct

Encoding: HLAfixedRecord

Semantics: Rotation state of a ship's propulsion shaft.

Name	Type	Semantic
CurrentShaftRate	RevolutionsPerMinuteInteger16	The current shaft speed. A positive shaft speed indicates that the shaft is rotating in a clockwise direction (when viewed from the stern to bow), and a negative shaft speed indicates an anti-clockwise direction.
OrderedShaftRate	RevolutionsPerMinuteInteger16	The ordered shaft speed. If the ordered shaft speed is different from the current shaft speed then the shaft will accelerate or decelerate at the shaft rate of change. A positive shaft speed indicates that the shaft is rotating in a clockwise direction (when viewed from the stern to bow), and a negative shaft speed indicates an anti-clockwise direction.
ShaftRateOfChange	SpeedChangeRateRPMPerSecondInteger32	The absolute value of the shaft's rotational acceleration. The shaft will change its rotational speed at this rate if the current shaft speed differs from the ordered shaft rate.

11. Module Warfare



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Warfare FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	Represents the firing of munitions and expendables and detonation, explosion, burst, or ignition of munitions, non-munitions, and expendables.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Base
Foundation
Enumerations

11.1. Object Classes



11.1.1. DirectedEnergyFire

Full Name: HLAObjectRoot.DirectedEnergyFire
Sharing: Publish/Subscribe
Semantics: *Communicates information associated with the firing of a DE weapon.*
Attributes:

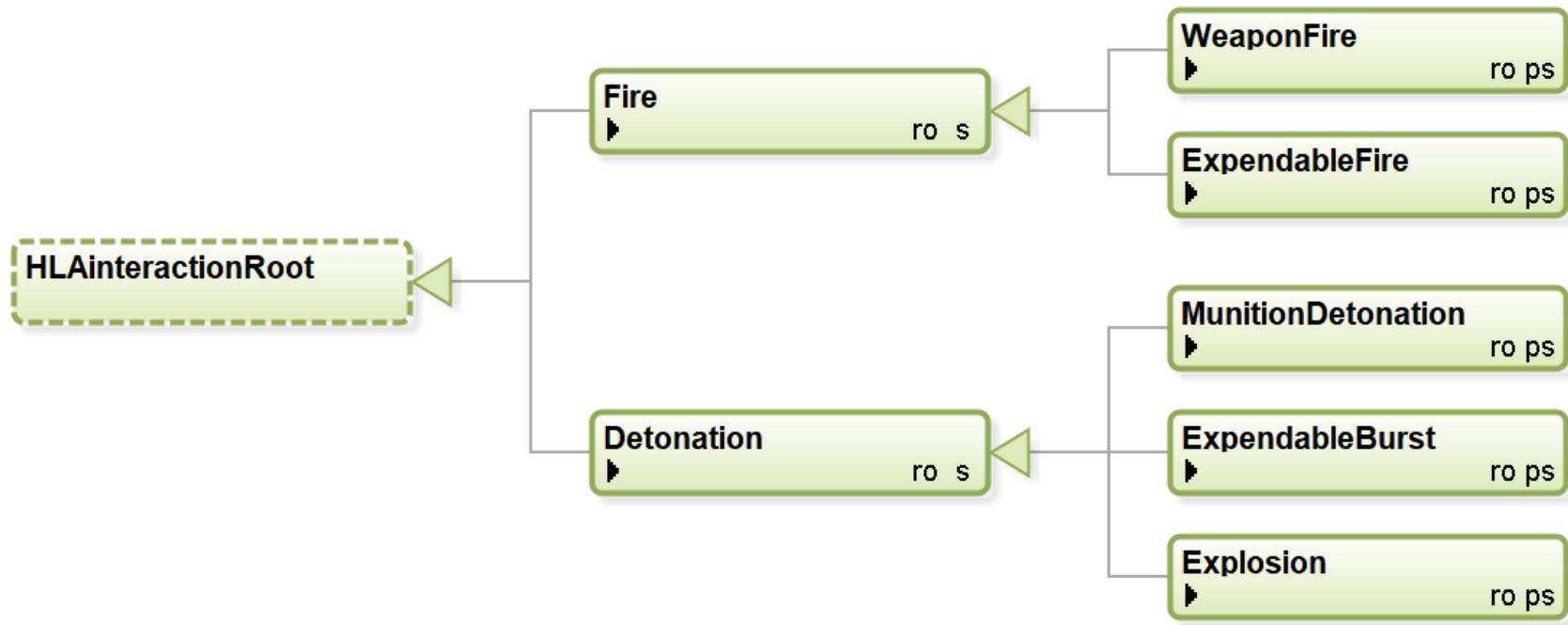
FiringObjectIdentifier RPRnoteWarfare1	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RTObjectId	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The object instance ID of the firing entity.					
EventIdentifier	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EventIdentifierStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	An ID, generated by the issuing federate, used to associate to related fire and detonation events.					

MunitionType	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	EntityTypeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The munition type enumeration for the DE weapon beam.					
ShotStartTime	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	ClockTimeStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The simulation time at start of the shot.					
CumulativeShotTime	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TimeSecondFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On object update				
	Semantics					
	The current cumulative duration of the shot, expressed in units of seconds.					
RelativeFiringLocation	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	RelativePositionStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The location of the DE weapon aperture/emitter in coordinates relative to the firing entity.					
ApertureDiameter	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MeterFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	The beam diameter at the aperture/emitter.					

Wavelength	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	MeterFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Static	NA				
	Semantics	<i>The emissions wavelength.</i>				
PulseRepetitionFrequency	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	FrequencyHertzFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The current pulse repetition frequency.</i>				
PulseWidth	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	TimeSecondFloat32	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The pulse width emissions.</i>				
PulseShape	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DEFirePulseShapeEnum8	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>The pulse shape.</i>				
Flags	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	DEFireFlagsDataStruct	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	<i>On change</i>				
	Semantics	<i>Flags indicating status information.</i>				

AreaAimpointData	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	AreaAimpointStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteWarfare2				
	Semantics					
	Information about the directed energy beam area targeting data.					
PrecisionAimpointData	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	PrecisionAimpointStructArray	PS	DA	RO	HLAbestEffort	
	Update type	Update Condition				
	Conditional	On change RPRnoteWarfare2				
	Semantics					
	Information about the directed energy beam precise targeting data.					
HLAprivilegeToDeleteObject	Datatype	Sharing	Ownership	Order	Transportation	Dimensions
	HLAtoken	PS	DA	TS	HLAreliable	
	Update type	Update Condition				
	Static	NA				
	Semantics					
	Inherited from HLAobjectRoot in MIM					

11.2. Interaction Classes



11.2.1. HLAinteractionRoot

Full Name: HLAinteractionRoot
 Sharing: Publish/Subscribe
 Transportation type: HLAREliable
 Order: Receive
 Dimensions:
 Semantics:
 Parameters: -

11.2.2. Fire

Full Name: HLAinteractionRoot.Fire

Sharing: Subscribe

Transportation
type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *Base class for information associated with the firing or launch of a munition or expendable.*

Parameters:

Name	Datatype	Semantics
FiringObjectIdentifier RPRnoteWarfare1	RTObjectId	<i>The object instance ID of the entity firing the munition or expendable.</i>
TargetObjectIdentifier RPRnoteWarfare3	RTObjectId	<i>The object instance ID of the intended target (if any).</i>
EventIdentifier	EventIdentifierStruct	<i>An ID, generated by the issuing federate, used to associate related fire and detonation events.</i>
FireMissionIndex	UnsignedInteger32	<i>A unique index to identify the fire mission (used to associate weapon fire interactions in a single fire mission).</i>
FiringLocation	LocationStruct	<i>The location, in world coordinates, from which the munition or expendable was launched.</i>
InitialVelocityVector	VelocityVectorStruct	<i>The velocity vector of the munition or expendable when fired.</i>
FireControlSolutionRange	LengthMeterFloat32	<i>The range used in the fire control solution. Zero if the range is unknown or inapplicable.</i>

11.2.3. WeaponFire

Full Name: HLAinteractionRoot.Fire.WeaponFire

Sharing: Publish/Subscribe

Transportation
type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *Communicates information associated with the firing or launch of a munition.*

Parameters:

Name	Datatype	Semantics
MunitionObjectIdentifier RPRnoteWarfare3	RTObjectId	<i>The object instance ID of the fired munition, if an object instance is registered.</i>
MunitionType	EntityTypeStruct	<i>The type of munition being fired.</i>
WarheadType	MunitionDescriptorWarheadEnum16	<i>The type of warhead fitted to the munition being fired.</i>
FuseType	MunitionDescriptorFuseEnum16	<i>The type of fuse on the munition.</i>
QuantityFired	UnsignedInteger16	<i>The number of rounds fired in the burst.</i>
RateOfFire	FireRateRoundPerMinuteUnsignedInteger16	<i>The rate of fire in rounds per minute when quantity > 1. Zero otherwise.</i>
FiringObjectIdentifier RPRnoteWarfare1 <i>Inherited from Fire in Warfare</i>	RTObjectId	<i>The object instance ID of the entity firing the munition or expendable.</i>
TargetObjectIdentifier RPRnoteWarfare3 <i>Inherited from Fire in Warfare</i>	RTObjectId	<i>The object instance ID of the intended target (if any).</i>
EventIdentifier <i>Inherited from Fire in Warfare</i>	EventIdentifierStruct	<i>An ID, generated by the issuing federate, used to associate related fire and detonation events.</i>
FireMissionIndex <i>Inherited from Fire in Warfare</i>	UnsignedInteger32	<i>A unique index to identify the fire mission (used to associate weapon fire interactions in a single fire mission).</i>
FiringLocation <i>Inherited from Fire in Warfare</i>	LocationStruct	<i>The location, in world coordinates, from which the munition or expendable was launched.</i>
InitialVelocityVector <i>Inherited from Fire in Warfare</i>	VelocityVectorStruct	<i>The velocity vector of the munition or expendable when fired.</i>

Name	Datatype	Semantics
FireControlSolutionRange <i>Inherited from Fire in Warfare</i>	LengthMeterFloat32	<i>The range used in the fire control solution. Zero if the range is unknown or inapplicable.</i>

11.2.4. ExpendableFire

Full Name: HLAinteractionRoot.Fire.ExpendableFire

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *Communicates information associated with the firing or launch of an expendable.*

Parameters:

Name	Datatype	Semantics
ExpendableObjectIdentifier RPRnoteWarfare3	RTObjectId	<i>The object instance ID of the fired expendable, if an object instance is registered.</i>
ExpendableType	EntityTypeStruct	<i>The type of expendable being fired.</i>
FiringObjectIdentifier RPRnoteWarfare1 <i>Inherited from Fire in Warfare</i>	RTObjectId	<i>The object instance ID of the entity firing the munition or expendable.</i>
TargetObjectIdentifier RPRnoteWarfare3 <i>Inherited from Fire in Warfare</i>	RTObjectId	<i>The object instance ID of the intended target (if any).</i>
EventIdentifier <i>Inherited from Fire in Warfare</i>	EventIdentifierStruct	<i>An ID, generated by the issuing federate, used to associate related fire and detonation events.</i>
FireMissionIndex <i>Inherited from Fire in Warfare</i>	UnsignedInteger32	<i>A unique index to identify the fire mission (used to associate weapon fire interactions in a single fire mission).</i>
FiringLocation <i>Inherited from Fire in Warfare</i>	LocationStruct	<i>The location, in world coordinates, from which the munition or expendable was launched.</i>

Name	Datatype	Semantics
InitialVelocityVector <i>Inherited from Fire in Warfare</i>	VelocityVectorStruct	<i>The velocity vector of the munition or expendable when fired.</i>
FireControlSolutionRange <i>Inherited from Fire in Warfare</i>	LengthMeterFloat32	<i>The range used in the fire control solution. Zero if the range is unknown or inapplicable.</i>

11.2.5. Detonation

Full Name: HLAinteractionRoot.Detonation

Sharing: Subscribe

Transportation
type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *Base class for information associated with the impact or detonation of a munition, a non-munition explosion, or the burst or ignition of an expendable.*

Parameters:

Name	Datatype	Semantics
EventIdentifier	EventIdentifierStruct	<i>An ID, generated by the issuing federate, used to associate related fire and detonation events.</i>
FinalVelocityVector	VelocityVectorStruct	<i>The velocity vector of the munition immediately before detonation/impact, the velocity of a non-munition entity immediately before exploding, or the velocity of an expendable immediately before a chaff burst or ignition of a flare.</i>
DetonationLocation	LocationStruct	<i>The location of the detonation in world coordinates.</i>
RelativeDetonationLocation	RelativePositionStruct	<i>The location of the munition detonation, the expendable detonation, or the non-munition explosion in respectively the target entity's coordinate system, the coordinate system of the entity that launched the expendable, or the exploded entity's coordinate system.</i>
DetonationResultCode	DetonationResultEnum8	<i>The type of detonation (including no detonation).</i>
VariableParametersArray	VariableParameterVariantStructArray	<i>Array of information on articulated parts, attached parts, and other data associated with a detonation or explosion.</i>

11.2.6. MunitionDetonation

Full Name: HLAinteractionRoot.Detonation.MunitionDetonation

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *Communicates information associated with the impact or detonation of a munition.*

Parameters:

Name	Datatype	Semantics
FiringObjectIdentifier RPRnoteWarfare1	RTLObjectId	<i>The object instance ID of the entity that fired the munition.</i>
TargetObjectIdentifier RPRnoteWarfare3	RTLObjectId	<i>The object instance ID of the target entity if known.</i>
MunitionObjectIdentifier RPRnoteWarfare3	RTLObjectId	<i>The Object instance ID of the fired munition, if an object instance is registered.</i>
MunitionType	EntityTypeStruct	<i>The type of munition that is detonating.</i>
WarheadType	MunitionDescriptorWarheadEnum16	<i>The type of warhead on the munition.</i>
FuseType	MunitionDescriptorFuseEnum16	<i>The type of fuse on the munition.</i>
QuantityFired	UnsignedInteger16	<i>The number of rounds fired in the burst.</i>
RateOfFire	FireRateRoundPerMinuteUnsignedInteger16	<i>The rate of fire in rounds per minute when quantity > 1. Zero otherwise.</i>
EventIdentifier Inherited from Detonation in Warfare	EventIdentifierStruct	<i>An ID, generated by the issuing federate, used to associate related fire and detonation events.</i>
FinalVelocityVector Inherited from Detonation in Warfare	VelocityVectorStruct	<i>The velocity vector of the munition immediately before detonation/impact, the velocity of a non-munition entity immediately before exploding, or the velocity of an expendable immediately before a chaff burst or ignition of a flare.</i>

Name	Datatype	Semantics
DetonationLocation <i>Inherited from Detonation in Warfare</i>	LocationStruct	<i>The location of the detonation in world coordinates.</i>
RelativeDetonationLocation <i>Inherited from Detonation in Warfare</i>	RelativePositionStruct	<i>The location of the munition detonation, the expendable detonation, or the non-munition explosion in respectively the target entity's coordinate system, the coordinate system of the entity that launched the expendable, or the exploded entity's coordinate system.</i>
DetonationResultCode <i>Inherited from Detonation in Warfare</i>	DetonationResultEnum8	<i>The type of detonation (including no detonation).</i>
VariableParametersArray <i>Inherited from Detonation in Warfare</i>	VariableParameterVariantStructArray	<i>Array of information on articulated parts, attached parts, and other data associated with a detonation or explosion.</i>

11.2.7. ExpendableBurst

Full Name: HLAinteractionRoot.Detonation.ExpendableBurst

Sharing: Publish/Subscribe

Transportation: HLABestEffort

type:

Order: Receive

Dimensions:

Semantics: *Communicates information associated with the burst or ignition of an expendable.*

Parameters:

Name	Datatype	Semantics
LaunchingObjectIdentifier RPRnoteWarfare1	RTObjectId	<i>The object instance ID of the entity that launched the expendable.</i>
ThreatObjectIdentifier RPRnoteWarfare3	RTObjectId	<i>The object instance ID of the target threat entity, if applicable and known, that caused the release of the expendable.</i>
ExpendableObjectIdentifier RPRnoteWarfare3	RTObjectId	<i>The Object instance ID of the fired expendable, if an object instance is registered.</i>
ExpendableType	EntityTypeStruct	<i>The type of expendable.</i>

Name	Datatype	Semantics
EventIdentifier <i>Inherited from Detonation in Warfare</i>	EventIdentifierStruct	<i>An ID, generated by the issuing federate, used to associate related fire and detonation events.</i>
FinalVelocityVector <i>Inherited from Detonation in Warfare</i>	VelocityVectorStruct	<i>The velocity vector of the munition immediately before detonation/impact, the velocity of a non-munition entity immediately before exploding, or the velocity of an expendable immediately before a chaff burst or ignition of a flare.</i>
DetonationLocation <i>Inherited from Detonation in Warfare</i>	LocationStruct	<i>The location of the detonation in world coordinates.</i>
RelativeDetonationLocation <i>Inherited from Detonation in Warfare</i>	RelativePositionStruct	<i>The location of the munition detonation, the expendable detonation, or the non-munition explosion in respectively the target entity's coordinate system, the coordinate system of the entity that launched the expendable, or the exploded entity's coordinate system.</i>
DetonationResultCode <i>Inherited from Detonation in Warfare</i>	DetonationResultEnum8	<i>The type of detonation (including no detonation).</i>
VariableParametersArray <i>Inherited from Detonation in Warfare</i>	VariableParameterVariantStructArray	<i>Array of information on articulated parts, attached parts, and other data associated with a detonation or explosion.</i>

11.2.8. Explosion

Full Name: HLAinteractionRoot.Detonation.Explosion

Sharing: Publish/Subscribe

Transportation
type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *Communicates information associated with a non-munition explosion.*

Parameters:

Name	Datatype	Semantics
SourceObjectIdentifier RPRnoteWarfare1	RTObjectId	<i>The object instance ID of the entity that caused the non-munition entity, or portion thereof, to explode.</i>
ExplodedObjectIdentifier RPRnoteWarfare3	RTObjectId	<i>The object instance ID of the entity, or portion thereof, that exploded.</i>

Name	Datatype	Semantics
ExplodingObjectType	EntityTypeStruct	<i>The type of object that exploded.</i>
ExplosiveMaterial	ExplosiveMaterialCategoriesEnum16	<i>The material that exploded.</i>
ExplosiveForce	ExplosiveForceKilogramTNFloat32	<i>The explosive force expressed as the equivalent kilograms of TNT.</i>
EventIdentifier <i>Inherited from Detonation in Warfare</i>	EventIdentifierStruct	<i>An ID, generated by the issuing federate, used to associate related fire and detonation events.</i>
FinalVelocityVector <i>Inherited from Detonation in Warfare</i>	VelocityVectorStruct	<i>The velocity vector of the munition immediately before detonation/impact, the velocity of a non-munition entity immediately before exploding, or the velocity of an expendable immediately before a chaff burst or ignition of a flare.</i>
DetonationLocation <i>Inherited from Detonation in Warfare</i>	LocationStruct	<i>The location of the detonation in world coordinates.</i>
RelativeDetonationLocation <i>Inherited from Detonation in Warfare</i>	RelativePositionStruct	<i>The location of the munition detonation, the expendable detonation, or the non-munition explosion in respectively the target entity's coordinate system, the coordinate system of the entity that launched the expendable, or the exploded entity's coordinate system.</i>
DetonationResultCode <i>Inherited from Detonation in Warfare</i>	DetonationResultEnum8	<i>The type of detonation (including no detonation).</i>
VariableParametersArray <i>Inherited from Detonation in Warfare</i>	VariableParameterVariantStructArray	<i>Array of information on articulated parts, attached parts, and other data associated with a detonation or explosion.</i>

11.3. Datatypes

11.3.1. Simple Datatypes

ExplosiveForceKilogramTNTFloat32

Representation: HLAfloat32BE

Units: kilogram of TNT (4.184×10^6 Joules per kilogram)

Resolution: NA

Accuracy: NA

Semantics: Explosive force expressed as the equivalent kilograms of TNT.

FireRateRoundPerMinuteUnsignedInteger16

Representation: [RPRunsignedInteger16BE](#)

Units: round/minute

Resolution: 1

Accuracy: perfect

Semantics: Rate of fire, in rounds per minute.

IrradianceWattPerMeterSquaredFloat32

Representation: HLAfloat32BE

Units: watt per meter squared ($W/(m^2)$)

Resolution: NA

Accuracy: NA

Semantics: Radiant flux (power) received by a surface per unit area.

11.3.2. Array Datatypes

AreaAimpointStructArray

Element [AreaAimpointStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Array with DE area aimpoint data.*

BeamAntennaStructArray

Element [BeamAntennaStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Array with beam antenna pattern data.*

PrecisionAimpointStructArray

Element [PrecisionAimpointStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Array with DE precision aimpoint data.*

TargetEnergyDepositionStructArray

Element [TargetEnergyDepositionStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Array with target energy deposition data.*

11.3.3. Fixed Record Datatypes

AreaAimpointStruct

Encoding: HLAfixedRecord

Semantics: *Specification of directed energy weapon targeting information when the target is an area.*

Name	Type	Semantic
BeamAntennaPatterns	BeamAntennaStructArray	<i>One or more non-self-identifying beam antenna patterns.</i>
TargetEnergyDepositions	TargetEnergyDepositionStructArray	<i>Zero or more non-self-identifying target energy deposition data sets.</i>

BeamSpotShapeStruct

Encoding: HLAfixedRecord

Semantics: *Parameters describing the ellipse shape approximating the irradiance region arriving at the target. The ellipse parameters are measured in the plane perpendicular to the direction of the beam. Refer to IEEE Std 1278.1-2012 Section A.2.4.3 for a detailed description.*

Name	Type	Semantic
CrossSectionSemiMajorAxis	MeterFloat32	<i>The length of the semi-major axis of an ellipse.</i>
CrossSectionSemiMinorAxis	MeterFloat32	<i>The length of the semi-minor axis of an ellipse.</i>
CrossSectionOrientationAngle	AngleRadianFloat32	<i>The orientation angle from the local 'down' vector to the semi-major axis of an ellipse.</i>

DEFireFlagsDataStruct

Encoding: HLAfixedRecord

Semantics: *DE Fire-Flags.*

Name	Type	Semantic
WeaponOn	RPRboolean	<i>Identifies the State of the DE Weapon.</i>

PrecisionAimpointStruct

Encoding: HLAfixedRecord

Semantics: *Specification of directed energy weapon targeting information when the target is a specific target entity.*

Name	Type	Semantic
TargetSpotLocation	LocationStruct	<i>Instantaneous location of the target spot in world coordinates.</i>
TargetSpotRelativeLocation	RelativePositionStruct	<i>Instantaneous location of the target spot in the target entity's coordinate frame.</i>
TargetSpotVelocityVector	VelocityVectorStruct	<i>Instantaneous target spot velocity.</i>
TargetSpotAccelerationVector	AccelerationVectorStruct	<i>Instantaneous target spot acceleration.</i>
TargetObjectIdentifier	RTLObjectId	<i>The object instance ID of the target affected by the weapon.</i>
TargetComponentIdentifier	EntityDamageStatusComponentIdentificationEnum8	<i>The entity component that is being targeted.</i>
BeamSpotType	DEPrecisionAimpointBeamSpotTypeEnum8	<i>The beam spot type specifies how the DE precision weapon energy is distributed on the target.</i>
BeamSpotShape	BeamSpotShapeStruct	<i>The beam spot ellipse shape approximating the irradiance region arriving at the target.</i>
PeakIrradiance	IrradianceWattPerMeterSquaredFloat32	<i>Peak irradiance of emissions.</i>

TargetEnergyDepositionStruct

Encoding: HLAfixedRecord

Semantics: *Directed energy deposition properties for a target entity.*

Name	Type	Semantic
TargetObjectIdentifier	RTLObjectId	<i>The object instance ID of the target.</i>
PeakIrradiance	IrradianceWattPerMeterSquaredFloat32	<i>Peak irradiance deposited at the target.</i>

11.4. Notes

RPRnoteWarfare1

Semantics: *This must reference a valid Object instance.*

RPRnoteWarfare2

Semantics: *Refer to SISO-STD-001 Section 7.13.1.1 for a detailed description of the update conditions of each of the alternatives possible as array element.*

RPRnoteWarfare3

Semantics: *If there is no object instance associated with the parameter, then this should be set to the empty string (no characters). Refer to SISO-STD-001 Section 7.8.6 for handling empty strings.*

12. Module Logistics



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Logistics FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	The Logistics module defines interactions that represent repair and resupply logistic services.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Base
Foundation
Enumerations

12.1. Interaction Classes

12.1.1. HLAinteractionRoot

Full Name: HLAinteractionRoot
Sharing: Publish/Subscribe
Transportation type: HLAreliable
Order: Receive
Dimensions:
Semantics:
Parameters: -

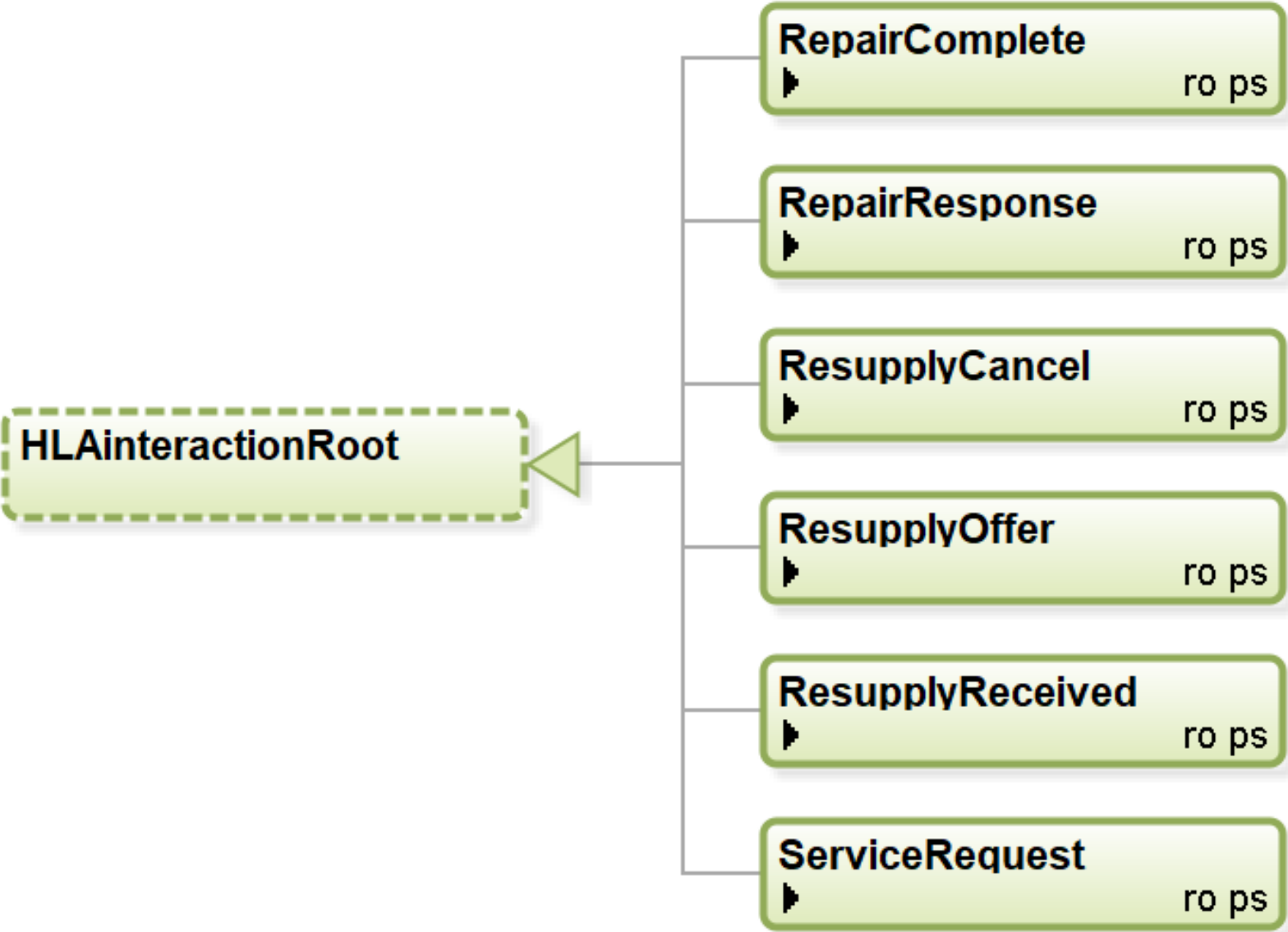
12.1.2. RepairComplete

Full Name: HLAinteractionRoot.RepairComplete
Sharing: Publish/Subscribe
Transportation type: HLABestEffort
Order: Receive
Dimensions:
Semantics: *Notifies the requesting entity that the requested repair has been completed.*
Parameters:

Name	Datatype	Semantics
ReceivingObject RPRnoteLogistics1	RTIobjectId	Object requesting repairs.
RepairingObject RPRnoteLogistics1	RTIobjectId	Object that is able to perform the requested repair.
RepairType	RepairCompleteRepairEnum16	Type of repair performed.

12.1.3. RepairResponse

Full Name: HLAinteractionRoot.RepairResponse
Sharing: Publish/Subscribe



Transportation type: HLAbestEffort
 Order: Receive
 Dimensions:
 Semantics: *Acknowledges the notification of the completion of a repair.*
 Parameters:

Name	Datatype	Semantics
ReceivingObject RPRnoteLogistics1	RTObjectId	<i>Object requesting repairs.</i>
RepairingObject RPRnoteLogistics1	RTObjectId	<i>Object that is able to perform the requested repair.</i>
RepairResultCode	RepairResponseRepairResultEnum8	<i>Result of repair.</i>

12.1.4. ResupplyCancel

Full Name: HLAinteractionRoot.ResupplyCancel
 Sharing: Publish/Subscribe
 Transportation type: HLAbestEffort
 Order: Receive
 Dimensions:
 Semantics: *Communicates the canceling of a service function by either the receiving or the supplying entity.*
 Parameters:

Name	Datatype	Semantics
ReceivingObject RPRnoteLogistics1	RTObjectId	<i>Object that is receiving supplies.</i>
SupplyingObject RPRnoteLogistics1	RTObjectId	<i>Object that has offered supplies.</i>

12.1.5. ResupplyOffer

Full Name: HLAinteractionRoot.ResupplyOffer
 Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *Communicates the offer of supplies from a supplying entity to a receiving entity.*

Parameters:

Name	Datatype	Semantics
ReceivingObject RPRnoteLogistics1	RTObjectId	<i>Object that the supplies are being offered to.</i>
SupplyingObject RPRnoteLogistics1	RTObjectId	<i>Object that is offering the supplies.</i>
SuppliesData	SupplyStructArray	<i>List of supplies that are offered.</i>

12.1.6. ResupplyReceived

Full Name: HLAinteractionRoot.ResupplyReceived

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *Acknowledge the receipt of supplies.*

Parameters:

Name	Datatype	Semantics
ReceivingObject RPRnoteLogistics1	RTObjectId	<i>Object that is receiving the supplies.</i>
SupplyingObject RPRnoteLogistics1	RTObjectId	<i>Object that is providing the supplies.</i>
SuppliesData	SupplyStructArray	<i>List of supplies taken by receiving object.</i>

12.1.7. ServiceRequest

Full Name: HLAinteractionRoot.ServiceRequest

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A request for logistics support. The requesting entity issues the interaction to the supplying entity asking for repair or specific supplies.*

Parameters:

Name	Datatype	Semantics
RequestingObject RPRnoteLogistics1	RTObjectId	Object requesting service.
ServicingObject RPRnoteLogistics1	RTObjectId	Object that is able to provide the requested service.
ServiceType	ServiceRequestServiceTypeRequestedEnum8	Type of requested service.
SuppliesData	SupplyStructArray	For a service of resupply, the list of supplies to be exchanged. If the service requested is not resupply, then this parameter shall be omitted.

12.2. Datatypes

12.2.1. Array Datatypes

SupplyStructArray

Element [SupplyStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *A list of supply types and the number of each being offered or requested.*

12.2.2. Fixed Record Datatypes

SupplyStruct

Encoding: HLAfixedRecord

Semantics: Represents a single supply type and the quantity being offered or requested.

Name	Type	Semantic
SupplyType	EntityTypeStruct	The type of supply being offered or requested.
Quantity	Float32	The number of units of the supply type. The unit measure depends on the supply type and shall use the SI units of measurement used for such supplies.

12.3. Notes

RPRnoteLogistics1

Semantics: *This must reference a valid Object instance.*

13. Module Simulation Management



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Simulation Management FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	Simulation Management interactions.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Base
Foundation
Enumerations

13.1. Interaction Classes

13.1.1. HLAinteractionRoot

Full Name: HLAinteractionRoot
Sharing: Publish/Subscribe
Transportation type: HLAreliable
Order: Receive
Dimensions:
Semantics:
Parameters: -

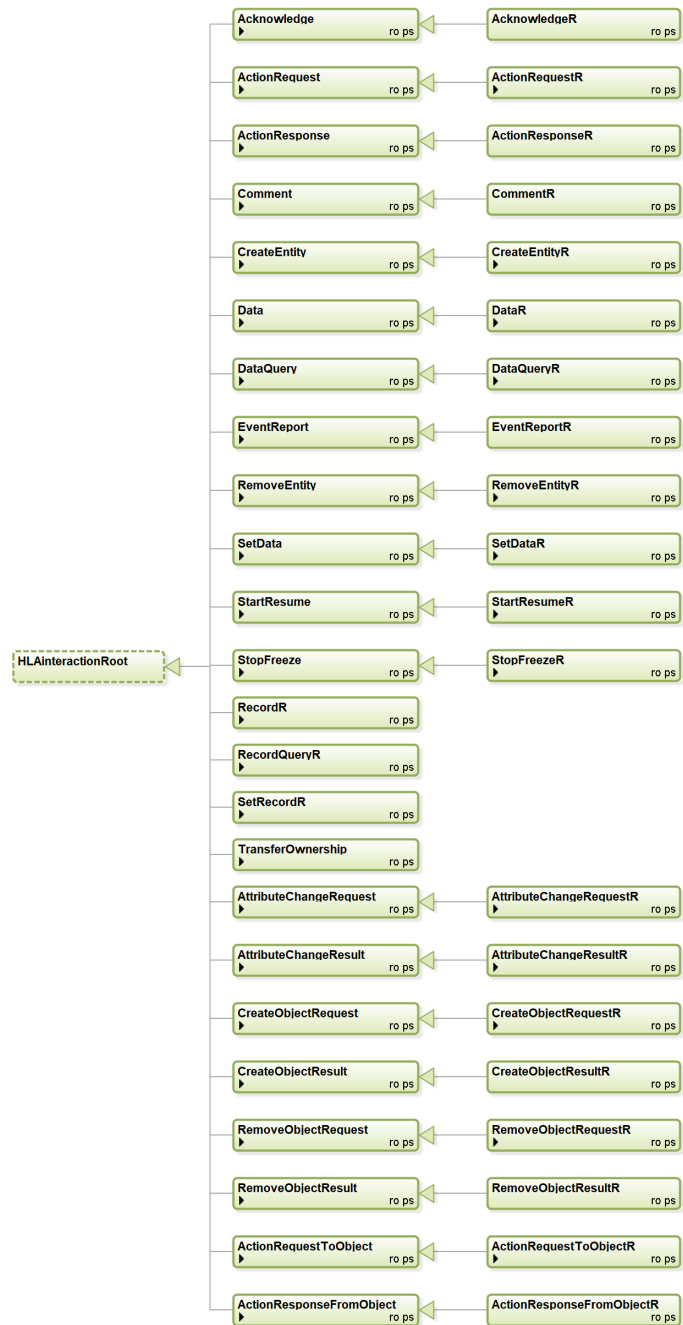
13.1.2. Acknowledge

[RPRnoteSIMAN1](#)

Full Name: HLAinteractionRoot.Acknowledge
Sharing: Publish/Subscribe
Transportation type: HLABestEffort
Order: Receive
Dimensions:
Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a SIMAN interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	The entity ID of the entity or application sending the interaction.
ReceivingEntity	EntityIdentifierStruct	The entity ID of the entity or application which is the intended recipient of the interaction.
RequestIdentifier	UnsignedInteger32	This field matches this response with the specific StartResume, StopFreeze, CreateEntity or RemoveEntity interaction sent by the simulation manager.



Name	Datatype	Semantics
AcknowledgeFlag	AcknowledgeAcknowledgeFlagEnum16	<i>The type of interaction being acknowledged.</i>
ResponseFlag	AcknowledgeResponseFlagEnum16	<i>The type of response made to the interaction by the recipient.</i>

13.1.3. AcknowledgeR

[RPRnoteSIMAN2](#)

Full Name: HLAinteractionRoot.Acknowledge.AcknowledgeR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a SIMAN interaction, with a specified acknowledgement protocol, from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity <i>Inherited from Acknowledge in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity <i>Inherited from Acknowledge in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient of the interaction.</i>
RequestIdentifier <i>Inherited from Acknowledge in Simulation Management</i>	UnsignedInteger32	<i>This field matches this response with the specific StartResume, StopFreeze, CreateEntity or RemoveEntity interaction sent by the simulation manager.</i>
AcknowledgeFlag <i>Inherited from Acknowledge in Simulation Management</i>	AcknowledgeAcknowledgeFlagEnum16	<i>The type of interaction being acknowledged.</i>

Name	Datatype	Semantics
ResponseFlag <i>Inherited from Acknowledge in Simulation Management</i>	AcknowledgeResponseFlagEnum16	<i>The type of response made to the interaction by the recipient.</i>

13.1.4. ActionRequest

[RPRnoteSIMAN3](#)

Full Name: HLAIinteractionRoot.ActionRequest

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more federates to request that they perform a specified action.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
RequestIdentifier	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>
ActionRequestCode	ActionRequestActionIDEnum32	<i>The action that the recipient(s) are requested to perform.</i>
FixedDatums	FixedDatumStructArray	<i>Optional additional data items (types and values).</i>
VariableDatumSet	VariableDatumStructArray	<i>Optional additional data items (types and values). These data items are not of fixed length.</i>

13.1.5. ActionRequestR[RPRnoteSIMAN4](#)

Full Name: HLAinteractionRoot.ActionRequest.ActionRequestR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more federates to request that they perform a specified action. The Simulation Manager federate specifies the acknowledgement protocol to be used.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol RPRnoteSIMAN5	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
OriginatingEntity <i>Inherited from ActionRequest in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity <i>Inherited from ActionRequest in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
RequestIdentifier <i>Inherited from ActionRequest in Simulation Management</i>	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>
ActionRequestCode <i>Inherited from ActionRequest in Simulation Management</i>	ActionRequestActionCodeEnum32	<i>The action that the recipient(s) are requested to perform.</i>
FixedDatums <i>Inherited from ActionRequest in Simulation Management</i>	FixedDatumStructArray	<i>Optional additional data items (types and values).</i>

Name	Datatype	Semantics
VariableDatumSet <i>Inherited from ActionRequest in Simulation Management</i>	VariableDatumStructArray	<i>Optional additional data items (types and values). These data items are not of fixed length.</i>

13.1.6. ActionResponse

[RPRnoteSIMAN3](#)

Full Name: HLAinteractionRoot.ActionResponse

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponse interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient of the interaction.</i>
RequestIdentifier	UnsignedInteger32	<i>This field matches this response with the specific ActionRequest interaction sent by the simulation manager.</i>
RequestStatus	ActionResponseRequestStatusEnum32	<i>The status of the request that the recipient has been asked to perform.</i>
FixedDatums	FixedDatumStructArray	<i>Additional, fixed-length data items (types and values).</i>
VariableDatumSet	VariableDatumStructArray	<i>Additional, non-fixed-length, data items (types and values).</i>

13.1.7. ActionResponseR[RPRnoteSIMAN4](#)**Full Name:** HLAIinteractionRoot.ActionResponse.ActionResponseR**Sharing:** Publish/Subscribe**Transportation type:** HLAReliable**Order:** Receive**Dimensions:****Semantics:** *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponseR interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.***Parameters:**

Name	Datatype	Semantics
OriginatingEntity <i>Inherited from ActionResponse in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity <i>Inherited from ActionResponse in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient of the interaction.</i>
RequestIdentifier <i>Inherited from ActionResponse in Simulation Management</i>	UnsignedInteger32	<i>This field matches this response with the specific ActionRequest interaction sent by the simulation manager.</i>
RequestStatus <i>Inherited from ActionResponse in Simulation Management</i>	ActionResponseRequestStatusEnum32	<i>The status of the request that the recipient has been asked to perform.</i>
FixedDatums <i>Inherited from ActionResponse in Simulation Management</i>	FixedDatumStructArray	<i>Additional, fixed-length data items (types and values).</i>

Name	Datatype	Semantics
VariableDatumSet <i>Inherited from ActionResponse in Simulation Management</i>	VariableDatumStructArray	<i>Additional, non-fixed-length, data items (types and values).</i>

13.1.8. Comment

Full Name: HLAinteractionRoot.Comment

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction used to insert messages and information into a log stream and closely matches the structures used by the Data interaction class. This information is usually unsolicited in nature. The information contained within the Interaction should be used for commenting purposes only.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
VariableDatumSet	VariableDatumStructArray	<i>The set of data items (types and values) associated with the interaction.</i>

13.1.9. CommentR

Full Name: HLAinteractionRoot.Comment.CommentR

Sharing: Publish/Subscribe

Transportation type: HLAreliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction used to insert messages and information into a log stream and closely matches the structures used by the Data interaction class. This information is usually unsolicited in nature. The information contained within the Interaction should be used for commenting purposes only.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity <i>Inherited from Comment in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity <i>Inherited from Comment in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
VariableDatumSet <i>Inherited from Comment in Simulation Management</i>	VariableDatumStructArray	<i>The set of data items (types and values) associated with the interaction.</i>

13.1.10. CreateEntity

[RPRnoteSIMAN1](#)

Full Name: HLAinteractionRoot.CreateEntity

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that an application creates an entity.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity RPRnoteSIMAN6	EntityIdentifierStruct	<i>The FederateIdentifier field identifies the federate that is responsible for creating the entity (if possible). The complete parameter defines the entity ID of the new entity (see also note [18]).</i>

Name	Datatype	Semantics
RequestIdentifier	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>

13.1.11. CreateEntityR

[RPRnoteSIMAN2](#)

Full Name: HLAinteractionRoot.CreateEntity.CreateEntityR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that an application creates an entity, using a specified acknowledgement service.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
OriginatingEntity <i>Inherited from CreateEntity in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity RPRnoteSIMAN6 <i>Inherited from CreateEntity in Simulation Management</i>	EntityIdentifierStruct	<i>The FederateIdentifier field identifies the federate that is responsible for creating the entity (if possible). The complete parameter defines the entity ID of the new entity (see also note [18]).</i>
RequestIdentifier <i>Inherited from CreateEntity in Simulation Management</i>	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>

13.1.12. Data

Full Name: HLAinteractionRoot.Data

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge either a) a DataQuery interaction (in which case the Data interaction contains the results of the query) or b) a SetData interaction (in which case the Data interaction contains the data that the federate was able to set).*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	The entity ID of the entity or application sending the interaction.
ReceivingEntity	EntityIdentifierStruct	The entity ID of the entity or application which is the intended recipient of the interaction.
RequestIdentifier	UnsignedInteger32	This field matches this response with the specific SetData or DataQuery interaction sent by the simulation manager.
FixedDatums	FixedDatumStructArray	The set of data items (types and values), of fixed length, that the recipient can return for this interaction.
VariableDatumSet	VariableDatumStructArray	The set of data items (types and values), of variable length, that the recipient can return for this interaction.

13.1.13. DataR

Full Name: HLAinteractionRoot.Data.DataR

Sharing: Publish/Subscribe

Transportation type: HLAreliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge either a) a DataQueryR interaction (in which case the DataR interaction contains the results of the query) or b) a SetDataR interaction (in which case the DataR interaction contains the data that the federate was able to set).*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
OriginatingEntity <i>Inherited from Data in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity <i>Inherited from Data in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient of the interaction.</i>
RequestIdentifier <i>Inherited from Data in Simulation Management</i>	UnsignedInteger32	<i>This field matches this response with the specific SetData or DataQuery interaction sent by the simulation manager.</i>
FixedDatums <i>Inherited from Data in Simulation Management</i>	FixedDatumStructArray	<i>The set of data items (types and values), of fixed length, that the recipient can return for this interaction.</i>
VariableDatumSet <i>Inherited from Data in Simulation Management</i>	VariableDatumStructArray	<i>The set of data items (types and values), of variable length, that the recipient can return for this interaction.</i>

13.1.14. DataQuery

[RPRnoteSIMAN7](#)

Full Name: HLAinteractionRoot.DataQuery

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate supply the current values of specified data.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>

Name	Datatype	Semantics
ReceivingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient of the interaction.</i>
RequestIdentifier	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>
TimeInterval	TimestampUnsignedInteger32	<i>The interval between regular updates of the requested data. If this field is zero then the recipient should only issue a single Data interaction in response to this interaction.</i>
FixedDatumIdentifiers	DatumIdentifierArray	<i>The set of fixed length data items (types) that the recipient is requested to supply data for.</i>
VariableDatumIdentifiers	DatumIdentifierArray	<i>The set of variable length data items (types) that the recipient is requested to supply data for.</i>

13.1.15. DataQueryR

[RPRnoteSIMAN8](#)

Full Name: HLInteractionRoot.DataQuery.DataQueryR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate supply the current values of specified data. The Simulation Manager federate specifies the acknowledgement protocol to be used.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
OriginatingEntity <i>Inherited from DataQuery in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>

Name	Datatype	Semantics
ReceivingEntity <i>Inherited from DataQuery in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient of the interaction.</i>
RequestIdentifier <i>Inherited from DataQuery in Simulation Management</i>	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>
TimeInterval <i>Inherited from DataQuery in Simulation Management</i>	TimestampUnsignedInteger32	<i>The interval between regular updates of the requested data. If this field is zero then the recipient should only issue a single Data interaction in response to this interaction.</i>
FixedDatumIdentifiers <i>Inherited from DataQuery in Simulation Management</i>	DatumIdentifierArray	<i>The set of fixed length data items (types) that the recipient is requested to supply data for.</i>
VariableDatumIdentifiers <i>Inherited from DataQuery in Simulation Management</i>	DatumIdentifierArray	<i>The set of variable length data items (types) that the recipient is requested to supply data for.</i>

13.1.16. EventReport

Full Name: HLAinteractionRoot.EventReport

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to allow a federate to alert a Simulation Manager federate that a particular event has occurred.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
EventType	EventReportEventTypeEnum32	<i>The type of event being reported.</i>

Name	Datatype	Semantics
FixedDatums	FixedDatumStructArray	<i>The set of fixed size data items (types and values) associated with this event.</i>
VariableDatumSet	VariableDatumStructArray	<i>The set of variable size data items (types and values) associated with this event.</i>

13.1.17. EventReportR

Full Name: HLAinteractionRoot.EventReport.EventReportR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to allow a federate to alert a Simulation Manager federate that a particular event has occurred.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity <i>Inherited from EventReport in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity <i>Inherited from EventReport in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
EventType <i>Inherited from EventReport in Simulation Management</i>	EventReportEventTypeEnum32	<i>The type of event being reported.</i>
FixedDatums <i>Inherited from EventReport in Simulation Management</i>	FixedDatumStructArray	<i>The set of fixed size data items (types and values) associated with this event.</i>
VariableDatumSet <i>Inherited from EventReport in Simulation Management</i>	VariableDatumStructArray	<i>The set of variable size data items (types and values) associated with this event.</i>

13.1.18. RemoveEntity

[RPRnoteSIMAN1](#)

Full Name: HLAinteractionRoot.RemoveEntity
 Sharing: Publish/Subscribe
 Transportation type: HLAbestEffort
 Order: Receive
 Dimensions:
 Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that a specified entity be removed from the simulation.*
 Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
RequestIdentifier	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>

13.1.19. RemoveEntityR

[RPRnoteSIMAN2](#)

Full Name: HLAinteractionRoot.RemoveEntity.RemoveEntityR
 Sharing: Publish/Subscribe
 Transportation type: HLAreliable
 Order: Receive
 Dimensions:
 Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that a specified entity be removed from the simulation.*
 Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
OriginatingEntity <i>Inherited from RemoveEntity in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity <i>Inherited from RemoveEntity in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
RequestIdentifier <i>Inherited from RemoveEntity in Simulation Management</i>	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>

13.1.20. SetData

[RPRnoteSIMAN9](#)

Full Name: HLAIinteractionRoot.SetData

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate sets the values of specified data to specified values.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient of the interaction.</i>

Name	Datatype	Semantics
RequestIdentifier	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>
FixedDatums	FixedDatumStructArray	<i>The set of fixed length data items (types and values) that the recipient is requested to set.</i>
VariableDatumSet	VariableDatumStructArray	<i>The set of variables length data items (types and values) that the recipient is requested to set.</i>

13.1.21. SetDataR

[RPRnoteSIMAN10](#)

Full Name: HLAinteractionRoot.SetData.SetDataR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate sets the values of specified data to specified values. The Simulation Manager federate specifies the acknowledgement protocol to be used.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
OriginatingEntity <i>Inherited from SetData in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity <i>Inherited from SetData in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient of the interaction.</i>
RequestIdentifier <i>Inherited from SetData in Simulation Management</i>	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>

Name	Datatype	Semantics
FixedDatums <i>Inherited from SetData in Simulation Management</i>	FixedDatumStructArray	<i>The set of fixed length data items (types and values) that the recipient is requested to set.</i>
VariableDatumSet <i>Inherited from SetData in Simulation Management</i>	VariableDatumStructArray	<i>The set of variables length data items (types and values) that the recipient is requested to set.</i>

13.1.22. StartResume

[RPRnoteSIMAN1](#)

Full Name: HLAIinteractionRoot.StartResume

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to either a) start simulating one or more entities or b) resume simulation of one or more entities after a pause.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
RealWorldTime	ClockTimeStruct	<i>The real world time (GMT) that the entity or entities should start/resume at.</i>
RequestIdentifier	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>
SimulationTime	ClockTimeStruct	<i>The simulation time that the entity or entities should use when they start/resume.</i>

13.1.23. StartResumeR[RPRnoteSIMAN2](#)

Full Name: HLAinteractionRoot.StartResume.StartResumeR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to either a) start simulating one or more entities or b) resume simulation of one or more entities after a pause. The Simulation Manager federate specifies the acknowledgement protocol to be used.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
OriginatingEntity <i>Inherited from StartResume in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity <i>Inherited from StartResume in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
RealWorldTime <i>Inherited from StartResume in Simulation Management</i>	ClockTimeStruct	<i>The real world time (GMT) that the entity or entities should start/resume at.</i>
RequestIdentifier <i>Inherited from StartResume in Simulation Management</i>	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>
SimulationTime <i>Inherited from StartResume in Simulation Management</i>	ClockTimeStruct	<i>The simulation time that the entity or entities should use when they start/resume.</i>

13.1.24. StopFreeze

[RPRnoteSIMAN1](#)

Full Name: HLAIinteractionRoot.StopFreeze

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that one or more entities either a) freeze (pause) their simulation or b) stop their simulation.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	The entity ID of the entity or application sending the interaction.
ReceivingEntity	EntityIdentifierStruct	The entity ID of the entity or application which is the intended recipient(s) of the interaction.
RequestIdentifier	UnsignedInteger32	The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.
RealWorldTime	ClockTimeStruct	The real world time (GMT) that the entity or entities should stop/freeze at.
Reason	StopFreezeReasonEnum8	The reason for the stop or freeze.
ReflectValues	RPRboolean	Whether the entity or entities being stopped/frozen should continue to reflect values when stopped/frozen.
RunInternalSimulationClock	RPRboolean	Whether the entity or entities being stopped/frozen should continue to run their internal simulation clock when stopped/frozen.
UpdateAttributes	RPRboolean	Whether the entity or entities being stopped/frozen should continue to update attributes when stopped/frozen.

13.1.25. StopFreezeR

[RPRnoteSIMAN2](#)

Full Name: HLAIinteractionRoot.StopFreeze.StopFreezeR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that one or more entities either a) pause their simulation or b) stop their simulation. The Simulation Manager federate specifies the acknowledgement protocol to be used.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
OriginatingEntity <i>Inherited from StopFreeze in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity <i>Inherited from StopFreeze in Simulation Management</i>	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
RequestIdentifier <i>Inherited from StopFreeze in Simulation Management</i>	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>
RealWorldTime <i>Inherited from StopFreeze in Simulation Management</i>	ClockTimeStruct	<i>The real world time (GMT) that the entity or entities should stop/freeze at.</i>
Reason <i>Inherited from StopFreeze in Simulation Management</i>	StopFreezeReasonEnum8	<i>The reason for the stop or freeze.</i>
ReflectValues <i>Inherited from StopFreeze in Simulation Management</i>	RPRboolean	<i>Whether the entity or entities being stopped/frozen should continue to reflect values when stopped/frozen.</i>

Name	Datatype	Semantics
RunInternalSimulationClock <i>Inherited from StopFreeze in Simulation Management</i>	RPRboolean	<i>Whether the entity or entities being stopped/frozen should continue to run their internal simulation clock when stopped/frozen.</i>
UpdateAttributes <i>Inherited from StopFreeze in Simulation Management</i>	RPRboolean	<i>Whether the entity or entities being stopped/frozen should continue to update attributes when stopped/frozen.</i>

13.1.26. RecordR

Full Name: HLAinteractionRoot.RecordR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a RecordQueryR or SetRecordR interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the federate has implemented the request.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient of the interaction.</i>
RequestIdentifier	UnsignedInteger32	<i>This field matches this response with the specific RecordQueryR or SetRecordR interaction sent by the simulation manager.</i>
EventType	RecordREventTypeEnum16	<i>The type of event that caused the RecordR interaction to be issued.</i>
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for the transaction.</i>
ResponseSerialNumber	UnsignedInteger32	<i>Used to identify the serial number of the RecordR interaction when more than one interaction is used to report record values.</i>

Name	Datatype	Semantics
RecordSetData	RecordSetStructArray1Plus	<i>The set of records containing the information requested.</i>

13.1.27. RecordQueryR

[RPRnoteSIMAN11](#)

Full Name: HLAinteractionRoot.RecordQueryR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to allow a Simulation Manager federate to request data, in record format, from another federate.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application sending the interaction.</i>
ReceivingEntity	EntityIdentifierStruct	<i>The entity ID of the entity or application which is the intended recipient(s) of the interaction.</i>
RequestIdentifier	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>
TimeInterval	TimestampUnsignedInteger32	<i>The interval between regular updates of the requested data. If this field is zero then the recipient should only issue a single RecordR interaction in response to this interaction.</i>
EventType	RecordQueryREventTypeEnum16	<i>Specifies the type of event that the receiving entity or application should use to trigger the issue of a RecordR interaction in response to this query. If this is zero, then reporting shall be periodic based upon the value of the TimeInterval parameter.</i>
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for the transaction.</i>
RecordIdentifiers	DatumIdentifierArray	<i>Identifies the records for which information is requested.</i>

13.1.28. SetRecordR[RPRnoteSIMAN12](#)

Full Name: HLAinteractionRoot.SetRecordR

Sharing: Publish/Subscribe

Transportation
type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to request that a federate sets the values of specified data to specified values (provided in record format).*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	The entity ID of the entity or application sending the interaction.
ReceivingEntity	EntityIdentifierStruct	The entity ID of the entity or application which is the intended recipient(s) of the interaction. ³
RequestIdentifier	UnsignedInteger32	The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	The acknowledgement protocol to be used for a transaction.
RecordSetData	RecordSetStructArray1Plus	Specifies the information, in record format, to be set by the receiving entity.

13.1.29. TransferOwnership

Full Name: HLAinteractionRoot.TransferOwnership

Sharing: Publish/Subscribe

Transportation
type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent to initiate the transfer of control of an entity.*

Parameters:

Name	Datatype	Semantics
OriginatingEntity	EntityIdentifierStruct	The entity ID of the entity or application sending the interaction.
ReceivingEntity	EntityIdentifierStruct	The entity ID of the entity or application which is the intended recipient(s) of the interaction.
RequestIdentifier	UnsignedInteger32	The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.
TransferType	TransferControlTransferTypeEnum8	The type of transfer to be performed.
TransferEntity	RTObjectId	The ID of the object to be transferred.
RecordSetData	RecordSetStructArray1Plus	Specifies the information, in record format, to be set by the receiving entity.

13.1.30. AttributeChangeRequest

[RPRnoteSIMAN13](#)

Full Name: HLAINteractionRoot.AttributeChangeRequest

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to ask that a specified attribute be set to a specified value.*

Parameters:

Name	Datatype	Semantics
ObjectIdentifiers	RTObjectIdArray	The list of objects that are the intended recipients of this interaction.

Name	Datatype	Semantics
AttributeValueSet	AttributeValuePairStructArray1Plus	<i>The set of attributes and their values, that the recipients are asked to update.</i>

13.1.31. AttributeChangeRequestR

[RPRnoteSIMAN14](#)

Full Name: HLAinteractionRoot.AttributeChangeRequest.AttributeChangeRequestR
Sharing: Publish/Subscribe
Transportation type: HLAReliable
Order: Receive
Dimensions:
Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to ask that a specified attribute be set to a specified value. The Simulation Manager federate specifies the acknowledgement protocol to be used.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol RPRnoteSIMAN5	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
ObjectIdentifiers <i>Inherited from AttributeChangeRequest in Simulation Management</i>	RTObjectIdArray	<i>The list of objects that are the intended recipients of this interaction.</i>
AttributeValueSet <i>Inherited from AttributeChangeRequest in Simulation Management</i>	AttributeValuePairStructArray1Plus	<i>The set of attributes and their values, that the recipients are asked to update.</i>

13.1.32. AttributeChangeResult

[RPRnoteSIMAN13](#)

Full Name: HLAinteractionRoot.AttributeChangeResult
Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an AttributeChangeRequest interaction from a Simulation Manager federate, and to inform the Simulation Manager federate whether the attribute was set to the specified value or not.*

Parameters:

Name	Datatype	Semantics
ObjectIdentifier	RTObjectId	<i>The ID of the object replying to the AttributeChangeRequest interaction.</i>
AttributeChangeResult	AcknowledgeResponseFlagEnum16	<i>Indicates ability to comply.</i>
AttributeValueSet	AttributeValuePairStructArray1Plus	<i>The set of attributes and their values that the recipient has been able to update.</i>

13.1.33. AttributeChangeResultR

[RPRnoteSIMAN14](#)

Full Name: HLInteractionRoot.AttributeChangeResult.AttributeChangeResultR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an AttributeChangeRequest interaction from a Simulation Manager federate, and to inform the Simulation Manager federate whether the attribute was set to the specified value or not. The Simulation Manager federate specifies the acknowledgement protocol to be used.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for this transaction.</i>

Name	Datatype	Semantics
ObjectIdentifier <i>Inherited from AttributeChangeResult in Simulation Management</i>	RTObjectId	<i>The ID of the object replying to the AttributeChangeRequest interaction.</i>
AttributeChangeResult <i>Inherited from AttributeChangeResult in Simulation Management</i>	AcknowledgeResponseFlagEnum16	<i>Indicates ability to comply.</i>
AttributeValueSet <i>Inherited from AttributeChangeResult in Simulation Management</i>	AttributeValuePairStructArray1Plus	<i>The set of attributes and their values that the recipient has been able to update.</i>

13.1.34. CreateObjectRequest

[RPRnoteSIMAN15](#)

Full Name: HLAinteractionRoot.CreateObjectRequest

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to another federate requesting that it create an object instance of a particular type.*

Parameters:

Name	Datatype	Semantics
ObjectClass	HLAunicodeString	<i>The fully qualified (as defined in the HLA standard) class name of the object to be created.</i>
AttributeValueSet	AttributeValuePairStructArray1Plus	<i>The set of attributes, and associated values, to be used to initialize the object.</i>
RequestIdentifier	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>

13.1.35. CreateObjectRequestR[RPRnoteSIMAN16](#)

Full Name: HLAinteractionRoot.CreateObjectRequest.CreateObjectRequestR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager federate to another federate requesting that it create an object instance of a particular type. The Simulation Manager federate specifies the acknowledgement protocol to be used.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
ObjectClass <i>Inherited from CreateObjectRequest in Simulation Management</i>	HLAunicodeString	<i>The fully qualified (as defined in the HLA standard) class name of the object to be created.</i>
AttributeValueSet <i>Inherited from CreateObjectRequest in Simulation Management</i>	AttributeValuePairStructArray1Plus	<i>The set of attributes, and associated values, to be used to initialize the object.</i>
RequestIdentifier <i>Inherited from CreateObjectRequest in Simulation Management</i>	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>

13.1.36. CreateObjectResult[RPRnoteSIMAN15](#)

Full Name: HLAinteractionRoot.CreateObjectResult

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a CreateObjectRequest from a Simulation Manager federate and to inform the Simulation Manager federate whether the object creation was successful or not.*

Parameters:

Name	Datatype	Semantics
CreateObjectResult	AcknowledgeResponseFlagEnum16	<i>Indicates ability to comply.</i>
RequestIdentifier	UnsignedInteger32	<i>This field matches this response with the specific CreateObject interaction sent by the simulation manager.</i>

13.1.37. CreateObjectResultR

[RPRnoteSIMAN16](#)

Full Name: HLAinteractionRoot.CreateObjectResult.CreateObjectResultR

Sharing: Publish/Subscribe

Transportation type: HLAReliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a CreateObjectRequest from a Simulation Manager federate and to inform the Simulation Manager federate whether the object creation was successful or not. The Simulation Manager federate specifies the acknowledgement protocol to be used.*

Parameters:

Name	Datatype	Semantics
CreateObjectResult <i>Inherited from CreateObjectResult in Simulation Management</i>	AcknowledgeResponseFlagEnum16	<i>Indicates ability to comply.</i>

Name	Datatype	Semantics
RequestIdentifier <i>Inherited from CreateObjectResult in Simulation Management</i>	UnsignedInteger32	<i>This field matches this response with the specific CreateObject interaction sent by the simulation manager.</i>

13.1.38. RemoveObjectRequest

[RPRnoteSIMAN17](#)

Full Name: HLAIinteractionRoot.RemoveObjectRequest

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that one or more specified objects be removed from the simulation.*

Parameters:

Name	Datatype	Semantics
ObjectIdentifiers	RTIobjectIdArray	<i>The IDs of the objects to be removed.</i>
RequestIdentifier	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>

13.1.39. RemoveObjectRequestR

[RPRnoteSIMAN18](#)

Full Name: HLAIinteractionRoot.RemoveObjectRequest.RemoveObjectRequestR

Sharing: Publish/Subscribe

Transportation type: HLAreliable

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction, sent from a Simulation Manager to request that one or more specified objects be removed from the simulation.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for a transaction.</i>
ObjectIdentifiers <i>Inherited from RemoveObjectRequest in Simulation Management</i>	RTObjectIdArray	<i>The IDs of the objects to be removed.</i>
RequestIdentifier <i>Inherited from RemoveObjectRequest in Simulation Management</i>	UnsignedInteger32	<i>The Request ID is a monotonically increasing integer identifier inserted by the Simulation Manager into all Simulation management interactions. It is used as a unique identifier to identify the latest in a series of competing requests and identifying acknowledgements.</i>

13.1.40. RemoveObjectResult

[RPRnoteSIMAN17](#)

Full Name: HLAIinteractionRoot.RemoveObjectResult

Sharing: Publish/Subscribe

Transportation type: HLAbestEffort

Order: Receive

Dimensions:

Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a RemoveObjectRequest interaction, and to inform the Simulation Manager federate whether the removal was successful or not.*

Parameters:

Name	Datatype	Semantics
RemoveObjectResult	AcknowledgeResponseFlagEnum16	<i>Indicates ability to comply.</i>
RequestIdentifier	UnsignedInteger32	<i>This field matches this response with the specific RemoveObject interaction sent by the simulation manager.</i>

13.1.41. RemoveObjectResultR

[RPRnoteSIMAN18](#)

Full Name: HLAinteractionRoot.RemoveObjectResult.RemoveObjectResultR
Sharing: Publish/Subscribe
Transportation type: HLAReliable
Order: Receive
Dimensions:
Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of a RemoveObjectRequestR interaction, and to inform the Simulation Manager federate whether the removal was successful or not.*

Parameters:

Name	Datatype	Semantics
RemoveObjectResult <i>Inherited from RemoveObjectResult in Simulation Management</i>	AcknowledgeResponseFlagEnum16	<i>Indicates ability to comply.</i>
RequestIdentifier <i>Inherited from RemoveObjectResult in Simulation Management</i>	UnsignedInteger32	<i>This field matches this response with the specific RemoveObject interaction sent by the simulation manager.</i>

13.1.42. ActionRequestToObject

[RPRnoteSIMAN19](#)

Full Name: HLAinteractionRoot.ActionRequestToObject
Sharing: Publish/Subscribe
Transportation type: HLAbestEffort
Order: Receive
Dimensions:
Semantics: *A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more specific object instances to request that they perform a specified action.*

Parameters:

Name	Datatype	Semantics
ObjectIdentifiers	RTObjectIdArray	<i>The list of objects that are the intended recipients of this interaction.</i>
ActionRequestCode	ActionRequestActionIDEnum32	<i>The action that the recipient(s) are intended to perform.</i>

13.1.43. ActionRequestToObjectR

[RPRnoteSIMAN20](#)

Full Name: HLAinteractionRoot.ActionRequestToObject.ActionRequestToObjectR
Sharing: Publish/Subscribe
Transportation type: HLAReliable
Order: Receive
Dimensions:
Semantics: *A Simulation Management (SIMAN) interaction sent from a Simulation Manager federate to one or more specific object instances to request that they perform a specified action. The Simulation Manager federate specifies the acknowledgement protocol to be used.*

Parameters:

Name	Datatype	Semantics
AcknowledgementProtocol RPRnoteSIMAN5	RequiredReliabilityServiceEnum8	<i>The acknowledgement protocol to be used for this transaction.</i>
ObjectIdentifiers <i>Inherited from ActionRequestToObject in Simulation Management</i>	RTObjectIdArray	<i>The list of objects that are the intended recipients of this interaction.</i>
ActionRequestCode <i>Inherited from ActionRequestToObject in Simulation Management</i>	ActionRequestActionIDEnum32	<i>The action that the recipient(s) are intended to perform.</i>

13.1.44. ActionResponseFromObject

[RPRnoteSIMAN19](#)

Full Name: HLAinteractionRoot.ActionResponseFromObject

Sharing: Publish/Subscribe
 Transportation type: HLAbestEffort
 Order: Receive
 Dimensions:
 Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponseToObject interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the object has implemented the request.*

Parameters:

Name	Datatype	Semantics
ActionResult	ActionResponseRequestStatusEnum32	<i>The status of the request that the recipient has been asked to perform.</i>

13.1.45. ActionResponseFromObjectR

[RPRnoteSIMAN20](#)

Full Name: HLAIinteractionRoot.ActionResponseFromObject.ActionResponseFromObjectR
 Sharing: Publish/Subscribe
 Transportation type: HLAReliable
 Order: Receive
 Dimensions:
 Semantics: *A Simulation Management (SIMAN) interaction designed to acknowledge receipt of an ActionResponseToObjectR interaction from a Simulation Manager federate and to inform the Simulation Manager federate whether the object has implemented the request.*

Parameters:

Name	Datatype	Semantics
ActionResult <i>Inherited from ActionResponseFromObject in Simulation Management</i>	ActionResponseRequestStatusEnum32	<i>The status of the request that the recipient has been asked to perform.</i>

13.2. Datatypes

13.2.1. Array Datatypes

AttributeValuePairStructArray1Plus

Element [AttributeValuePairStruct](#)
Type:
Cardinality: [1..2147483647]
Encoding: HLAVariableArray
Semantics: *Array of AttributeValuePairStruct.*

DatumIdentifierArray

Element [VariableRecordTypesEnum32](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Array of DatumIdentifierEnum32.*

FixedDatumStructArray

Element [FixedDatumStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Array of FixedDatumStructs.*

RecordSetStructArray1Plus

Element [RecordSetStruct](#)
Type:
Cardinality: [1..2147483647]
Encoding: HLAVariableArray
Semantics: *Array of RecordSetStruct.*

RecordStructArray

Element [RecordStruct](#)
Type:
Cardinality: Dynamic
Encoding: HLAVariableArray
Semantics: *Array of RecordStruct.*

13.2.2. Fixed Record Datatypes

ArticulatedPartParameterTypeStruct

Encoding: HLAfixedRecord

Semantics: *Specifies the parameter type of an attached part, including the type class and type metric.*

Name	Type	Semantic
ParameterTypeClass	ArticulatedPartsTypeClassEnum32	<i>The type class uniquely identifies a particular articulated part on a given entity type. Guidance for uniquely assigning type classes to an entity's articulated parts is given in Section 4.8 of the enumeration document (SISO-REF-010).</i>
ParameterTypeMetric	ArticulatedPartsTypeMetricEnum8	<i>The type metric uniquely identifies the transformation to be applied to the articulated part.</i>

AssociationStruct

Encoding: HLAfixedRecord

Semantics: *Represents an entity's association with another entity or location.*

Name	Type	Semantic
AssociationType	EntityAssociationAssociationTypeEnum8	<i>Indicates the type of association.</i>
AssociatedObjectIdentifier	RTObjectId	<i>Identifies the associated object, if any.</i>
AssociatedLocation	LocationStruct	<i>Specifies the associated location in world coordinates, if any.</i>

AttributeValuePairStruct

Encoding: HLAfixedRecord

Semantics: *Pair of an attribute name identifying an attribute and data for that attribute.*

Name	Type	Semantic
AttributeName	HLAunicodeString	<i>The attribute name.</i>
AttributeValue	OctetArray	<i>Value for the specified attribute.</i>

EngineFuelReloadStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying an entity's engine fuel reload information.*

Name	Type	Semantic
StandardQuantity	UnsignedInteger32	The standard quantity of this fuel type normally loaded at this fuel location.
MaximumQuantity	UnsignedInteger32	The maximum quantity of this fuel type that this fuel location is capable of holding.
StandardQuantityReloadTime	TimeSecondInteger32	The number of seconds of simulation time normally required to reload the standard quantity of this fuel type at this specific fuel location.
MaximumQuantityReloadTime	TimeSecondInteger32	The number of seconds of simulation time normally required to reload the maximum possible quantity of this fuel type at this fuel location.
FuelMeasurementUnits	FuelMeasurementUnitsEnum8	Identifies the fuel measurement units.
FuelType	SupplyFuelTypeEnum8	Identifies the type of fuel.
FuelLocation	FuelLocationEnum8	Indicates the location of the fuel as related to the entity.

EngineFuelStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying an entity's engine fuel information.*

Name	Type	Semantic
FuelQuantity	UnsignedInteger32	The quantity of fuel remaining, in increments specified by the FuelMeasurementUnits field.
FuelMeasurementUnits	FuelMeasurementUnitsEnum8	Identifies the fuel measurement units.
FuelType	SupplyFuelTypeEnum8	Identifies the type of fuel.
FuelLocation	FuelLocationEnum8	Indicates the location of the fuel as related to the entity.

ExpendableReloadStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the reload information for an entity expendable.*

Name	Type	Semantic
ExpendableType	EntityTypeStruct	Identifies the entity type of the expendable.

Name	Type	Semantic
StationOrLauncher	AttachedOrArticulatedPartParameterTypeVariantStruct	Indicates the station or launcher to which the expendable is assigned.
StandardQuantity	UnsignedInteger16	The standard quantity of this expendable type normally loaded at this station/launcher.
MaximumQuantity	UnsignedInteger16	The maximum quantity of this expendable type that this station/launcher is capable of holding.
StandardQuantityReloadTime	TimeSecondInteger32	The number of seconds of simulation time normally required to reload the standard quantity of this expendable type at this station/launcher.
MaximumQuantityReloadTime	TimeSecondInteger32	The number of seconds of simulation time normally required to reload the maximum possible quantity of this expendable type at this station/launcher.

ExpendableStruct

Encoding: HLAfixedRecord

Semantics: Record specifying the information regarding an entity's expendable.

Name	Type	Semantic
ExpendableType	EntityTypeStruct	Identifies the entity type of the expendable.
StationOrLauncher	AttachedOrArticulatedPartParameterTypeVariantStruct	Indicates the station or launcher to which the expendable is assigned.
Quantity	UnsignedInteger16	The quantity remaining of this expendable.
ExpendableStatus	MunitionExpendableStatusEnum8	Specifies the status of the expendable.

FixedDatumStruct

Encoding: HLAfixedRecord

Semantics: Identifier and value for a fixed datum.

Name	Type	Semantic
FixedDatumIdentifier	VariableRecordTypesEnum32	The identifier for this fixed datum.
FixedDatumValue RPRnoteSIMAN21	OpaqueDatum32	The value of the fixed datum. The field may be used to convey one 8-, 16-, or 32-bit data value as specified by the FixedDatumIdentifier. Padding bits follow an 8- or 16-bit value to fill the 32-bit field.

LaunchedMunitionStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying a launched munition entity's firing and targeting information.*

Name	Type	Semantic
FireEventIdentifier	EventIdentifierStruct	Contains the same identifier specified in the EventIdentifier parameter of the Fire interaction that communicated the launch of the munition.
FiringObjectIdentifier	RTObjectId	Contains the same identifier as specified in the FiringObjectIdentifier parameter of the Fire interaction that communicated the launch of the munition.
TargetObjectIdentifier	RTObjectId	Identifies the munition's target entity, if any.
TargetLocation	LocationStruct	Specifies the munition's geographic target in world coordinates.

MunitionReloadStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the reload information for an entity munition.*

Name	Type	Semantic
MunitionType	EntityTypeStruct	Identifies the entity type of the munition.
StationOrLauncher	AttachedOrArticulatedPartParameterTypeVariantStruct	Indicates the station or launcher to which the munition is assigned.
StandardQuantity	UnsignedInteger16	The standard quantity of this munition type normally loaded at this station/launcher.
MaximumQuantity	UnsignedInteger16	The maximum quantity of this munition type that this station/launcher is capable of holding.
StandardQuantityReloadTime	TimeSecondInteger32	The number of seconds of simulation time normally required to reload the standard quantity of this munition type at this station/launcher.
MaximumQuantityReloadTime	TimeSecondInteger32	The number of seconds of simulation time normally required to reload the maximum possible quantity of this munition type at this station/launcher.

MunitionStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying the information regarding an entity's munition.*

Name	Type	Semantic
MunitionType	EntityTypeStruct	Identifies the entity type of the munition.
StationOrLauncher	AttachedOrArticulatedPartParameterTypeVariantStruct	Indicates the station or launcher to which the munition is assigned.
Quantity	UnsignedInteger16	The quantity remaining of this munition.
MunitionStatus	MunitionExpendableStatusEnum8	Specifies the status of the munition.

OwnershipStatusStruct

Encoding: HLAfixedRecord

Semantics: *Conveys ownership and conflict status information associated with an object.*

Name	Type	Semantic
ObjectIdentifier	RTIObjectId	Identifier for the object that is the subject of this record.
OwnershipStatus	OwnershipStatusEnum8	The ownership and/or ownership conflict status of the object.

RecordSetStruct RPRnoteSIMAN22

Encoding: HLAfixedRecord

Semantics: *A set of records and record details.*

Name	Type	Semantic
RecordSetIdentifier	VariableRecordTypesEnum32	This field shall specify the data structure used to convey the parameter values of the record for each record.
RecordSetSerialNumber	UnsignedInteger32	This field shall specify the serial number of the first record in the array.
RecordValues	RecordStructArray	This field contains the records of the format specified by the Record ID field.

RecordStruct

Encoding: HLAfixedRecord

Semantics: *Record data.*

Name	Type	Semantic
NumberOfBits	BitsUnsignedInteger16	<i>Number of bits of data.</i>
RecordData	OctetArray	<i>Data.</i>

SensorStruct

Encoding: HLAfixedRecord

Semantics: *Represents an entity's sensor information.*

Name	Type	Semantic
SensorType	SensorTypeVariantStruct	<i>Specifies the sensor type.</i>
SensorOnOffStatus	SensorOnOffStatusEnum8	<i>Specifies the on/off status of the sensor.</i>
Station	AttachedOrArticulatedPartParameterTypeVariantStruct	<i>Indicates the station to which the sensor is assigned.</i>
Quantity	UnsignedInteger16	<i>Specifies the quantity of this sensor.</i>

StorageFuelReloadStruct

Encoding: HLAfixedRecord

Semantics: *Record specifying an entity's engine fuel reload information.*

Name	Type	Semantic
StandardQuantity	UnsignedInteger32	<i>The standard quantity of this fuel type normally loaded at this fuel location.</i>
MaximumQuantity	UnsignedInteger32	<i>The maximum quantity of this fuel type that this fuel location is capable of holding.</i>
StandardQuantityReloadTime	TimeSecondInteger32	<i>The number of seconds of simulation time normally required to reload the standard quantity of this fuel type at this specific fuel location.</i>
MaximumQuantityReloadTime	TimeSecondInteger32	<i>The number of seconds of simulation time normally required to reload the maximum possible quantity of this fuel type at this fuel location.</i>
FuelMeasurementUnits	FuelMeasurementUnitsEnum8	<i>Identifies the fuel measurement units.</i>
FuelType	SupplyFuelTypeEnum8	<i>Identifies the type of fuel.</i>
FuelLocation	FuelLocationEnum8	<i>Indicates the location of the fuel as related to the entity.</i>

StorageFuelStruct

Encoding: HLAfixedRecord
Semantics: *Record specifying an entity's engine fuel information.*

Name	Type	Semantic
FuelQuantity	UnsignedInteger32	<i>The quantity of fuel remaining, in increments specified by the FuelMeasurementUnits field.</i>
FuelMeasurementUnits	FuelMeasurementUnitsEnum8	<i>Identifies the fuel measurement units.</i>
FuelType	SupplyFuelTypeEnum8	<i>Identifies the type of fuel.</i>
FuelLocation	FuelLocationEnum8	<i>Indicates the location of the fuel as related to the entity.</i>

13.2.3. Variant Record Datatypes

AttachedOrArticulatedPartParameterTypeVariantStruct

Encoding: RPRextendedVariantRecord

Discriminant name: AttachedOrArticulatedPart

Discriminant type: [VariableParameterRecordTypeEnum8](#)

Semantics: *Specifies the parameter type for either an attached or articulated part.*

Name	Enumerator	Type	Semantics
AttachedPartParameterType	Attached_Part	AttachedPartsEnum32	<i>Variant for representing the parameter type of an attached part.</i>
ArticulatedPartParameterType	Articulated_Part	ArticulatedPartParameterTypeStruct	<i>Variant for representing the parameter type of an articulated part.</i>

SensorTypeVariantStruct

Encoding: RPRextendedVariantRecord

Discriminant name: SensorTypeSource

Discriminant type: [SensorTypeSourceEnum8](#)

Semantics: *Specifies the sensor type.*

Name	Enumerator	Type	Semantics
OtherActiveSensors	Other_Active_Sensors	SensorRecordSensorTypeOtherActiveSensorsEnum16	<i>Variant for representing other active sensor types.</i>
Electromagnetic	Electromagnetic	EmitterNameEnum16	<i>Variant for representing electromagnetic sensor types.</i>
PassiveSensors	Passive_Sensors	SensorRecordSensorTypePassiveSensorsEnum16	<i>Variant for representing passive sensor types.</i>
MinefieldSensors	Minefield_Sensors	MinefieldSensorTypeVariantStruct	<i>Variant for representing minefield sensor types.</i>
UnderwaterAcoustics	Underwater_Acoustics	UAAcousticSystemNameEnum16	<i>Variant for representing underwater acoustics sensor types.</i>
Lasers	Lasers	DesignatorSystemNameEnum16	<i>Variant for representing laser sensor types.</i>

13.3. Notes

RPRnoteSIMAN1

Semantics: *The Acknowledge interaction is issued in response to the CreateEntity, RemoveEntity, StartResume, and StopFreeze interactions. See IEEE Std 1278.1-20128 Sections 5.6.5.6 and 7.5.6.*

RPRnoteSIMAN2

Semantics: *The AcknowledgeR interaction is issued in response to the CreateEntityR, RemoveEntityR, StartResumeR, and StopFreezeR interactions. See IEEE Std 1278.1-2012 Sections 5.12.4.6 and 7.11.6.*

RPRnoteSIMAN3

Semantics: *The ActionResponse interaction should be sent in response to an ActionRequest interaction by the receiving federate.*

RPRnoteSIMAN4

Semantics: *The ActionResponseR interaction should be sent in response to an ActionRequestR interaction by the receiving federate.*

RPRnoteSIMAN5

Semantics: *See IEEE Std 1278.1-2012 Section 5.12 for a description of the acknowledgement protocols (called reliability services in that document).*

RPRnoteSIMAN6

Semantics: *If the EntityNumber field is set to RQST_ASSIGN_ID (fixed value: FFFE (H)) then the receiving application should assign the entity number. Entity identifications shall be in accordance with IEEE Std 1278.1-2012 Section 6.2.28*

RPRnoteSIMAN7

Semantics: *The Data interaction should be sent in response to a DataQuery interaction by the receiving federate.*

RPRnoteSIMAN8

Semantics: *The DataR interaction should be sent in response to a DataQueryR interaction by the receiving federate.*

RPRnoteSIMAN9

Semantics: *The Data interaction should be sent in response to a SetData interaction by the receiving federate.*

RPRnoteSIMAN10

Semantics: *The DataR interaction should be sent in response to a SetDataR interaction by the receiving federate.*

RPRnoteSIMAN11

Semantics: *The RecordR interaction should be sent in response to a RecordQueryR interaction by the receiving federate.*

RPRnoteSIMAN12

Semantics: *The RecordR interaction should be sent in response to a SetRecordR interaction by the receiving federate.*

RPRnoteSIMAN13

Semantics: *The AttributeChangeResult interaction should be sent in response to an AttributeChangeRequest interaction by any object that is mentioned in the associated object list. The AttributeChangeResult parameter should indicate the success or failure of the object to implement the requested changes. If the object can implement the changes then the AttributeChangeResult parameter should be set to successful and the AttributeValueSet shall be empty.*

If the object cannot immediately comply with the request, but is likely to be able to comply in the future then the AttributeChangeRequest parameter shall be set to pending and the AttributeValueSet shall be empty. When the object complies with the request (or fails to implement the request) then another AttributeChangeResult interaction shall be sent informing the initiating federate of the result of the request.

If the object cannot comply with the request then the AttributeChangeResult parameter shall indicate the reason why the object cannot comply with the request. The AttributeValueSet shall contain the attributes which have been not been set and the current values of those attributes.

RPRnoteSIMAN14

Semantics: *The AttributeChangeResultR interaction should be sent in response to an AttributeChangeRequestR interaction by any object that is mentioned in the associated object list. The AttributeChangeResult parameter should indicate the success or failure of the object to implement the requested changes. If the object can implement the changes then the AttributeChangeResult parameter should be set to successful and the AttributeSet shall be empty.*

If the object cannot immediately comply with the request, but is likely to be able to comply in the future then the AttributeChangeRequest parameter shall be set to pending and the AttributeSet shall be empty. When the object complies with the request (or fails to implement the request) then another AttributeChangeResultR interaction shall be sent informing the initiating federate of the result of the request.

If the object cannot comply with the request then the AttributeChangeResult parameter shall indicate the reason why the object cannot comply with the request. The AttributeSet shall contain the attributes which have been not been set and the current values of those attributes.

RPRnoteSIMAN15

Semantics: *The CreateObjectResult interaction should be sent in response to an CreateObjectRequest interaction by the receiving federate.*

RPRnoteSIMAN16

Semantics: *The CreateObjectResultR interaction should be sent in response to an CreateObjectRequestR interaction by the receiving federate.*

RPRnoteSIMAN17

Semantics: *The RemoveObjectResult interaction should be sent in response to an RemoveObjectRequest interaction by the receiving federate.*

RPRnoteSIMAN18

Semantics: *The standard acknowledgement protocol is identical to the requirements of SIMAN in IEEE Std 1278.1-2012 Section 5.6 (i.e. it acts identically to the parent interaction). The other acknowledgement protocols are described in IEEE Std 1278.1-2012 Section 5.12 (they are called reliability services in that document).*

RPRnoteSIMAN19

Semantics: *The ActionResponseFromObject interaction should be sent in response to an ActionRequestToObject interaction by the receiving federate.*

RPRnoteSIMAN20

Semantics: *The ActionResponseFromObjectR interaction should be sent in response to an ActionRequestToObjectR interaction by the receiving federate.*

RPRnoteSIMAN21

Semantics: *The units of the FixedDatumValue field is determined by the value of the FixedDatumIdentifier field. The units, etc., for each of the FixedDatumIdentifier enumeration values are to be derived from the federation agreements, which could refer to SISO-REF-010.*

RPRnoteSIMAN22

Semantics: *The type of the RecordStruct is determined by the value of the RecordSetIdentifier field. The types and associated units, etc., for each of the RecordSetIdentifier enumeration values The values for the individual fields are to be derived from the federation agreements, which could refer to SISO-REF-010.*

14. Module Information Operations



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Information Operations FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	Information operations interactions.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

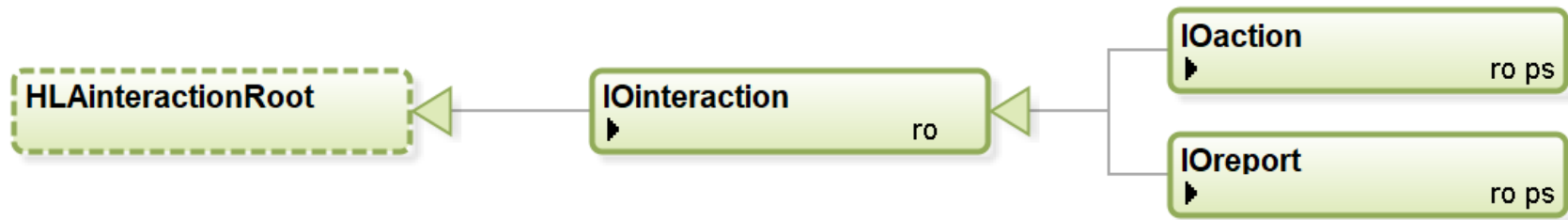
References

Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Foundation FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Enumerations FOM Module
Dependency	SISO-STD-001.1-2025 - Real-time Platform Reference Base FOM Module
Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

Dependencies

Base
Foundation
Enumerations

14.1. Interaction Classes



14.1.1. HLAInteractionRoot

Full Name: HLAinteractionRoot
 Sharing: Publish/Subscribe
 Transportation type: HLAREliable
 Order: Receive
 Dimensions:
 Semantics:
 Parameters: -

14.1.2. IOinteraction

Full Name: HLAinteractionRoot.IOinteraction
 Sharing:
 Transportation type: HLABestEffort
 Order: Receive
 Dimensions:
 Semantics: *Base class for interactions associated with information operations.*
 Parameters:

Name	Datatype	Semantics
OriginatingFederate	FederateIdentifierStruct	<i>This field shall identify the simulation that is issuing the interaction.</i>
SimulationSource	IOActionIOSimulationSourceEnum16	<i>This field shall identify the name of the simulation model issuing this interaction.</i>
AttackerObject	RTObjectId	<i>This field shall identify the IO attacker object.</i>
PrimaryTargetObject	RTObjectId	<i>This field shall identify the IO primary target object.</i>
CommunicationsNodes	IOCommunicationsNodeStructArray	<i>Information about the IO communications nodes.</i>
Effects	IOEffectStructArray	<i>Information about the IO effects.</i>

14.1.3. IOAction

Full Name: HLAinteractionRoot.IOinteraction.IOAction

Sharing: Publish/Subscribe

Transportation
type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *Actions initiated by an IO simulation to support interactions with other IO simulations shall be communicated using the IO Action interaction.*

Parameters:

Name	Datatype	Semantics
ReceivingFederate	FederateIdentifierStruct	<i>This field shall identify the simulation to which this interaction is addressed, if applicable.</i>
RequestIdentifier	UnsignedInteger32	<i>This field shall identify the request number for this IO Action.</i>
WarfareType	IOActionIOWarfareTypeEnum16	<i>This field shall identify the type of IO warfare.</i>

Name	Datatype	Semantics
ActionType	IOActionIOActionTypeEnum16	<i>This field shall identify the type of IO action.</i>
ActionPhase	IOActionIOActionPhaseEnum16	<i>This field shall identify the phase of the IO action.</i>
OriginatingFederate <i>Inherited from IOinteraction in Information Operations</i>	FederateIdentifierStruct	<i>This field shall identify the simulation that is issuing the interaction.</i>
SimulationSource <i>Inherited from IOinteraction in Information Operations</i>	IOActionIOSimulationSourceEnum16	<i>This field shall identify the name of the simulation model issuing this interaction.</i>
AttackerObject <i>Inherited from IOinteraction in Information Operations</i>	RTObjectId	<i>This field shall identify the IO attacker object.</i>
PrimaryTargetObject <i>Inherited from IOinteraction in Information Operations</i>	RTObjectId	<i>This field shall identify the IO primary target object.</i>
CommunicationsNodes <i>Inherited from IOinteraction in Information Operations</i>	IOCommunicationsNodeStructArray	<i>Information about the IO communications nodes.</i>
Effects <i>Inherited from IOinteraction in Information Operations</i>	IOEffectStructArray	<i>Information about the IO effects.</i>

14.1.4. IOreport

Full Name: HLAinteractionRoot.IOinteraction.IOreport

Sharing: Publish/Subscribe

Transportation type: HLABestEffort

Order: Receive

Dimensions:

Semantics: *The information operations status of an entity shall be conveyed using the IO Report interaction.*

Parameters:

Name	Datatype	Semantics
ReportType	IOReportIOReportTypeEnum8	<i>This field shall identify the type of IO report.</i>
OriginatingFederate <i>Inherited from IOinteraction in Information Operations</i>	FederateIdentifierStruct	<i>This field shall identify the simulation that is issuing the interaction.</i>
SimulationSource <i>Inherited from IOinteraction in Information Operations</i>	IOActionIOSimulationSourceEnum16	<i>This field shall identify the name of the simulation model issuing this interaction.</i>
AttackerObject <i>Inherited from IOinteraction in Information Operations</i>	RTObjectId	<i>This field shall identify the IO attacker object.</i>
PrimaryTargetObject <i>Inherited from IOinteraction in Information Operations</i>	RTObjectId	<i>This field shall identify the IO primary target object.</i>
CommunicationsNodes <i>Inherited from IOinteraction in Information Operations</i>	IOCommunicationsNodeStructArray	<i>Information about the IO communications nodes.</i>
Effects <i>Inherited from IOinteraction in Information Operations</i>	IOEffectStructArray	<i>Information about the IO effects.</i>

14.2. Datatypes

14.2.1. Simple Datatypes

TimeSecondUnsignedInteger16

Representation: [RPRunsignedInteger16BE](#)

Units: second (s)

Resolution: 1

Accuracy: perfect

Semantics: Time, based on SI base unit second, unit symbol s.

14.2.2. Array Datatypes

IOCommunicationsNodeStructArray

Element [IOCommunicationsNodeStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Array with information about the IO communications nodes.*

IOEffectStructArray

Element [IOEffectStruct](#)

Type:

Cardinality: Dynamic

Encoding: HLAVariableArray

Semantics: *Array with information about the IO effects.*

14.2.3. Fixed Record Datatypes

CommunicationsNodeIDStruct

Encoding: HLAfixedRecord

Semantics: *Specification of a Communications Node ID record. See IEEE Std 1278.1-2012 Section 6.2.48.4 for more information.*

Name	Type	Semantic
ObjectID	RTLObjectId	<i>This field shall identify the object containing the communications node.</i>
ElementID	UnsignedInteger16	<i>This field shall identify a particular communications element (e.g., radio, network component, or modem) within a given entity. If this element is a radio for which a RadioTransmitter object is being published, this value shall correspond to the RadioIndex of that object.</i>

IOCommunicationsNodeStruct

Encoding: HLAfixedRecord

Semantics: *Specification of an IO Communications Node record. See IEEE Std 1278.1-2012 Section 6.2.48.2 for more information.*

Name	Type	Semantic
CommunicationsNodeType	IOCommsNodeRecordCommsNodeTypeEnum8	<i>This field shall indicate the present type of communications node (sender, receiver, etc.).</i>
CommunicationsNodeID	CommunicationsNodeIDStruct	<i>This field shall indicate the communications node identifier for the entity that is part of a simulated communications network.</i>

IOEffectStruct

Encoding: HLAfixedRecord

Semantics: *Specification of an IO Effect record. See IEEE Std 1278.1-2012 Section 6.2.48.3 for more information.*

Name	Type	Semantic
Status	IOEffectsRecordIOStatusEnum8	<i>This field shall indicate whether the IO effect has an effect on the sender, receiver, message(s), or some combination of them.</i>

Name	Type	Semantic
LinkType	IOEffectsRecordIOLinkTypeEnum8	<i>This field shall identify the IO link type as a logical or physical link or node.</i>
Effect	IOEffectsRecordIOEffectEnum8	<i>This field shall indicate the IO effect associated with this IO attack.</i>
EffectDutyCycle	PercentUnsignedInteger8	<i>This field shall indicate the IO effect duty cycle represented as a percentage in the range of 0% to 100% in 1% increments.</i>
EffectDuration	TimeSecondUnsignedInteger16	<i>This field shall indicate the duration of the IO effect in seconds, from 1 s to 65 534 s. It shall be set to indicate IO_UNTIL_FURTHER_NOTICE (fixed value: 65 535) if the duration has no fixed simulation time interval. It shall be set to zero only if the Effect field is set to Terminate Effect (255).</i>
Process	IOEffectsRecordIOProcessEnum16	<i>This field shall indicate the IO process being performed.</i>

15. Module Switches



Information

Name:	SISO-STD-001.1-2025 - Real-time Platform Reference Switches FOM Module
Type:	FOM
Version:	3.0
Modification Date:	2025-11-04
Security Classification:	Unclassified
Purpose:	The RPR FOM supports interoperability for real-time, platform oriented defense simulation.
Application Domain:	All domains
Description:	This module contains the switches, required by the HLA standard to be part of a complete FOM.
Use Limitation:	

Other:	Copyright © 2025 by the Simulation Interoperability Standards Organization, Inc. 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA All rights reserved. Schema and API: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for all purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” Should a reader require additional information, contact the SISO Inc. Board of Directors. Documentation: SISO hereby grants a general, royalty-free license to copy, distribute, display, and make derivative works from this material, for noncommercial purposes, provided that any use of the material contains the following attribution: “Reprinted with permission from SISO Inc.” The material may not be used for a commercial purpose without express written permission from the SISO Inc. Board of Directors. SISO Inc. Board of Directors 7901 4th St N, Suite 300-4043 St Petersburg FL 33702 USA
---------------	--

Primary author Point Of Contact

Name:	RPR FOM Product Development Group
Organization:	SISO - Simulation Interoperability Standards Organization
Telephone:	
Email:	prowe@sisostandards.org

References

Text Document	Standard for Guidance, Rationale, and Interoperability Modalities for the Real-time Platform Reference Federation Object Model (RPR FOM) SISO-STD-001-2025 4 November 2025
Text Document	IEEE Standard for Distributed Interactive Simulation - Application Protocols IEEE Std 1278.1-2012 19 December 2012

15.2. Switches

Auto Provide	Disabled
Convey Region Designator Sets	Disabled
Convey Producing Federate	Disabled
Attribute Scope Advisory	Disabled
Attribute Relevance Advisory	Disabled
Object Class Relevance Advisory	Disabled
Interaction Relevance Advisory	Disabled
Service Reporting	Disabled
Exception Reporting	Disabled
Delay Subscription Evaluation	Disabled
Automatic Resign Action	CancelThenDeleteThenDivest